ORDER NO. 00AC-G913,00AC-G915

MFG NO. B1700734,B1700737

ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

COVER				SH. No.	TC0001
SIGNATURE		DATE	TITLE		
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM		
CHKD.	T. Kawata	2014-04-18			
APPD.	Y. Fukushima	2014-04-18			
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CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
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DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
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CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
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DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
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TC0828	EXTERNAL OUTPUT COM DO(8)	352	3
TC0829	EXTERNAL OUTPUT COM DO(9)	353	3
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TC0831	EXTERNAL OUTPUT COM DO(11)	355	3
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INDEX 6		SH. No.	TC0002E
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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REVISION MANAGEMENT SHEET

REV.NO. / DATE	REVISIONS	DWN.	CHKD.	APPD.	RELATED SH.NO.	REMARKS

REVISION MANAGEMENT SHEET			SH. No.	TC0003
SIGNATURE		DATE	TITLE	
DWN.	(S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T.Kawata	2014-04-18		
APPD.	Y.Fukushima	2014-04-18		
REGD.	Hitachi,Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 8	REV. 3

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NO.	DEVICE NO.	SH. NO.
1	DDC1TC0403	TC0403
2	DDC2TC0403	TC0403
3	DDC3TC0403	TC0403
4	DDC4TC0403	TC0403
5	DDC5TC0402	TC0402
6	DDC6TC0401	TC0401
7	DDC7TC0401	TC0401
8	DFB1TC0421	TC0421
9	DFB2TC0421	TC0421
10	DFB3TC0421	TC0421
11	DFB4TC0421	TC0421
12	DFC1TC0420A	TC0420A
13	DFC2TC0419	TC0419
14	DFC3TC0419	TC0419
15	DFC4TC0419	TC0419
16	DFC5TC0419	TC0419
17	DFD1TC0420A	TC0420A
18	DFD2TC0420	TC0420
19	DFD3TC0420	TC0420
20	DFD4TC0420	TC0420
21	DFD5TC0420	TC0420
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NO.	DEVICE NO.	SH. NO.
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NO.	DEVICE NO.	SH. NO.
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NO.	DEVICE NO.	SH. NO.
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NO.	DEVICE NO.	SH. NO.
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NO.	DEVICE NO.	SH. NO.
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RELAY LIST

SH. No. TC0005

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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SYMBOL LIST

No.	NAME	SYMBOL
1	Normally open contact (Make)	
2	Normally close contact (Break)	
3	Limit switch(Make)	
4	Limit switch(Break)	
5	Make contact delayed When closing	
6	Break contact delayed When opening	
7	Make contact delayed When re-opening	
8	Break contact delayed When re-closing	
9	Push-button switch(Make) (non-locking)	E
10	Push-button switch(Break) (non-locking)	E
11	Manually operated switch (locking)	
12	Turn switch (spring return)	
13	Turn switch (non-spring return)	
14	Control switch (Manual return type)	
15	Control switch (Self return type)	

No.	NAME	SYMBOL
16	Circuit breaker control switch with lamp (Rotation after pull, self return type)	G R
17	Make contact without spring return(Make)	
18	Break contact without spring return(break)	
19	Circuit breaker	
20	Change-over break before contack	
21	Shielded conductors	
22	Twisted pair conductors	
23	Relay coil (slow-operating relay)	
24	Relay coil (slow-releasing relay)	
25	Oparating coil (Relay coil)	
26	Keep relay coil	
27	Resistor	
28	Fuse	
29	Aux. PT	U V u v

No.	NAME	SYMBOL
30 -1	Voltage transformer(VT)	
30 -2	Transformer	
31	Voltage test terminal	
32	Current test terminal & Link (BAR)	
33	Current transformer (General)	
34	Free wheeling diode	
35	Test block	
36	AND	
37	OR	
38	NOT	
39	Time delay close	TDC A B TIME
40	Crossing of conductor not connected	
41	Junction	
42	Junction of connected wire	

No.	NAME	SYMBOL
43	Circuit terminal	
44	Capacition	
45	Ground connection PE-BUS: Panel earth bus IE-BUS: Instrument earth bus	
46	Indicator V:Voltmeter A:Amperemeter W:Active power meter var:Reactive power meter F,Hz:Frequency meter cos or PF:Power Factor meter Wh:Watt Hour Meter varh:Var Hour Meter	W var Hz V
47	Panel or Rack	ABBREVIATION OF PANEL (TCS) PANEL No.
48	Instrument,Device	
49	Lamp with resistor R:Red G:Green W:White	R
50	LED,Lamp	
51	Disconnect switch Earting switch	M
52	Manual operated and residual contact(Make)	

PROJECT DWG.No.	AK-0-EE-AC-42-001		
SYMBOL LIST		SH. No.	TC0006
SIGNATURE		DATE	TITLE
DWN.	(S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T.Kawata	2014-04-18	
APPD.	Y.Fukushima	2014-04-18	
REGD.	Hitachi,Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 10 / REV. 3

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DELETED

CONTROLLER CONSTRUCTION (1)		SH. No.	TC0020
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 11	REV. 3

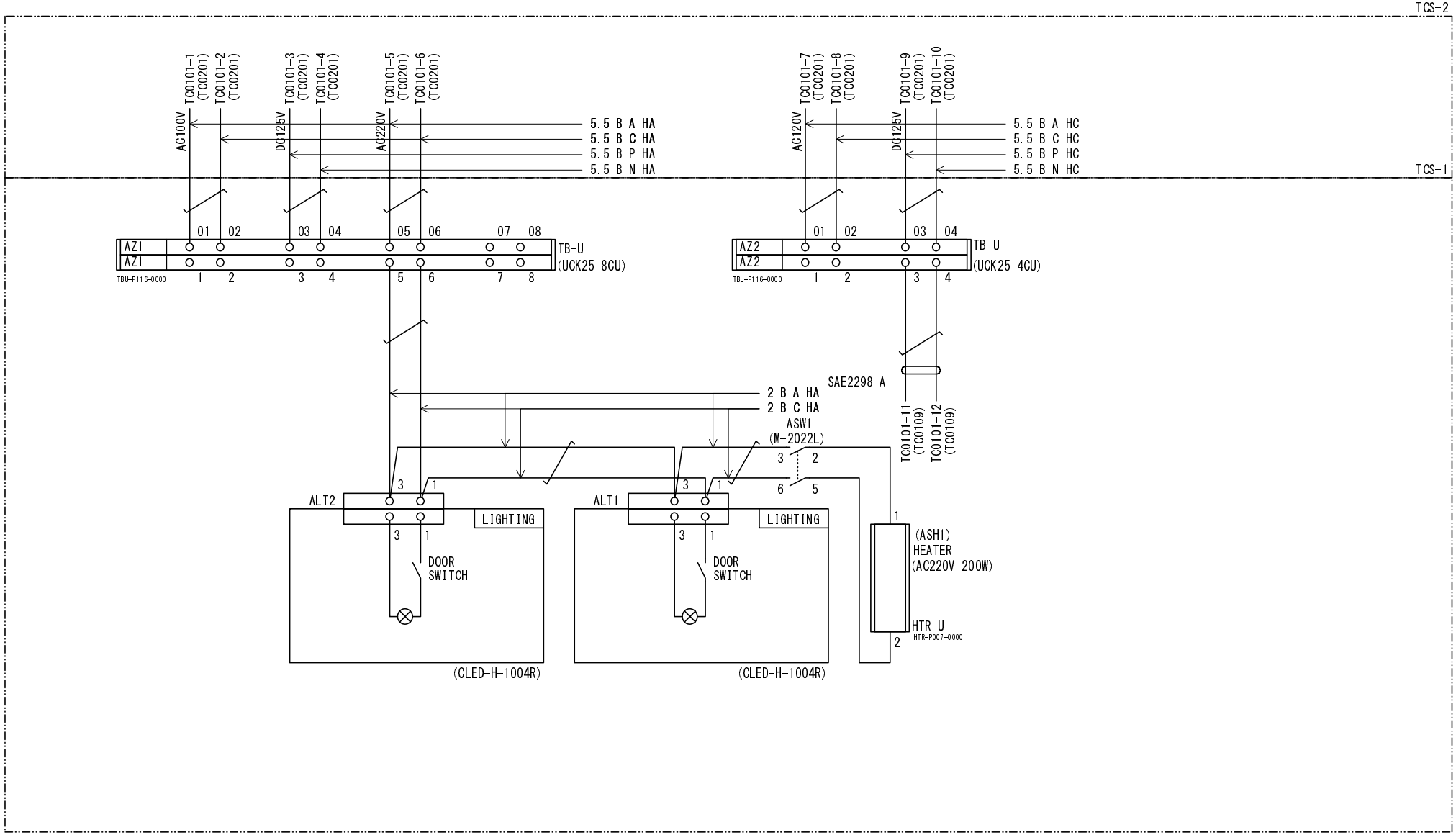
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SIGNATURE	DATE	TITLE	
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CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 12	REV. 3

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DELETED

CONTROLLER CONSTRUCTION (3)		SH. No.	TC0022
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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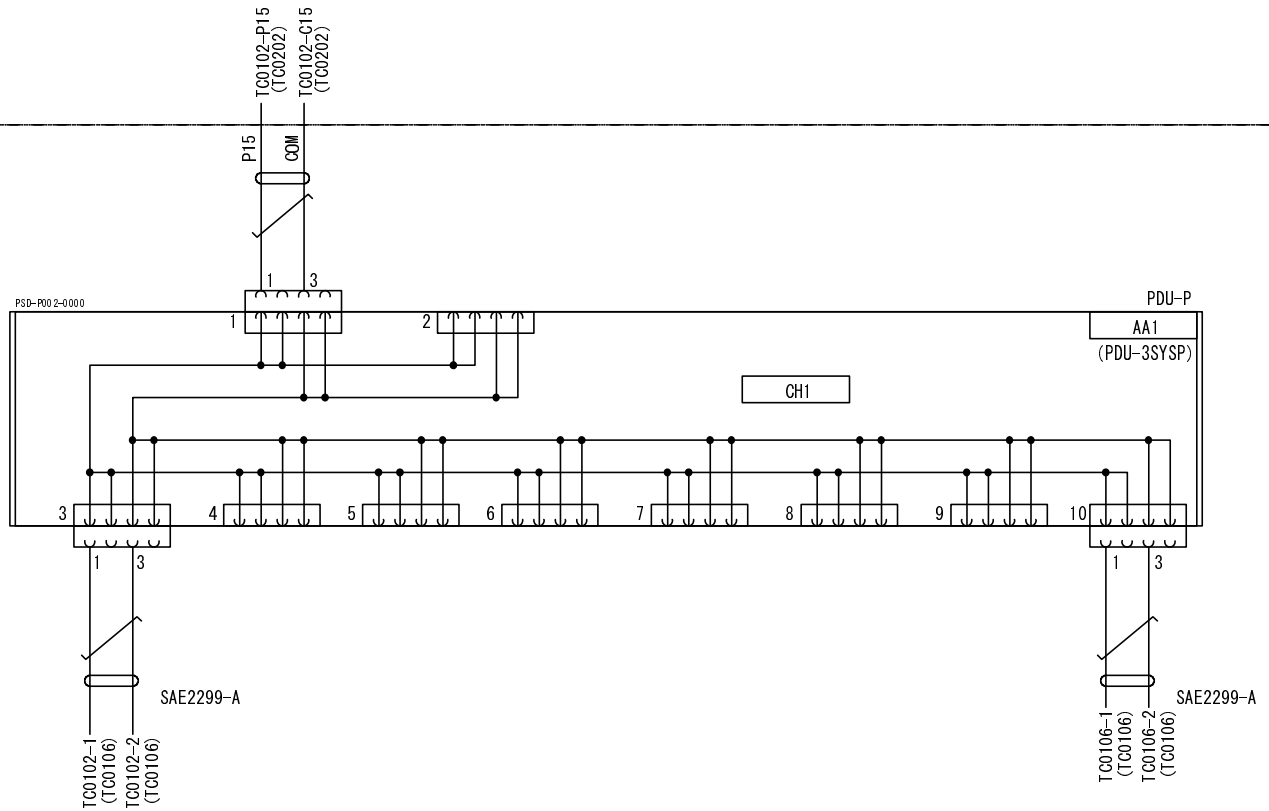


POWER SUPPLY SYSTEM(1)		SH. No.	TC0101
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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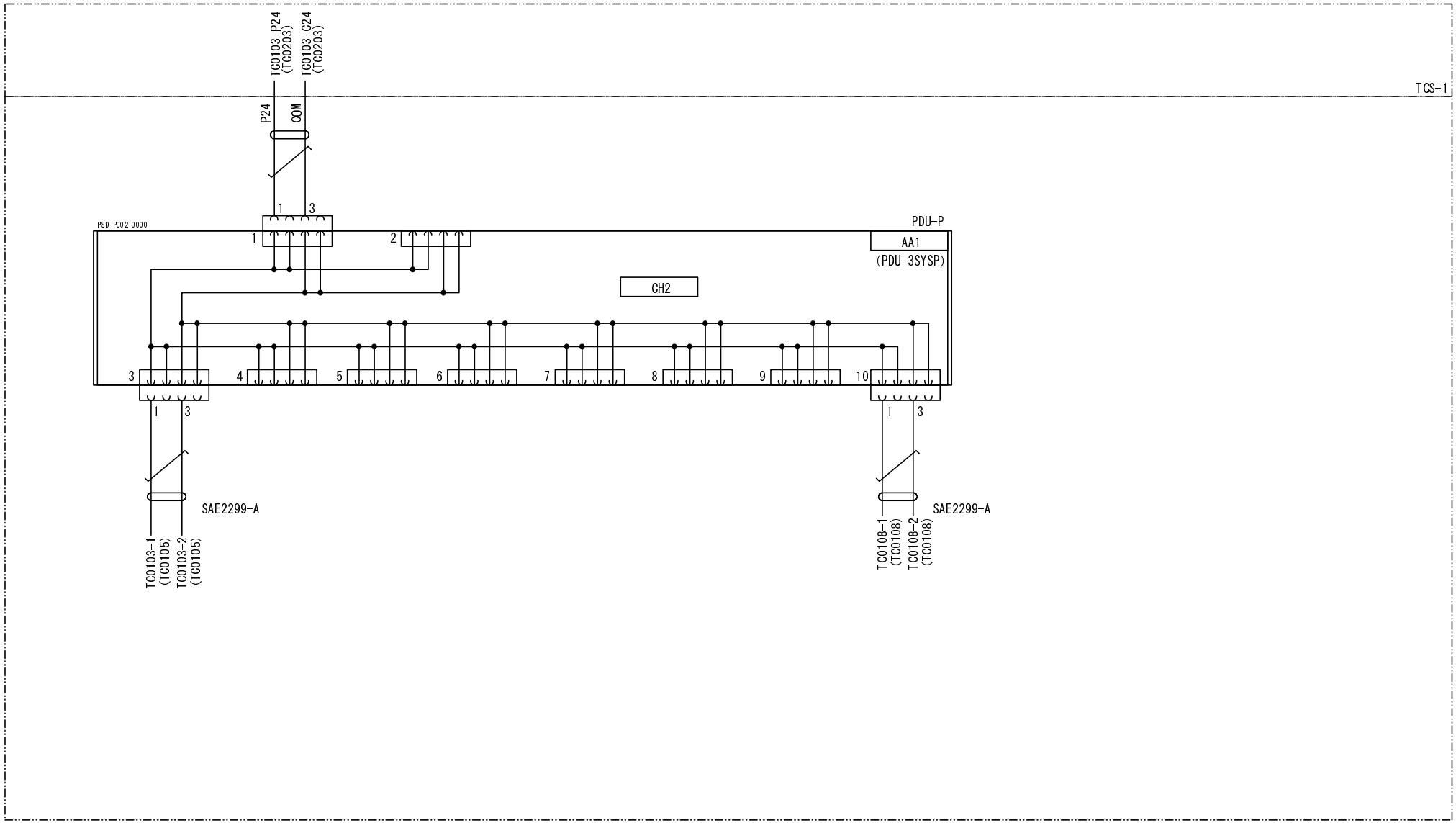
TCS-2

TCS-1



POWER SUPPLY SYSTEM(2)		SH. No.	TC0102
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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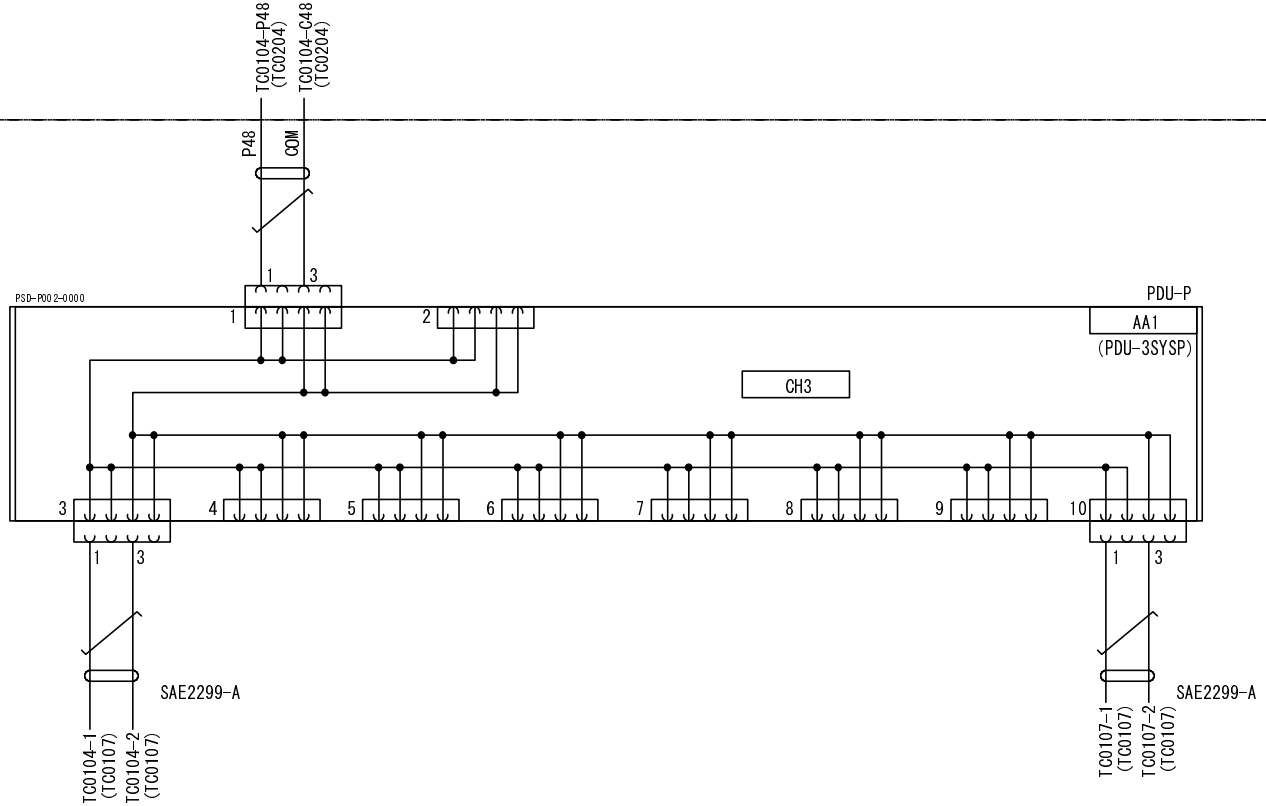
2016/07/28 TC0102 XSMM-TC



POWER SUPPLY SYSTEM(3)		SH. No.	TC0103
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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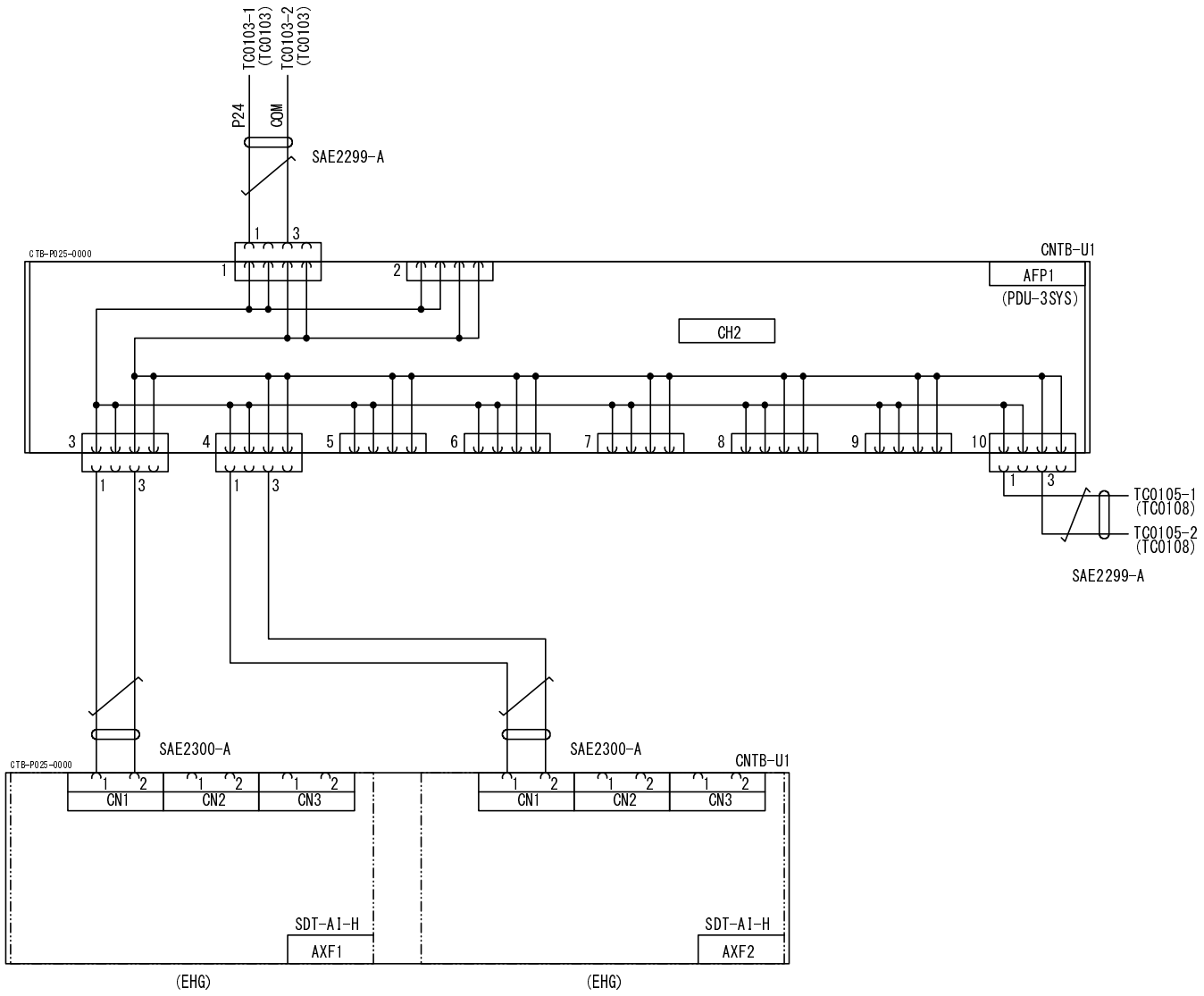
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TCS-1



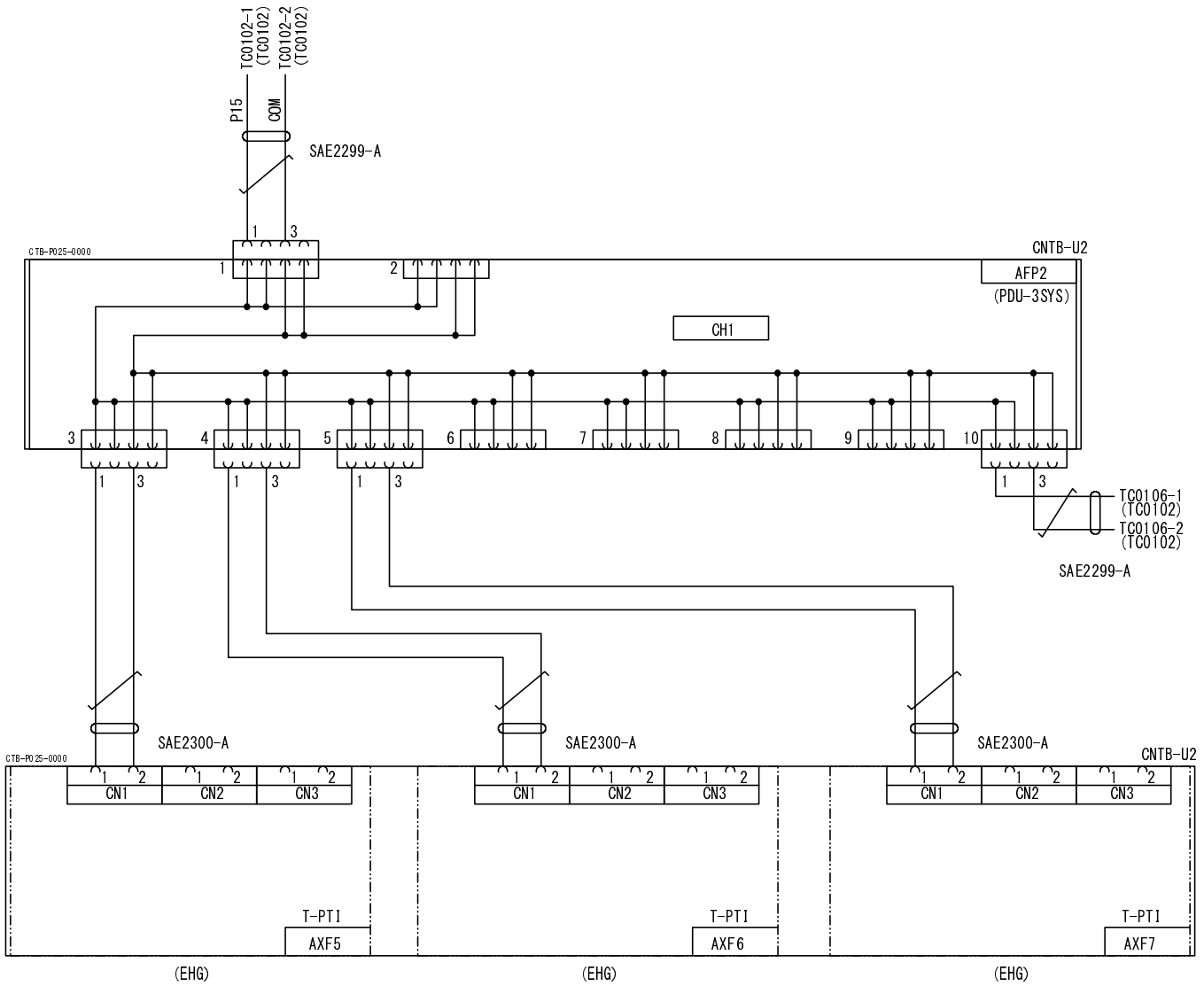
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SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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TCS-1



POWER SUPPLY SYSTEM(5)		SH. No.	TC0105
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
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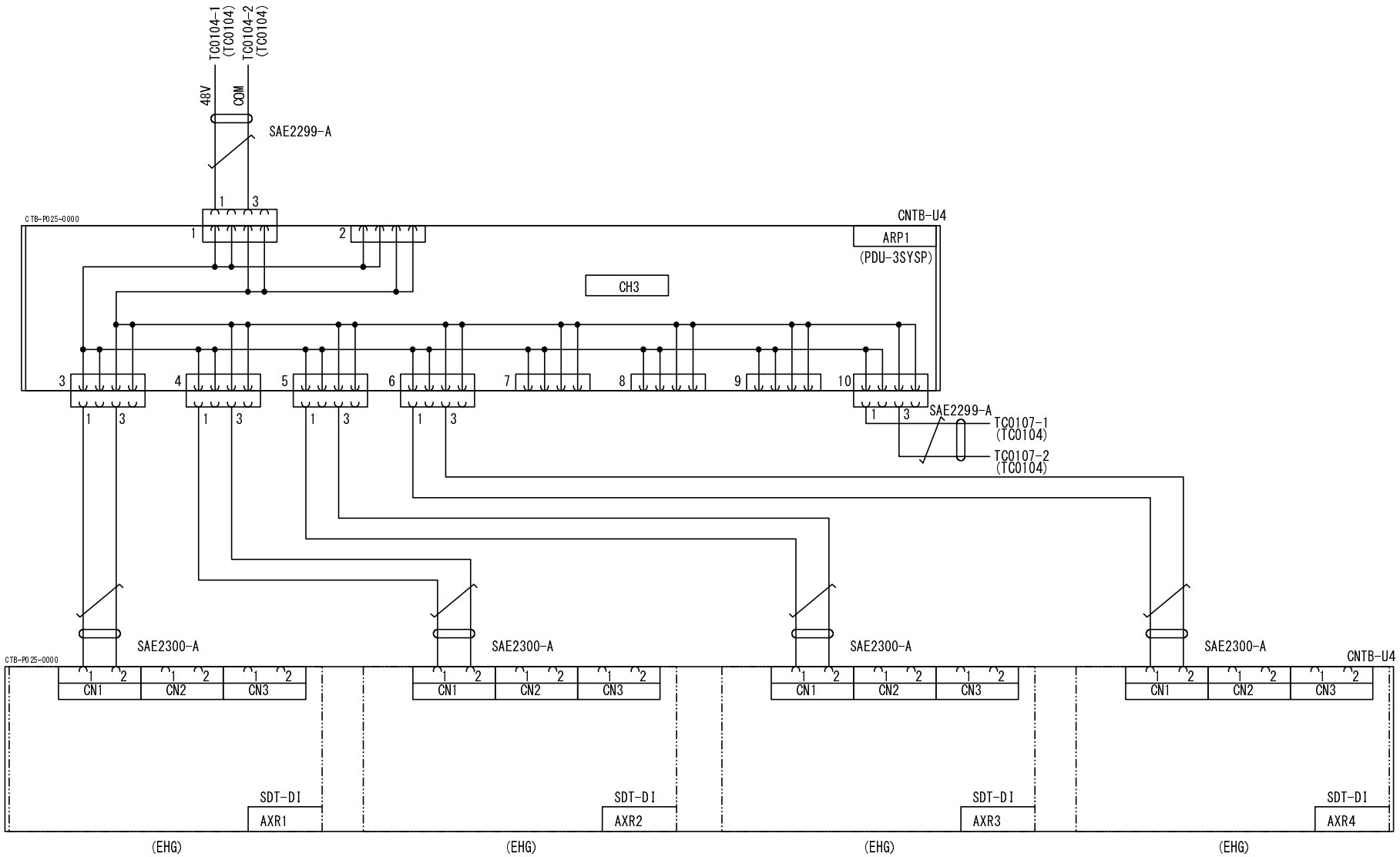
TCS-1



POWER SUPPLY SYSTEM(6)			SH. No.	TC0106
SIGNATURE		DATE	TITLE	
DWN.	(S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T.Kawata	2014-04-18		
APPD.	Y.Fukushima	2014-04-18		
REGD.	Hitachi,Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 19	REV. 3

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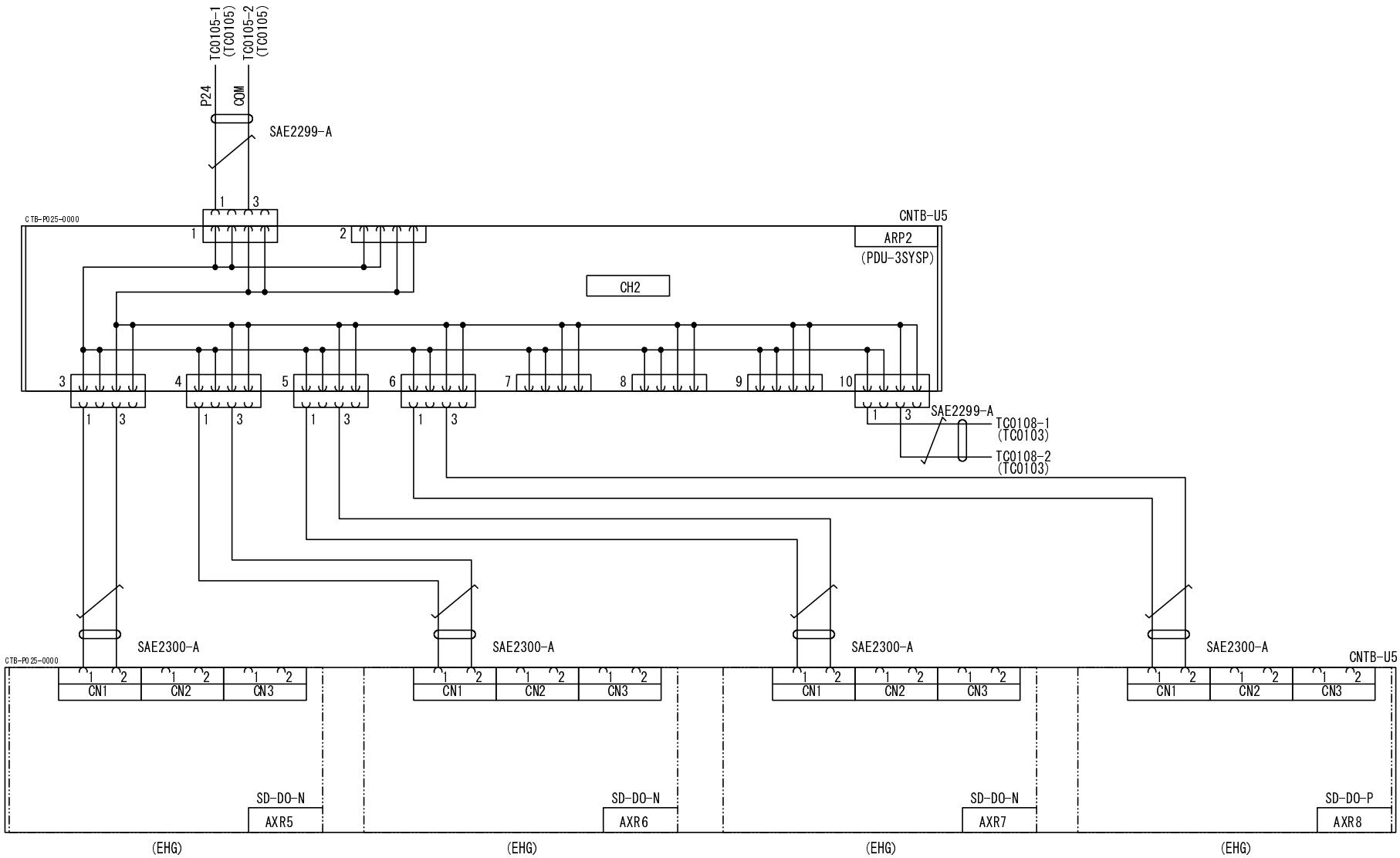
TCS-1



POWER SUPPLY SYSTEM(7)		SH. No.	TC0107
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
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2016/07/28 TC0107 XSMM-TC

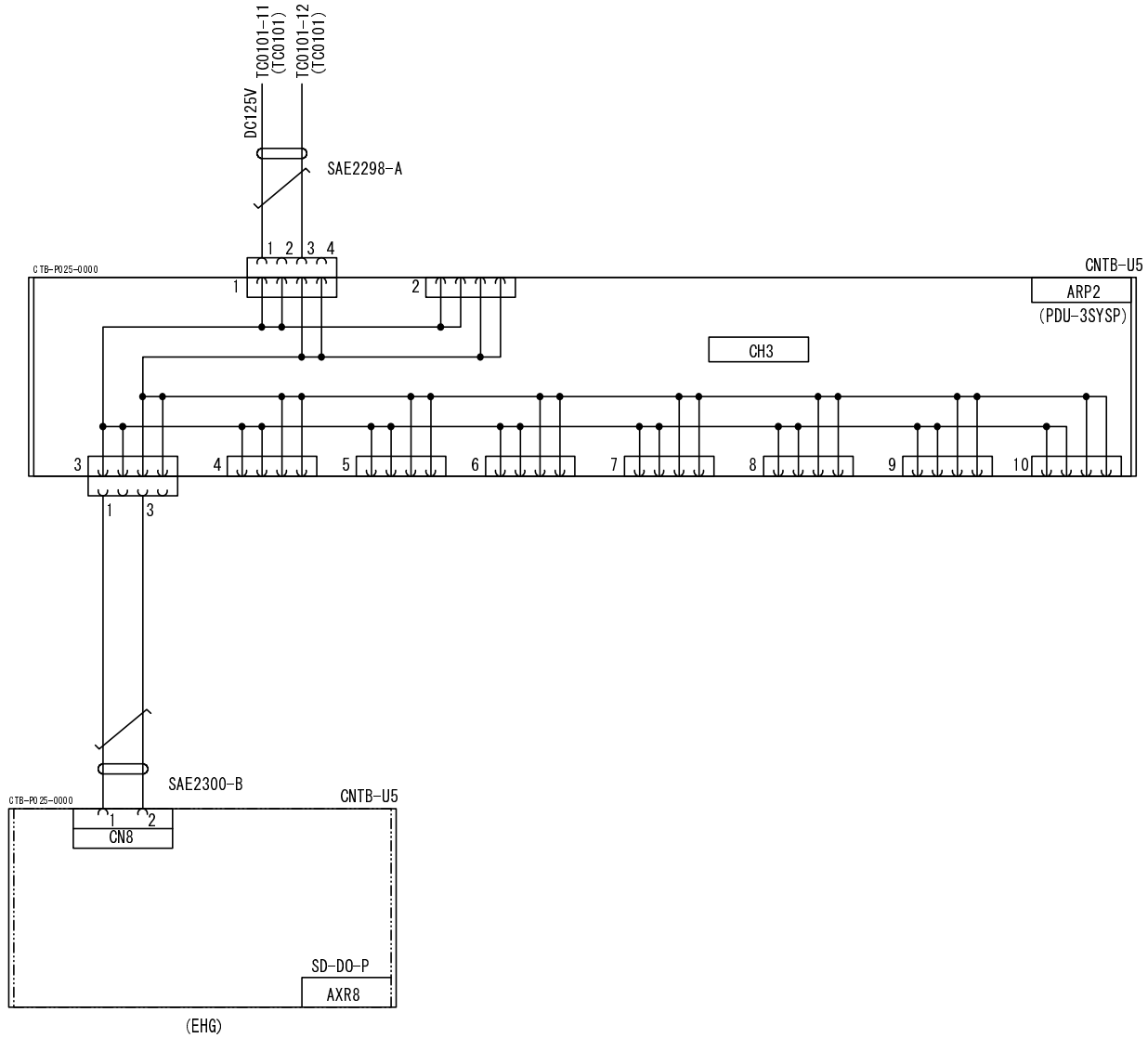
TCS-1



POWER SUPPLY SYSTEM(8)		SH. No.	TC0108
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
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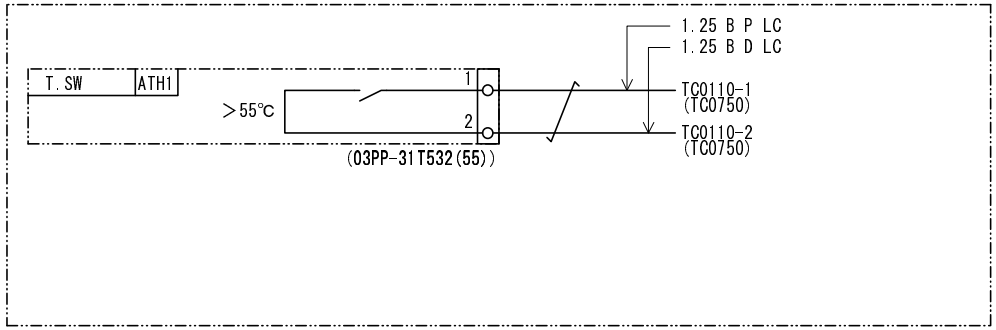
2016/07/28 TC0108 XSMM-TC

TCS-1



POWER SUPPLY SYSTEM(9)		SH. No.	TC0109
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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TCS-1

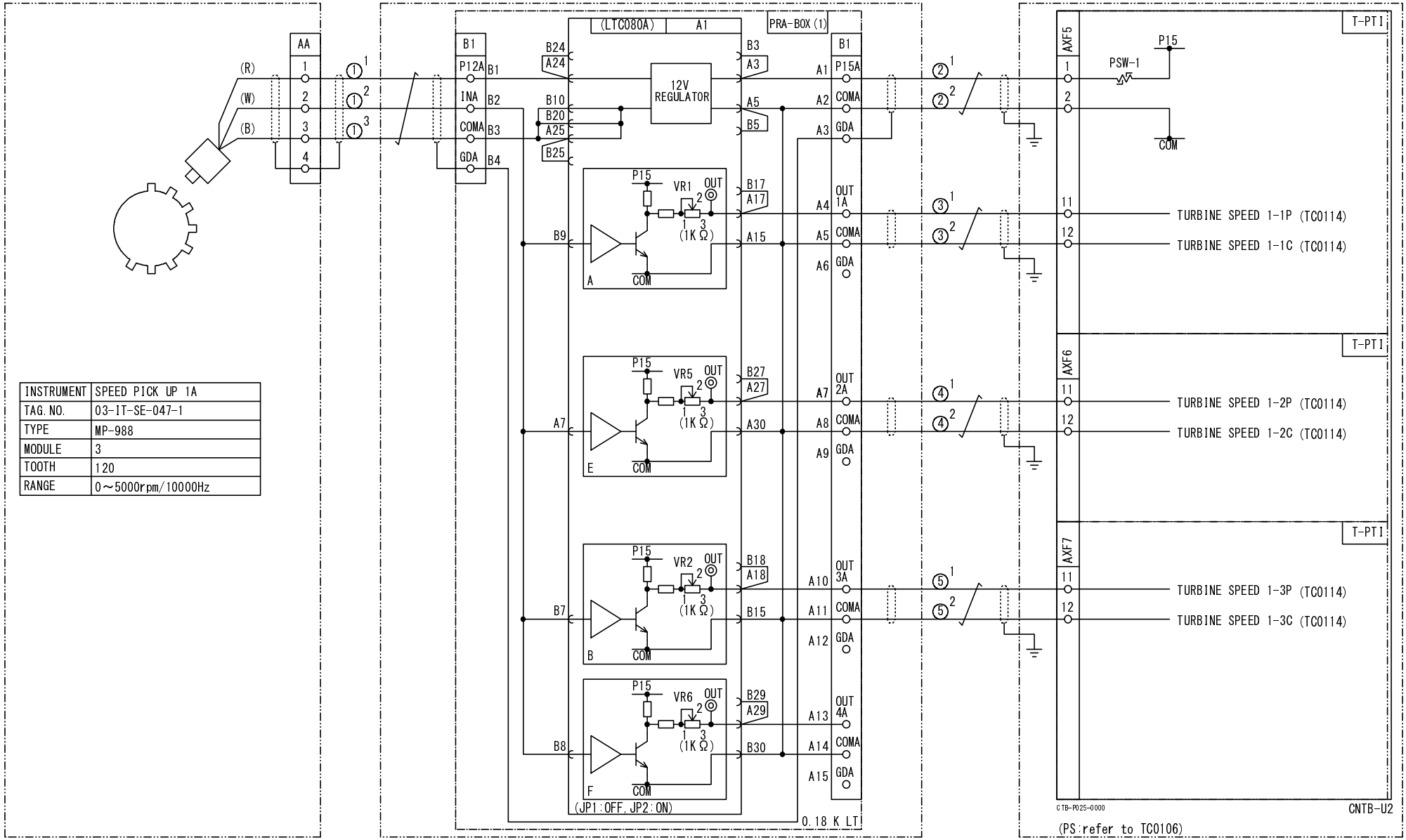


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SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 23 / REV. 3

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TERMINAL BOX

TCS-1



INSTRUMENT	SPEED PICK UP 1A
TAG.NO.	03-IT-SE-047-1
TYPE	MP-988
MODULE	3
TOOTH	120
RANGE	0～5000rpm/10000Hz

INDEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK

TURBINE SPEED INPUT(1)

SH. No. TC0111

	SIGNATURE	DATE
DWN.	(S.Kikuchi)	2014-04-18
CHKD.	T.Kawata	2014-04-18
APPD.	Y.Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

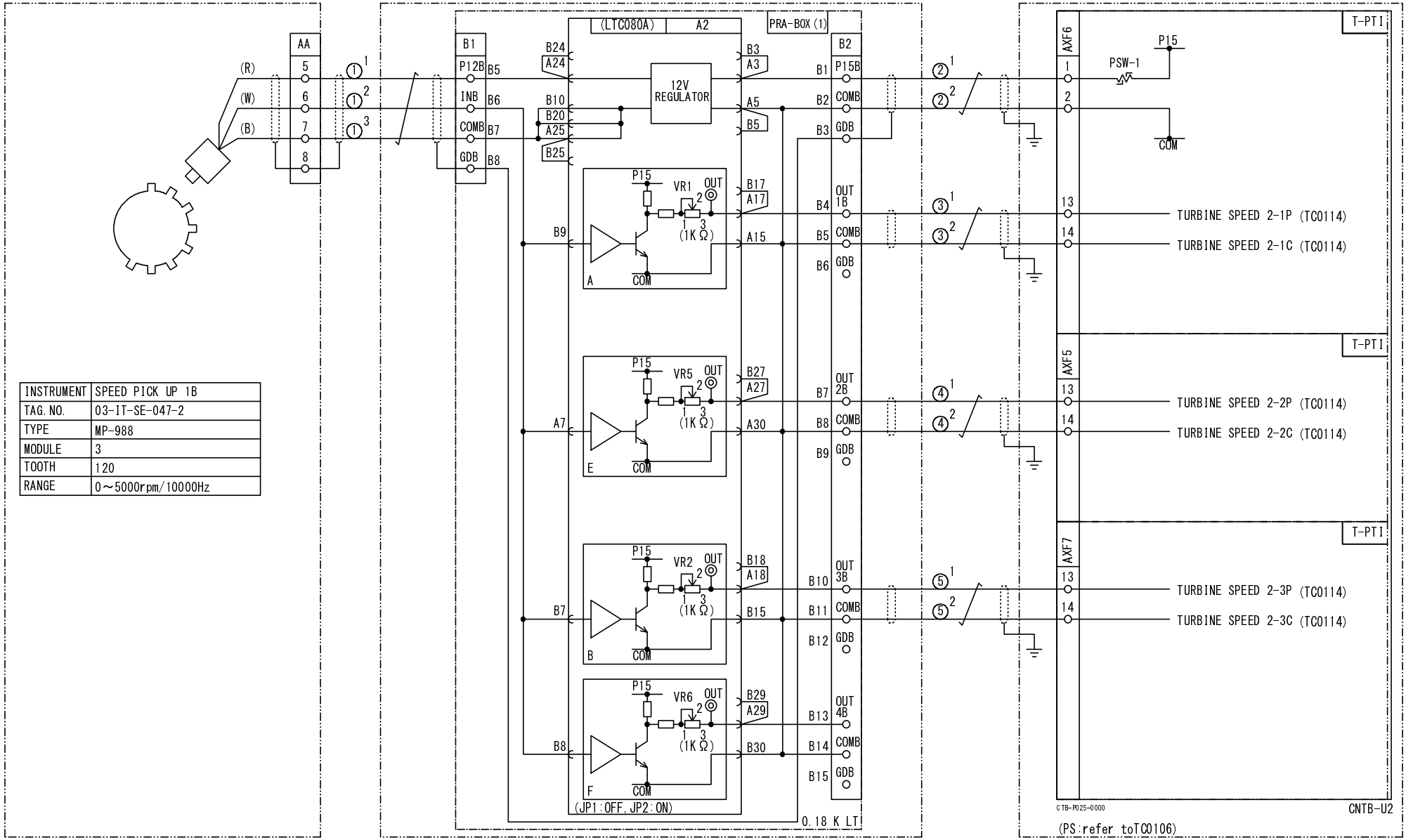
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TERMINAL BOX

TCS-1



INSTRUMENT	SPEED PICK UP 1B
TAG.NO.	03-IT-SE-047-2
TYPE	MP-988
MODULE	3
TOOTH	120
RANGE	0～5000rpm/10000Hz

INDEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK

TURBINE SPEED INPUT(2)

SH. No. TC0112

	SIGNATURE	DATE
DWN.	(S.Kikuchi)	2014-04-18
CHKD.	T.Kawata	2014-04-18
APPD.	Y.Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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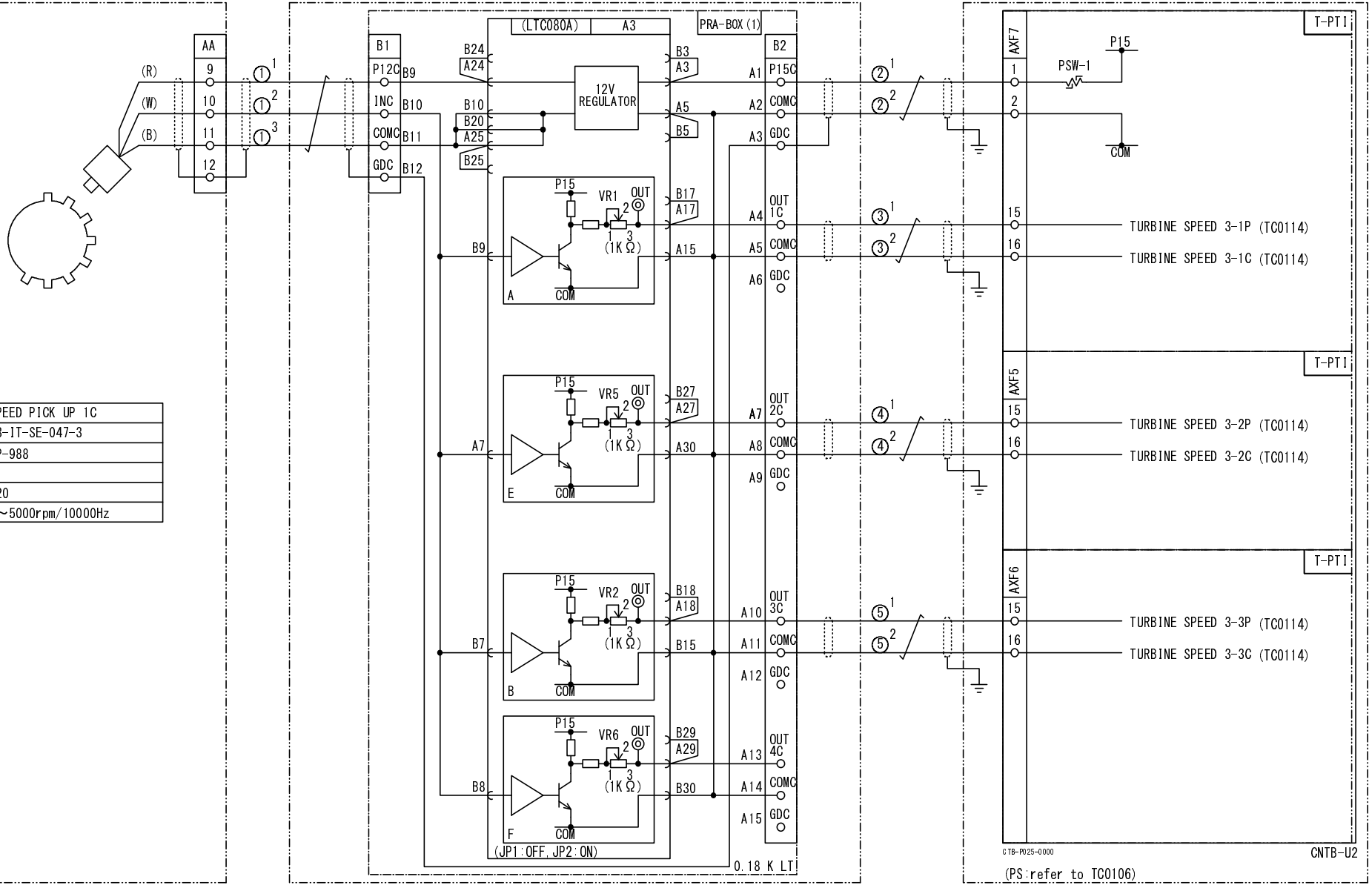
REV. 3

2016/07/28 TC0112 XSMM-TC

TERMINAL BOX

TCS-1

INSTRUMENT	SPEED PICK UP 1C
TAG. NO.	03-IT-SE-047-3
TYPE	MP-988
MODULE	3
TOOTH	120
RANGE	0～5000rpm/10000Hz



INDEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK

TURBINE SPEED INPUT(3)

SH. No. TC0113

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

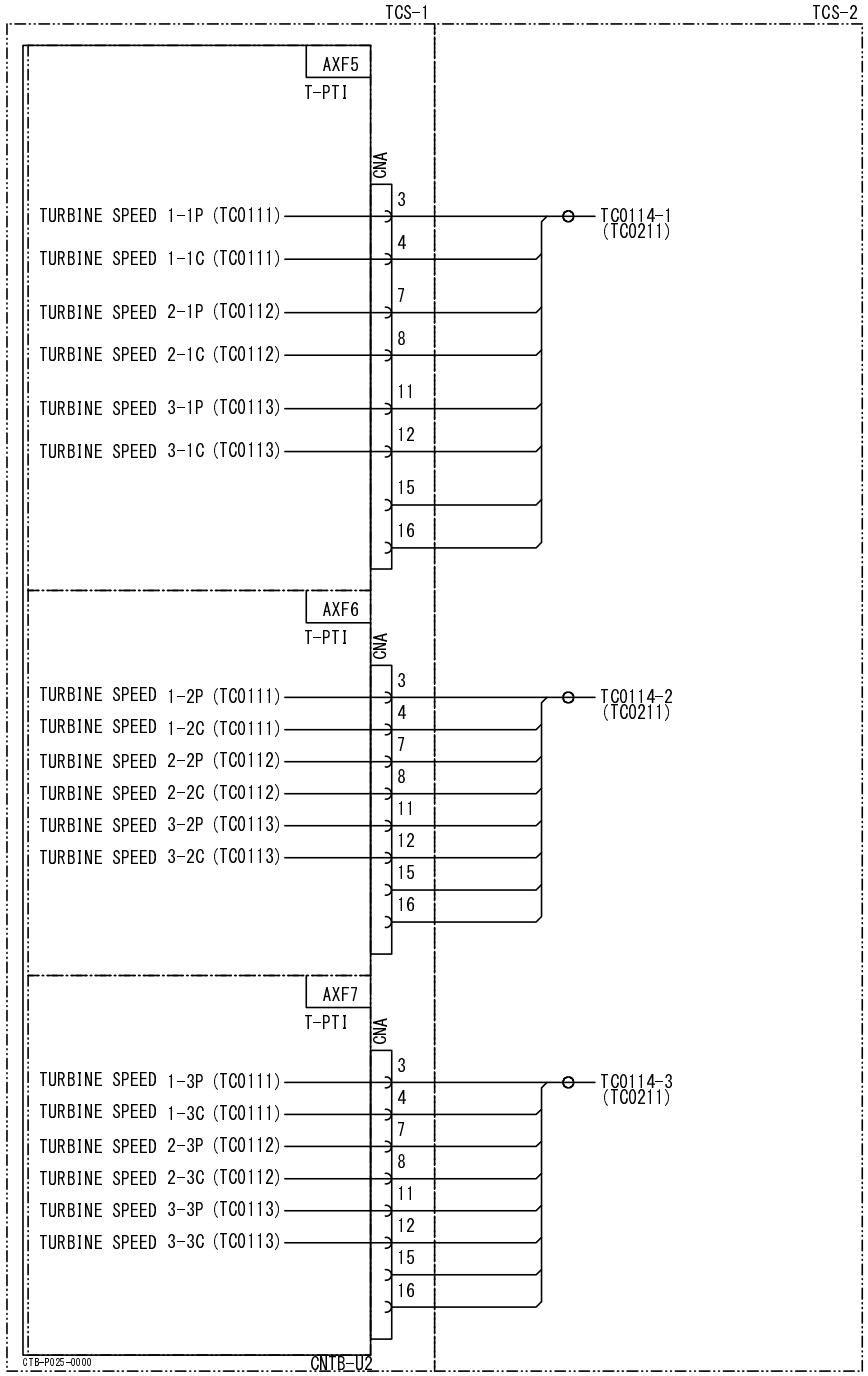
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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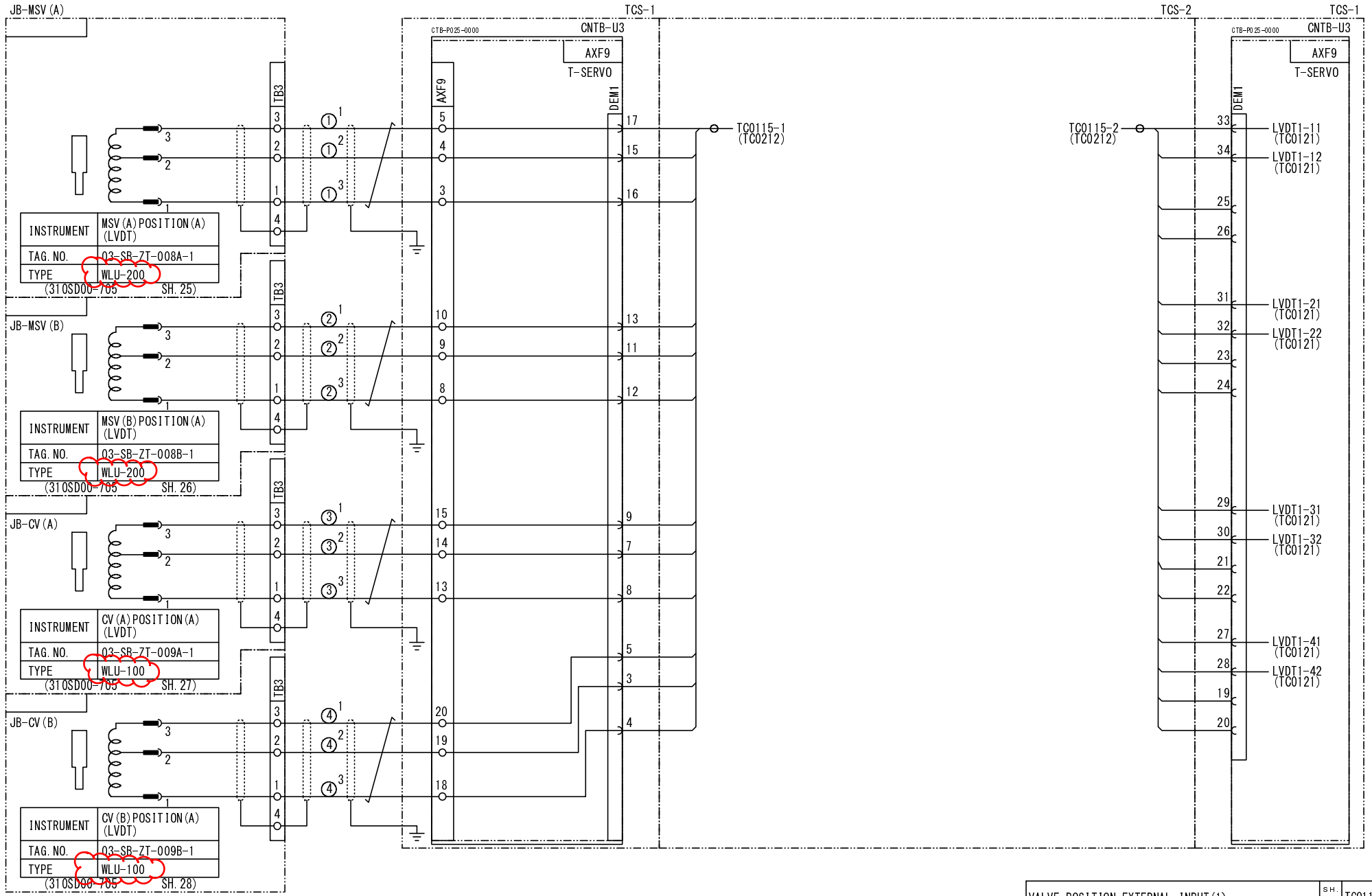
REV. 3

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TURBINE SPEED INPUT(4)				SH. No.	TC0114
SIGNATURE		DATE	TITLE		
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM		
CHKD.	T. Kawata	2014-04-18			
APPD.	Y. Fukushima	2014-04-18			
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IN-DEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK

VALVE POSITION EXTERNAL INPUT (1)

SH. No. TC0115

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

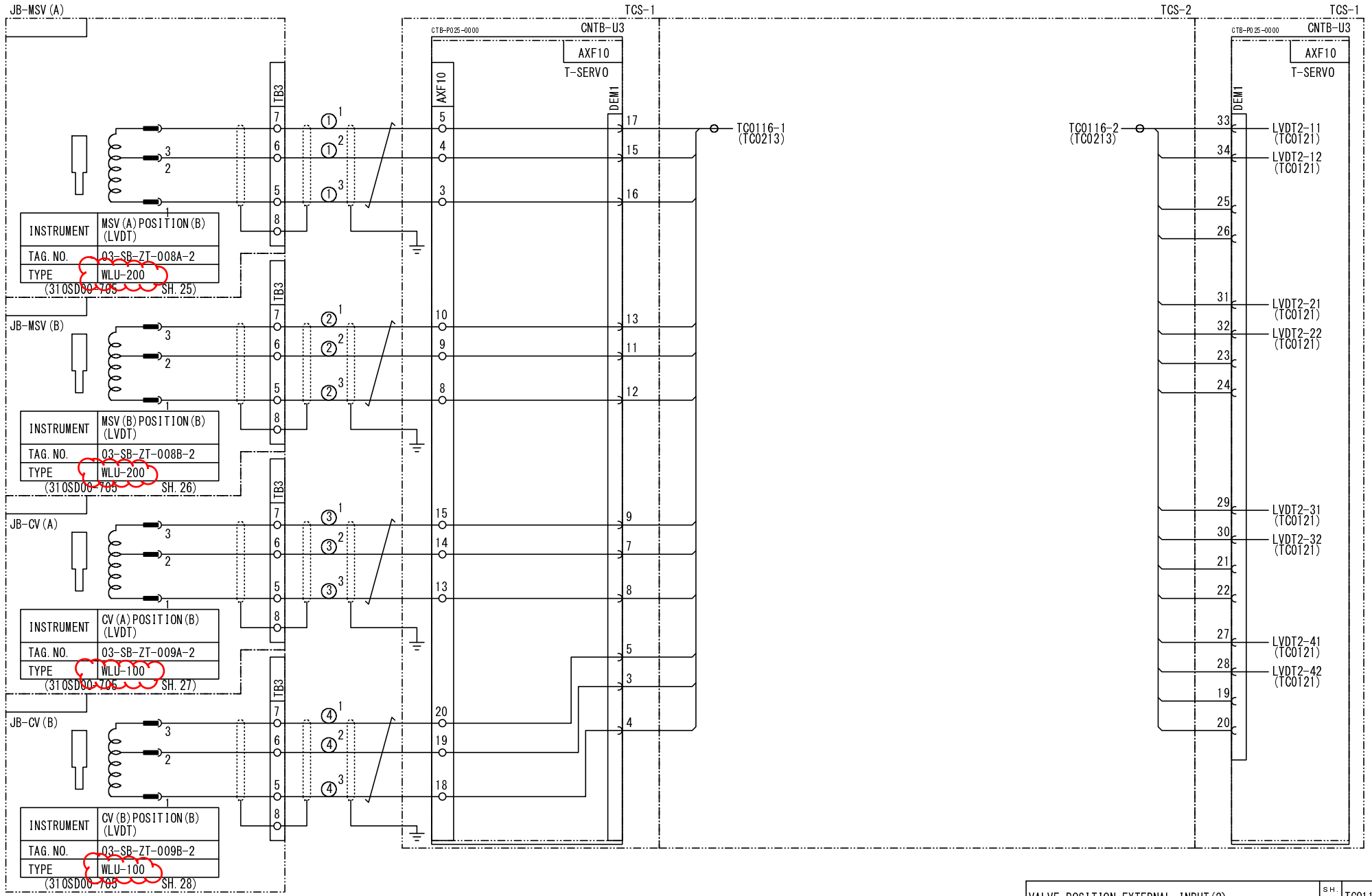
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DWG. No. 331DM07733

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IN-DEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK

VALVE POSITION EXTERNAL INPUT (2)

SH. No. TC0116

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

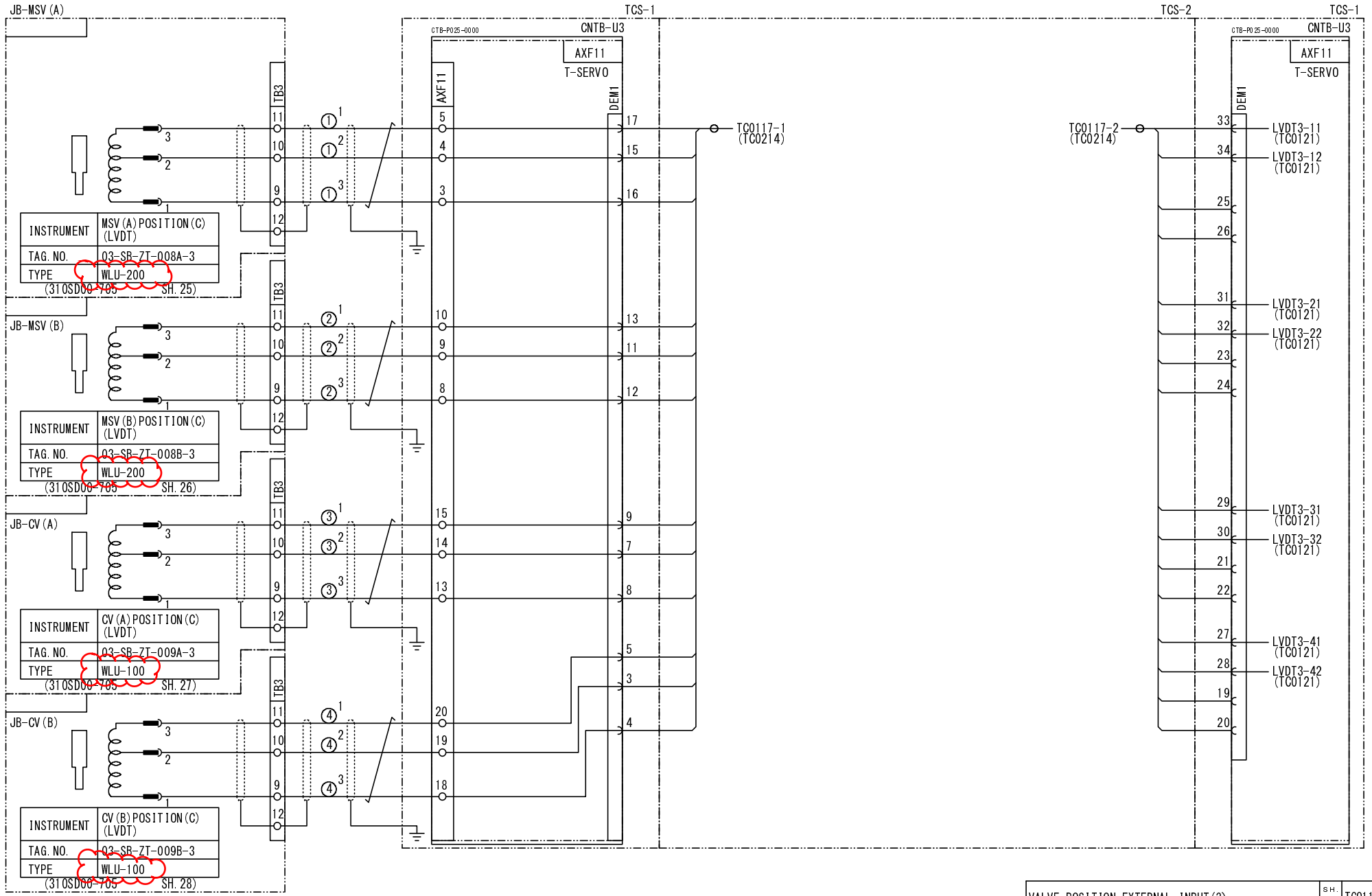
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0116 XSMM-TC



IN-DEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK

VALVE POSITION EXTERNAL INPUT (3)

SH. No. TC0117

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

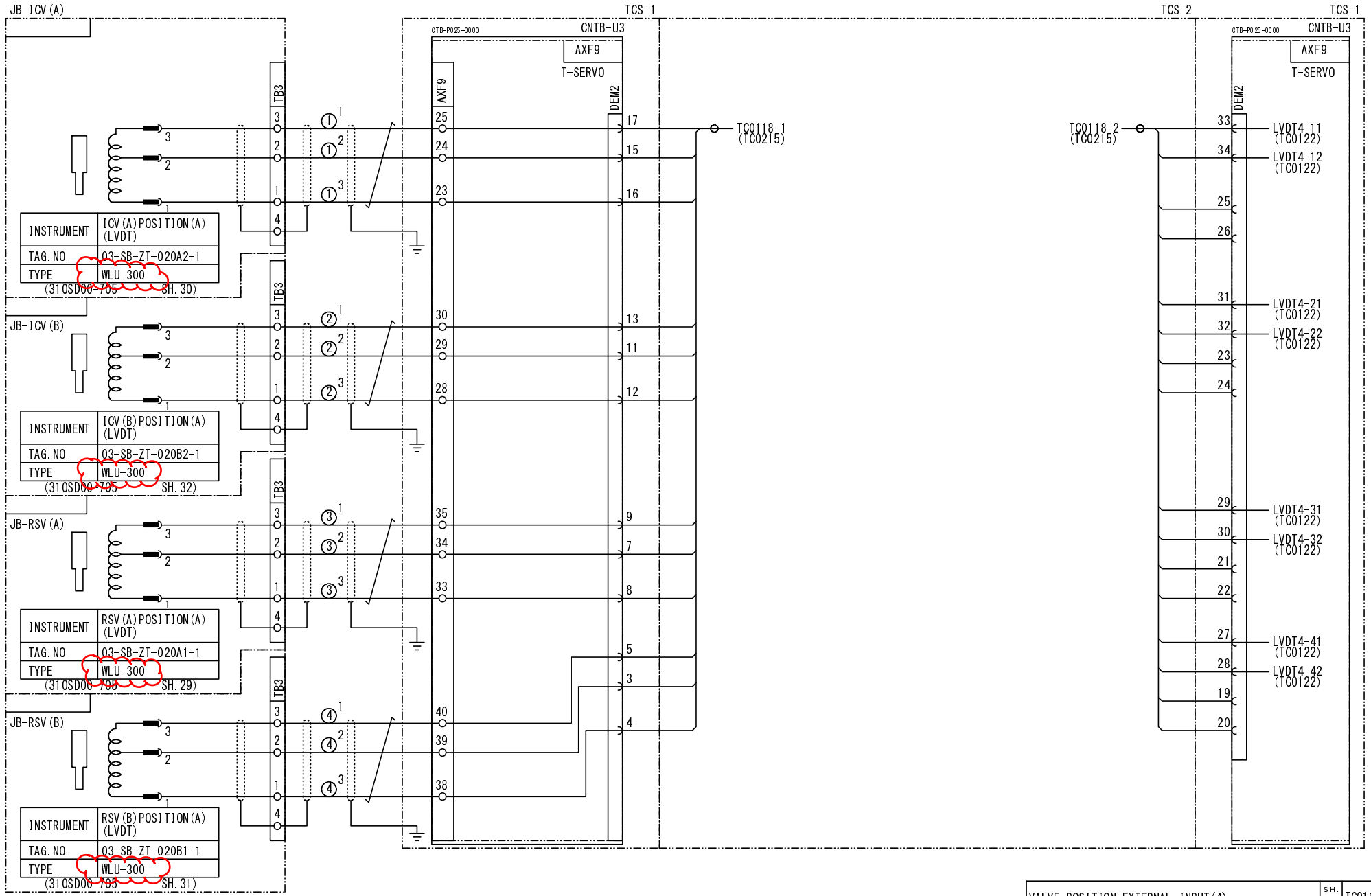
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0117 XSMM-TC



IN-DEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK

VALVE POSITION EXTERNAL INPUT(4)

SH. No. TC0118

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

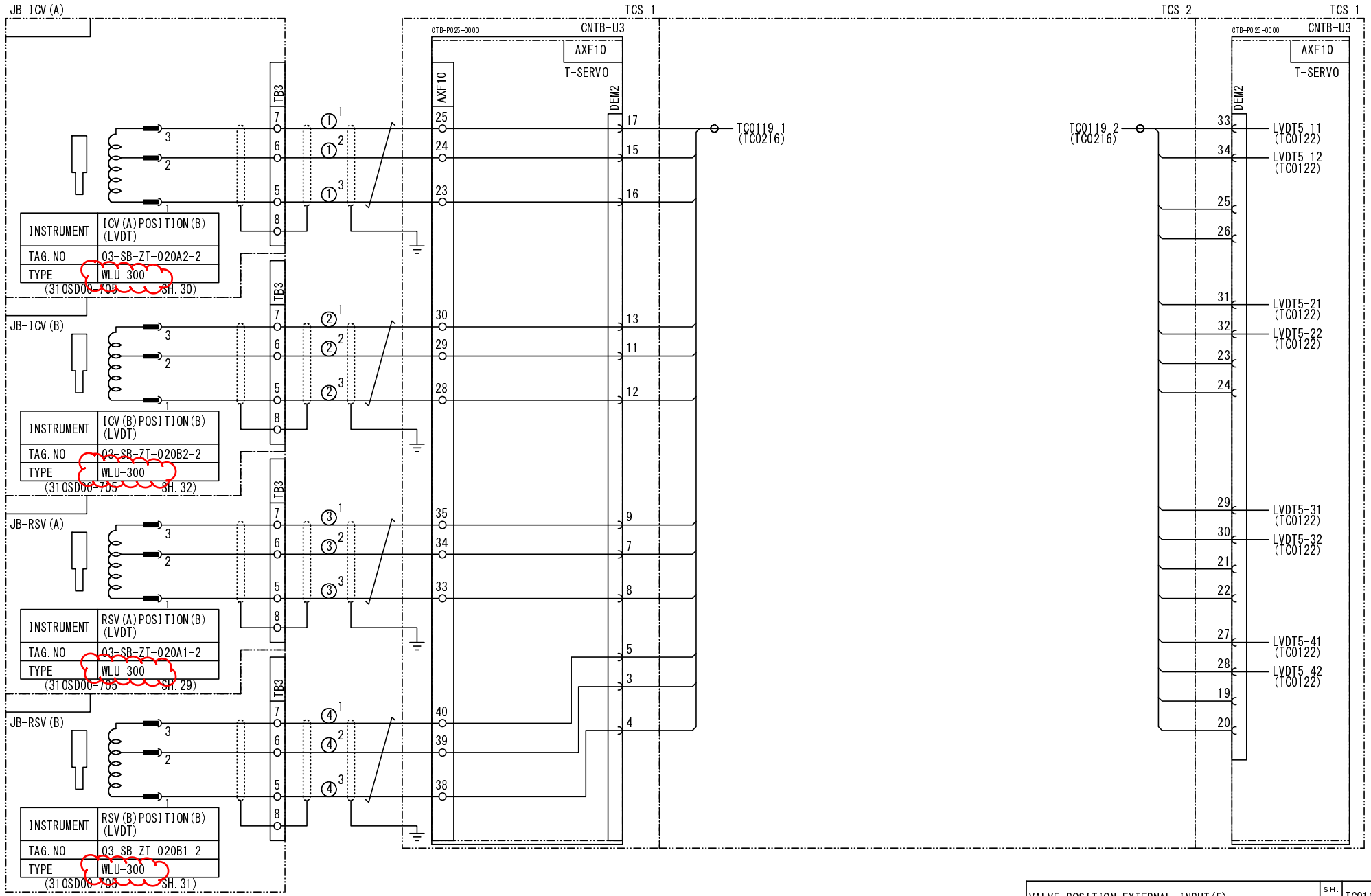
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0118 XSMM-TC



IN-DEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK

VALVE POSITION EXTERNAL INPUT(5)

SH. No. TC0119

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

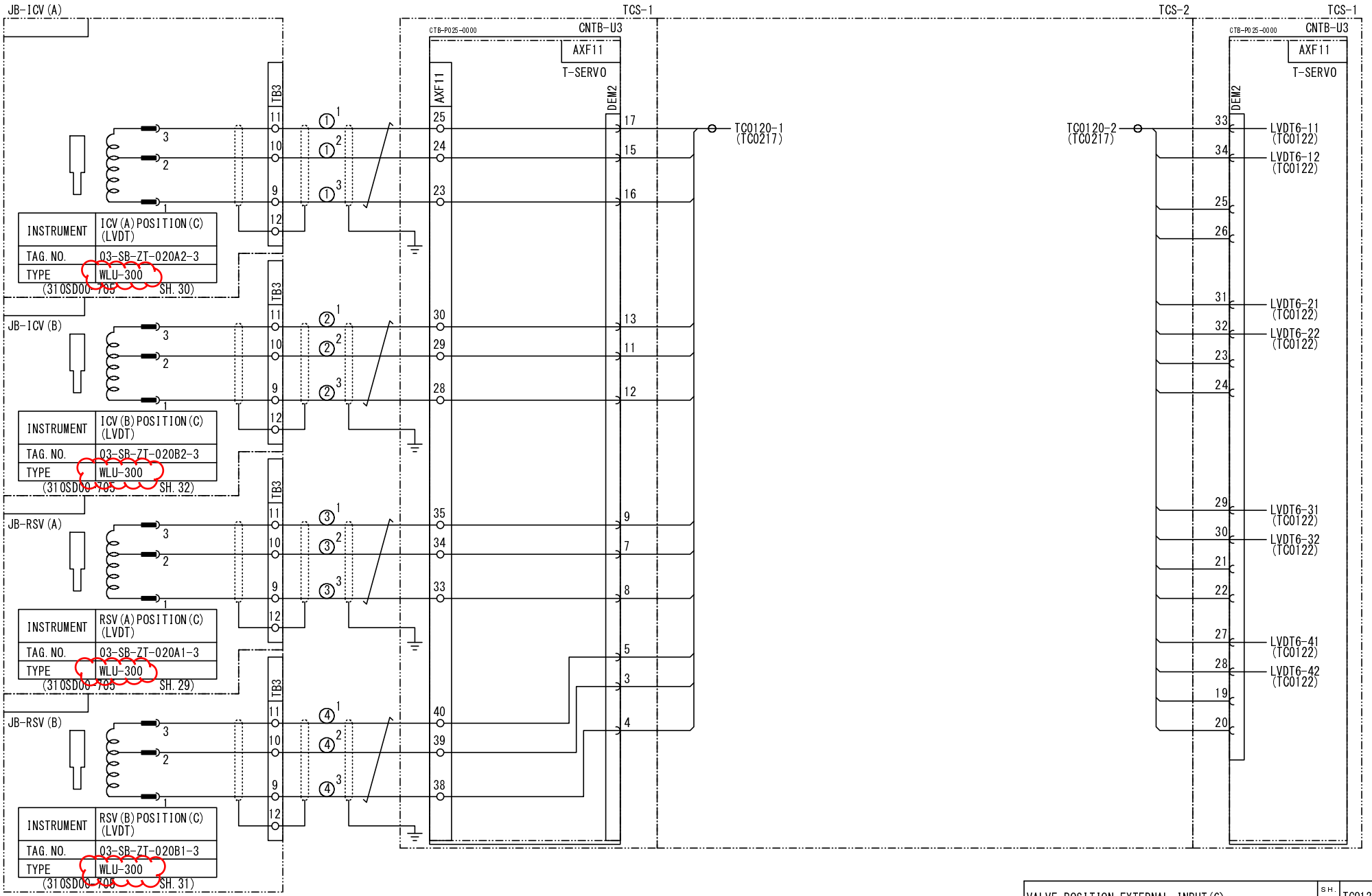
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0119 XSMM-TC



IN-DEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK

VALVE POSITION EXTERNAL INPUT(6)

SH. No. TC0120

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

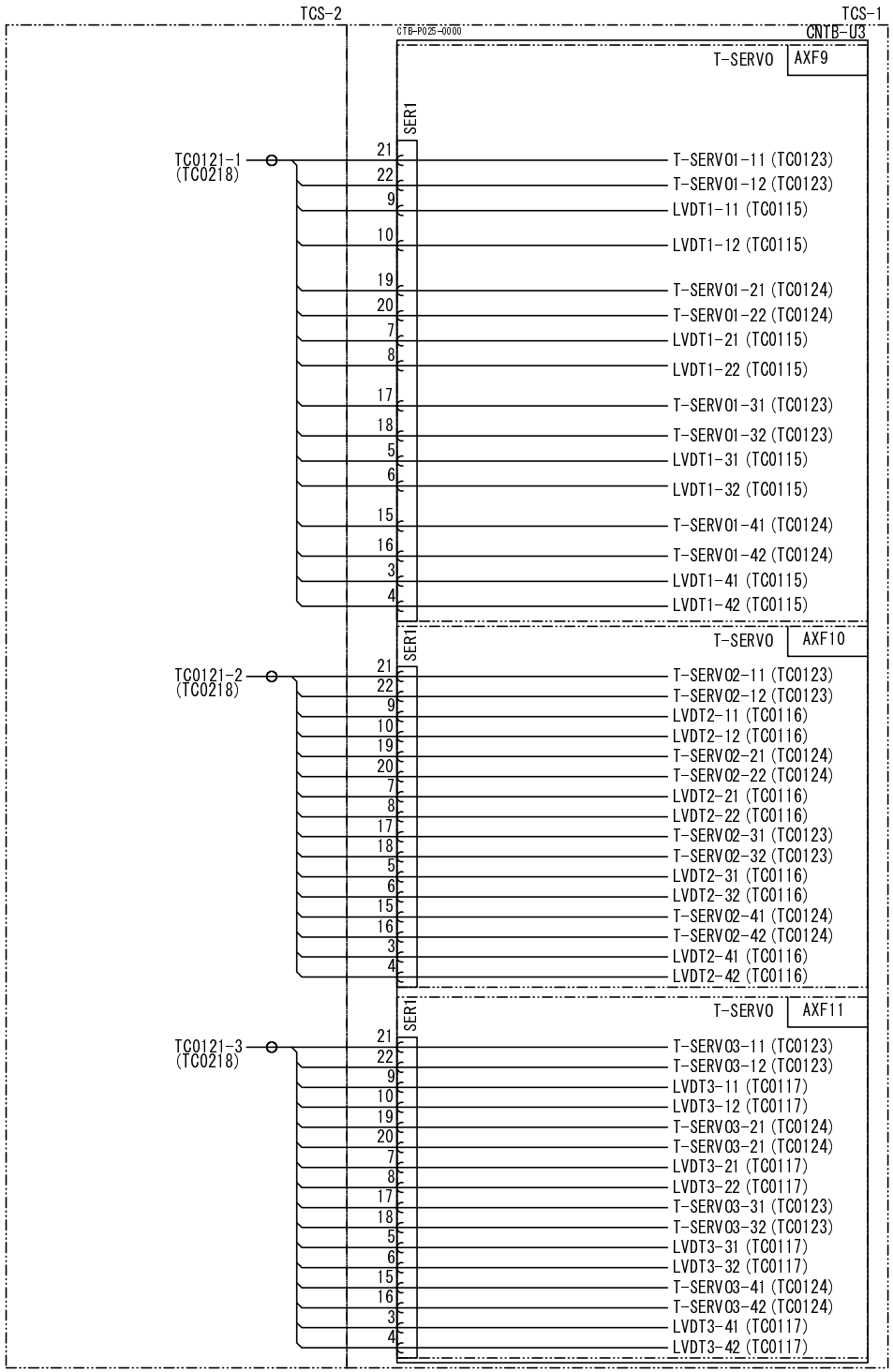
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0120 XSMM-TC



OUTPUT EXTERNAL SERVO VALVE(1)

SH. No. TC0121

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE
ELEMENTARY DIAGRAM
OF
TURBINE CONTROL SYSTEM

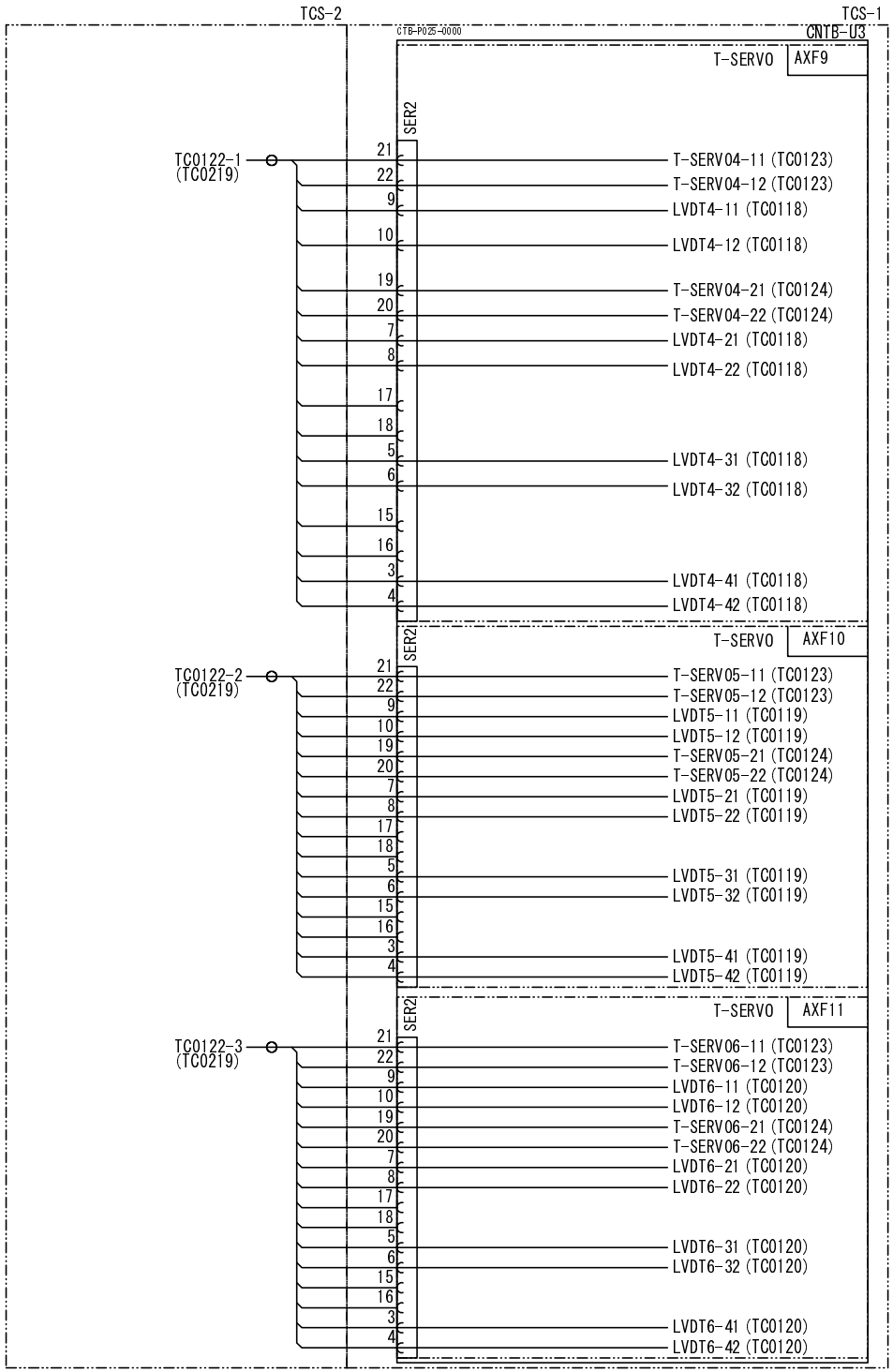
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

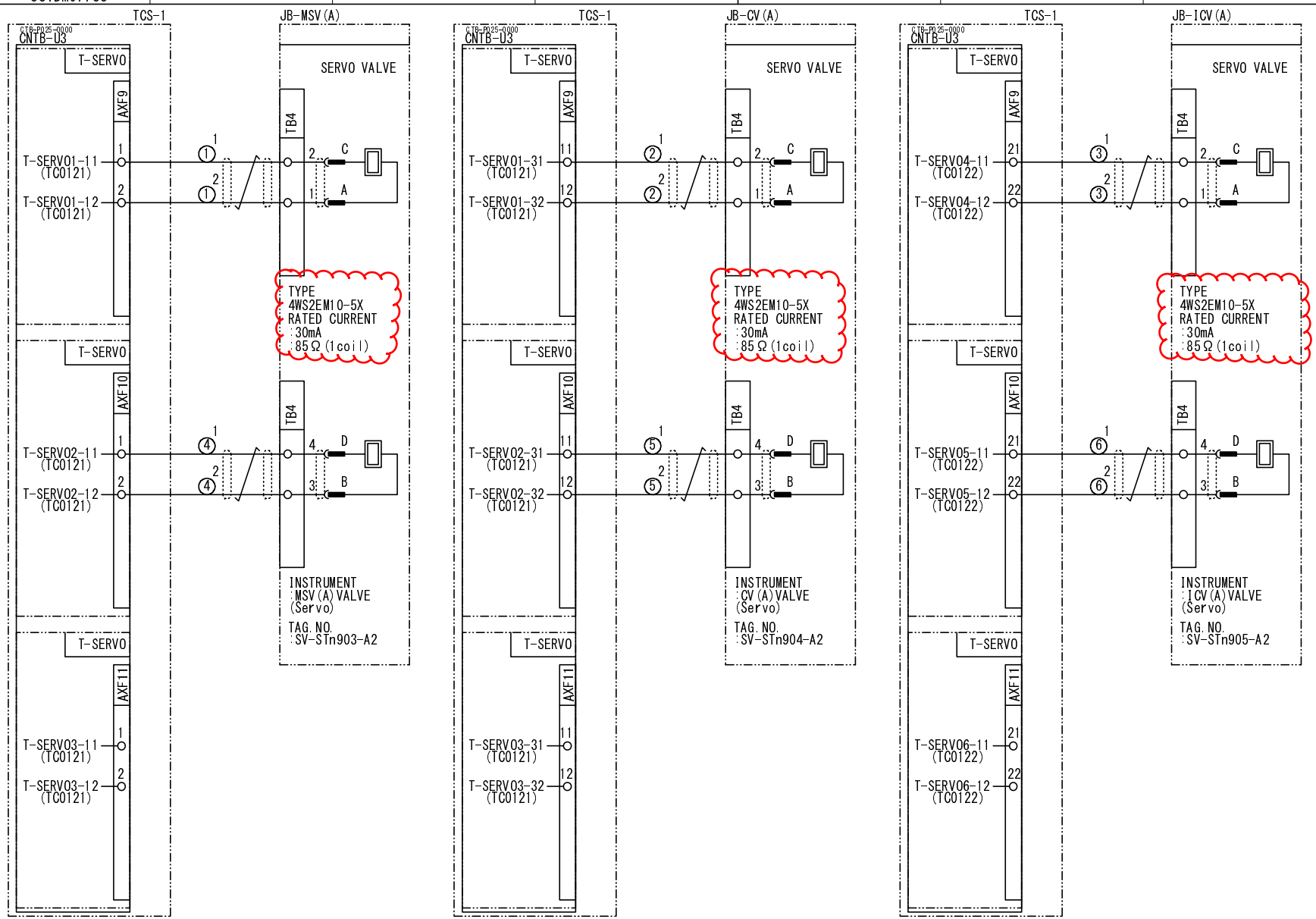
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OUTPUT EXTERNAL SERVO VALVE(2)		SH. No.	TC0122
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 35	REV. 3



INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
③		6/6	TCS-1				
②		6/6	TCS-1				
①		6/6	TCS-1				

OUTPUT EXTERNAL SERVO VALVE(3)

SH. No. TC0123

	SIGNATURE	DATE
DWN.	(S.Kikuchi)	2014-04-18
CHKD.	T.Kawata	2014-04-18
APPD.	Y.Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

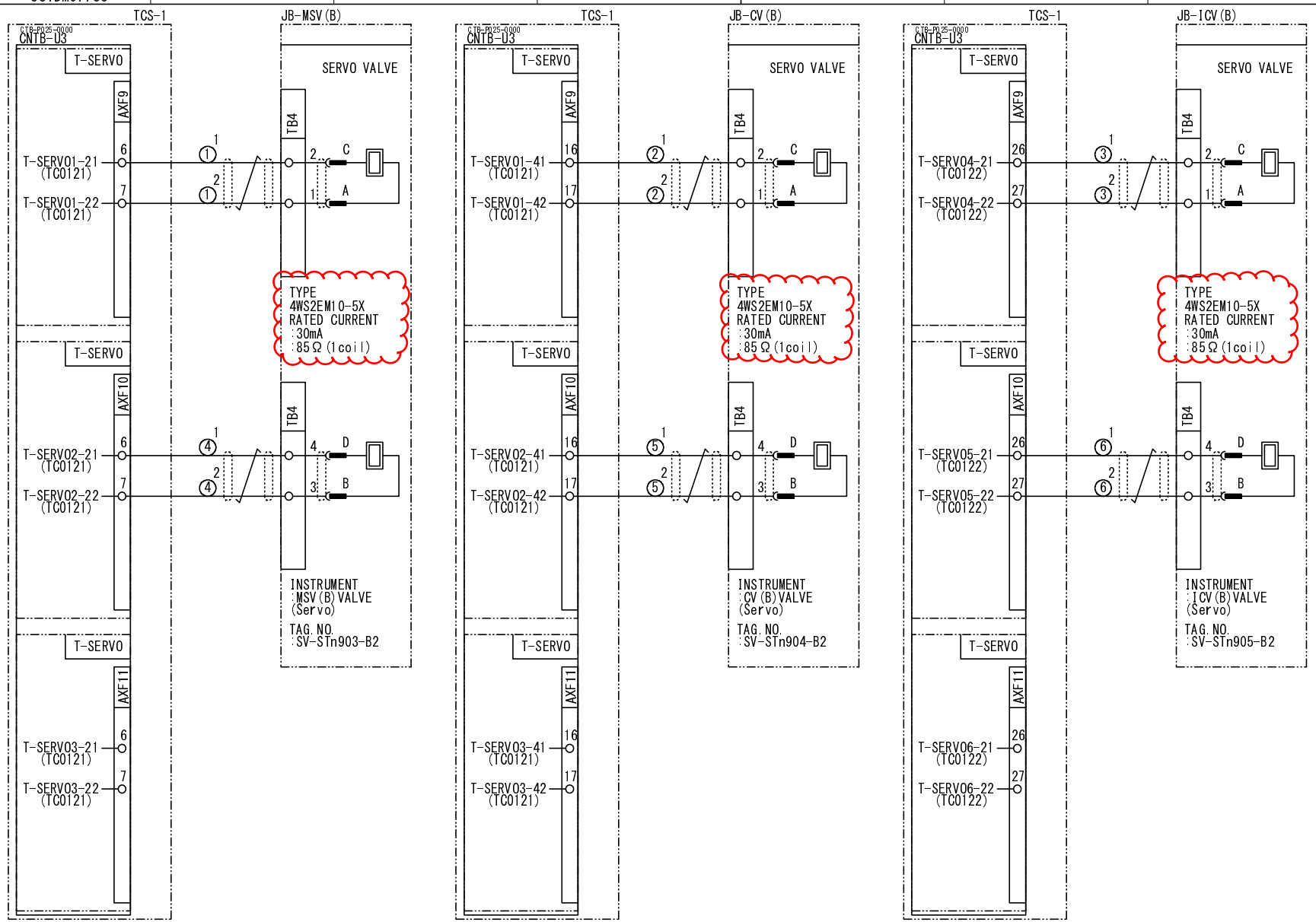
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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IN-DEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
③		6/6	TCS-1				
②		6/6	TCS-1				
①		6/6	TCS-1				

OUTPUT EXTERNAL SERVO VALVE(4)

SH. No. TC0124

	SIGNATURE	DATE
DWN.	(S.Kikuchi)	2014-04-18
CHKD.	T.Kawata	2014-04-18
APPD.	Y.Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

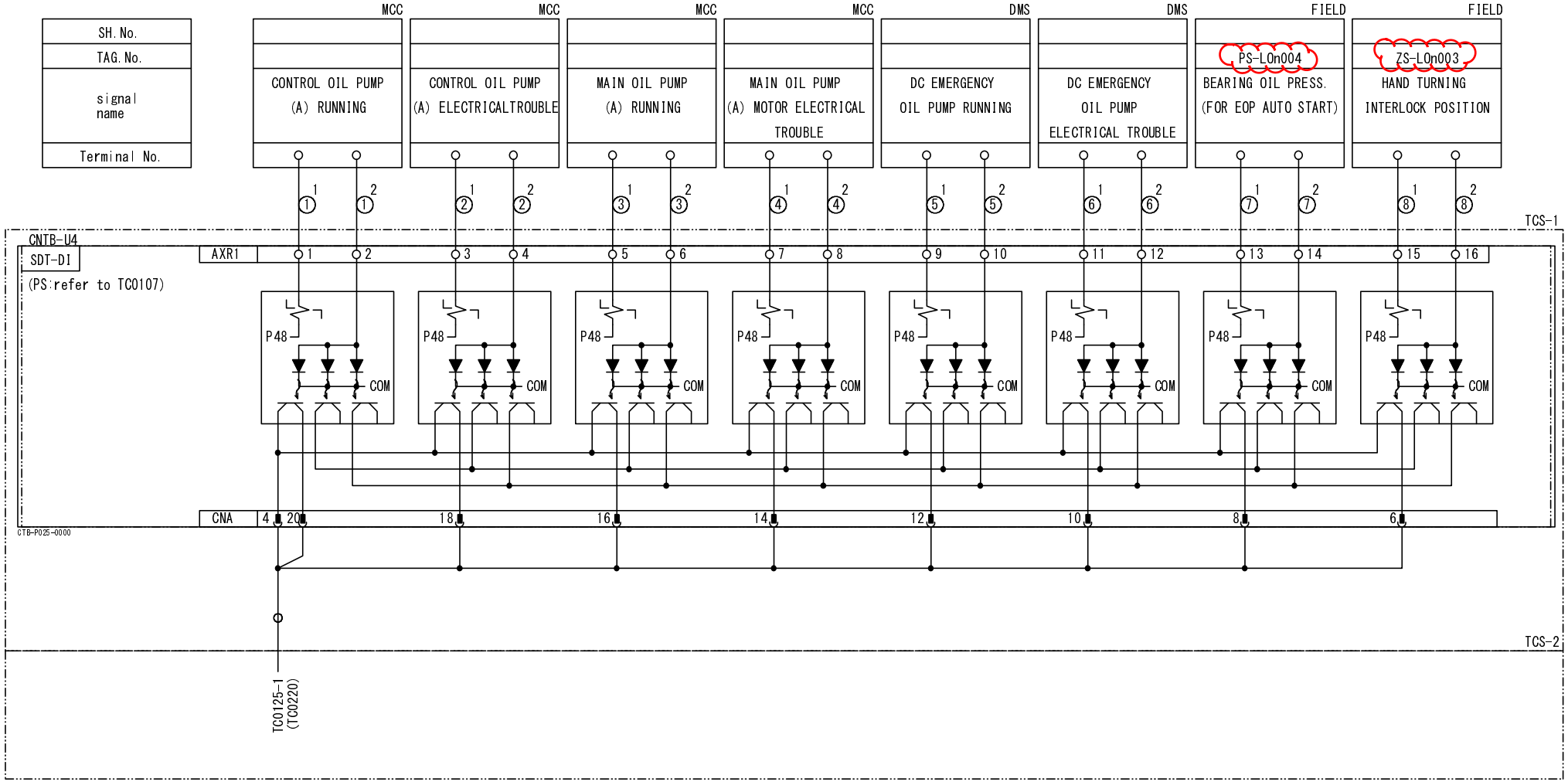
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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IN-DEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	FIELD	TCS-1			
⑦		2/2	FIELD	TCS-1			
⑥		2/2	DMC	TCS-1			
⑤		2/2	DMC	TCS-1			
④		2/2	MCC	TCS-1			
③		2/2	MCC	TCS-1			
②		2/2	MCC	TCS-1			
①		2/2	MCC	TCS-1			

EXTERNAL INPUT EHG DI(1)

SH. No. TC0125

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

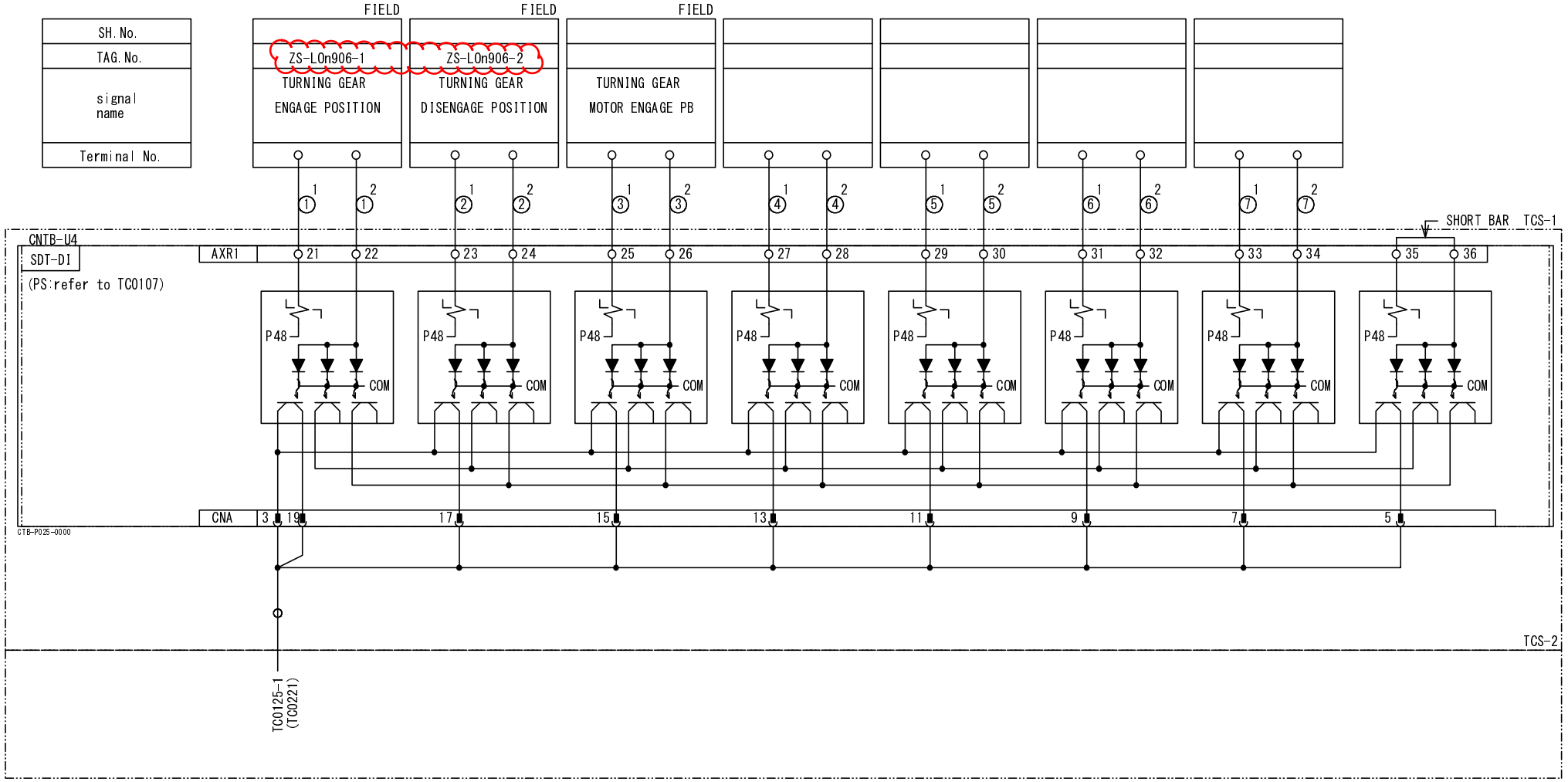
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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IN-DEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		2/2		TCS-1			
⑦		2/2		TCS-1			
⑥		2/2		TCS-1			
⑤		2/2		TCS-1			
④		2/2		TCS-1			
③		2/2	FIELD	TCS-1			
②		2/2	FIELD	TCS-1			
①		2/2	FIELD	TCS-1			

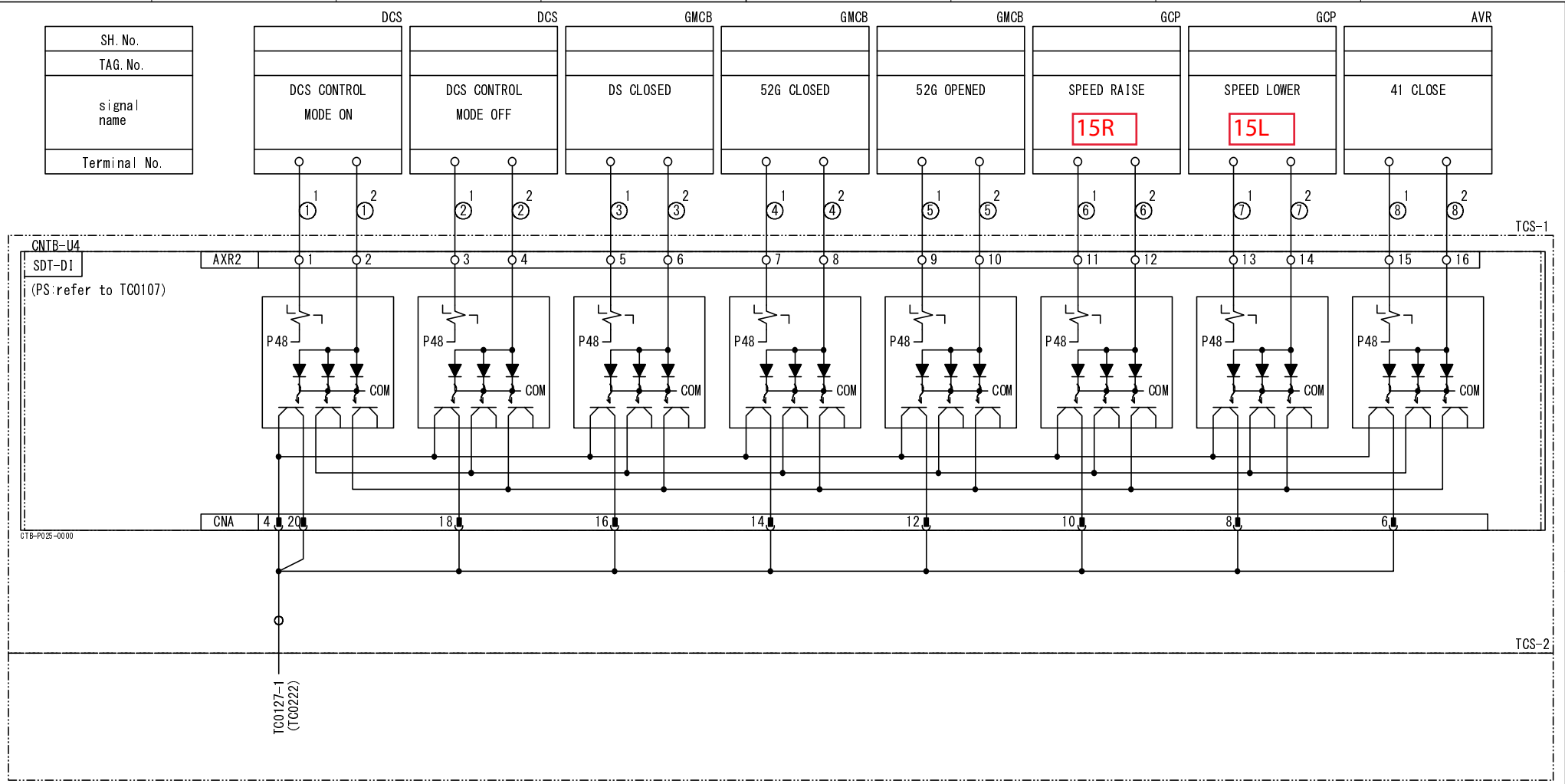
EXTERNAL INPUT EHG DI(2) | SH. No. TC0126

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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2016/07/28 TC0126 XSMM-TC



INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		2/2	AVR	TCS-1			
⑦		2/2	GCP	TCS-1			
⑥		2/2	GCP	TCS-1			
⑤		2/2	GCP	TCS-1			
④		2/2	GMCB	TCS-1			
③		2/2	GMCB	TCS-1			
②		2/2	DCS	TCS-1			
①		2/2	DCS	TCS-1			

EXTERNAL INPUT EHG DI(3)

SH. No. TC0127

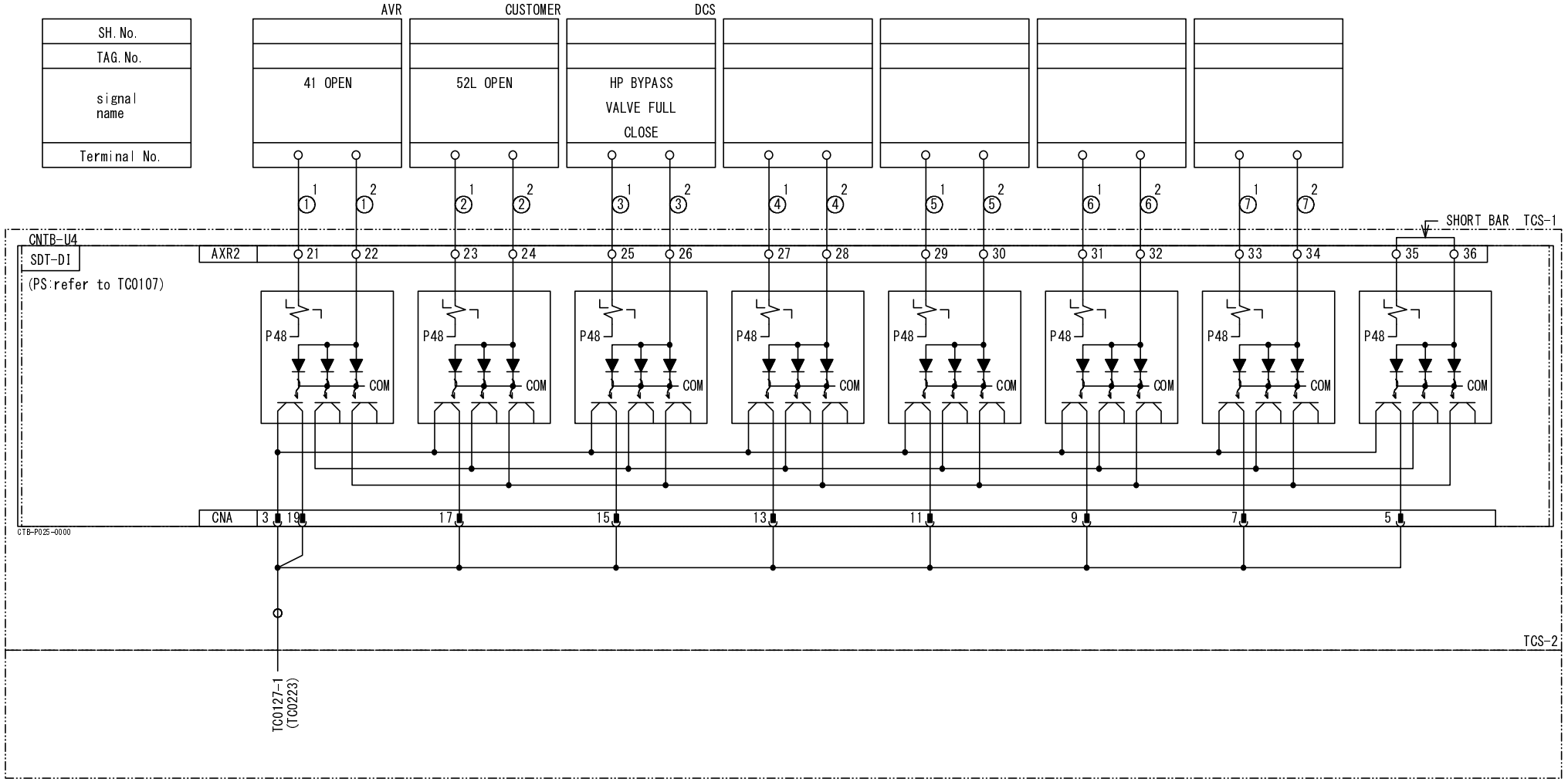
	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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○							
⑦		2/2		TCS-1			
⑥		2/2		TCS-1			
⑤		2/2		TCS-1			
④		2/2		TCS-1			
③		2/2	DCS	TCS-1			
②		2/2	CUSTOMER	TCS-1			
①		2/2	AVR	TCS-1			
INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		

EXTERNAL INPUT EHG DI(4)

SH. No. TC0128

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

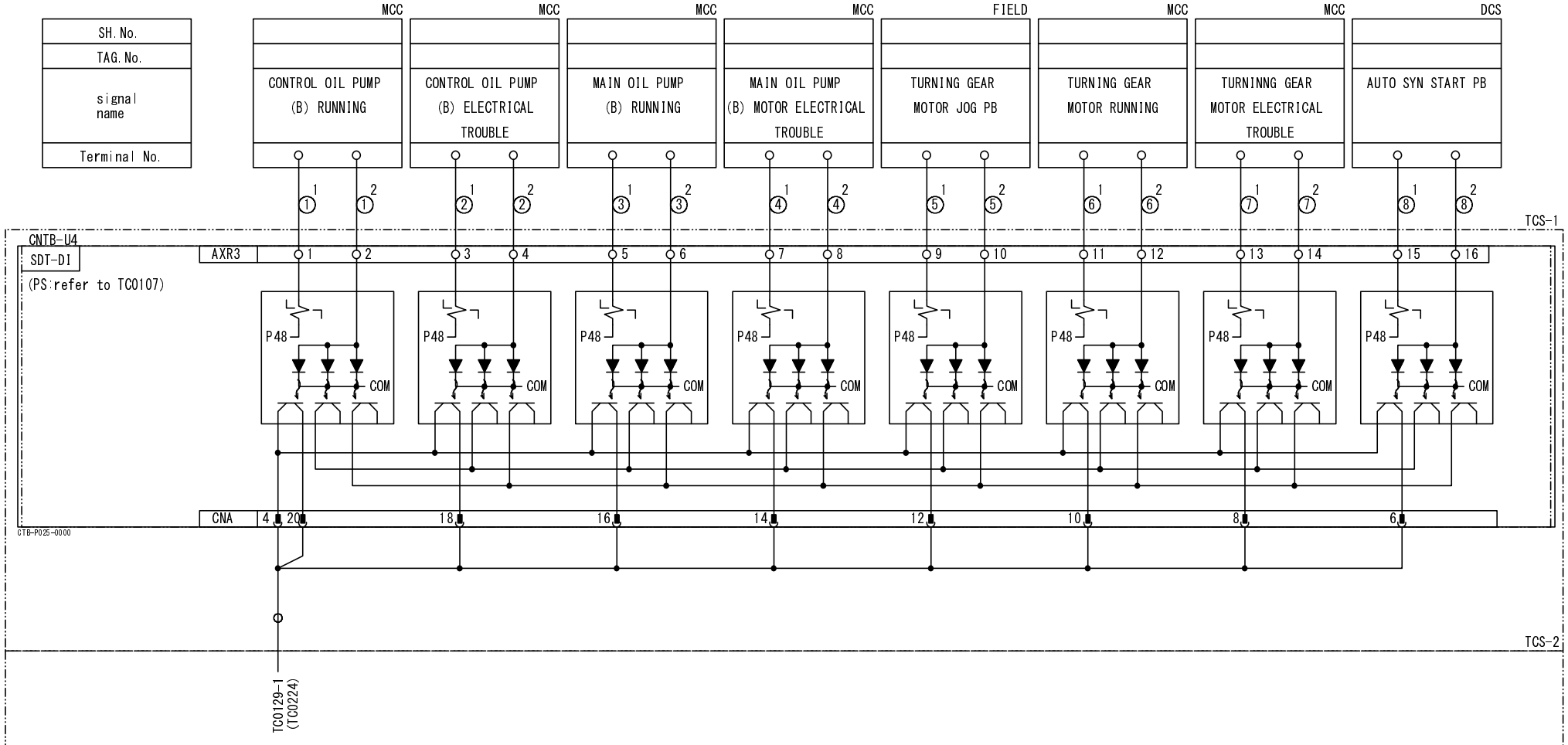
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		2/2	DCS	TCS-1			
⑦		2/2	MCC	TCS-1			
⑥		2/2	MCC	TCS-1			
⑤		2/2	FIELD	TCS-1			
④		2/2	MCC	TCS-1			
③		2/2	MCC	TCS-1			
②		2/2	MCC	TCS-1			
①		2/2	MCC	TCS-1			

EXTERNAL INPUT EHG DI(5)

SH. No. TC0129

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

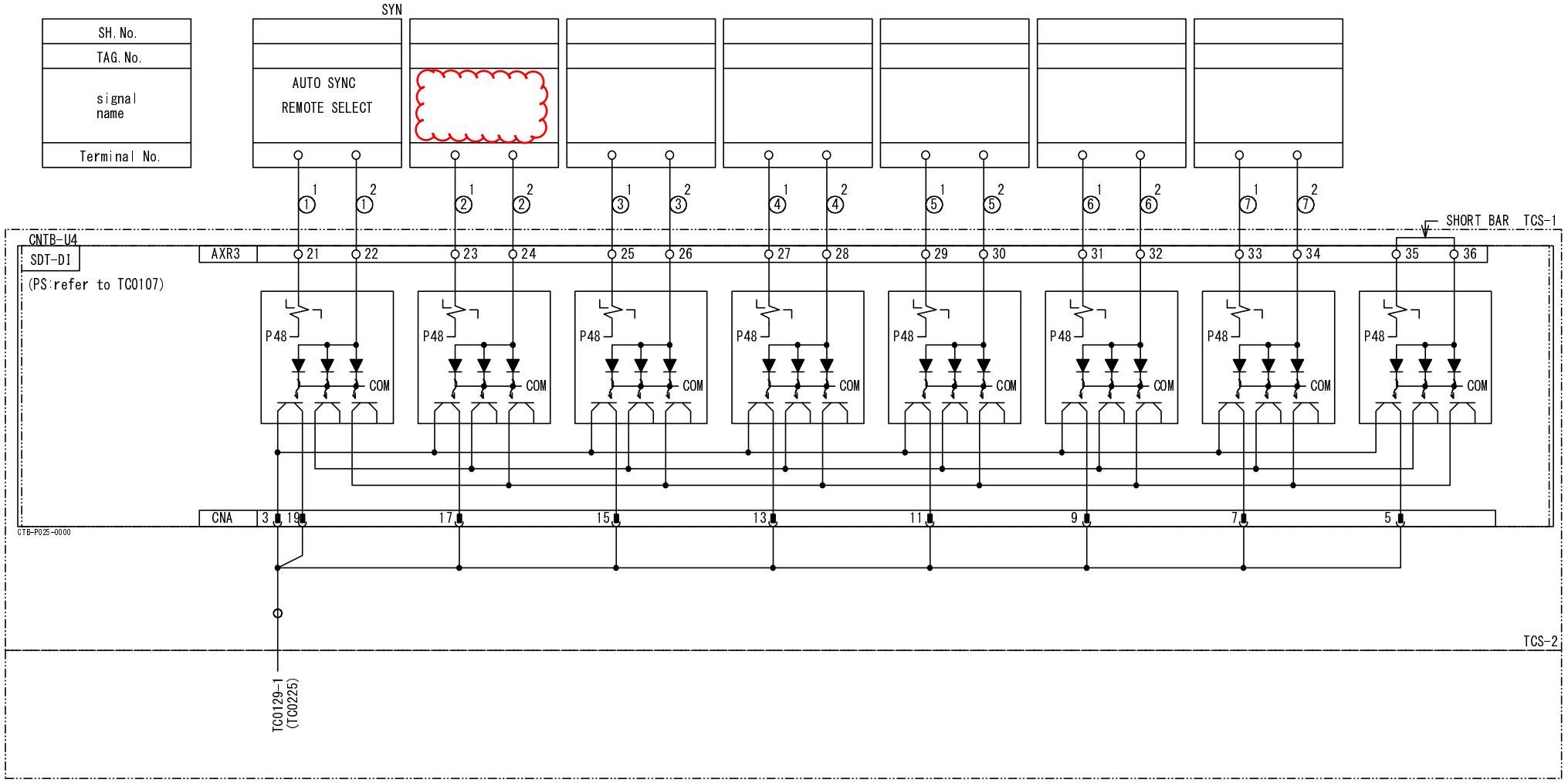
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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○							
⑦		2/2		TCS-1			
⑥		2/2		TCS-1			
⑤		2/2		TCS-1			
④		2/2		TCS-1			
③		2/2		TCS-1			
②		2/2		TCS-1			
①		2/2	SYN	TCS-1			
INDEX	CABLE NO.	CORE NO.	CABLE CONNECTION FROM	CABLE CONNECTION TO	RELATED SH. NO. FROM	RELATED SH. NO. TO	REMARK

EXTERNAL INPUT EHG DI(6)

SH. No. TC0130

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

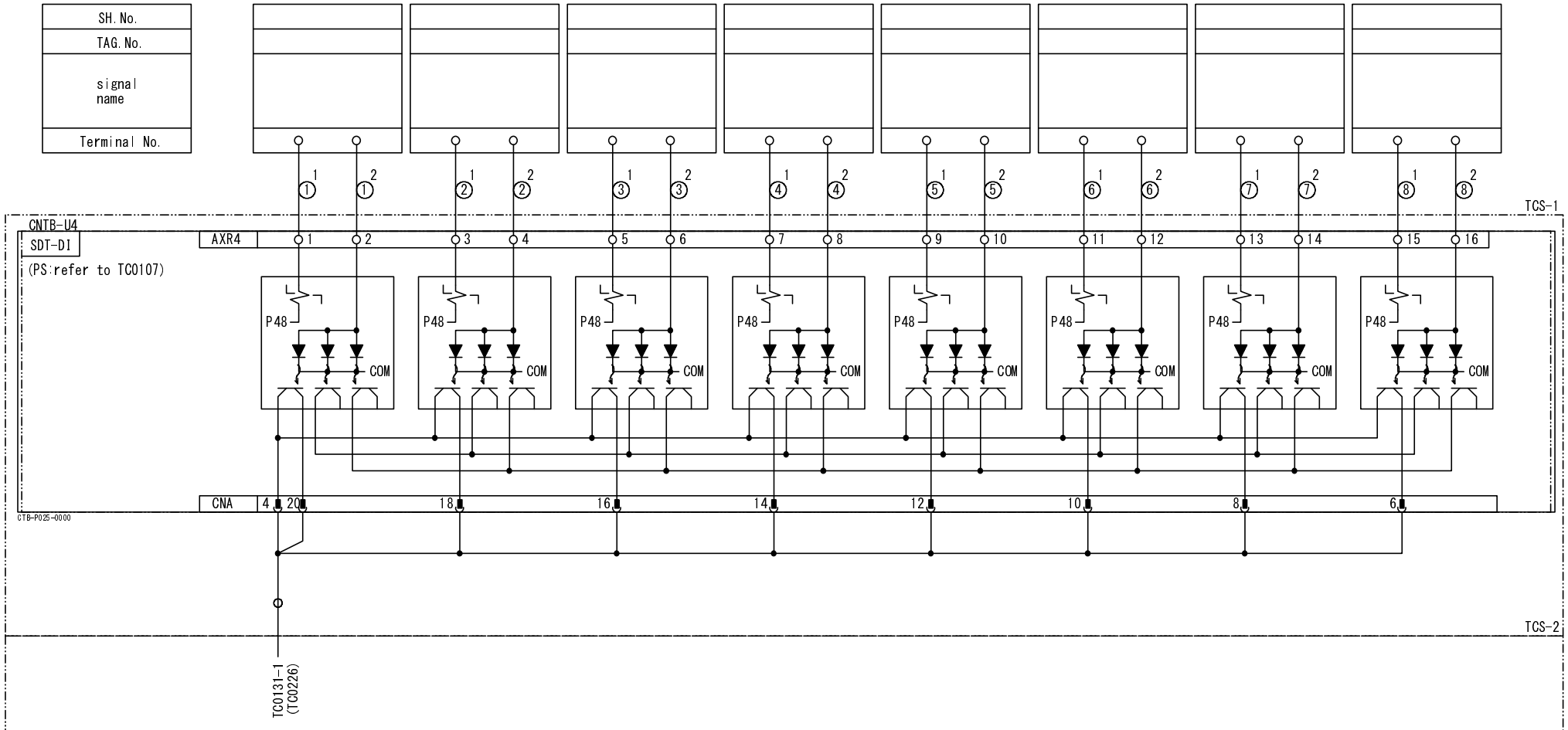
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		2/2		TCS-1			
⑦		2/2		TCS-1			
⑥		2/2		TCS-1			
⑤		2/2		TCS-1			
④		2/2		TCS-1			
③		2/2		TCS-1			
②		2/2		TCS-1			
①		2/2		TCS-1			

EXTERNAL INPUT EHG DI(7)

SH. No. TC0131

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

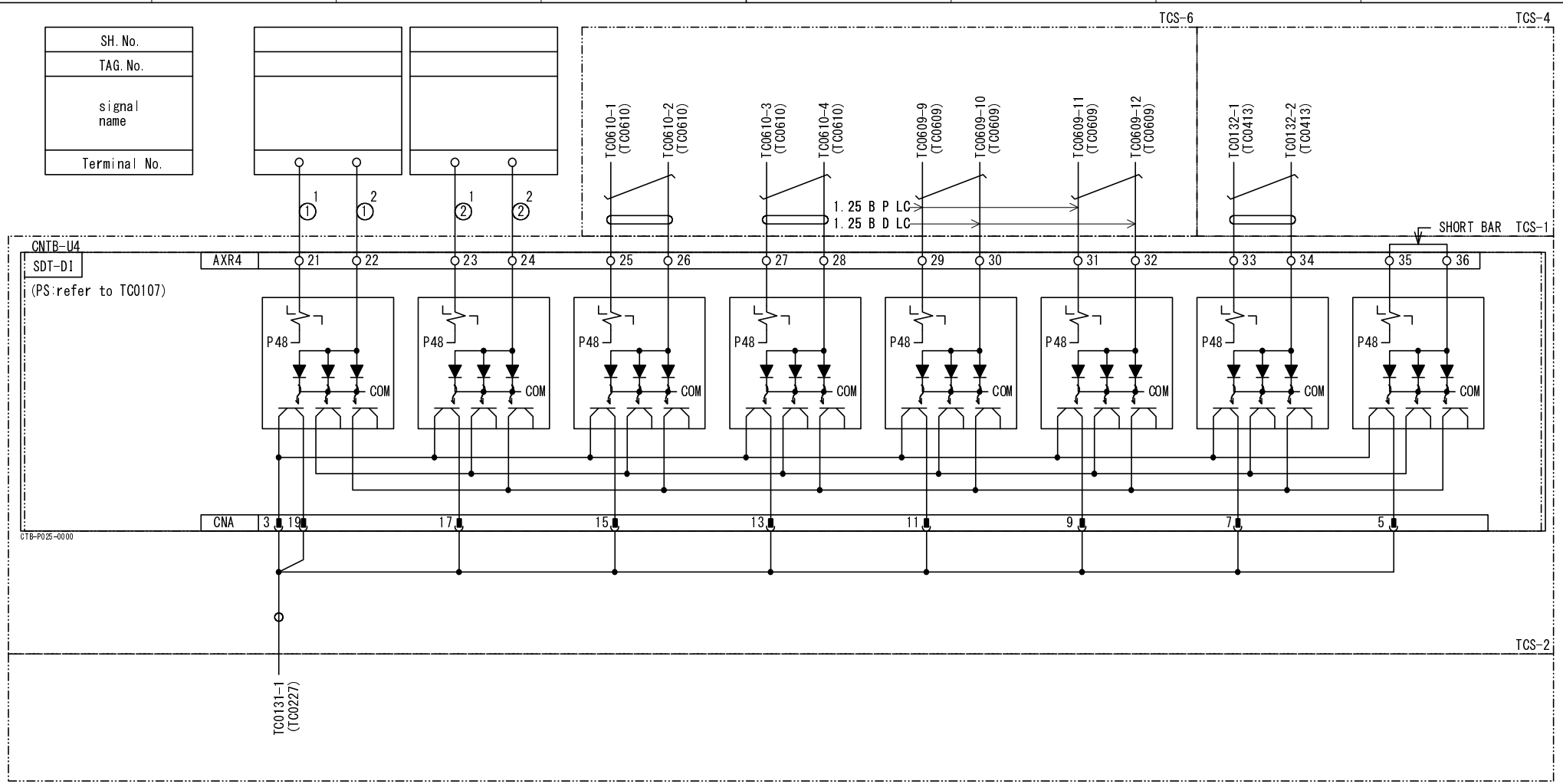
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0131 XSMM-TC



INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
②		2/2		TCS-1			
①		2/2		TCS-1			

EXTERNAL INPUT EHG DI(8)

SH. No. TC0132

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

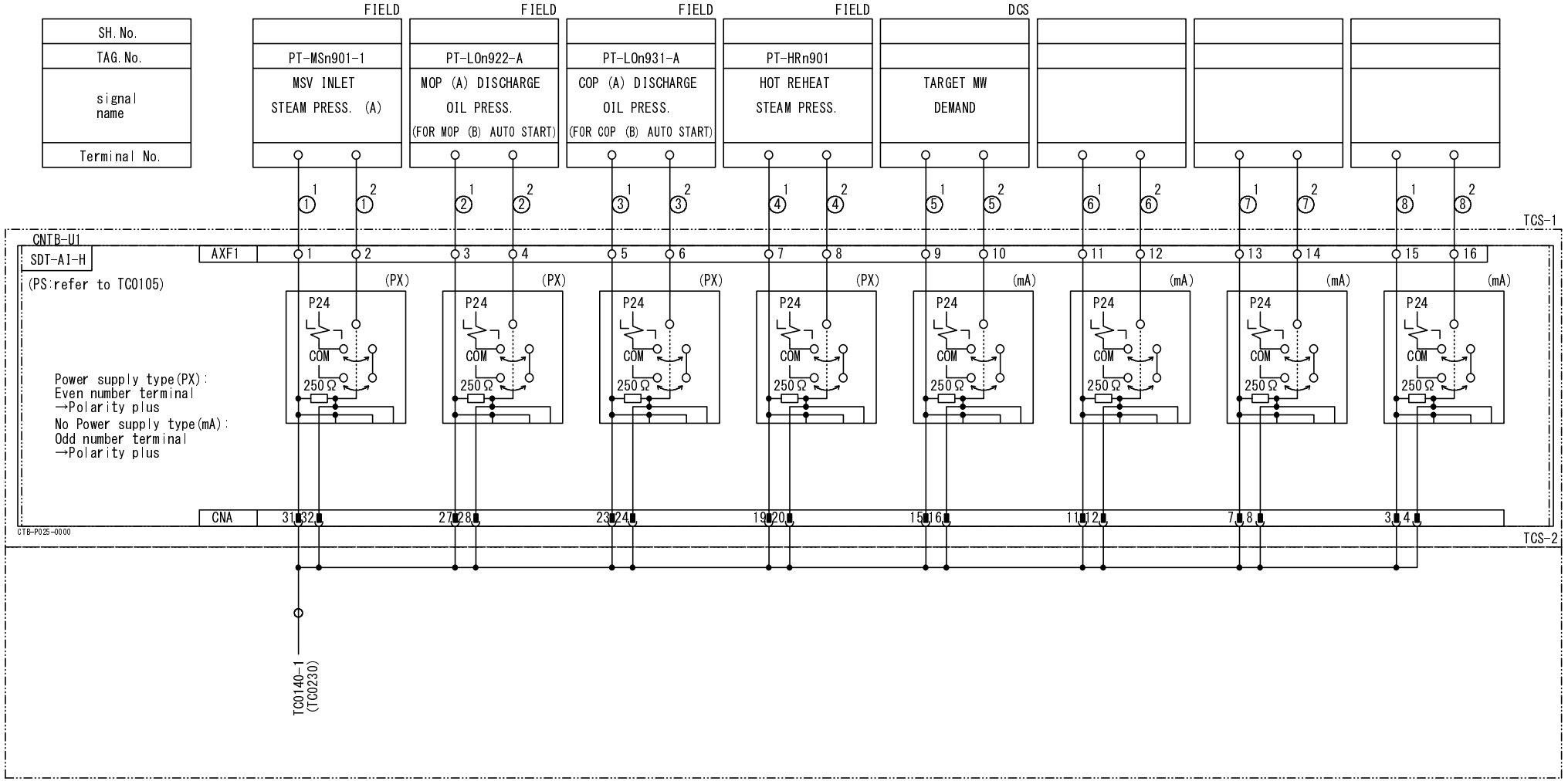
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0132 XSMM-TC



IN-DEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2		TCS-1			
⑦		2/2		TCS-1			
⑥		2/2		TCS-1			
⑤		2/2	DCS	TCS-1			
④		2/2	FIELD	TCS-1			
③		2/2	FIELD	TCS-1			
②		2/2	FIELD	TCS-1			
①		2/2	FIELD	TCS-1			

EXTERNAL INPUT EHG AI(1)

SH. No. TC0140

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

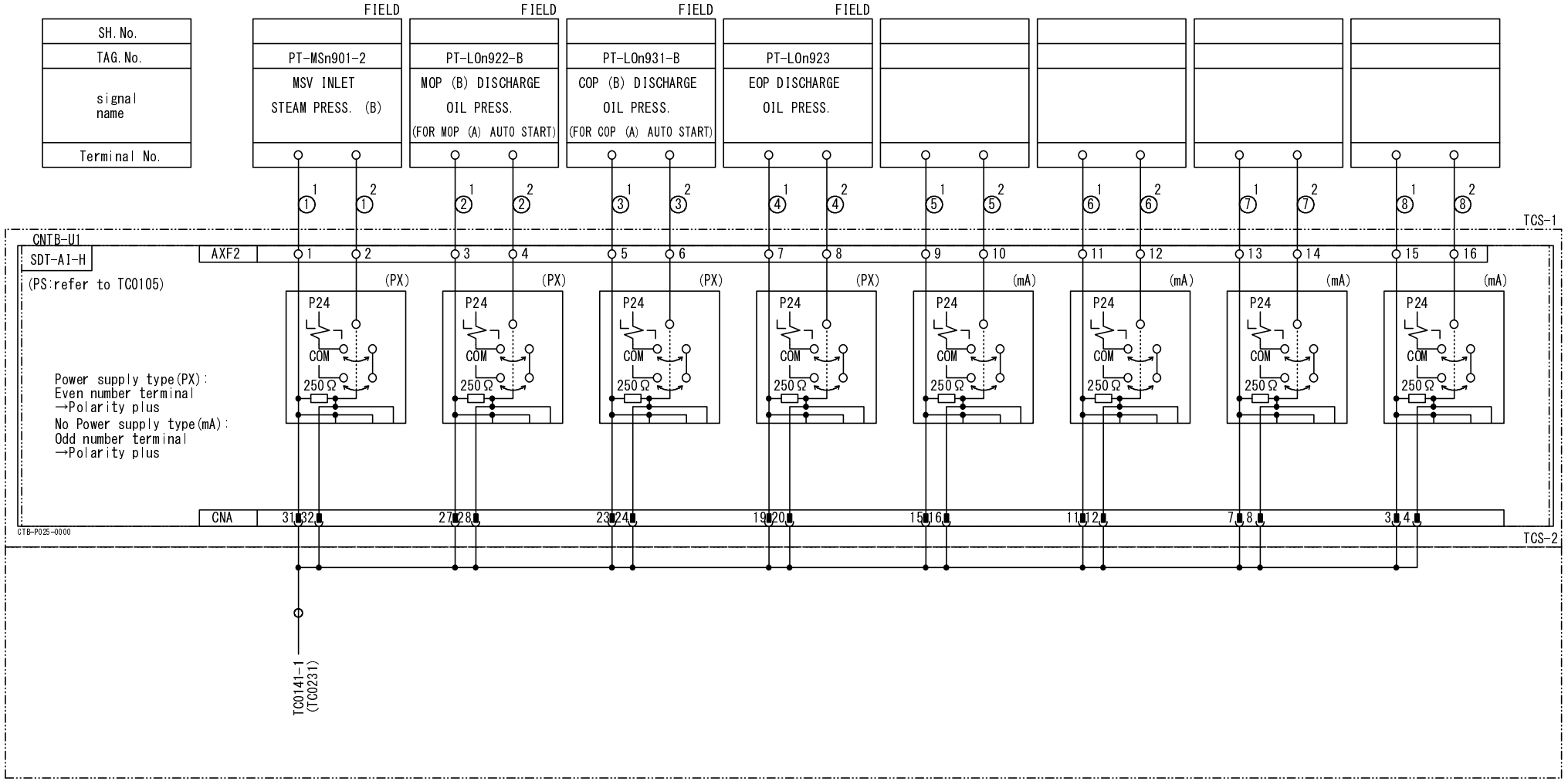
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0140 XSMM-TC



INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		2/2		TCS-1			
⑦		2/2		TCS-1			
⑥		2/2		TCS-1			
⑤		2/2		TCS-1			
④		2/2	FIELD	TCS-1			
③		2/2	FIELD	TCS-1			
②		2/2	FIELD	TCS-1			
①		2/2	FIELD	TCS-1			

EXTERNAL INPUT EHG AI(2)

SH. No. TC0141

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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2016/07/28 TC0141 XSMM-TC

TCS-2

TCS-1

TC0150-1
(TC0240)

CNTB-U5
SD-DO-N
(PS:refer to TC0108)

CNA 21 3 20 18 16 14 12 10 8 6

P24
C24

P24 P24 P24 P24 P24 P24 P24 P24

AXR5 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

CTB-P025-0000

Terminal No.	①	①	②	②	③	③	④	④	⑤	⑤	⑥	⑥	⑦	⑦	⑧	⑧
signal name	CONTROL OIL PUMP (A) START COMMAND		MAIN OIL PUMP (A) START COMMAND		DC EMERGENCY OIL PUMP START		DC EMERGENCY OIL PUMP MANUAL STOP				TURNING GEAR MOTOR START COMMAND				TURNING GEAR ENGAGED	
TAG.No.																
SH.No.																
	MCC		MCC		DMS		DMS				MCC				DCS	

INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	DCM	TCS-1			
⑦		2/2		TCS-1			
⑥		2/2	MCC	TCS-1			
⑤		2/2		TCS-1			
④		2/2	DCM	TCS-1			
③		2/2	DCM	TCS-1			
②		2/2	MCC	TCS-1			
①		2/2	MCC	TCS-1			

EXTERNAL OUTPUT EHG DO(1)

SH. No. TC0150

	SIGNATURE	DATE
DWN.	(S.Kikuchi)	2014-04-18
CHKD.	T.Kawata	2014-04-18
APPD.	Y.Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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DWG. No. 331DM07733

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2016/07/28 TC0150 XSMM-TC

TCS-2

TCS-1

TC0150-1
(TC0241)

CNTB-U5
SD-DO-N
(PS:refer to TC0108)

CNA 22 21 19 17 15 13 11 9 7 5

P24
C24

P24 P24 P24 P24 P24 P24 P24 P24

AXR5 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36

CTB-P025-0000

TCS-4

1 2 1 2 1 2 1 2

Terminal No.							
signal name							
TAG. No.							
SH. No.							

TC0151-5 (TC0424)
TC0151-6 (TC0424)
TC0151-7 (TC0424)
TC0151-8 (TC0424)
TC0151-1 (TC0419)
TC0151-2 (TC0419)

2 B D LC

INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
④		2/2	TCS-1				
③		2/2	TCS-1				
②		2/2	TCS-1				
①		2/2	TCS-1				

EXTERNAL OUTPUT EHG DO(2)

SH. No. TC0151

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE
ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

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2016/07/28 TC0151 XSMM-TC

TCS-2

TCS-1

TC0152-1 (TC0242)

CNTB-U5
SD-DO-N
(PS:refer to TC0108)

CNA 21 3 20 18 16 14 12 10 8 6

P24
C24

P24 P24 P24 P24 P24 P24 P24 P24

AXR6 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

CTB-P025-0000

Terminal No.	① 1	① 2	② 1	② 2	③ 1	③ 2	④ 1	④ 2	⑤ 1	⑤ 2	⑥ 1	⑥ 2	⑦ 1	⑦ 2	⑧ 1	⑧ 2
signal name	52L OPENED FOR HOUSE LOAD OPERATION		TURBINE SPEED UNDER 2400RPM		ACCELERATION RELAY OPERATED		DCS MODE ON FEEDBACK								ASD2 START SIGNAL (52L)	
TAG. No.																
SH. No.																
	DCS		DCS		DCS		DCS								GCP	

INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	TCS-1	GCP			
⑦		2/2	TCS-1				
⑥		2/2	TCS-1				
⑤		2/2	TCS-1				
④		2/2	TCS-1	DCS			
③		2/2	TCS-1	DCS			
②		2/2	TCS-1	DCS			
①		2/2	TCS-1	DCS			

EXTERNAL OUTPUT EHG DO(3)

SH. No. TC0152

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0152 XSMM-TC

TCS-2

TCS-1

TC0152-1 (TC0243)

SD-DO-N

(PS:refer to TC0108)

CNA 22 4 19 17 15 13 11 9 7 5

P24

C24

AXR6 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36

Terminal No.	① 1	① 2	② 1	② 2	③ 1	③ 2	④ 1	④ 2	⑤ 1	⑤ 2	⑥ 1	⑥ 2	⑦ 1	⑦ 2	⑧ 1	⑧ 2
signal name	HP-CV FULL CLOSE (BOTH CV(A)&(B))		TURBINE TRIP (MSV, RSV, ICV CLOSE)		MSV/CV(A) SEQUENTIAL FULL CLOSE TEST		MSV/CV(B) SEQUENTIAL FULL CLOSE TEST		41 CLOSED		GEN. LOCKOUT RELAY (86G) A OPERATED		GEN. LOCKOUT RELAY (86G) B OPERATED		ICV TO CV TRANSFER COMPLETE	
TAG.No.																
SH.No.																
	DCS		DCS		DCS		DCS		DCS		DCS		DCS		DCS	

INDEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK
⑧		2/2	TCS-1	DCS			
⑦		2/2	TCS-1	DCS			
⑥		2/2	TCS-1	DCS			
⑤		2/2	TCS-1	DCS			
④		2/2	TCS-1	DCS			
③		2/2	TCS-1	DCS			
②		2/2	TCS-1	DCS			
①		2/2	TCS-1	DCS			

EXTERNAL OUTPUT EHG DO(4)

SH. No. TC0153

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/08/23 TC0153 XSMM-TC

TCS-2

TCS-1

TC0154-1 (TC0244)

CNTB-U5
SD-DO-N
(PS:refer to TC0108)

CNA 21 3 20 18 16 14 12 10 8 6

P24
C24

P24 P24 P24 P24 P24 P24 P24 P24

AXR7 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

CTB-P025-0000

Terminal No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
signal name	CONTROL OIL PUMP (B) START COMMAND		MAIN OIL PUMP (B) START COMMAND		ASD1 START SIGNAL (52G)				DCS CONTROL MODE PERMISSIVE						POWER LOAD UNBALANCE RELAY OPERATED	
TAG. No.																
SH. No.																
	MCC		MCC		GCP				DCS						DCS	

INDEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK
⑧		2/2	TCS-1	DCS			
⑦		2/2	TCS-1				
⑥		2/2	TCS-1				
⑤		2/2	TCS-1	DCS			
④		2/2	TCS-1				
③		2/2	TCS-1	GCP			
②		2/2	TCS-1	MCC			
①		2/2	TCS-1	MCC			

EXTERNAL OUTPUT EHG DO(5)

SH. No. TC0154

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0154 XSMM-TC

TCS-2

TCS-1

TC0154-1
(TC0245)

CNTB-U5
SD-DO-N
(PS:refer to TC0108)

CNA 22 21 19 17 15 13 11 9 7 5

P24
C24

P24 P24 P24 P24 P24 P24 P24 P24

AXR7 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36

CTB-P025-0000

TCS-4

1 2 1 2 1 2 1 2

Terminal No.							
signal name							
TAG. No.							
SH. No.							

TC0155-7 (TC0424)
TC0155-8 (TC0424)
TC0155-9 (TC0424)
TC0155-10 (TC0424)
TC0155-3 (TC0420A)
TC0155-4 (TC0420A)
TC0155-1 (TC0420)
TC0155-2 (TC0420)

IN-DEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
④		2/2	TCS-1				
③		2/2	TCS-1				
②		2/2	TCS-1				
①		2/2	TCS-1				

EXTERNAL OUTPUT EHG DO(6)

SH. No. TC0155

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE
ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0155 XSMM-TC

TCS-2

TCS-1

TC0156-1
(TC0246)

CNTB-U5
SD-DO-P
(PS:refer to TC0109)

CNA 21 3 20 18 16 14 12 10 8 6

P24
C24

OUTPUT
VOLTAGE
DC125V

P(A) N(C)

AXR8 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

CTB-P025-0000

Terminal No.								
signal name	TURNING GEAR ENGAGE SOLENOID VALVE	TURNING GEAR ENGAGE (LAMP)	TURNING GEAR MOTOR START (LAMP)					
TAG. No.	SV-L0n906							
SH. No.								
	FIELD	FIELD	FIELD					

INDEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK
⑧		2/2	TCS-1				
⑦		2/2	TCS-1				
⑥		2/2	TCS-1				
⑤		2/2	TCS-1				
④		2/2	TCS-1				
③		2/2	TCS-1	FIELD			
②		2/2	TCS-1	FIELD			
①		2/2	TCS-1	FIELD			

EXTERNAL OUTPUT EHG DO(7)

SH. No. TC0156

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0156 XSMM-TC

TCS-2

TC0156-1
(TC0247)

TCS-1

CNTB-U5
SD-DO-P

(PS:refer to TC0109)

CNA 22 4 19 17 15 13 11 9 7 5

P24
C24

OUTPUT VOLTAGE

P24 P(A) N(C)

AXR8 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36

CTB-P025-0000

Terminal No.															
signal name															
TAG. No.															
SH. No.															

INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	TCS-1				
⑦		2/2	TCS-1				
⑥		2/2	TCS-1				
⑤		2/2	TCS-1				
④		2/2	TCS-1				
③		2/2	TCS-1				
②		2/2	TCS-1				
①		2/2	TCS-1				

EXTERNAL OUTPUT EHG DO(8)

SH. No. TC0157

SIGNATURE		DATE
DWN.	(S.Kikuchi)	2014-04-18
CHKD.	T.Kawata	2014-04-18
APPD.	Y.Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

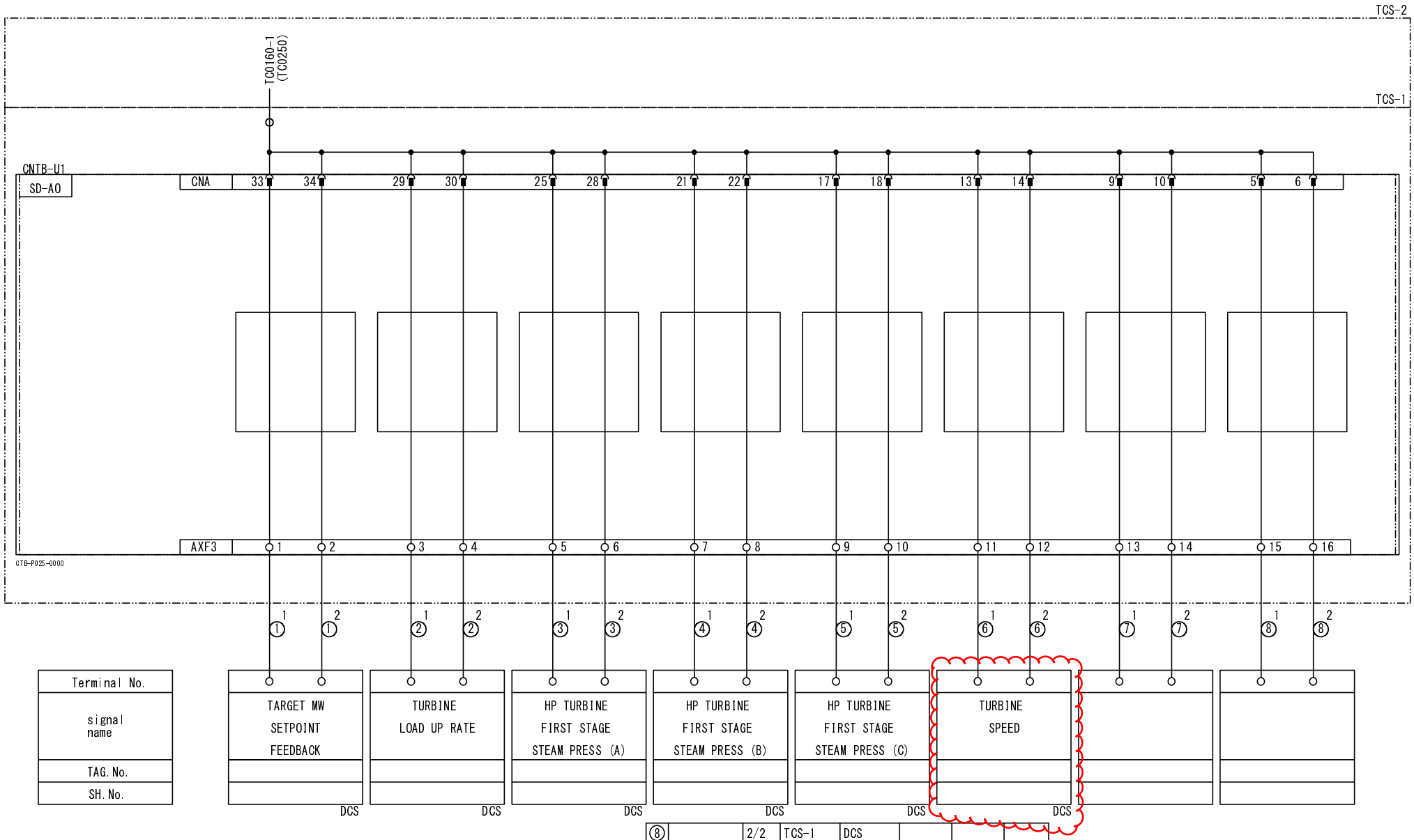
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	TCS-1	DCS			
⑦		2/2	TCS-1	DCS			
⑥		2/2	TCS-1	DCS			
⑤		2/2	TCS-1	DCS			
④		2/2	TCS-1	DCS			
③		2/2	TCS-1	DCS			
②		2/2	TCS-1	DCS			
①		2/2	TCS-1	DCS			

EXTERNAL OUTPUT EHG AO

SH. No. TC0160

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

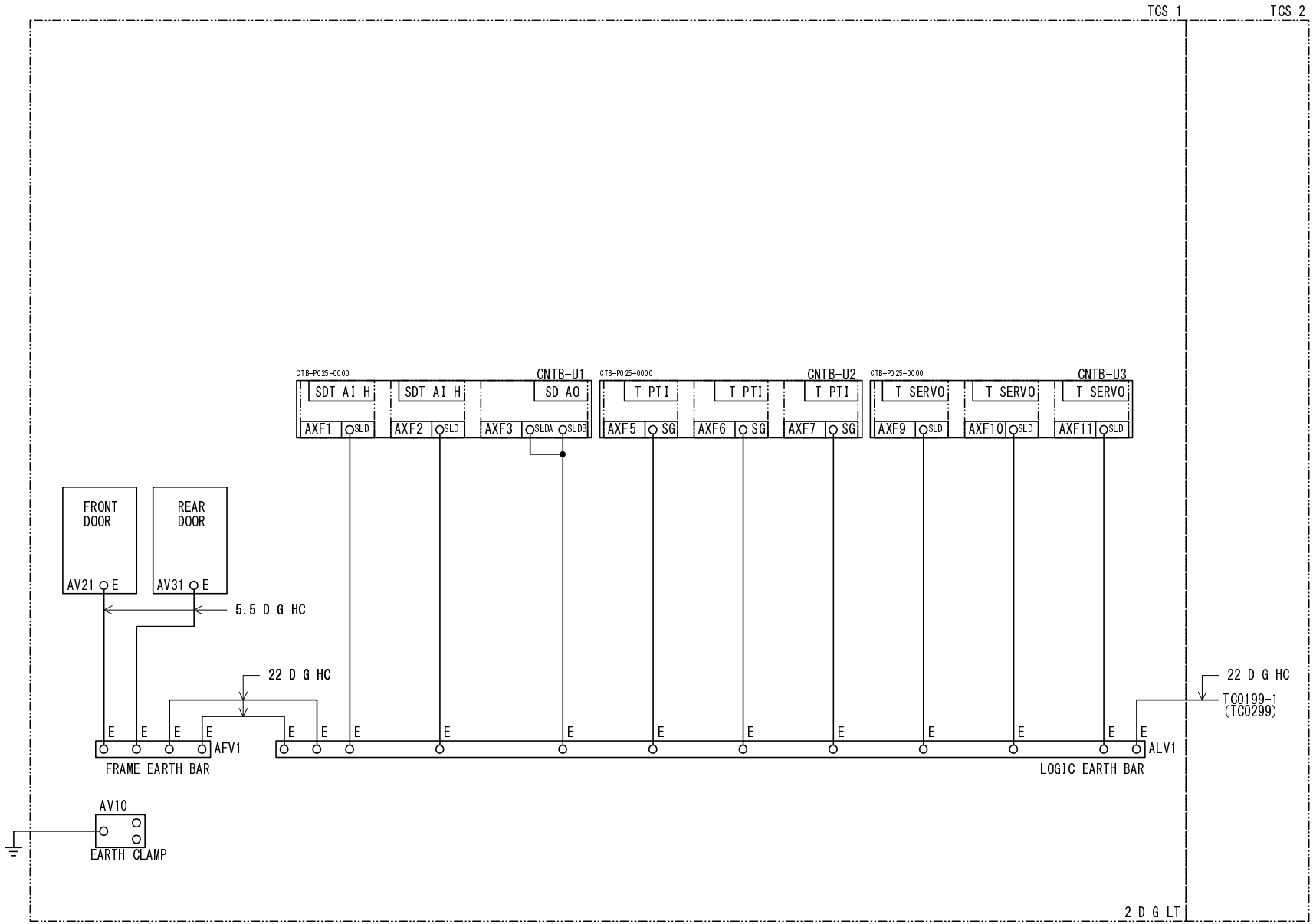
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0160 XSMM-TC



POWER SUPPLY SYSTEM EARTH CIRCUIT		SH. No.	TC0199
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 57 / REV. 3

2016/07/28 TC0199 XSMM-TC

TCS-3

TCS-2

TCS-1

AC100V TC0201-11 (TC0301)
TC0201-12 (TC0301)
DC125V TC0201-13 (TC0301)
TC0201-14 (TC0301)
AC220V TC0201-15 (TC0301)
TC0201-16 (TC0301)
AC120V TC0201-17 (TC0301)
TC0201-18 (TC0301)
DC125V TC0201-19 (TC0301)
TC0201-20 (TC0301)

5.5 B A HA
5.5 B C HA
5.5 B P HA
5.5 B N HA
5.5 B A HC
5.5 B C HC
5.5 B P HC
5.5 B N HC

BZ1 01 02 03 04 05 06 07 08 (UCK25-8CU)
BZ2 01 02 03 04 (UCK25-4CU)

TC0201-1 (TC0205)
TC0201-2 (TC0205)
TC0201-3 (TC0205)
TC0201-4 (TC0205)

TC0101-1 (TC0101)
TC0101-2 (TC0101)
TC0101-3 (TC0101)
TC0101-4 (TC0101)
TC0101-5 (TC0101)
TC0101-6 (TC0101)
TC0101-7 (TC0101)
TC0101-8 (TC0101)
TC0101-9 (TC0101)
TC0101-10 (TC0101)

2 B A HA
2 B C HA

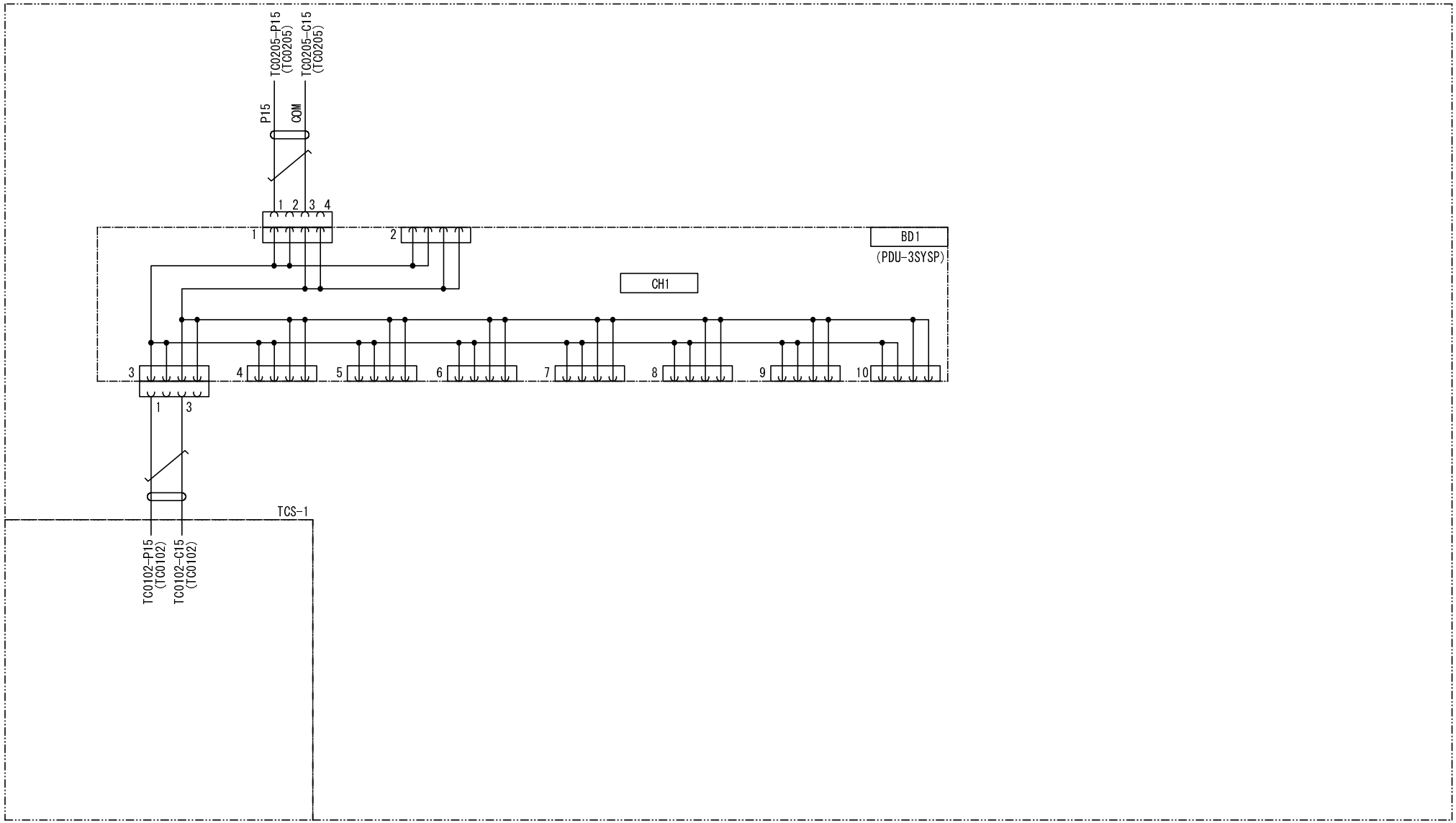
BLT1
LIGHTING
DOOR SWITCH
BSW1 (M-2022L)
(CLED-H-1004R)

BLT2
LIGHTING
DOOR SWITCH
(CLED-H-1004R)

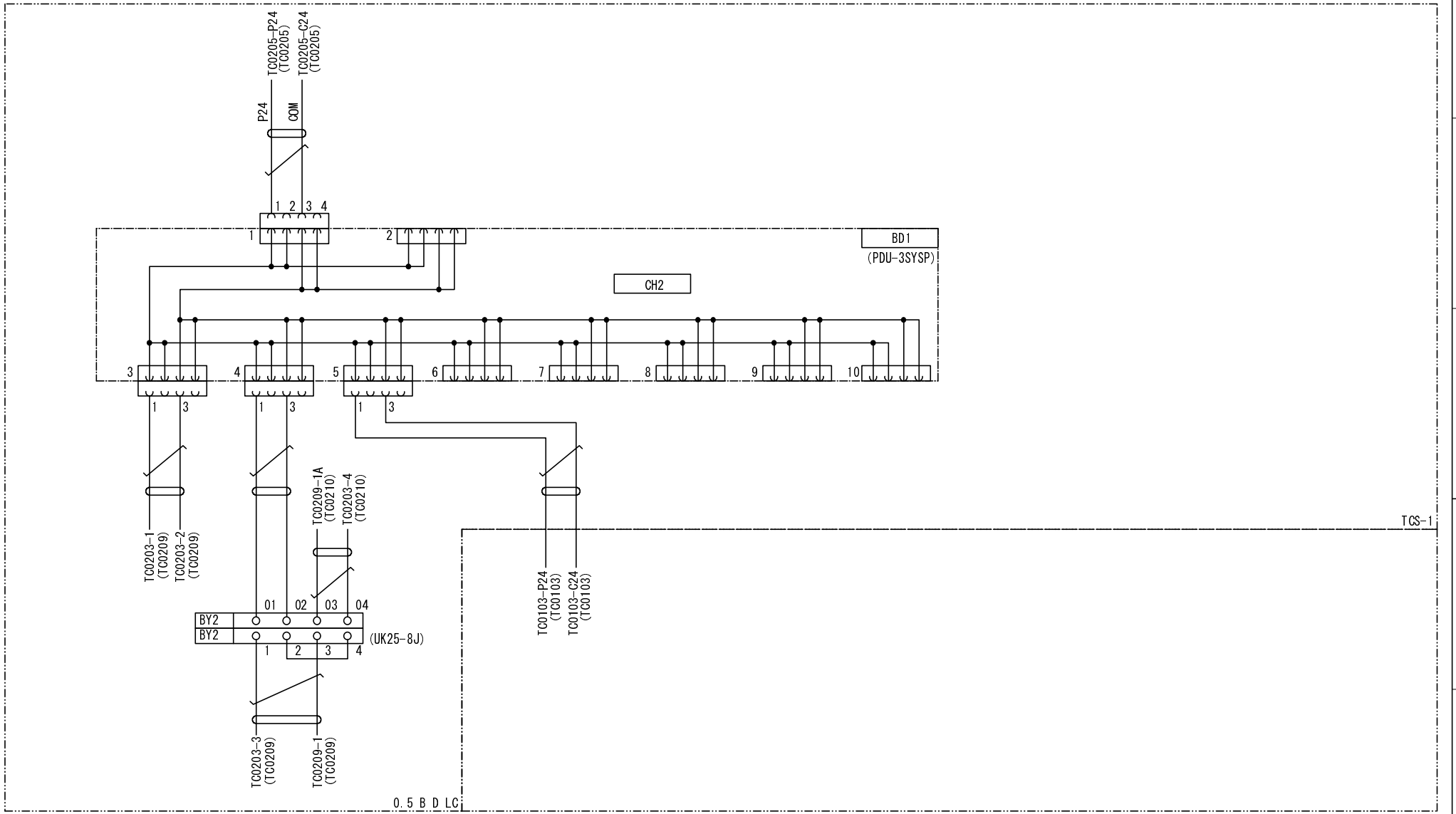
(BSH1) HEATER (AC220V 200W)

POWER SUPPLY SYSTEM(1)		SH. No.	TC0201
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 58 / REV. 3

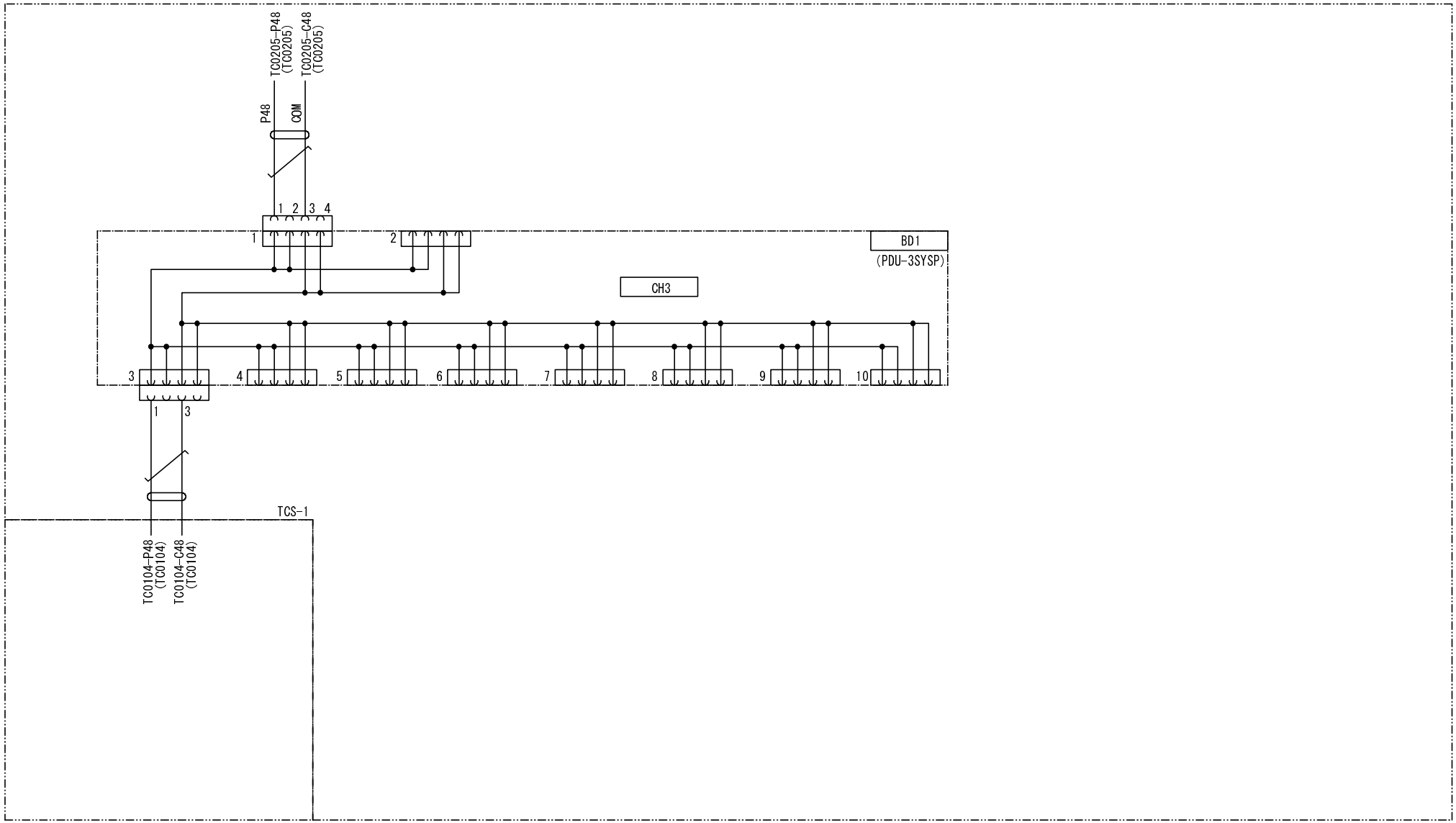
2016/07/28 TC0201 XSMM-TC



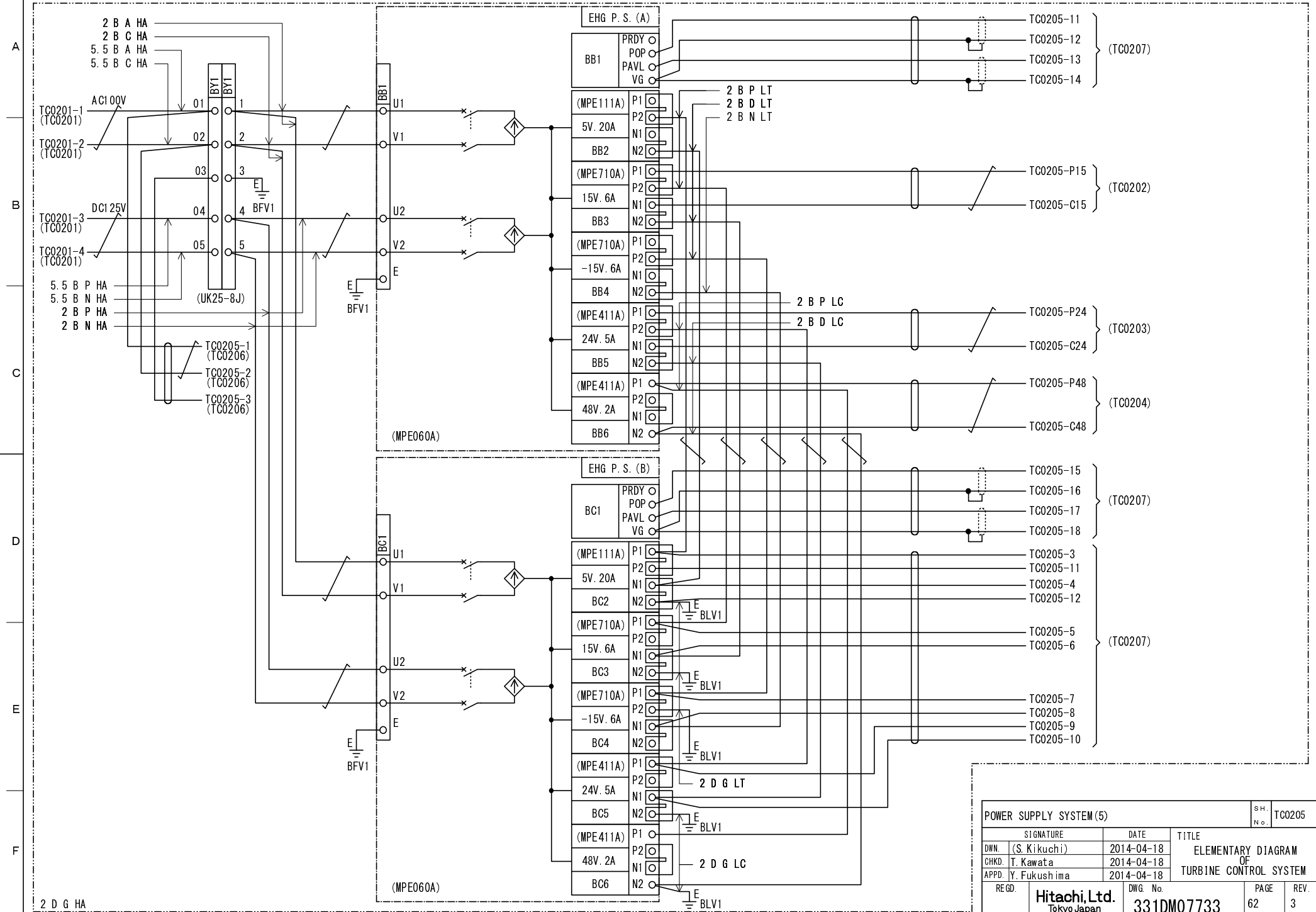
POWER SUPPLY SYSTEM(2)		SH. No.	TC0202
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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POWER SUPPLY SYSTEM(3)		SH. No.	TC0203
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
REGD. Hitachi,Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 60	REV. 3

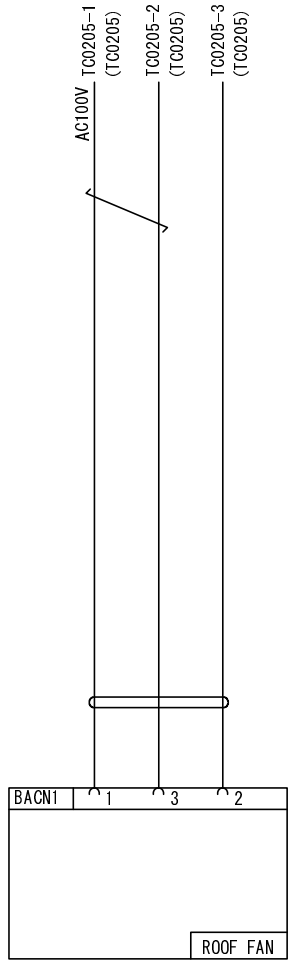


POWER SUPPLY SYSTEM(4)		SH. No.	TC0204
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 61	REV. 3

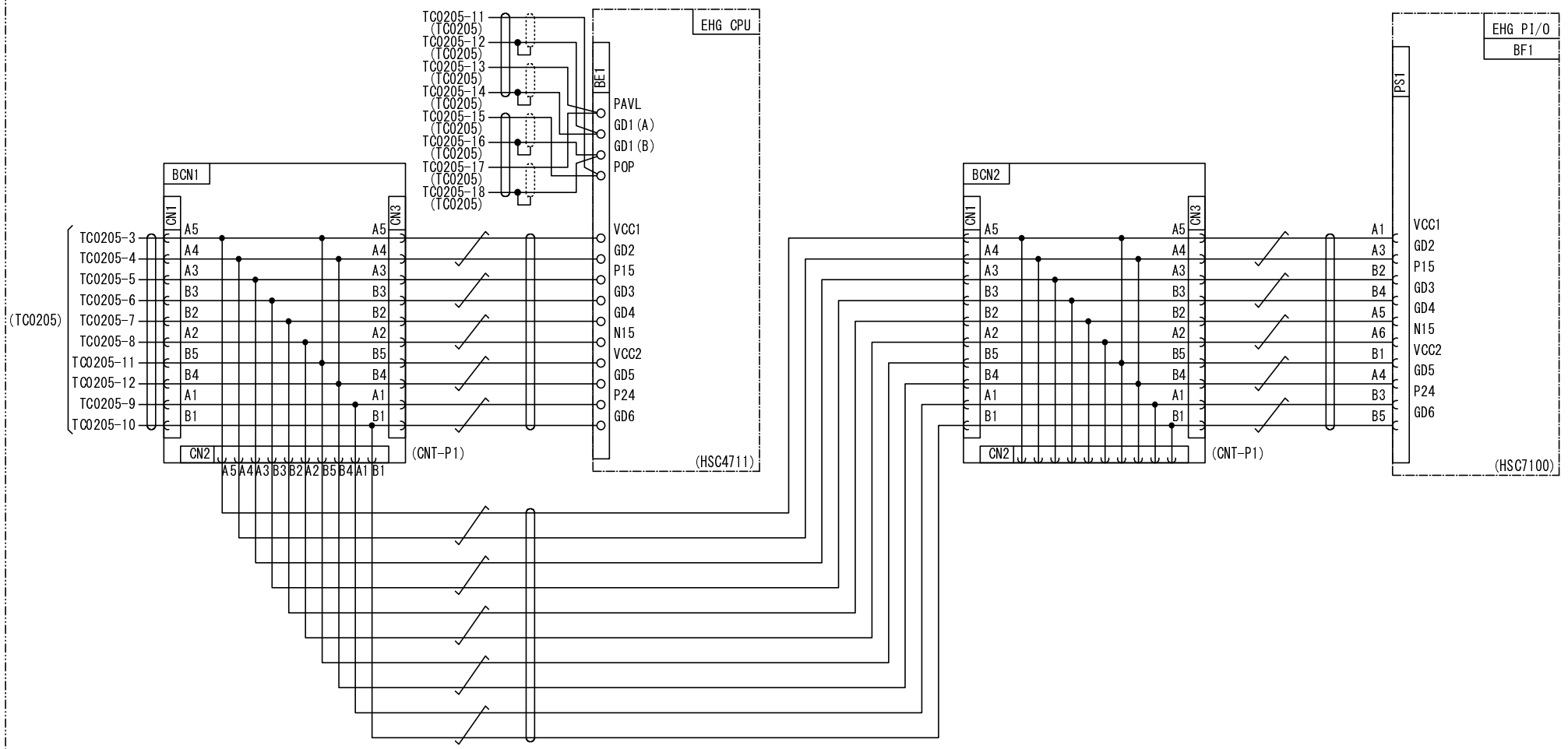


POWER SUPPLY SYSTEM(5)		SH. No.	TC0205
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 62	REV. 3

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POWER SUPPLY SYSTEM(6)		SH. No.	TC0206
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 63	REV. 3



POWER SUPPLY SYSTEM(7)		SH. No.	TC0207
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 64 / REV. 3

2016/07/28 TC0207 XSMM-TC

TCS-2

ROOF FAN	BACN2

PWR LOST

＞55℃

(4P-X)

3 → TC0208-1 (TC0750)
4 → TC0208-2 (TC0750)
1 → TC0208-3 (TC0750)
2 → TC0208-4 (TC0750)

TCS-7

PI NAME	CPU
TYPE	LPU652A
SLOT	0,1
UNIT No.	BE1

CPU RUN

(DO)
6 → TC0208-5 (TC0209)
5 → TC0208-6 (TC0209)

PI NAME	CPU
TYPE	LPU652A
SLOT	2,3
UNIT No.	BE1

CPU RUN

(DO)
6 → TC0208-7 (TC0209)
5 → TC0208-8 (TC0209)

TCS-7

EHG P.S. (A)	BB1

A → TC0208-9 (TC0750)
C → TC0208-10 (TC0750)

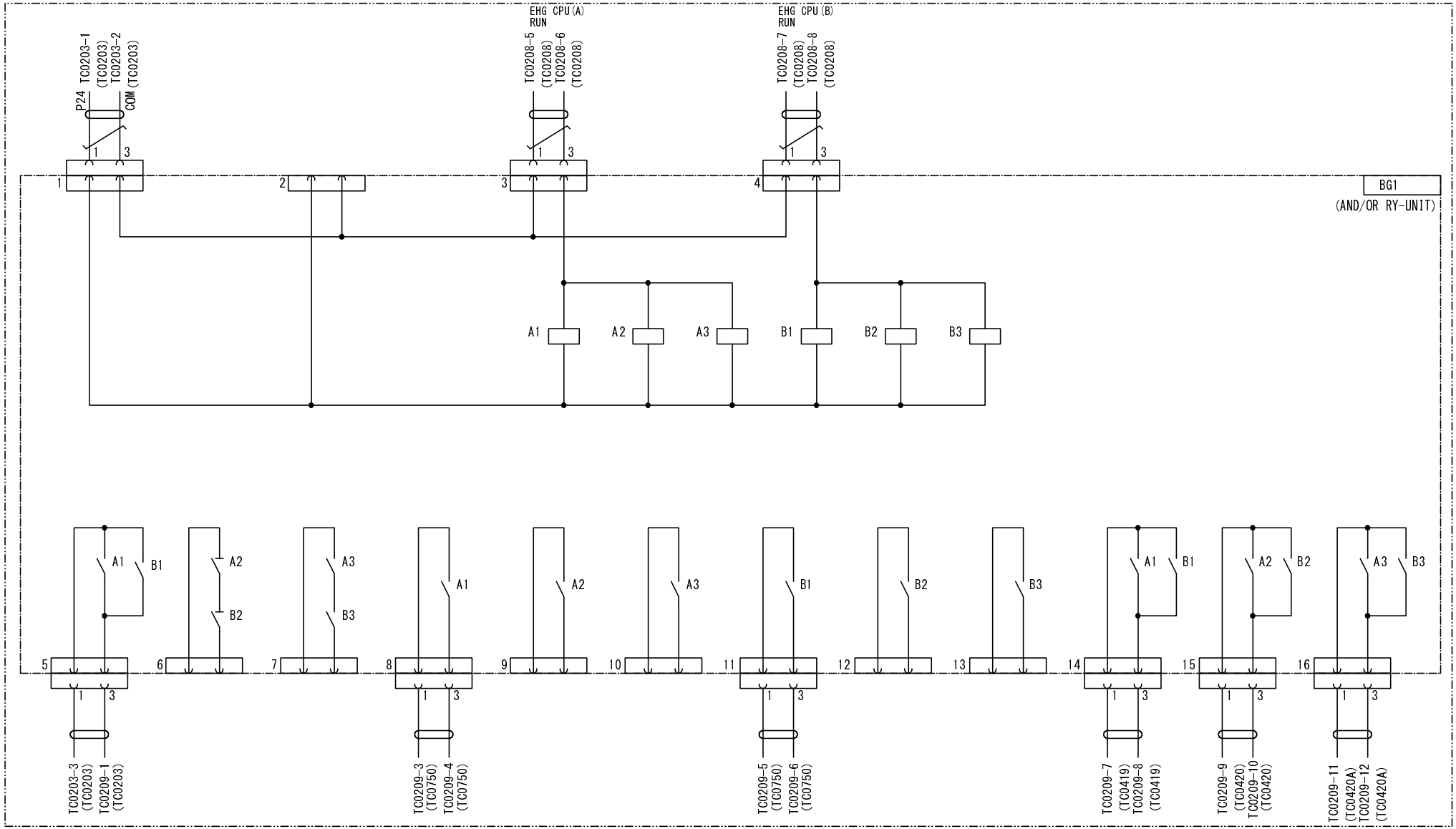
EHG P.S. (B)	BC1

A → TC0208-11 (TC0750)
C → TC0208-12 (TC0750)

1.25 B P LC
1.25 B D LC

DIAGNOSIS CIRCUIT(1)			SH. No.	TC0208
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 65	REV. 3

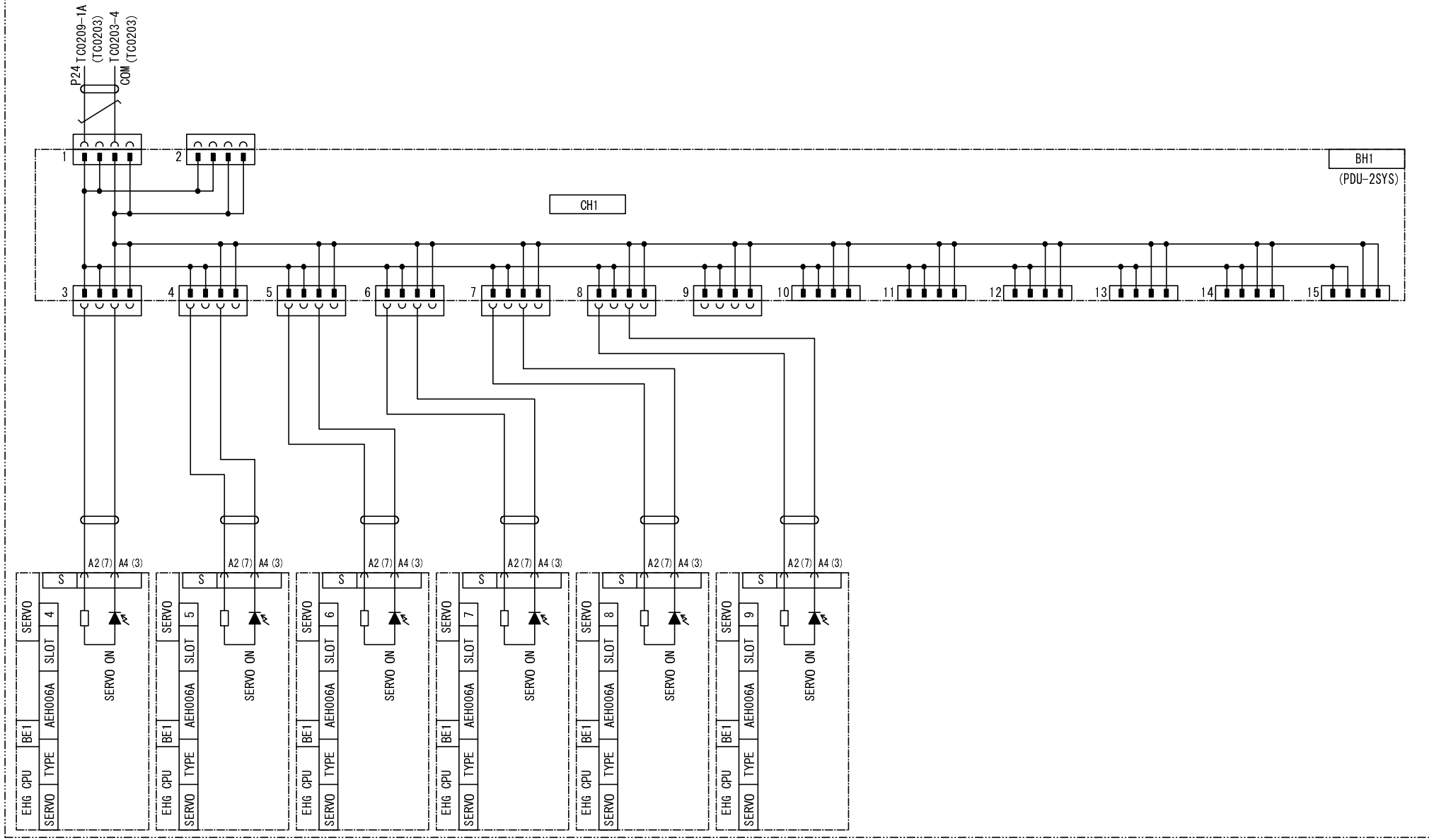
2016/07/28 TC0208 XSMM-TC



DIAGNOSIS CIRCUIT(2)		SH. No.	TC0209
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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TCS-2



DIAGNOSIS CIRCUIT(3)		SH. No.	TC0210
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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2016/07/28 TC0210 XSMM-TC

TCS-1 TCS-2

EHG CPU
BE1

PTI | TYPE | LYT000A | SLOT | A

(T)	CH. NO.	SIGNAL RANGE	SIGNAL NAME
3,4	0	0～5000rpm 0～10kHz	TURBINE SPEED(1)
7,8	1	0～5000rpm 0～10kHz	TURBINE SPEED(2)
11,12	2	0～5000rpm 0～10kHz	TURBINE SPEED(3)
15,16	3		(SPARE)

TC0114-1 (TC0114)

PTI | TYPE | LYT000A | SLOT | B

TC0114-2 (TC0114) 3,4,7,8,11,12,15,16 (DITTO)

PTI | TYPE | LYT000A | SLOT | C

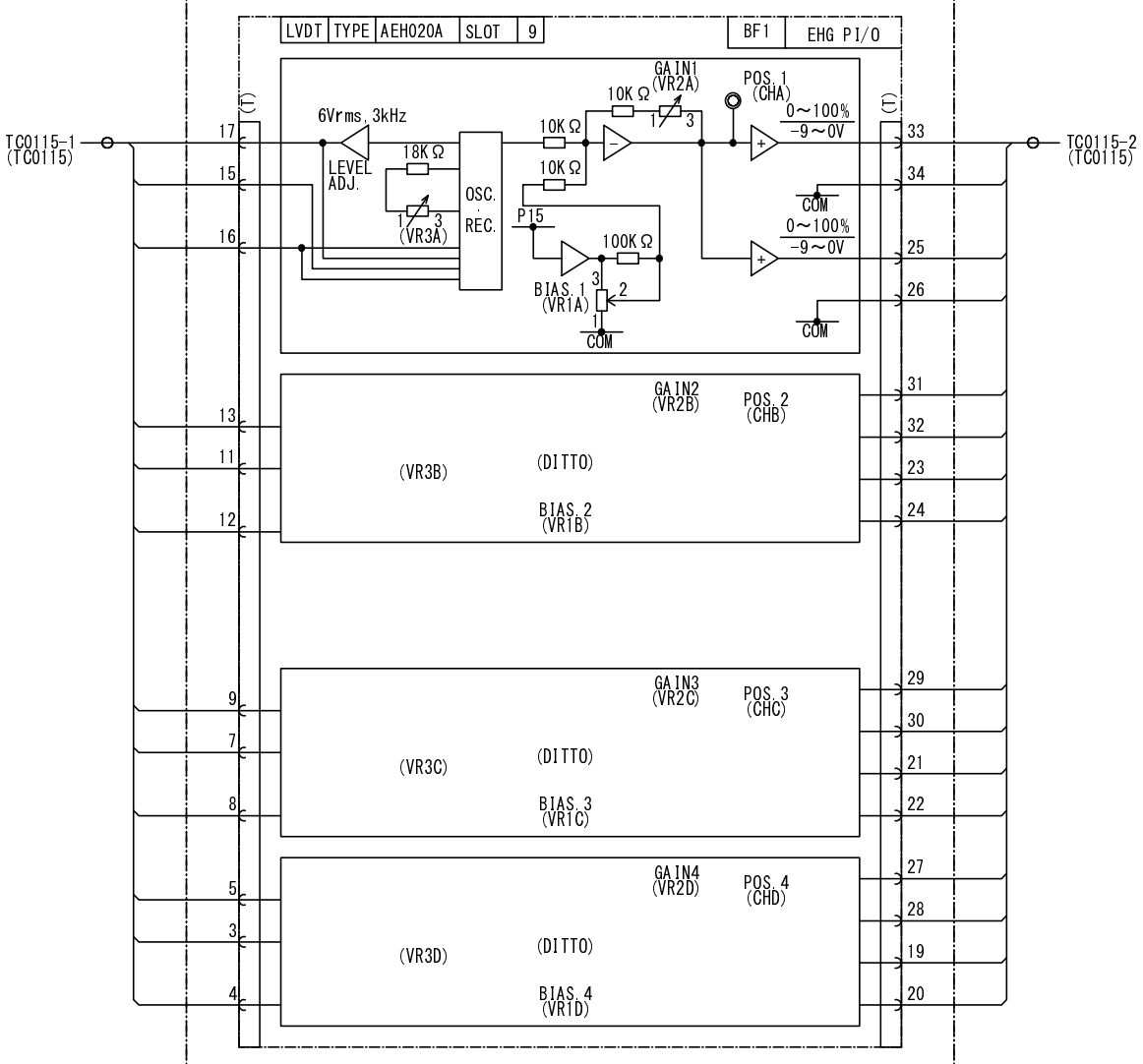
TC0114-3 (TC0114) 3,4,7,8,11,12,15,16 (DITTO)

LOGIC DIAGRAM SH. NO. | TCE980

TURBINE SPEED INPUT		SH. No.	TC0211
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 68 / REV. 3

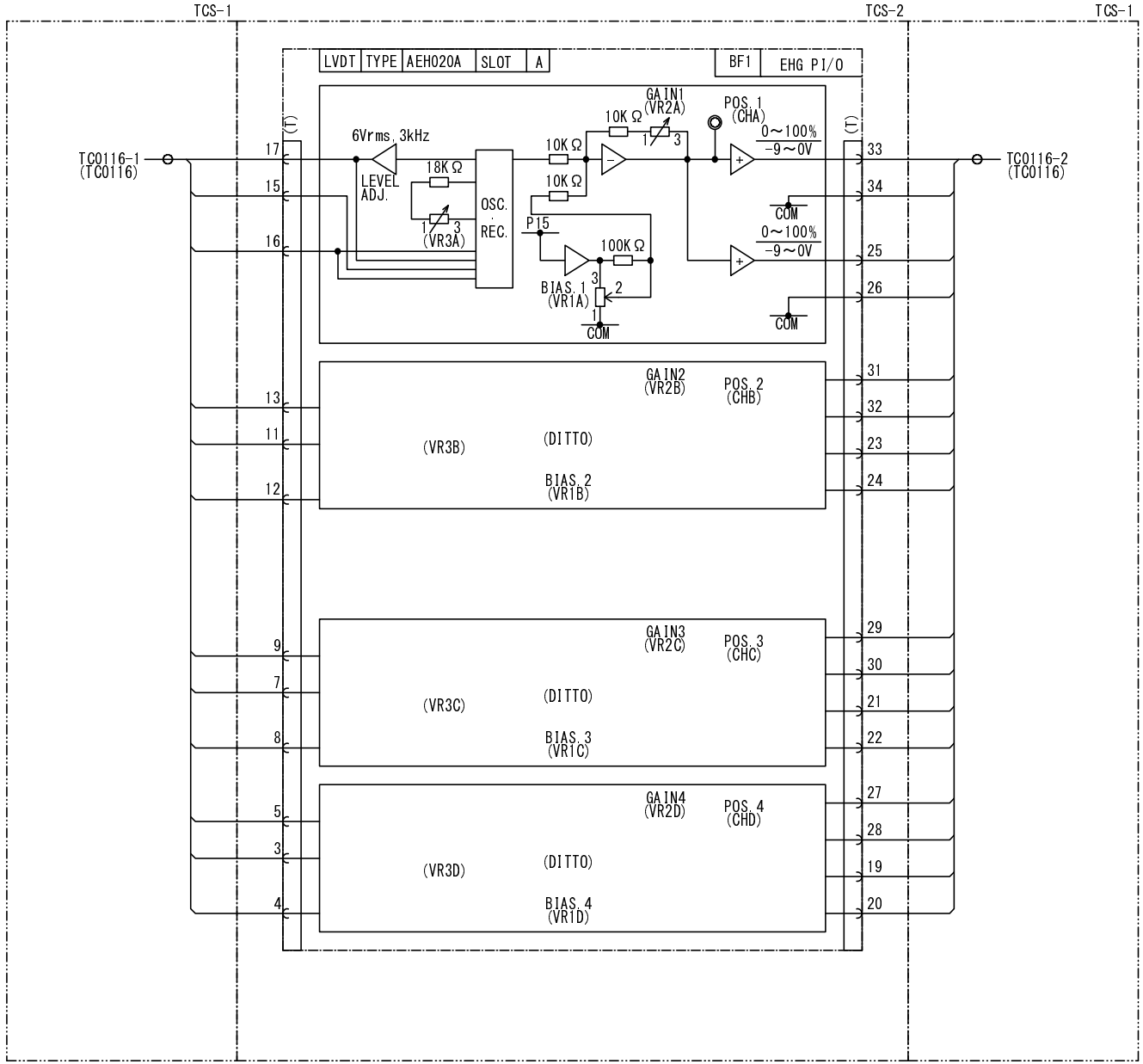
2016/07/28 TC0211 XSMM-TC

TCS-1 TCS-2 TCS-1



VALVE POSITION INPUT(1)		SH. No.	TC0212
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 69 / REV. 3

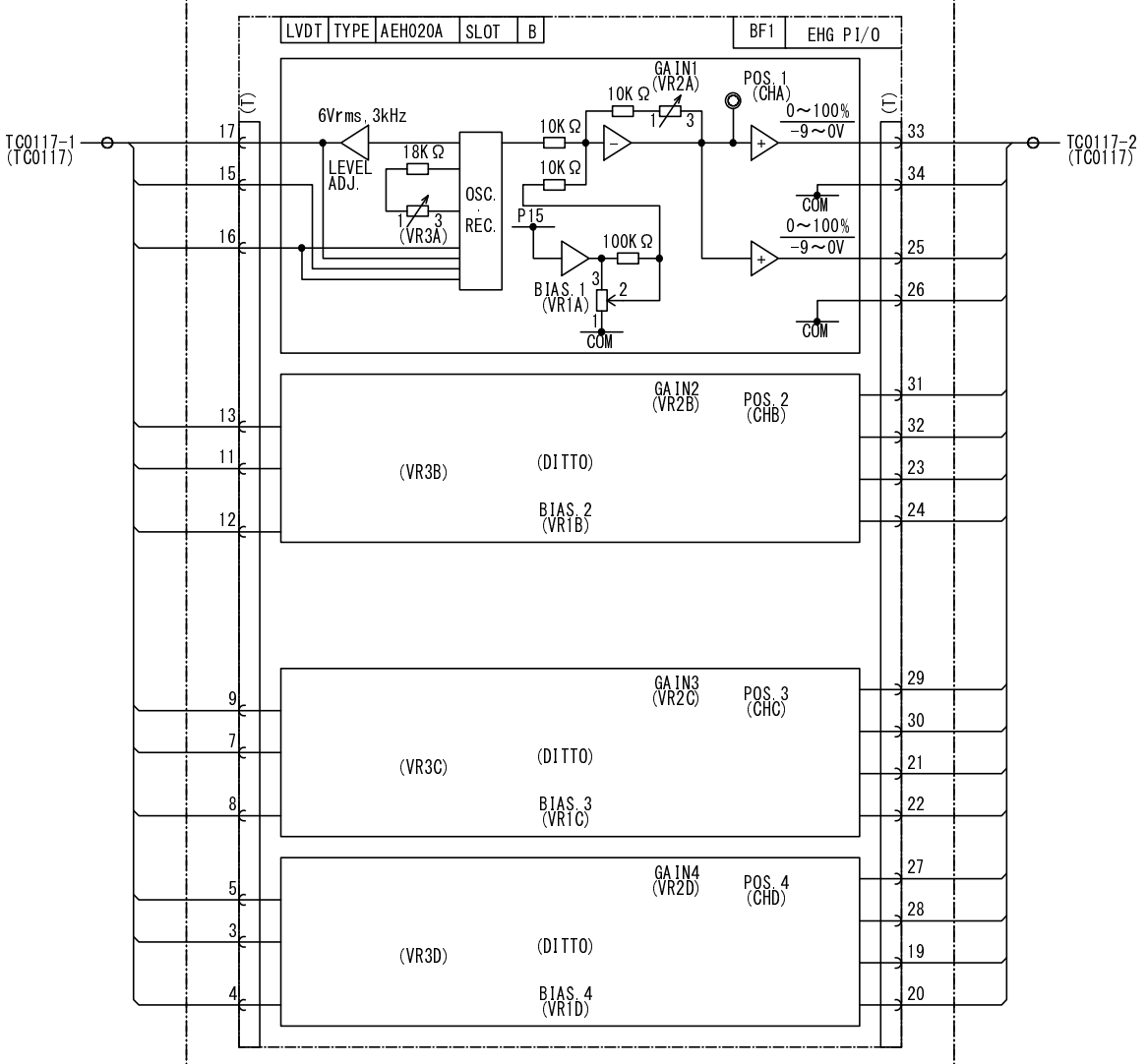
2016/07/28 TC0212 XSMM-TC



VALVE POSITION INPUT(2)		SH. No.	TC0213
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 70 / REV. 3

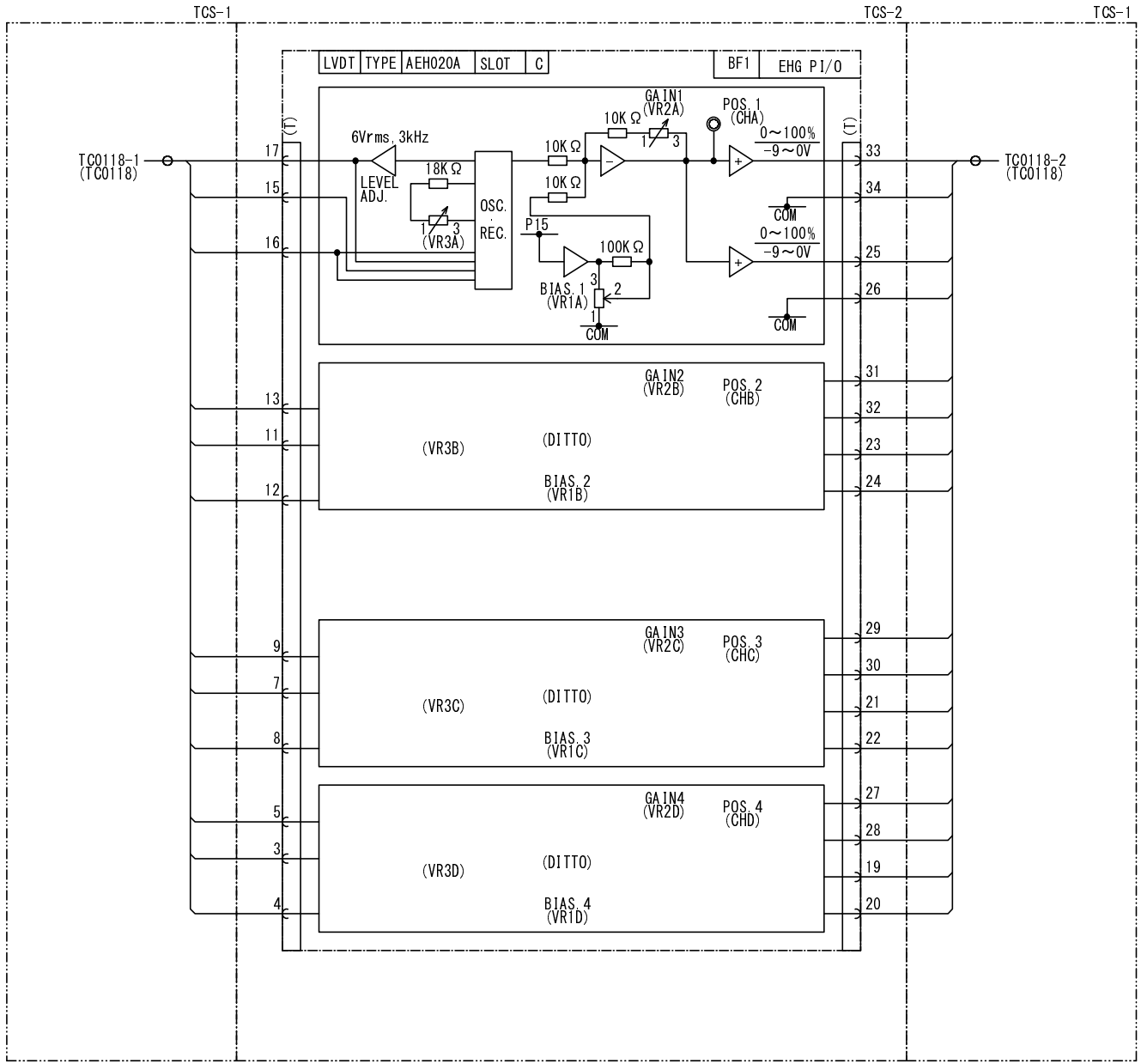
2016/07/28 TC0213 XSMM-TC

TCS-1 TCS-2 TCS-1



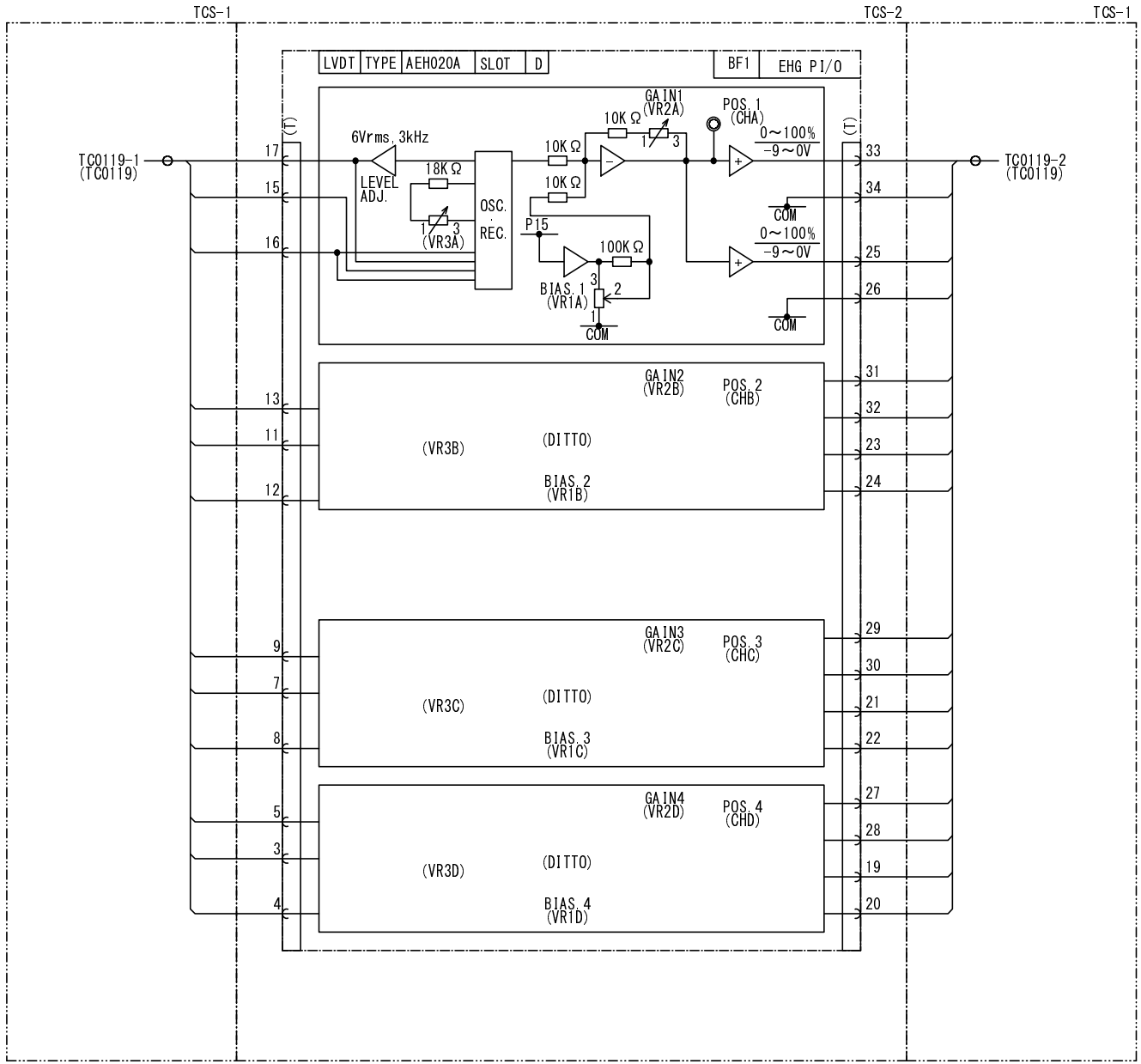
VALVE POSITION INPUT(3)		SH. No.	TC0214
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 71 / REV. 3

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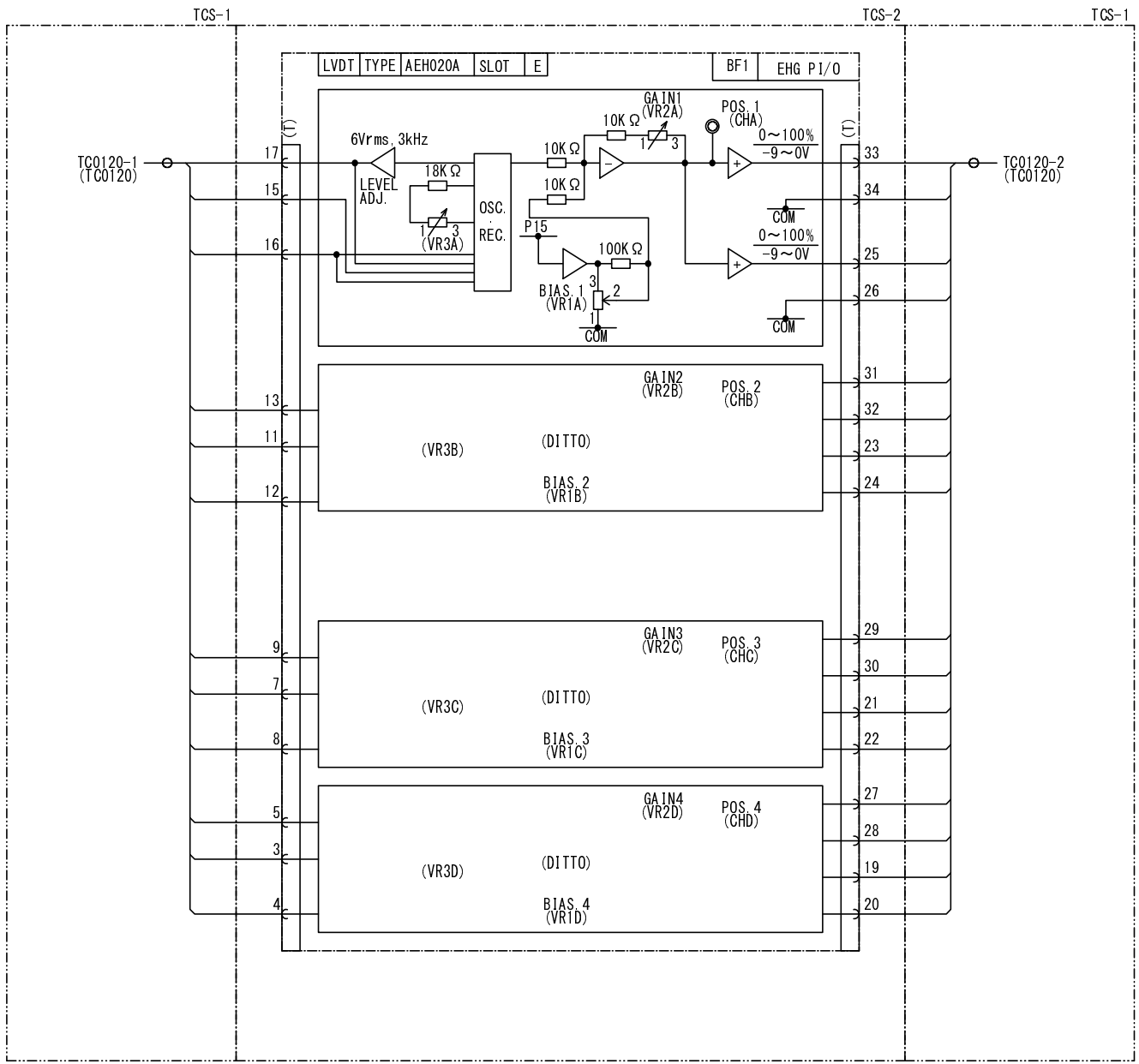
VALVE POSITION INPUT(4)		SH. No.	TC0215
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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VALVE POSITION INPUT(5)		SH. No.	TC0216
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 73 / REV. 3

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VALVE POSITION INPUT(6)		SH. No.	TC0217
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 74 / REV. 3

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TCS-2 TCS-1

EHG CPU

SERVO	TYPE	AEH006A	SLOT	4	UNIT No	BE1

SIGNAL NAME	RANGE	CH. NO.
MSV(A) SERVO VALVE	0～100% / 7～0V	O-0
MSV(A) POSITION	0～100% / -7～0V	I-0
MSV(A) SERVO VALVE	-100～100% / -9.56～9.56V	I-4
MSV(B) SERVO VALVE	0～100% / 7～0V	O-1
MSV(B) POSITION	0～100% / -7～0V	I-1
MSV(B) SERVO VALVE	-100～100% / -9.56～9.56V	I-5
CV(A) SERVO VALVE	0～100% / 5～0V	O-2
CV(A) POSITION	0～100% / -5～0V	I-2
CV(A) SERVO VALVE	-100～100% / -9.56～9.56V	I-6
CV(B) SERVO VALVE	0～100% / 5～0V	O-3
CV(B) POSITION	0～100% / -5～0V	I-3
CV(B) SERVO VALVE	-100～100% / -9.56～9.56V	I-7

24KΩ DIPS1 47KΩ GAIN.1 (VR2A) 3 1 50KΩ 390Ω DIPS2 1.0KΩ 1.2KΩ 1.77KΩ 1.77KΩ

COM COM

GAIN.2 (VR2B) (DITTO)

GAIN.3 (VR2C) (DITTO)

GAIN.4 (VR2D) (DITTO)

(T) 21 22 9 10 19 20 7 8 17 18 5 6 15 16 3 4

TC0121-1 (TC0121)

SERVO	TYPE	AEH006A	SLOT	5	UNIT No	BE1

(DITTO)

(T) 21 22 9 10 19 20 7 8 17 18 5 6 15 16 3 4

TC0121-2 (TC0121)

SERVO	TYPE	AEH006A	SLOT	6	UNIT No	BE1

(DITTO)

(T) 21 22 9 10 19 20 7 8 17 18 5 6 15 16 3 4

TC0121-3 (TC0121)

LOGIC DIAGRAM SH. NO.	TCE990 TCE991

OUTPUT TO SERVO VALVE(1)	SH. No.	TC0218

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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TCS-2 TCS-1

EHG CPU

SERVO	TYPE	AEH006A	SLOT	7	UNIT No	BE1

SIGNAL NAME	RANGE	CH. NO.
ICV(A) SERVO VALVE	0～100% / 9～0V	O-0
ICV(A) POSITION	0～100% / -9～0V	I-0
ICV(A) SERVO VALVE	-100～100% / -9.56～9.56V	I-4
ICV(B) SERVO VALVE	0～100% / 9～0V	O-1
ICV(B) POSITION	0～100% / -9～0V	I-1
ICV(B) SERVO VALVE	-100～100% / -9.56～9.56V	I-5
(SPARE)	/ 9～0V	O-2
RSV(A) POSITION	0～100% / -9～0V	I-2
(SPARE)	/ -7～7V	I-6
(SPARE)	/ 9～0V	O-3
RSV(B) POSISION	0～100% / -9～0V	I-3
(SPARE)	/ -7～7V	I-7

24KΩ DIPS1 47KΩ 1.77KΩ 1.77KΩ
GAIN.1 (VR2A) 3 1 50KΩ
390Ω DIPS2 1.0KΩ 1.2KΩ
COM
COM
GAIN.2 (VR2B) (DITTO)
GAIN.3 (VR2C) (DITTO)
GAIN.4 (VR2D) (DITTO)

(T) 21 22 9 10 19 20 7 8 17 18 5 6 15 16 3 4 → TC0122-1 (TC0122)

SERVO	TYPE	AEH006A	SLOT	8	UNIT No	BE1

(DITTO) → TC0122-2 (TC0122)

SERVO	TYPE	AEH006A	SLOT	9	UNIT No	BE1

(DITTO) → TC0122-3 (TC0122)

LOGIC DIAGRAM SH. NO.	TCE991 TCE992

OUTPUT TO SERVO VALVE(2)	SH. No.	TC0219

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD.	DWG. No.	PAGE	REV.
Hitachi, Ltd. Tokyo Japan	331DM07733	76	3

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TCS-1

TCS-2

TC125-1
(TC0125)

(LYD050A)

unit NO.	slot NO.	4 20	18	16	14	12	10	8	6 (S)
BF1	0								
CTL	CH. NO.	DI 0	DI 1	DI 2	DI 3	DI 4	DI 5	DI 6	DI 7
EHG+MD	signal range/level								
logic SH. NO.									
TCE900 DI	signal name	CONTROL OIL PUMP (A) RUNNING	CONTROL OIL PUMP (A) ELECTRICAL TROUBLE	MAIN OIL PUMP (A) RUNNING	MAIN OIL PUMP (A) MOTOR ELECTRICAL TROUBLE	DC EMERGENCY OIL PUMP RUNNING	DC EMERGENCY OIL PUMP ELECTRICAL TROUBLE	BEARING OIL PRESS. (FOR EOP AUTO START)	HAND TURNING INTERLOCK POSITION

EHG DIGITAL INPUT(1)		SH. No.	TC0220
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
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2016/07/28 TC0220 XSMM-TC

TCS-1

TCS-2

TC0125-1
(TC0126)

(LYD050A)

unit NO.	slot NO.	3 19	17	15	13	11	9	7	5 (S)
BF1	0								
CTL	CH. NO.	DI 8	DI 9	DI A	DI B	DI C	DI D	DI E	DI F
EHG+MD	signal range/level								
logic SH. NO.									
TCE900 DI	signal name	TURNING GEAR ENGAGE POSITION	TURNING GEAR DISENGAGE POSITION	TURNING GEAR MOTOR ENGAGE PB					P.S.NORMAL

EHG DIGITAL INPUT(2)		SH. No.	TC0221
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
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TCS-1

TCS-2

TC0127-1
(TC0127)

(LYD050A)

unit NO.	slot NO.	4 20	18	16	14	12	10	8	6 (T)
BF1	0								
CTL	CH. NO.	DI 10	DI 11	DI 12	DI 13	DI 14	DI 15	DI 16	DI 17
EHG+MD	signal range/level								
logic SH. NO.									
TCE900	signal name	DCS CONTROL	DCS CONTROL	DS CLOSED	52G CLOSED	52G OPENED	SPEED RAISE	SPEED LOWER	41 CLOSE
DI		MODE ON	MODE OFF						

EHG DIGITAL INPUT(3)			SH. No.	TC0222
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
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TCS-1

TCS-2

TC0127-1
(TC0128)

(LYD050A)

unit NO.	slot NO.	3 19	17	15	13	11	9	7	5 (T)
BF1	0								
CTL	CH. NO.	DI 18	DI 19	DI 1A	DI 1B	DI 1C	DI 1D	DI 1E	DI 1F
EHG+MD	signal range/level								
logic SH. NO.									
TCE900	signal name	41 OPEN	52L OPEN	HP BYPASS VALVE FULL CLOSE					P.S.NORMAL
DI									

EHG DIGITAL INPUT(4)		SH. No.	TC0223
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
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2016/07/28 TC0223 XSMM-TC

TCS-1

TCS-2

TC129-1
(TC0129)

(LYD050A)

unit NO.	slot NO.	4 20	18	16	14	12	10	8	6	(S)
BF1	1									
CTL	CH. NO.	DI 0	DI 1	DI 2	DI 3	DI 4	DI 5	DI 6	DI 7	
EHG+MD	signal range/level									
logic SH. NO.										
TCE901	signal name	CONTROL OIL PUMP	CONTROL OIL PUMP	MAIN OIL PUMP	MAIN OIL PUMP	TTURNING GEAR	TURNING GEAR	TURNINNG GEAR	AUTO SYN START PB	
DI		(B) RUNNING	(B) ELECTRICAL TROUBLE	(B) RUNNING	(B) MOTOR ELECTRICAL TROUBLE	MOTOR JOG PB	MOTOR RUNNING	MOTOR ELECTRICAL TROUBLE		

EHG DIGITAL INPUT(5)		SH. No.	TC0224
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 81	REV. 3

TCS-1

TCS-2

TC0129-1
(TC0130)

(LYD050A)

(S)

unit NO.	slot NO.	3	19	17	15	13	11	9	7	5
BF1	1									
CTL	CH. NO.	DI 8		DI 9	DI A	DI B	DI C	DI D	DI E	DI F
EHG+MD	signal range/level									
logic SH. NO.										
TCE901	signal name	AUTO SYNC								P.S.NORMAL
DI		REMOTE SELECT								

EHG DIGITAL INPUT(6)				SH. No.	TC0225
SIGNATURE		DATE	TITLE		
DWN.	(S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM		
CHKD.	T.Kawata	2014-04-18			
APPD.	Y.Fukushima	2014-04-18			
REGD.	Hitachi, Ltd. Tokyo Japan		DWG. No. 331DM07733	PAGE 82	REV. 3

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TCS-1

TCS-2

TC0131-1
(TC0131)

(LYD050A)

unit NO.	slot NO.	4 20	18	16	14	12	10	8	6	(T)
BF1	1									
CTL	CH. NO.	DI 10	DI 11	DI 12	DI 13	DI 14	DI 15	DI 16	DI 17	
EHG+MD	signal range/level									
logic SH. NO.										
TCE901	signal name									
DI										

EHG DIGITAL INPUT(7)			SH. No.	TC0226
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 83	REV. 3

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TCS-1

TCS-2

TC0131-1
(TC0132)

(LYD050A)

unit NO.	slot NO.	3 19	17	15	13	11	9	7	5 (T)
BF1	1								
CTL	CH. NO.	DI 18	DI 19	DI 1A	DI 1B	DI 1C	DI 1D	DI 1E	DI 1F
EHG+MD	signal range/level								
logic SH. NO.									
TCE901	signal name			COM CPU(A) NORMAL	COM CPU(B) NORMAL	COM P.S.(A) ABNORMAL	COM P.S.(B) ABNORMAL	PLU OPERATED	P.S.NORMAL
DI									

EHG DIGITAL INPUT(8)			SH. No.	TC0227
SIGNATURE		DATE	TITLE	
DWN.	(S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T.Kawata	2014-04-18		
APPD.	Y.Fukushima	2014-04-18		
REGD.	Hitachi,Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 84	REV. 3

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TCS-1

TCS-2

TC0140-1
(TC0140)

(LYA000A)

unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
BF1	5								(T)
CTL	CH.NO.	AI 0	AI 1	AI 2	AI 3	AI 4	AI 5	AI 6	AI 7
EHG+MD	signal range/level	0～150kg/cm2	0～10kg/cm2	0～250kg/cm2	0～250kg/cm2	0～200MW			
logic SH.NO.		1～5V	1～5V	1～5V	1～5V	1～5V			
TCE930 AI	signal name	MSV INLET STEAM PRESS. (A)	MOP (A) DISCHARGE OIL PRESS. (FOR MOP (B) AUTO STAR	COP (A) DISCHARGE OIL PRESS. (FOR COP (B) AUTO STAR	HOT REHEAT STEAM PRESS.	TARGET MW DEMAND			

EHG ANALOG INPUT(1)		SH. No.	TC0230
SIGNATURE		DATE	TITLE
DWN.	(S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T.Kawata	2014-04-18	
APPD.	Y.Fukushima	2014-04-18	
REGD.	Hitachi,Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 85 / REV. 3

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TCS-1

TCS-2

TC0141-1
(TC0141)

(LYA000A)

unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
BF1	6								(T)
CTL	CH.NO.	AI 0	AI 1	AI 2	AI 3	AI 4	AI 5	AI 6	AI 7
EHG+MD	signal range/level	0～150kg/cm2	0～10kg/cm2	0～250kg/cm2	0～5kg/cm2				
logic SH.NO.		1～5V	1～5V	1～5V	1～5V				
TCE930 AI	signal name	MSV INLET STEAM PRESS. (B)	MOP (B) DISCHARGE OIL PRESS. (FOR MOP (A) AUTO STAR	COP (B) DISCHARGE OIL PRESS. (FOR COP (A) AUTO STAR	EOP DISCHARGE OIL PRESS.				

EHG ANALOG INPUT(2)			SH. No.	TC0231
SIGNATURE		DATE	TITLE	
DWN.	(S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T.Kawata	2014-04-18		
APPD.	Y.Fukushima	2014-04-18		
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TCS-2

(LYD105A)

DO / logic SH. NO.	signal name	CONTROL OIL PUMP (A) START COMMAND	MAIN OIL PUMP (A) START COMMAND	DC EMERGENCY OIL PUMP START	DC EMERGENCY OIL PUMP MANUAL STOP		TURNING GEAR MOTOR START COMMAND		TURNING GEAR ENGAGED
TCE960	signal range/level								
CTL									
EHG+MD	CH. NO.	DO 0	DO 1	DO 2	DO 3	DO 4	DO 5	DO 6	DO 7
unit NO.	slot NO.								
BF1	2	21 3 20	18	16	14	12	10	8	6 (S)

TCS-1

TC0150-1 (TC0150)

EHG DIGITAL OUTPUT(1)			SH. No.	TC0240
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
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TCS-2

(LYD105A)

DO / logic SH. NO.	signal name					RSV&ICV FULL CLOSE	MSV&ICV FULL CLOSE		EHG SOFT MAJOR TROUBLE OR BUG (0:TRIP 1:NORMAL)
TCE960	signal range/level								
CTL									
EHG+MD	CH. NO.	DO 8	DO 9	DO A	DO B	DO C	DO D	DO E	DO F
unit NO.	slot NO.								
BF1	2	22 4 19	17	15	13	11	9	7	5 (S)

TCS-1

TC0150-1 (TC0151)

EHG DIGITAL OUTPUT(2)

SH. No. TC0241

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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TCS-2

(LYD105A)

DO / logic SH. NO.	signal name	52L OPEN FOR HOUSE LOAD OPERATION	TURBINE SPEED UNDER 2400rpm	ACCELERATION RELAY OPERATED	DCS MODE ON FEEDBACK			ASD2 START SIGNAL (52L)
TCE960	signal range/level							
CTL								
EHG+MD	CH. NO.	DO 10	DO 11	DO 12	DO 13	DO 14	DO 15	DO 16	DO 17
unit NO.	slot NO.							
BF1	2	21 3 20	18	16	14	12	10	8	6

(T)

TCS-1

TC0152-1 (TC0152)

EHG DIGITAL OUTPUT(3)			SH. No.	TC0242
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
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TCS-2

(LYD105A)

DO / logic SH. NO.	signal name	HP-CV FULL CLOSE (BOTH CV(A)&(B))	TURBINE TRIP (MSV, RSV, ICV CLOSE)	MSV/CV(A) SEQUENTIAL FULL CLOSE TEST	MSV/CV(B) SEQUENTIAL FULL CLOSE TEST	41 CLOSED	GEN. LOCKOUT RELAY (86G) A OPERATED	GEN. LOCKOUT RELAY (86G) B OPERATED	ICV TO CV TRANSFER COMPLETE
TCE960	signal range/level								
CTL									
EHG+MD	CH. NO.	DO 18	DO 19	DO 1A	DO 1B	DO 1C	DO 1D	DO 1E	DO 1F
unit NO.	slot NO.								
BF1	2	22 4 19	17	15	13	11	9	7	5

(T)

TCS-1

TC0152-1
(TC0153)

EHG DIGITAL OUTPUT(4)		SH. No.	TC0243
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 90 / REV. 3

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TCS-2

(LYD105A)

DO / logic SH. NO.	signal name	CONTROL OIL PUMP (B) START COMMAND	MAIN OIL PUMP (B) START COMMAND	ASD1 START SIGNAL (52G)		DCS MODE PERMISSIVE CONDITION			POWER LOAD UNBALANCE RELAY OPERATED
TCE961	signal range/level								
CTL									
EHG+MD	CH. NO.	DO 0	DO 1	DO 2	DO 3	DO 4	DO 5	DO 6	DO 7
unit NO.	slot NO.								
BF1	3	21 3 20	18	16	14	12	10	8	6 (S)

TCS-1

TC0154-1 (TC0154)

EHG DIGITAL OUTPUT(5)			SH. No.	TC0244
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
REGD.	Hitachi, Ltd. Tokyo Japan		DWG. No. 331DM07733	PAGE 91 / REV. 3

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TCS-2

(LYD105A)

DO / logic SH. NO.	signal name					RSV&ICV FULL CLOSE	MSV&ICV FULL CLOSE	EHG SOFT NOT MAJOR TROUBLE	EHG SOFT MAJOR TROUBLE OR BUG (0:TRIP 1:NORMAL)
TCE961	signal range/level								
CTL									
EHG+MD	CH. NO.	DO 8	DO 9	DO A	DO B	DO C	DO D	DO E	DO F
unit NO.	slot NO.								
BF1	3	22 4 19	17	15	13	11	9	7	5

(S)

TCS-1

TC0154-1 (TC0155)

EHG DIGITAL OUTPUT(6)

SH. No. TC0245

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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TCS-2

(LYD105A)

DO / logic SH. NO.	signal name	TURNING GEAR ENGAGE SOLENOID VALVE	TURNING GEAR ENGAGE (LAMP)	TURNING GEAR MOTOR START (LAMP)					
TCE961	signal range/level								
CTL									
EHG+MD	CH. NO.	DO 10	DO 11	DO 12	DO 13	DO 14	DO 15	DO 16	DO 17
unit NO.	slot NO.								
BF1	3	21 3 20	18	16	14	12	10	8	6

(T)

TCS-1

TC0156-1
(TC0156)

EHG DIGITAL OUTPUT(7)			SH. No.	TC0246
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
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TCS-2

(LYD105A)

DO / logic SH. NO.	signal name									
TCE961	signal range/level									
CTL										
EHG+MD	CH. NO.	DO 18	DO 19	DO 1A	DO 1B	DO 1C	DO 1D	DO 1E	DO 1F	
unit NO.	slot NO.									
BF1	3	22 4 19	17	15	13	11	9	7	5	(T)

TCS-1

TC0156-1
(TC0157)

EHG DIGITAL OUTPUT(8)			SH. No.	TC0247
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 94	REV. 3

TCS-2

(LYA100A)

AO / logic SH. NO.	signal name	TARGET MW SET POINT FEED BACK	MAX TURBINE LOAD UP RATE	HP TURBINE FIRST STAGE STEAM PRESS (A)	HP TURBINE FIRST STAGE STEAM PRESS (B)	HP TURBINE FIRST STAGE STEAM PRESS (C)	TURBINE SPEED		
TCE970	signal range/level	0～200MW	0～150%/min	0～150kg/cm2	0～150kg/cm2	0～150kg/cm2	0～4000rpm		
CTL		4～20mA	4～20mA	4～20mA	4～20mA	4～20mA	4～20mA		
EHG+MD	CH. NO.	AO 0	AO 1	AO 2	AO 3	AO 4	AO 5	AO 6	AO 7
unit NO.	slot NO.								
BF1	7	33 34	29 30	25 26	21 22	17 18	13 14	9 10	5 6

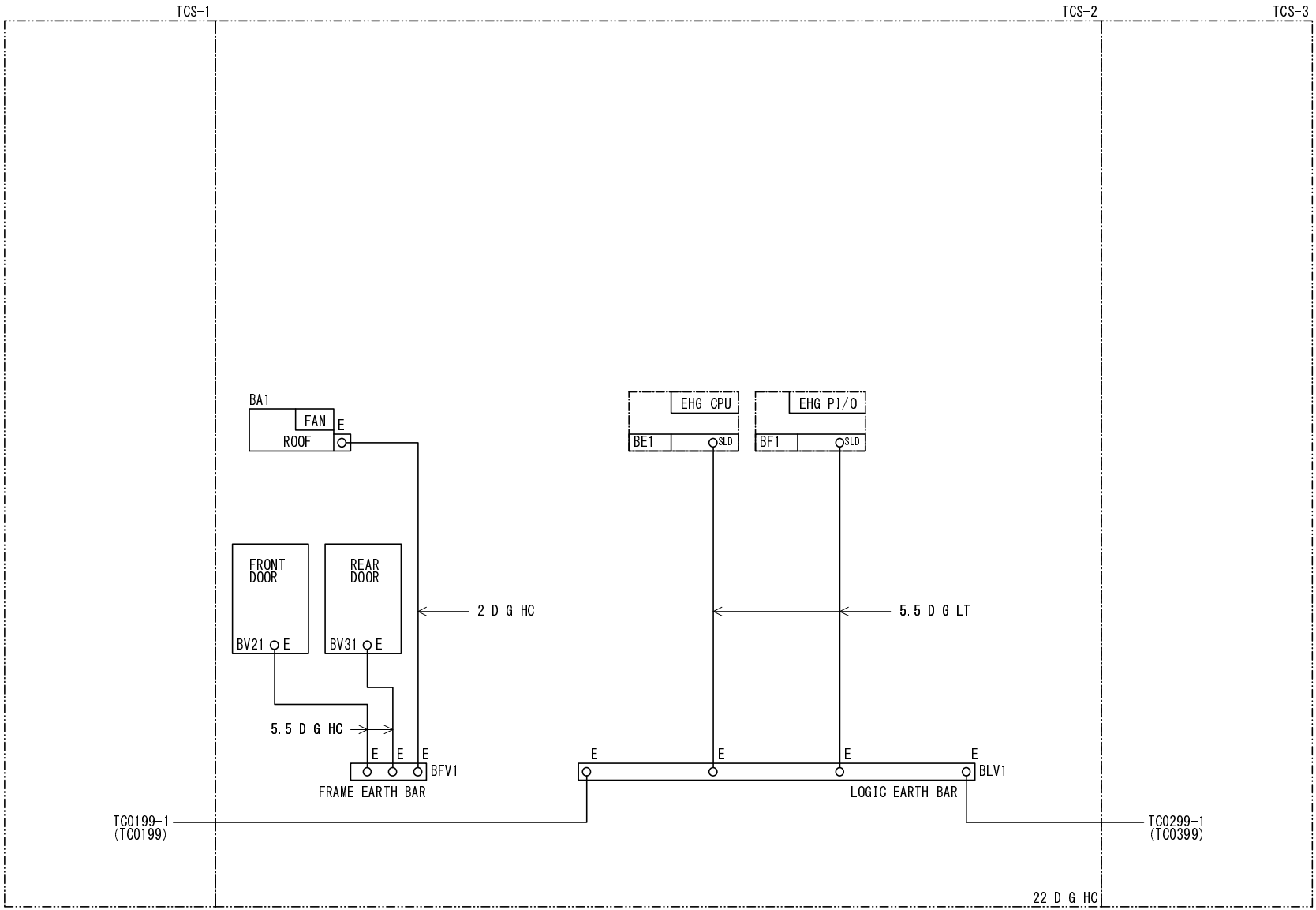
(T)

TCS-1

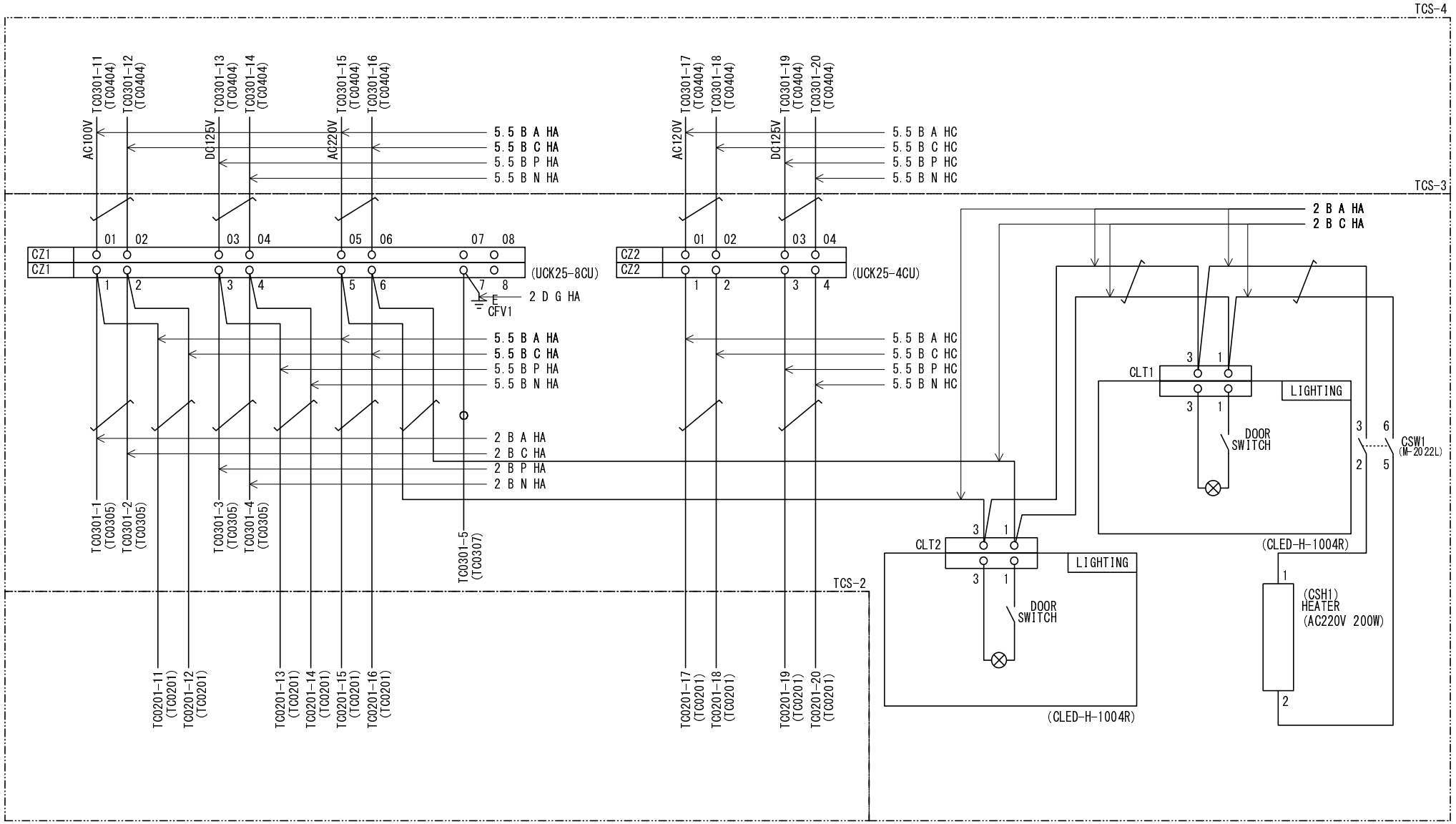
TC0160-1
(TC0160)

EHG ANALOG OUTPUT			SH. No.	TC0250
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
REGD.	Hitachi, Ltd. Tokyo Japan		DWG. No. 331DM07733	PAGE 95 / REV. 3

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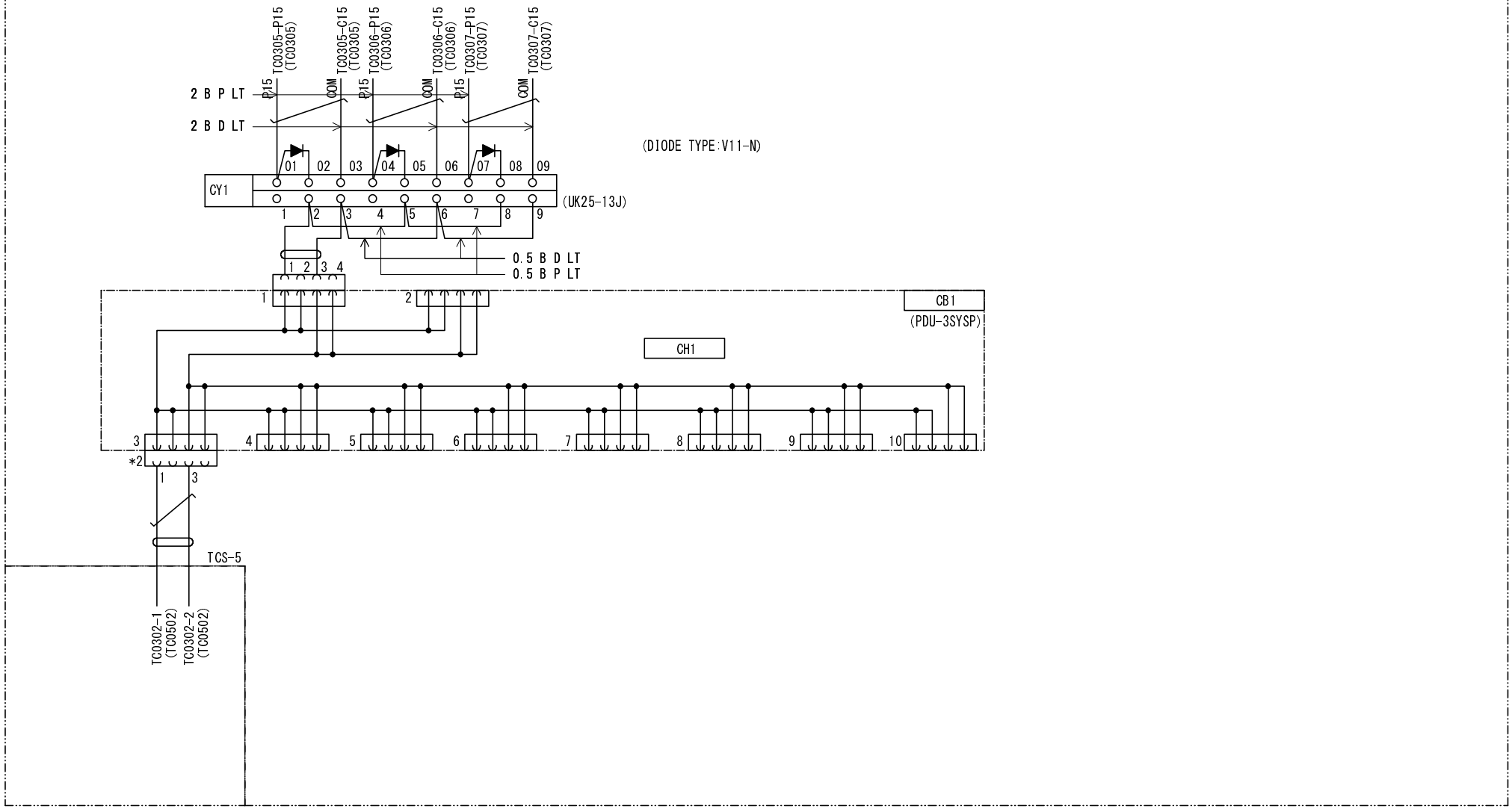


POWER SUPPLY SYSTEMEARTH CIRCUIT		SH. No.	TC0299
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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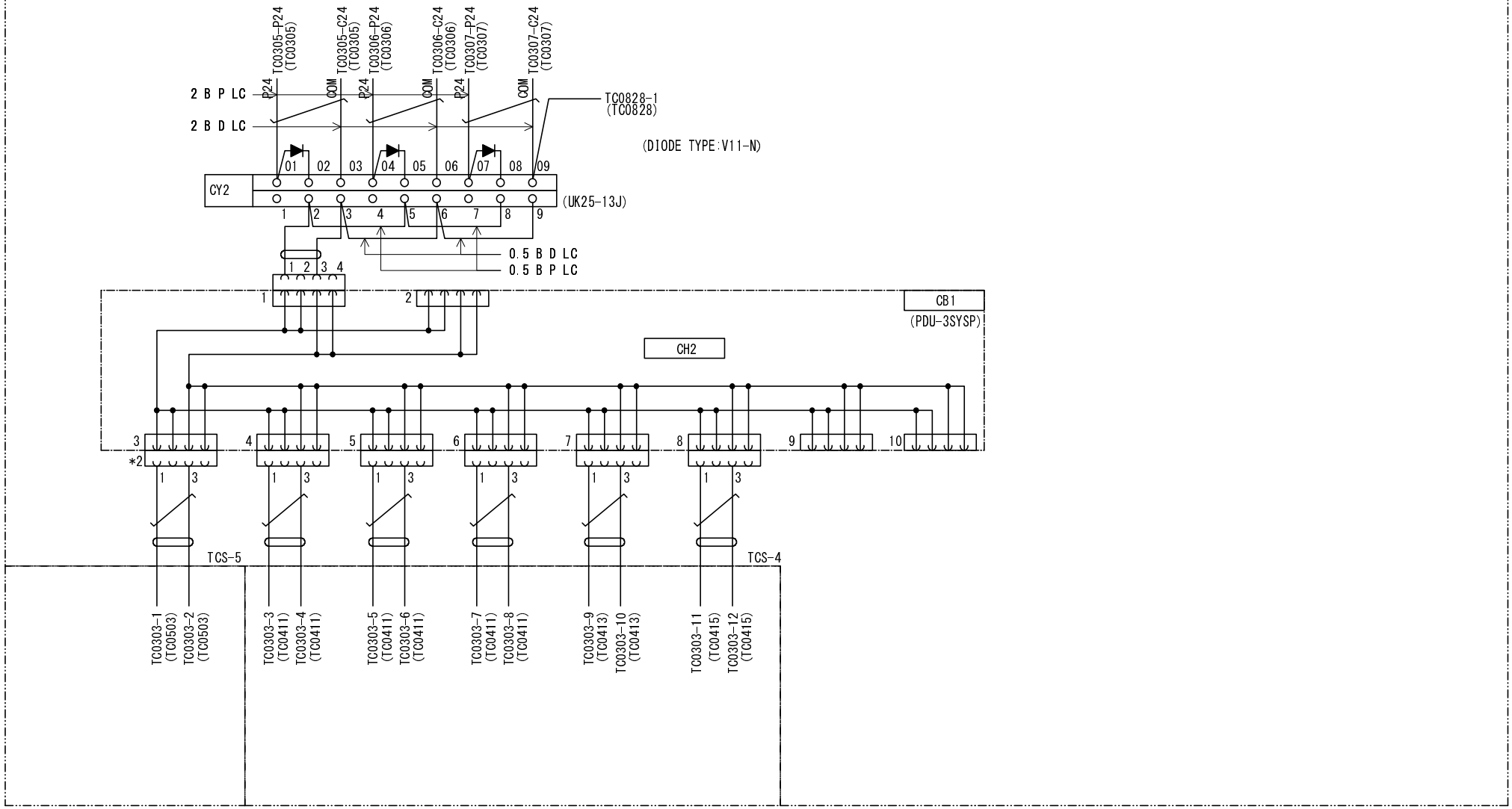


POWER SUPPLY SYSTEM(1)		SH. No.	TC0301
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 97	REV. 3

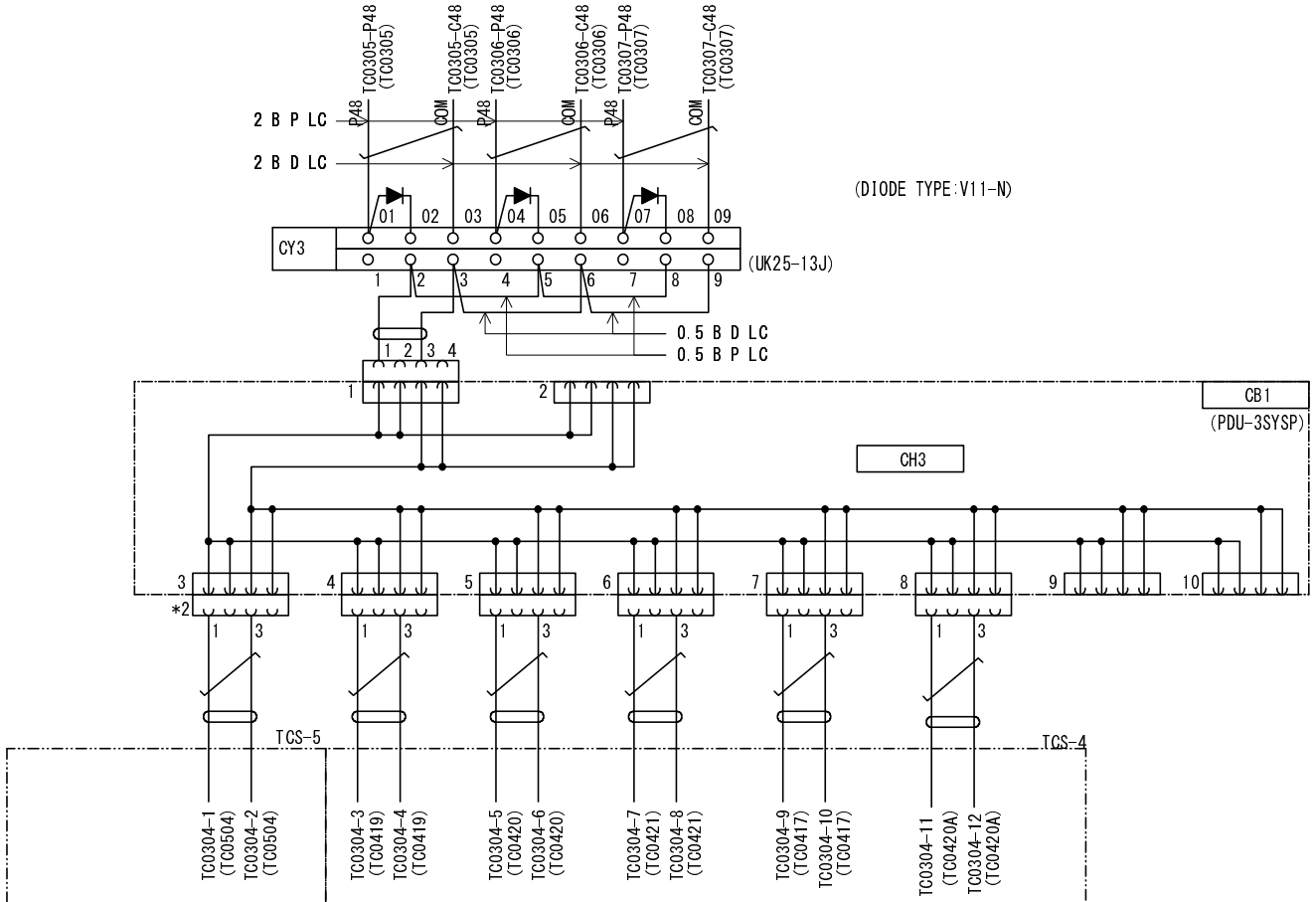
2016/07/28 TC0301 XSMM-TC



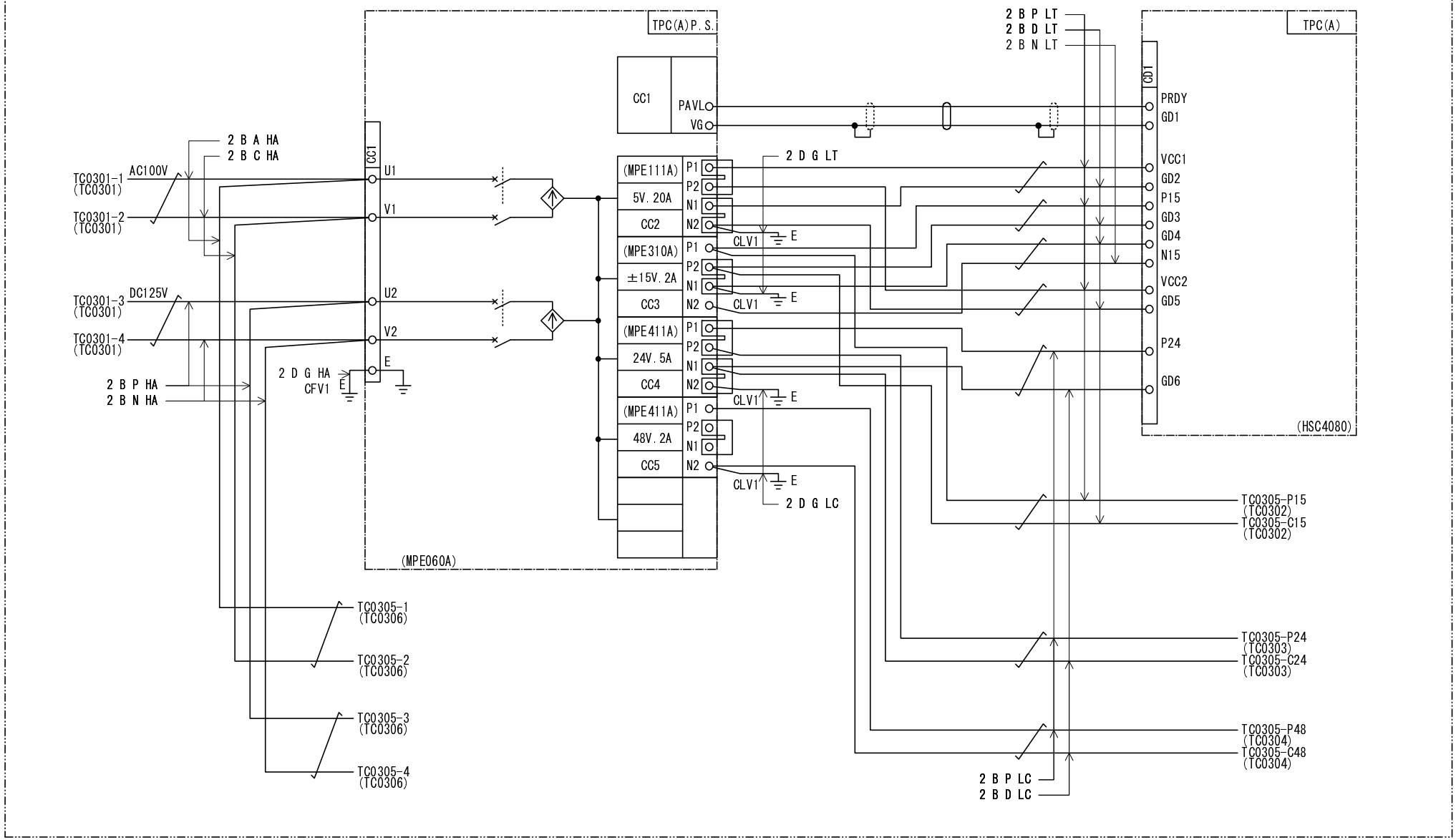
POWER SUPPLY SYSTEM(2)		SH. No.	TC0302
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
REGD. Hitachi,Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 98	REV. 3



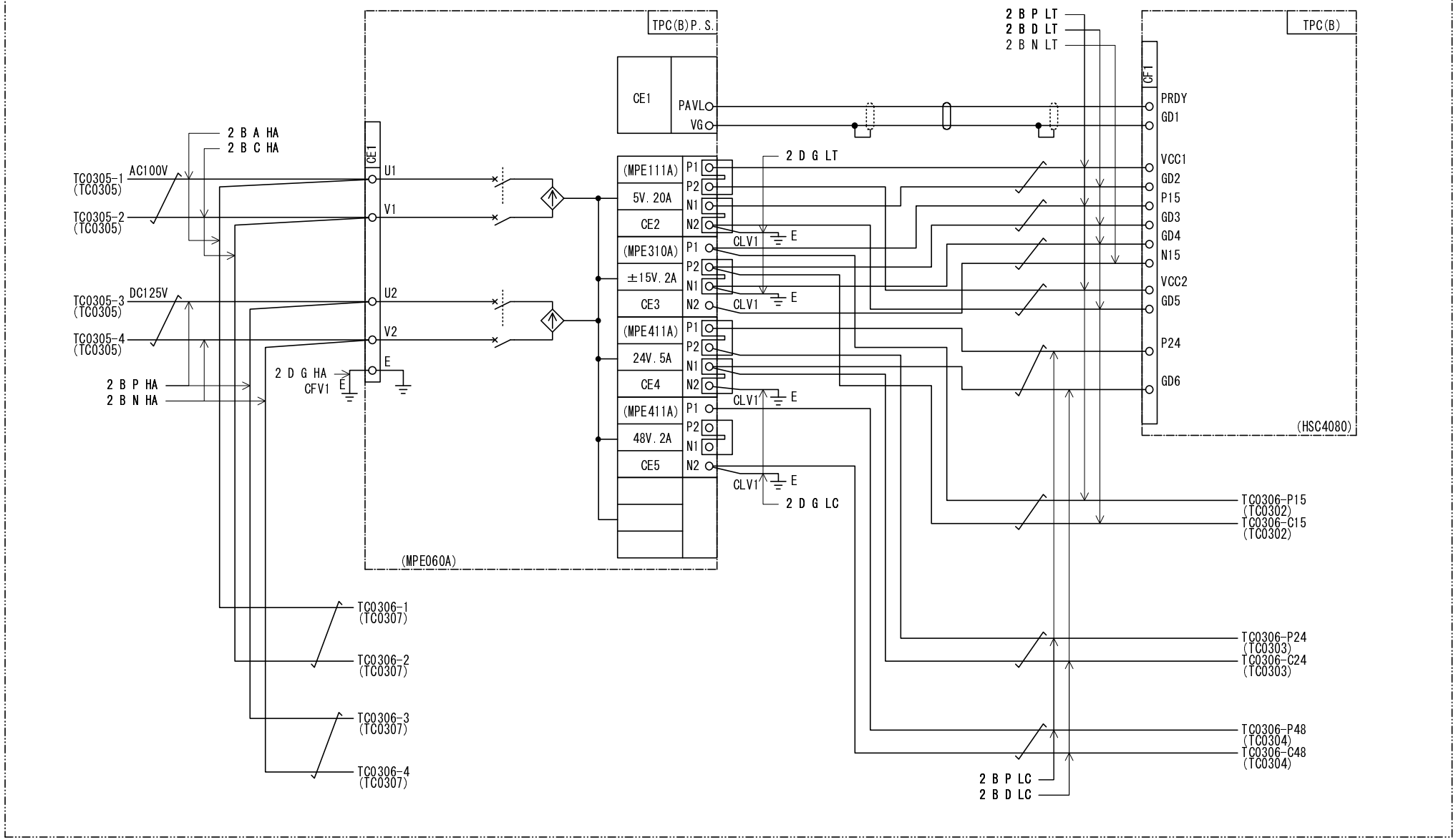
POWER SUPPLY SYSTEM(3)		SH. No.	TC0303
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
REGD. Hitachi,Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 99	REV. 3



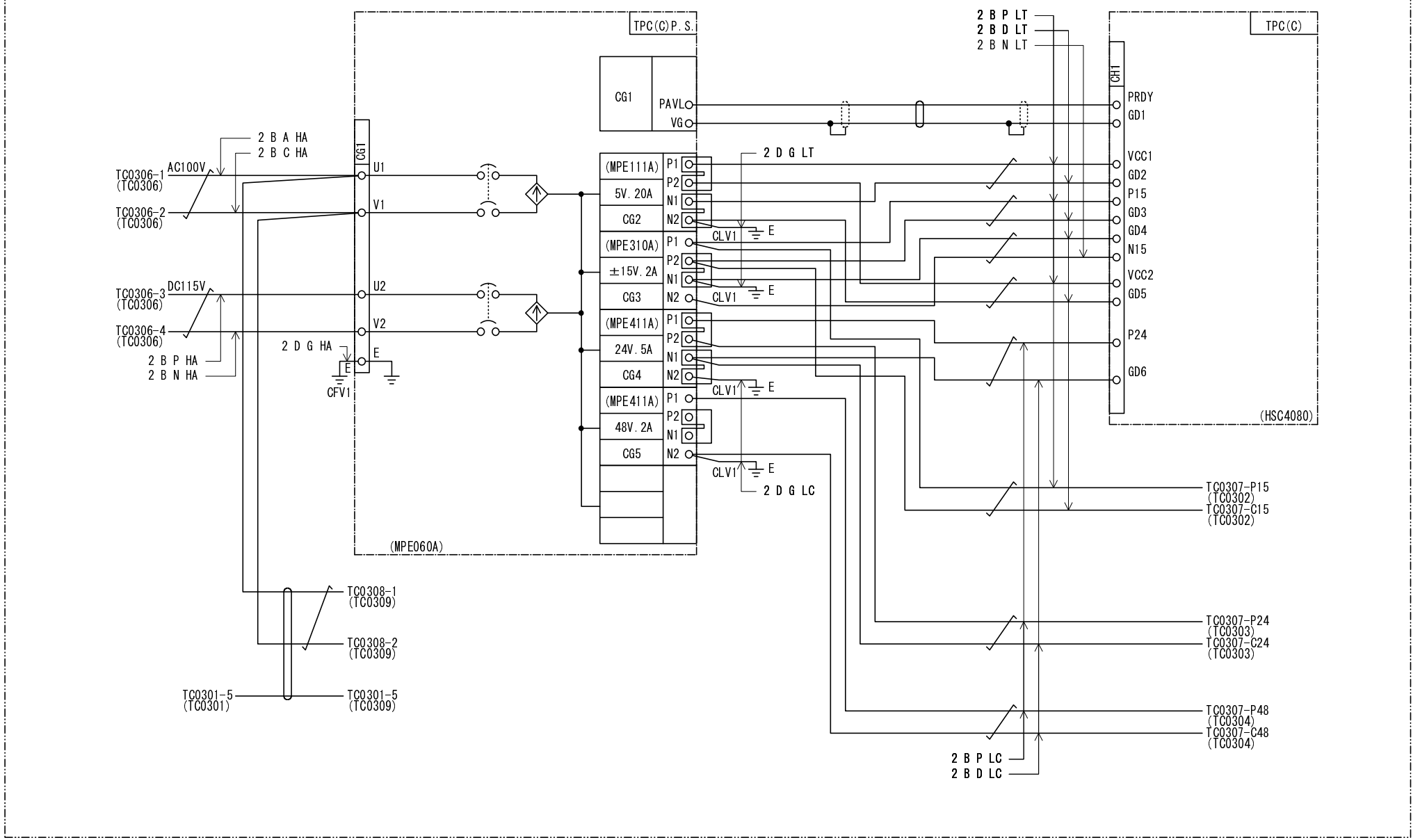
POWER SUPPLY SYSTEM(4)				SH. No.	TC0304
SIGNATURE		DATE	TITLE		
DWN.	(S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM		
CHKD.	T.Kawata	2014-04-18			
APPD.	Y.Fukushima	2014-04-18			
REGD.	Hitachi,Ltd. Tokyo Japan		DWG. No. 331DM07733	PAGE 100	REV. 3



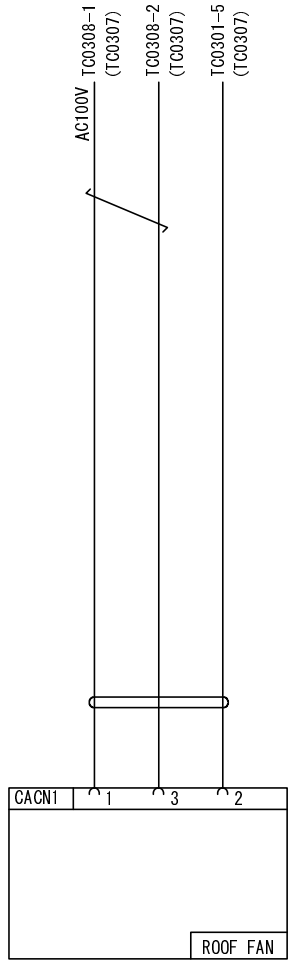
POWER SUPPLY SYSTEM(5)		SH. No.	TC0305
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 101 / REV. 3



POWER SUPPLY SYSTEM(6)		SH. No.	TC0306
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 102 / REV. 3

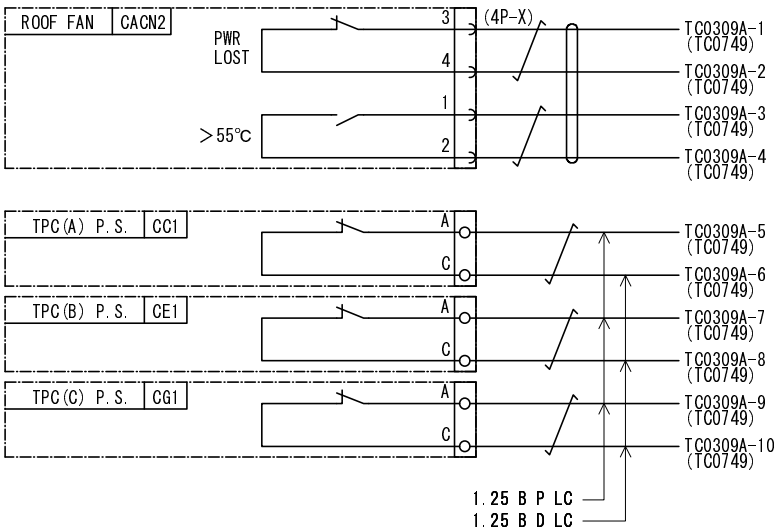


POWER SUPPLY SYSTEM(7)		SH. No.	TC0307
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
REGD. Hitachi,Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 103	REV. 3



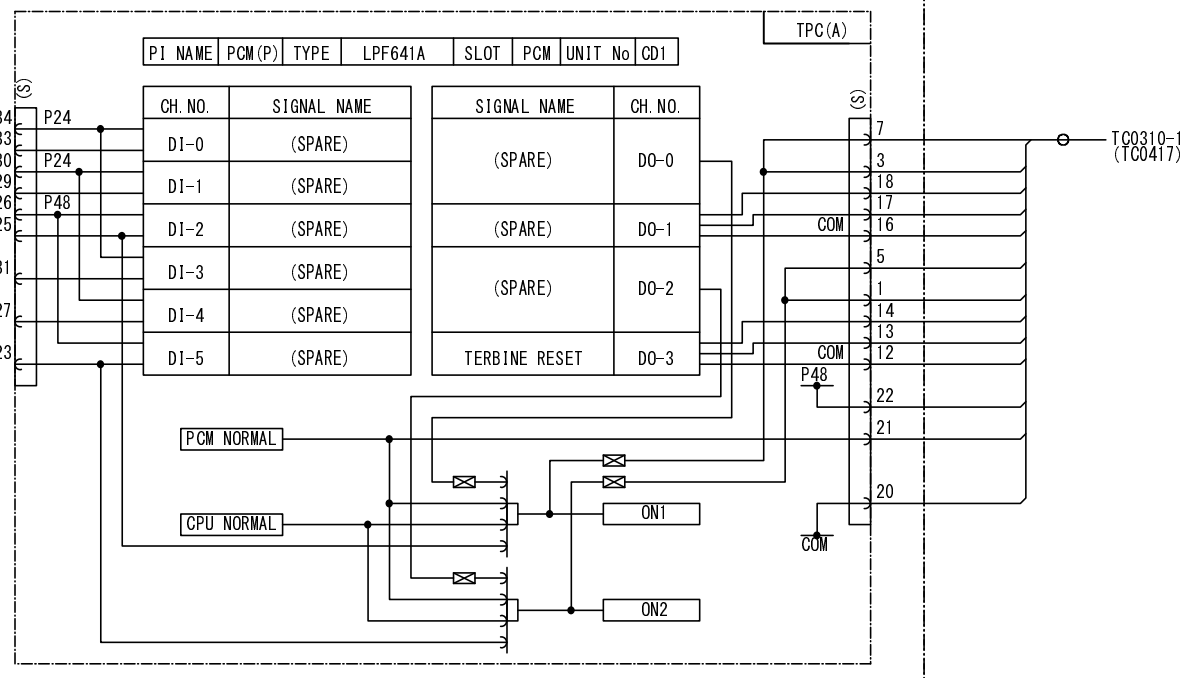
POWER SUPPLY SYSTEM(8)				SH. No.	TC0309
SIGNATURE		DATE	TITLE		
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM		
CHKD.	T. Kawata	2014-04-18			
APPD.	Y. Fukushima	2014-04-18			
REGD.	Hitachi, Ltd. Tokyo Japan		DWG. No. 331DM07733	PAGE 104	REV. 3

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DIAGNOSIS CIRCUIT		SH. No.	TC0309A
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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TRIP PCM INPUT/OUTPUT(1)	※	SH. No.	TC0310
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 106	REV. 3

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TCS-4 TCS-3 TCS-4

TPC(B)

PI NAME	PCM(P)	TYPE	LPF641A	SLOT	PCM	UNIT No	CF1

(S)

CH. NO.	SIGNAL NAME
DI-0	(SPARE)
DI-1	(SPARE)
DI-2	(SPARE)
DI-3	(SPARE)
DI-4	(SPARE)
DI-5	(SPARE)

SIGNAL NAME	CH. NO.
(SPARE)	DO-0
(SPARE)	DO-1
(SPARE)	DO-2
TERBINE RESET	DO-3

(S)

34 P24
33
30 P24
29
26 P48
25
31
27
23

7 TC0311-1 (TC0417)
3
18
17
COM 16
5
1
14
13
COM 12
P48
22
21
20
COM

PCM NORMAL
CPU NORMAL
ON1
ON2

TRIP PCM INPUT/OUTPUT(2)	※	SH. No.	TC0311
SIGNATURE		DATE	TITLE
DWN. (S. Kikuchi)		2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD. T. Kawata		2014-04-18	
APPD. Y. Fukushima		2014-04-18	
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TCS-4 TCS-3 TCS-4

TPC(C)

PI NAME	PCM(P)	TYPE	LPF641A	SLOT	PCM	UNIT No	CH1

CH. NO.	SIGNAL NAME
DI-0	(SPARE)
DI-1	(SPARE)
DI-2	(SPARE)
DI-3	(SPARE)
DI-4	(SPARE)
DI-5	(SPARE)

SIGNAL NAME	CH. NO.
(SPARE)	DO-0
(SPARE)	DO-1
(SPARE)	DO-2
TURBINE RESET	DO-3

(S) 34 P24, 33, 30 P24, 29, 26 P48, 25, 31, 27, 23

(S) 7, 3, 18, 17, COM 16, 5, 1, 14, 13, COM 12, P48, 22, 21, 20, COM

TC0312-1 (TC0417)

PCM NORMAL

CPU NORMAL

ON1

ON2

TRIP PCM INPUT/OUTPUT(3)	※	SH. No.	TC0312

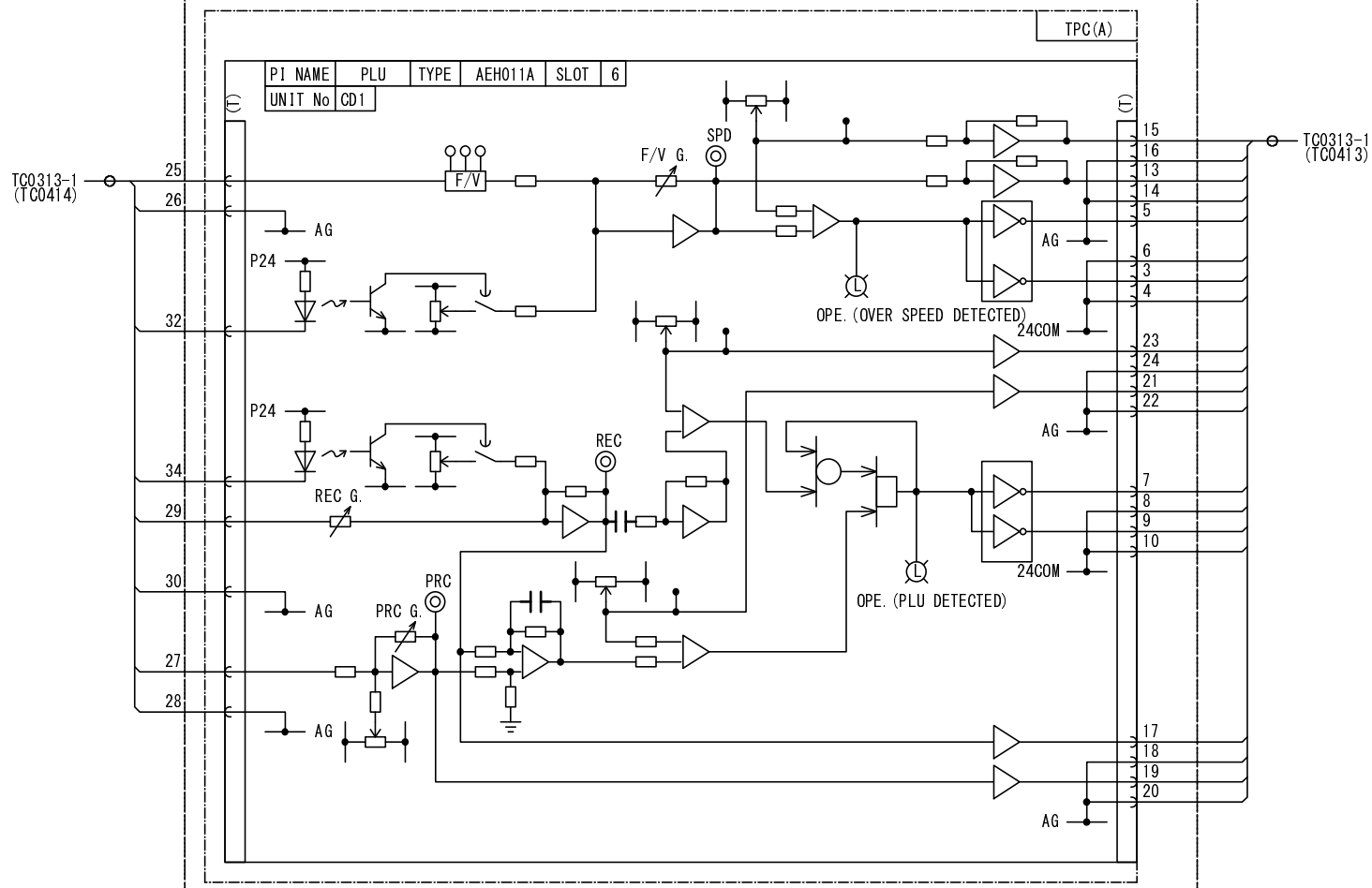
	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No.	PAGE	REV.
		331DM07733	108	3

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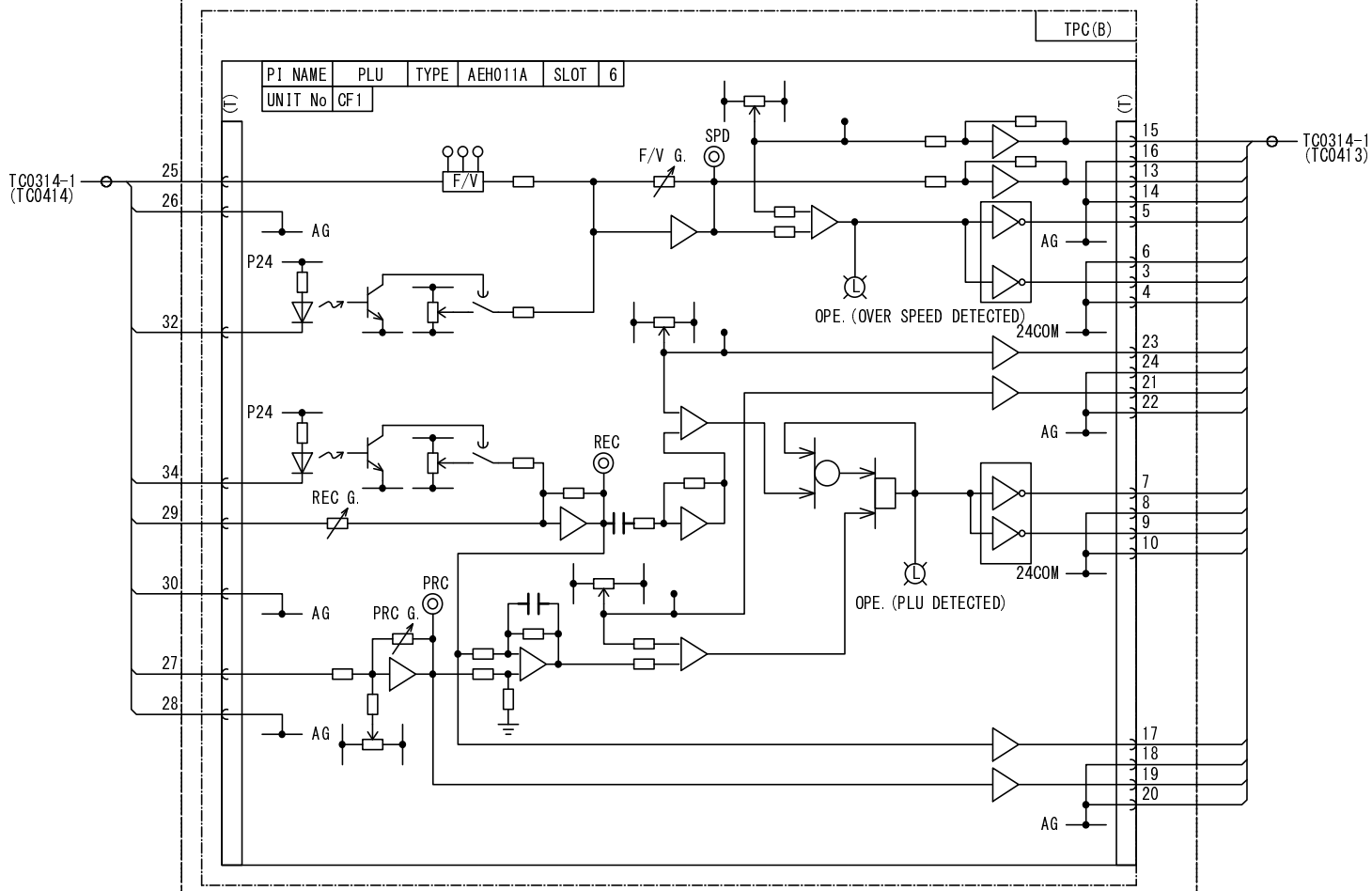
TCS-4
TCS-3
TCS-4



PLU/EOS DETECTOR(1)		※ OSP3	SH. No. TC0313
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 109 / REV. 3

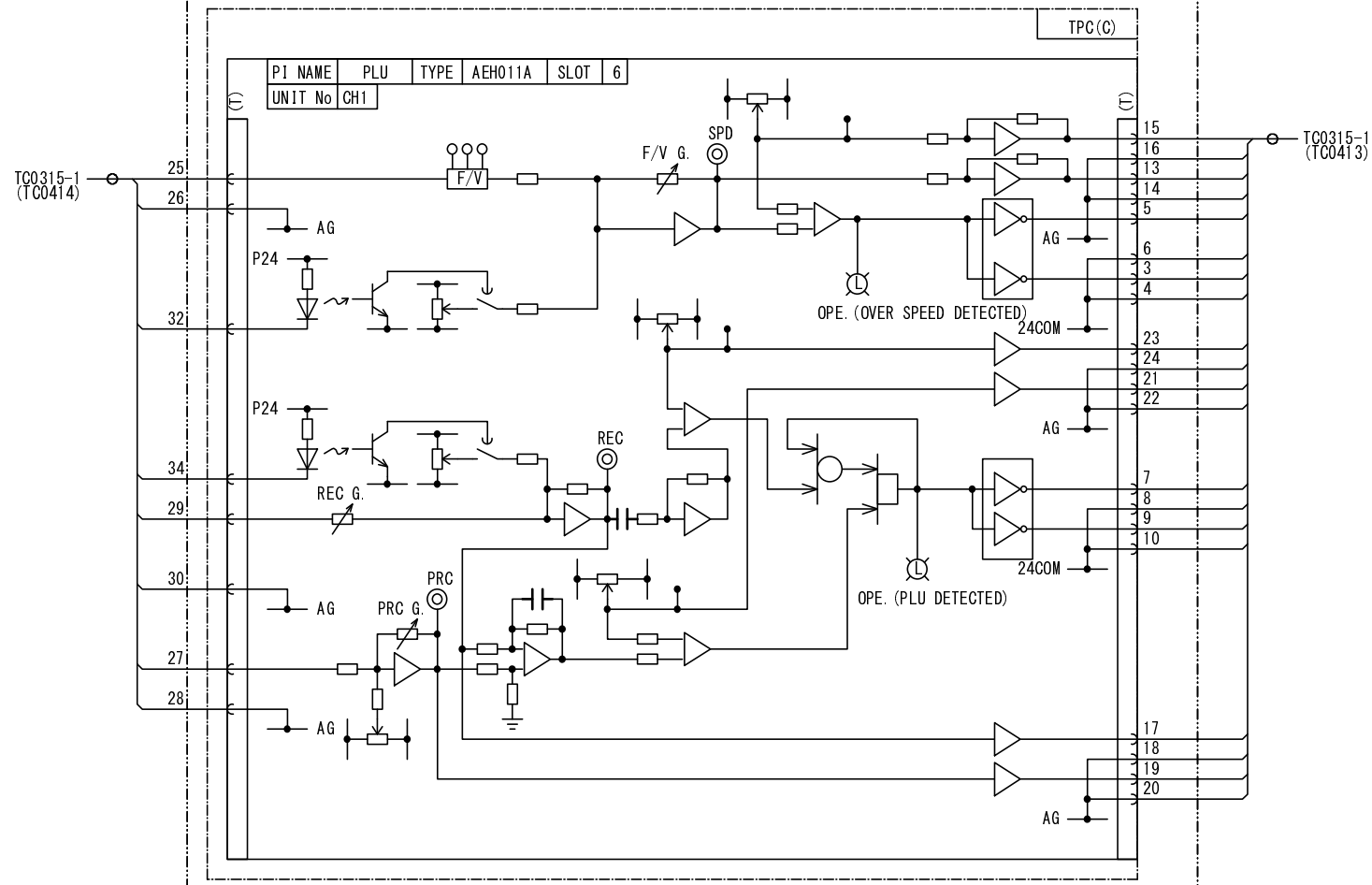
2016/07/28 TC0313 XSMM-TC

TCS-4
TCS-3
TCS-4



PLU/EOS DETECTOR(2)	※ OSP4	SH. No.	TC0314
SIGNATURE		DATE	TITLE
DWN. (S. Kikuchi)		2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD. T. Kawata		2014-04-18	
APPD. Y. Fukushima		2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 110 / REV. 3

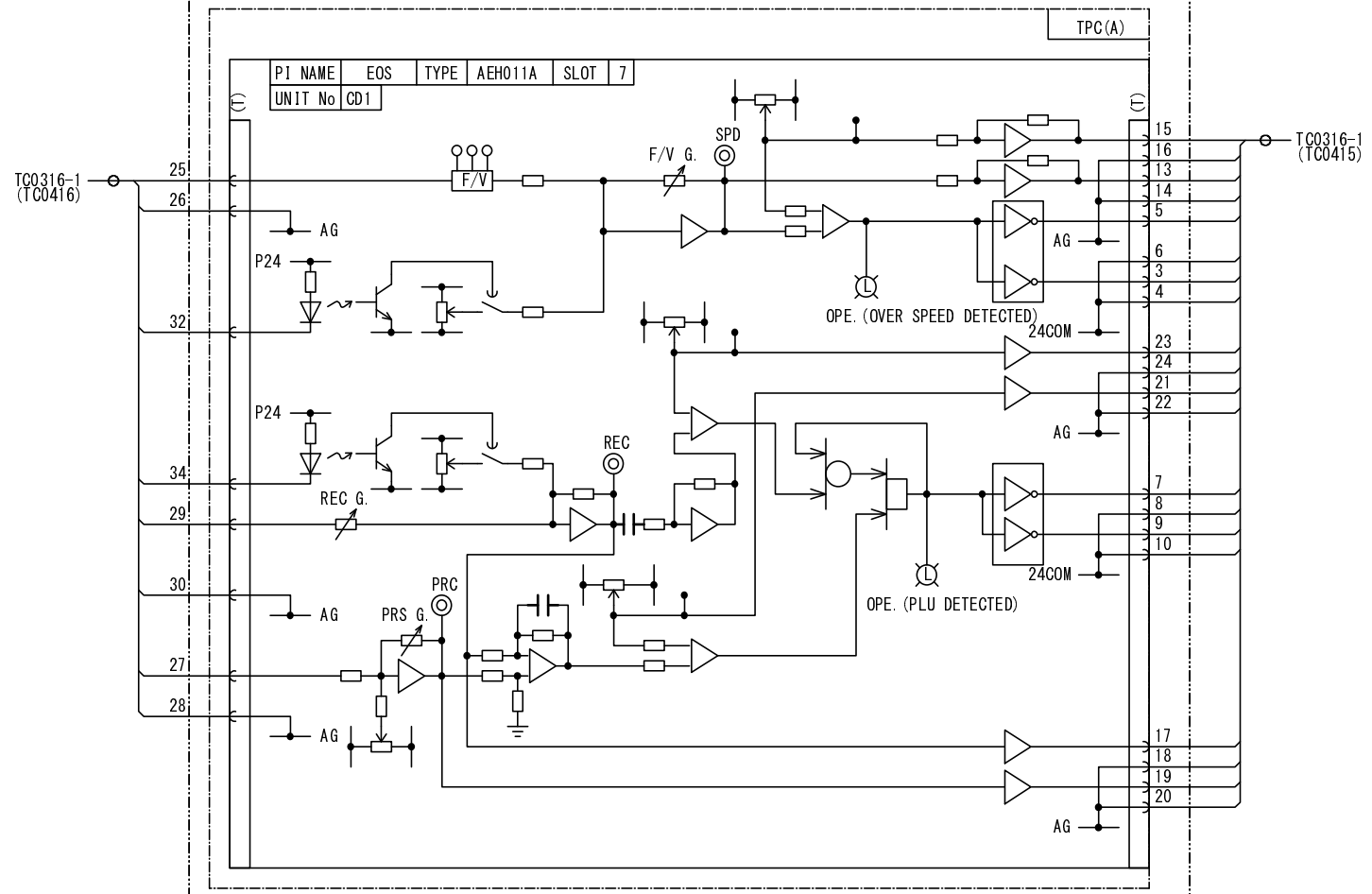
TCS-4 TCS-3 TCS-4



PLU/EOS DETECTOR(3)		※ OSP5	SH. No. TC0315
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 111 / REV. 3

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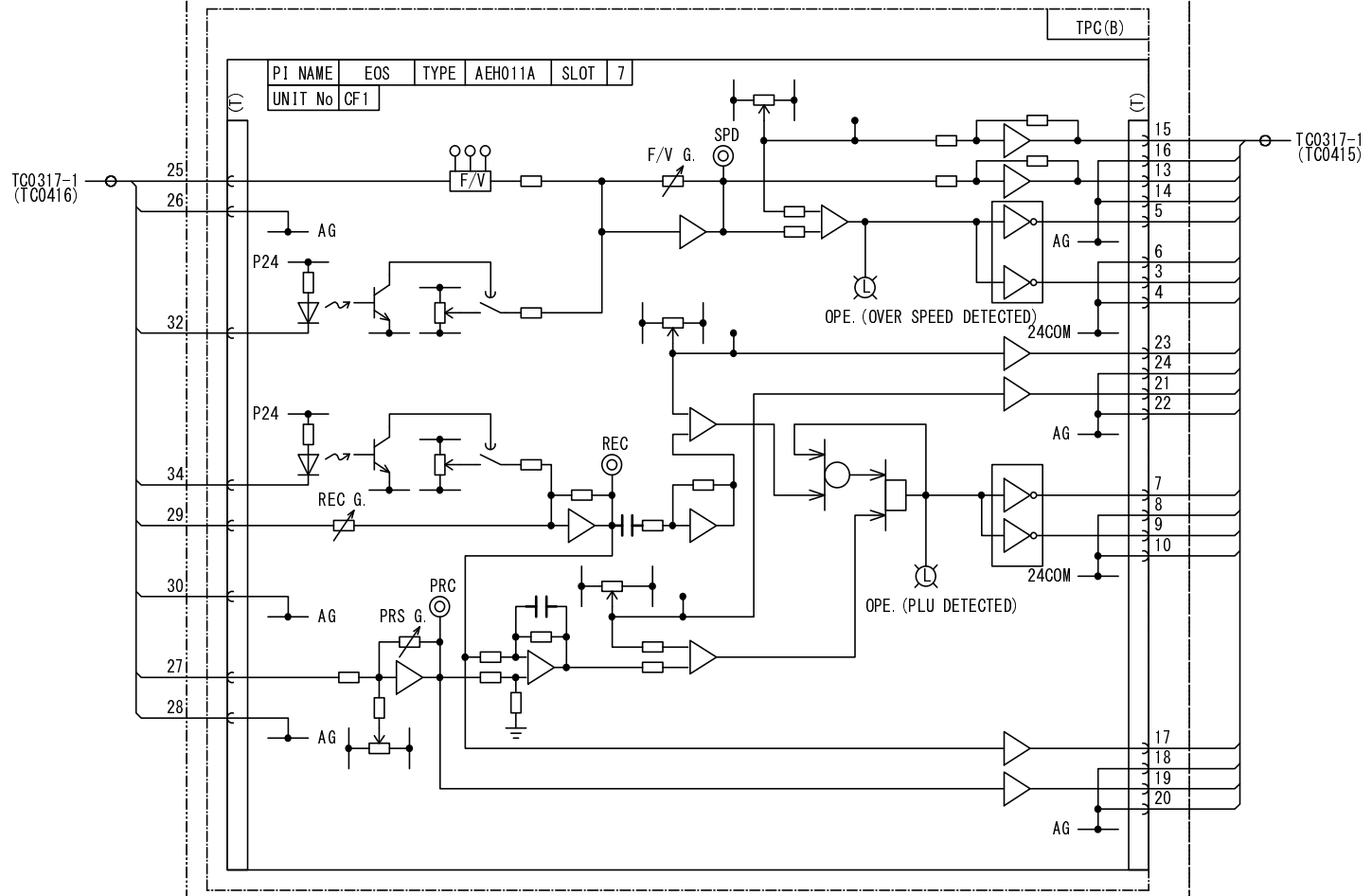
TCS-4 TCS-3 TCS-4



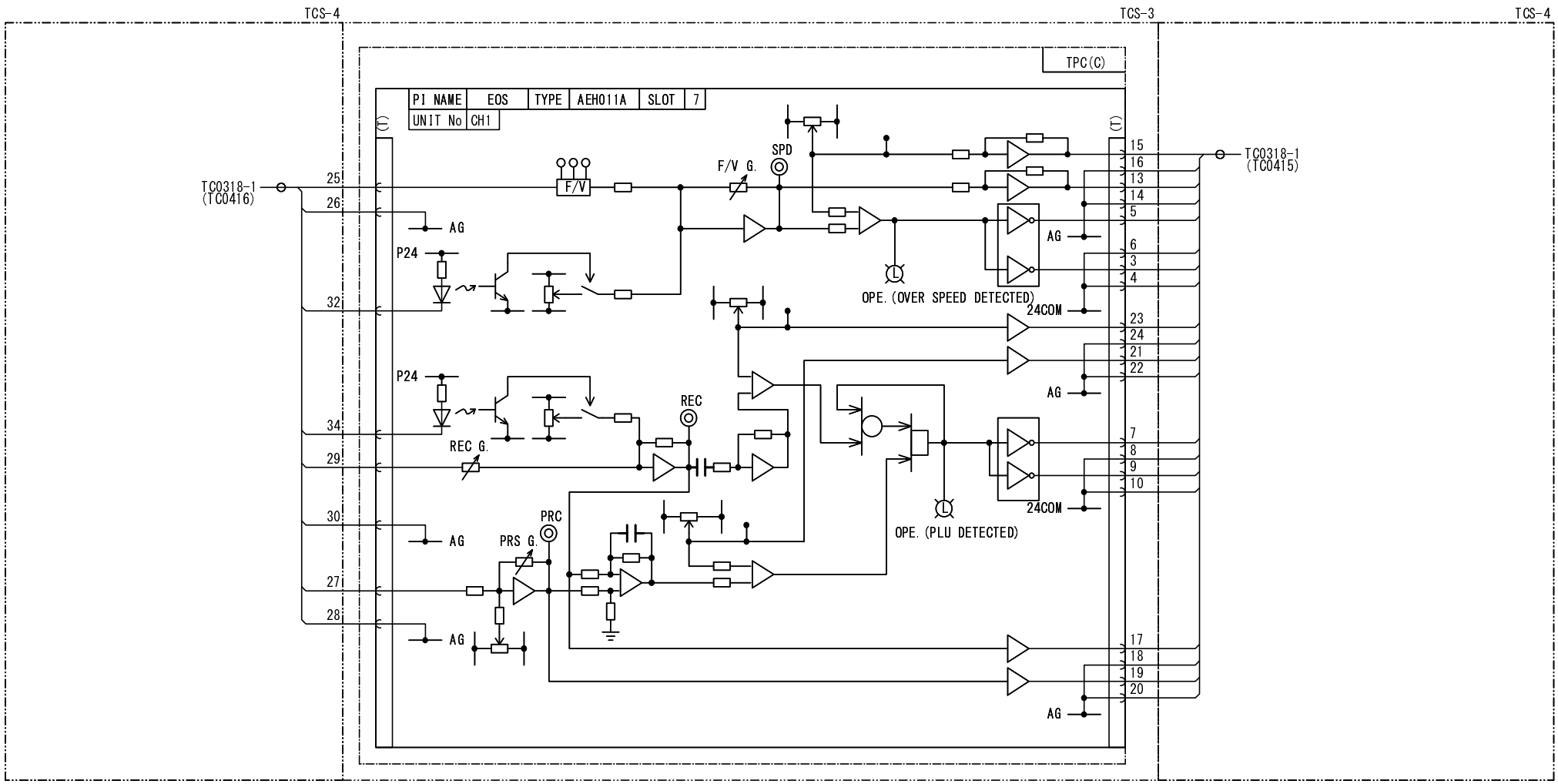
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SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 112 / REV. 3

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TCS-4
TCS-3
TCS-4



PLU/EOS DETECTOR(5)		※ OSP4	SH. No. TC0317
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 113 / REV. 3



PLU/EOS DETECTOR(6)			※ OSP5	SH. No.	TC0318
SIGNATURE		DATE	TITLE		
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM		
CHKD.	T. Kawata	2014-04-18			
APPD.	Y. Fukushima	2014-04-18			
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TCS-5

TCS-3

TC0319-1 (TC0520)
TC0319-2 (TC0520)
TC0319-3 (TC0520)

(LYD050A)

unit NO.	slot NO.	3 4 20	18	16	14	12	10	8	6
CH1	0								(S)
unit NO.	slot NO.	3 4 20	18	16	14	12	10	8	6
CF1	0								(S)
unit NO.	slot NO.	3 4 20	18	16	14	12	10	8	6
CD1	0								(S)
CTL	CH. NO.	DI 0	DI 1	DI 2	DI 3	DI 4	DI 5	DI 6	DI 7
EHG*MD@90. OA0. OB0	signal range/level								
logic SH. NO.									
TCP1900. 2900. 3900 DI	signal name	NO. 1 BEARING SHAFT VIBRATION VERY HIGH (A)	NO. 2 BEARING SHAFT VIBRATION VERY HIGH (A)	NO. 3 GEN. SHAFT VIBRATION VERY HIGH (A)	NO. 4 GEN. SHAFT VIBRATION VERY HIGH (A)	GEN TRANSF. PROTECTION OPERATED A	GEN TRANSF. PROTECTION OPERATED B	BOILER MFT 1	86G-A GEN LOCK OUT RELAY OPERATED

TPC DIGITAL INPUT(1)	※	SH. No.	TC0319
SIGNATURE		DATE	TITLE
DWN. (S. Kikuchi)		2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD. T. Kawata		2014-04-18	
APPD. Y. Fukushima		2014-04-18	
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TC0319-1 (TC0521)
TC0319-2 (TC0521)
TC0319-3 (TC0521)

(LYD050A)

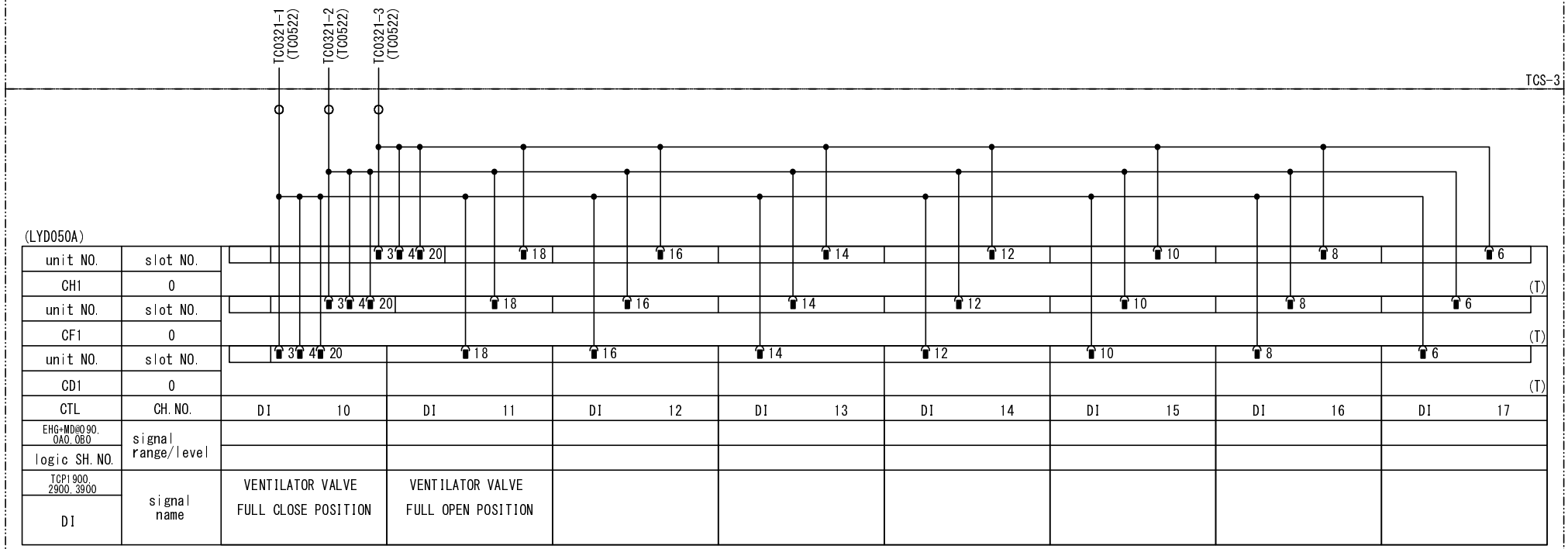
unit NO.	slot NO.	19	17	15	13	11	9	7	5
CH1	0								(S)
unit NO.	slot NO.	19	17	15	13	11	9	7	5
CF1	0								(S)
unit NO.	slot NO.	19	17	15	13	11	9	7	5
CD1	0								(S)
CTL	CH. NO.	DI 8	DI 9	DI A	DI B	DI C	DI D	DI E	DI F
EHG*MD@090. 0A0. 0B0	signal range/level								
logic SH. NO.									
TCP1900. 2900. 3900 DI	signal name	TURBINE TRIP PB-1							P.S.NORMAL

TPC DIGITAL INPUT(2)		※	SH. No.	TC0320
SIGNATURE		DATE	TITLE	
DWN.	(S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T.Kawata	2014-04-18		
APPD.	Y.Fukushima	2014-04-18		
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TPC DIGITAL INPUT(3)		※	SH. No.	TC0321	
SIGNATURE		DATE	TITLE		
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM		
CHKD.	T. Kawata	2014-04-18			
APPD.	Y. Fukushima	2014-04-18			
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TC0321-1 (TC0523)
TC0321-2 (TC0523)
TC0321-3 (TC0523)

(LYD050A)

unit NO.	slot NO.			19		17		15		13		11		9		7		5			
CH1	0																			(T)	
unit NO.	slot NO.			19		17		15		13		11		9		7		5			
CF1	0																			(T)	
unit NO.	slot NO.			19		17		15		13		11		9		7		5			
CD1	0																			(T)	
CTL	CH. NO.	DI	18	DI	19	DI	1A	DI	1B	DI	1C	DI	1D	DI	1E	DI	1F				
EHG+MD@090, 0A0, 0B0	signal range/level																				
logic SH. NO.																					
TCP1900, 2900, 3900 DI	signal name													MASTER TRIP SOLENOID (A1) TRIP POSITION		MASTER TRIP SOLENOID (B1) TRIP POSITION		P.S.NORMAL			

TPC DIGITAL INPUT(4)	※	SH. No.	TC0322

SIGNATURE		DATE
DWN.	(S.Kikuchi)	2014-04-18
CHKD.	T.Kawata	2014-04-18
APPD.	Y.Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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TC0323-1 (TC0524)

TC0323-2 (TC0524)

TC0323-3 (TC0524)

(LYD050A)

unit NO.	slot NO.	3 4 20	18	16	14	12	10	8	6
CH1	1								(S)
unit NO.	slot NO.	3 4 20	18	16	14	12	10	8	6
CF1	1								(S)
unit NO.	slot NO.	3 4 20	18	16	14	12	10	8	6
CD1	1								(S)
CTL	CH. NO.	DI 0	DI 1	DI 2	DI 3	DI 4	DI 5	DI 6	DI 7
EHG*MD@090. 0A0. 0B0	signal range/level								
logic SH. NO.									
TCP1901. 2901. 3901 DI	signal name	NO. 1 BEARING SHAFT VIBRATION VERY HIGH (B)	NO. 2 BEARING SHAFT VIBRATION VERY HIGH (B)	NO. 3 GEN. SHAFT VIBRATION VERY HIGH (B)	NO. 4 GEN. SHAFT VIBRATION VERY HIGH (B)	UNIT TRANSF. PROTECTION OPERATED A	UNIT TRANSF. PROTECTION OPERATED B	BOILER MFT 2	86G-B GEN LOCK OUT RELAY OPERATED

TPC DIGITAL INPUT(5)		※	SH. No.	TC0323
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
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TC0323-1 (TC0525)
TC0323-2 (TC0525)
TC0323-3 (TC0525)

(LYD050A)

unit NO.	slot NO.	19		17		15		13		11		9		7		5	
CH1	1														(S)
unit NO.	slot NO.	19		17		15		13		11		9		7		5	
CF1	1														(S)
unit NO.	slot NO.	19		17		15		13		11		9		7		5	
CD1	1														(S)
CTL	CH. NO.	DI	8	DI	9	DI	A	DI	B	DI	C	DI	D	DI	E	DI	F
EHG*MD@090. 0A0. 0B0	signal range/level																
logic SH. NO.																	
TCP1901. 2901. 3901 DI	signal name	TURBINE TRIP PB-2														P. S. NORMAL	

TPC DIGITAL INPUT(6)		※	SH. No.	TC0324
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
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TC0325-1 (TC0526)

TC0325-2 (TC0526)

TC0325-3 (TC0526)

(LYD050A)

unit NO.	slot NO.	3	4	20	18	16	14	12	10	8	6						
CH1	1																(T)
unit NO.	slot NO.	3	4	20	18	16	14	12	10	8	6						
CF1	1																(T)
unit NO.	slot NO.	3	4	20	18	16	14	12	10	8	6						
CD1	1																(T)
CTL	CH. NO.	DI		10	DI	11	DI	12	DI	13	DI	14	DI	15	DI	16	DI	17
EHG*MD@Q90. OA0. OB0	signal range/level																
logic SH. NO.																	
TCP1901. 2901. 3901 DI	signal name	BLOWDOWN VALVE FULL CLOSE POSITION			BLOWDOWN VALVE FULL OPEN POSITION												

TPC DIGITAL INPUT(7)		※	SH. No.	TC0325	
SIGNATURE		DATE	TITLE		
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM		
CHKD.	T. Kawata	2014-04-18			
APPD.	Y. Fukushima	2014-04-18			
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 121	REV. 3	

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TC0325-1 (TC0527)
TC0325-2 (TC0527)
TC0325-3 (TC0527)

(LYD050A)

unit NO.	slot NO.			19		17		15		13		11		9		7		5			
CH1	1																			(T)	
unit NO.	slot NO.			19		17		15		13		11		9		7		5			
CF1	1																			(T)	
unit NO.	slot NO.			19		17		15		13		11		9		7		5			
CD1	1																			(T)	
CTL	CH. NO.	DI	18	DI	19	DI	1A	DI	1B	DI	1C	DI	1D	DI	1E	DI	1F				
EHG+MD@090. 0A0. 0B0	signal range/level																				
logic SH. NO.																					
TCP1901. 2901. 3901 DI	signal name											MASTER TRIP SOLENOID (A2) TRIP POSITION		MASTER TRIP SOLENOID (B2) TRIP POSITION		P.S.NORMAL					

TPC DIGITAL INPUT(8)		※	SH. No. TC0326
SIGNATURE		DATE	TITLE
DWN.	(S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T.Kawata	2014-04-18	
APPD.	Y.Fukushima	2014-04-18	
REGD.	Hitachi,Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 122 / REV. 3

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TC0327-1 (TC0528)
TC0327-2 (TC0528)
TC0327-3 (TC0528)

(LYD050A)

unit NO.	slot NO.	3 4 20	18	16	14	12	10	8	6
CH1	2								(S)
unit NO.	slot NO.	3 4 20	18	16	14	12	10	8	6
CF1	2								(S)
unit NO.	slot NO.	3 4 20	18	16	14	12	10	8	6
CD1	2								(S)
CTL	CH. NO.	DI 0	DI 1	DI 2	DI 3	DI 4	DI 5	DI 6	DI 7
EHG*MD@90. OA0. OB0	signal range/level								
logic SH. NO.									
TCP1902. 2902. 3902 DI	signal name	NO. 1 BEARING SHAFT VIBRATION VERY HIGH (C)	NO. 2 BEARING SHAFT VIBRATION VERY HIGH (C)	NO. 3 GEN. SHAFT VIBRATION VERY HIGH (C)	NO. 4 GEN. SHAFT VIBRATION VERY HIGH (C)	50BF - 86BF PROTECTION OPERATED A	50BF - 86BF PROTECTION OPERATED B		

TPC DIGITAL INPUT(9)		※	SH. No.	TC0327
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
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TC0327-1 (TC0529)
TC0327-2 (TC0529)
TC0327-3 (TC0529)

(LYD050A)

unit NO.	slot NO.	19	17	15	13	11	9	7	5
CH1	2								(S)
unit NO.	slot NO.	19	17	15	13	11	9	7	5
CF1	2								(S)
unit NO.	slot NO.	19	17	15	13	11	9	7	5
CD1	2								(S)
CTL	CH. NO.	DI 8	DI 9	DI A	DI B	DI C	DI D	DI E	DI F
EHG*MD@Q90, QA0, QB0	signal range/level								
logic SH. NO.									
TCP1902, 2902, 3902	signal name								P.S.NORMAL
DI									

TPC DIGITAL INPUT(10)	※	SH. No.	TC0328

SIGNATURE		DATE
DWN.	(S.Kikuchi)	2014-04-18
CHKD.	T.Kawata	2014-04-18
APPD.	Y.Fukushima	2014-04-18

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TC0329-1 (TC0530)

TC0329-2 (TC0530)

TC0329-3 (TC0530)

(LYD050A)

unit NO.	slot NO.	3	4	20	18		16		14		12		10		8		6		
CH1	2																		(T)
unit NO.	slot NO.	3	4	20	18		16		14		12		10		8		6		
CF1	2																		(T)
unit NO.	slot NO.	3	4	20	18		16		14		12		10		8		6		
CD1	2																		(T)
CTL	CH. NO.	DI		10	DI		11	DI		12	DI		13	DI		14	DI		15	DI		16	DI		17
EHG*MD@Q90, QA0, QB0	signal range/level																		
logic SH. NO.																			
TCP1902, 2902, 3902	signal name																		
DI																			

TPC DIGITAL INPUT(11)		※		SH. No.	TC0329
SIGNATURE		DATE	TITLE		
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM		
CHKD.	T. Kawata	2014-04-18			
APPD.	Y. Fukushima	2014-04-18			
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TC0329-1 (TC0531)
TC0329-2 (TC0531)
TC0329-3 (TC0531)

(LYD050A)

unit NO.	slot NO.			19		17		15		13		11		9		7		5	
CH1	2																		(T)
unit NO.	slot NO.			19		17		15		13		11		9		7		5	
CF1	2																		(T)
unit NO.	slot NO.			19		17		15		13		11		9		7		5	
CD1	2																		(T)
CTL	CH. NO.	DI	18	DI	19	DI	1A	DI	1B	DI	1C	DI	1D	DI	1E	DI	1F		
EHG*MD@090. 0A0. 0B0	signal range/level																		
logic SH. NO.																			
TCP1902. 2902. 3902 DI	signal name																	P. S. NORMAL	

TPC DIGITAL INPUT(12)		※	SH. No.	TC0330	
SIGNATURE		DATE	TITLE		
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM		
CHKD.	T. Kawata	2014-04-18			
APPD.	Y. Fukushima	2014-04-18			
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TC0331-1 (TC0532)
TC0331-2 (TC0532)
TC0331-3 (TC0532)

(LYA000A)

unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
CH1	8								(T)
unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
CF1	8								(T)
unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
CD1	8								(T)

CTL	CH. NO.	AI 0	AI 1	AI 2	AI 3	AI 4	AI 5	AI 6	AI 7
EHG*MD@090. 0A0. 0B0	signal range/level	0～50kg/cm2	0～5kg/cm2	0～250kg/cm2	0～250kg/cm2	0～150kg/cm2	0～240MW	0～0.4mm	
logic SH. NO.		1～5V	1～5V	1～5V	1～5V	1～5V	1～5V	1～5V	
TCP1910. 2910. 3910 AI	signal name	HP TURBINE EXHAUST STEAM PRESS. (A)	BEARING OIL PRESS. (A)	CONTROL OIL PRESS. (A)	TURBINE TRIP OIL PRESS. (A)	HP TURBINE FIRST STAGE STEAM PRESS. (A)	GEN. MW (A)	NO.1 TURBINE SHAFT VIBRATION (X)	

TPC ANALOG INPUT(1)	※	SH. No.	TC0331

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

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TC0332-1 (TC0533)
TC0332-2 (TC0533)
TC0332-3 (TC0533)

(LYA000A)

unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
CH1	9								(T)
unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
CF1	9								(T)
unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
CD1	9								(T)

CTL	CH. NO.	AI 0	AI 1	AI 2	AI 3	AI 4	AI 5	AI 6	AI 7
EHG*MD@090. 0A0. 0B0	signal range/level	0～1kg/cm2	-2～2mm					0～0.4mm	
logic SH. NO.		1～5V	1～5V					1～5V	
TCP1910. 2910. 3910. AI	signal name	LP TURBINE EXHAUST HOOD STEAM PRESS. (CONDENSER VACUUM) (A	TURBINE ROT OR POSITION (A)					NO.1 TURBINE SHAFT VIBRATION (Y)	

TPC ANALOG INPUT(2)	※	SH. No.	TC0332

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

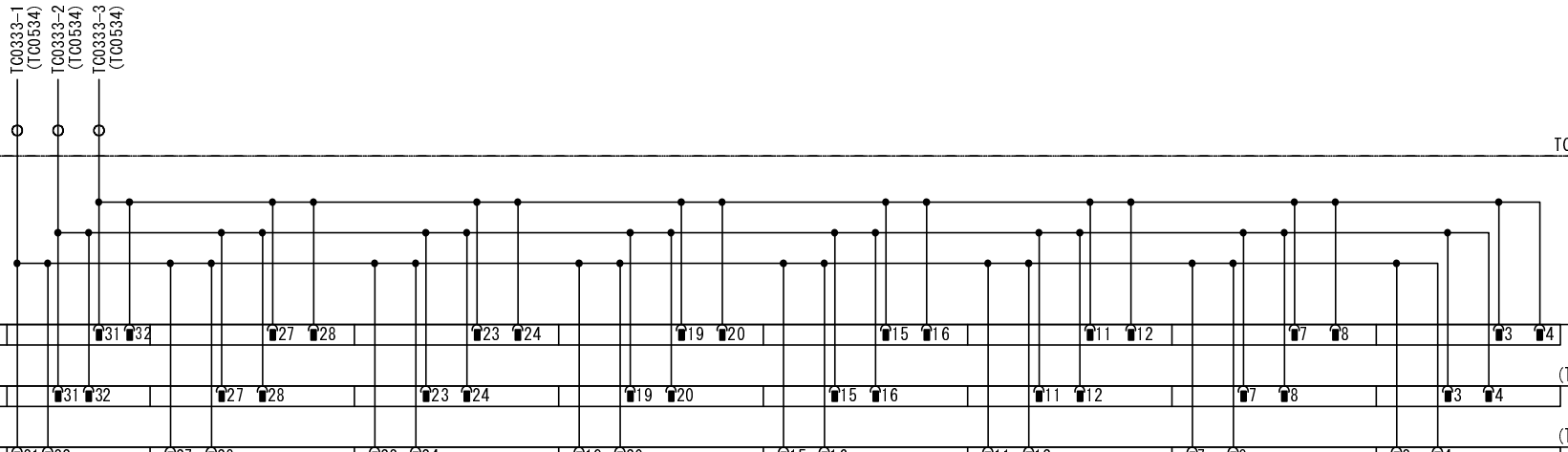
TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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(LYA000A)

unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
CH1	A								(T)
unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
CF1	A								(T)
unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
CD1	A								(T)
CTL	CH. NO.	AI 0	AI 1	AI 2	AI 3	AI 4	AI 5	AI 6	AI 7
EHG*MD@090. 0A0. 0B0	signal range/level	0～50kg/cm2	0～5kg/cm2	0～250kg/cm2	0～250kg/cm2	0～150kg/cm2	0～240MW	0～0.4mm	
logic SH. NO.		1～5V	1～5V	1～5V	1～5V	1～5V	1～5V	1～5V	
TCP1910. 2910. 3910. AI	signal name	HP TURBINE EXHAUST STEAM PRESS. (B)	BEARING OIL PRESS. (B)	CONTROL OIL PRESS. (B)	TURBINE TRIP OIL PRESS. (B)	HP TURBINE FIRST STAGE STEAM PRESS. (B)	GEN. MW (B)	NO.2 TURBINE SHAFT VIBRATION (X)	

TPC ANALOG INPUT(3)		※	SH. No.	TC0333
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
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TC0334-1 (TC0535)
TC0334-2 (TC0535)
TC0334-3 (TC0535)

(LYA000A)

unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
CH1	B								(T)
unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
CF1	B								(T)
unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
CD1	B								(T)

CTL	CH. NO.	AI 0	AI 1	AI 2	AI 3	AI 4	AI 5	AI 6	AI 7
EHG*MD@090. OA0. OB0	signal range/level	0～1kg/cm2	-2～2mm					0～0.4mm	
logic SH. NO.		1～5V	1～5V					1～5V	
TCP1911, 2911, 3911 AI	signal name	LP TURBINE EXHAUST HOOD STEAM PRESS. (CONDENSER VACUUM) (B	TURBINE ROTOR POSITION (B)					NO.2 TURBINE SHAFT VIBRATION (Y)	

TPC ANALOG INPUT(4)	※	SH. No.	TC0334

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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(LYA000A)

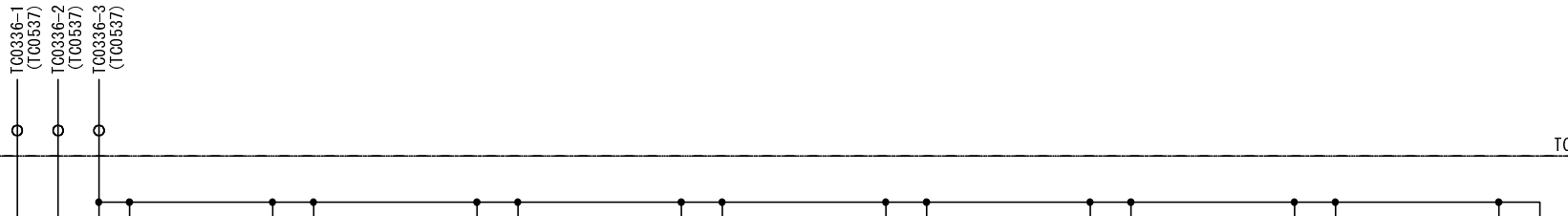
unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
CH1	C								(T)
unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
CF1	C								(T)
unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
CD1	C								(T)
CTL	CH. NO.	AI 0	AI 1	AI 2	AI 3	AI 4	AI 5	AI 6	AI 7
EHG*MD@Q90. OAO. OBO	signal range/level	0～50kg/cm2	0～5kg/cm2	0～250kg/cm2	0～250kg/cm2	0～150kg/cm2	0～0.4mm	0～0.4mm	
logic SH. NO.		1～5V	1～5V	1～5V	1～5V	1～5V	1～5V	1～5V	
TCP1911. 2911. 3911 AI	signal name	HP TURBINE EXHAUST STEAM PRESS. (C)	BEARING OIL PRESS. (C)	CONTROL OIL PRESS. (C)	TURBINE TRIP OIL PRESS. (C)	HP TURBINE FIRST STAGE STEAM PRESS. (C)	NO.4 GENERATOR SHAFT VIBRATION (X)	NO.3 GENERATOR SHAFT VIBRATION (X)	

TPC ANALOG INPUT(5)		※	SH. No.	TC0335
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
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(LYA000A)

unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
CH1	D								(T)
unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
CF1	D								(T)
unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
CD1	D								(T)
CTL	CH. NO.	AI 0	AI 1	AI 2	AI 3	AI 4	AI 5	AI 6	AI 7
EHG*MD@090. OA0. OB0	signal range/level	0～1kg/cm2	-2～2mm				0～0.4mm	0～0.4mm	
logic SH. NO.		1～5V	1～5V				1～5V	1～5V	
TCP1911. 2911. 3911 AI	signal name	LP TURBINE EXHAUST HOOD STEAM PRESS. (CONDENSER VACUUM) (C	TURBINE ROTOR POSITION (C)				NO.4 GENERATOR SHAFT VIBRATION (Y)	NO.3 GENERATOR SHAFT VIBRATION (Y)	

TPC ANALOG INPUT(6)		※	SH. No.	TC0336
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 132	REV. 3

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TCS-3

(LYD105A)

DO / logic SH. NO.	signal name	HP TURBINE INLET/EXHAUST STEAM DIFF. PRESS. VERY HIGH	CONTROL OIL PRESS. VERY LOW TRIP	CONDENSER VACUUM VERY LOW TRIP	MAIN STEAM TEMP. VERY LOW	HP TURBINE EXHAUST STEAM TEMP. VERY HIGH TRIP		MASTER TRIP SOLENOID VALVE(A1) ENERGIZE COMMAND	MASTER TRIP SOLENOID VALVE(B1) ENERGIZE COMMAND
TCP1930, 2930, 3930	signal range/level								
CTL									
EHG+MD@090, 0A0, 0B0	CH. NO.	DO 0	DO 1	DO 2	DO 3	DO 4	DO 5	DO 6	DO 7
unit NO.	slot NO.								
CD1	3	21 3 20	18	16	14	12	10	8	6 (S)
unit NO.	slot NO.								
CF1	3	21 3 20	18	16	14	12	10	8	6 (S)
unit NO.	slot NO.								
CH1	3	21 3 20	18	16	14	12	10	8	6 (S)

TCS-5

TC0340-1 (TC0540)
TC0340-2 (TC0540)
TC0340-3 (TC0540)

TPC DIGITAL OUTPUT(1)		※	SH. No.	TC0340
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 133	REV. 3

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TCS-3

(LYD105A)

DO / logic SH. NO.	signal name	TURBINE TRIP OIL PRESS. VERY LOW TRIP	HP TURBINE EXHAUST STEAM PRESS. VERY HIGH TRIP	LP TURBINE EXHAUST HOOD STEAM TEMP. VERY HIGH TRIP	BEARING OIL PRESS. VERY LOW	TURBINE BACK-UP OVERSPEED TRIP			
TCP1930, 2930, 3930	signal range/level								
CTL									
EHG+MD@090, 0A0, 0B0	CH. NO.	DO 8	DO 9	DO A	DO B	DO C	DO D	DO E	DO F
unit NO.	slot NO.								
CD1	3	19	17	15	13	11	9	7	5 (S)
unit NO.	slot NO.								
CF1	3	19	17	15	13	11	9	7	5 (S)
unit NO.	slot NO.								
CH1	3	19	17	15	13	11	9	7	5 (S)

TCS-5

TC0340-1 (TC0541)
TC0340-2 (TC0541)
TC0340-3 (TC0541)

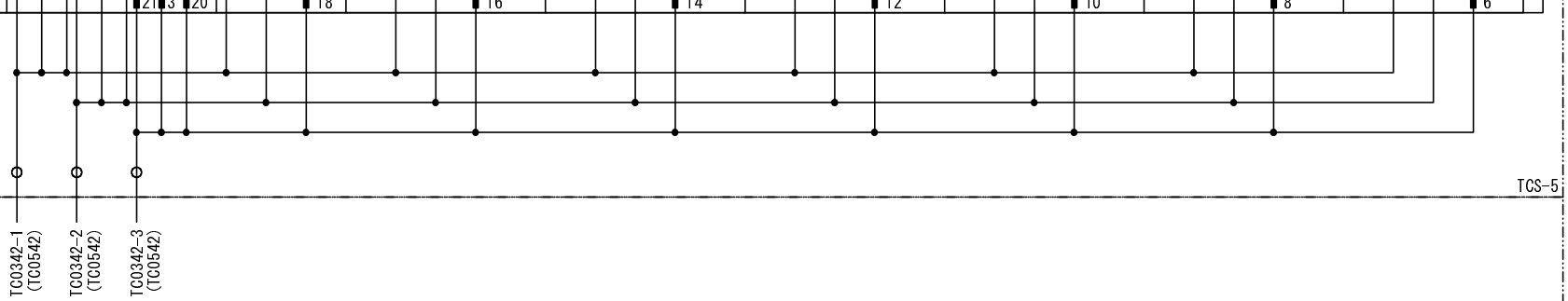
TPC DIGITAL OUTPUT(2)		※	SH. No.	TC0341
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 134	REV. 3

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TCS-3

(LYD105A)

DO / logic SH. NO.	signal name	TURBINE OVERSPEED TRIP	TURBINE MANUAL TRIP		THRUST BEARING FAILURE (FRONT) TRIP	THRUST BEARING FAILURE (REAR) TRIP	NO.1 TURBINE SHAFT VIBRATION VERY HIGH	NO.2 TURBINE SHAFT VIBRATION VERY HIGH	NO.3 GEN. SHAFT VIBRATION VERY HIGH
TCP1930, 2930, 3930	signal range/level								
CTL									
EHG+MD@090, 0A0, 0B0	CH. NO.	DO 10	DO 11	DO 12	DO 13	DO 14	DO 15	DO 16	DO 17
unit NO.	slot NO.								
CD1	3	21 3 20	18	16	14	12	10	8	6 (T)
unit NO.	slot NO.								
CF1	3	21 3 20	18	16	14	12	10	8	6 (T)
unit NO.	slot NO.								
CH1	3	21 3 20	18	16	14	12	10	8	6 (T)



TCS-5

TC0342-1 (TC0542)

TC0342-2 (TC0542)

TC0342-3 (TC0542)

TPC DIGITAL OUTPUT(3)		※	SH. No.	TC0342
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 135	REV. 3

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TCS-3

(LYD105A)

DO / logic SH. NO.	signal name	NO.4 GEN. SHAFT VIBRATION VERY HIGH	TURBINE RESET	TURBINE TRIP		ST TRIP (GROUP A)	ST TRIP (GROUP B)	MASTER TRIP SOLENOID VALVE(B2) ENERGIZE COMMAND	MASTER TRIP SOLENOID VALVE(A2) ENERGIZE COMMAND
TCP1930, 2930, 3930	signal range/level								
CTL									
EHG+MD@090, 0A0, 0B0	CH. NO.	DO 18	DO 19	DO 1A	DO 1B	DO 1C	DO 1D	DO 1E	DO 1F
unit NO.	slot NO.								
CD1	3	22 24 19	17	15	13	11	9	7	5 (T)
unit NO.	slot NO.								
CF1	3	22 24 19	17	15	13	11	9	7	5 (T)
unit NO.	slot NO.								
CH1	3	22 24 19	17	15	13	11	9	7	5 (T)

TCS-5

TC0342-1 (TC0543)

TC0342-2 (TC0543)

TC0342-3 (TC0543)

TPC DIGITAL OUTPUT(4)	※	SH. No.	TC0343
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 136 / REV. 3

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TCS-3

(LYD105A)

DO / logic SH. NO.	signal name	VENTILATOR VALVE SOLENOID VALVE ENERGIZE COMMAND	BLOWDOWN VALVE ENERGIZE COMMAND						
TCP1931, 2931, 3931	signal range/level								
CTL									
EHG+MD@090, 0A0, 0B0	CH. NO.	DO 0	DO 1	DO 2	DO 3	DO 4	DO 5	DO 6	DO 7
unit NO.	slot NO.								
CD1	4	21 3 20	18	16	14	12	10	8	6 (S)
unit NO.	slot NO.								
CF1	4	21 3 20	18	16	14	12	10	8	6 (S)
unit NO.	slot NO.								
CH1	4	21 3 20	18	16	14	12	10	8	6 (S)

TCS-5

TC0344-1 (TC0544)

TC0344-2 (TC0544)

TC0344-3 (TC0544)

TPC DIGITAL OUTPUT(5)		※	SH. No.	TC0344
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 137	REV. 3

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TCS-3

(LYD105A)

DO / logic SH. NO.	signal name																
TCP1931, 2931, 3931 / CTL	signal range/level																
EHG+MD@090, 0A0, 0B0	CH. NO.	DO	8	DO	9	DO	A	DO	B	DO	C	DO	D	DO	E	DO	F
unit NO.	slot NO.																
CD1	4	19		17		15		13		11		9		7		5	(S)
unit NO.	slot NO.																
CF1	4	19		17		15		13		11		9		7		5	(S)
unit NO.	slot NO.																
CH1	4	19		17		15		13		11		9		7		5	(S)

TCS-5

TC0344-1 (TC0545)
TC0344-2 (TC0545)
TC0344-3 (TC0545)

TPC DIGITAL OUTPUT(6)		※	SH. No.	TC0345	
SIGNATURE		DATE	TITLE		
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM		
CHKD.	T. Kawata	2014-04-18			
APPD.	Y. Fukushima	2014-04-18			
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 138	REV. 3	

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TCS-3

(LYD105A)

DO / logic SH. NO.	signal name	MSV (A) FAST ACTING SOLENOID VALVE	CV (A) FAST ACTING SOLENOID VALVE	ICV (A) FAST ACTING SOLENOID VALVE	RSV (A) FAST ACTING SOLENOID VALVE	RSV (A) TEST SOLENOID VALVE	MSV (B) FAST ACTING SOLENOID VALVE	CV (B) FAST ACTING SOLENOID VALVE	ICV (B) FAST ACTING SOLENOID VALVE
TCP1931, 2931, 3931	signal range/level								
CTL									
EHG+MD@Q90, 0A0, 0B0	CH. NO.	DO 10	DO 11	DO 12	DO 13	DO 14	DO 15	DO 16	DO 17
unit NO.	slot NO.								
CD1	4	21 3 20	18	16	14	12	10	8	6 (T)
unit NO.	slot NO.								
CF1	4	21 3 20	18	16	14	12	10	8	6 (T)
unit NO.	slot NO.								
CH1	4	21 3 20	18	16	14	12	10	8	6 (T)

TCS-5

TC0346-1 (TC0546)

TC0346-2 (TC0546)

TC0346-3 (TC0546)

TPC DIGITAL OUTPUT(7)	※	SH. No.	TC0346
SIGNATURE		DATE	TITLE
DWN. (S. Kikuchi)		2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD. T. Kawata		2014-04-18	
APPD. Y. Fukushima		2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 139 / REV. 3

TCS-3

(LYD105A)

DO / logic SH. NO.	signal name	RSV (B) FAST ACTING SOLENOID VALVE	RSV (B) TEST SOLENOID VALVE						
TCP1931, 2931, 3931	signal range/level								
CTL									
EHG+MD@Q90, OA0, OB0	CH. NO.	DO 18	DO 19	DO 1A	DO 1B	DO 1C	DO 1D	DO 1E	DO 1F
unit NO.	slot NO.								
CD1	4	22 24 19	17	15	13	11	9	7	5 (T)
unit NO.	slot NO.								
CF1	4	22 24 19	17	15	13	11	9	7	5 (T)
unit NO.	slot NO.								
CH1	4	22 24 19	17	15	13	11	9	7	5 (T)

TCS-5

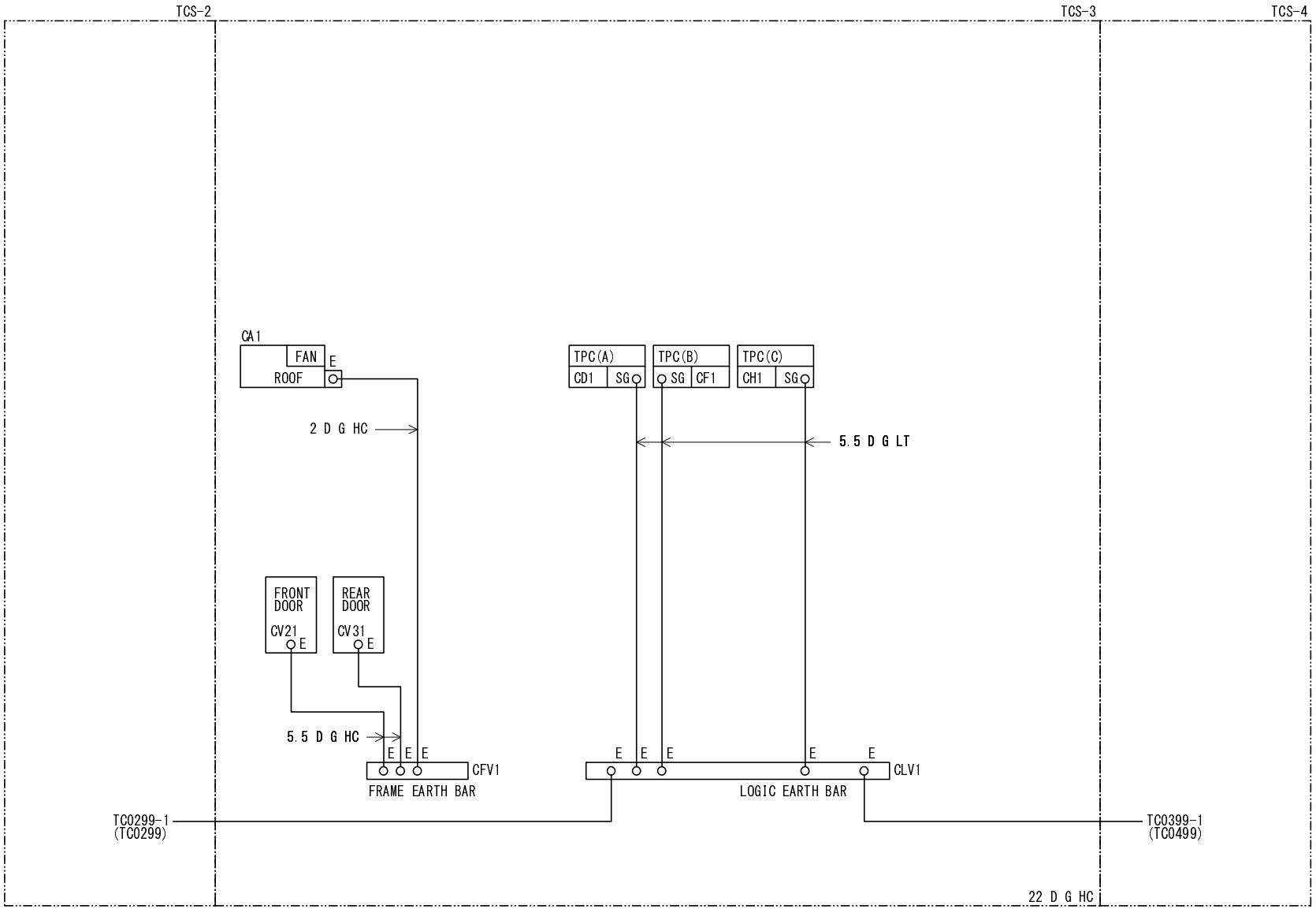
TC0346-1 (TC0547)

TC0346-2 (TC0547)

TC0346-3 (TC0547)

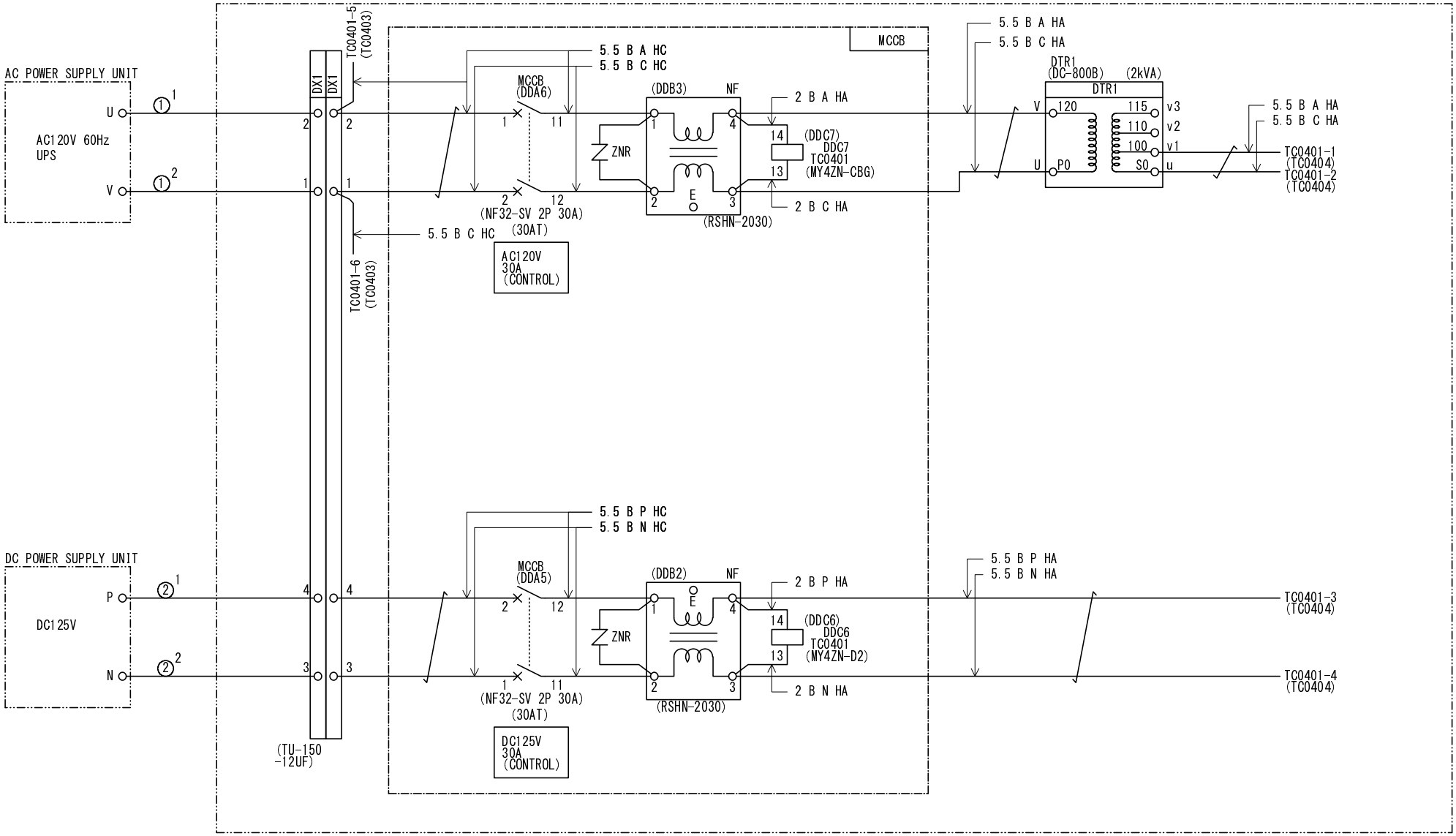
TPC DIGITAL OUTPUT(8)		※	SH. No.	TC0347
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 140	REV. 3

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POWER SUPPLY SYSTEM EARTH CIRCUIT		SH. No.	TC0399
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 141	REV. 3

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DEV.	a			b		
DDC7	9	5		9	1	TC0410 2
TC04	10	6		10	2	
01	11	7		11	3	
	12	8		12	4	
DDC6	9	5		9	1	TC0410 2
TC04	10	6		10	2	
01	11	7		11	3	
	12	8		12	4	

INDEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK
○							
○							
○							
○							
○							
○							
②		2/2		TCS-4			
①		2/2		TCS-4			

POWER SUPPLY SYSTEM(1)(POWER INPUT)

SH. No. TC0401

SIGNATURE		DATE
DWN.	(S.Kikuchi)	2014-04-18
CHKD.	T.Kawata	2014-04-18
APPD.	Y.Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

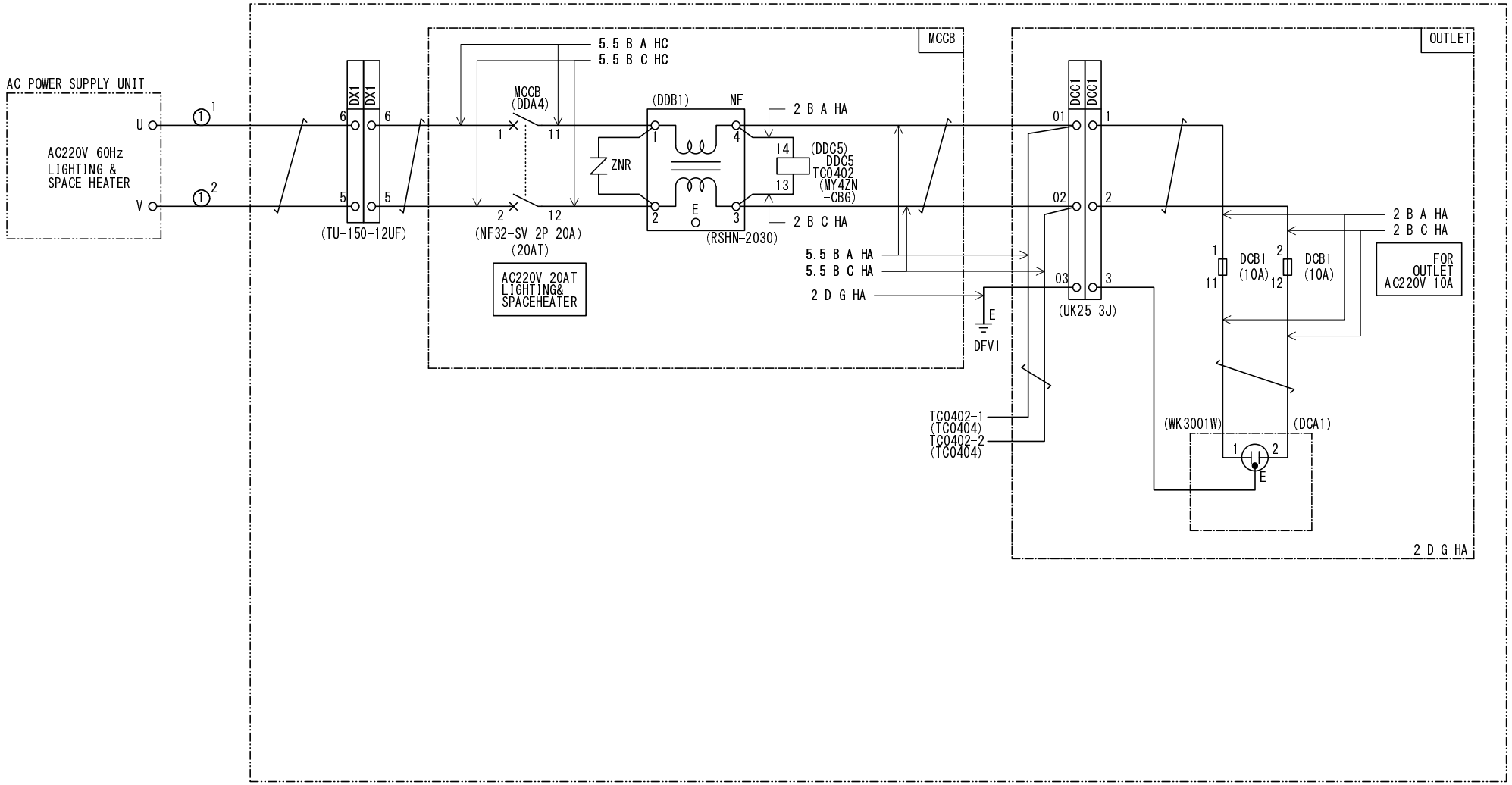
DWG. No. 331DM07733

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TCS-4

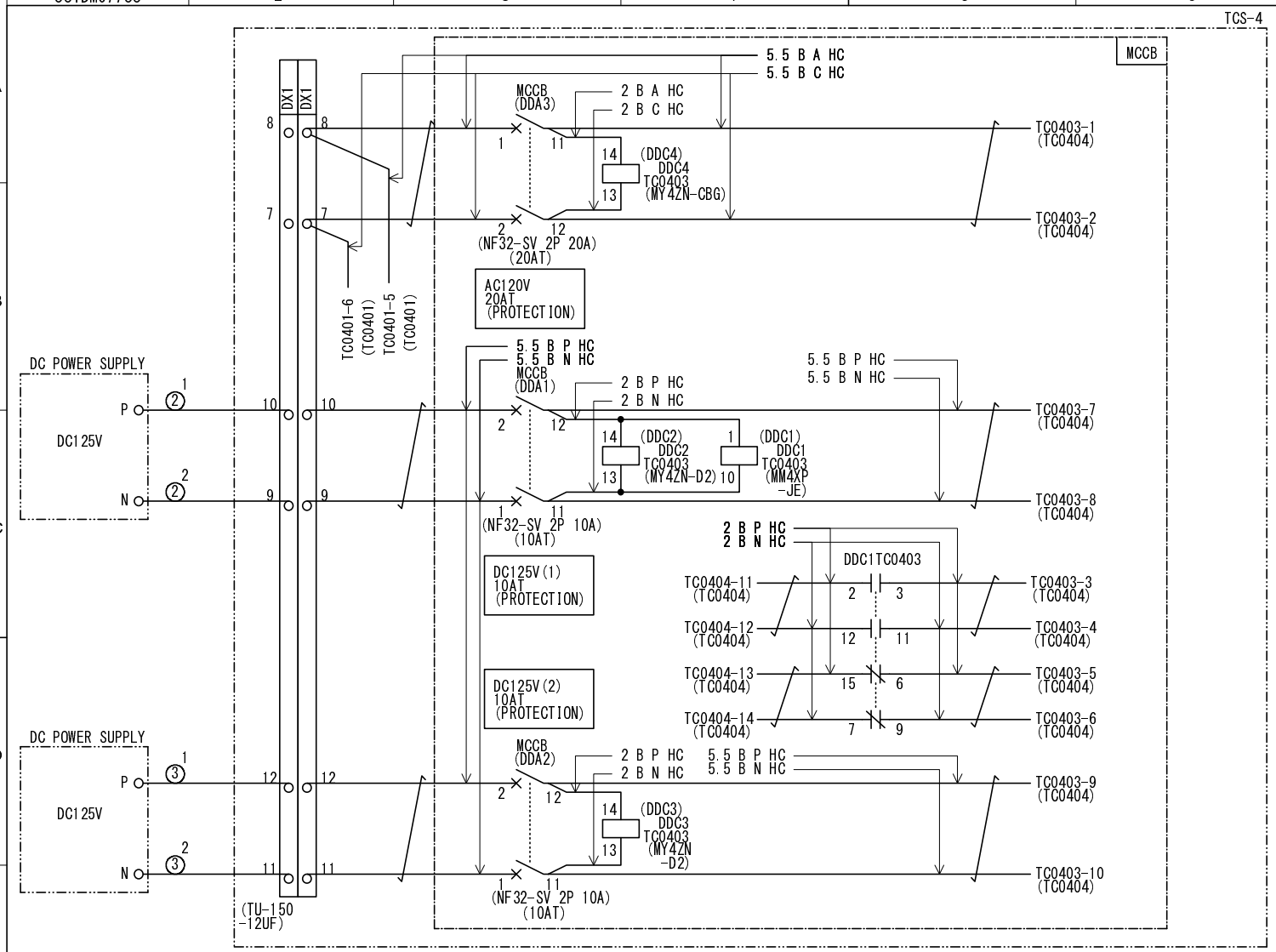


DEV.	a		b	
DDC5	9 5		9 1	TC0410 2
TC04	10 6		10 2	
02	11 7		11 3	
	12 8		12 4	

○								
○								
○								
○								
○								
③		2/2		TCS-4				
②		2/2		TCS-4				
①		2/2		TCS-4				
IN-DEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO	FROM RELATED SH. NO.	TO	REMARK	

POWER SUPPLY SYSTEM(2)(POWER INPUT for OUTLET)		SH. No.	TC0402
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 143 / REV. 3

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DEV.	a			b		DEV.	a			b	
DDC4 TC0403	9 5			9 1	TC0410: 2	DDC3 TC0403	9 5			9 1	TC0410: 2
	10 6			10 2			10 6			10 2	
	11 7			11 3			11 7	TC0420A: 6		11 3	
	12 8			12 4			12 8			12 4	
DDC2 TC0403	9 5			9 1	TC0410: 2	DDC1 TC0403	2 3	*	5	2 4	
	10 6			10 2			11 12	*	5	11 5	
	11 7	TC0420A: 5		11 3			15 14			15 6	* 5
	12 8			12 4			9 8			9 7	* 5

INDEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK

TRIP POWER SUPPLY SYSTEM ※ SH. No. TC0403

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

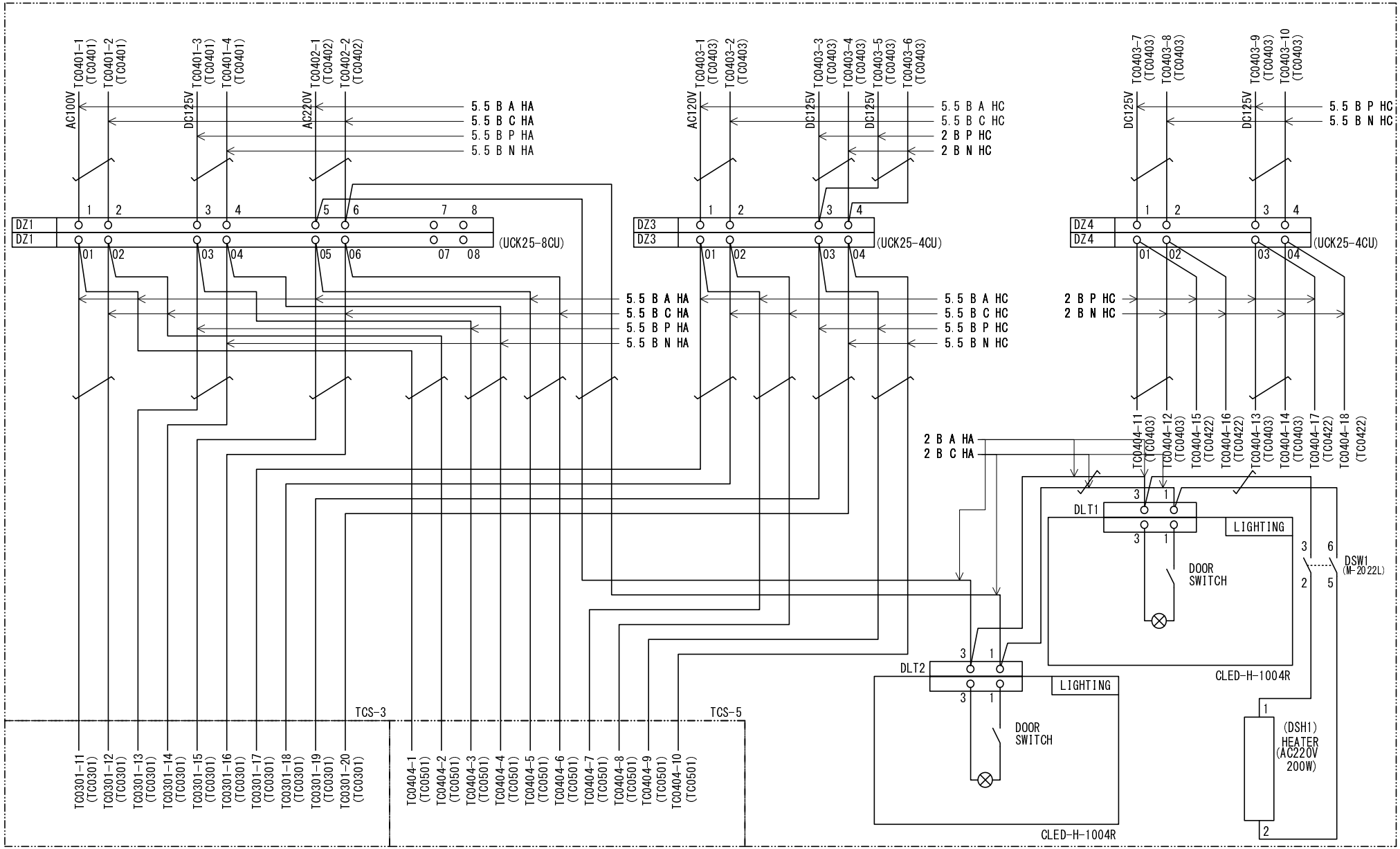
TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

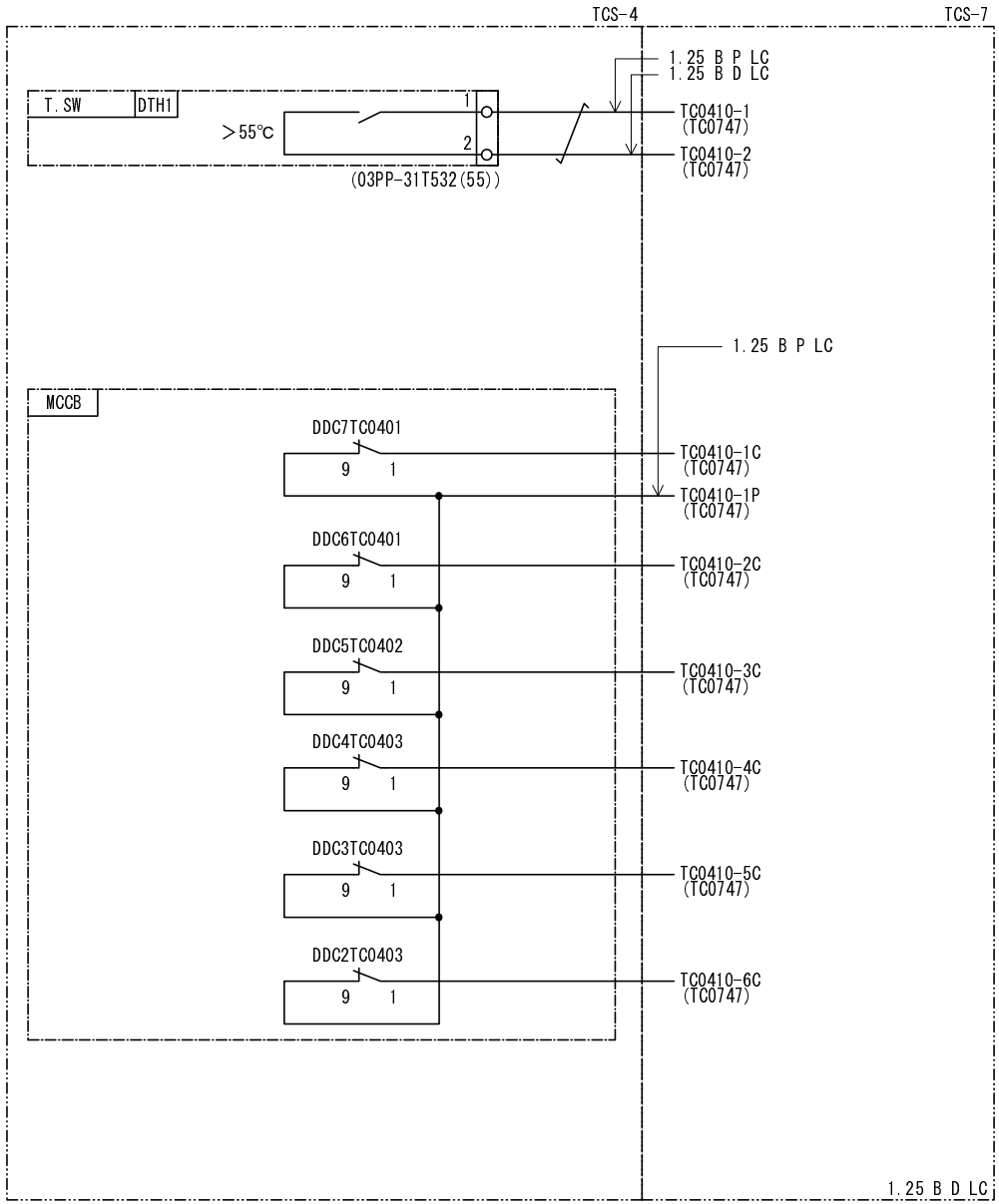
DWG. No. 331DM07733 PAGE 144 REV. 3

2016/07/28 TC0403 XSMM-TC

TCS-4



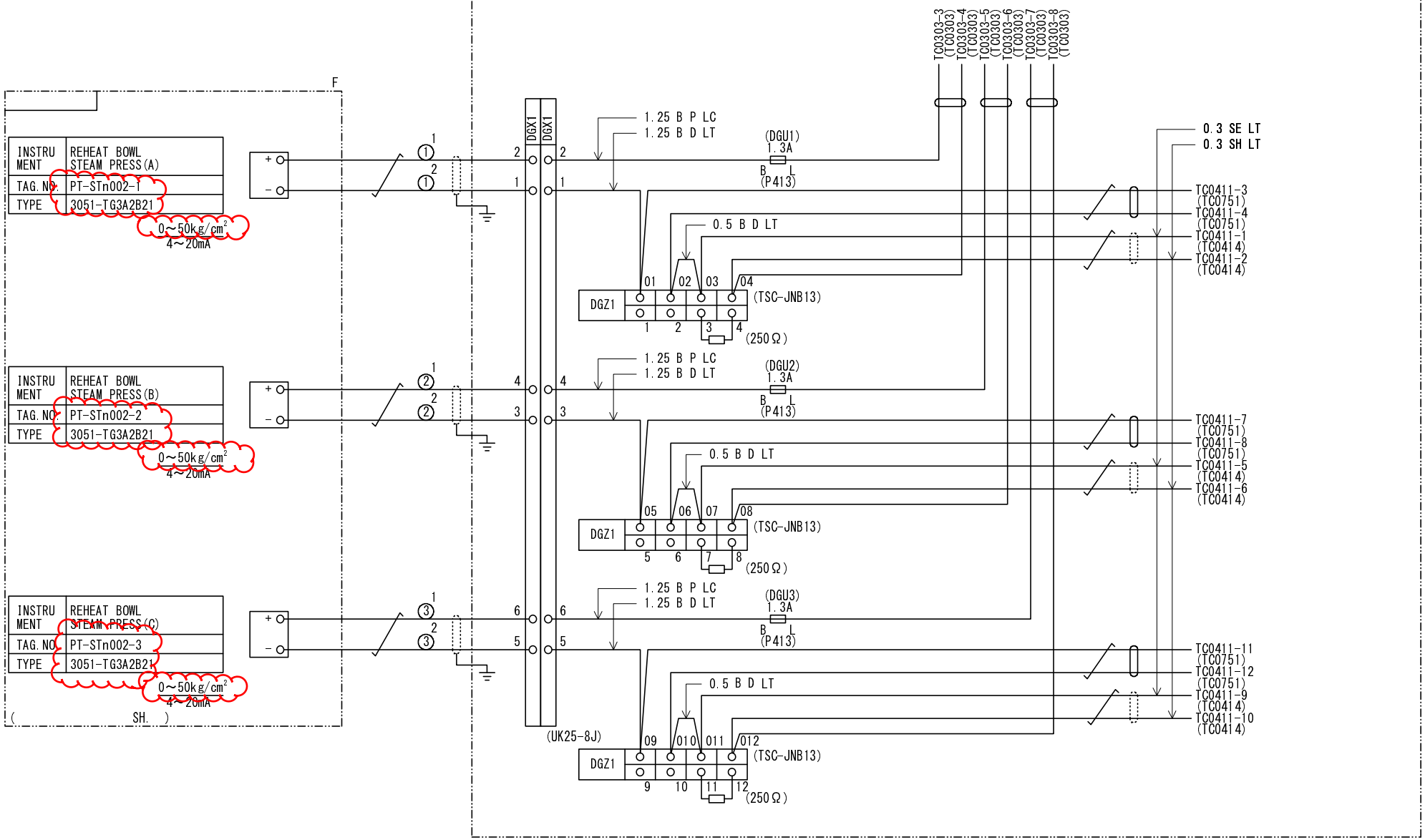
POWER SUPPLY SYSTEM		SH. No.	TC0404
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 145 / REV. 3



DIAGNOSIS CIRUCUIT		SH. No.	TC0410
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 146	REV. 3

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TCS-4



IN-DEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
③		2/2	F	TCS-4			
②		2/2	F	TCS-4			
①		2/2	F	TCS-4			

STEAM PRESS INPUT ※

SH. No. TC0411

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

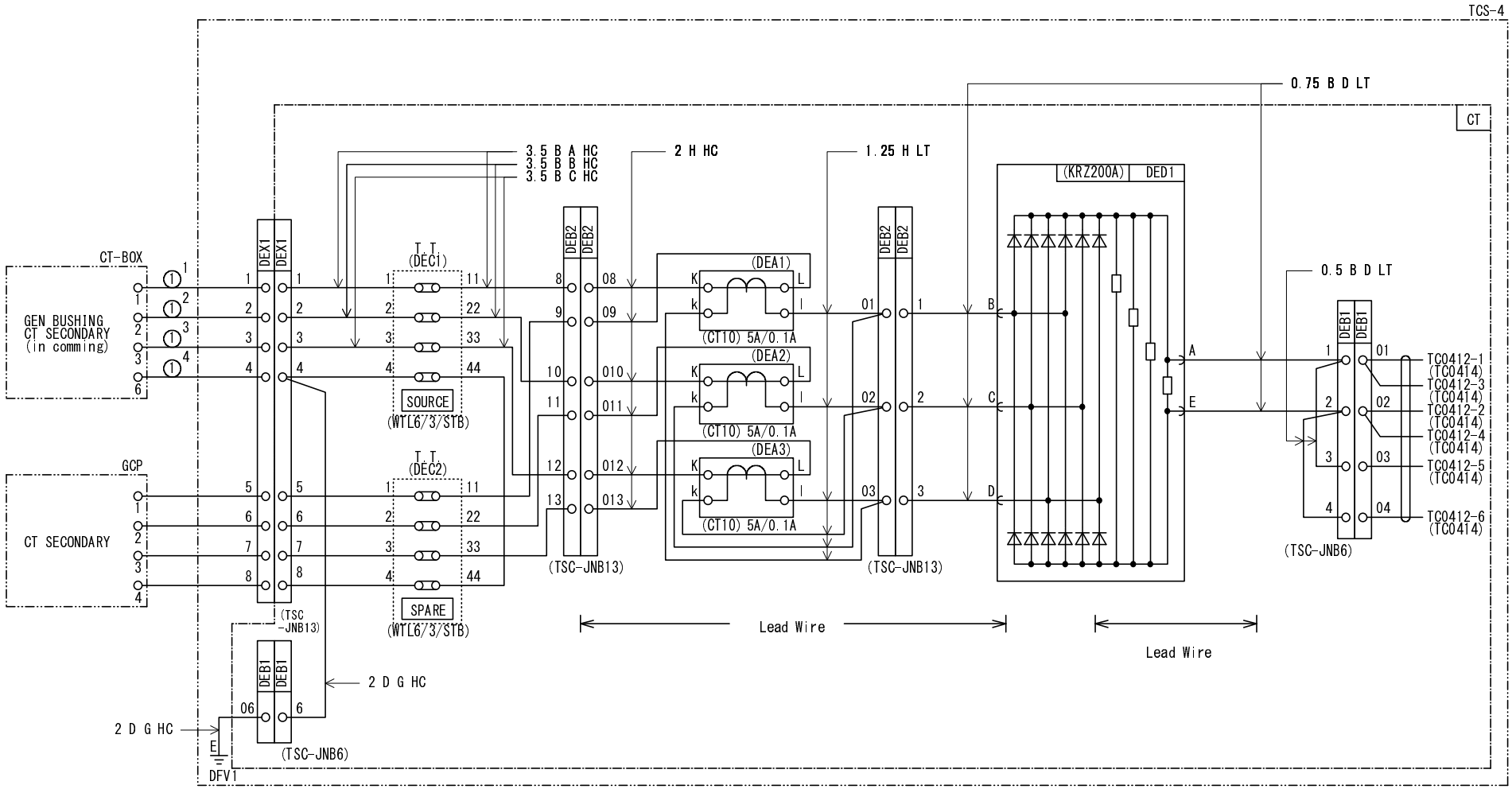
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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IN-DEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK

PLU/EOS DETECTOR ※ OSP2 SH. No. TC0412

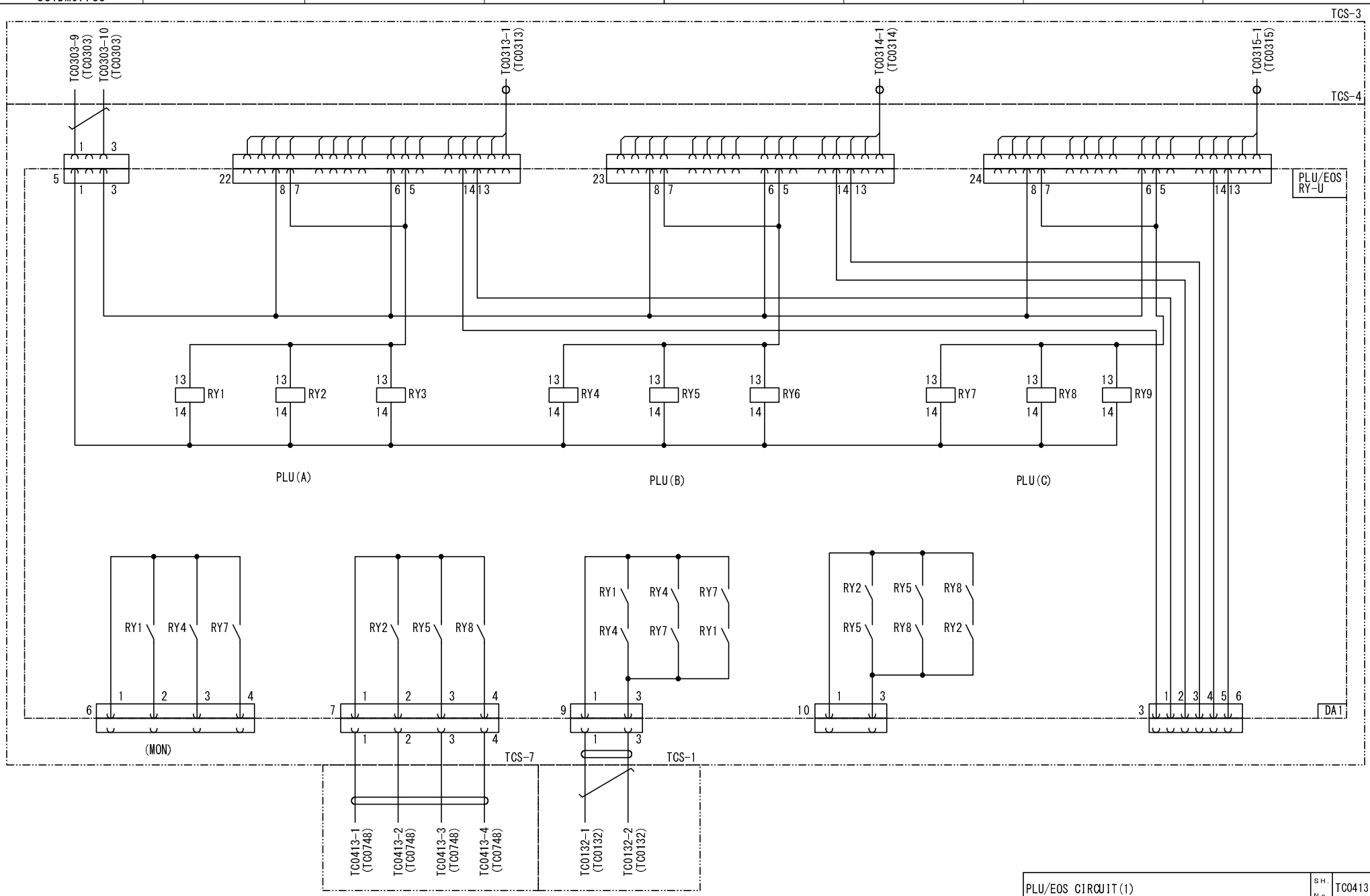
SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

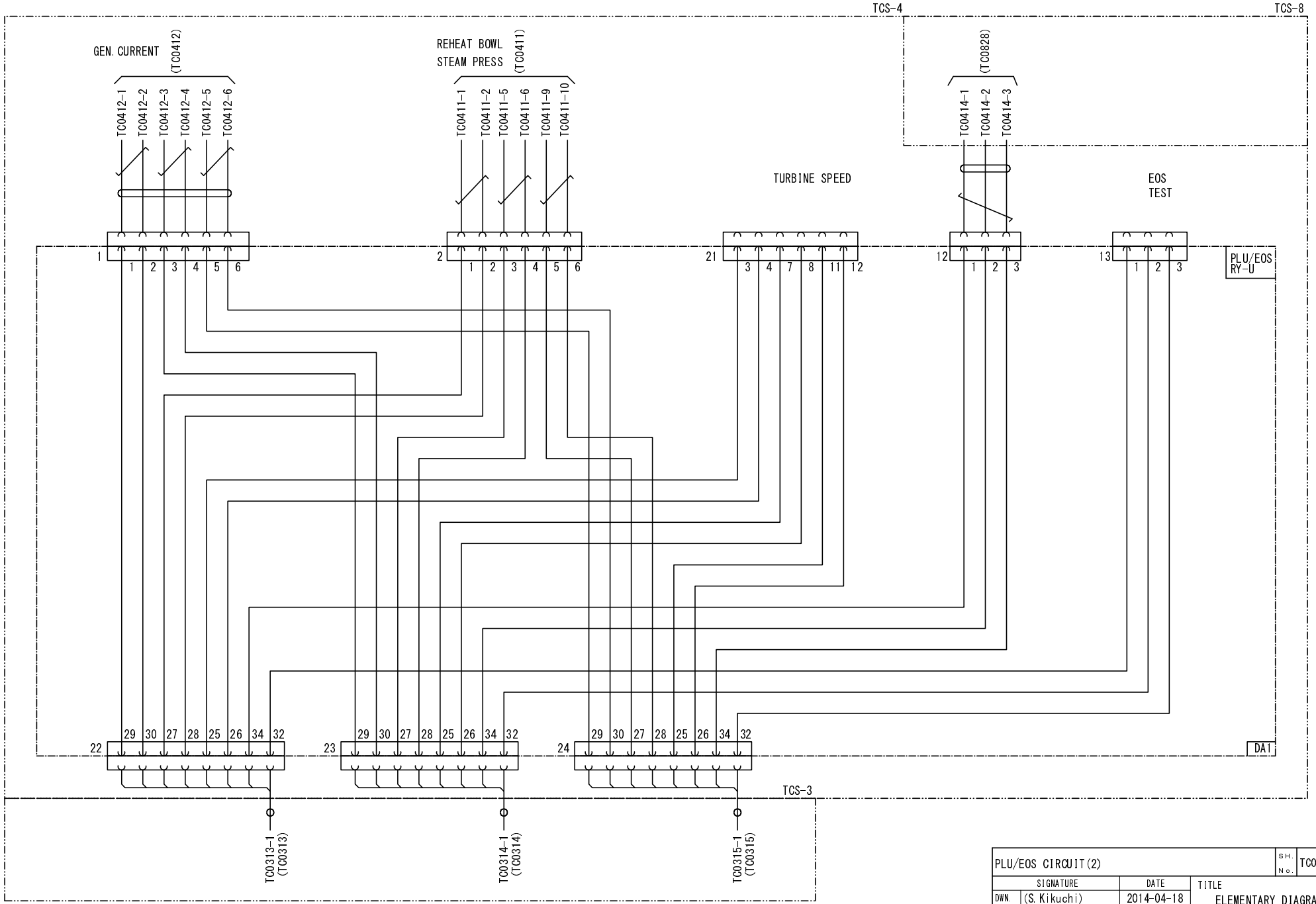
REGD. Hitachi, Ltd. Tokyo Japan

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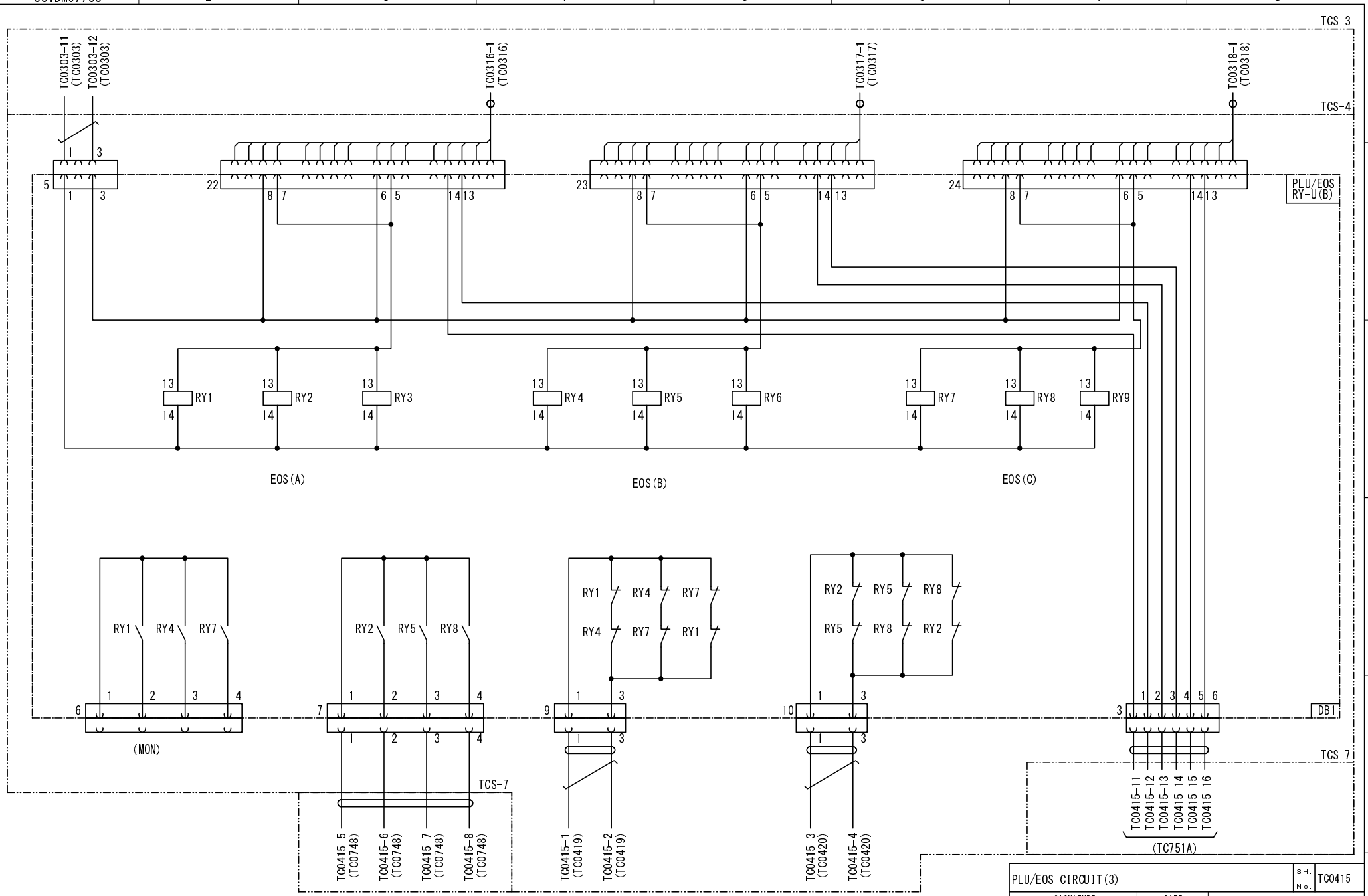
2016/07/28 TC0412 XSMM-TC



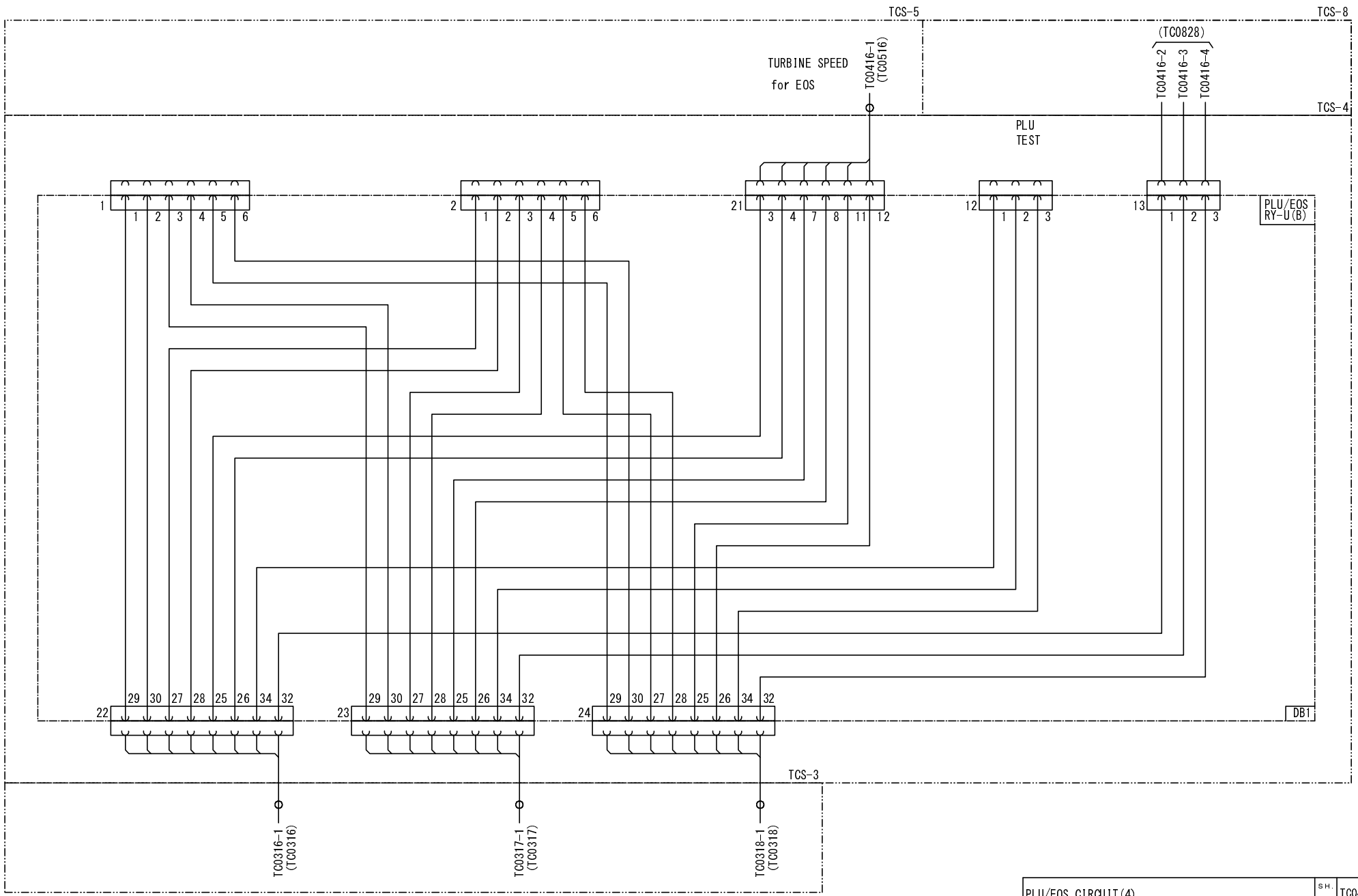
PLU/EOS CIRCUIT(1)		SH. No.	TC0413
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 149 / REV. 3



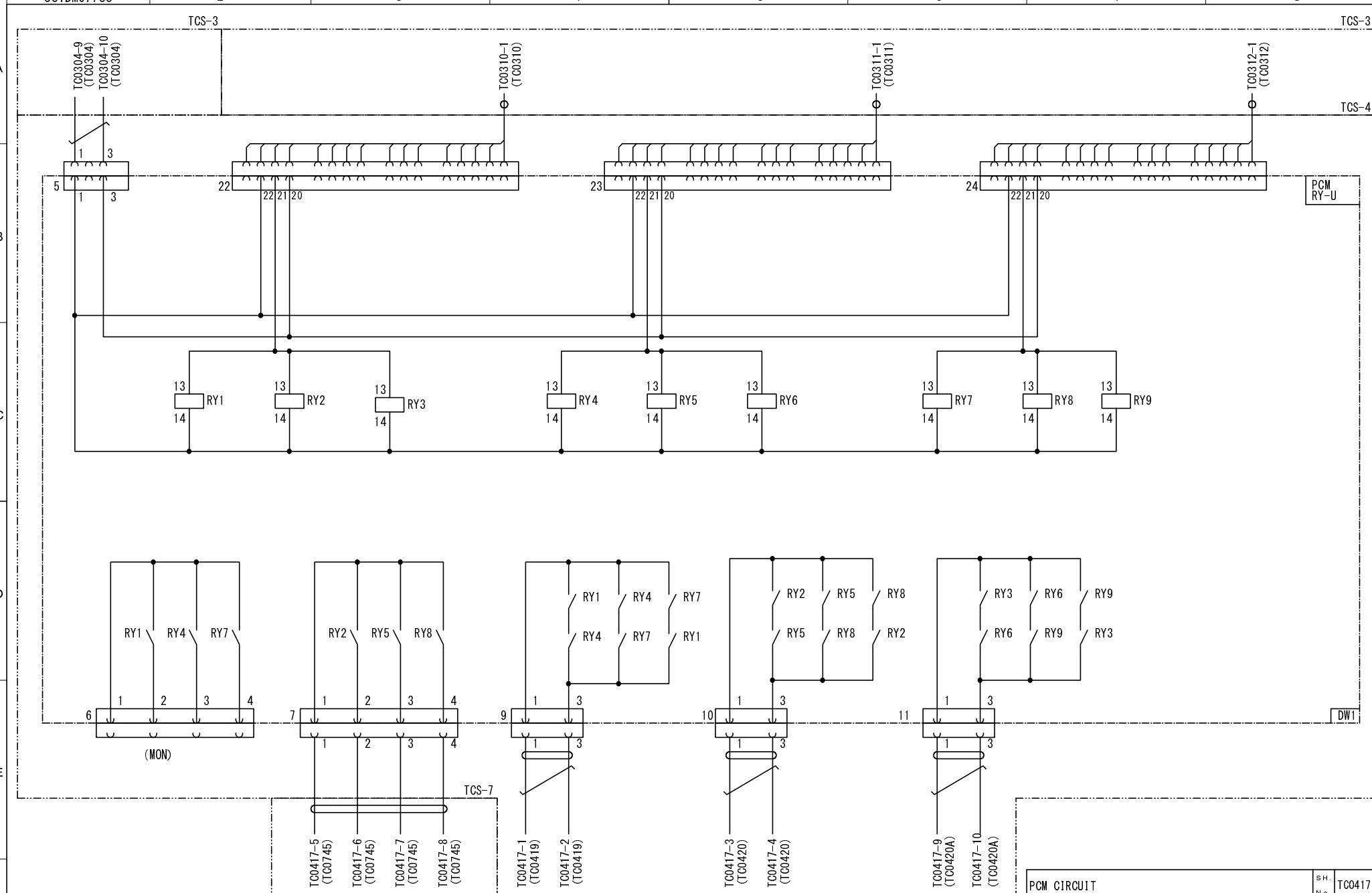
PLU/EOS CIRCUIT(2)		SH. No.	TC0414
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 150	REV. 3



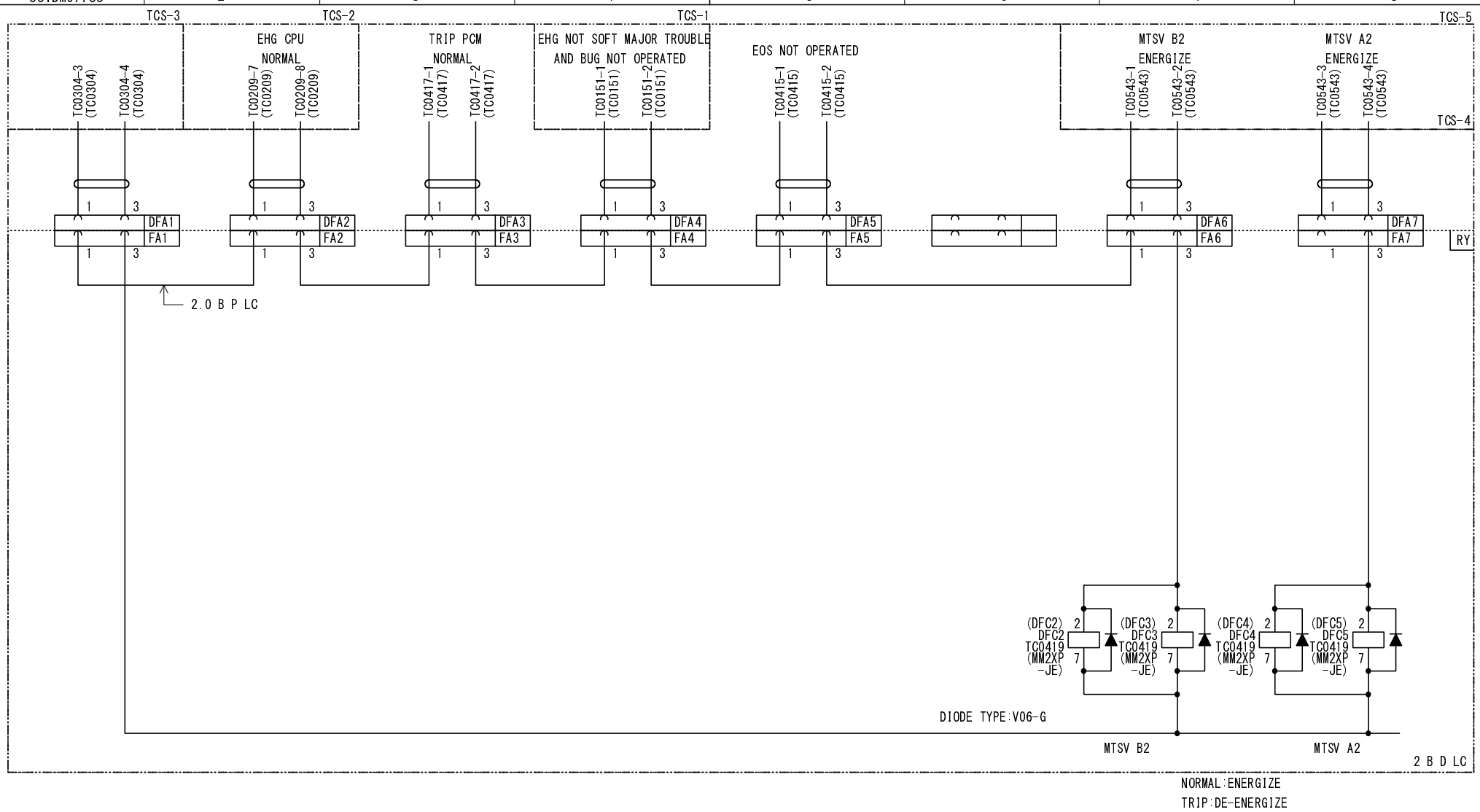
PLU/EOS CIRCUIT(3)		SH. No.	TC0415
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 151 / REV. 3



PLU/EOS CIRCUIT(4)		SH. No.	TC0416
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 152 / REV. 3



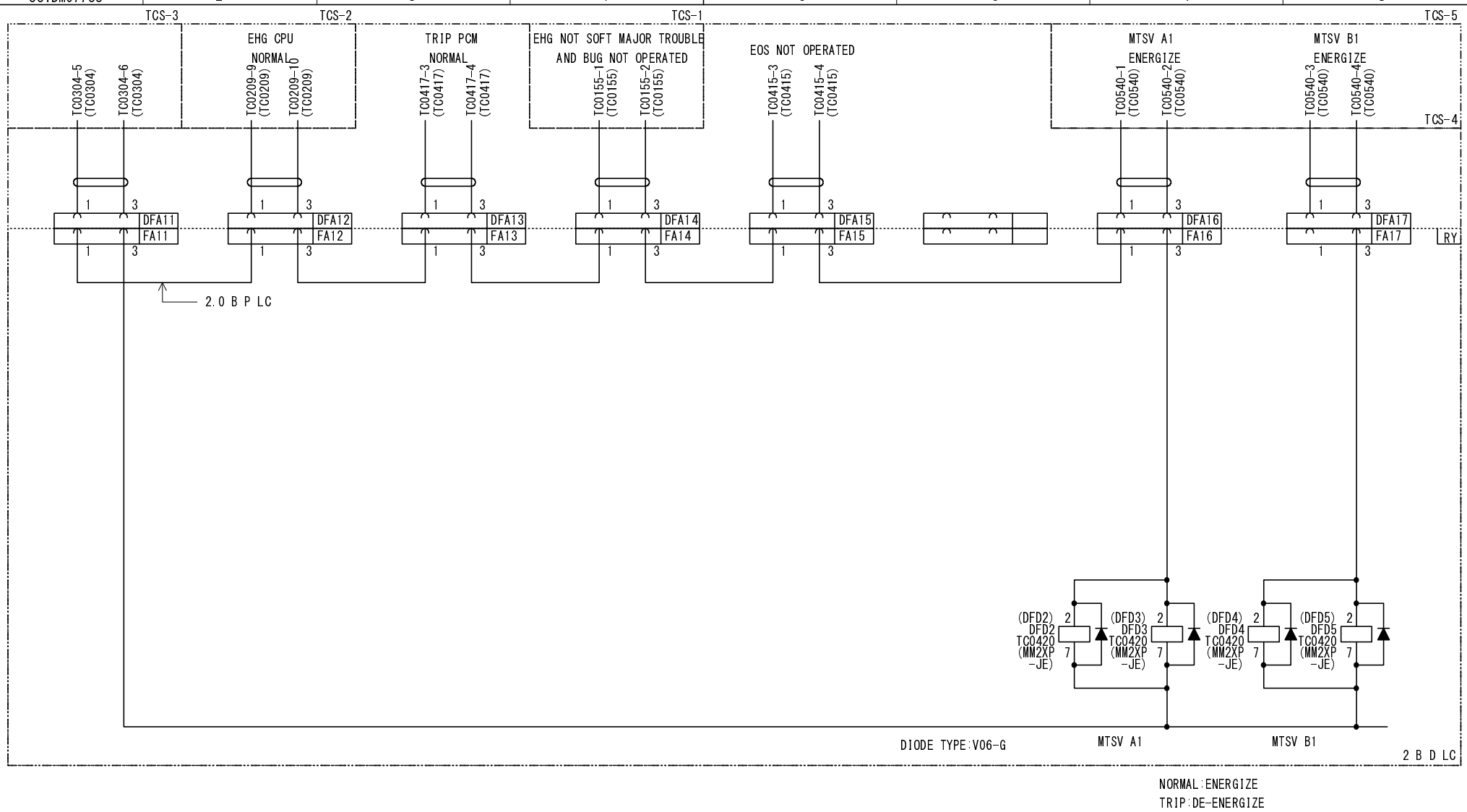
PCM CIRCUIT		SH. No.	TC0417
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 153 / REV. 3



DEV.	a			b			DEV.	a			b		
DFC2	1 3	TC0422	2	1 4			DFC4	1 3	TC0422	5	1 4		
TC0419	8 6	TC0422	2	8 5			TC0419	8 6	TC0422	5	8 5		
DFC3	1 3	TC0422	2	1 4			DFC5	1 3	TC0422	5	1 4		
TC0419	8 6	TC0422	2	8 5			TC0419	8 6	TC0422	5	8 5		

MTSV RELAY CRICUIT(1)		SH. No.	TC0419
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
REGD.	Hitachi,Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 154 / REV. 3

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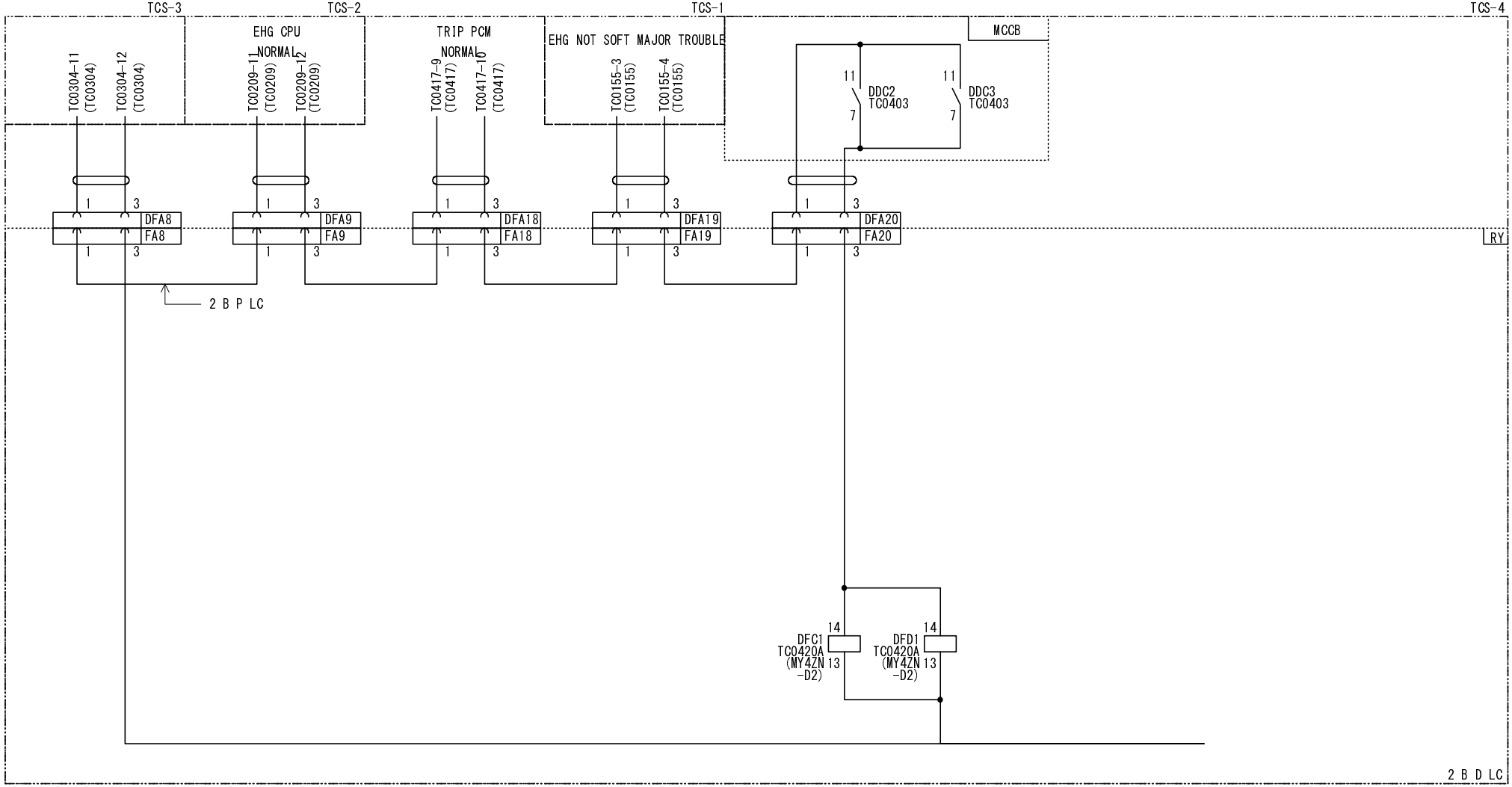


NORMAL:ENERGIZE
TRIP:DE-ENERGIZE

DEV.	a			b			DEV.	a			b		
DFD2 TC0420	1	3	TC0423:2	1	4		DFD4 TC0420	1	3	TC0423:5	1	4	
	8	6	TC0423:2	8	5			8	6	TC0423:5	8	5	
DFD3 TC0420	1	3	TC0423:2	1	4		DFD5 TC0420	1	3	TC0423:5	1	4	
	8	6	TC0423:2	8	5			8	6	TC0423:5	8	5	

MTSV RELAY CRICUIT(2)		SH. No.	TC0420
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
REGD. Hitachi,Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 155 (1)	REV. 3

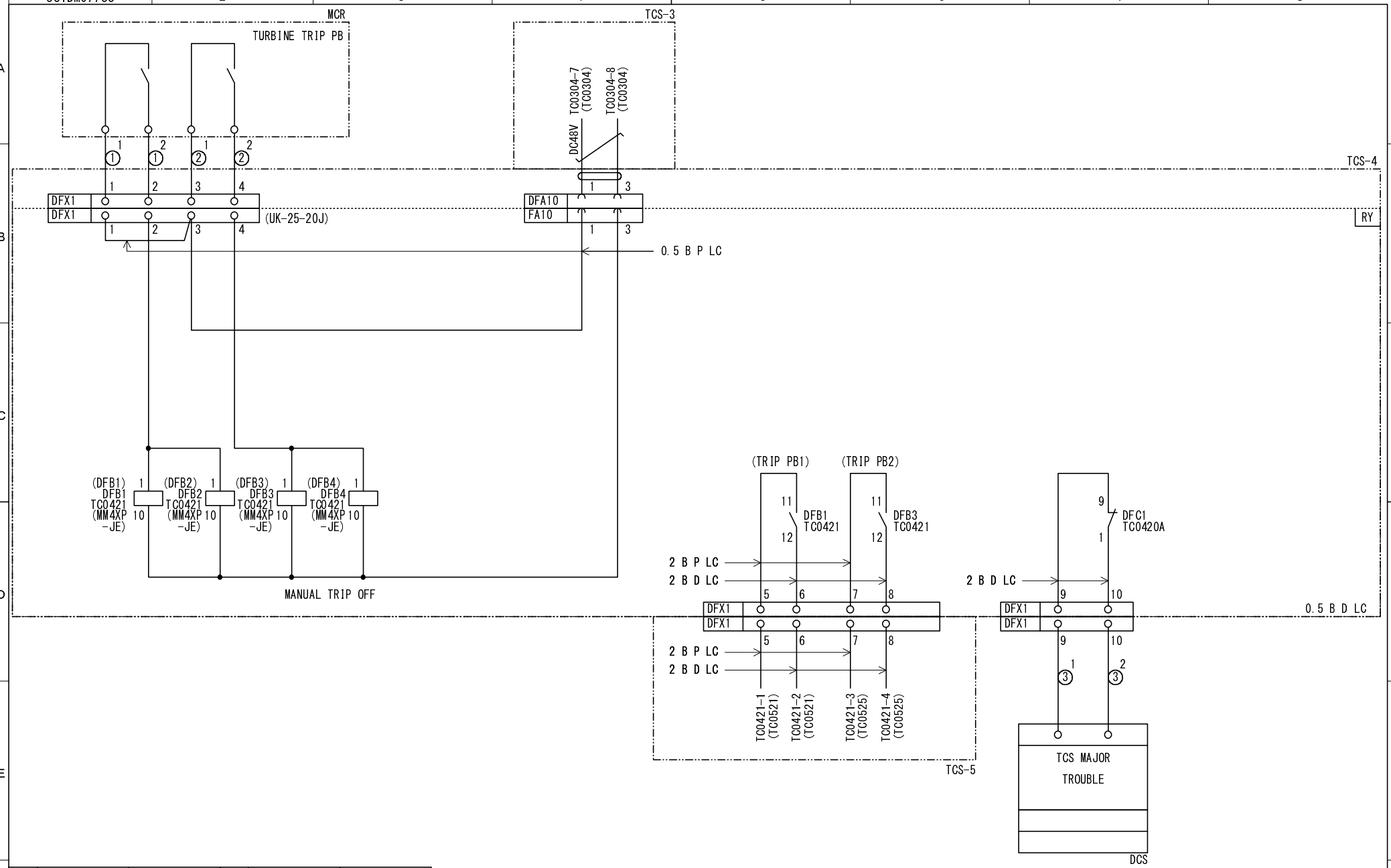
2016/07/28 TC0420 XSMM-TC



DEV.	a		b		DEV.	a		b	
DFC1 TC0420A	9 5		9 1	TC0421 7					
	10 6		10 2	TC0424 5					
	11 7		11 3						
	12 8		12 4	TC0424 4					
DFD1 TC0420A	9 5		9 1						
	10 6		10 2	TC0424 6					
	11 7		11 3						
	12 8		12 4	TC0424 5					

RY CRICUIT		SH. No.	TC0420A
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 155A	REV. 3

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DEV.	a			b		DEV.	a			b	
DFB1 TC0421	2 3			2 4	TC0422: 2	DFB3 TC0421	2 3			2 4	TC0422: 2
	11 12	*	5	11 5			11 12	*	6	11 5	
	15 14			15 6			15 14			15 6	
	9 8			9 7	TC0423: 2		9 8			9 7	TC0423: 2
DFB2 TC0421	2 3			2 4	TC0422: 5	DFB4 TC0421	2 3			2 4	TC0422: 5
	11 12			11 5			11 12			11 5	
	15 14			15 6			15 14			15 6	
	9 8			9 7	TC0423: 5		9 8			9 7	TC0423: 5

INDEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK
○							
○							
③		2/2	TCS-4	DCS			
○							
○							

TURBINE TRIPPING RELAY CRICUIT ※

SH. No. TC0421

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

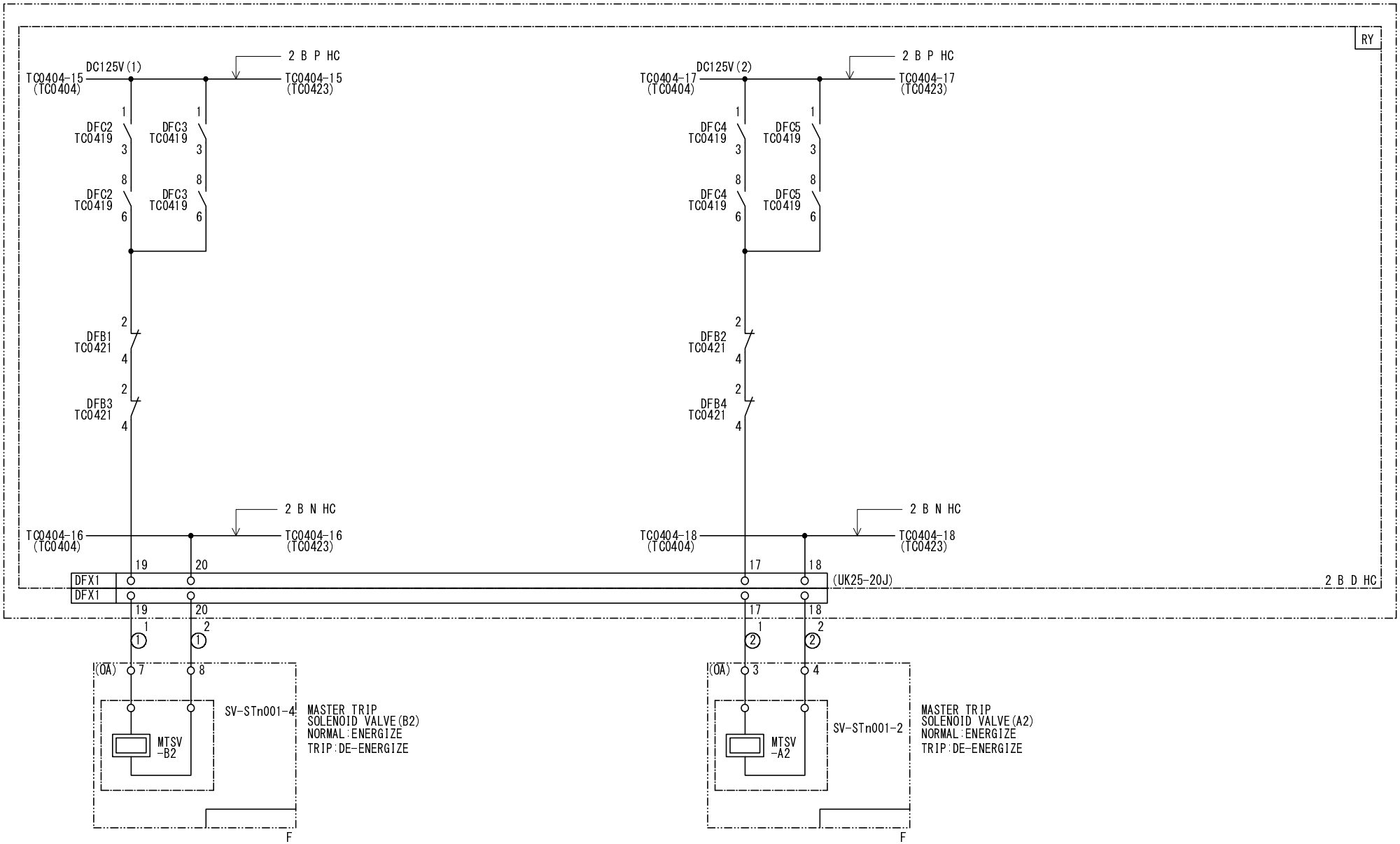
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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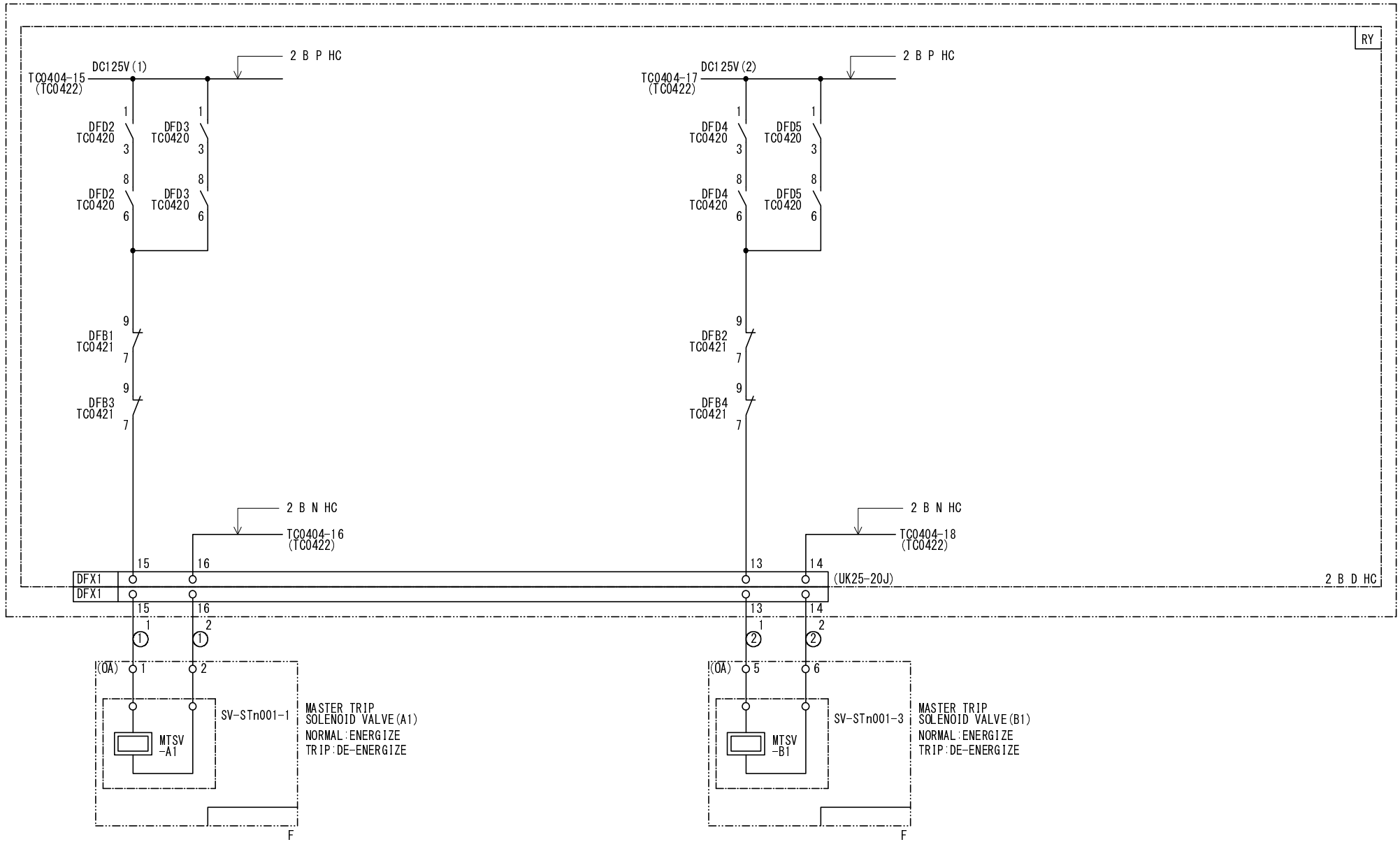
REV. 3

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INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
②		2/2	TCS-4	F			
①		2/2	TCS-4	F			

MTSV CRICUIT(1)		SH. No.	TC0422
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 157 / REV. 3

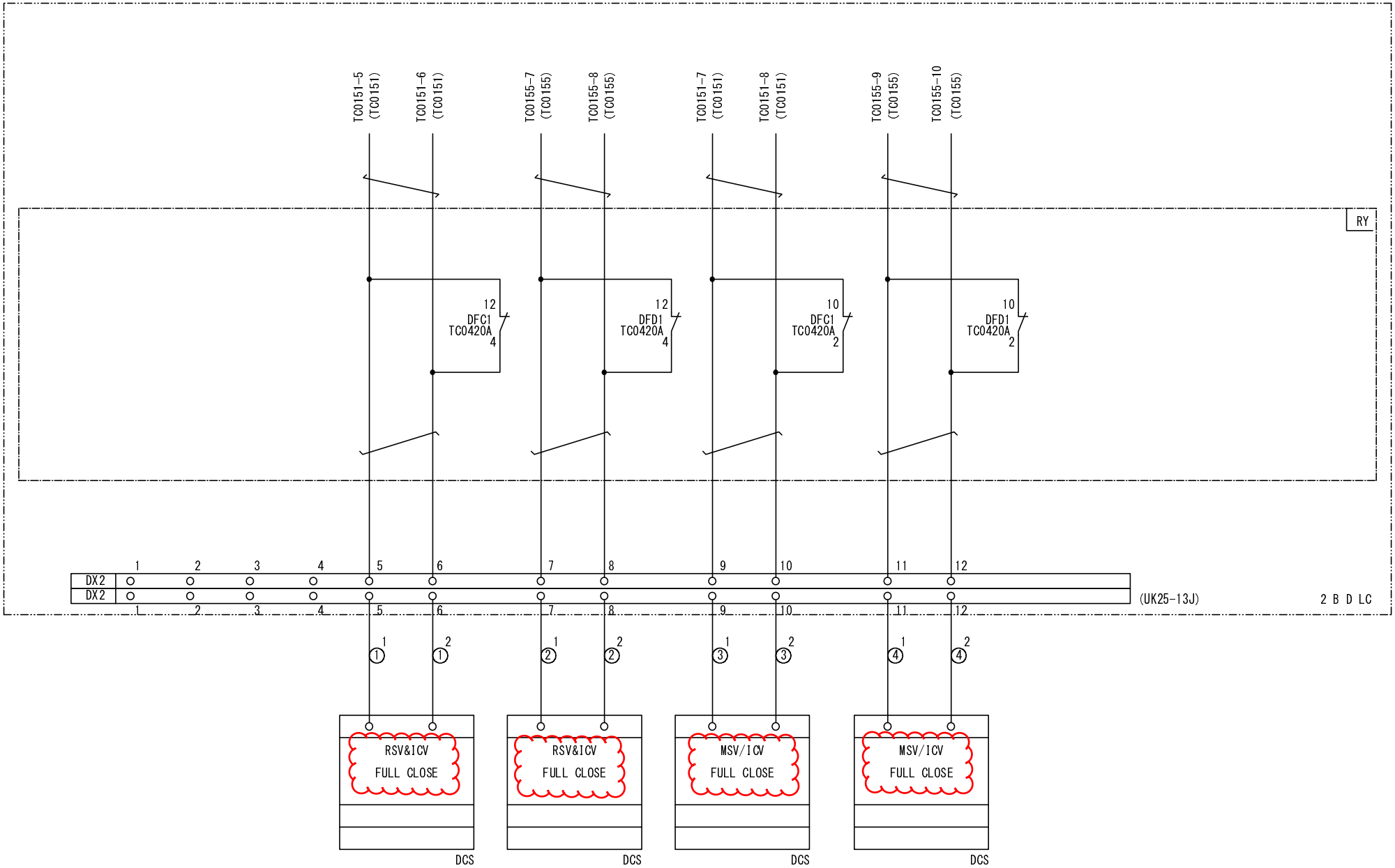


INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
②		2/2	TCS-4	F			
①		2/2	TCS-4	F			

MTSV CRICUIT(2)		SH. No.	TC0423
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 158 (1) / REV. 3

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TCS-4



INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
④		2/2	TCS-4	DCS			
③		2/2	TCS-4	DCS			
②		2/2	TCS-4	DCS			
①		2/2	TCS-4	DCS			

EXTENAL OUTPUT

SH. No. TC0424

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

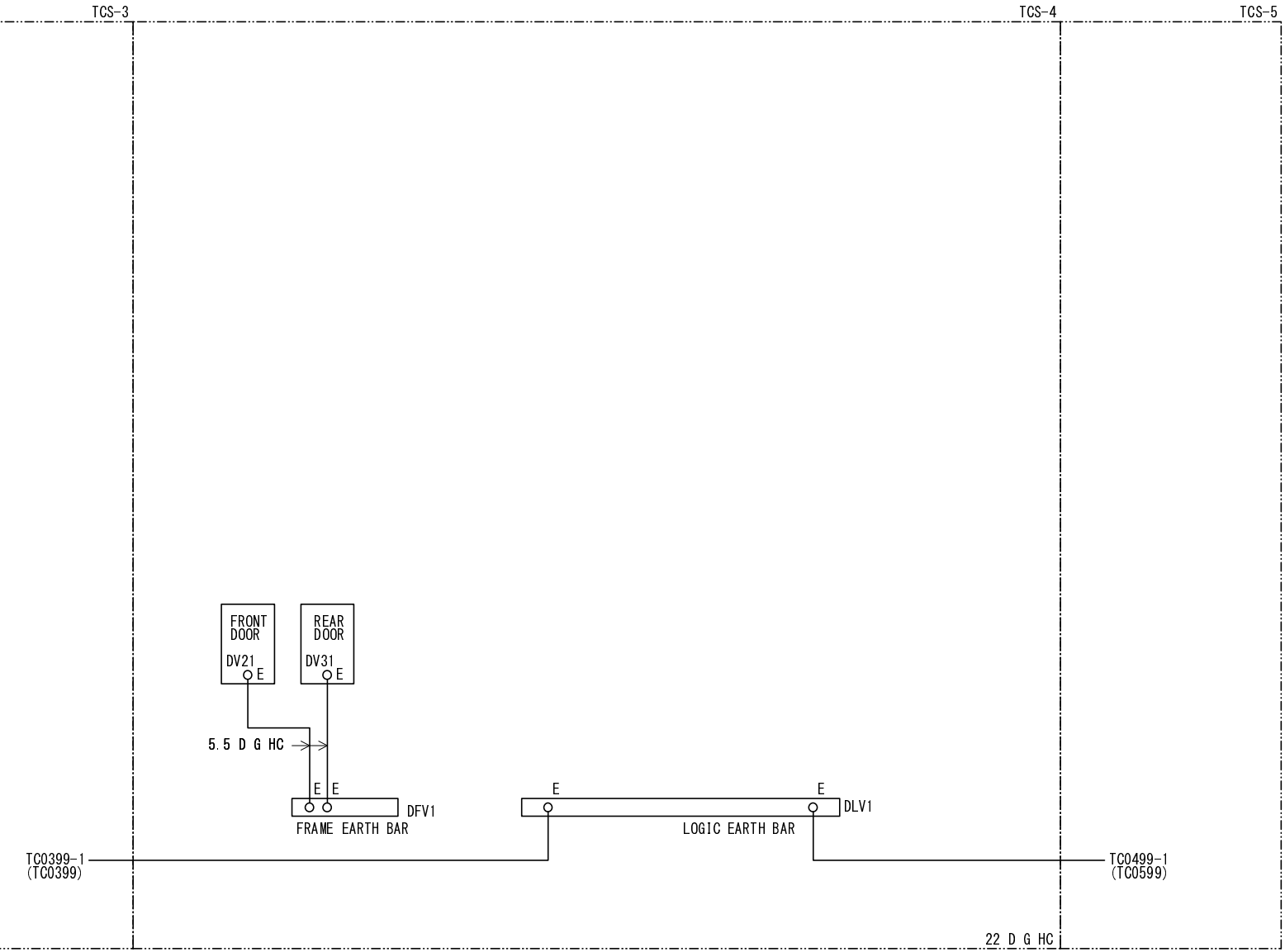
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

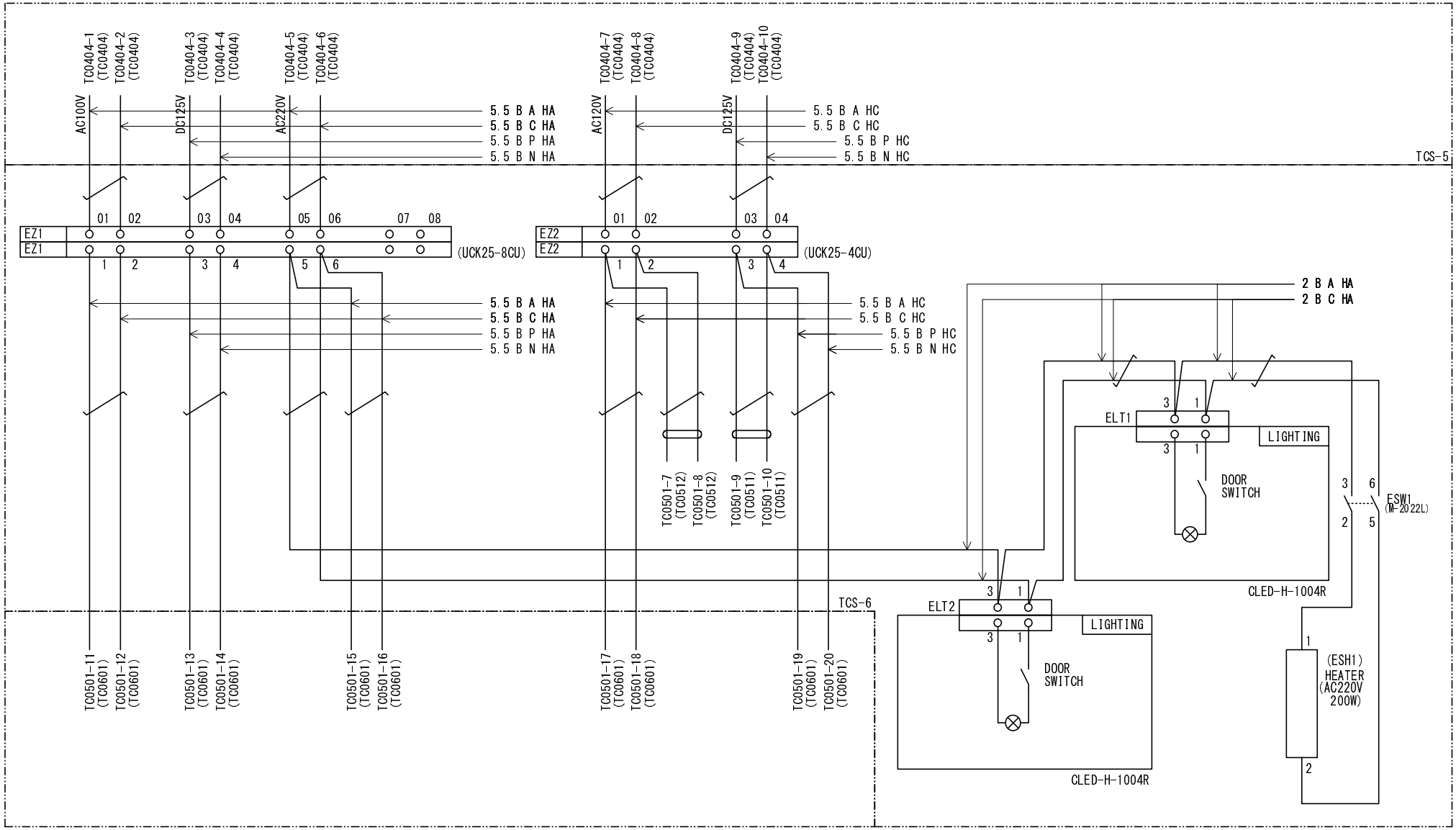
PAGE 158A

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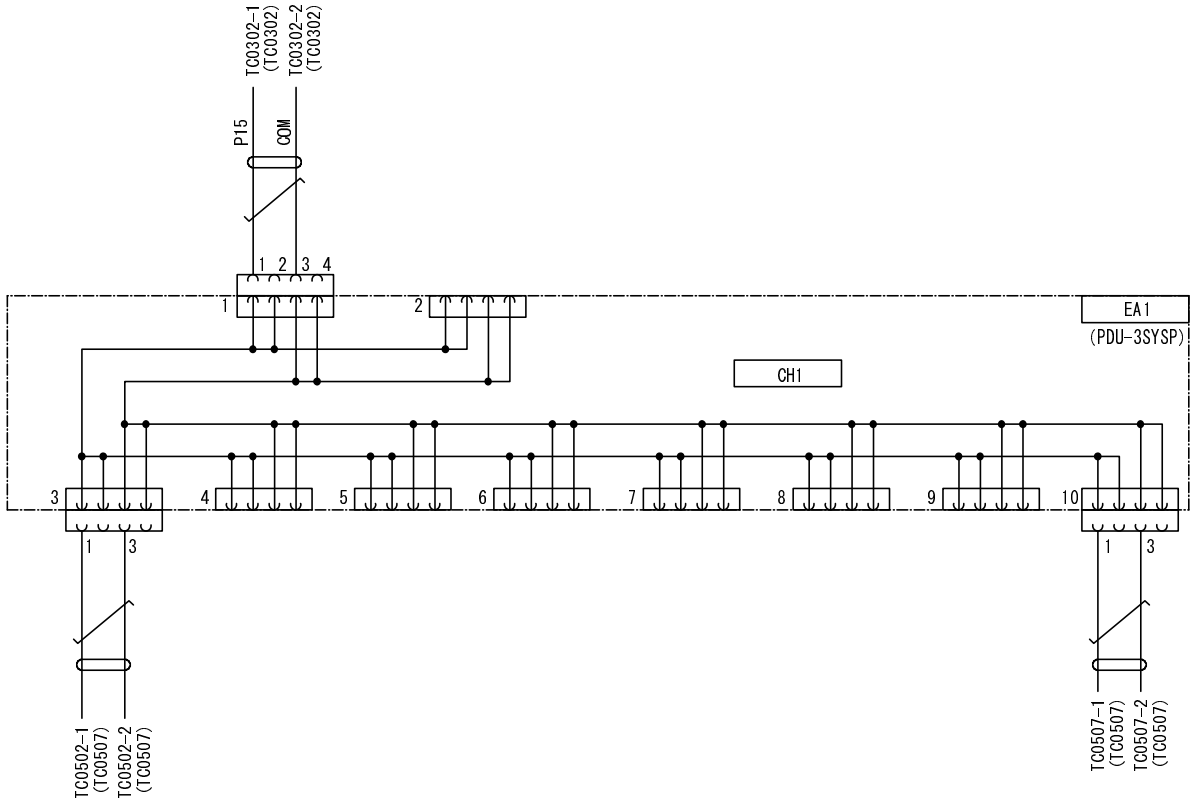
POWER SUPPLY SYSTEM EARTH CIRCUIT		SH. No.	TC0499
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 159 / REV. 3



POWER SUPPLY SYSTEM(1)		SH. No.	TC0501
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 160 / REV. 3

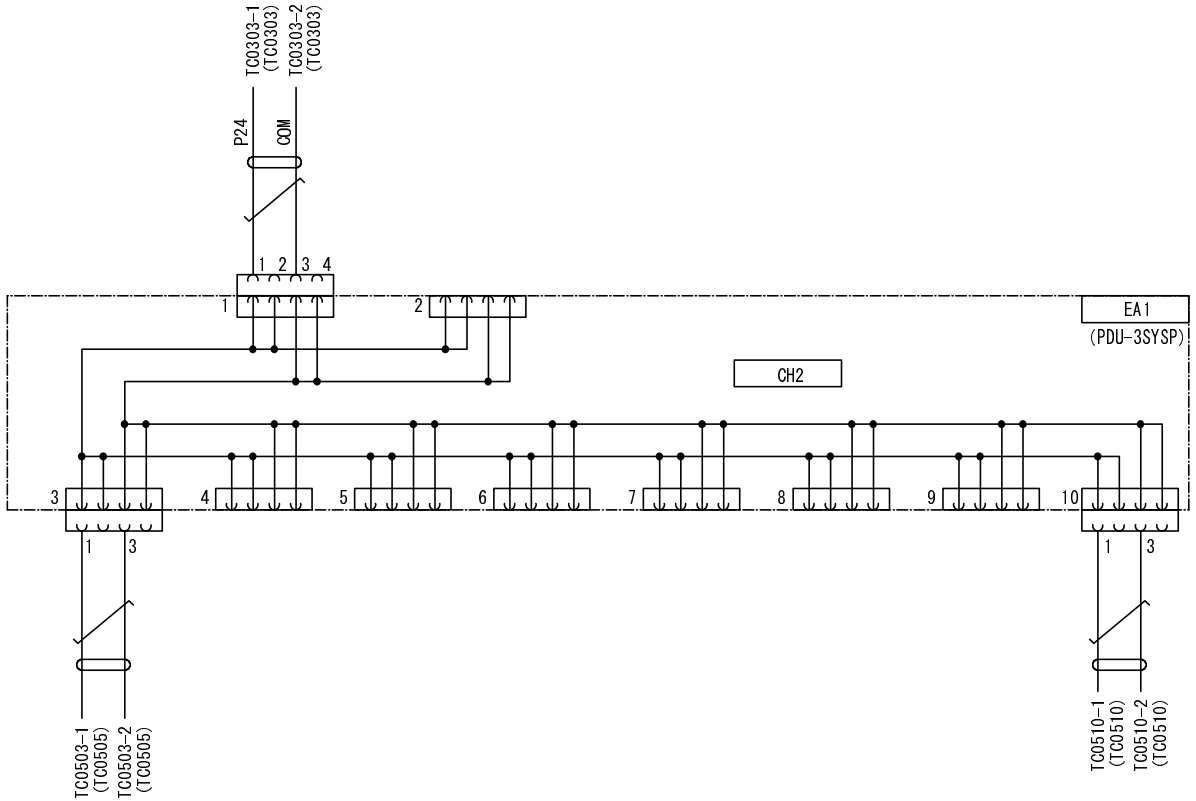
2016/07/28 TC0501 XSMM-TC

TCS-5



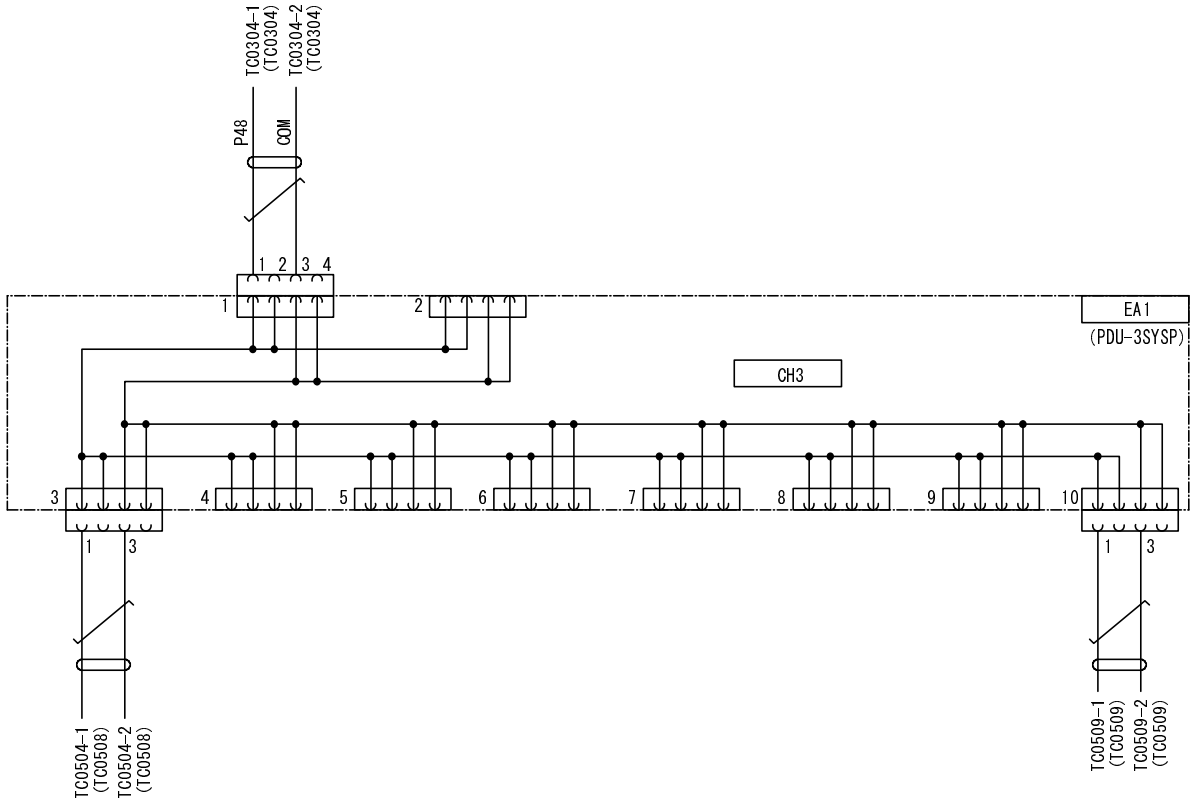
POWER SUPPLY SYSTEM(2)		SH. No.	TC0502
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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POWER SUPPLY SYSTEM(3)		SH. No.	TC0503
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 162	REV. 3

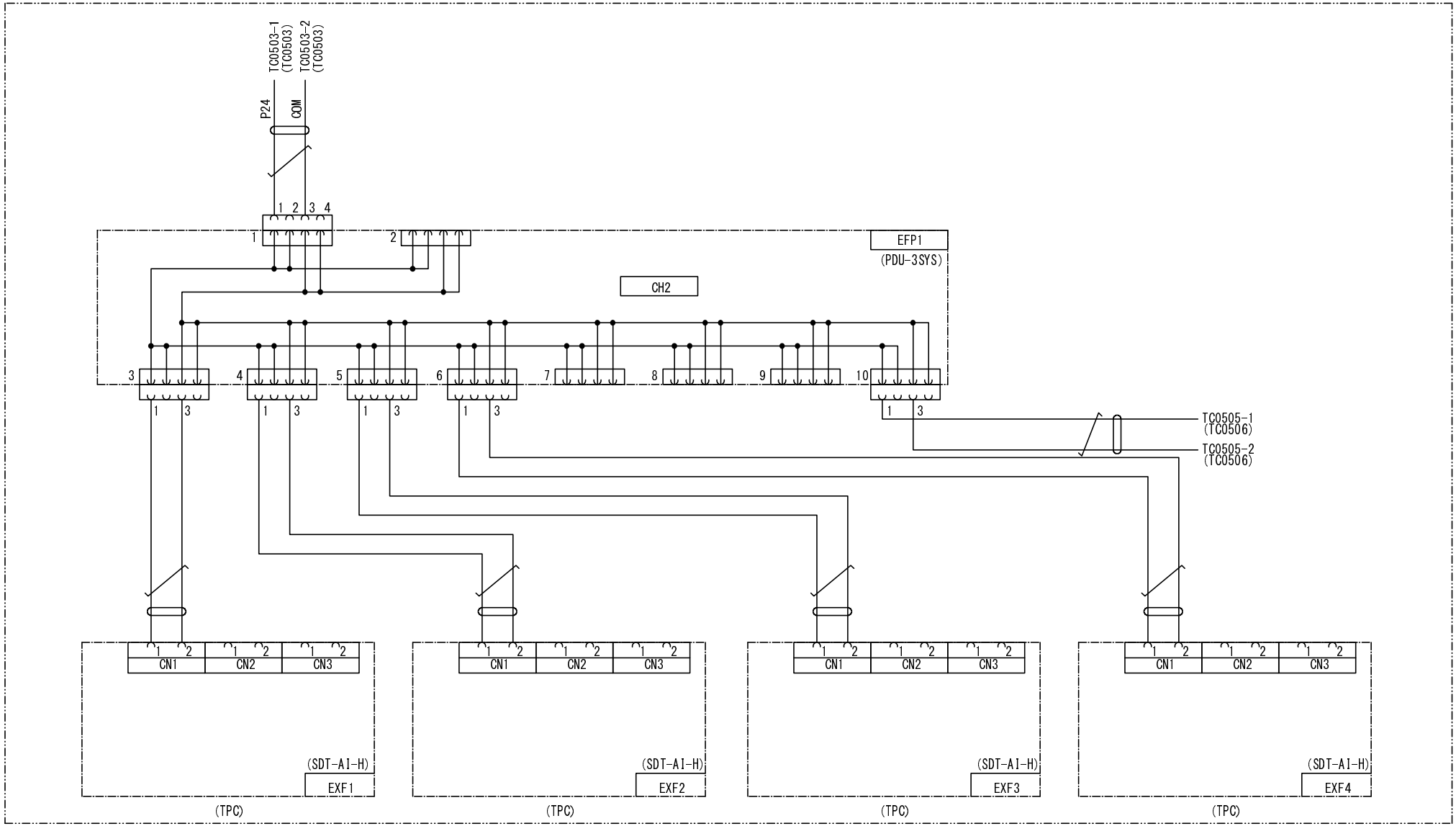
TCS-5



POWER SUPPLY SYSTEM(4)		SH. No.	TC0504
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 163	REV. 3

2016/07/28 TC0504 XSMM-TC

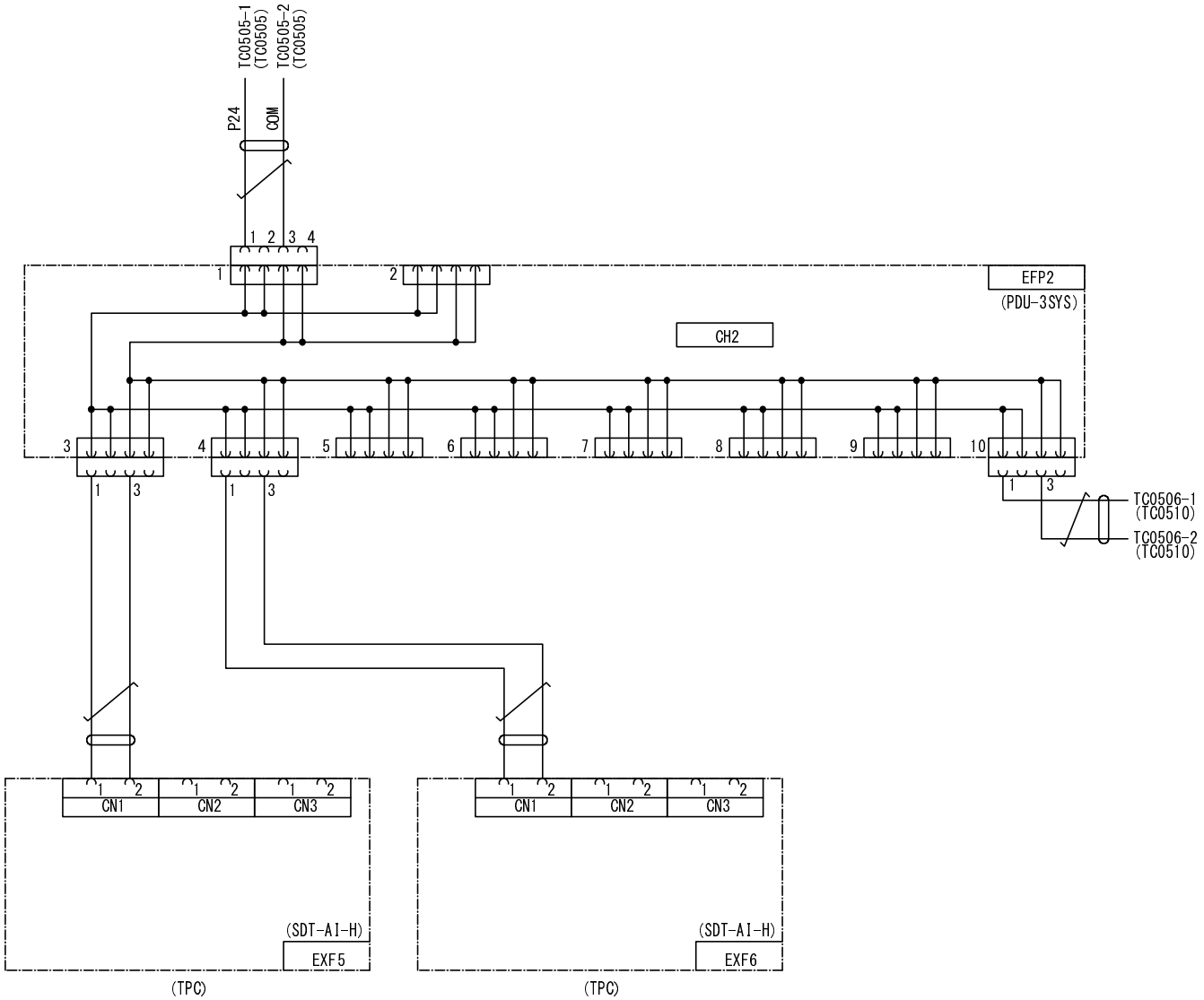
TCS-5



POWER SUPPLY SYSTEM(5)				SH. No.	TC0505
SIGNATURE		DATE	TITLE		
DWN.	(S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM		
CHKD.	T.Kawata	2014-04-18			
APPD.	Y.Fukushima	2014-04-18			
REGD.	Hitachi, Ltd. Tokyo Japan		DWG. No. 331DM07733	PAGE 164	REV. 3

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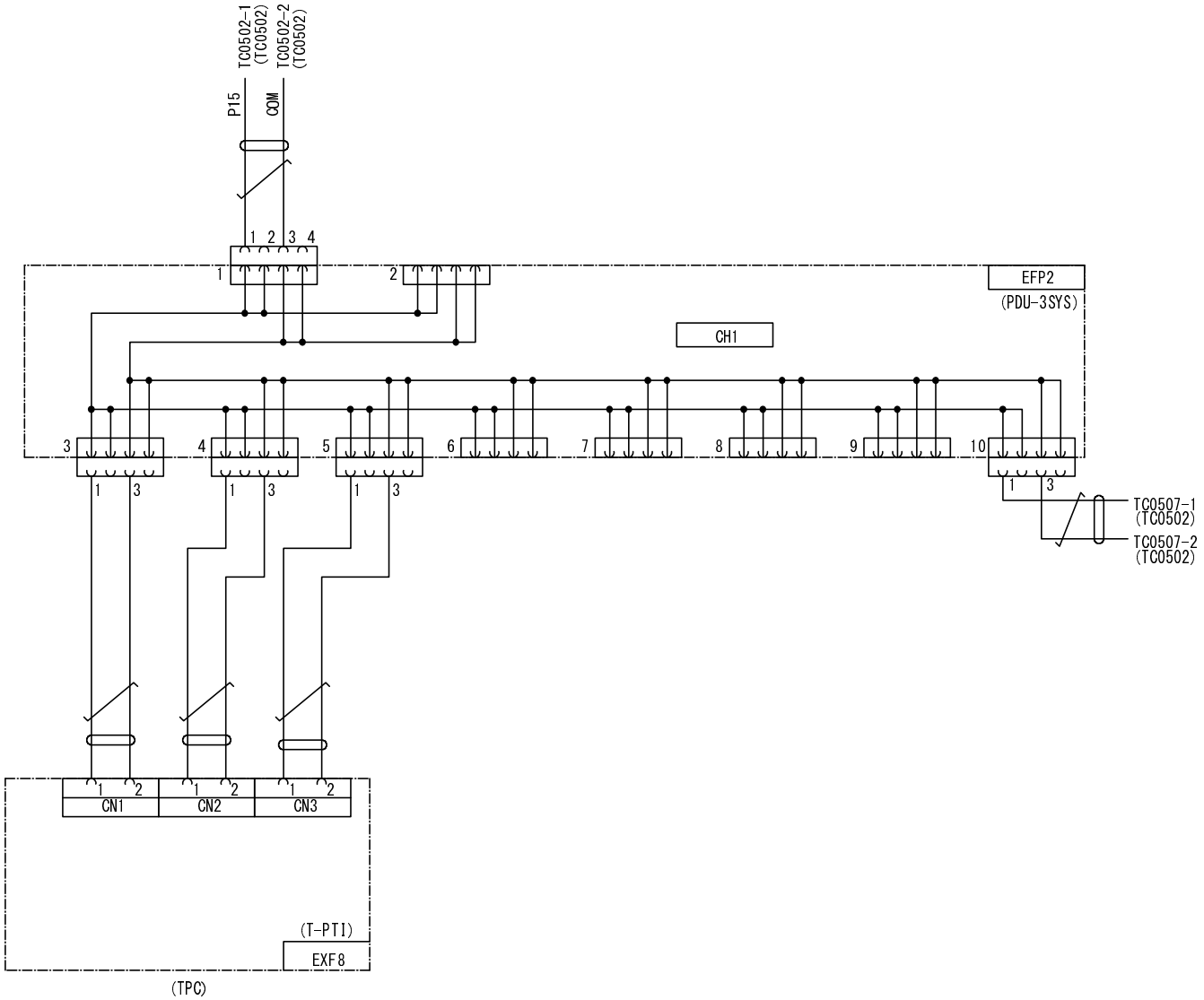
TCS-5



POWER SUPPLY SYSTEM(6)		SH. No.	TC0506
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 165	REV. 3

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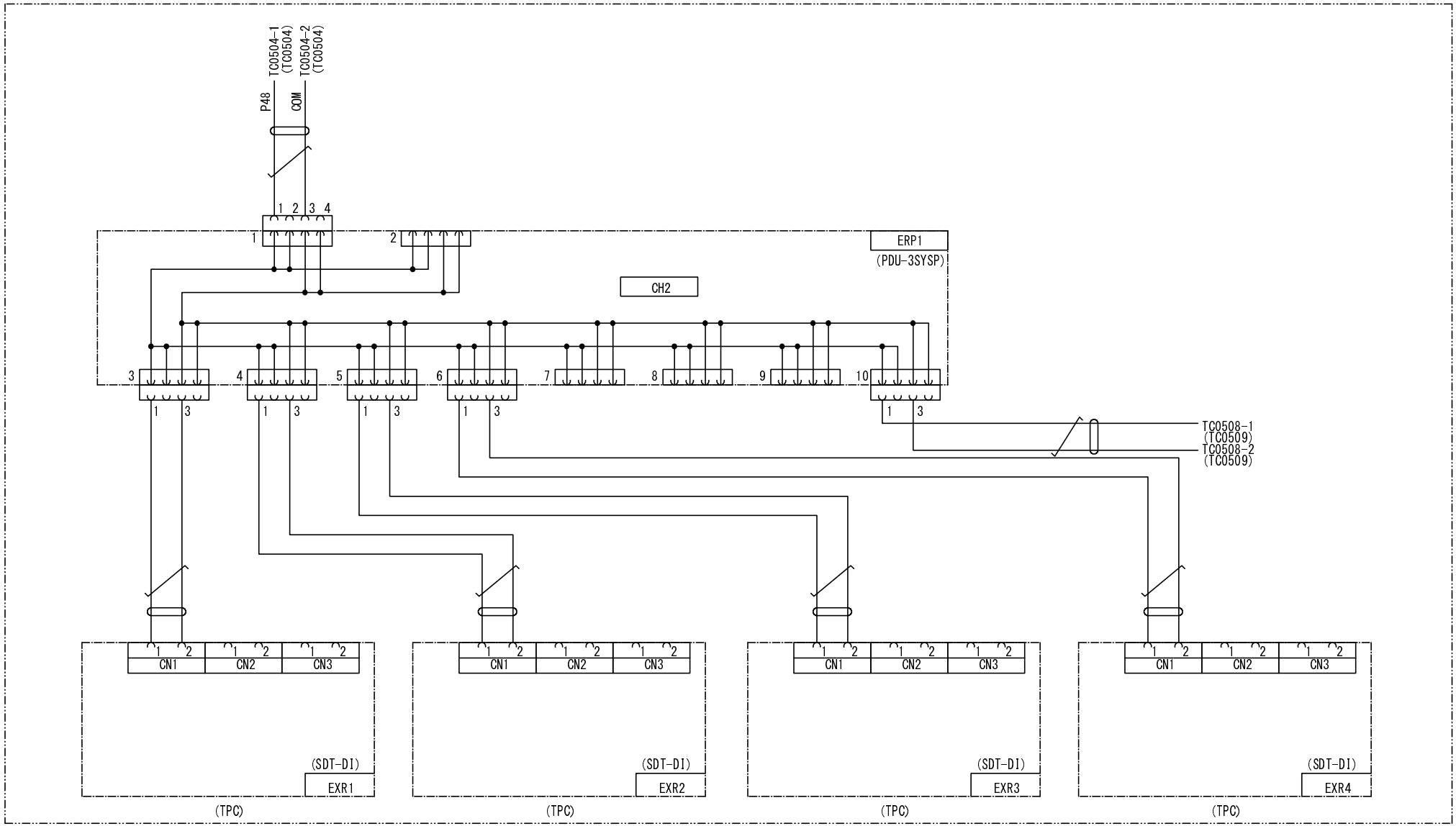
TCS-5



POWER SUPPLY SYSTEM(7)		SH. No.	TC0507
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 166 / REV. 3

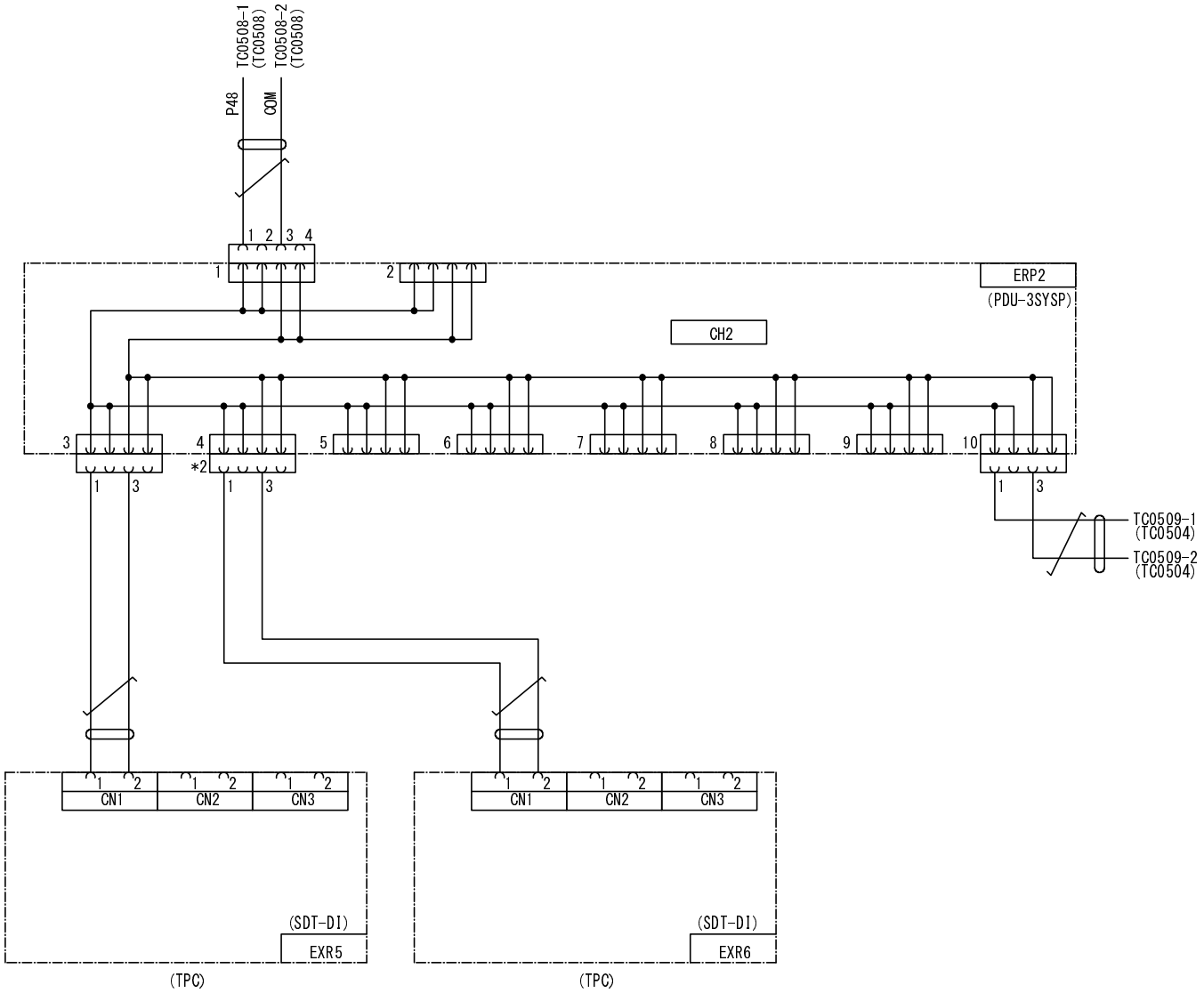
2016/07/28 TC0507 XSMM-TC

TCS-5



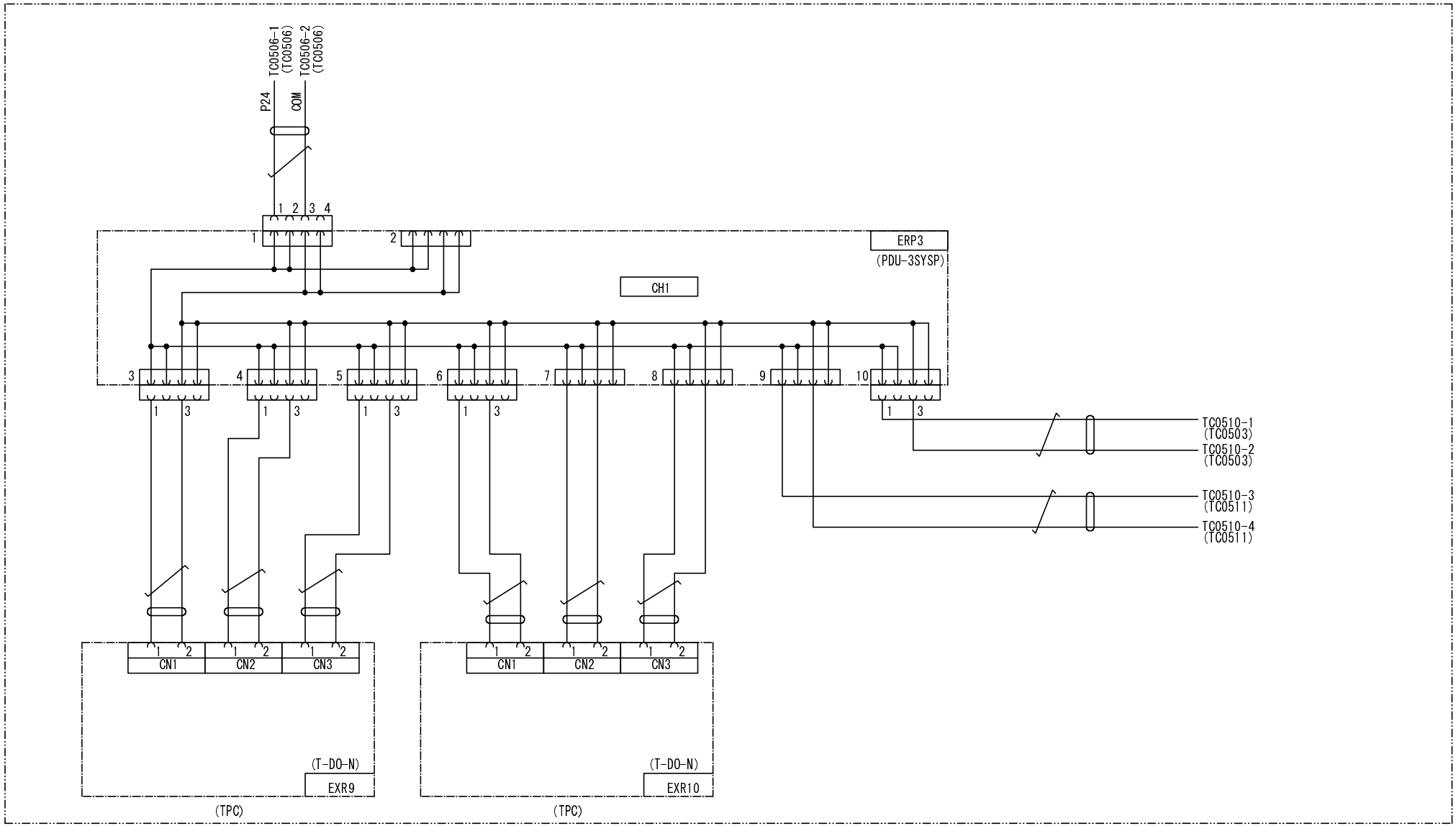
POWER SUPPLY SYSTEM(8)			SH. No.	TC0508
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
REGD.	Hitachi, Ltd. Tokyo Japan		DWG. No. 331DM07733	PAGE 167 / REV. 3

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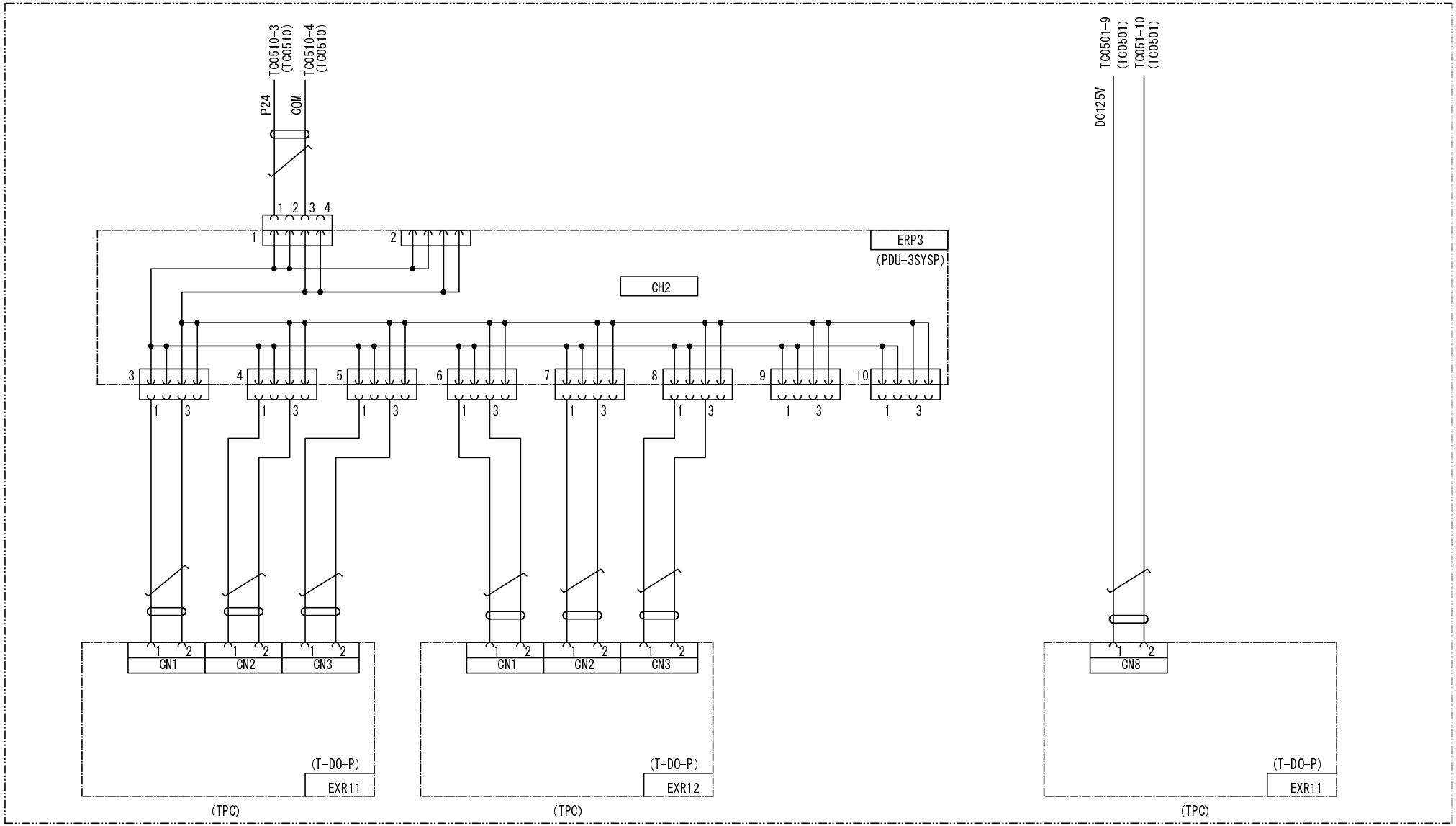
POWER SUPPLY SYSTEM(9)				SH. No.	TC0509
SIGNATURE		DATE	TITLE		
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM		
CHKD.	T. Kawata	2014-04-18			
APPD.	Y. Fukushima	2014-04-18			
REGD.	Hitachi, Ltd. Tokyo Japan		DWG. No. 331DM07733	PAGE 168	REV. 3

2016/07/28 TC0509 XSMM-TC



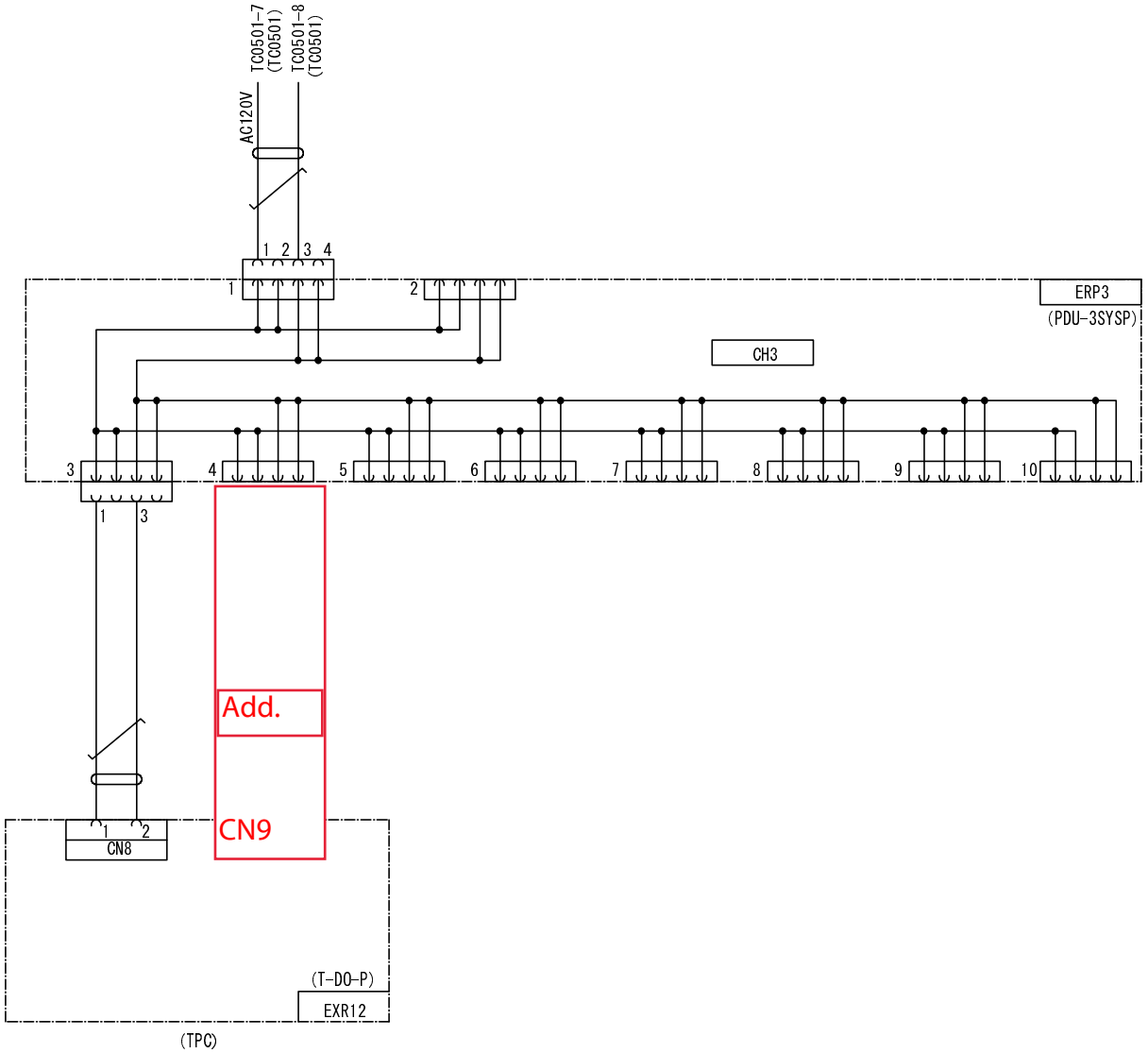
POWER SUPPLY SYSTEM(10)		SH. No.	TC0510
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
REGD. Hitachi,Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 169	REV. 3

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POWER SUPPLY SYSTEM(11)		SH. No.	TC0511
SIGNATURE		DATE	TITLE
DWN.	(S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T.Kawata	2014-04-18	
APPD.	Y.Fukushima	2014-04-18	
REGD.	Hitachi,Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 170 / REV. 3

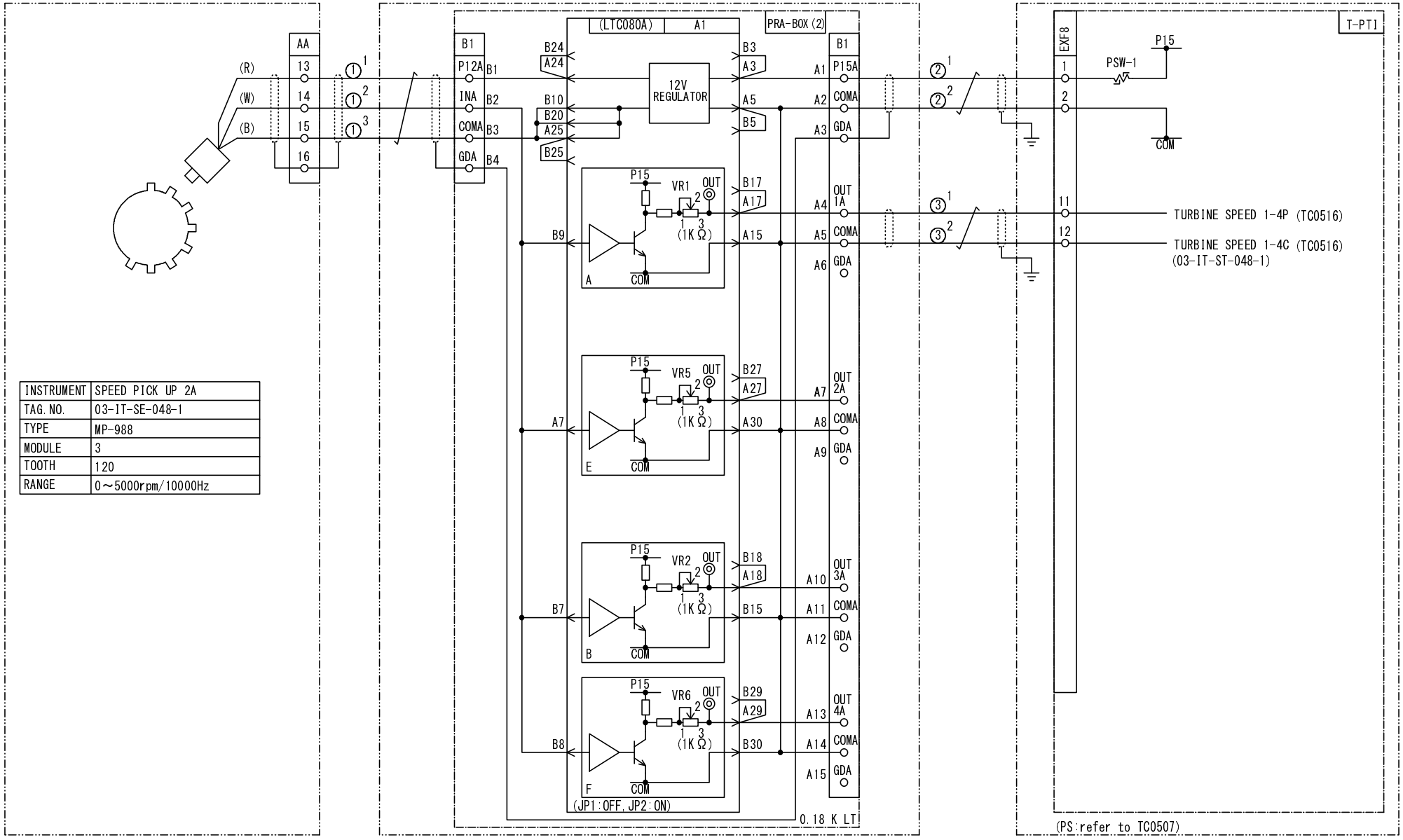
2016/07/28 TC0511 XSMM-TC



POWER SUPPLY SYSTEM(12)		SH. No.	TC0512
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 171	REV. 3

TERMINAL BOX

TCS-5



INSTRUMENT	SPEED PICK UP 2A
TAG. NO.	03-IT-SE-048-1
TYPE	MP-988
MODULE	3
TOOTH	120
RANGE	0～5000rpm/10000Hz

INDEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK

TURBINE SPEED INPUT(1)

SH. No. TC0513

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

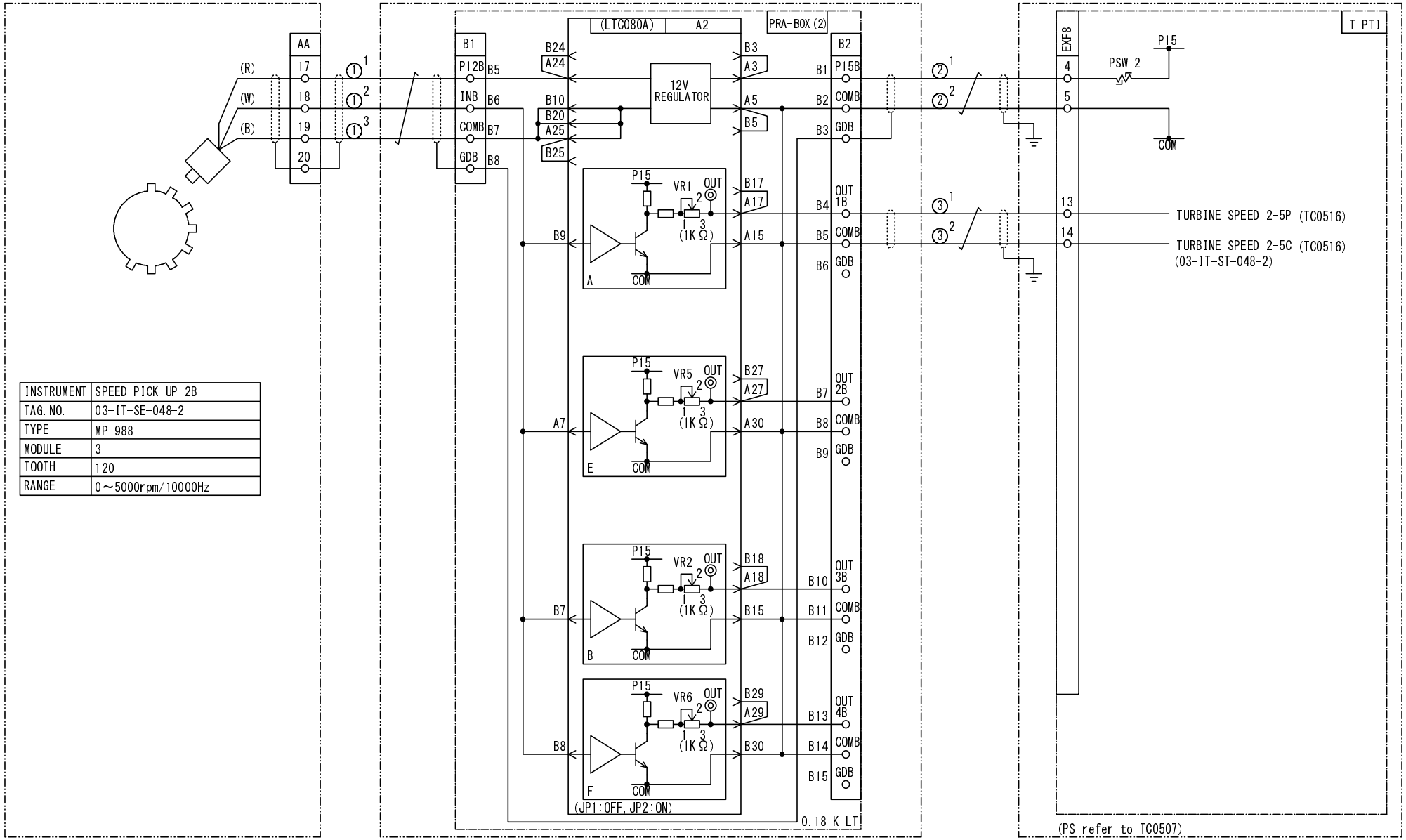
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TERMINAL BOX

TCS-5



INSTRUMENT	SPEED PICK UP 2B
TAG. NO.	03-IT-SE-048-2
TYPE	MP-988
MODULE	3
TOOTH	120
RANGE	0～5000rpm/10000Hz

INDEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK

TURBINE SPEED INPUT(2)

SH. No. TC0514

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

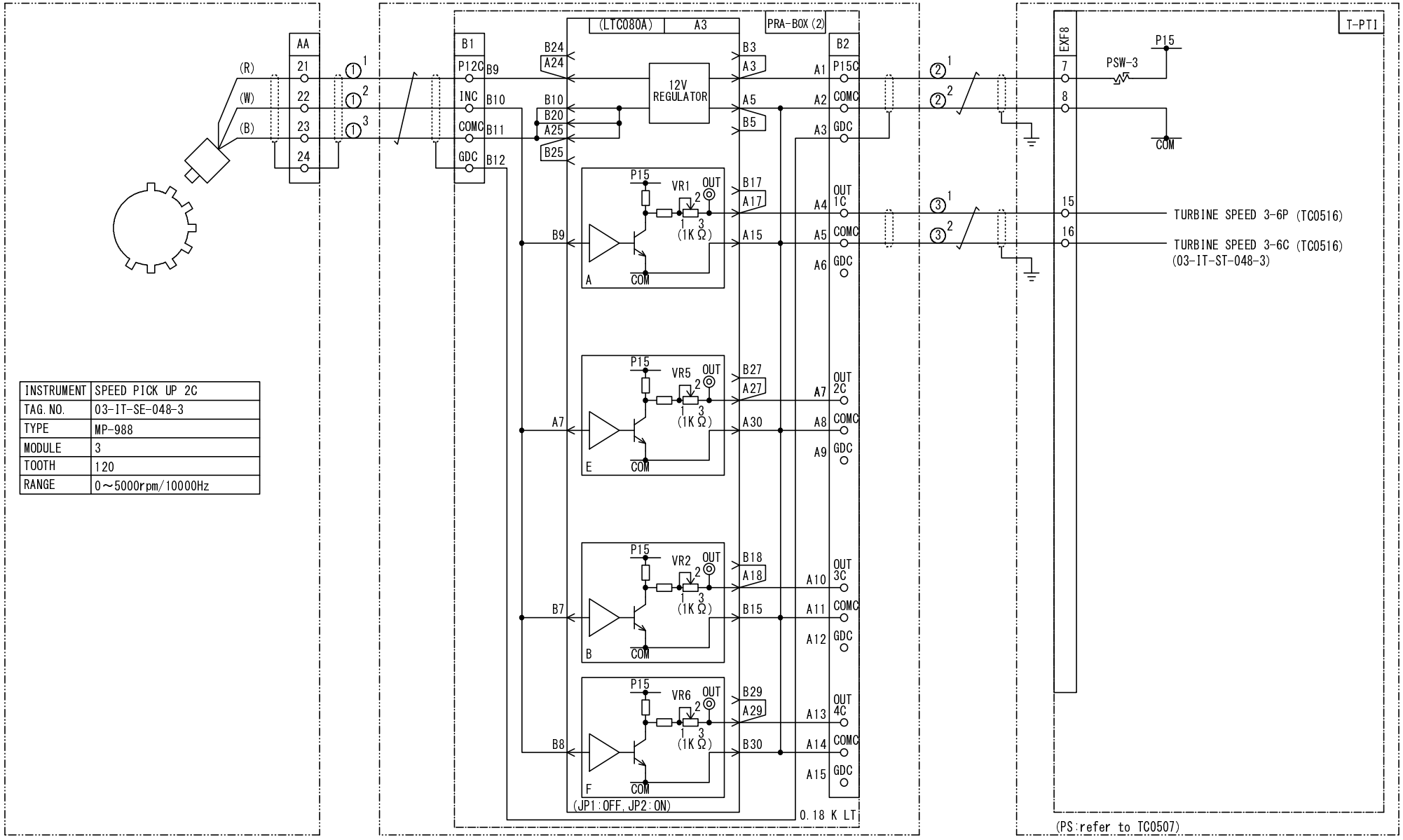
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TERMINAL BOX

TCS-5



INSTRUMENT	SPEED PICK UP 2C
TAG. NO.	03-IT-SE-048-3
TYPE	MP-988
MODULE	3
TOOTH	120
RANGE	0～5000rpm/10000Hz

INDEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK

TURBINE SPEED INPUT(3)

SH. No. TC0515

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

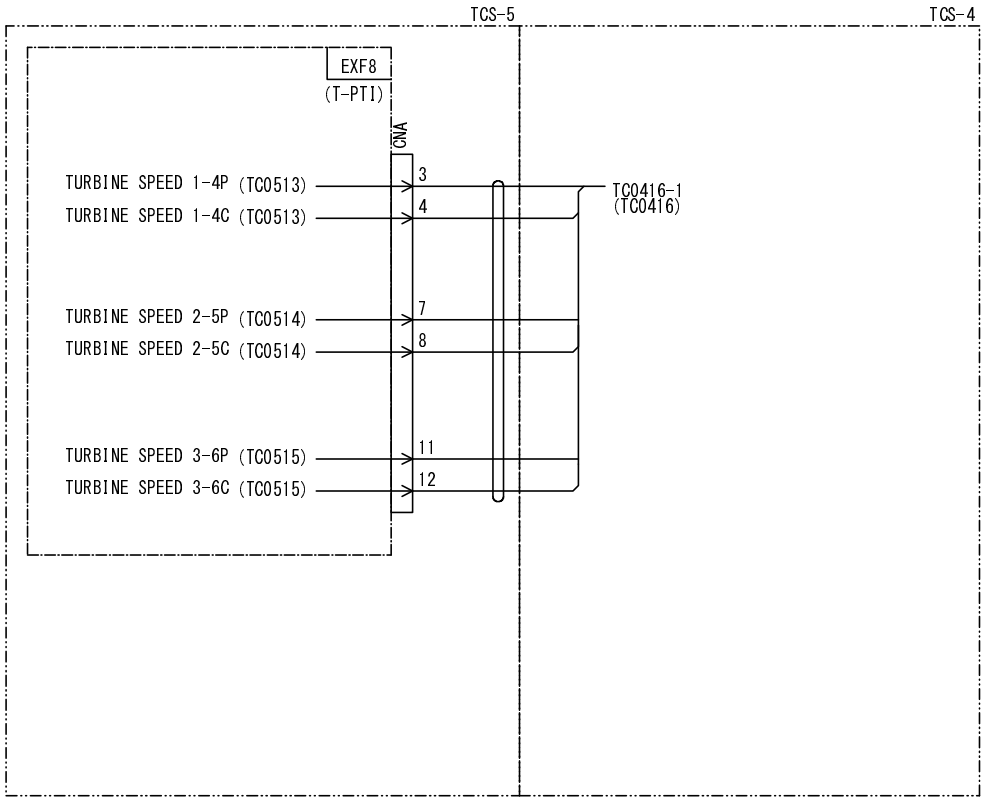
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0515 XSMM-TC

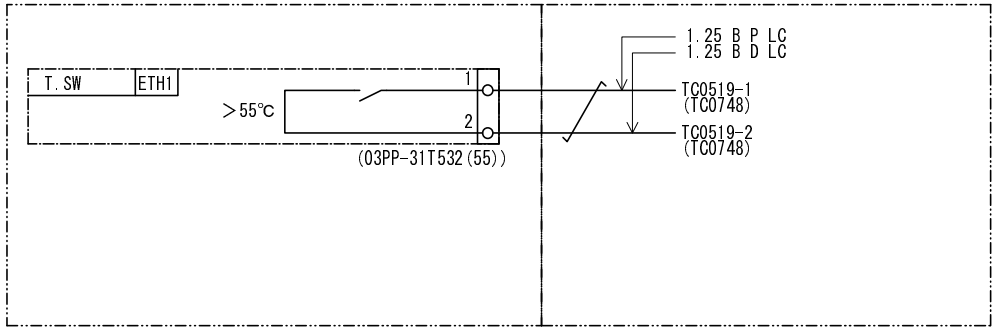


TURBINE SPEED INPUT(4)		SH. No.	TC0516
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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LOGIC DIAGRAM SH. NO.	TCA950

2016/07/28 TC0516 XSMM-TC

TCS-5 TCS-7



DIAGNOSIS CIRUCUIT		SH. No.	TC0519
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 176 / REV. 3

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SH. No.	TAG. No.	signal name	Terminal No.
TSI	NO.1 BEARING SHAFT VIBRATION VERY HIGH (A)		1 / 2
TSI	NO.2 BEARING SHAFT VIBRATION VERY HIGH (A)		1 / 2
TSI	NO.3 GEN. SHAFT VIBRATION VERY HIGH (A)		1 / 2
TSI	NO.4 GEN. SHAFT VIBRATION VERY HIGH (A)		1 / 2
CUSTOMER	GEN. TRANSF. PROTECTION OPERATED A		1 / 2
CUSTOMER	GEN. TRANSF. PROTECTION OPERATED B		1 / 2
DCS	BOILER MFT 1		1 / 2
GPP	86G-A GEN. LOCK OUT RELAY OPERATED		1 / 2

TCS-5

SDT-DI

(PS:refer to TC0508)

EXR1 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

P48 COM

CNA 4 20 18 16 14 12 10 8 6

CNB 4 20 18 16 14 12 10 8 6

CNC 4 20 18 16 14 12 10 8 6

TCS-3

TC0319-1 (TC0319)

TC0319-2 (TC0319)

TC0319-3 (TC0319)

INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		2/2	GPP	TCS-5			
⑦		2/2	DCS	TCS-5			
⑥		2/2	CUSTOMER	TCS-5			
⑤		2/2	CUSTOMER	TCS-5			
④		2/2	TSI	TCS-5			
③		2/2	TSI	TCS-5			
②		2/2	TSI	TCS-5			
①		2/2	TSI	TCS-5			

EXTERNAL INPUT TPC DI(1) SH. No. TC0520

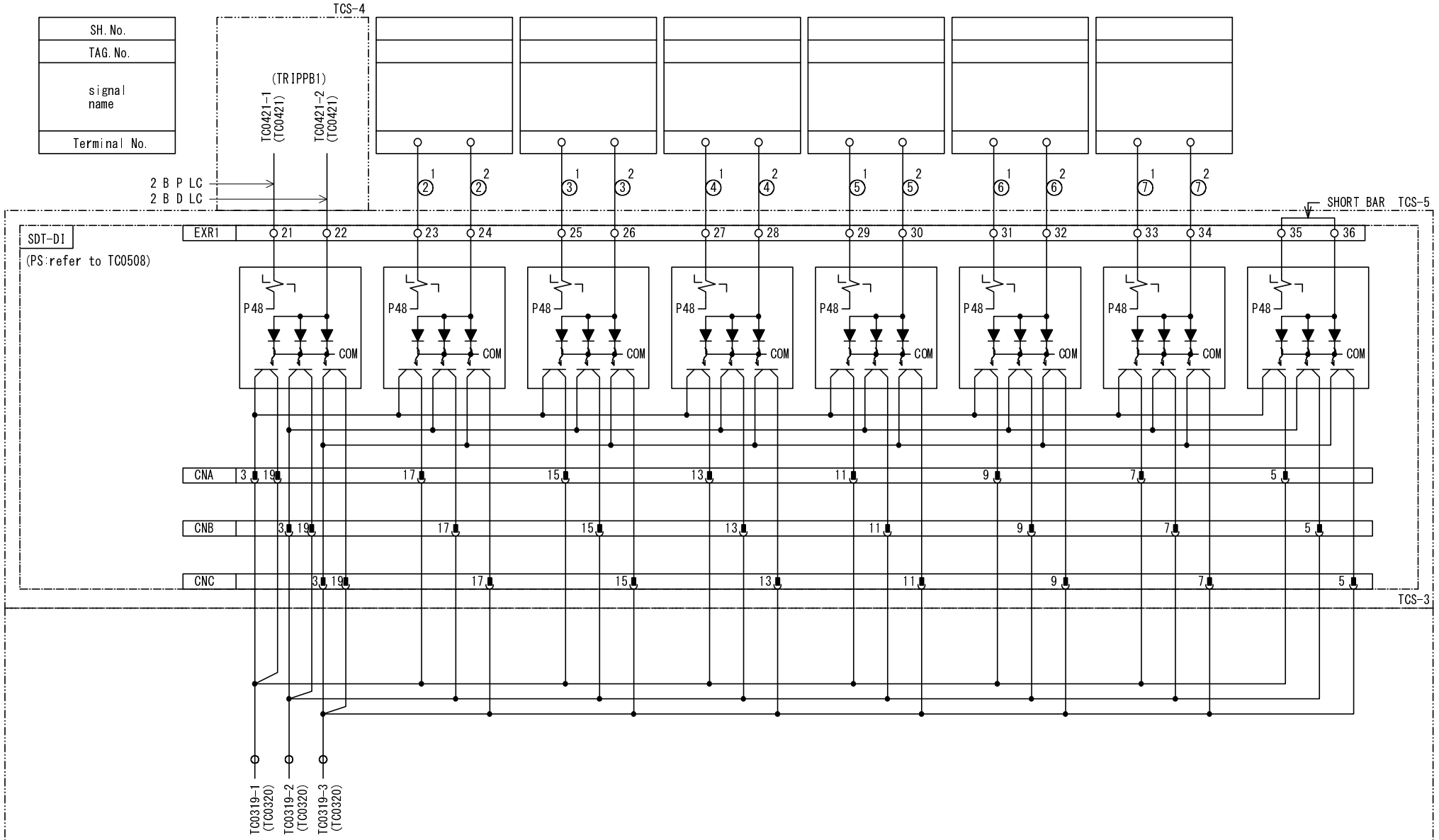
	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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○							
⑦		2/2		TCS-5			
⑥		2/2		TCS-5			
⑤		2/2		TCS-5			
④		2/2		TCS-5			
③		2/2		TCS-5			
②		2/2		TCS-5			
○							
INDEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK

EXTERNAL INPUT TPC DI(2) SH. No. TC0521

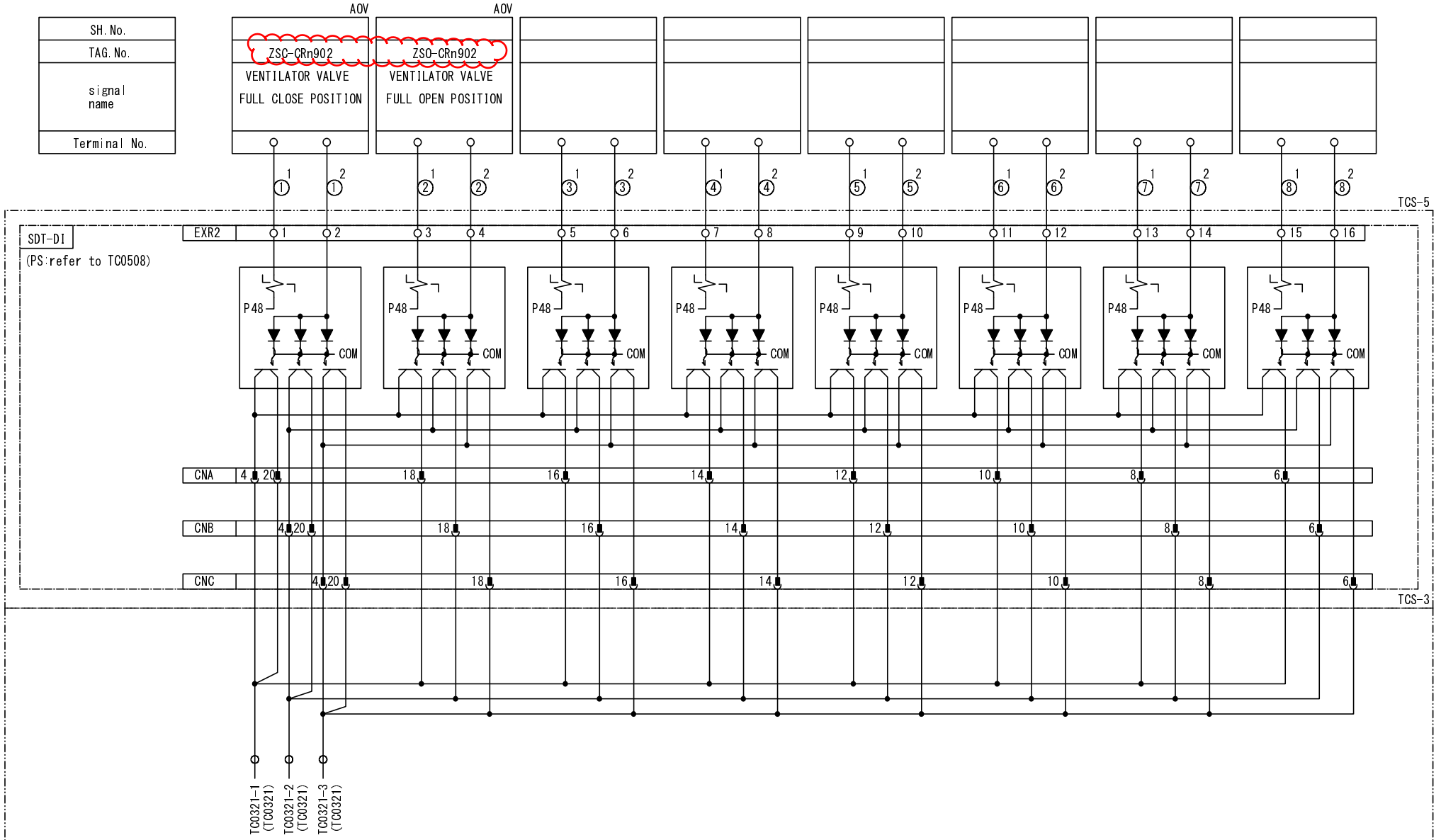
SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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2016/07/28 TC0521 XSMM-TC



INDEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK
⑧		2/2		TCS-5			
⑦		2/2		TCS-5			
⑥		2/2		TCS-5			
⑤		2/2		TCS-5			
④		2/2		TCS-5			
③		2/2		TCS-5			
②		2/2	AOV	TCS-5			
①		2/2	AOV	TCS-5			

EXTERNAL INPUT TPC DI(3)

SH. No. TC0522

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

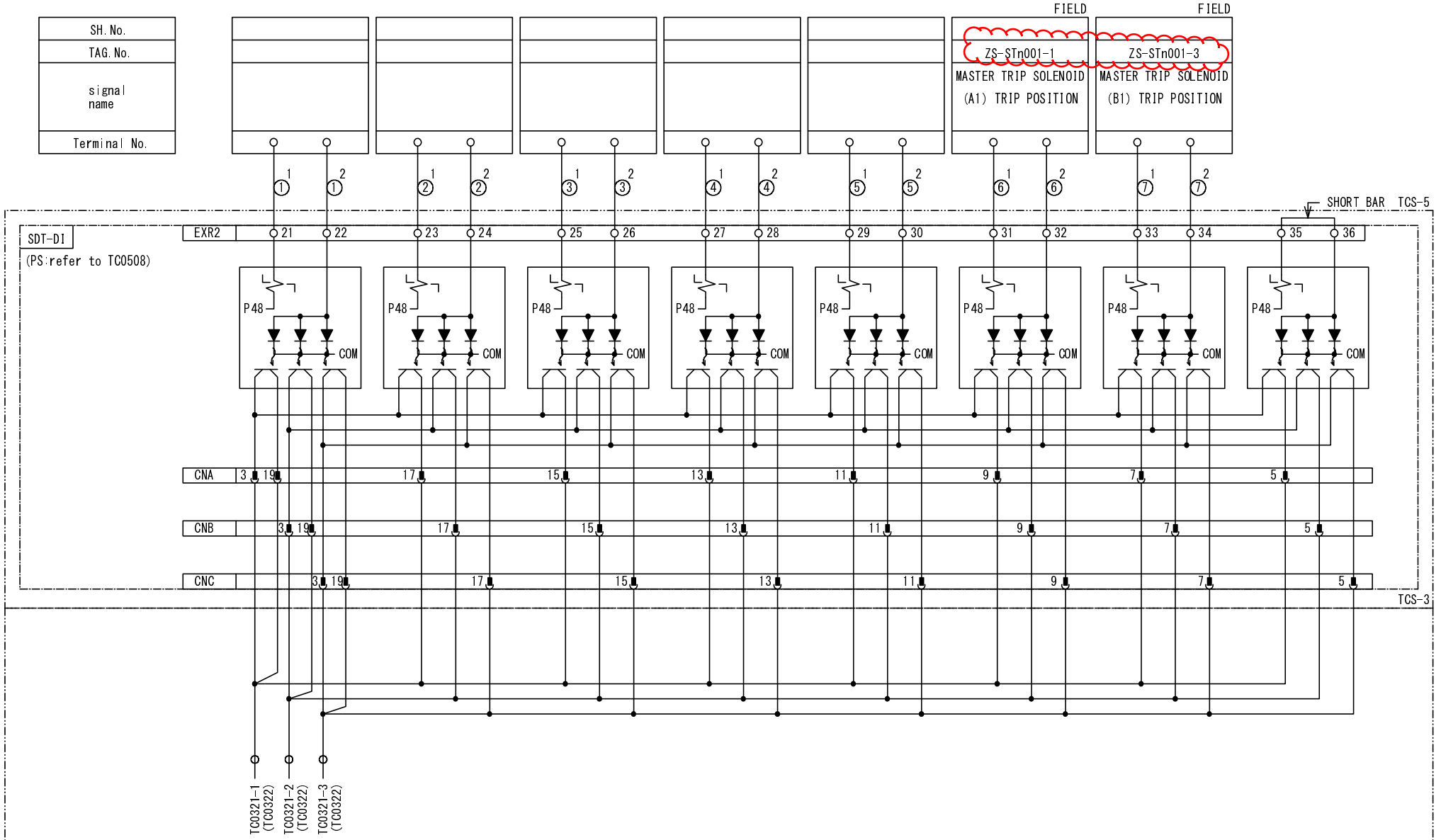
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INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑦		2/2		TCS-5			
⑥		2/2		TCS-5			
⑤		2/2		TCS-5			
④		2/2		TCS-5			
③		2/2		TCS-5			
②		2/2	FIELD	TCS-5			
①		2/2	FIELD	TCS-5			

EXTERNAL INPUT TPC DI(4)

SH. No. TC0523

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

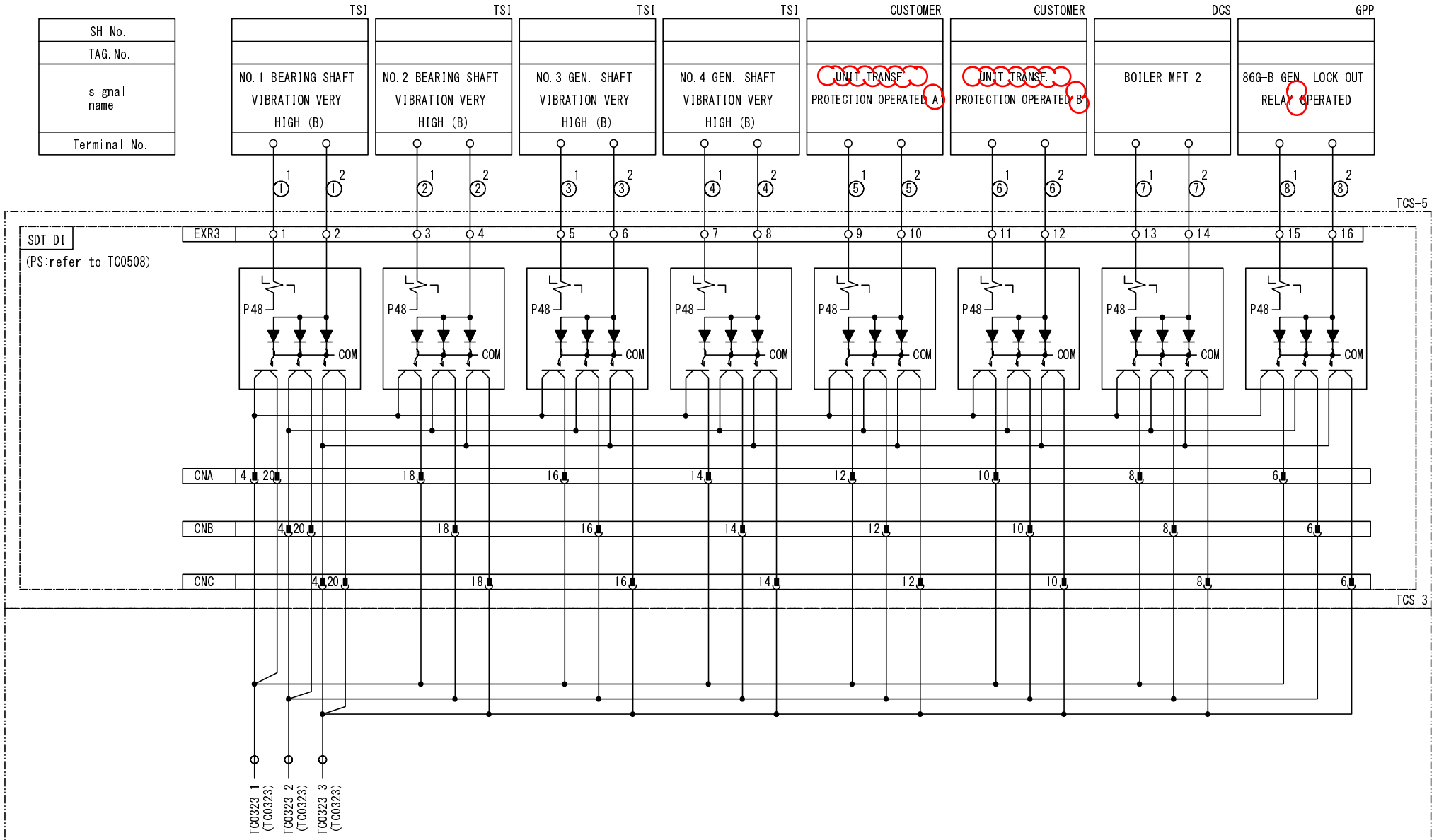
REGD. Hitachi, Ltd. Tokyo Japan

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INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		2/2	GPP	TCS-5			
⑦		2/2	DCS	TCS-5			
⑥		2/2	CUSTOMER	TCS-5			
⑤		2/2	CUSTOMER	TCS-5			
④		2/2	TSI	TCS-5			
③		2/2	TSI	TCS-5			
②		2/2	TSI	TCS-5			
①		2/2	TSI	TCS-5			

EXTERNAL INPUT TPC DI(5)

SH. No. TC0524

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

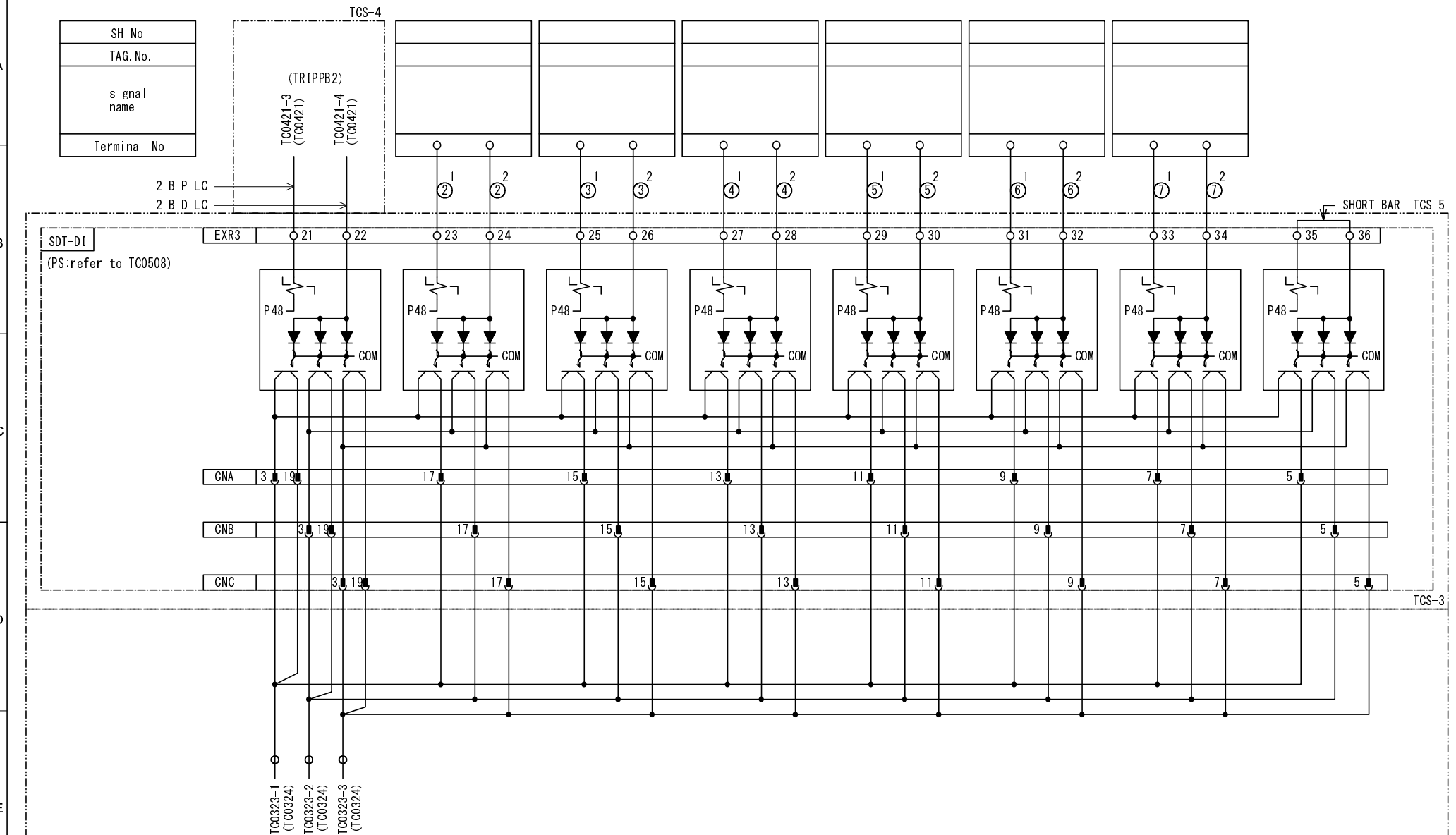
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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IN-DEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK
⑦		2/2		TCS-5			
⑥		2/2		TCS-5			
⑤		2/2		TCS-5			
④		2/2		TCS-5			
③		2/2		TCS-5			
②		2/2		TCS-5			

EXTERNAL INPUT TPC DI(6)

SH. No. TC0525

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

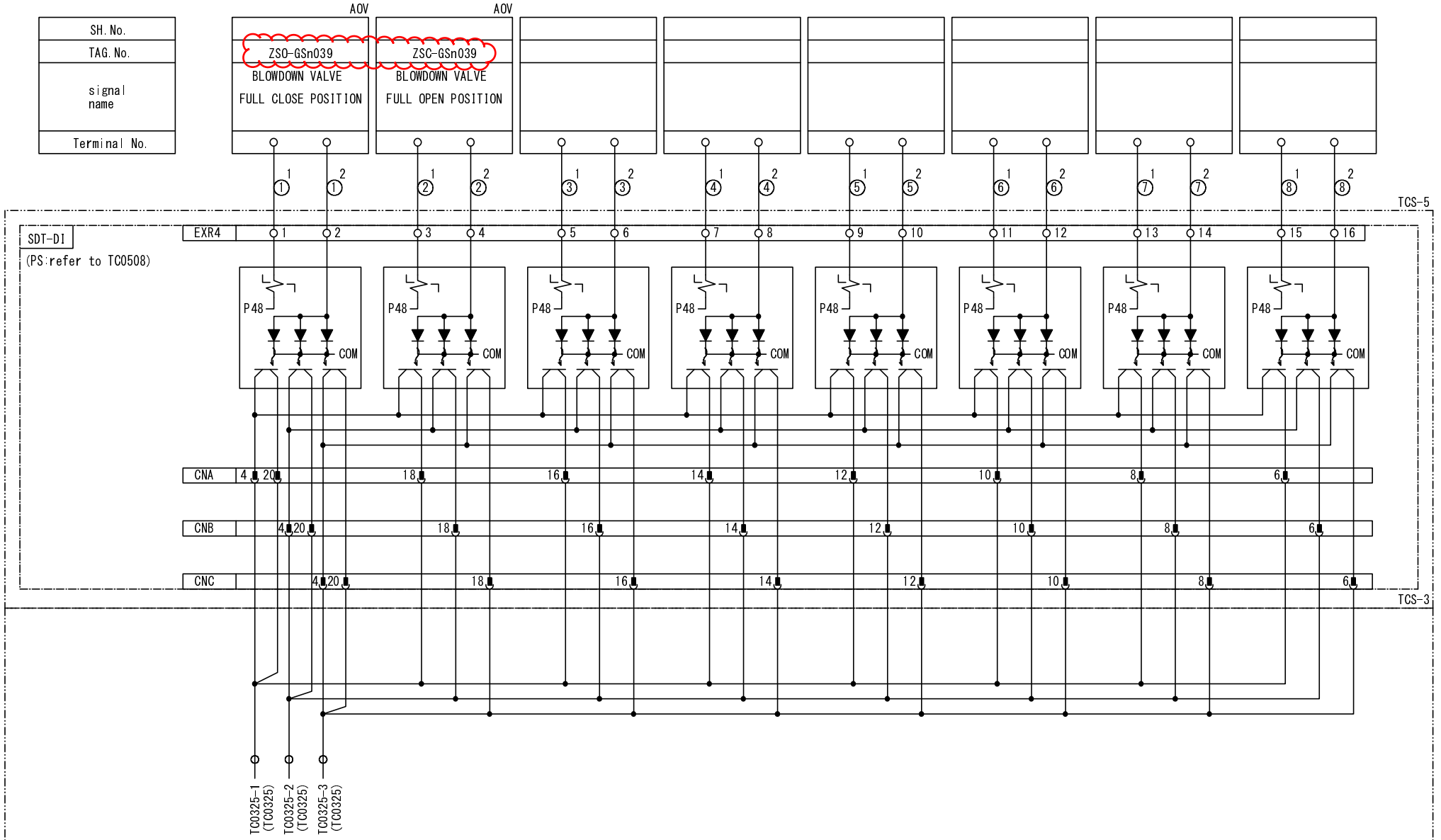
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2		TCS-5			
⑦		2/2		TCS-5			
⑥		2/2		TCS-5			
⑤		2/2		TCS-5			
④		2/2		TCS-5			
③		2/2		TCS-5			
②		2/2	AOV	TCS-5			
①		2/2	AOV	TCS-5			

EXTERNAL INPUT TPC DI(7)

SH. No. TC0526

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

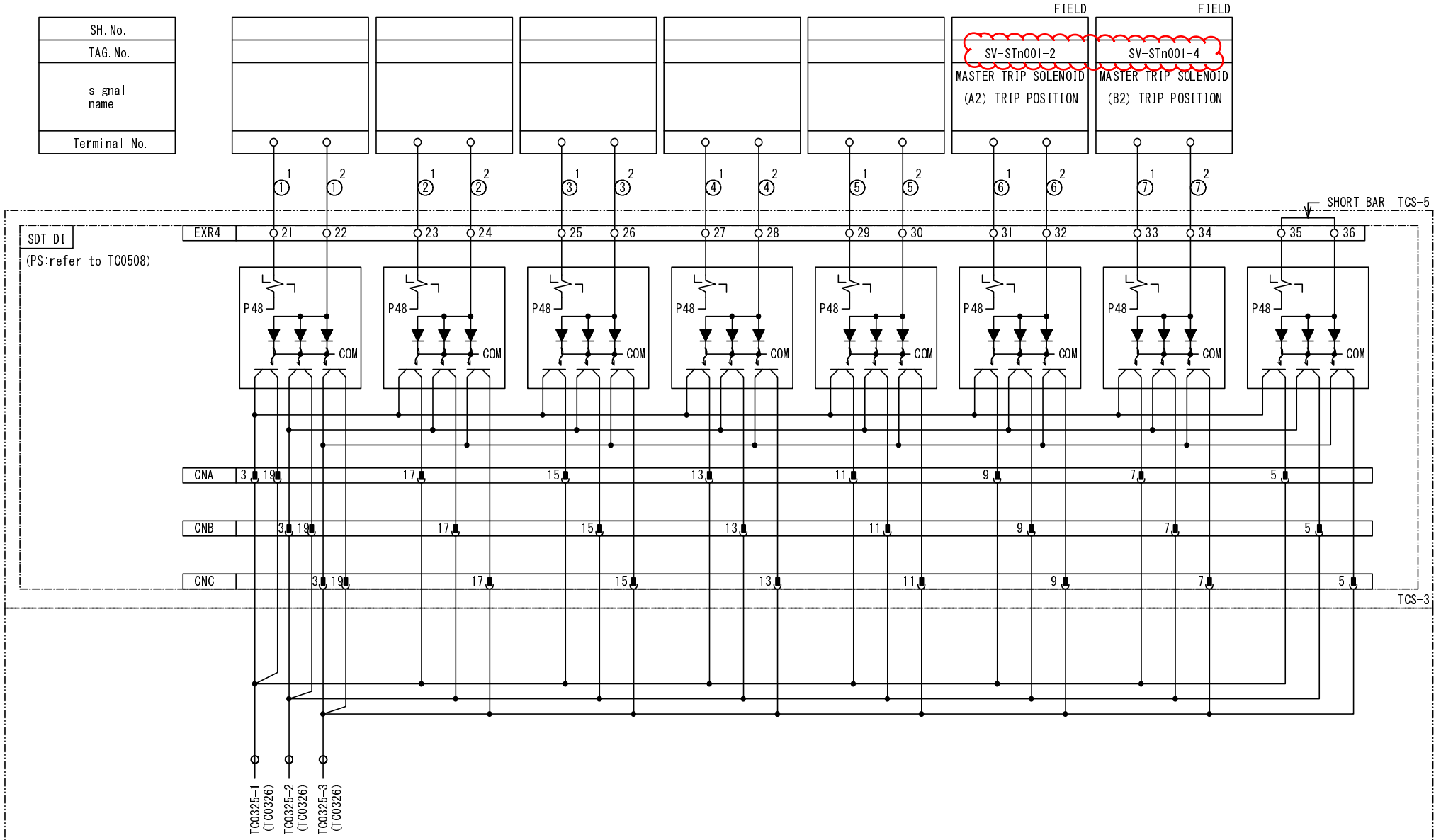
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DWG. No. 331DM07733

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⑦		2/2	FIELD	TCS-5				
⑥		2/2	FIELD	TCS-5				
⑤		2/2		TCS-5				
④		2/2		TCS-5				
③		2/2		TCS-5				
②		2/2		TCS-5				
①		2/2		TCS-5				
INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK	
			CABLE CONNECTION		RELATED SH. NO.			

EXTERNAL INPUT TPC DI(8)

SH. No. TC0527

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

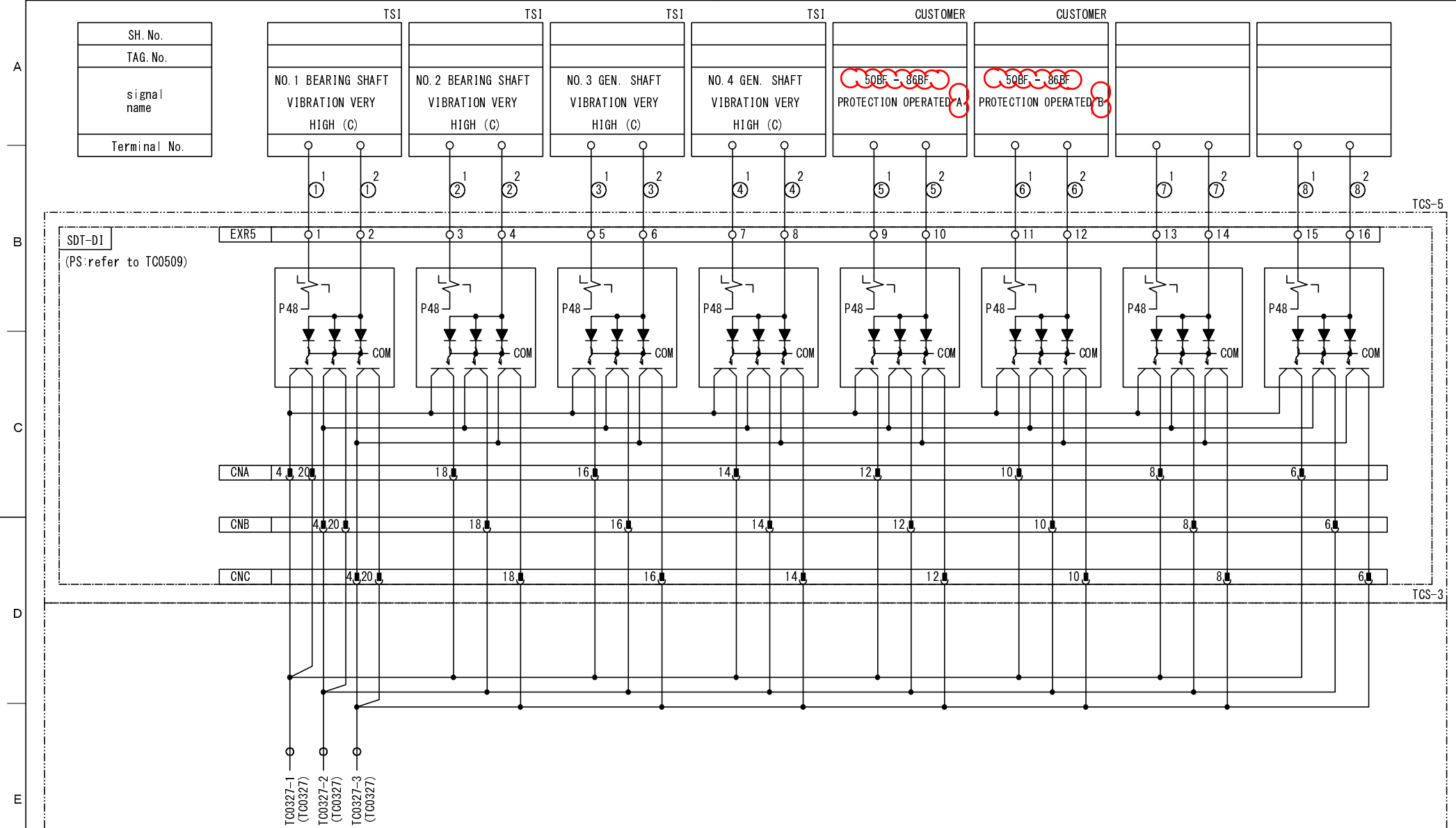
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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IN-DEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		2/2		TCS-5			
⑦		2/2		TCS-5			
⑥		2/2	CUSTOMER	TCS-5			
⑤		2/2	CUSTOMER	TCS-5			
④		2/2	TSI	TCS-5			
③		2/2	TSI	TCS-5			
②		2/2	TSI	TCS-5			
①		2/2	TSI	TCS-5			

EXTERNAL INPUT TPC DI(9)

SH. No. TC0528

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

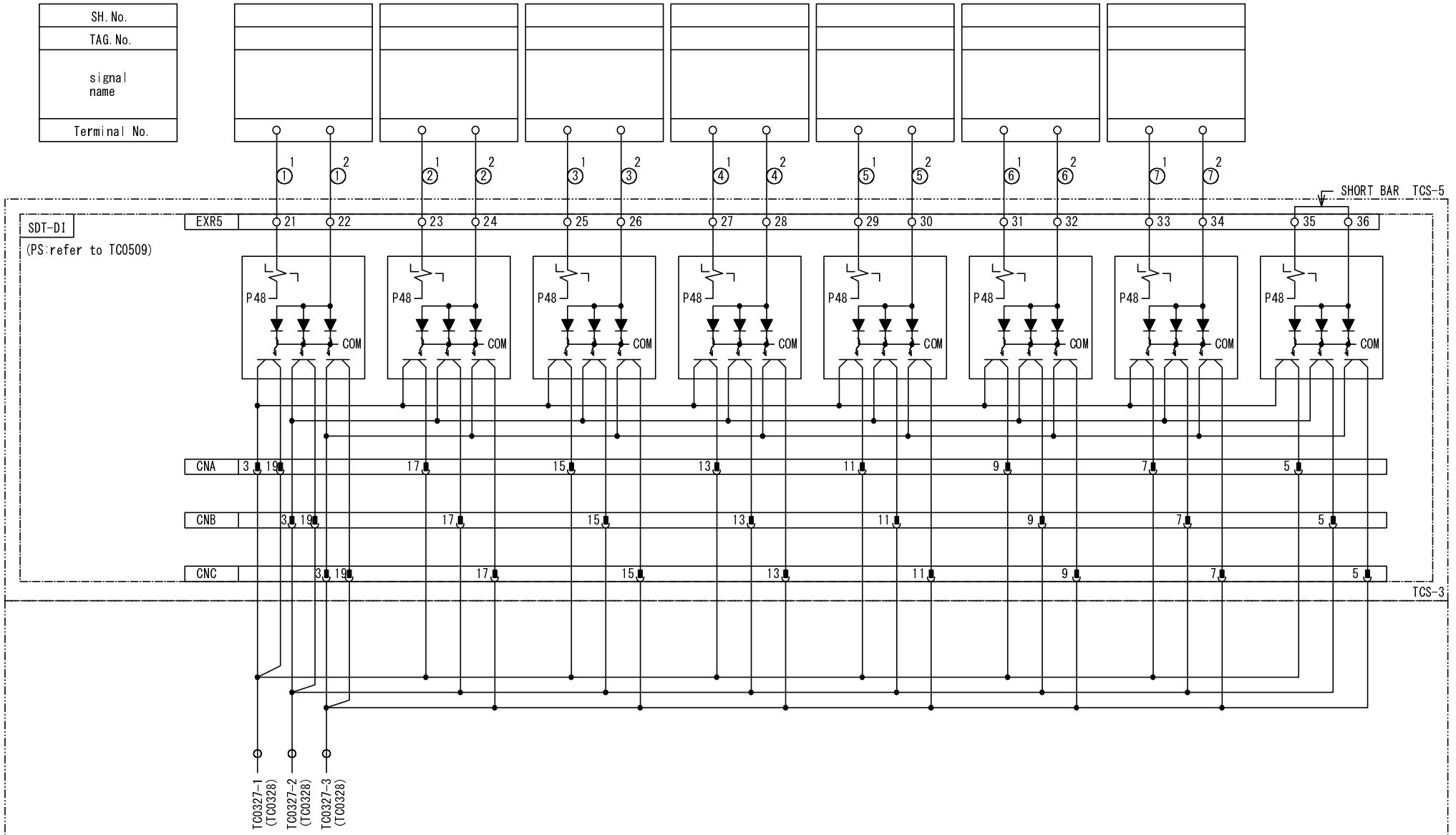
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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INDEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK
◯		2/2		TCS-5			
⑦		2/2		TCS-5			
⑥		2/2		TCS-5			
⑤		2/2		TCS-5			
④		2/2		TCS-5			
③		2/2		TCS-5			
②		2/2		TCS-5			
①		2/2		TCS-5			

EXTERNAL INPUT TPC DI(10)

SH. No. TC0529

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

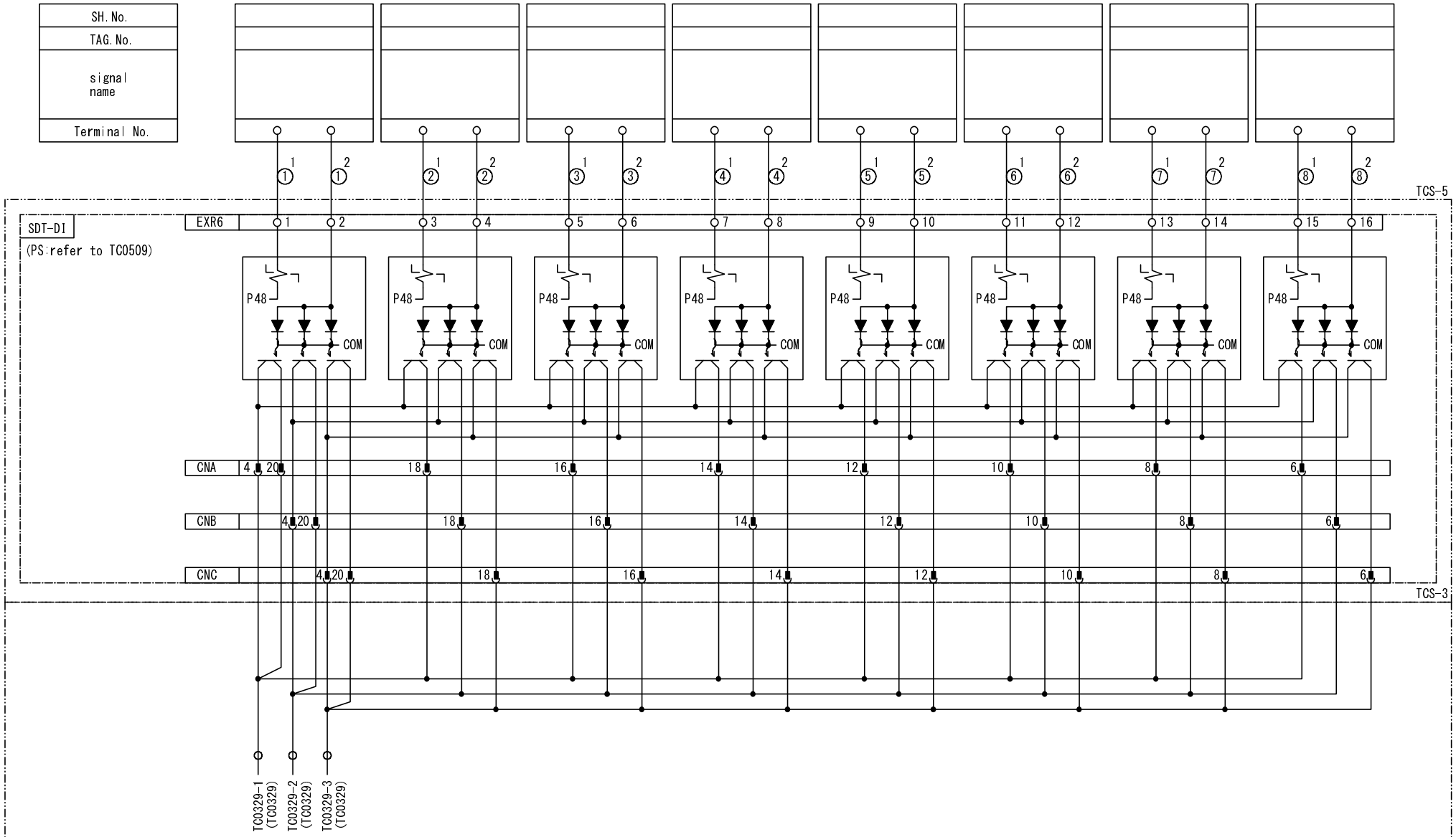
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DWG. No. 331DM07733

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INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2		TCS-5			
⑦		2/2		TCS-5			
⑥		2/2		TCS-5			
⑤		2/2		TCS-5			
④		2/2		TCS-5			
③		2/2		TCS-5			
②		2/2		TCS-5			
①		2/2		TCS-5			

EXTERNAL INPUT TPC DI(11)

SH. No. TC0530

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

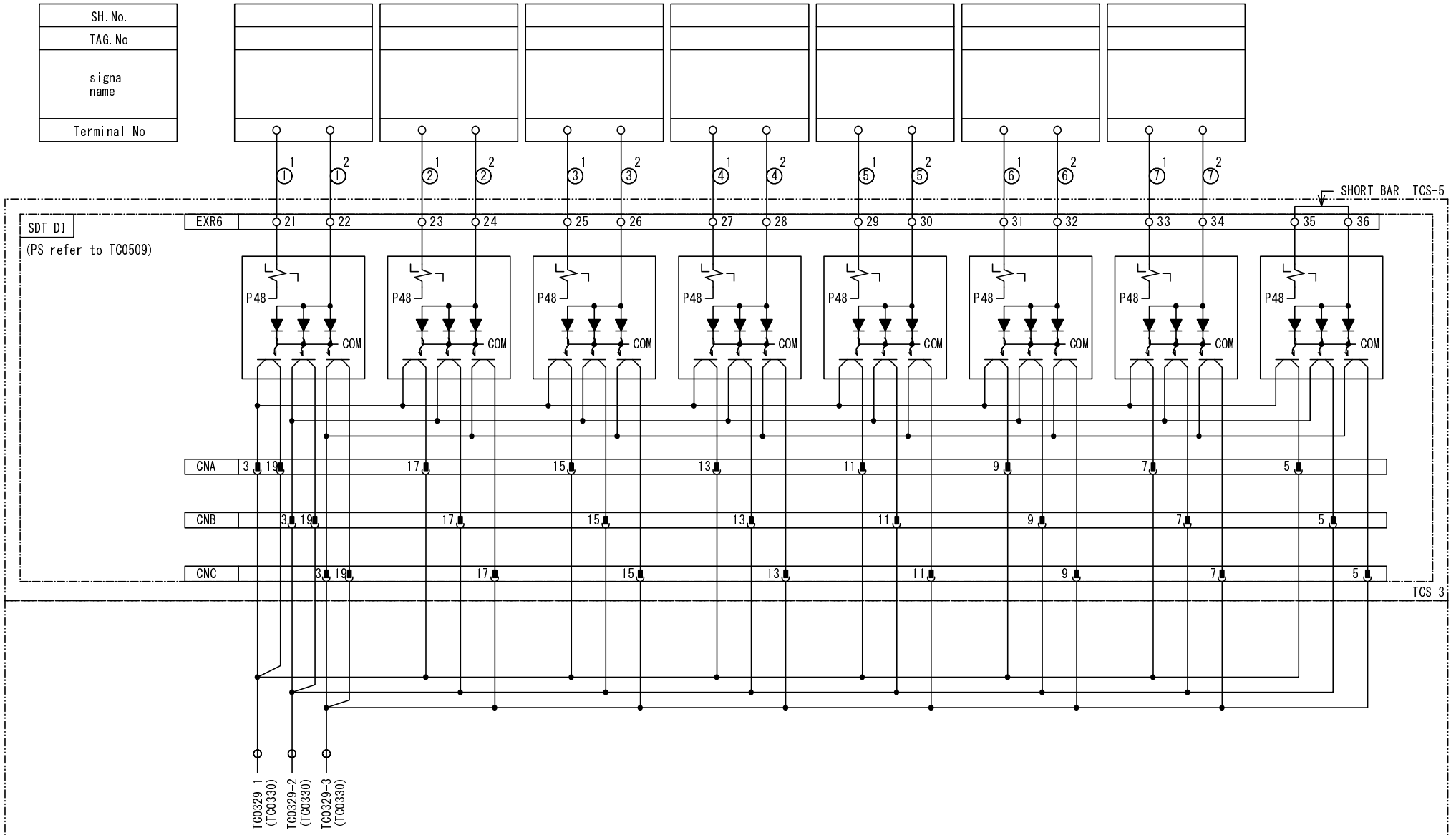
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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IN-DEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK
⑦		2/2		TCS-5			
⑥		2/2		TCS-5			
⑤		2/2		TCS-5			
④		2/2		TCS-5			
③		2/2		TCS-5			
②		2/2		TCS-5			
①		2/2		TCS-5			

EXTERNAL INPUT TPC DI(12)

SH. No. TC0531

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

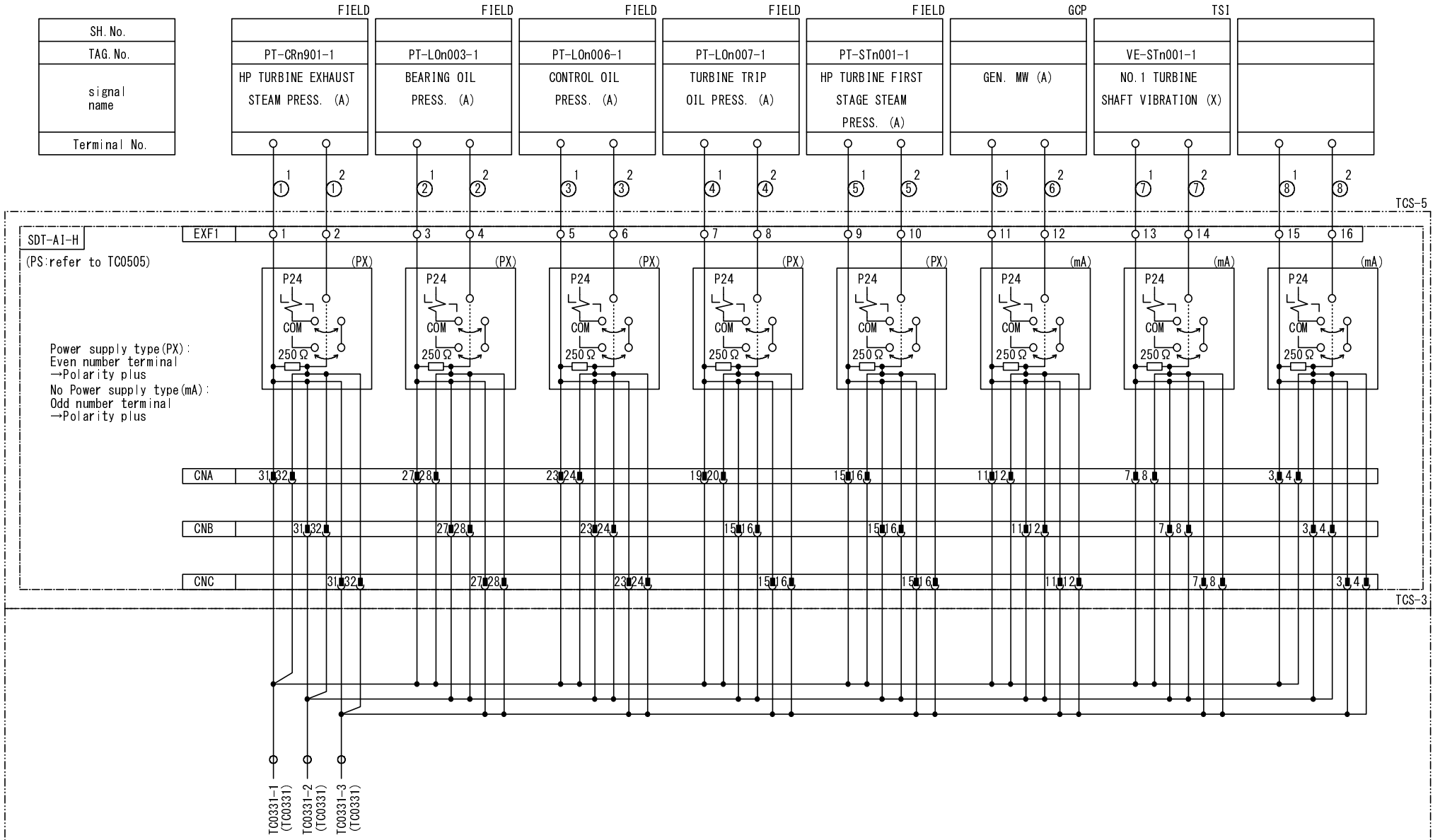
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DWG. No. 331DM07733

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INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2		TCS-5			
⑦		2/2	TSI	TCS-5			
⑥		2/2	GCP	TCS-5			
⑤		2/2	FIELD	TCS-5			
④		2/2	FIELD	TCS-5			
③		2/2	FIELD	TCS-5			
②		2/2	FIELD	TCS-5			
①		2/2	FIELD	TCS-5			

EXTERNAL INPUT TPC AI(1) ※

SH. No. TC0532

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

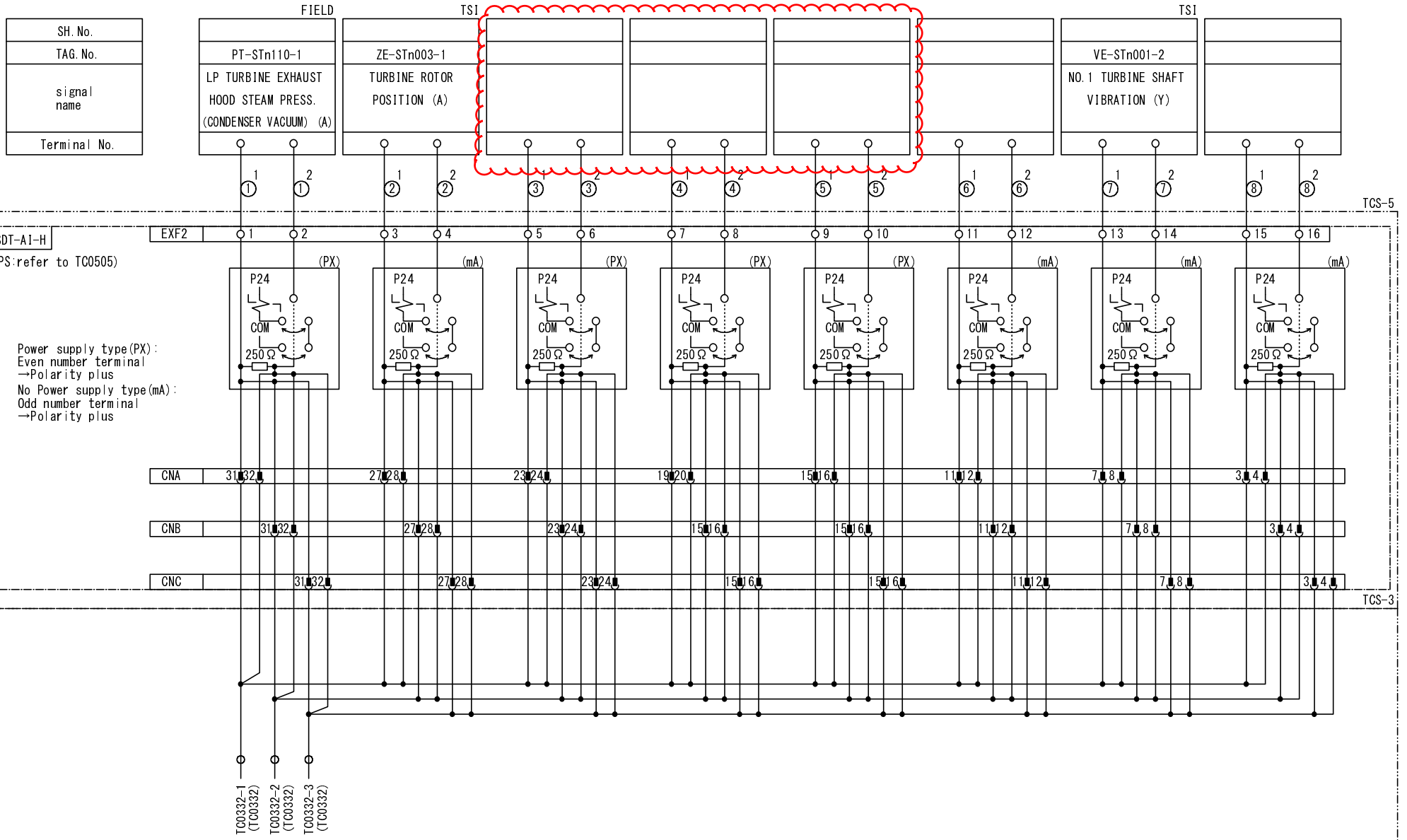
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		2/2		TCS-5			
⑦		2/2	TSI	TCS-5			
⑥		2/2		TCS-5			
⑤		2/2		TCS-5			
④		2/2		TCS-5			
③		2/2		TCS-5			
②		2/2	TSI	TCS-5			
①		2/2	FIELD	TCS-5			

EXTERNAL INPUT TPC AI(2) ※

SH. No. TC0533

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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SH. No.		FIELD	FIELD	FIELD	FIELD	FIELD	GCP	TSI	
TAG. No.		PT-CRn901-2	PT-LOn003-2	PT-LOn006-2	PT-LOn007-2	PT-STn001-2		VE-STn002-1	
signal name		HP TURBINE EXHAUST STEAM PRESS. (B)	BEARING OIL PRESS. (B)	CONTROL OIL PRESS. (B)	TURBINE TRIP OIL PRESS. (B)	HP TURBINE FIRST STAGE STEAM PRESS. (B)	GEN. MW (B)	NO.2 TURBINE SHAFT VIBRATION (X)	
Terminal No.		① ①	② ②	③ ③	④ ④	⑤ ⑤	⑥ ⑥	⑦ ⑦	⑧ ⑧

TCS-5

SDT-AI-H

(PS:refer to TC0505)

Power supply type(PX):
Even number terminal
→Polarity plus
No Power supply type(mA):
Odd number terminal
→Polarity plus

EXF3 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

(PX) (PX) (PX) (PX) (PX) (mA) (mA) (mA)

P24 COM 250Ω

CNA 31 32 27 28 23 24 19 20 15 16 11 12 7 8 3 4

CNB 31 32 27 28 23 24 15 16 15 16 11 12 7 8 3 4

CNC 31 32 27 28 23 24 15 16 15 16 11 12 7 8 3 4

TCS-3

TC0333-1 (TC0333)
TC0333-2 (TC0333)
TC0333-3 (TC0333)

INDEX	CABLE NO.	CORE NO.	CABLE CONNECTION FROM	CABLE CONNECTION TO	RELATED SH. NO. FROM	RELATED SH. NO. TO	REMARK
⑧		2/2		TCS-5			
⑦		2/2	TSI	TCS-5			
⑥		2/2	GCP	TCS-5			
⑤		2/2	FIELD	TCS-5			
④		2/2	FIELD	TCS-5			
③		2/2	FIELD	TCS-5			
②		2/2	FIELD	TCS-5			
①		2/2	FIELD	TCS-5			

EXTERNAL INPUT TPC AI(3) ※

SH. No. TC0534

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

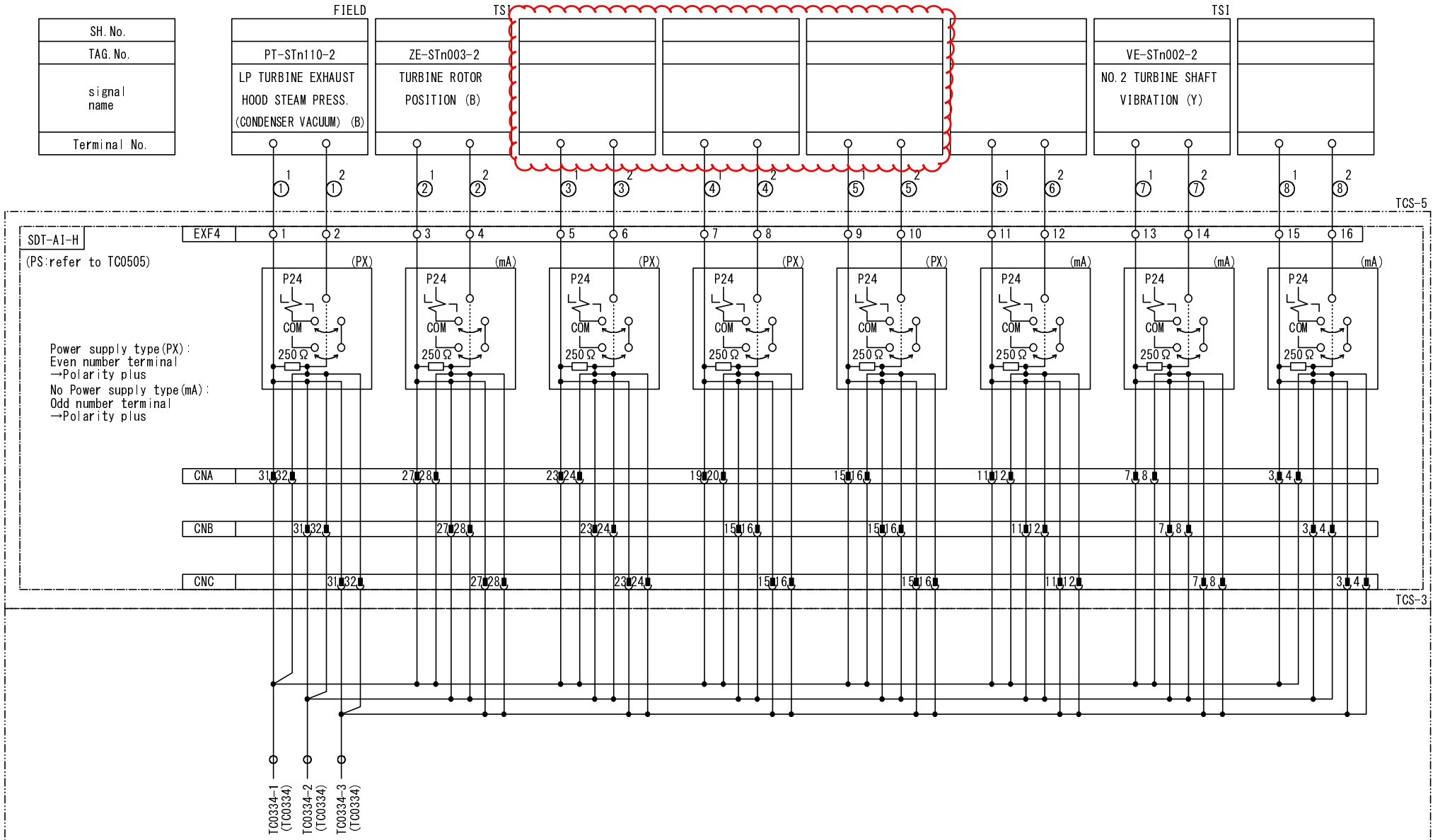
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		2/2		TCS-5			
⑦		2/2	TSI	TCS-5			
⑥		2/2		TCS-5			
⑤		2/2		TCS-5			
④		2/2		TCS-5			
③		2/2		TCS-5			
②		2/2	TSI	TCS-5			
①		2/2	FIELD	TCS-5			

EXTERNAL INPUT TPC AI(4) ※

SH. No. TC0535

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

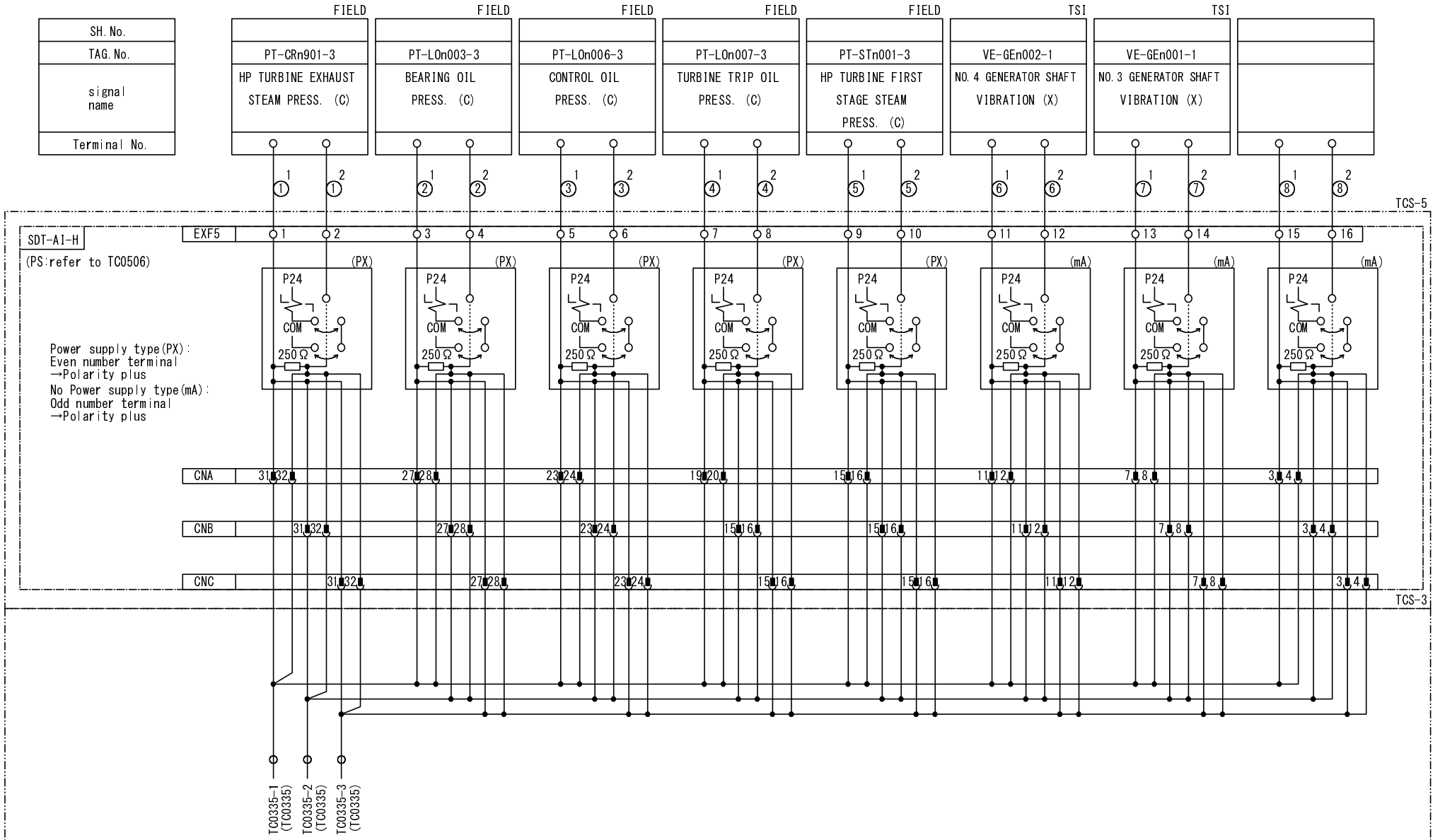
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	TSI	TCS-5			
⑦		2/2	TSI	TCS-5			
⑥		2/2	FIELD	TCS-5			
⑤		2/2	FIELD	TCS-5			
④		2/2	FIELD	TCS-5			
③		2/2	FIELD	TCS-5			
②		2/2	FIELD	TCS-5			
①		2/2	FIELD	TCS-5			

EXTERNAL INPUT TPC AI(5) ※

SH. No. TC0536

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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SH. No.		PT-STn110-3	ZE-STn003-3				VE-GEn002-2	VE-GEn001-2	
TAG. No.									
signal name		LP TURBINE EXHAUST HOOD STEAM PRESS. (CONDENSER VACUUM) (C)	TURBINE ROTOR POSITION (C)				NO.4 GENERATOR SHAFT VIBRATION (Y)	NO.3 GENERATOR SHAFT VIBRATION (Y)	
Terminal No.									
		FIELD	TSI				TSI	TSI	

TCS-5

SDT-AI-H

(PS:refer to TC0506)

EXF6

Power supply type(PX):
Even number terminal
→Polarity plus
No Power supply type(mA):
Odd number terminal
→Polarity plus

CNA

CNB

CNC

TCS-3

TC0336-1 (TC0336)
TC0336-2 (TC0336)
TC0336-3 (TC0336)

INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		2/2		TCS-5			
⑦		2/2	TSI	TCS-5			
⑥		2/2	TSI	TCS-5			
⑤		2/2		TCS-5			
④		2/2		TCS-5			
③		2/2		TCS-5			
②		2/2	TSI	TCS-5			
①		2/2	FIELD	TCS-5			

EXTERNAL INPUT TPC AI(6) ※

SH. No. TC0537

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

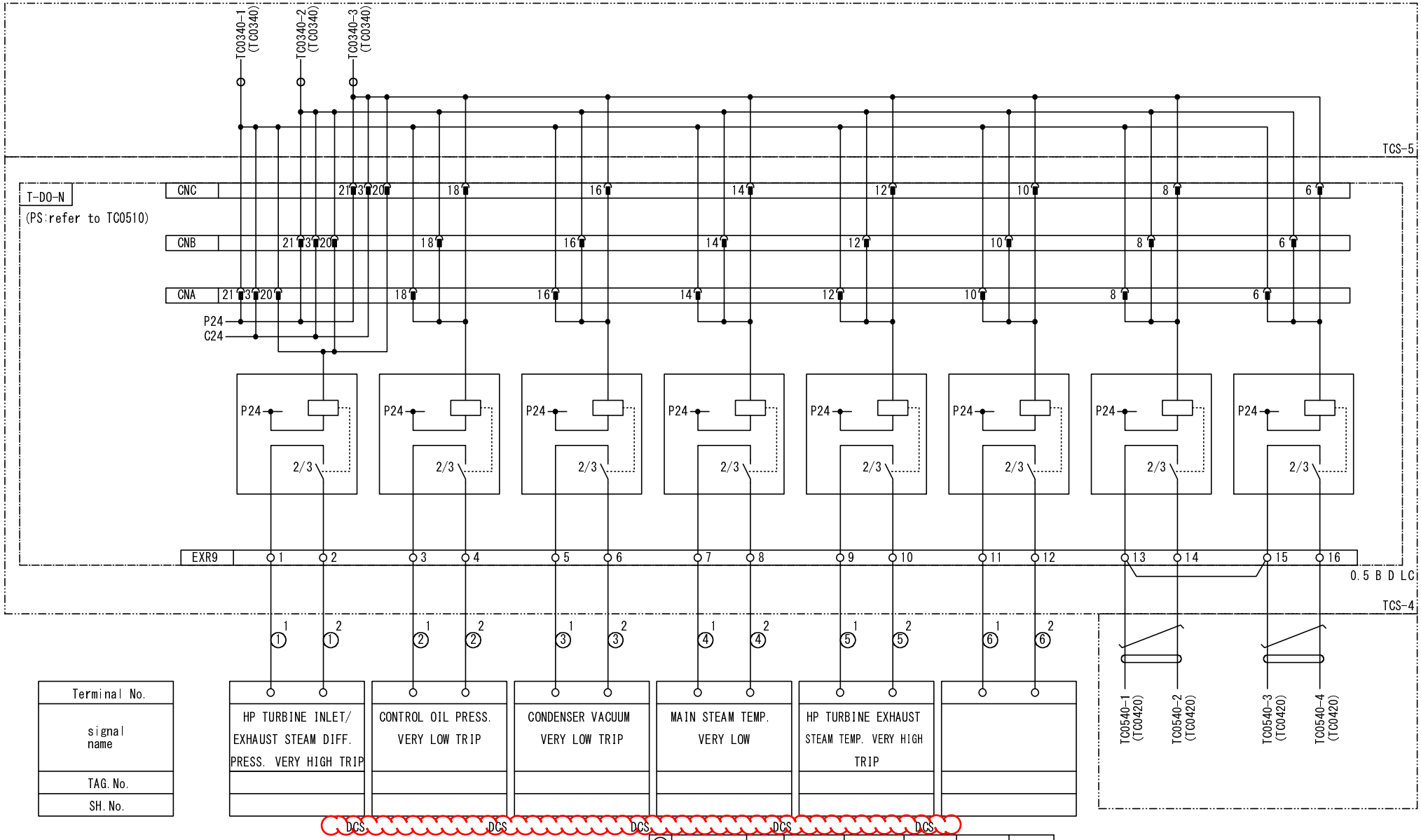
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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IN-DEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK
⑥		2/2	TCS-5	DCS			
⑤		2/2	TCS-5	DCS			
④		2/2	TCS-5	DCS			
③		2/2	TCS-5	DCS			
②		2/2	TCS-5	DCS			
①		2/2	TCS-5	DCS			

EXTERNAL OUTPUT TPC DO(1) ※

SH. No. TC0540

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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TCS-3

TCS-5

TC0340-1 (TC0341)
TC0340-2 (TC0341)
TC0340-3 (TC0341)

T-DO-N
(PS:refer to TC0510)

CNC 22 4 19 17 15 13 11 9 7 5
CNB 22 4 19 17 15 13 11 9 7 5
CNA 22 4 19 17 15 13 11 9 7 5

P24
C24

P24 2/3 ×8

EXR9 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36

1 2 ① ① ② ② ③ ③ ④ ④ ⑤ ⑤ ⑥ ⑥ ⑦ ⑦ ⑧ ⑧

Terminal No.							
signal name	TURBINE TRIP OIL PRESS. VERY LOW TRIP	HP TURBINE EXHAUST STEAM PRESS. VERY HIGH TRIP	LP TURBINE EXHAUST HOOD STEAM TEMP. VERY HIGH TRIP	BEARING OIL PRESS. VERY LOW	TURBINE BACK-UP OVERSPEED TRIP		
TAG. No.							
SH. No.							

DCS DCS DCS DCS DCS DCS DCS DCS

INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	TCS-5	DCS			
⑦		2/2	TCS-5	DCS			
⑥		2/2	TCS-5	DCS			
⑤		2/2	TCS-5	DCS			
④		2/2	TCS-5	DCS			
③		2/2	TCS-5	DCS			
②		2/2	TCS-5	DCS			
①		2/2	TCS-5	DCS			

EXTERNAL OUTPUT TPC DO(2) ※

SH. No. TC0541

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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TCS-3

TCS-5

TC0342-1 (TC0342)
TC0342-2 (TC0342)
TC0342-3 (TC0342)

T-DO-N
(PS:refer to TC0510)

CNC 21 3 20 18 16 14 12 10 8 6
CNB 21 3 20 18 16 14 12 10 8 6
CNA 21 3 20 18 16 14 12 10 8 6

P24
C24

P24 2/3 ×8

EXR10 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

1 2 ×8

Terminal No.								
signal name	TURBINE OVERSPEED TRIP	TURBINE MANUAL TRIP		THRUST BEARING FAILURE (FRONT) TRIP	THRUST BEARING FAILURE (REAR) TRIP	NO.1 TURBINE SHAFT VIBRATION VERY HIGH	NO.2 TURBINE SHAFT VIBRATION VERY HIGH	NO.3 GEN. SHAFT VIBRATION VERY HIGH
TAG.No.								
SH.No.								
	DCS DCS			DCS DCS	DCS DCS	DCS DCS	DCS DCS	DCS DCS

INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	TCS-5	DCS			
⑦		2/2	TCS-5	DCS			
⑥		2/2	TCS-5	DCS			
⑤		2/2	TCS-5	DCS			
④		2/2	TCS-5	DCS			
③		2/2	TCS-5				
②		2/2	TCS-5	DCS			
①		2/2	TCS-5	DCS			

EXTERNAL OUTPUT TPC DO(3) ※

SH. No. TC0542

	SIGNATURE	DATE
DWN.	(S.Kikuchi)	2014-04-18
CHKD.	T.Kawata	2014-04-18
APPD.	Y.Fukushima	2014-04-18

TITLE
ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

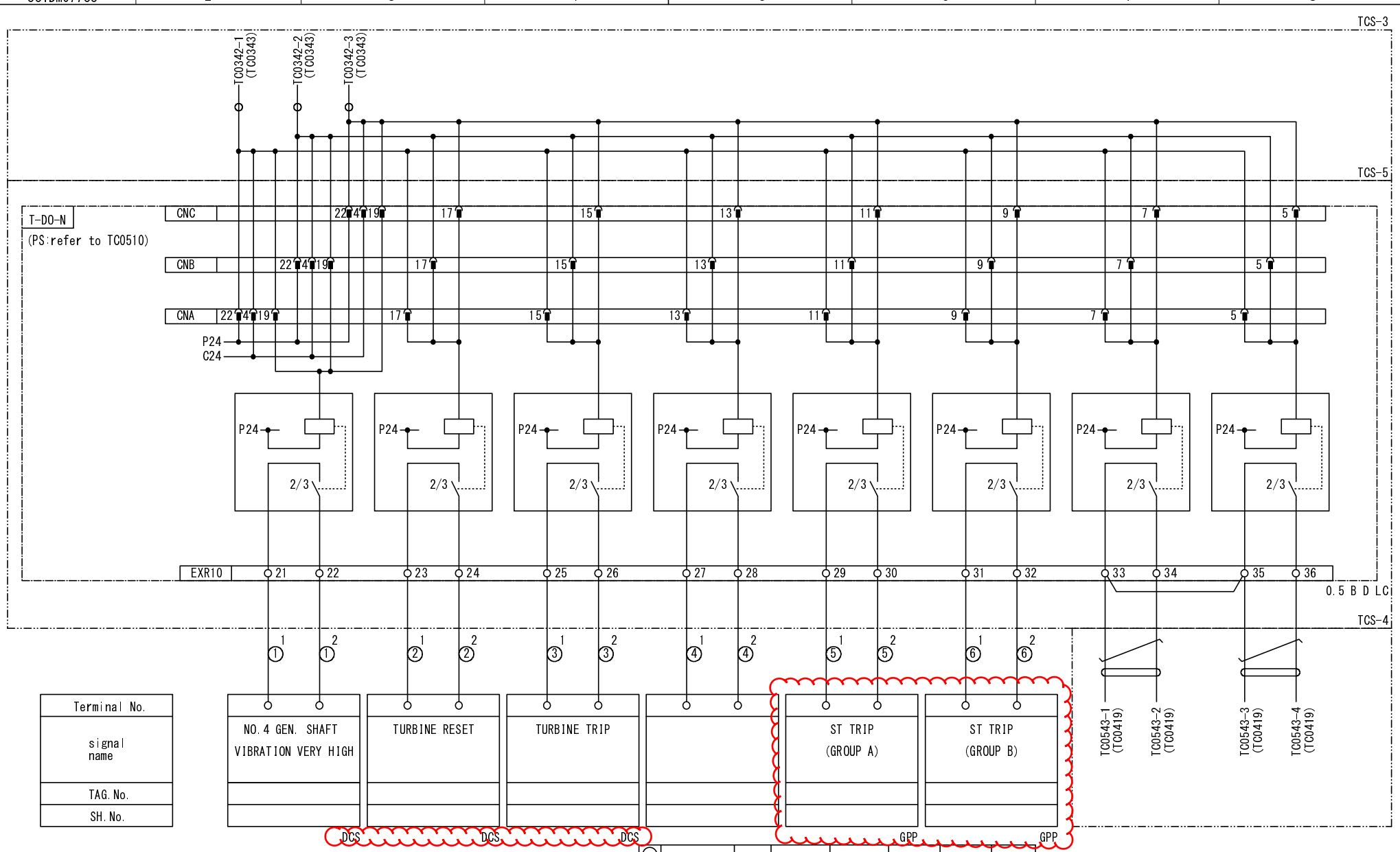
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DWG. No. 331DM07733

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INDEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK
⑥		2/2	TCS-5	DCS			
⑤		2/2	TCS-5	DCS			
④		2/2	TCS-5				
③		2/2	TCS-5	DCS			
②		2/2	TCS-5	DCS			
①		2/2	TCS-5	DCS			

EXTERNAL OUTPUT TPC DO(4) ※

SH. No. TC0543

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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TCS-3

TCS-5

TC0344-1 (TC0344)
TC0344-2 (TC0344)
TC0344-3 (TC0344)

T-DO-P
(PS:refer to TC0511)

CNC 21 3 20 18 16 14 12 10 8 6
CNB 21 3 20 18 16 14 12 10 8 6
CNA 21 3 20 18 16 14 12 10 8 6

P24
C24

OUTPUT VOLTAGE DC125V

P24 P(A) N(C) 2/3

EXR11 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Terminal No.								
signal name	VENTILATOR VALVE SOLENOID VALVE ENERGIZE COMMAND	BLOWDOWN VALVE ENERGIZE COMMAND						
TAG. No.	SV-CRn902	SV-GSn039						
SH. No.								

SV SV

INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	TCS-5				
⑦		2/2	TCS-5				
⑥		2/2	TCS-5				
⑤		2/2	TCS-5				
④		2/2	TCS-5				
③		2/2	TCS-5				
②		2/2	TCS-5	SV			
①		2/2	TCS-5	SV			

EXTERNAL OUTPUT TPC DO(5) ※

SH. No. TC0544

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

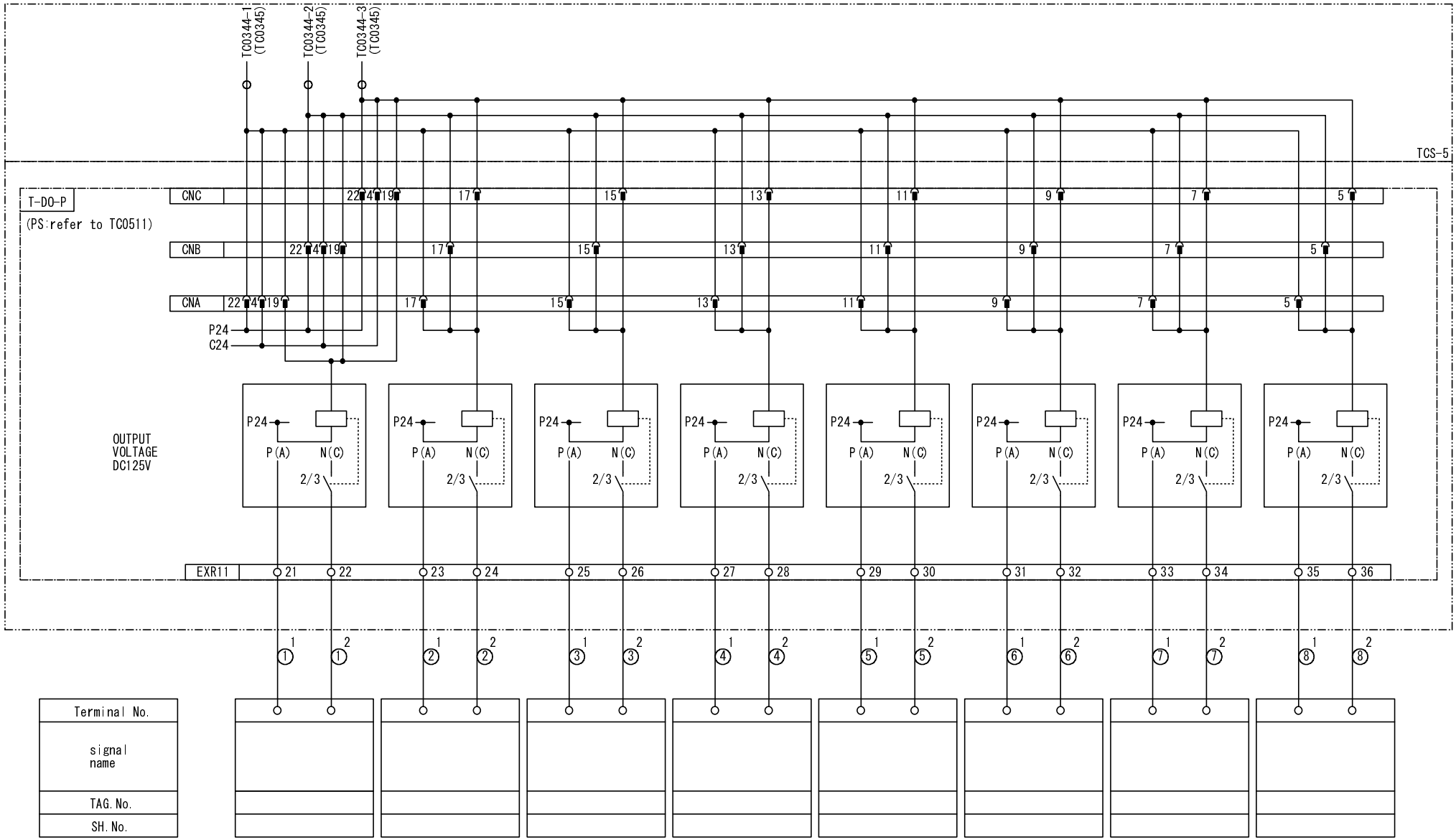
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	TCS-5				
⑦		2/2	TCS-5				
⑥		2/2	TCS-5				
⑤		2/2	TCS-5				
④		2/2	TCS-5				
③		2/2	TCS-5				
②		2/2	TCS-5				
①		2/2	TCS-5				

EXTERNAL OUTPUT TPC DO(6)

SH. No. TC0545

SIGNATURE		DATE
DWN.	(S.Kikuchi)	2014-04-18
CHKD.	T.Kawata	2014-04-18
APPD.	Y.Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi,Ltd. Tokyo Japan

DWG. No. 331DM07733

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TCS-3

TCS-5

TC0346-1 (TC0346)
TC0346-2 (TC0346)
TC0346-3 (TC0346)

T-DO-P
(PS:refer to TC0512)

CNC 21 3 20 18 16 14 12 10 8 6
CNB 21 3 20 18 16 14 12 10 8 6
CNA 21 3 20 18 16 14 12 10 8 6

P24
C24

OUTPUT VOLTAGE AC120V

P24 P(A) N(C) 2/3

EXR12 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Terminal No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
signal name	MSV (A) FAST ACTING SOLENOID VALVE		CV (A) FAST ACTING SOLENOID VALVE		ICV (A) FAST ACTING SOLENOID VALVE		RSV (A) FAST ACTING SOLENOID VALVE		RSV (A) TEST SOLENOID VALVE		MSV (B) FAST ACTING SOLENOID VALVE		CV (B) FAST ACTING SOLENOID VALVE		ICV (B) FAST ACTING SOLENOID VALVE	
TAG. No.	SV-STn903-A1		SV-STn904-A1		SV-STn905-A1		SV-STn906-A1		SV-STn906-A2		SV-STn903-B1		SV-STn904-B1		SV-STn905-B1	
SH. No.	SV		SV		SV		SV		SV		SV		SV		SV	

INDEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK
⑧		2/2	TCS-5	SV			
⑦		2/2	TCS-5	SV			
⑥		2/2	TCS-5	SV			
⑤		2/2	TCS-5	SV			
④		2/2	TCS-5	SV			
③		2/2	TCS-5	SV			
②		2/2	TCS-5	SV			
①		2/2	TCS-5	SV			

EXTERNAL OUTPUT TPC DO(7) ※

SH. No. TC0546

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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TCS-3

TC0346-1 (TC0347)
TC0346-2 (TC0347)
TC0346-3 (TC0347)

TCS-5

T-DO-P
(PS:refer to TC0512)

CNC 22 4 19 17 15 13 11 9 7 5
CNB 22 4 19 17 15 13 11 9 7 5
CNA 22 4 19 17 15 13 11 9 7 5

P24
C24

OUTPUT VOLTAGE AC120V

P24 P(A) N(C) 2/3

EXR12 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36

Terminal No.	① 1	① 2	② 1	② 2	③ 1	③ 2	④ 1	④ 2	⑤ 1	⑤ 2	⑥ 1	⑥ 2	⑦ 1	⑦ 2	⑧ 1	⑧ 2
signal name	RSV (B) FAST ACTING SOLENOID VALVE		RSV (B) TEST SOLENOID VALVE													
TAG. No.	SV-STn906-B1		SV-STn906-B2													
SH. No.																

SV SV

INDEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK
⑧		2/2	TCS-5				
⑦		2/2	TCS-5				
⑥		2/2	TCS-5				
⑤		2/2	TCS-5				
④		2/2	TCS-5				
③		2/2	TCS-5				
②		2/2	TCS-5	SV			
①		2/2	TCS-5	SV			

EXTERNAL OUTPUT TPC DO(8)

SH. No. TC0547

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

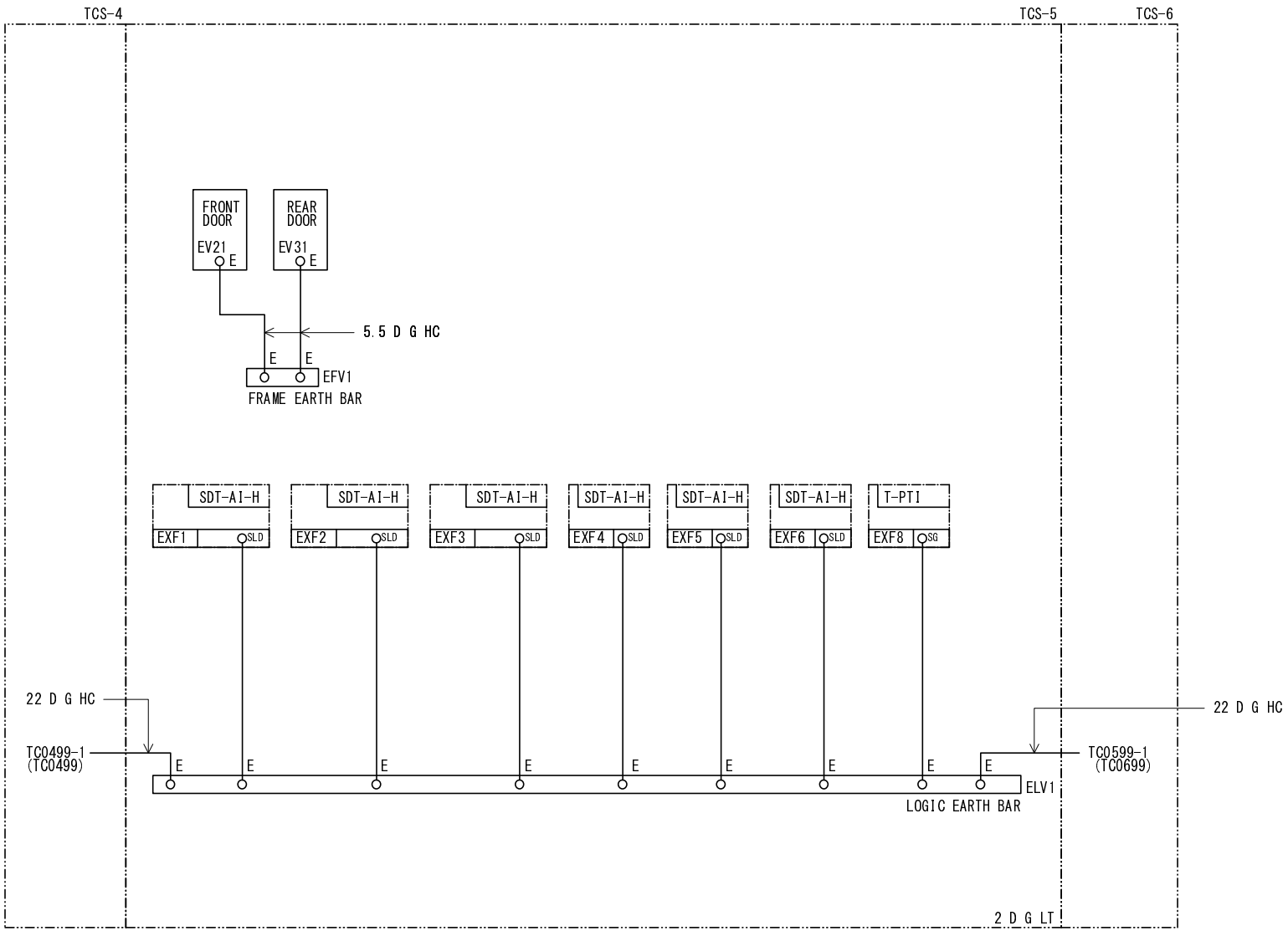
DWG. No. 331DM07733

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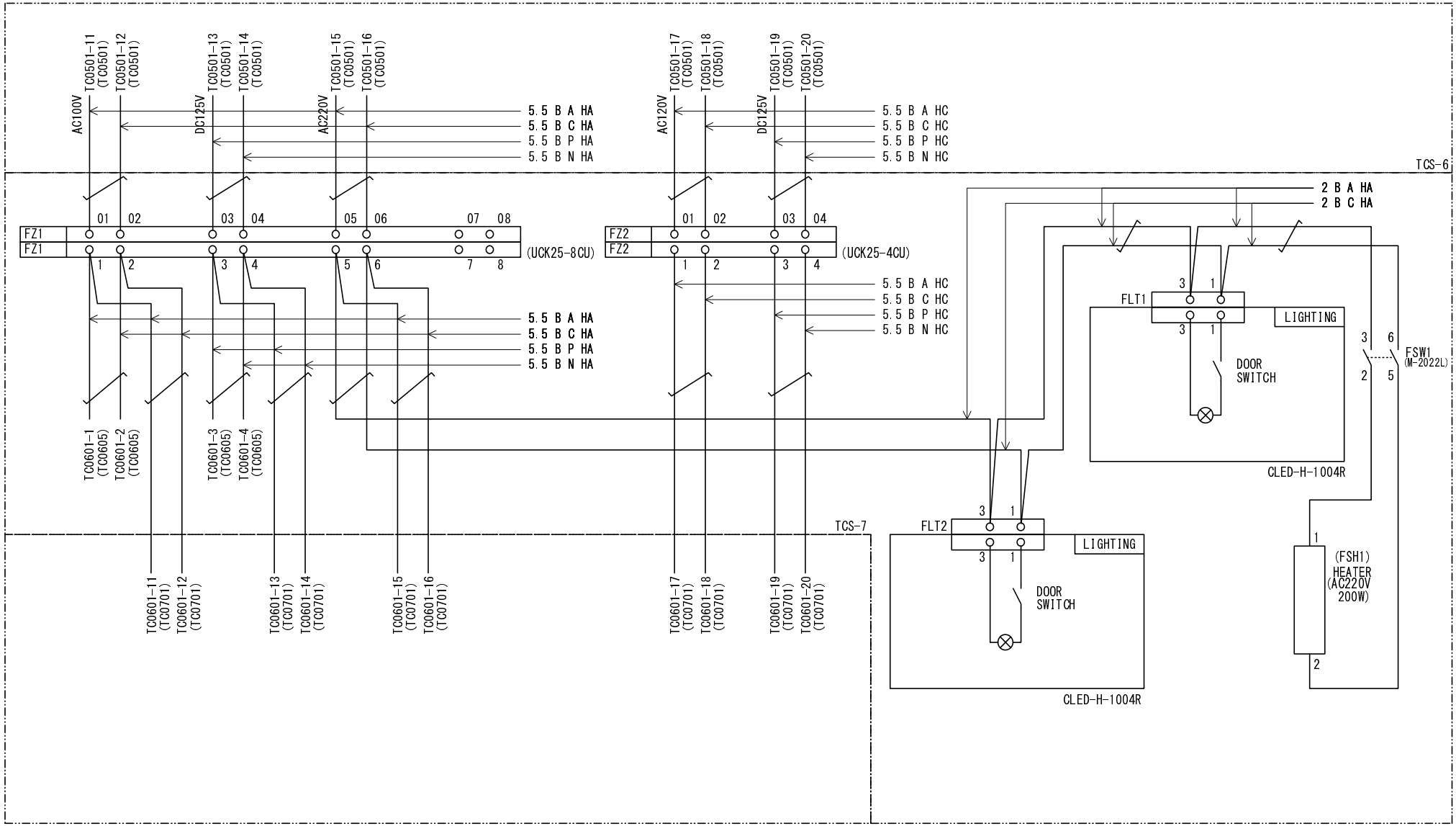
2016/07/28 TC0547 XSMM-TC

TCS-4 TCS-5 TCS-6



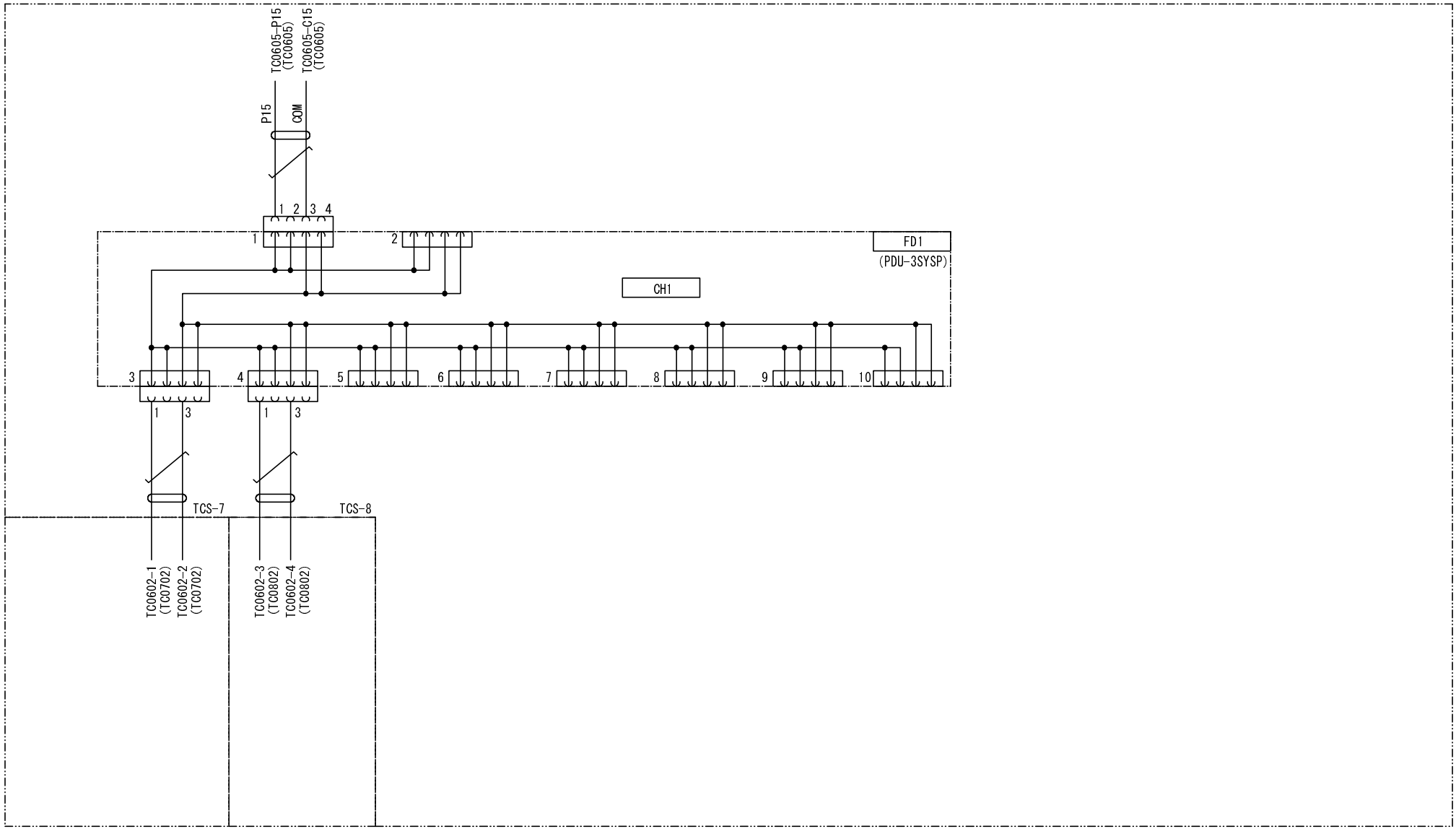
POWER SUPPLY SYSTEM EARTH CIRCUIT		SH. No.	TC0599
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 203	REV. 3

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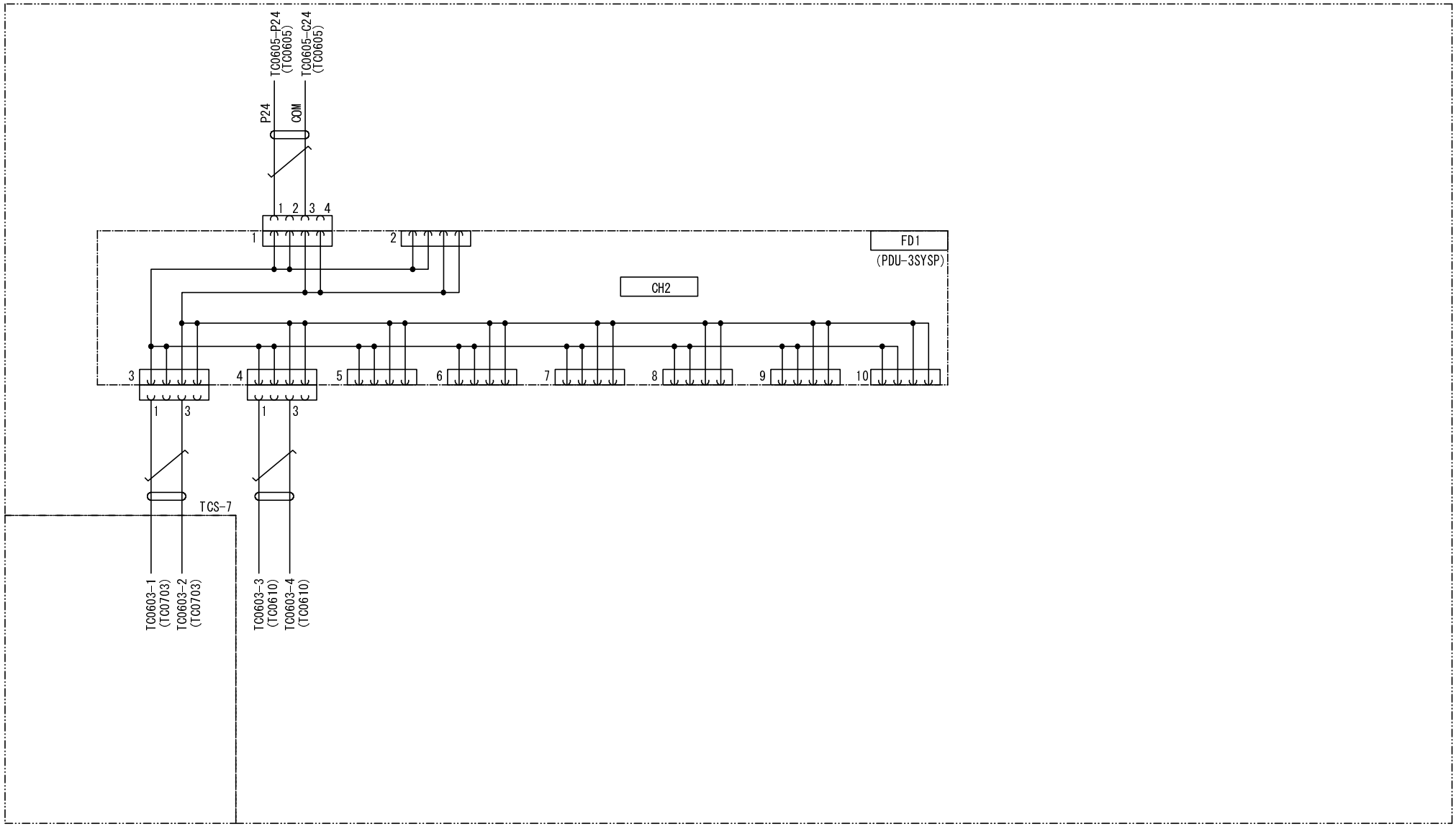
POWER SUPPLY SYSTEM(1)		SH. No.	TC0601
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 204 / REV. 3

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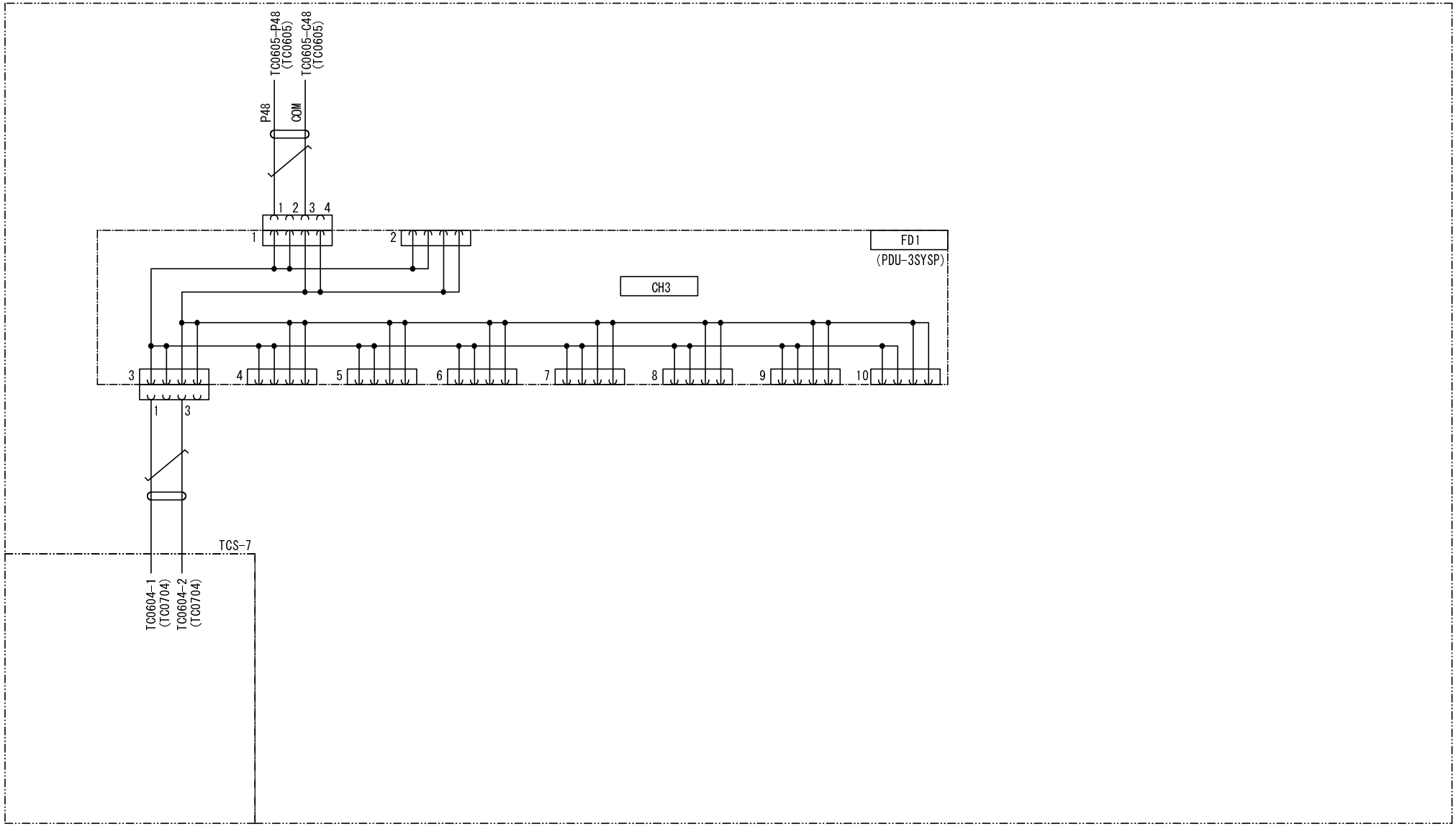
POWER SUPPLY SYSTEM(2)		SH. No.	TC0602
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 205	REV. 3

TCS-6

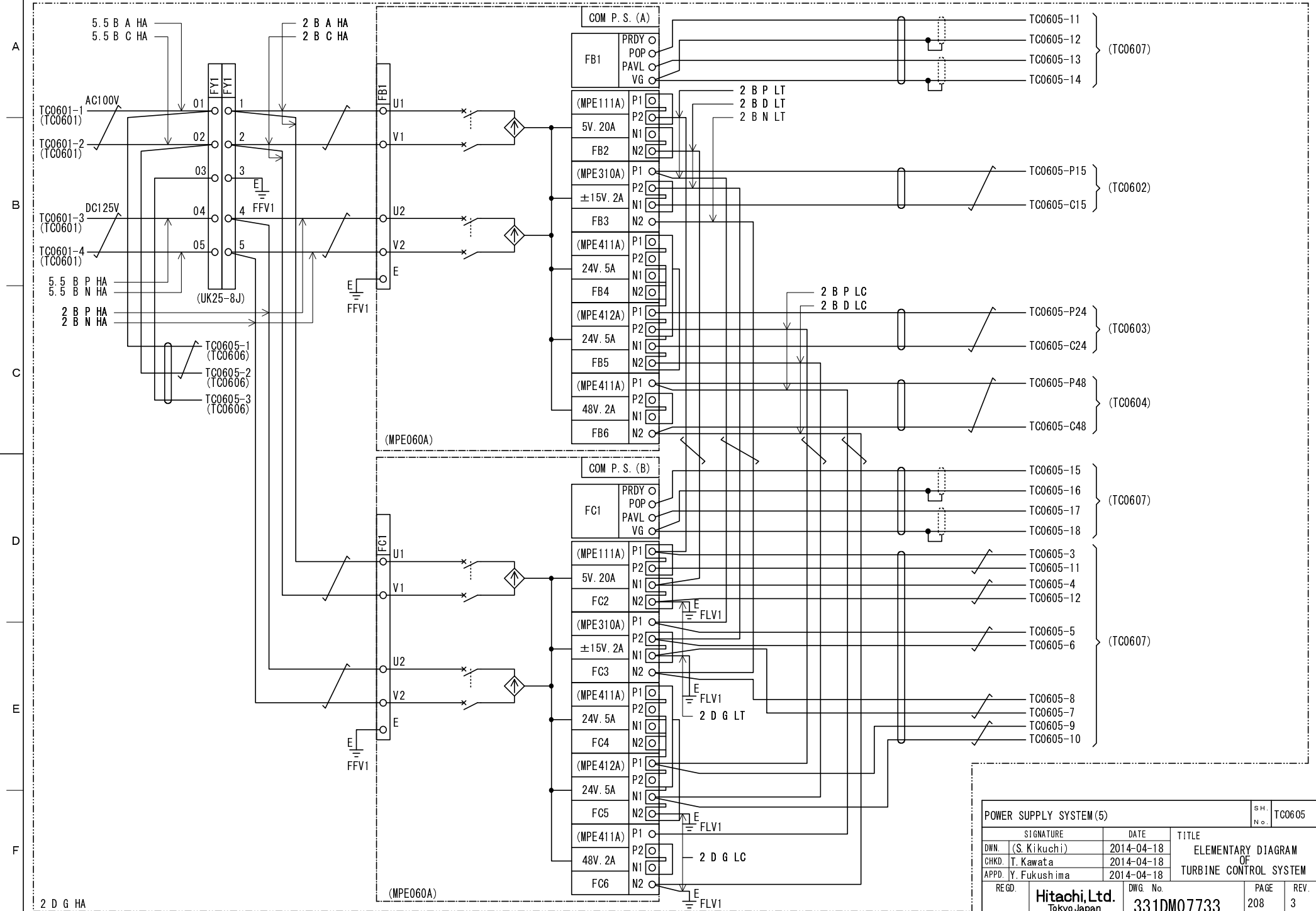


POWER SUPPLY SYSTEM(3)		SH. No.	TC0603
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 206	REV. 3

TCS-6

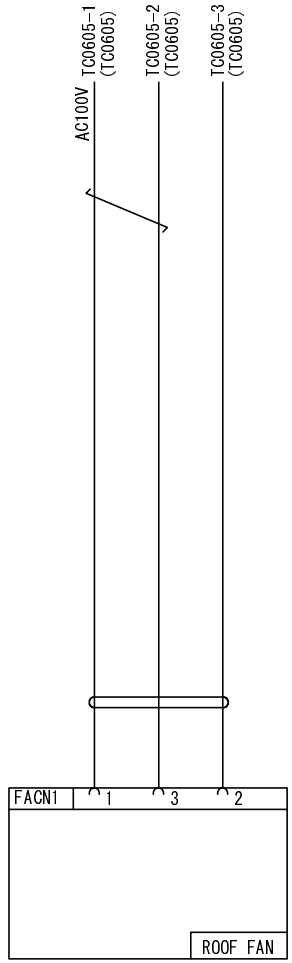


POWER SUPPLY SYSTEM(4)		SH. No.	TC0604
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
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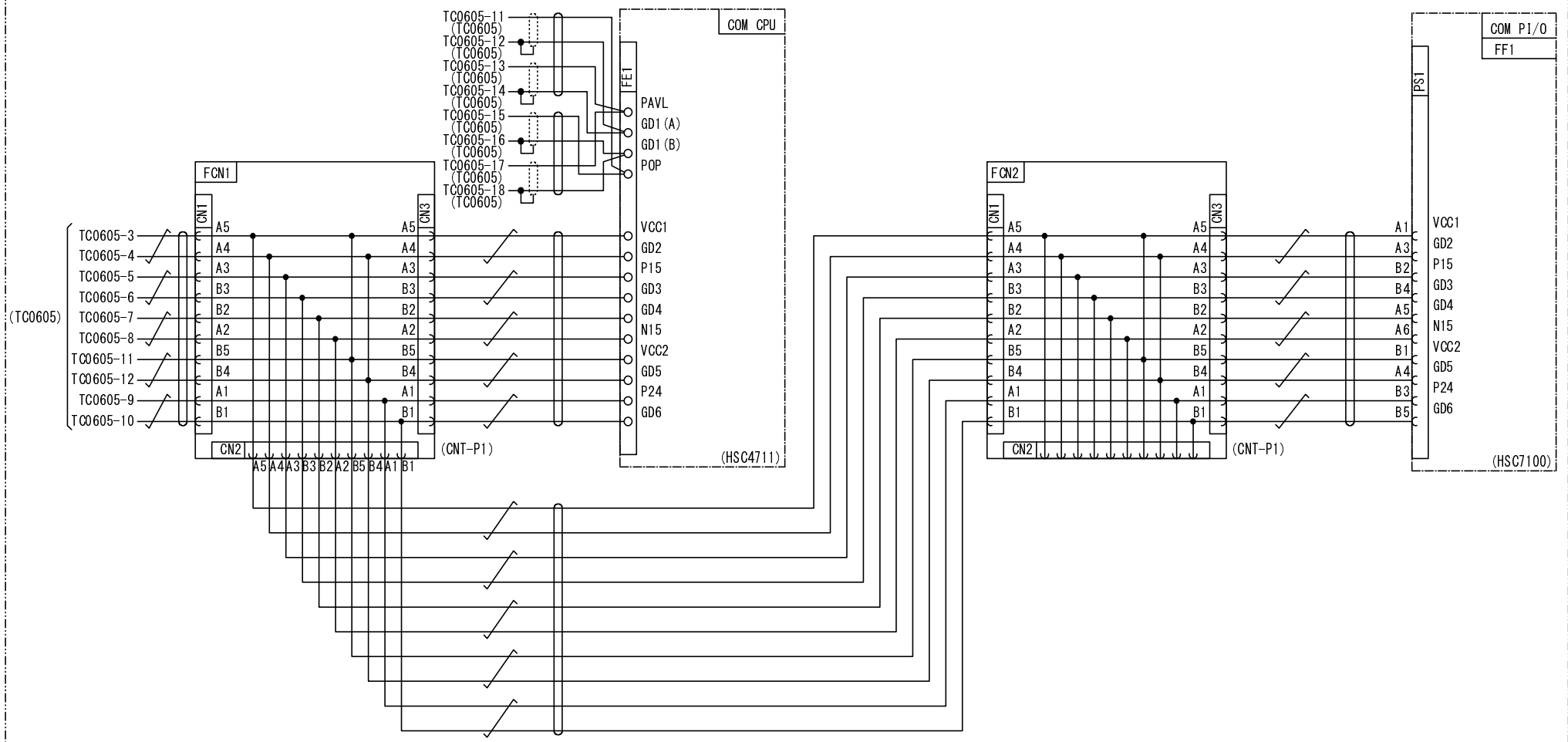


POWER SUPPLY SYSTEM(5)		SH. No.	TC0605
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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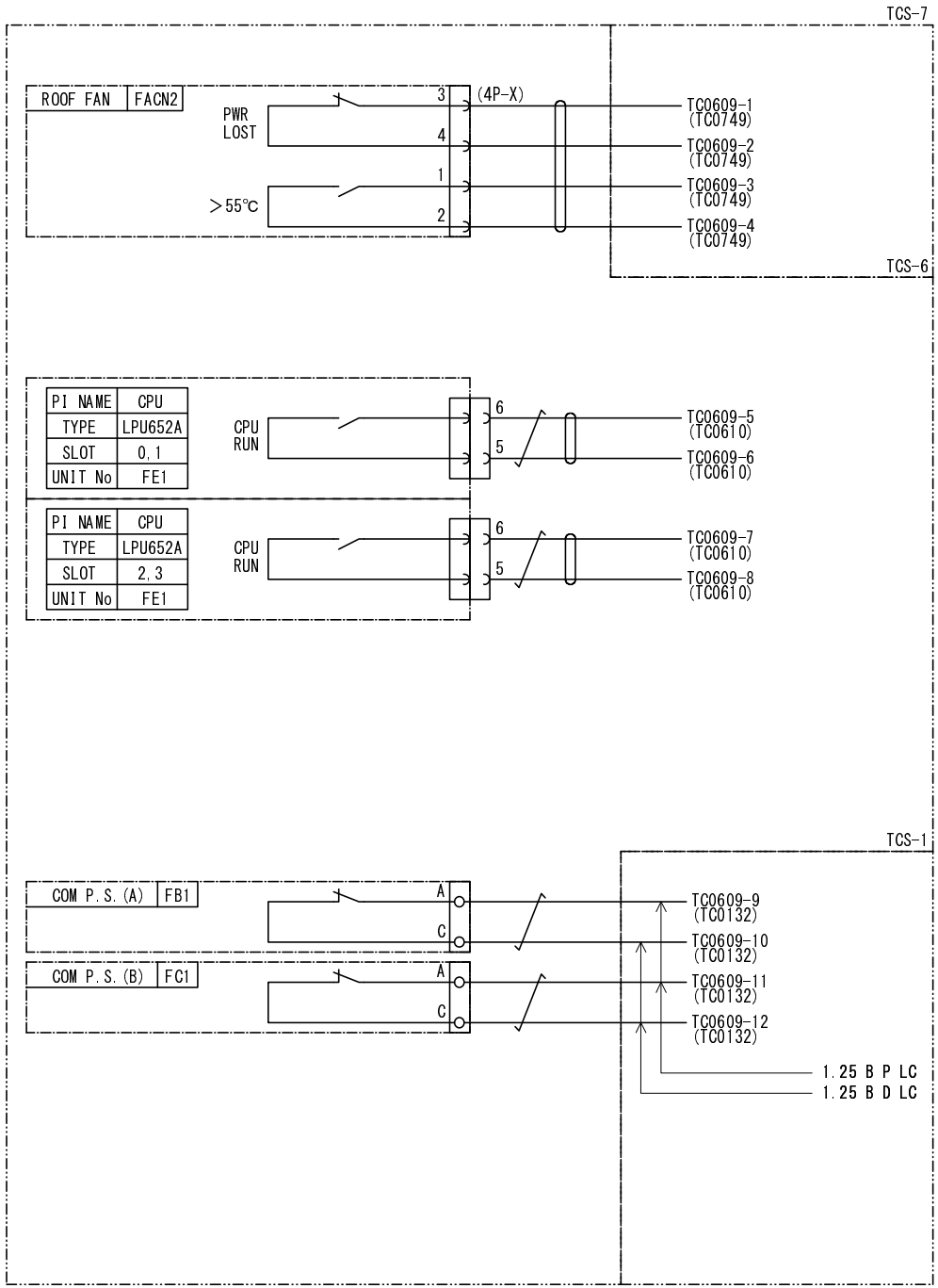
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POWER SUPPLY SYSTEM(6)		SH. No.	TC0606
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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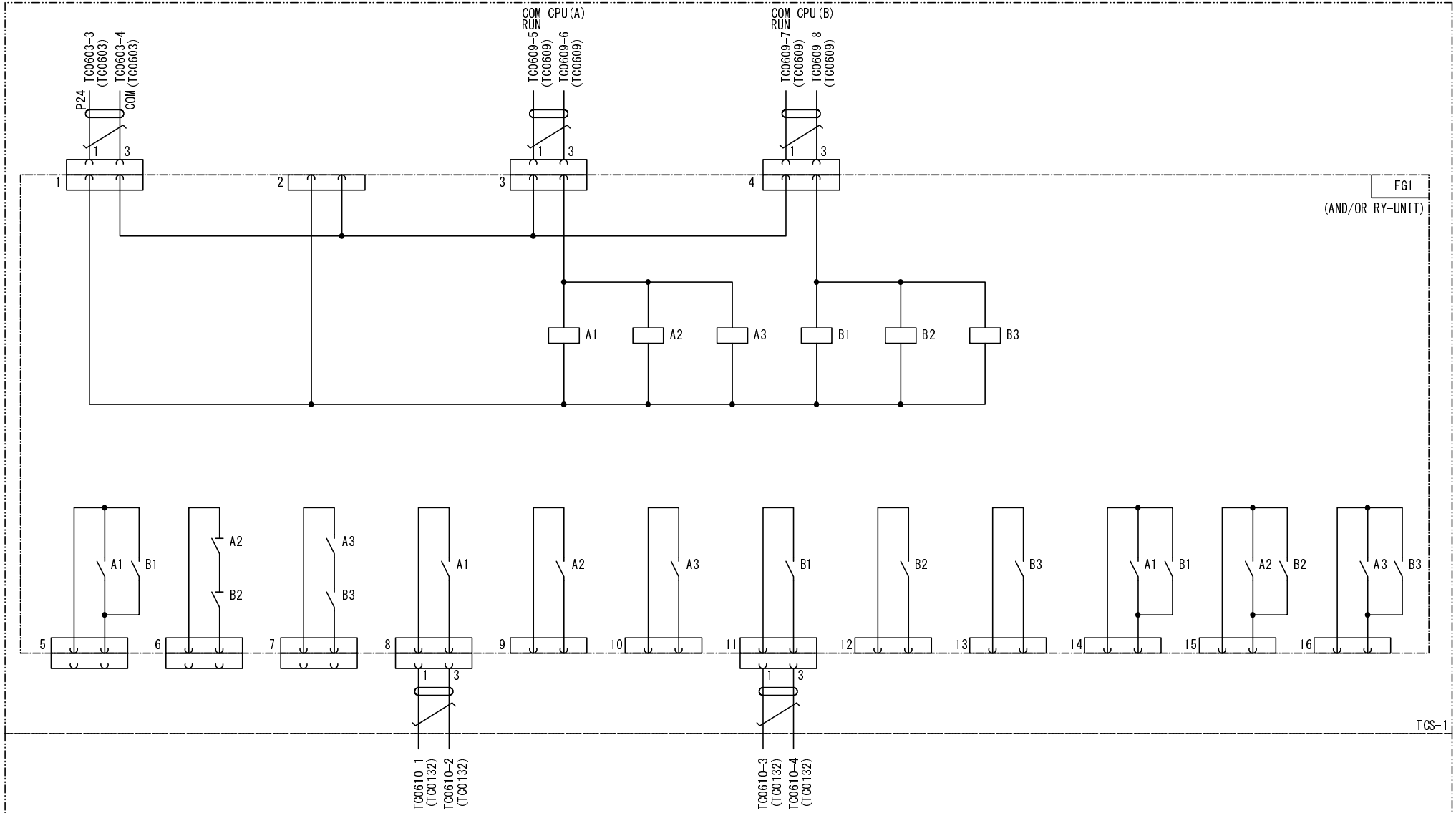


POWER SUPPLY SYSTEM(7)		SH. No.	TC0607
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 210 / REV. 3



DIAGNOSIS CIRCUIT(1)		SH. No.	TC0609
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 211 / REV. 3

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DEV.	a	b	DEV.	a	b

DIAGNOSIS CIRCUIT(2)		SH. No.	TC0610
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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TC0621-1
(TC0721)

(LYD050A) (S)

unit NO.	slot NO.	4 20	18	16	14	12	10	8	6
FE1	5								
CTL	CH. NO.	DI 0	DI 1	DI 2	DI 3	DI 4	DI 5	DI 6	DI 7
COMM+MD	signal range/level								
logic SH. NO.									
TCA900 DI	signal name	MSV (A) ABOVE SEAT SECONDRY DRAIN VALVE OPENED	MSV (A) ABOVE SEAT SECONDRY DRAIN VALVE CLOSED	MSV (A) ABOVE SEAT SECONDRY DRAIN VALVE ELECTRICAL TROUB	CV (A) ABOVE SEAT SECONDRY DRAIN VALVE OPENED	CV (A) ABOVE SEAT SECONDRY DRAIN VALVE CLOSED	CV (A) ABOVE SEAT SECONDRY DRAIN VALVE ELECTRICAL TROUB	CRV (A) BELLOW SEAT SECONDARY DRAIN VALVE OPENED	CRV (A) BELLOW SEAT SECONDARY DRAIN VALVE CLOSED

COM DIGITAL INPUT(1)			SH. No.	TC0621
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
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TC0621-1
(TC0722)

(LYD050A)

unit NO.	slot NO.	3	19	17	15	13	11	9	7	5	(S)
FE1	5										
CTL	CH. NO.	DI	8	DI	9	DI	A	DI	B	DI	C	DI	D	DI	E	DI	F
COMM+MD	signal range/level																
logic SH. NO.																	
TCA900	signal name	CRV (A) BELOW SEAT SECONDRY DRAIN VALVE ELECTRICAL TROUB		AC EXCITER EARTH FAULT		AUTO/MANUAL START READY		CONTROL LOCAL		MSV (A) ABOVE SEAT SECONDRY DRAIN VALVE TORQUE OPEN		CRV (A) BELOW SEAT SECONDRY DRAIN VALVE TORQUE OPEN		CV (A) ABOVE SEAT SECONDRY DRAIN VALVE TORQUE OPEN		P.S.NORMAL	
DI																	

COM DIGITAL INPUT(2)		SH. No.	TC0622
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
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TC0623-1
(TC0723)

TCS-6

(LYD050A)

unit NO.	slot NO.	4 20	18	16	14	12	10	8	6	(T)
FE1	5									
CTL	CH. NO.	DI 10	DI 11	DI 12	DI 13	DI 14	DI 15	DI 16	DI 17	
COMM+MD	signal range/level									
logic SH. NO.										
TCA900	signal name	STEAM SEAL	STEAM SEAL	NO.1 LP HEATER		NO.4 EXTRACTIONSTEAM	NO.4 EXTRACTION	NO.3 LP HEATER	GEN FIELD	
DI		DIVERTING VALVE CONDENSER SIDE	DIVERTING VALVE HEATER SIDE	LEVEL VERY HIGH		NON RETURN VALVE FULL CLOSE POSITION	STEAM NON RETURN VALVE FULL OPEN POSITI	LEVEL HIGH HIGH	EARTH FAULT	

COM DIGITAL INPUT(3)		SH. No.	TC0623
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
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TC0623-1
(TC0724)

(LYD050A)

unit NO.	slot NO.	3 19	17	15	13	11	9	7	5 (T)
FE1	5								
CTL	CH. NO.	DI 18	DI 19	DI 1A	DI 1B	DI 1C	DI 1D	DI 1E	DI 1F
COMM+MD	signal range/level								
logic SH. NO.									
TCA900	signal name	AVR IN USE	AVR MAJOR TROUBLE	AVR MINOR TROUBLE		MSV (A) ABOVE SEAT SECONDRY DRAIN VALVE TORQUE CLOSE	CRV (A) BELOW SEAT SECONDRY DRAIN VALVE TORQUE CLOSE	CV (A) ABOVE SEAT SECONDRY DRAIN VALVE TORQUE CLOSE	P.S.NORMAL
DI									

COM DIGITAL INPUT(4)		SH. No.	TC0624
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
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TC0625-1
(TC0725)

(LYD050A)

unit NO.	slot NO.	4 20	18	16	14	12	10	8	6 (S)
FE1	6								
CTL	CH. NO.	DI 0	DI 1	DI 2	DI 3	DI 4	DI 5	DI 6	DI 7
COMM+MD	signal range/level								
logic SH. NO.									
TCA901 DI	signal name	MSV (B) ABOVE SEAT SECONDRY DRAIN VALVE OPENED	MSV (B) ABOVE SEAT SECONDRY DRAIN VALVE CLOSED	MSV (B) ABOVE SEAT SECONDRY DRAIN VALVE ELECTRICAL TROUBLE	CV (B) ABOVE SEAT SECONDARY DRAIN VALVE OPENED	CV (B) ABOVE SEAT SECONDARY DRAIN VALVE CLOSED	CV (B) ABOVE SEAT SECONDRY DRAIN VALVE ELECTRICAL TROUB	CRV (B) BELLOW SEAT SECONDRY DRAIN VALVE OPENED	CRV (B) BELLOW SEAT SECONDRY DRAIN VALVE CLOSED

COM DIGITAL INPUT(5)		SH. No.	TC0625
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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(LYD050A) (S)

unit NO.	slot NO.	3 19	17	15	13	11	9	7	5
FE1	6								
CTL	CH. NO.	DI 8	DI 9	DI A	DI B	DI C	DI D	DI E	DI F
COMM+MD	signal range/level								
logic SH. NO.									
TCA901 DI	signal name	CRV (B) BELOW SEAT SECONDRY DRAIN VALVE ELECTRICAL TROUBLE	TURBINE GLAND STEAM LINE DRAIN VALVE CLOSED	BACK UP AVR TROUBLE	PRIMARY AVR TROUBLE	PSS ON	CV (B) ABV SEAT SECONDRY DRAIN VALVE TORQUE OPEN	CRV (B) BELOW SEAT SECONDRY DRAIN VALVE TORQUE OPEN	P.S.NORMAL

COM DIGITAL INPUT(6)		SH. No.	TC0626
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
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(LYD050A) (T)

unit NO.	slot NO.	4 20	18	16	14	12	10	8	6
FE1	6								
CTL	CH. NO.	DI 10	DI 11	DI 12	DI 13	DI 14	DI 15	DI 16	DI 17
COMM+MD	signal range/level								
logic SH. NO.									
TCA901 DI	signal name	NO. 2 EXTRACTION STEAM NON RETURN VALVE FULL CLOSE POSITION	NO. 2 EXTRACTION STEAM NON RETURN VALVE FULL OPEN POSITION	NO. 1 HP HEATER LEVEL VERY HIGH	NO. 5 EXTRACTION STEAM NON RETURN VALVE FULL CLOSE POSITION	NO. 5 EXTRACTION STEAM NON RETURN VALVE FULL OPEN POSITION	NO. 2 LP HEATER LEVEL HIGH HIGH		HP/LP BYPASS CONTROL IN AUTO

COM DIGITAL INPUT(7) SH. No. TC0627

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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TC0627-1
(TC0728)

(LYD050A)

(T)

unit NO.	slot NO.	3 19	17	15	13	11	9	7	5
FE1	6								
CTL	CH. NO.	DI 18	DI 19	DI 1A	DI 1B	DI 1C	DI 1D	DI 1E	DI 1F
COMM+MD	signal range/level								
logic SH. NO.									
TCA901	signal name	OIL PLRIFIER	OIL PURIFIER PUMP MOTO		MSV (B) ABOVE SEAT	MSV (B) ABOVE SEAT	CV (B) ABOVE SEAT	CRV (B) BELOW SEAT	P.S.NORMAL
DI		PUMP RUNNING	R ELECTRICAL TROUBLE		SECONDRY DRAIN VALVE TORQUE OPEN	SECONDRY DRAIN VALVE TORQUE CLOSE	SECONDRY DRAIN VALVE TORQUE CLOSE	SECONDRY DRAIN VALVE TORQUE CLOSE	

COM DIGITAL INPUT(8)			SH. No.	TC0628
SIGNATURE		DATE	TITLE	
DWN.	(S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T.Kawata	2014-04-18		
APPD.	Y.Fukushima	2014-04-18		
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TC0629-1
(TC0729)

(LYD050A)

unit NO.	slot NO.	4 20	17	15	13	11	9	7	6 (S)
FE1	7								
CTL	CH. NO.	DI 0	DI 1	DI 2	DI 3	DI 4	DI 5	DI 6	DI 7
COMM+MD	signal range/level								
logic SH. NO.									
TCA902 DI	signal name	CV (B) BELLOW SEAT SECONDRY WARMING DRAIN VALVE OPENED	CV (B) BELLOW SEAT SECONDRY WARMING DRAIN VALVE CLOSED	CV (B) BELLOW SEAT SEC. WARMING DRAIN VLV ELECTRICAL TROUBL	HP TURBINE WARMING VALVE FULL CLOSE POSITION	HP TURBINE WARMING VALVE FULL OPEN POSITION	COLD REHEAT DRAIN VALVE CLOSE FEED BACK	COLD REHEAT DRAIN VALVE FULL OPEN FEED BACK	

COM DIGITAL INPUT(9)			SH. No.	TC0629
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
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TC0629-1
(TC0730)

(LYD050A)

unit NO.	slot NO.	3	19	17	15	13	11	9	7	5	(S)
FE1	7										
CTL	CH. NO.	DI 8		DI 9	DI A	DI B	DI C	DI D	DI E	DI F	
COMM+MD	signal range/level										
logic SH. NO.											
TCA902	signal name					CV(B) BEL SEC	CV(B) BEL SEC	HP TURBINE	HP TURBINE	P.S.NORMAL	
DI						DRAIN VALVE TORUQE SW OPEN	DRAIN VALVE TORUQE SW CLOSE	WARMING VALVE TORUQE SW OPEN	WARMING VALVE TORUQE SW CLOSE		

COM DIGITAL INPUT(10)			SH. No.	TC0630
SIGNATURE		DATE	TITLE	
DWN.	(S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T.Kawata	2014-04-18		
APPD.	Y.Fukushima	2014-04-18		
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TC0631-1
(TC0731)

(LYD050A)

unit NO.	slot NO.	4 20	18	16	14	12	10	8	6 (T)
FE1	7								
CTL	CH. NO.	DI 10	DI 11	DI 12	DI 13	DI 14	DI 15	DI 16	DI 17
COMM+MD	signal range/level								
logic SH. NO.									
TCA902	signal name	NO.3 EXTRACTION STM	NO.3 EXTRACTION STM	DEAERATOR STRAGE	DUPLEX OIL	GEN CONTROL	GEN TRANSF.	GEN VOLTAGE OK	52L CLOSE
DI		PRI. NON RETURN VLV FULL CLOSE POSITIO	PRI. NON RETURN VLV FULL OPEN POSITION	TANK LEVEL VERY HIGH	FILTER DIFF. PRESS.	PANEL TROUBLE	PROTECTION OPERATED		

COM DIGITAL INPUT(11)		SH. No.	TC0631
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
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TC0631-1
(TC0732)

(LYD050A)

unit NO.	slot NO.	3	19	17	15	13	11	9	7	5	(T)
FE1	7										
CTL	CH. NO.	DI	18	DI 19	DI	1A	DI 1B	DI 1C	DI	1D	DI 1E	DI 1F
COMM+MD	signal range/level										
logic SH. NO.											
TCA902	signal name	52L OPEN		ASD1 IN USE	AVR FAILURE TO BACUP STAUS		AVR IN AUTO	AVR IN MANUAL	AVR CH.1 SELECT	AVR CH.2 SELECT	P.S.NORMAL
DI											

COM DIGITAL INPUT(12)				SH. No.	TC0632
SIGNATURE		DATE	TITLE		
DWN.	(S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM		
CHKD.	T.Kawata	2014-04-18			
APPD.	Y.Fukushima	2014-04-18			
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TC0633-1
(TC0733)

(LYD050A)

unit NO.	slot NO.	4	20	18	16	14	12	10	8	6	(S)
FE1	8										
CTL	CH. NO.	DI 0		DI 1	DI 2	DI 3	DI 4	DI 5	DI 6	DI 7	
COMM+MD	signal range/level										
logic SH. NO.											
TCA903	signal name	SET POINT MAX		SET POINT MIN	OVER EXCITATION	UNDER EXCITATION	PF CONTROL ON	REACTIVE POWER ON	AVR VT FUSE NORMAL		
DI											

COM DIGITAL INPUT(13)			SH. No.	TC0633
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
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TC0633-1
(TC0734)

(LYD050A)

unit NO.	slot NO.	3	19	17	15	13	11	9	7	5	(S)
FE1	8										
CTL	CH. NO.	DI 8		DI 9	DI A	DI B	DI C	DI D	DI E	DI F	
COMM+MD	signal range/level										
logic SH. NO.											
TCA903	signal name									PSNORMAL	
DI											

COM DIGITAL INPUT(14)				SH. No.	TC0634
SIGNATURE		DATE	TITLE		
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM		
CHKD.	T. Kawata	2014-04-18			
APPD.	Y. Fukushima	2014-04-18			
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(LYD050A) (T)

unit NO.	slot NO.	4 20	18	16	14	12	10	8	6
FE1	8								
CTL	CH. NO.	DI 10	DI 11	DI 12	DI 13	DI 14	DI 15	DI 16	DI 17
COMM+MD	signal range/level								
logic SH. NO.									
TCA903	signal name	NO.3 EXTRACTION STM	NO.3 EXTRACTION STM	HP TURBINE WARMING	HP TURBINE WARMING	HP TURBINE WARMING	HP TURBINE WARMING	MAIN OIL TANK VAPOUR	MAIN OIL TANK VAPOUR
DI		SEC. NON RETURN VLV FULL CLOSE POSITIO	SEC. NON RETURN VLV FULL OPEN POSITION	LINE DRAIN VALVE AUTO SELECTED	LINE DRAIN VALVE OPEN FEEED BACK	LINE DRAIN VALVE CLOSE FEED BACK	VALVE ELECTRICAL TROUBLE	EXTRACTOR RUNNING	EXTRACTOR ELECTRICAL TROUBLE

COM DIGITAL INPUT(15)		SH. No.	TC0635
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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(LYD050A)

unit NO.	slot NO.	3 19	17	15	13	11	9	7	5 (T)
FE1	8								
CTL	CH. NO.	DI 18	DI 19	DI 1A	DI 1B	DI 1C	DI 1D	DI 1E	DI 1F
COMM+MD	signal range/level								
logic SH. NO.									
TCA903	signal name								P.S.NORMAL
DI									

COM DIGITAL INPUT(16)		SH. No.	TC0636
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
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TC0637-1
(TC0737)

(LYD050A)

unit NO.	slot NO.	4 20	18	16	14	12	10	8	6	(S)
FE1	9									
CTL	CH. NO.	DI 0	DI 1	DI 2	DI 3	DI 4	DI 5	DI 6	DI 7	
COMM+MD	signal range/level									
logic SH. NO.										
TCA904 DI	signal name	NO.1 TURBINE SHAFT VIBRATION (X) HIGH	NO.3 GEN. SHAFT VIBRATION (Y) HIGH	LP EXHAUST HOOD WATER SPRAY VALVE FULL CLOSE POSITION	LP EXHAUST HOOD WATER SPRAY VALVE FULL OPEN POSITION	COLD REHEAT NON RETURN VALVE FULL CLOSE POSITION	COLD REHEATNON RETURN VALVE FULL OPEN POSITION	GEN. WATER AND OIL LEAKAGE LEVEL HIGH	LUBE OIL CONDITIONER COALESCER FILTER DIFF. PRESS. HIGH	

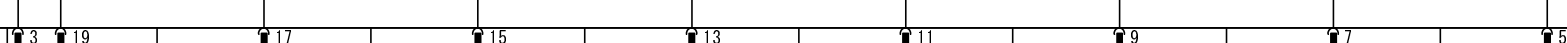
COM DIGITAL INPUT(17)		SH. No.	TC0637
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
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TC0637-1
(TC0738)



(LYD050A)

unit NO.	slot NO.	3	19	17	15	13	11	9	7	5	(S)
FE1	9										
CTL	CH. NO.	DI	8	DI	9	DI	A	DI	B	DI	C	DI	D	DI	E	DI	F
COMM+MD	signal range/level																
logic SH. NO.																	
TCA904	signal name	TURBINE SUPERVISORY		CONDENSER VACUUM		GLAND STEAM		GLAND STEAM								P.S.NORMAL	
DI		INSTRUMENT TROUBBLE		BREAK COMPLETE		EXHAUSTER ELECTRICAL TROUBLE		EXHAUSTER RUNNING									

COM DIGITAL INPUT(18)		SH. No.	TC0638
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
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TC0639-1
(TC0739)

(LYD050A)

unit NO.	slot NO.	4 20	18	16	14	12	10	8	6	(T)
FE1	9									
CTL	CH. NO.	DI 10	DI 11	DI 12	DI 13	DI 14	DI 15	DI 16	DI 17	
COMM+MD	signal range/level									
logic SH. NO.										
TCA904 DI	signal name	NO.1 TURBINE SHAFT VIBRATION (Y) HIGH	NO.3 GEN. SHAFT VIBRATION (X) HIGH	RSR DIODE FAULT	STEAM SEAL FEED VALVE FULL CLOSE POSITION	STEAM SEAL FEED INLET VALVE FULL CLOSE POSITION	STEAM SEAL FEED INLET VALVE FULL OPEN POSITION	STEAM SEAL FEED INLET VALVE ELECTRICAL TROUBLE	SSFV BYPASS VALVE FULL CLOSE POSITION	

COM DIGITAL INPUT(19)			SH. No.	TC0639
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
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TC0639-1
(TC0740)

(LYD050A)

unit NO.	slot NO.	3 19	17	15	13	11	9	7	5 (T)
FE1	9								
CTL	CH. NO.	DI 18	DI 19	DI 1A	DI 1B	DI 1C	DI 1D	DI 1E	DI 1F
COMM+MD	signal range/level								
logic SH. NO.									
TCA904	signal name	BOTH GEN PROTECTION	30DF/64F	30DF/64F	STEAM SEAL FEED	STEAM SEAL FEED			P.S.NORMAL
DI		RELAY TROUBLE	RELAY TROUBLE	LOSS OF CONTROL POWER	INLET VALVE TORQUE SW OPEN	INLET VALVE TORQUE SW CLOSE			

COM DIGITAL INPUT(20)		SH. No.	TC0640
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
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TC0641-1
(TC0741)

(LYD050A)

unit NO.	slot NO.	4 20	18	16	14	12	10	8	6	(S)
FE1	A									
CTL	CH. NO.	DI 0	DI 1	DI 2	DI 3	DI 4	DI 5	DI 6	DI 7	
COMM+MD	signal range/level									
logic SH. NO.										
TCA905	signal name	NO. 2 TURBINE SHAFT	NO. 4 GEN. SHAFT	86G-A RELAY COIL OPEN	GEN PROTECTION	MAIN OIL TANK	CONTROL OIL	SPACE HEATER	SPACE HEATER
DI		VIBRATION (X) HIGH	VIBRATION (Y) HIGH		RELAY (A) TROUBLE	LEVEL LOW	LINE FILTER (A) DIFF. PRESS. HIGH	RUNNING (GEN)	RUNNING (AC EXCITER)

COM DIGITAL INPUT(21)				SH. No.	TC0641
SIGNATURE		DATE	TITLE		
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM		
CHKD.	T. Kawata	2014-04-18			
APPD.	Y. Fukushima	2014-04-18			
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TC0641-1
(TC0742)

(LYD050A)

unit NO.	slot NO.	3 19	17	15	13	11	9	7	5 (S)
FE1	A								
CTL	CH. NO.	DI 8	DI 9	DI A	DI B	DI C	DI D	DI E	DI F
COMM+MD	signal range/level								
logic SH. NO.									
TCA905	signal name	SPACE HEATER FOR AC EXCITER. ELECTRICAL TROUBLE	SPACE HEATER FOR GEN. ELECTRICAL TROUBLE						P.S.NORMAL
DI									

COM DIGITAL INPUT(22)		SH. No.	TC0642
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
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TC0643-1
(TC0743)

TCS-6

(LYD050A) (T)

unit NO.	slot NO.	4 20	18	16	14	12	10	8	6
FE1	A								
CTL	CH. NO.	DI 10	DI 11	DI 12	DI 13	DI 14	DI 15	DI 16	DI 17
COMM+MD	signal range/level								
logic SH. NO.									
TCA905 DI	signal name	NO. 2 TURBINE SHAFT VIBRATION (Y) HIGH	NO. 4 GEN. SHAFT VIBRATION (X) HIGH	86G-B RELAY COIL OPEN	GEN PROTECTION RELAY (B) TROUBLE	MAIN OIL TANK LEVEL HIGH	CONTROL OIL LINE FILTER (B) DIFF. PRESS. HIGH	TURBINE GLAND STEAM LINE DRAIN VALVE FULL OPENED	TURBINE GLAND STEAM LINE DRAIN VALVE AUTO

COM DIGITAL INPUT(23)		SH. No.	TC0643
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
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TC0643-1
(TC0744)

(LYD050A)

unit NO.	slot NO.	3 19	17	15	13	11	9	7	5 (T)
FE1	A								
CTL	CH. NO.	DI 18	DI 19	DI 1A	DI 1B	DI 1C	DI 1D	DI 1E	DI 1F
COMM+MD	signal range/level								
logic SH. NO.									
TCA905	signal name	UNIT TRANSF.							P.S.NORMAL
DI		PROTECTION OPERATED							

COM DIGITAL INPUT(24)			SH. No.	TC0644
SIGNATURE		DATE	TITLE	
DWN.	(S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T.Kawata	2014-04-18		
APPD.	Y.Fukushima	2014-04-18		
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TC0645-1
(TC0745)

(LYD050A)

unit NO.	slot NO.	4 20	18	16	14	12	10	8	6	(S)
FE1	B									
CTL	CH. NO.	DI 0	DI 1	DI 2	DI 3	DI 4	DI 5	DI 6	DI 7	
COMM+MD	signal range/level									
logic SH. NO.										
TCA906	signal name					TRIP PCM (A) NORMAL	TRIP PCM (B) NORMAL	TRIP PCM (C) NORMAL		
DI										

COM DIGITAL INPUT(25)		SH. No.	TC0645
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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TC0645-1
(TC0746)

(LYD050A)

unit NO.	slot NO.	3 19	17	15	13	11	9	7	5 (S)
FE1	B								
CTL	CH. NO.	DI 8	DI 9	DI A	DI B	DI C	DI D	DI E	DI F
COMM+MD	signal range/level								
logic SH. NO.									
TCA906	signal name			TCS-8 TEMP HIGH	MODBUS PS(A) ABNORMAL	MODBUS PS(B) NORMAL	MODBUS CPU(A) NORMAL	MODBUS CPU(B) NORMAL	PS NORMAL
DI									

COM DIGITAL INPUT(26)			SH. No.	TC0646
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
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TC0647-1
(TC0747)

(LYD050A)

unit NO.	slot NO.	4 20	18	16	14	12	10	8	6 (T)
FE1	B								
CTL	CH. NO.	DI 10	DI 11	DI 12	DI 13	DI 14	DI 15	DI 16	DI 17
COMM+MD	signal range/level								
logic SH. NO.									
TCA906 DI	signal name		AC CONTROL P.S. ABNORMAL	DC CONTROL P.S. ABNORMAL	LIGHTING& SPACE HEATER P.S. ABNORMAL	AC PROTECTION P.S.ABNORMAL	DC PROTECTION P.S.(2)ABNORMAL	DC PROTECTION P.S.(1)ABNORMAL	TCS-4 TEMP HIGH

COM DIGITAL INPUT(27)		SH. No.	TC0647
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
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(LYD050A) (T)

unit NO.	slot NO.	3 19	17	15	13	11	9	7	5
FE1	B								
CTL	CH. NO.	DI 18	DI 19	DI 1A	DI 1B	DI 1C	DI 1D	DI 1E	DI 1F
COMM+MD	signal range/level								
logic SH. NO.									
TCA906	signal name	TCS-5 TEMP. HIGH	EOS OPERATED(A)	EOS OPERATED(B)	EOS OPERATED(C)	PLU OPERATED (A)	PLU OPERATED (B)	PLU OPERATED (C)	PS NORMAL
DI									

COM DIGITAL INPUT(28)		SH. No.	TC0648
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
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TCS-6

TC0651-1
(TC0751)

(LYA000A)

(T)

unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
FF1	6								
CTL	CH. NO.	AI 0	AI 1	AI 2	AI 3	AI 4	AI 5	AI 6	AI 7
COMM+MD	signal range/level	-1～2kg/cm2	0～18kV	0～10kV			0～50kg/cm2	0～50kg/cm2	0～50kg/cm2
logic SH. NO.		1～5V	1～5V	1～5V			1～5V	1～5V	1～5V
TCA930	signal name	GLAND STEAM	GEN. VOLTAGE (A-B)	GEN. CURRENT (A)			REHEAT BOWL	REHEAT BOWL	REHEAT BOWL
AI		SEAL HEADER PRESS. (A)					STEAM PRESS. (A)	STEAM PRESS. (B)	STEAM PRESS. (C)

COM ANALOG INPUT		SH. No.	TC0651
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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TCS-6

TC0655-1
(TC0751A)

(LYA000A)

(T)

unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
FF1	7								
CTL	CH. NO.	AI 0	AI 1	AI 2	AI 3	AI 4	AI 5	AI 6	AI 7
COMM+MD	signal range/level	-1～2kg/cm2	0～18kV	0～10kV	0～120VDC	0～40A			
logic SH. NO.		1～5V	1～5V	1～5V	1～5V	1～5V			
TCA930	signal name	GLAND STEAM SEAL HEADER PRESS. (B)	GEN. VOLTAGE (B-C)	GEN. CURRENT (B)	GEN. FIELD VOLTAGE	GEN. FIELD CURRENT			
AI									

COM ANALOG INPUT(1)				SH. No.	TC0655
SIGNATURE		DATE	TITLE		
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM		
CHKD.	T. Kawata	2014-04-18			
APPD.	Y. Fukushima	2014-04-18			
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TC0656-1
(TC0751B)

(LYA000A)

unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
FF1	8								
CTL	CH. NO.	AI 0	AI 1	AI 2	AI 3	AI 4	AI 5	AI 6	AI 7
COMM+MD	signal range/level	0～120VDC	0～18kV						
logic SH. NO.		1～5V	1～5V						
TCA930	signal name	GEN. VOLTAGE (C-A)	GEN. CURRET (C)						
AI									

(T)

COM ANALOG INPUT(2)				SH. No.	TC0656
SIGNATURE		DATE	TITLE		
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM		
CHKD.	T. Kawata	2014-04-18			
APPD.	Y. Fukushima	2014-04-18			
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TC0657-1
(TC0751C)

(LYA000A)

unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
FF1	9								
CTL	CH. NO.	AI 0	AI 1	AI 2	AI 3	AI 4	AI 5	AI 6	AI 7
COMM+MD	signal range/level		0～50kg/cm2	0-300degC					
logic SH. NO.			1～5V	1-5V					
TCA931	signal name		AUX STEAM	GLAND STEAM					
AI			HEADER PRESS. (LOWER)	DRAINE SEPARATOR TURBINE SIDE TEMP.					

(T)

COM ANALOG INPUT(3)		SH. No.	TC0657
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
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TC0658-1
(TC0751D)

(LYA000A)

unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4 (T)
FF1	A								
CTL	CH. NO.	AI 0	AI 1	AI 2	AI 3	AI 4	AI 5	AI 6	AI 7
COMM+MD	signal range/level	-11～14mm	0～50mm	0～0.5mm					
logic SH. NO.		1～5V	1～5V	1～5V					
TCA931	signal name	HP TURBINE DIFF. EXPANSION	TURBINE SHELL EXPANSION (HP TURBINE SIDE)	LOW SPEED ECCENTRICITY DETECTOR					
AI									

COM ANALOG INPUT(4)		SH. No.	TC0658
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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TC0659-1
(TC0751E)

(LYA000A) (T)

unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
FF1	B								
CTL	CH. NO.	AI 0	AI 1	AI 2	AI 3	AI 4	AI 5	AI 6	AI 7
COMM+MD	signal range/level	-240～240Var	-0.5～0.5	0～100Wh	0～100Varh	55～65Hz			
logic SH. NO.		1～5V	1～5V	1～5V	1～5V	1～5V			
TCA931 AI	signal name	GEN. MVAR	GEN. POWER FACTOR	GEN. WH	GEN. VARH	GEN. FREQUENCY			

COM ANALOG INPUT(5)			SH. No.	TC0659
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
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TC0660-1
(TC0751F)

(LYA000A)

unit NO.	slot NO.	31 32	27 28	23 24	19 20	15 16	11 12	7 8	3 4
FF1	C								
CTL	CH. NO.	AI 0	AI 1	AI 2	AI 3	AI 4	AI 5	AI 6	AI 7
COMM+MD	signal range/level	0～100%	0～100%	0～100%	0～100%				
logic SH. NO.		1～5V	1～5V	1～5V	1～5V				
TCA932	signal name	STEAM SEAL	STEAM PACKING	CV (B) BELOW SEAT	HP TURBINE WARMING				
AI		FEED VALVE POSITION	UNLOADING VALVE POSITION	SECONDARY WARMING DRAIN VALVE POSITION	VALVE POSITION				

(T)

COM ANALOG INPUT(6)				SH. No.	TC0660
SIGNATURE		DATE	TITLE		
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM		
CHKD.	T. Kawata	2014-04-18			
APPD.	Y. Fukushima	2014-04-18			
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TC0661-1
(TC0755)

(LYA210A)

unit NO.	slot NO.	A1 A2	A3 A4	A5 A6	A7 A8	A9 A10	A11 A12	A13 A14	A15 A16 A17 (S)
FF1	0								
CTL	CH. NO.	AI 0	AI 1	AI 2	AI 3	AI 4	AI 5	AI 6	AI 7
COMM+MD	signal range/level								
logic SH. NO.									
TCA960	signal name								
THC AI									

COM THC INPUT (1)		SH. No.	TC0661
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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TC0662-1
(TC0756)

(LYA210A)

(T)

unit NO.	slot NO.	A1 A2	A3 A4	A5 A6	A7 A8	A9 A10	A11 A12	A13 A14	A15 A16 B16 A17
FF1	0								
CTL	CH. NO.	AI 8	AI 9	AI A	AI B	AI C	AI D	AI E	AI F
COMM+MD	signal range/level								
logic SH. NO.									
TCA960	signal name								RTD
THC AI									

COM THC INPUT(2)		SH. No.	TC0662
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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TC0663-1
(TC0757)

(LYA210A)

(S)

unit NO.	slot NO.	A1 A2	A3 A4	A5 A6	A7 A8	A9 A10	A11 A12	A13 A14	A15 A16 A17
FF1	1								
CTL	CH. NO.	AI 0	AI 1	AI 2	AI 3	AI 4	AI 5	AI 6	AI 7
COMM+MD	signal range/level	0～150degC	0～150degC	0～150degC	0～150degC			0～500degC	0～500degC
logic SH. NO.		TYPE K	TYPE K	TYPE K	TYPE K			TYPE K	TYPE K
TCA960	signal name	NO.1 TURBINE JOURNAL	NO.2 TURBINE JOURNAL	NO.3 BEARING	NO.4 BEARING			GLAND STEAM SEAL	GLAND STEAM SEAL
THC AI		BEARING METAL TEMP.	BEARING METAL TEMP.	METAL TEMP.	METAL TEMP.			HEADER TEMP.	LP INLET TEMP.

COM THC INPUT (3)		SH. No.	TC0663
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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TC0664-1
(TC0758)

(LYA210A) (T)

unit NO.	slot NO.	A1 A2	A3 A4	A5 A6	A7 A8	A9 A10	A11 A12	A13 A14	A15 A16 B16 A17
FF1	1								
CTL	CH. NO.	AI 8	AI 9	AI A	AI B	AI C	AI D	AI E	AI F
COMM+MD	signal range/level	0～700degC	0～700degC	0～700degC	0～400degC	0～700degC	0～500degC	0～150degC	
logic SH. NO.		TYPE K	TYPE K	TYPE K	TYPE K	TYPE K	TYPE K	TYPE K	
TCA960	signal name	HOT REHEAT STEAM TEMP.	MSV INLET STEAM TEMP. (D)	REHEAT BOWL OUTER METAL TEMP.	LP TURBINE EXTRACTION STEAM TEMP.	MSV INLET STEAM TEMP. (A)	HP TURBINE EXHAUST STEAM TEMP. (A)	LP TURBINE EXHAUST HOOD STEAM TEMP. (A)	RTD
THC AI									

COM THC INPUT (4)		SH. No.	TC0664
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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TC0665-1
(TC0759)

(LYA210A)

(S)

unit NO.	slot NO.	A1 A2	A3 A4	A5 A6	A7 A8	A9 A10	A11 A12	A13 A14	A15 A16 A17
FF1	2								
CTL	CH. NO.	AI 0	AI 1	AI 2	AI 3	AI 4	AI 5	AI 6	AI 7
COMM+MD	signal range/level	0～700degC	0～500degC	0～500degC	0～150degC	0～150degC	0～700degC	0～700degC	0～500degC
logic SH. NO.		TYPE K	TYPE K	TYPE K	TYPE K	TYPE K	TYPE K	TYPE K	TYPE K
TCA961	signal name	REHEAT BOWL INNER	IP TURBINE AFTER 15	IP TURBINE AFTER 18	THRUST BEARING	THRUST BEARING	MSV (B) INNER	CV (B) INNER	HP TURBINE
THC AI		WALL METAL TEMP. (A)	STAGE INNER WALL METAL TEMP. (UPPER)	STAGE INNER WALL METAL TEMP. (UPPER)	FRONT-EN. DEMBEDDED METAL TEMP. (A)	REAR-EN. DEMBEDDED METAL TEMP. (A)	WALL METAL TEMP.	WALL METAL TEMP.	EXHAUST INNER WALL METAL TEMP. (UPPER)

COM THC INPUT (5)			SH. No.	TC0665
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
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TC0666-1
(TC0760)

(LYA210A)

unit NO.	slot NO.	A1 A2	A3 A4	A5 A6	A7 A8	A9 A10	A11 A12	A13 A14	A15 A16 B16 A17 (T)
FF1	2								
CTL	CH. NO.	AI 8	AI 9	AI A	AI B	AI C	AI D	AI E	AI F
COMM+MD	signal range/level	0～700degC	0～500degC	0～500degC		0～700degC	0～500degC	0～150degC	
logic SH. NO.		TYPE K	TYPE K	TYPE K		TYPE K	TYPE K	TYPE K	
TCA961	signal name	HP TURBINE FIRST STAGE INNER WALL METAL TEMP.	IP-T AFTER 21 STAGE INNER WALL METAL TEMP. (UPPER)	IP-T AFTER 21 STAGE INNER WALL METAL TEMP. (LOWER)		MSV INLET STEAM TEMP. (B)	HP TURBINE EXHAUST STEAM TEMP. (B)	LP TURBINE EXHAUST HOOD STEAM TEMP. (B)	RTD
THC AI									

COM THC INPUT (6)			SH. No.	TC0666
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
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TC0667-1
(TC0761)

(LYA210A)

(S)

unit NO.	slot NO.	A1 A2	A3 A4	A5 A6	A7 A8	A9 A10	A11 A12	A13 A14	A15 A16 A17
FF1	3								
CTL	CH. NO.	AI 0	AI 1	AI 2	AI 3	AI 4	AI 5	AI 6	AI 7
COMM+MD	signal range/level	0～700degC	0～500degC	0～500degC	0～150degC	0～510degC	0～700degC	0～700degC	0～500degC
logic SH. NO.		TYPE K	TYPE K	TYPE K	TYPE K	TYPE K	TYPE K	TYPE K	TYPE K
TCA961	signal name	REHEAT BOWL INNER	IP TURBINE AFTER 15	IP TURBINE AFTER 18	THRUST BEARING	THRUST BEARING	MSV (B) OUTER	CV (B) OUTER	HP TURBINE EXHAUST
THC AI		WALL METAL TEMP. (B)	STAGE INNER WALL METAL TEMP. (LOWER)	STAGE INNER WALL METAL TEMP. (LOWER)	FRONT-EN. DEMBEDDED METAL TEMP. (B)	REAR-EN. DEMBEDDED METAL TEMP. (B)	WALL METAL TEMP.	WALL METAL TEMP.	INNER WALL METAL TEMP. (LOWER)

COM THC INPUT (7)		SH. No.	TC0667
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 254 / REV. 3

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TC0668-1
(TC0762)

(LYA210A)

(T)

unit NO.	slot NO.	A1 A2	A3 A4	A5 A6	A7 A8	A9 A10	A11 A12	A13 A14	A15 A16 B16 A17
FF1	3								
CTL	CH. NO.	AI 8	AI 9	AI A	AI B	AI C	AI D	AI E	AI F
COMM+MD	signal range/level	0～700degC	0～150degC	0～150degC		0～700degC	0～500degC	0～150degC	
logic SH. NO.		TYPE K	TYPE K	TYPE K		TYPE K	TYPE K	TYPE K	
TCA961	signal name	HP TURBINE FIRST STAGE OUTER WALL METAL TEMP.	NO.3 BEARING OUTLET OIL TEMP.	NO.4 BEARING OUTLET OIL TEMP.		MSV INLET STEAM TEMP. (C)	HP TURBINE EXHAUST STEAM TEMP. (C)	LP TURBINE EXHAUST HOOD STEAM TEMP. (C)	RTD
THC AI									

COM THC INPUT (8)		SH. No.	TC0668
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 255 / REV. 3

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TC0669-1
(TC0763)

(LYA220A)

unit NO.	slot NO.	A1 A2 B1	A3 A4 B3	A5 A6 B5	A7 A8 B7	A9 A10 B9	A11 A12 B11	A13 A14 B13	A15 A16 B15 (S)
FF1	4								
CTL	CH. NO.	AI 0	AI 1	AI 2	AI 3	AI 4	AI 5	AI 6	AI 7
COMM+MD	signal range/level	0～100degC	0～100degC	0～100degC	0～120degC	0～120degC	0～120degC	0～120degC	0～120degC
logic SH. NO.		PT100	PT100	PT100	PT100	PT100	PT100	PT100	PT100
TCA970	signal name	LUBE OIL COOLER	LUBE OIL COOLER	LUBE OIL COOLER	GEN. STATOR	GEN. STATOR	GEN. STATOR	GEN. STATOR	GEN. STATOR
RTD AI		INLET OIL TEMP.	OUTLET OIL TEMP(A)	OUTLET OIL TEMP(B)	COIL TEMP. (A)	COIL TEMP. (B)	COIL TEMP. (C)	COIL TEMP. (D)	COIL TEMP. (E)

COM THC INPUT (9)			SH. No.	TC0669
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
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TC0670-1
(TC0764)

(LYA220A)

unit NO.	slot NO.	A1 A2 B1	A3 A4 B3	A5 A6 B5	A7 A8 B7	A9 A10 B9	A11 A12 B11	A13 A14 B13	A15 A16 B15 (T)
FF1	4								
CTL	CH. NO.	AI 8	AI 9	AI A	AI B	AI C	AI D	AI E	AI F
COMM+MD	signal range/level	0～200degC	0～200degC	0～200degC	0～200degC	0～150degC	0～120degC		
logic SH. NO.		PT100	PT100	PT100	PT100	PT100	PT100		
TCA970	signal name	GEN. COOLING GAS	GEN. COOLING GAS	GEN. COOLING GAS	GEN. COOLING GAS	AC EXCITER	GEN. STATOR		
RTD AI		AIR COOLER OUTLET TEMP. (A)	AIR COOLER OUTLET TEMP. (B)	AIR COOLER OUTLET TEMP. (C)	AIR COOLER OUTLET TEMP. (D)	OUTLET AIR	COIL TEMP (F)		

COM THC INPUT (10)		SH. No.	TC0670
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 257 / REV. 3

TCS-6

(LYD105A)

DO / logic SH. NO.	signal name	CRV (A) BELLOW SEAT SECONDARY DRAIN VALVE CLOSE COMMAND	CRV (A) BELOW SEAT SECONDRY DRAIN VALVE CLOSE COMMAND	CV (A) ABOVE SEAT SECONDRY DRAIN VALVE OPEN COMMAND	CV (A) ABOVE SEAT SECONDRY DRAIN VALVE CLOSE COMMAND	CV (B) BELLOW SEAT SEC. WARMING DRAIN VLV CLOSE CMD.	CV (B) BELLOW SEAT SEC. WARMING DRAIN VLV OPEN CMD.	MSV (A) ABOVE SEAT SECONDRY DRAIN VALVE OPEN COMMAND	MSV (A) ABOVE SEAT SECONDRY DRAIN VALVE CLOSE COMMAND
TCA980	signal range/level								
CTL									
COMM+MD	CH. NO.	DO 0	DO 1	DO 2	DO 3	DO 4	DO 5	DO 6	DO 7
unit NO.	slot NO.								
FE1	C	21 3 20	18	16	14	12	10	8	6

(S)

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TC0671-1
(TC0821)

COM DIGITAL OUTPUT(1)			SH. No.	TC0671
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
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TCS-6

(LYD105A)

DO / logic SH. NO.	signal name	STEAM SEAL FEAD INLET VALVE CLOSE COMMAND	STEAM SEAL FEAD INLET VALVE OPEN COMMAND	TURBINE GLAND STEAM LINE DRAIN VALVE OPEN COMMAND	TURBINE GLAND STEAM LINE DRAIN VALVE CLOSE COMMAND	STEAM SEAL FEED VALVE CLOSE		AMB. COLD MODE TEMP.	MODE JUGEMENT INSTRUMENT NORMAL
TCA980	signal range/level								
CTL									
COMM+MD	CH. NO.	DO 8	DO 9	DO A	DO B	DO C	DO D	DO E	DO F
unit NO.	slot NO.								
FE1	C	22 4 19	17	15	13	11	9	7	5

(S)

TC0671-1
(TC0822)

TCS-8

COM DIGITAL OUTPUT(2)		SH. No.	TC0672
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 259 / REV. 3

TCS-6

(LYD105A)

DO / logic SH. NO.	signal name	TURBINE START-UP AMBIENT COLD MODE	TURBINE START-UP COLD 1 MODE	TURBINE START-UP COLD 2 MODE	TURBINE START-UP HOT MODE	TURBINE START-UP WARM MODE	TURBINE LOAD HOLD	[redacted]	3-41 OPEN COMMAND
TCA980	signal range/level								
CTL									
COMM+MD	CH. NO.	DO 10	DO 11	DO 12	DO 13	DO 14	DO 15	DO 16	DO 17
unit NO.	slot NO.								
FE1	C	21 3 20	18	16	14	12	10	8	6 (T)

TCS-8

TC0673-1 (TC0823)

COM DIGITAL OUTPUT(3)		SH. No.	TC0673
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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TCS-6

(LYD105A)

DO / logic SH. NO.	signal name		41 OPEN COMAND FROM ATS	PSS ON	PSS OFF	PF CONTROL ON	PF CONTROL OFF	TURBINE RATED SPEED	ST TRIP
TCA980	signal range/level								
CTL									
COMM+MD	CH. NO.	DO 18	DO 19	DO 1A	DO 1B	DO 1C	DO 1D	DO 1E	DO 1F
unit NO.	slot NO.								
FE1	C	22 4 19	17	15	13	11	9	7	5

(T)

TCS-8

TC0673-1 (TC0824)

COM DIGITAL OUTPUT(4)				SH. No.	TC0674
SIGNATURE		DATE	TITLE		
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM		
CHKD.	T. Kawata	2014-04-18			
APPD.	Y. Fukushima	2014-04-18			
REGD.	Hitachi, Ltd. Tokyo Japan		DWG. No. 331DM07733	PAGE 261	REV. 3

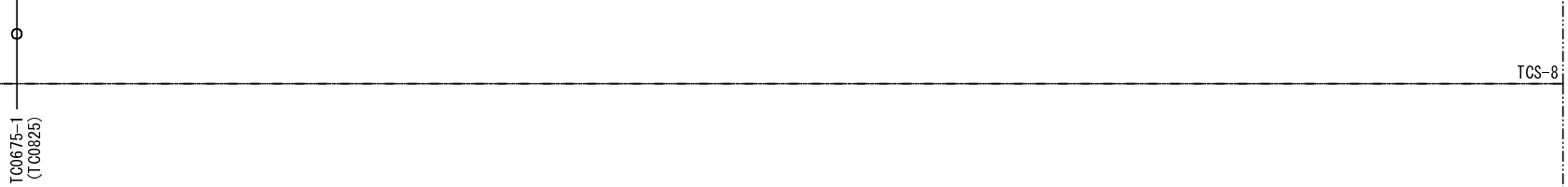
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TCS-6

(LYD105A)

DO / logic SH. NO.	signal name	CRV (B) BELLOW SEAT SECONDARY DRAIN VALVE OPEN COMMAND	CRV (B) BELOW SEAT SECONDRY DRAIN VALVE CLOSE COMMAND				MSV (B) ABOVE SEAT SECONDRY DRAIN VALVE OPEN COMMAND	MSV (B) ABOVE SEAT SECONDRY DRAIN VALVE CLOSE COMMAND	HP TURBINE WARMING LINE DRAIN VALVE CLOSE COMMAND
TCA981	signal range/level								
CTL									
COMM+MD	CH. NO.	DO 0	DO 1	DO 2	DO 3	DO 4	DO 5	DO 6	DO 7
unit NO.	slot NO.								
FE1	D	21 3 20	18	16	14	12	10	8	6

(S)



TC0675-1 (TC0825)

TCS-8

COM DIGITAL OUTPUT(5)		SH. No.	TC0675
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 262 / REV. 3

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(LYD105A)

DO / logic SH. NO.	signal name	HP TURBINE WARMING LINE DRAIN VALVE OPEN COMMAND	HP TURBINE WARMING VALVE CLOSE COMMAND	HP TURBINE WARMING VALVE OPEN COMMAND	CV (B) ABOVE SEAT SECONDARY DRAIN VALVE OPEN COMMAND	CV (B) ABOVE SEAT SECONDRY DRAIN VALVE CLOSE COMMAND	EXCITATION START	AUTO MODE	MANUAL MODE
TCA981	signal range/level								
CTL									
COMM+MD	CH. NO.	DO 8	DO 9	DO A	DO B	DO C	DO D	DO E	DO F
unit NO.	slot NO.								
FE1	D	22 4 19	17	15	13	11	9	7	5 (S)

TC0675-1 (TC0826)

TCS-8

COM DIGITAL OUTPUT(6)		SH. No.	TC0676
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 263 / REV. 3

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(LYD105A)

DO / logic SH. NO.	signal name	OIL PURIFIER PUMP START COMMAND	SPACE HEATER START COMMAND (AC EXCITER)	SPACE HEATER START COMMAND (GEN)	MAIN OIL TANK VAPUR EXTRACTOR START COMMAND	GLAND STEAM EXHAUSTER START COMMAND	REACTIVE POWER ON	REACTIVE POWER OFF	EXCITATION START
TCA981	signal range/level								
CTL									
COMM+MD	CH. NO.	DO 10	DO 11	DO 12	DO 13	DO 14	DO 15	DO 16	DO 17
unit NO.	slot NO.								
FE1	D	21 3 20	18	16	14	12	10	8	6

(T)

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TC0677-1
(TC0827)

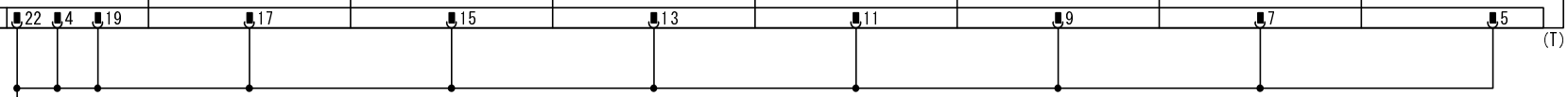
COM DIGITAL OUTPUT(7)			SH. No.	TC0677
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
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TCS-6

(LYD105A)

DO / logic SH. NO.	signal name			PLU1 TEST	PLU2 TEST	PLU3 TEST	EOS1 TEST	EOS2 TEST	EOS3 TEST
TCA981	signal range/level								
CTL									
COMM+MD	CH. NO.	DO 18	DO 19	DO 1A	DO 1B	DO 1C	DO 1D	DO 1E	DO 1F
unit NO.	slot NO.								
FE1	D	22 4 19	17	15	13	11	9	7	5 (T)



TCS-8

TC0677-1
(TC0828)

COM DIGITAL OUTPUT(8)			SH. No.	TC0678
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
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TCS-6

(LYD105A)

DO / logic SH. NO.	signal name	COLD REHEAT STEAM NON RETURN VALVE ENERGIZE COMMAND	NO.3 EXTRACTION STEAM NON RETURN VALVE ENERGIZE COMMAND	NO.2 EXTRACTION STEAM NON RETURN VALVE ENERGIZE COMMAND	STEAM SEAL DIVERTING VALVE SOLENOID VALVE ENERGIZE COMMAND				
TCA982	signal range/level								
CTL									
COMM+MD	CH. NO.	DO 0	DO 1	DO 2	DO 3	DO 4	DO 5	DO 6	DO 7
unit NO.	slot NO.								
FE1	E	21 3 20	18	16	14	12	10	8	6

(S)

TCS-8

TC0679-1
(TC0829)

COM DIGITAL OUTPUT(9)			SH. No.	TC0679
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
REGD.	Hitachi, Ltd. Tokyo Japan		DWG. No. 331DM07733	PAGE 266 / REV. 3

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(LYD105A)

DO / logic SH. NO.	signal name									
TCA982	signal range/level									
CTL										
COMM+MD	CH. NO.	DO 8	DO 9	DO A	DO B	DO C	DO D	DO E	DO F	
unit NO.	slot NO.									
FE1	E	22 4 19	17	15	13	11	9	7	5	(S)

TCS-8

TC0679-1
(TC0830)

COM DIGITAL OUTPUT(10) | SH. No. TC0680

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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TCS-6

(LYD105A)

DO / logic SH. NO.	signal name	LP EXHAUST HOOD WATER SPRAY VLV SOLENOID VLV ENERGIZE CMD.	NO.5 EXTRACTION STM NON RETURN VLV ENERGIZE COMMAND	NO.3 EXTRACTION STM PRI. NON RETURN VLV ENERGIZE CMD.	NO.4 EXTRACTION STEAM NON RETURN VALVE ENERGIZE COMMAND				
TCA982	signal range/level								
CTL									
COMM+MD	CH. NO.	DO 10	DO 11	DO 12	DO 13	DO 14	DO 15	DO 16	DO 17
unit NO.	slot NO.								
FE1	E	21 3 20	18	16	14	12	10	8	6

(T)

TCS-8

TC0681-1
(TC0831)

COM DIGITAL OUTPUT(11)			SH. No.	TC0681
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 268	REV. 3

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(LYD105A)

DO / logic SH. NO.	signal name									
TCA982	signal range/level									
CTL										
COMM+MD	CH. NO.	DO 18	DO 19	DO 1A	DO 1B	DO 1C	DO 1D	DO 1E	DO 1F	
unit NO.	slot NO.									
FE1	E	22 4 19	17	15	13	11	9	7	5	(T)

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TC0681-1
(TC0832)

COM DIGITAL OUTPUT(12)			SH. No.	TC0682
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 269	REV. 3

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TCS-6

(LYA100A)

AO / logic SH. NO.	signal name	LUBE OIL COOLER OUTLET OIL TEMP. (A)	STEAM SEEL FEED VALVE POSITION DEMAND	HP TURBINE WARMING VALVE POSITION	STEAM SEAL FEED VALVE MANIPULATION VALUE	GEN. CURRENT (A)	GEN. VOLTAGE (A-B)		
TCA990	signal range/level	0～100degC	100～0%	0～100%	0～100%	0～10kA	0～18kV		
CTL		4～20mA	4～20mA	4～20mA	4～20mA	4～20mA	4～20mA		
COMM+MD	CH. NO.	AO 0	AO 1	AO 2	AO 3	AO 4	AO 5	AO 6	AO 7
unit NO.	slot NO.								
FF1	D	33 34	29 30	25 26	21 22	17 18	13 14	9 10	5 6

(T)

TCS-7

TC791-1
(TC0791)

COM ANALOG OUTPUT(1)		SH. No.	TC0691
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 270 / REV. 3

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TCS-6

(LYA100A)

AO / logic SH. NO.	signal name	LUBE OIL COOLER OUTLET OIL TEMP. (B)	SPUV POSITION DEMAND	GEN. MVAR	GEN. CURRENT (B)	GEN. VOLTAGE (B-C)	GEN. POWER FACTOR	FREQUENCY	
TCA990	signal range/level	0～100degC	0～100%	Lead240～0～Lag240	0～10kA	0～18kV	Lead0.5～1～Lag0.5	55～65Hz	
CTL		4～20mA	4～20mA	4～20mA	4～20mA	4～20mA	4～20mA	4～20mA	
COMM+MD	CH. NO.	AO 0	AO 1	AO 2	AO 3	AO 4	AO 5	AO 6	AO 7
unit NO.	slot NO.								
FF1	E	33 34	29 30	25 26	21 22	17 18	13 14	9 10	5 6

(T)

TC0692-1
(TC0792)

TCS-7

COM ANALOG OUTPUT(2)			SH. No.	TC0692
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
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TCS-6

(LYA100A)

AO / logic SH. NO.	signal name	GEN. CURRENT (C)	GEN. CURRENT (C-A)						
TCA990	signal range/level	0～10kA	0～18kV						
CTL		4～20mA	4～20mA						
COMM+MD	CH. NO.	AO 0	AO 1	AO 2	AO 3	AO 4	AO 5	AO 6	AO 7
unit NO.	slot NO.								
FE1	F	33 34	29 30	25 26	21 22	17 18	13 14	9 10	5 6

(T)

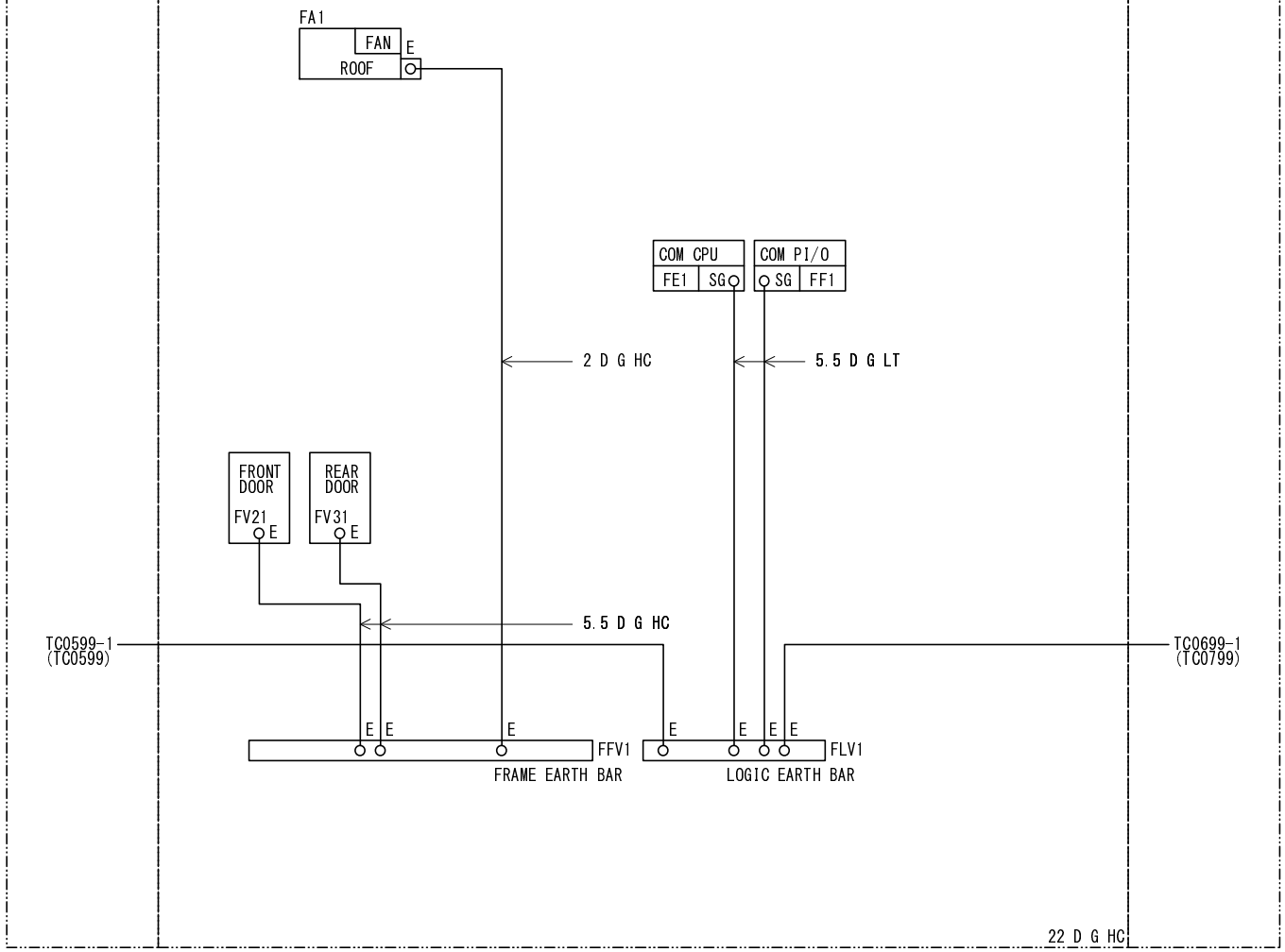
TCS-7

TC0693-1
(TC0793)

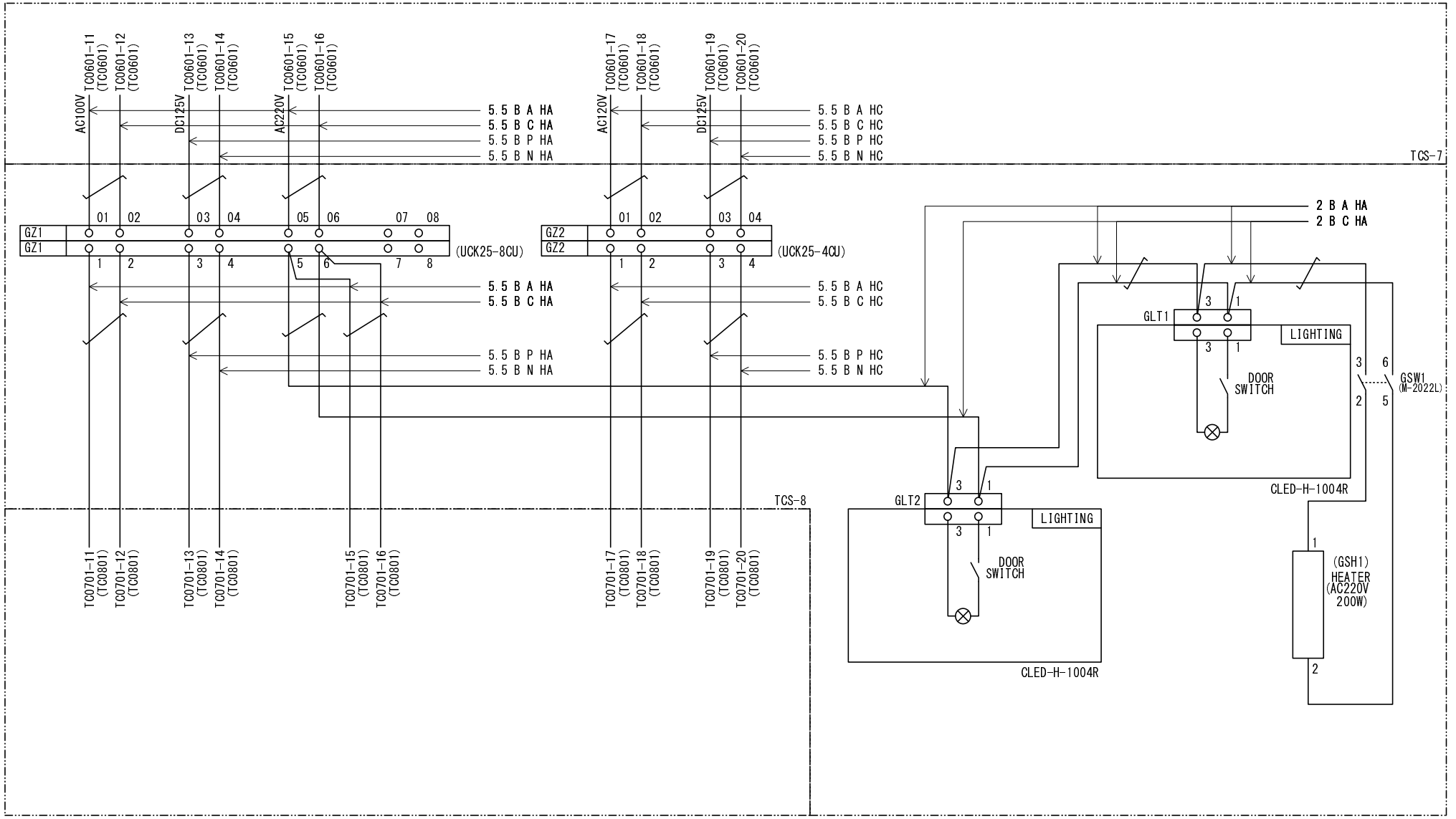
COM ANALOG OUTPUT(3)			SH. No.	TC0693
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 271A	REV. 3

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TCS-5 TCS-6 TCS-7



POWER SUPPLY SYSTEM EARTH CIRCUIT		SH. No.	TC0699
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 272 / REV. 3

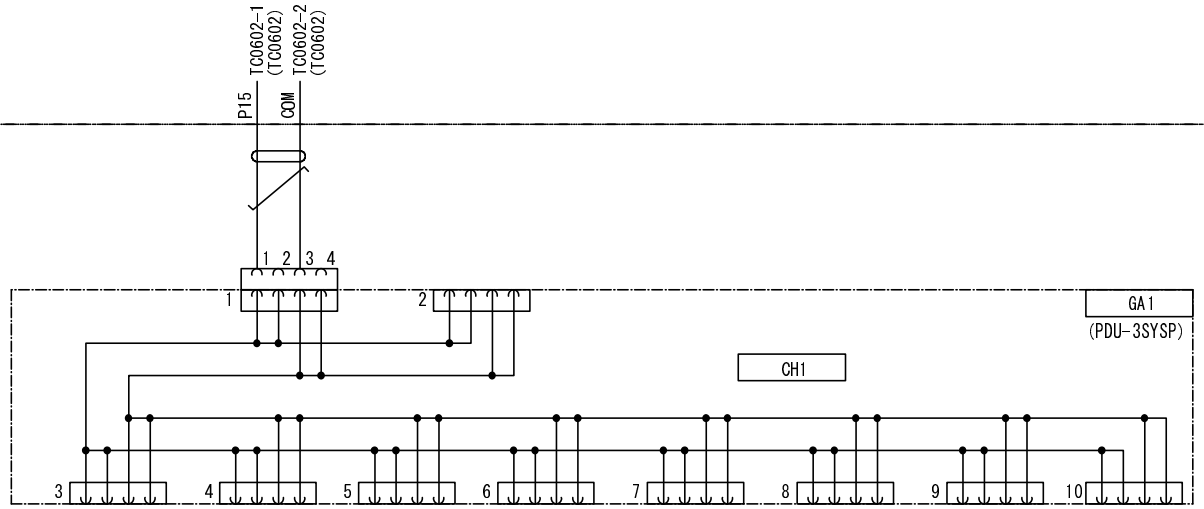


POWER SUPPLY SYSTEM(1)		SH. No.	TC0701
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 273 / REV. 3

2016/07/28 TC0701 XSMM-TC

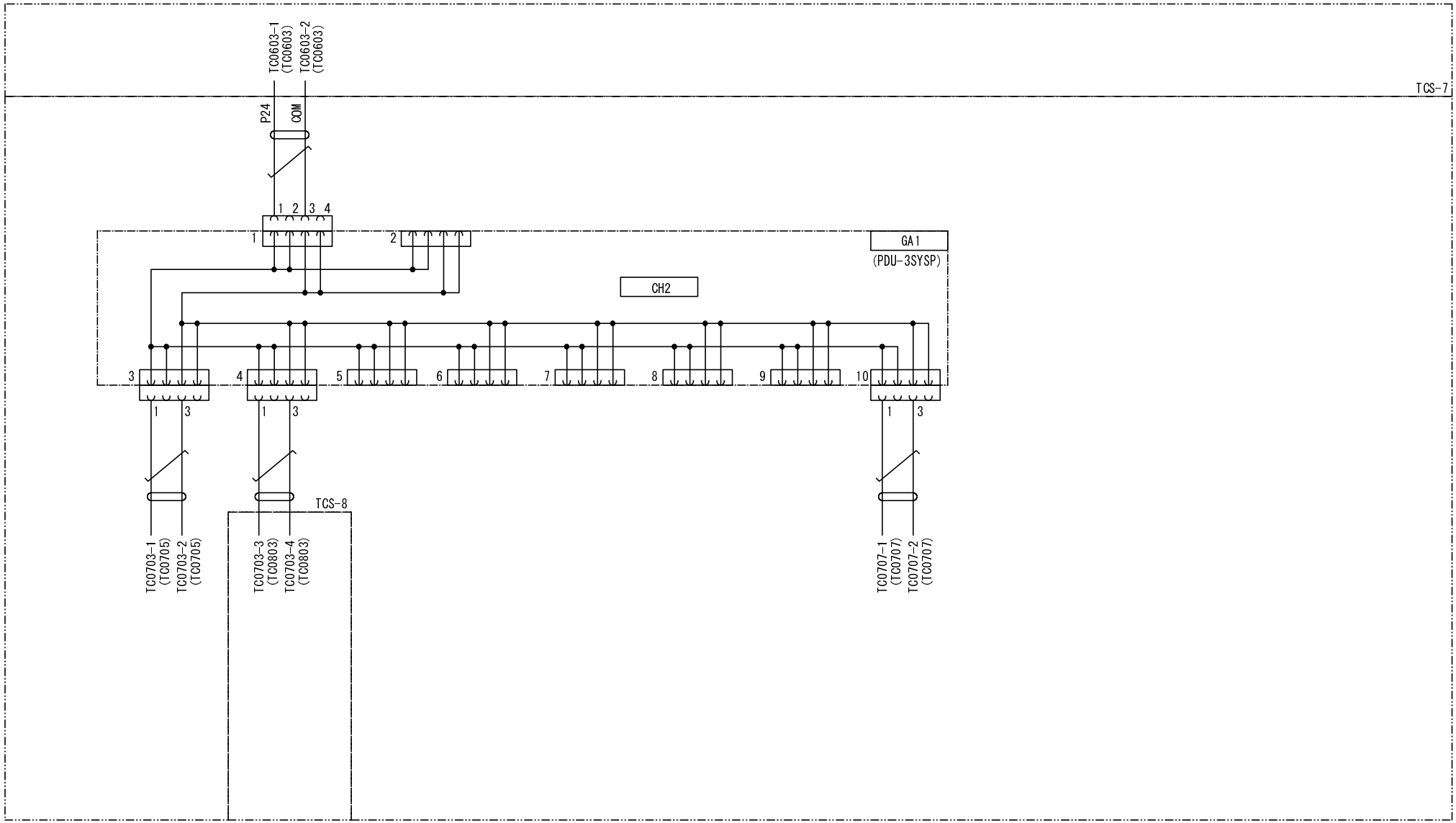
TCS-6

TCS-7

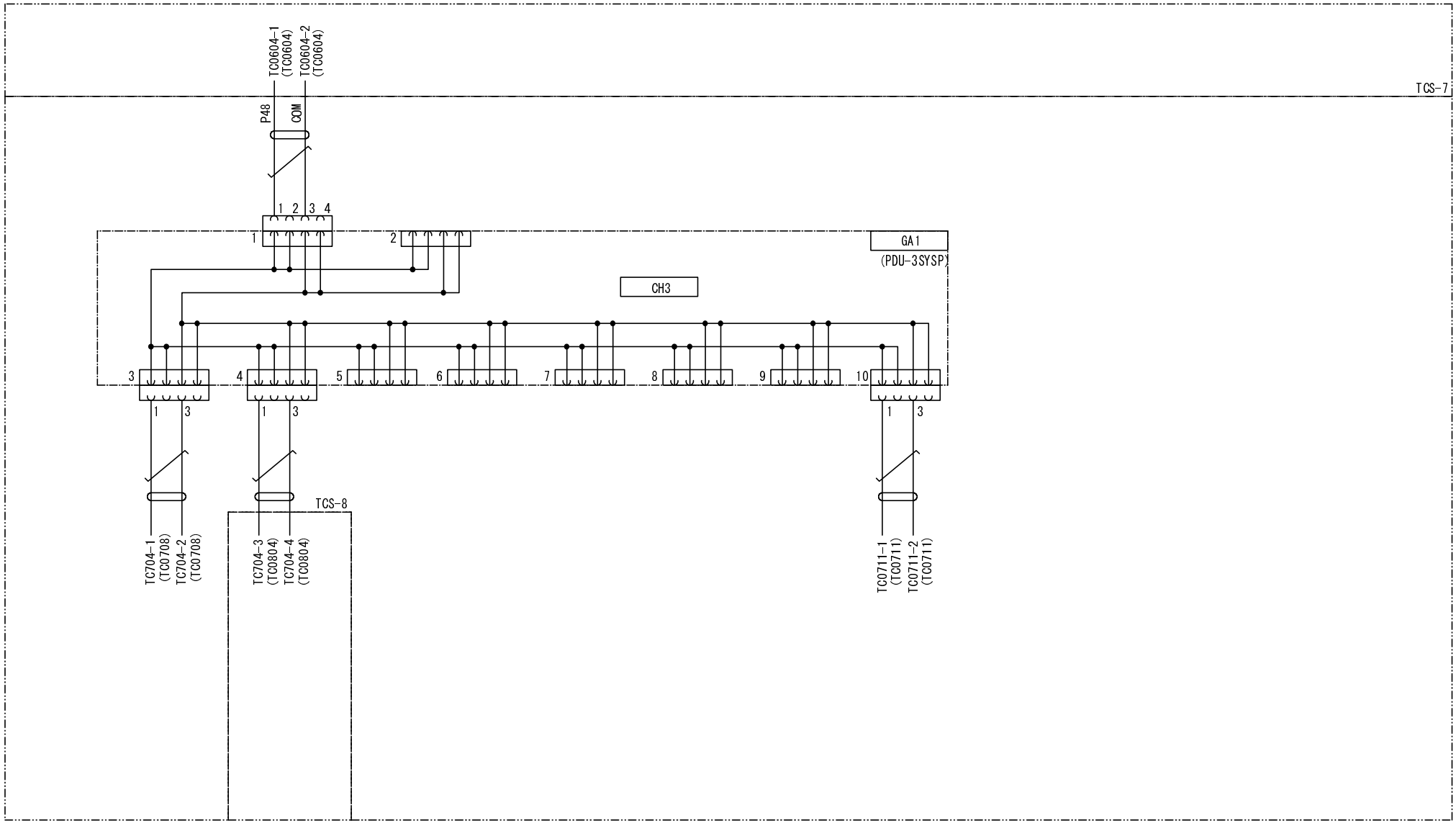


POWER SUPPLY SYSTEM(2)		SH. No.	TC0702
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 274	REV. 3

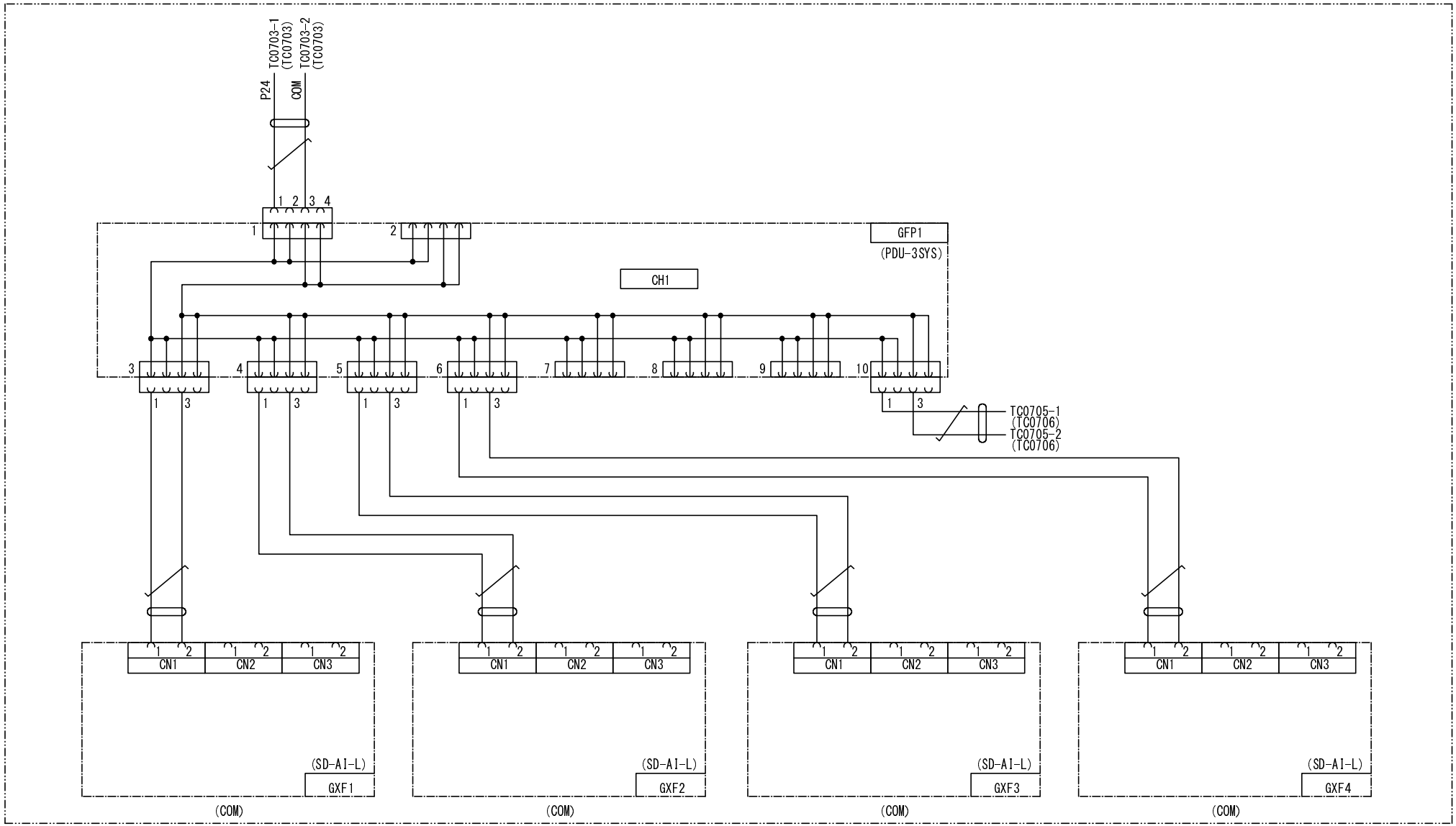
2016/07/28 TC0702 XSMM-TC



POWER SUPPLY SYSTEM(3)			SH. No.	TC0703
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 275	REV. 3

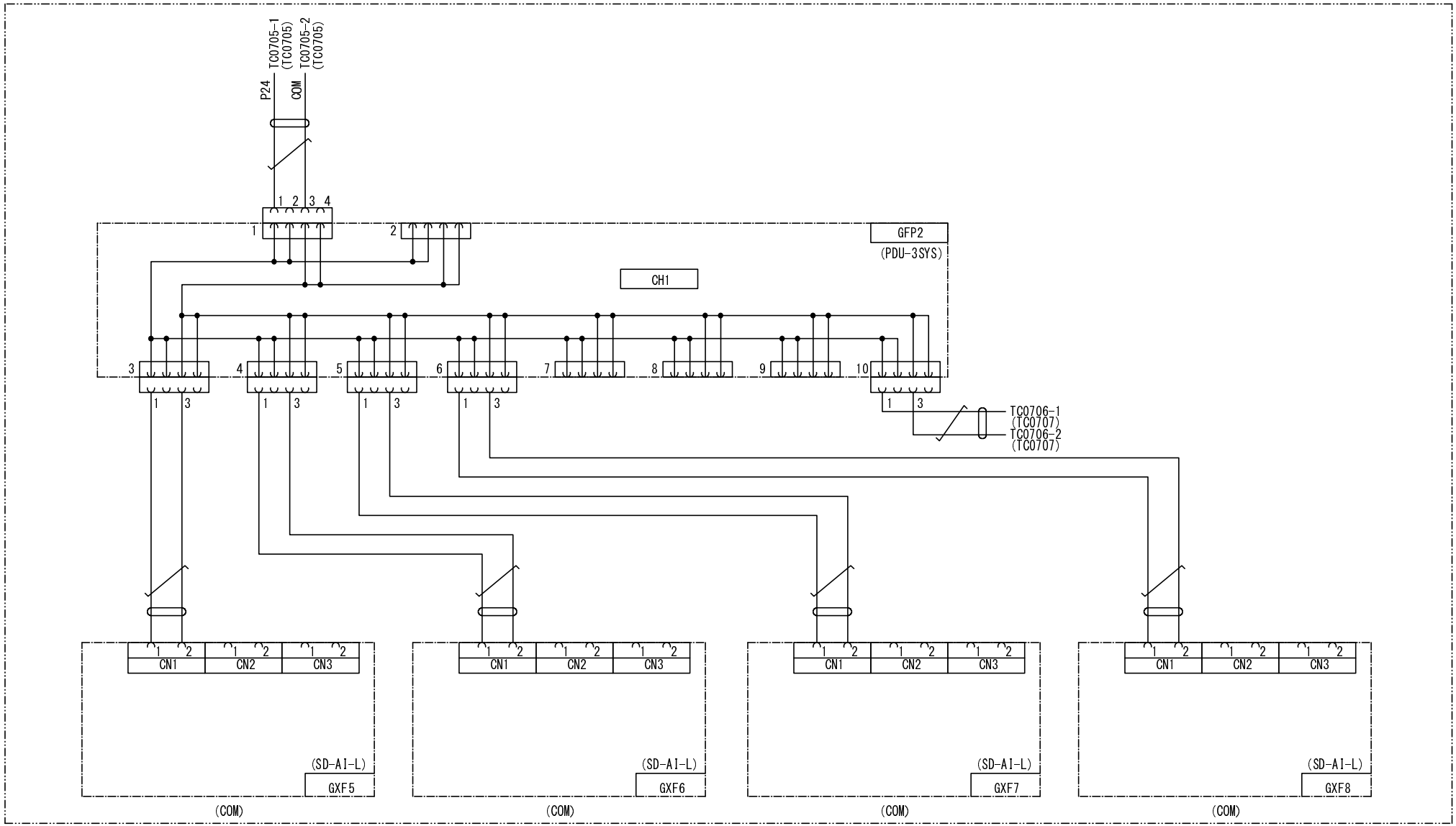


POWER SUPPLY SYSTEM(4)		SH. No.	TC0704
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 276	REV. 3



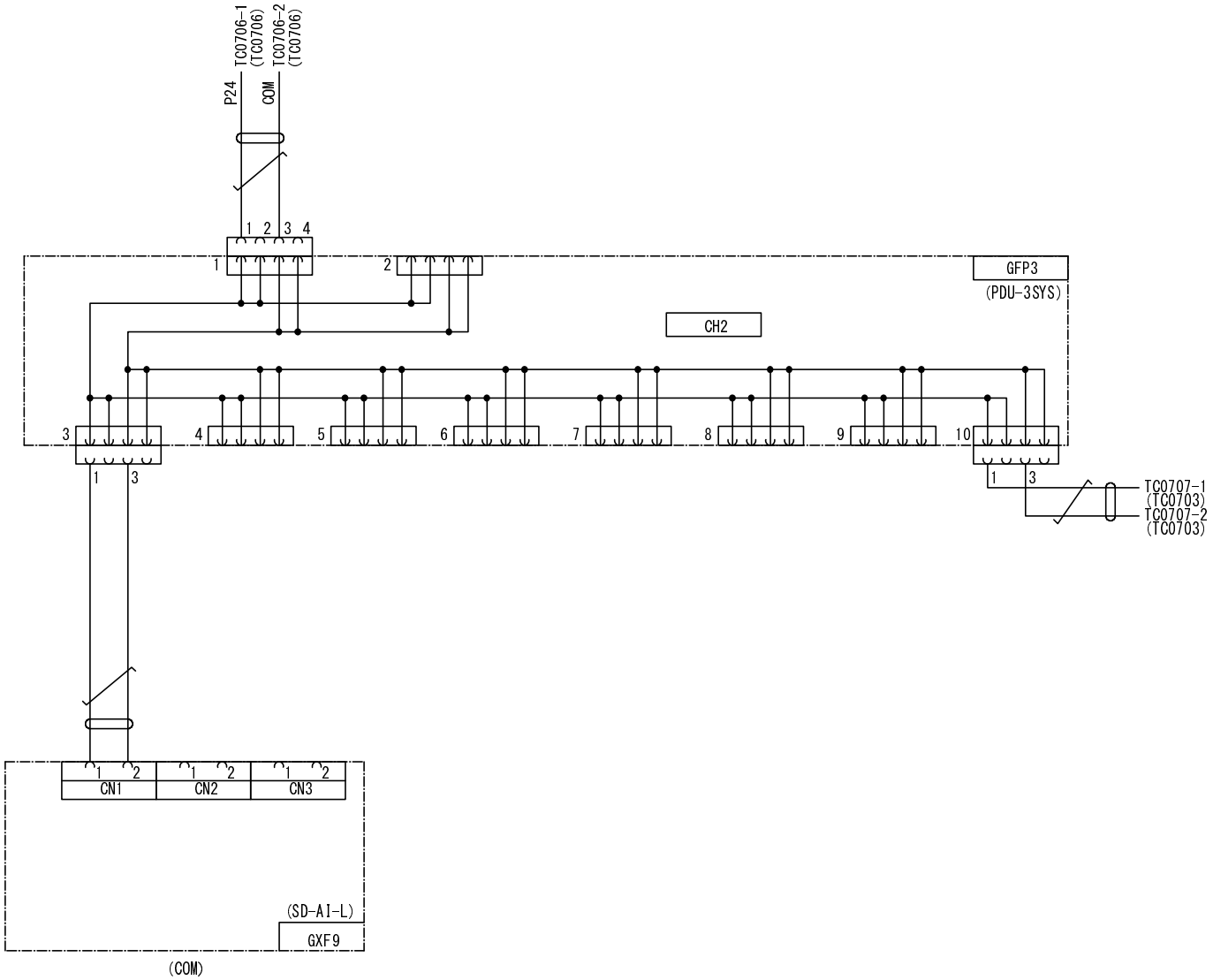
POWER SUPPLY SYSTEM(5)		SH. No.	TC0705
SIGNATURE		DATE	TITLE
DWN.	(S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T.Kawata	2014-04-18	
APPD.	Y.Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 277 / REV. 3

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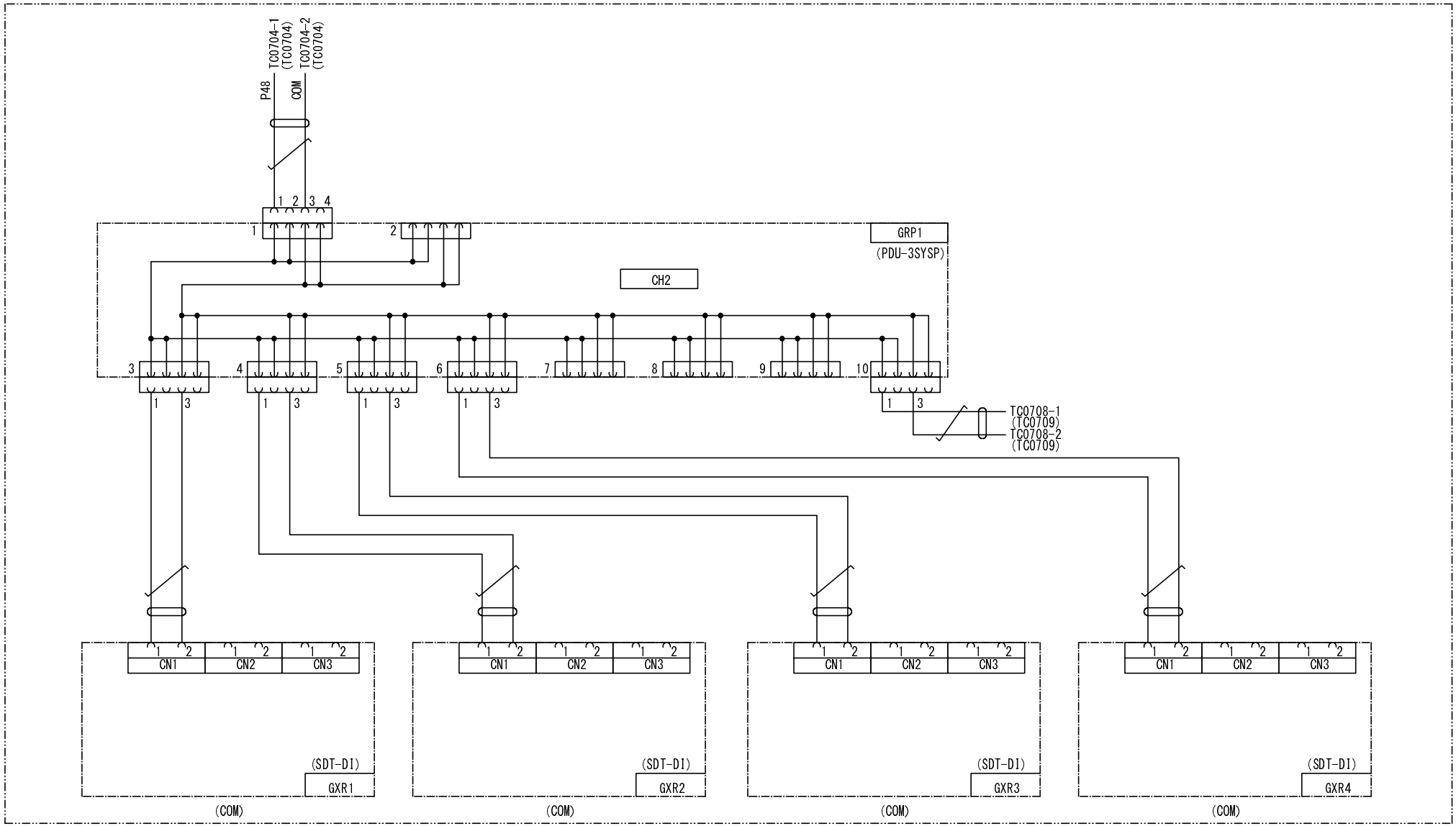


POWER SUPPLY SYSTEM(6)		SH. No.	TC0706
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
REGD. Hitachi,Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 278	REV. 3

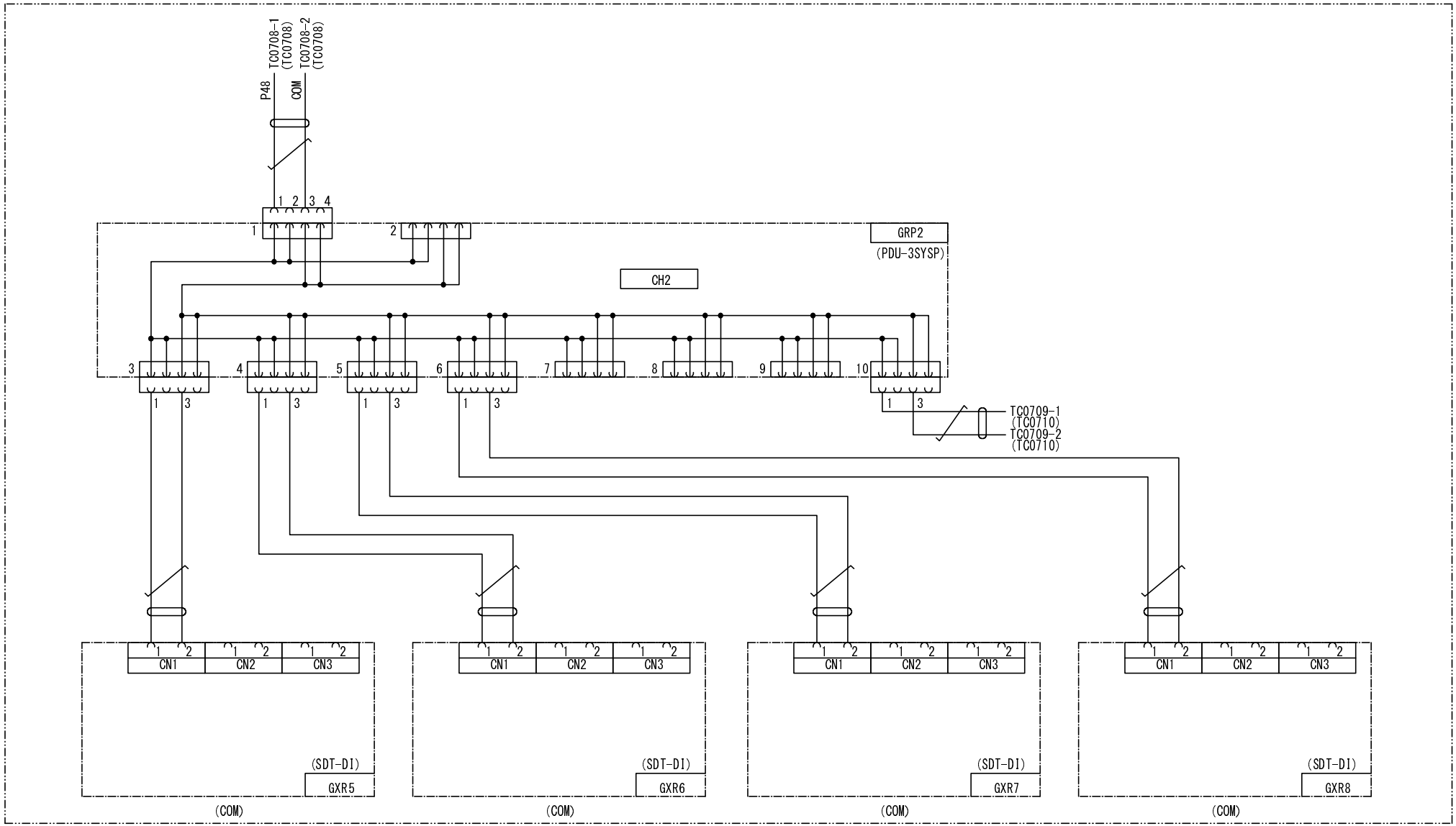
TCS-7



POWER SUPPLY SYSTEM(7)		SH. No.	TC0707
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
REGD. Hitachi,Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 279	REV. 3

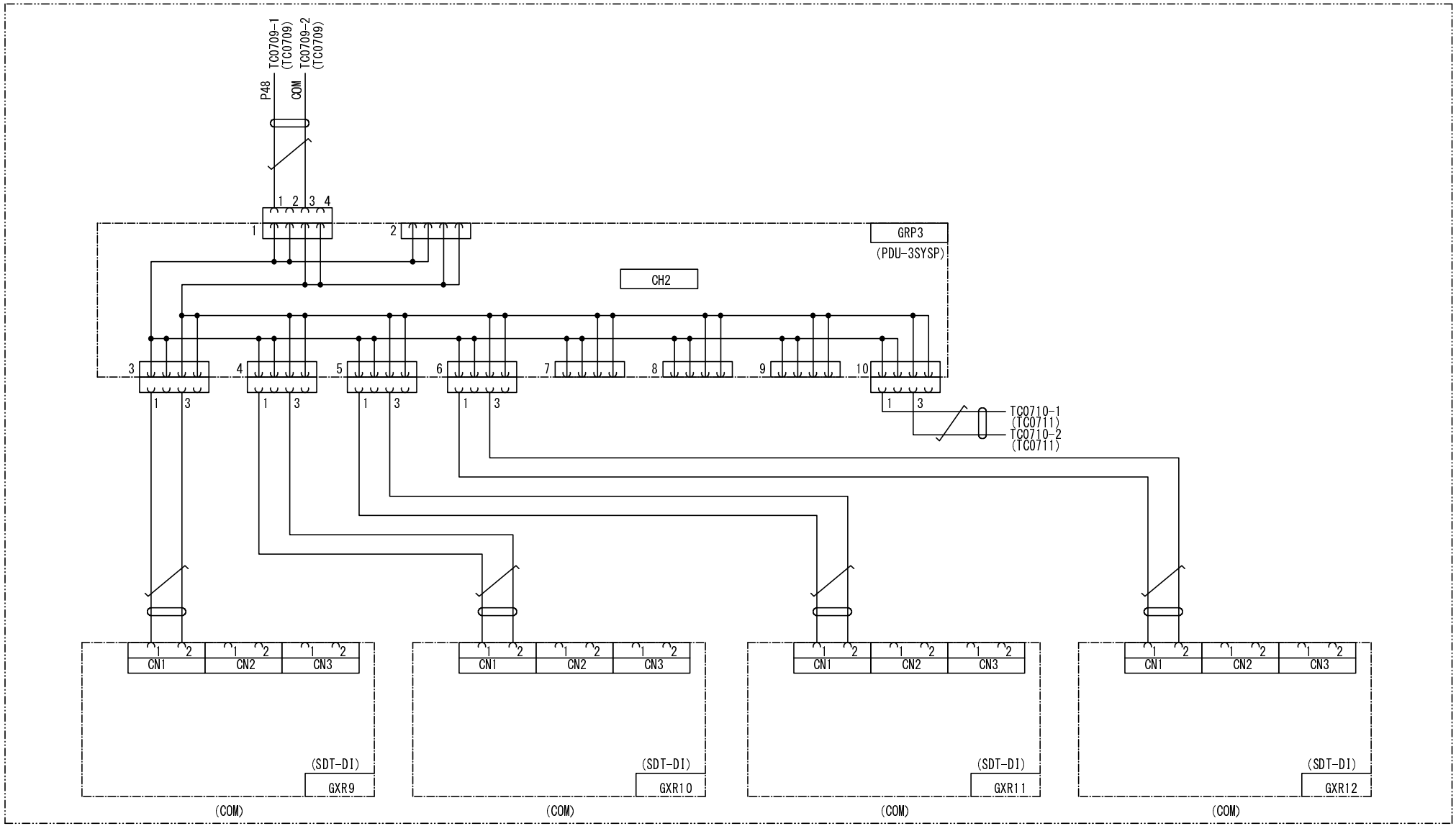


POWER SUPPLY SYSTEM(8)		SH. No.	TC0708
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
REGD. Hitachi,Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 280	REV. 3



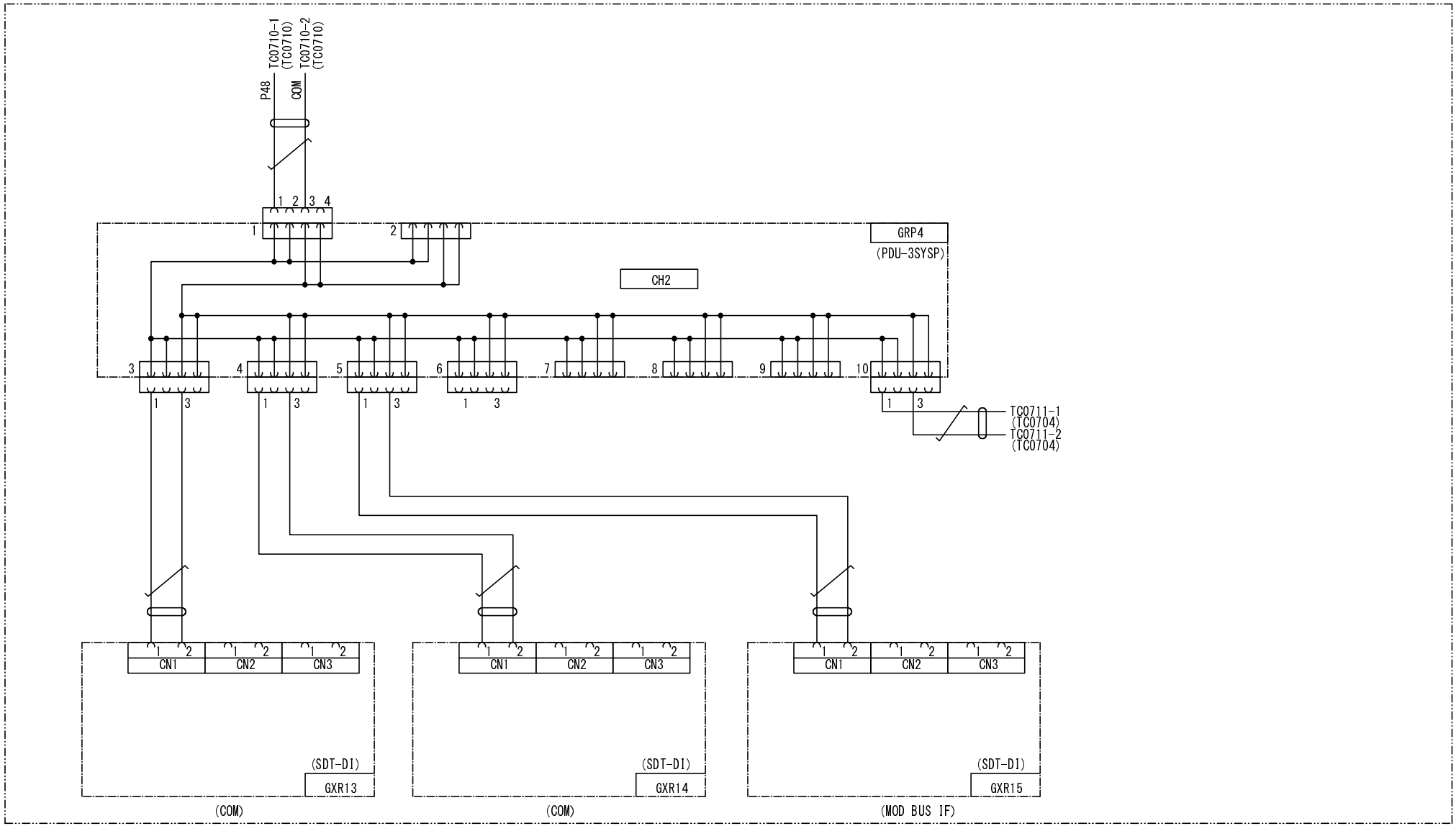
POWER SUPPLY SYSTEM(9)		SH. No.	TC0709
SIGNATURE		DATE	TITLE
DWN.	(S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T.Kawata	2014-04-18	
APPD.	Y.Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 281 / REV. 3

TCS-7



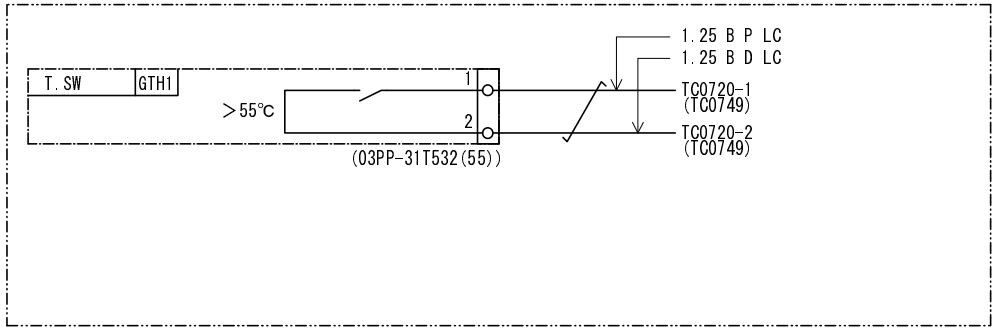
POWER SUPPLY SYSTEM(10)		SH. No.	TC0710
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
REGD. Hitachi,Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 282	REV. 3

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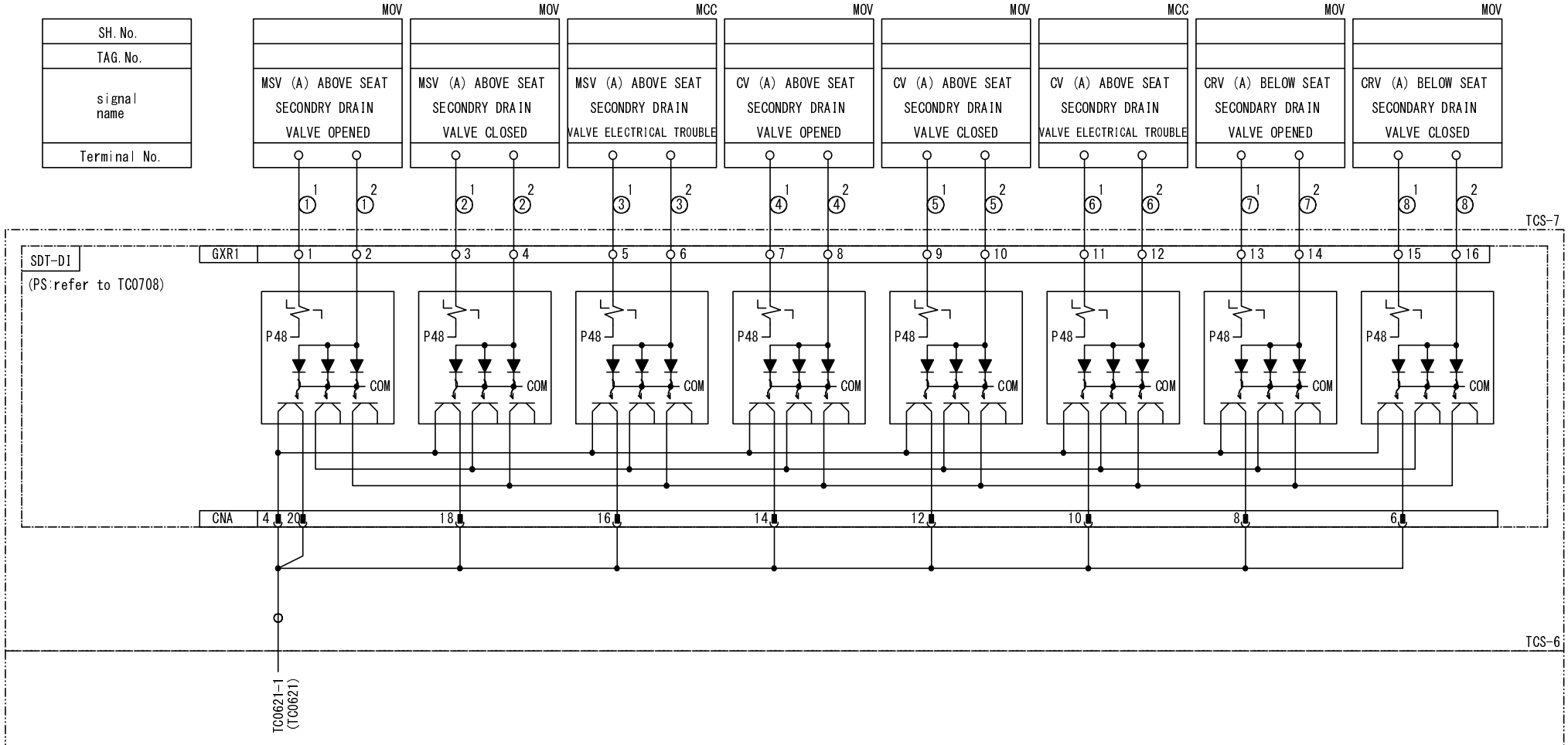


POWER SUPPLY SYSTEM(11)		SH. No.	TC0711
SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 283 / REV. 3

TCS-7



DIAGNOSIS CIRUCUIT		SH. No.	TC0720
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 284	REV. 3



IN-DEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	MOV	TCS-7			
⑦		2/2	MOV	TCS-7			
⑥		2/2	MCC	TCS-7			
⑤		2/2	MOV	TCS-7			
④		2/2	MOV	TCS-7			
③		2/2	MCC	TCS-7			
②		2/2	MOV	TCS-7			
①		2/2	MOV	TCS-7			

EXTERNAL INPUT COM DI(1)

SH. No. TC0721

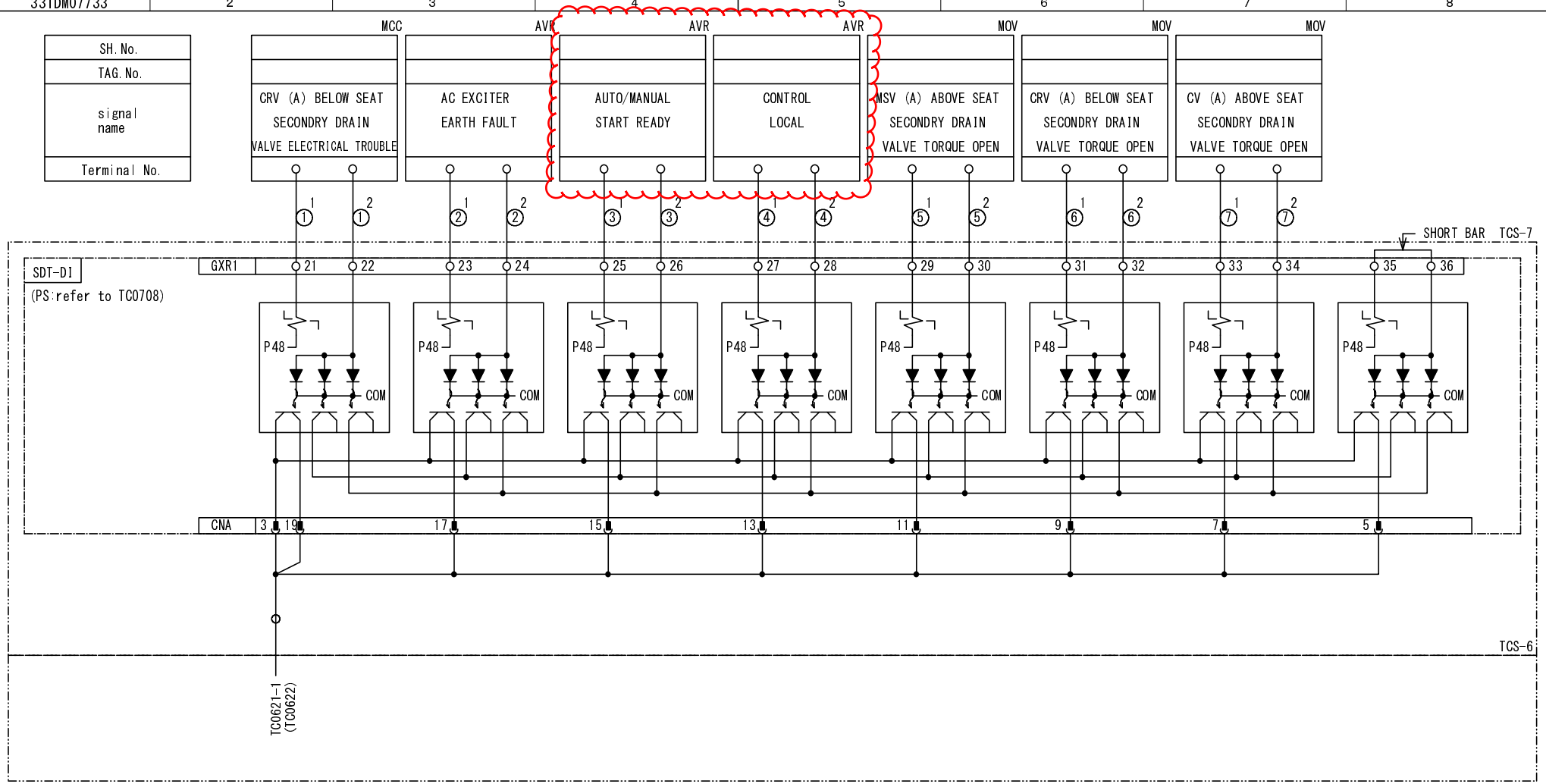
	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733 | PAGE 285 | REV. 3

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⑦		2/2	MOV	TCS-7			
⑥		2/2	MOV	TCS-7			
⑤		2/2	MOV	TCS-7			
④		2/2	AVR	TCS-7			
③		2/2	AVR	TCS-7			
②		2/2	AVR	TCS-7			
①		2/2	MCC	TCS-7			
INDEX	CABLE NO.	CORE NO.	CABLE CONNECTION FROM	CABLE CONNECTION TO	RELATED SH. NO. FROM	RELATED SH. NO. TO	REMARK

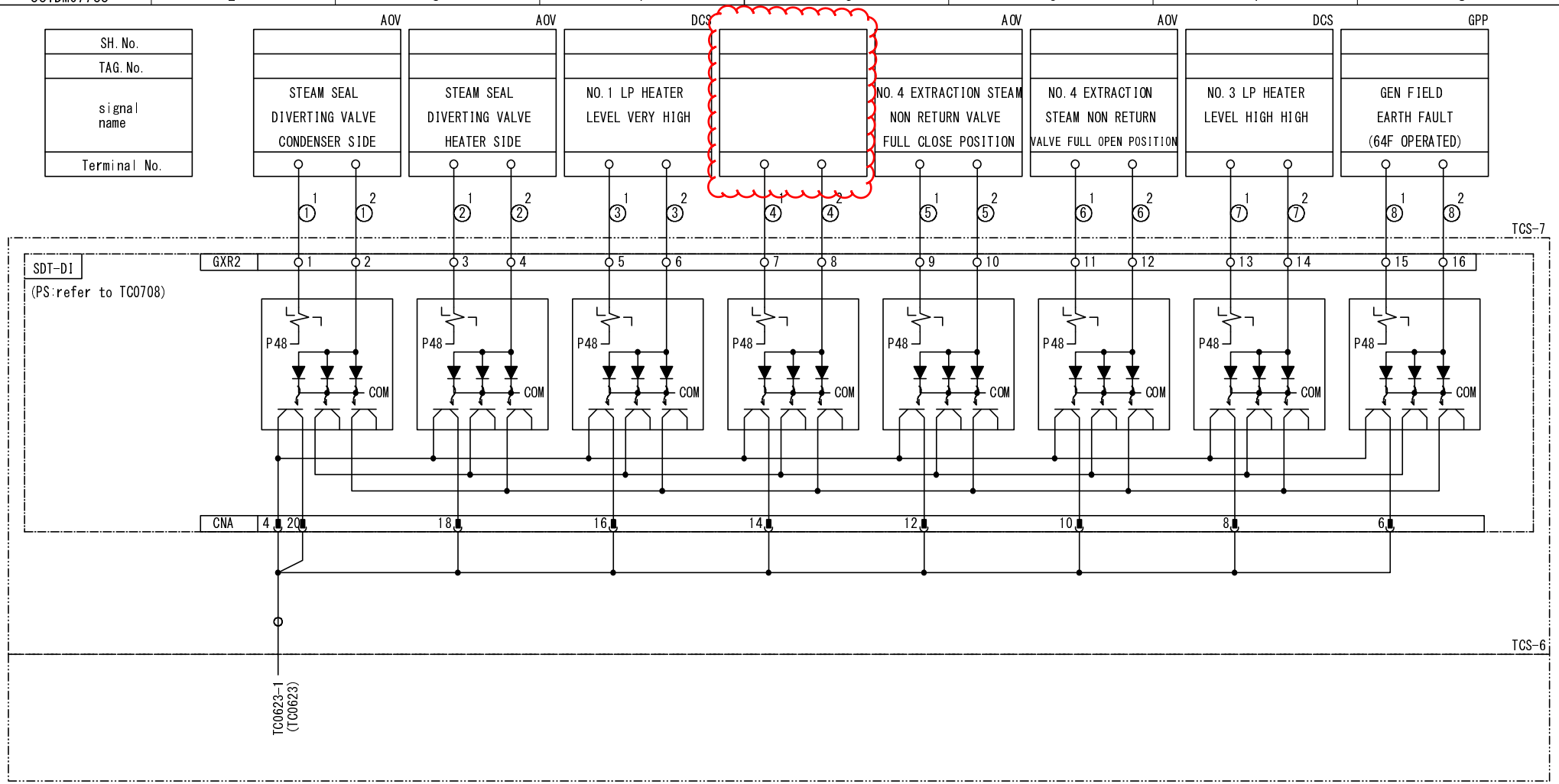
EXTERNAL INPUT COM DI(2) — SH. No. TC0722

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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IN-DEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	GPP	TCS-7			
⑦		2/2	DCS	TCS-7			
⑥		2/2	AOV	TCS-7			
⑤		2/2	AOV	TCS-7			
④		2/2		TCS-7			
③		2/2	DCS	TCS-7			
②		2/2	AOV	TCS-7			
①		2/2	AOV	TCS-7			

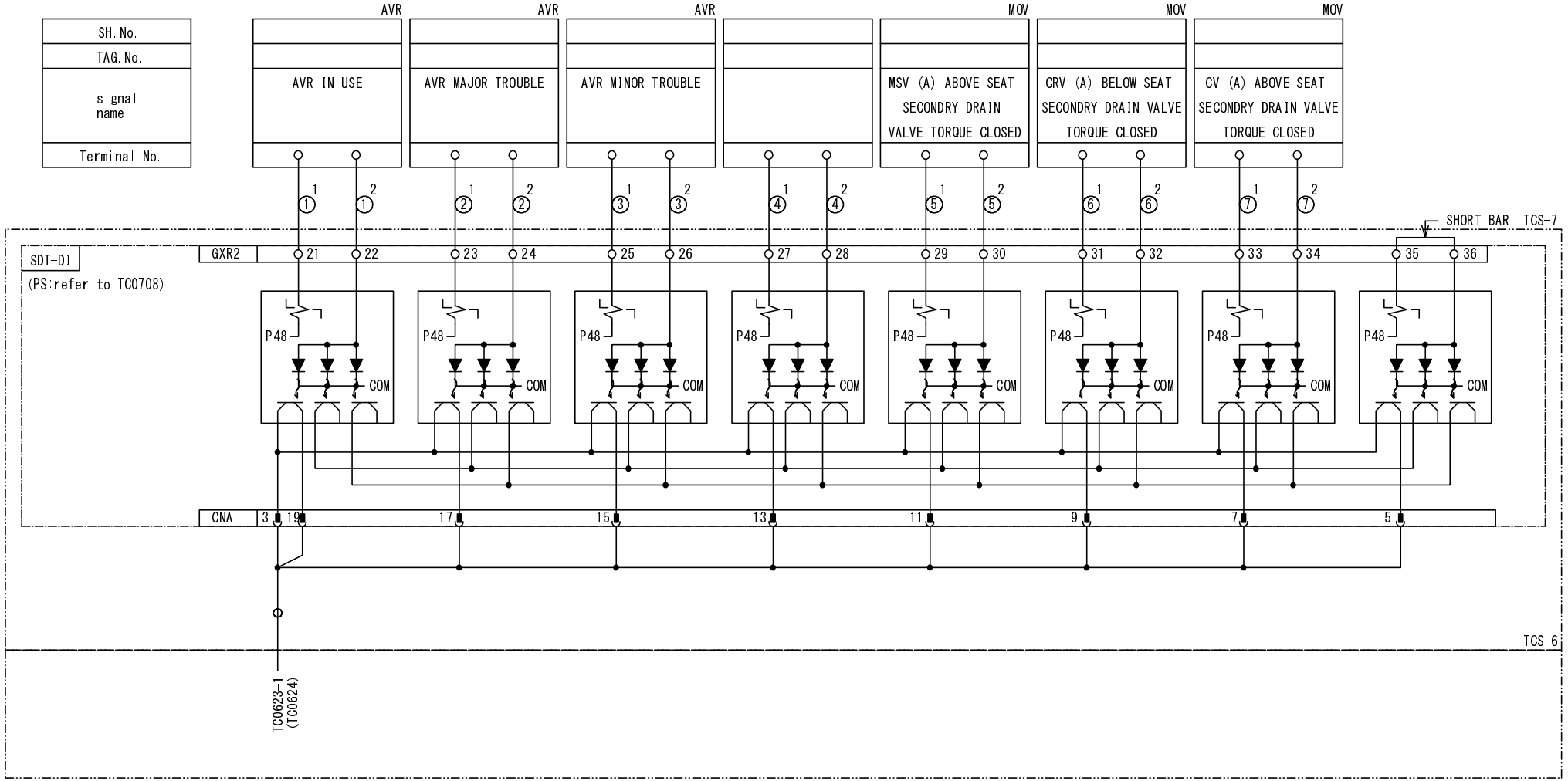
EXTERNAL INPUT COM DI(3) | SH. No. TC0723

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑦		2/2	MOV	TCS-7			
⑥		2/2	MOV	TCS-7			
⑤		2/2	MOV	TCS-7			
④		2/2		TCS-7			
③		2/2	AVR	TCS-7			
②		2/2	AVR	TCS-7			
①		2/2	AVR	TCS-7			

EXTERNAL INPUT COM DI(4)

SH. No. TC0724

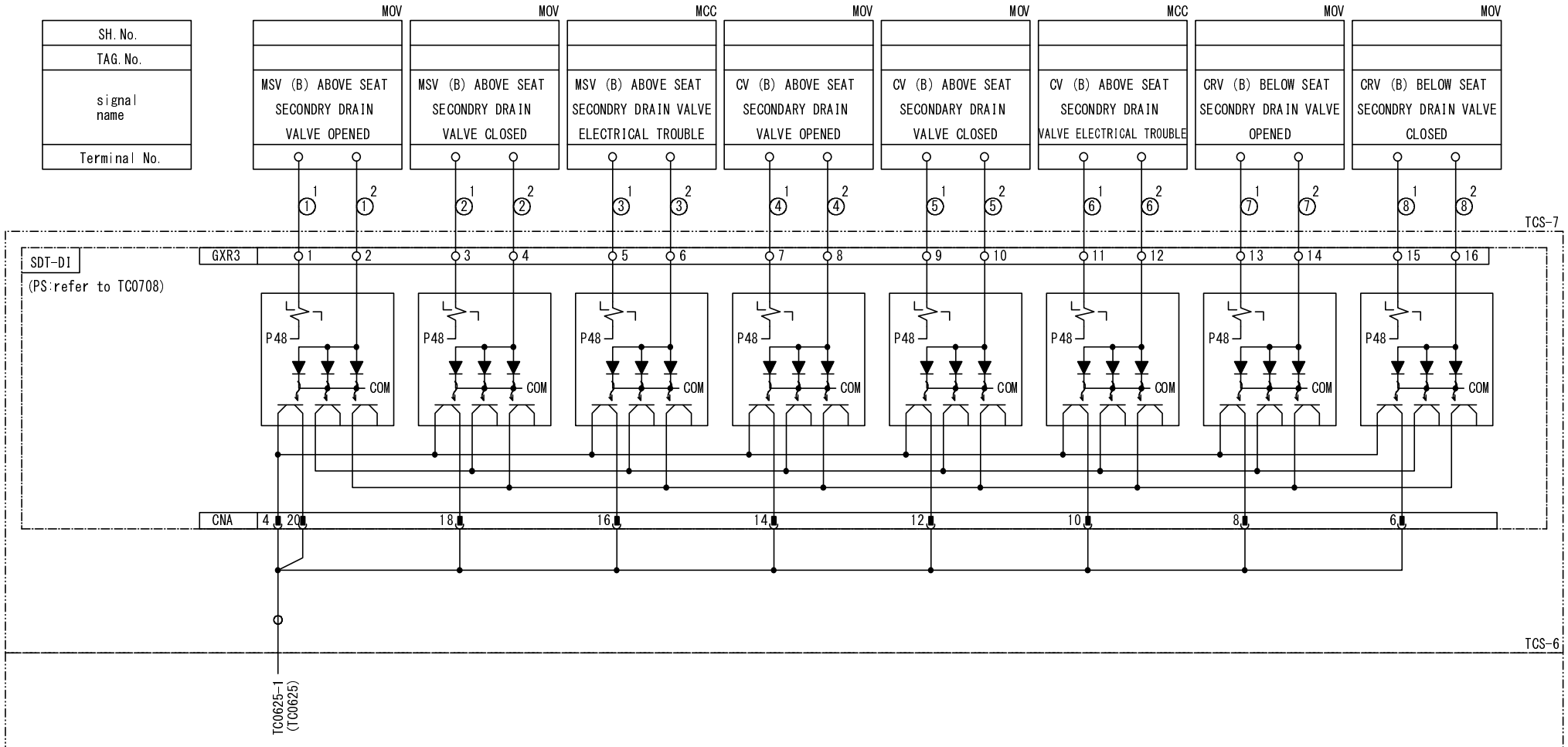
	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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IN-DEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	MOV	TCS-7			
⑦		2/2	MOV	TCS-7			
⑥		2/2	MCC	TCS-7			
⑤		2/2	MOV	TCS-7			
④		2/2	MOV	TCS-7			
③		2/2	MCC	TCS-7			
②		2/2	MOV	TCS-7			
①		2/2	MOV	TCS-7			

EXTERNAL INPUT COM DI(5)

SH. No. TC0725

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

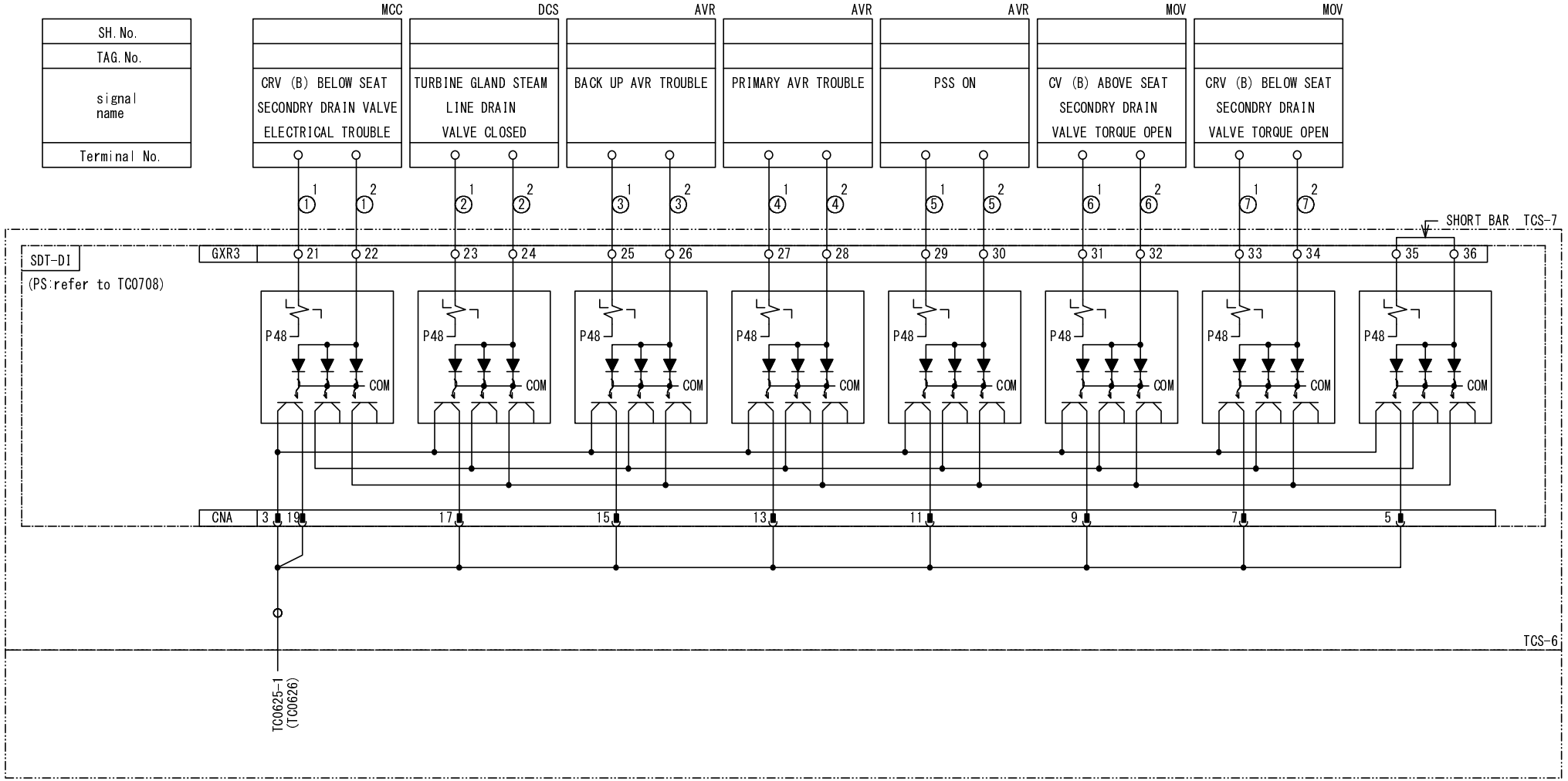
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑦		2/2	MOV	TCS-7			
⑥		2/2	MOV	TCS-7			
⑤		2/2	AVR	TCS-7			
④		2/2	AVR	TCS-7			
③		2/2	AVR	TCS-7			
②		2/2	DCS	TCS-7			
①		2/2	MCC	TCS-7			

EXTERNAL INPUT COM DI(6)

SH. No. TC0726

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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SH. No.	AOV	AOV	DCS	AOV	AOV	DCS	DCS
TAG. No.	ZSC-TSn700	ZSO-TSn700		ZSC-TSn730	ZSO-TSn730		
signal name	NO. 2 EXTRACTION STEAM NON RETURN VALVE FULL CLOSE POSITION	NO. 2 EXTRACTION STEAM NON RETURN VALVE FULL OPEN POSITION	NO. 1 HP HEATER LEVEL VERY HIGH	NO. 5 EXTRACTION STEAM NON RETURN VALVE FULL CLOSE POSITION	NO. 5 EXTRACTION STEAM NON RETURN VALVE FULL OPEN POSITION	NO. 2 LP HEATER LEVEL HIGH HIGH	HP/LP BYPASS CONTROL IN AUTO
Terminal No.	①-1 ①-2	②-1 ②-2	③-1 ③-2	④-1 ④-2	⑤-1 ⑤-2	⑥-1 ⑥-2	⑦-1 ⑦-2	⑧-1 ⑧-2

TCS-7

SDT-DI

(PS:refer to TC0708)

GXR4: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

P48 / COM

CNA: 4 20 18 16 14 12 10 8 6

TCS-6

TC0627-1 (TC0627)

INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		2/2	DCS	TCS-7			
⑦		2/2		TCS-7			
⑥		2/2	DCS	TCS-7			
⑤		2/2	AOV	TCS-7			
④		2/2	AOV	TCS-7			
③		2/2	DCS	TCS-7			
②		2/2	AOV	TCS-7			
①		2/2	AOV	TCS-7			

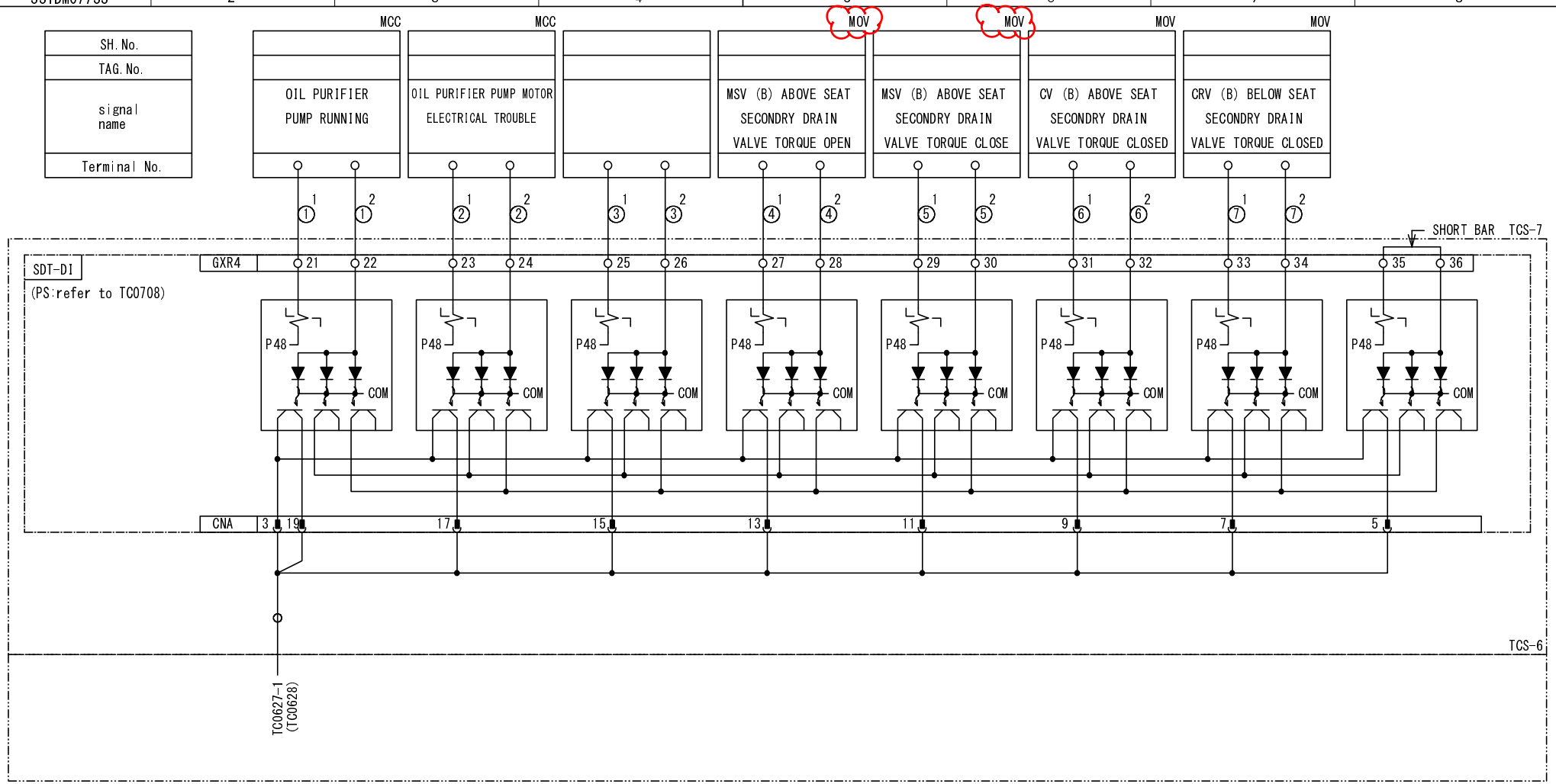
EXTERNAL INPUT COM DI(7) — SH. No. TC0727

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑦		2/2		TCS-7			
⑥		2/2		TCS-7			
⑤		2/2	MOV	TCS-7			
④		2/2	MOV	TCS-7			
③		2/2		TCS-7			
②		2/2	MCC	TCS-7			
①		2/2	MCC	TCS-7			

EXTERNAL INPUT COM DI(8)

SH. No. TC0728

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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SH. No.	MOV	MOV	MCC	MOV	MOV	DCS	DCS	
TAG. No.								
signal name	CV (B) BELOW SEAT SECONDRY WARMING DRAIN VALVE OPENED	CV (B) BELOW SEAT SECONDRY WARMING DRAIN VALVE CLOSED	CV (B) BELOW SEAT SEC. WARMING DRAIN VLV ELECTRICAL TROUBLE	HP TURBINE WARMING VALVE FULL CLOSE POSITION	HP TURBINE WARMING VALVE FULL OPEN POSITION	COLD REHEAT DRAIN VALVE CLOSE FEED BACK	COLD REHEAT DRAIN VALVE FULL OPEN FEED BACK	
Terminal No.	① 1 2	② 1 2	③ 1 2	④ 1 2	⑤ 1 2	⑥ 1 2	⑦ 1 2	⑧ 1 2

TCS-7

SDT-DI
(PS:refer to TC0709)

GXR5: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

P48 / COM

CNA: 4 20 18 16 14 12 10 8 6

TC0629-1 (TC0629)

TCS-6

INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		2/2		TCS-7			
⑦		2/2	DCS	TCS-7			
⑥		2/2	DCS	TCS-7			
⑤		2/2	MOV	TCS-7			
④		2/2	MOV	TCS-7			
③		2/2	MCC	TCS-7			
②		2/2	MOV	TCS-7			
①		2/2	MOV	TCS-7			

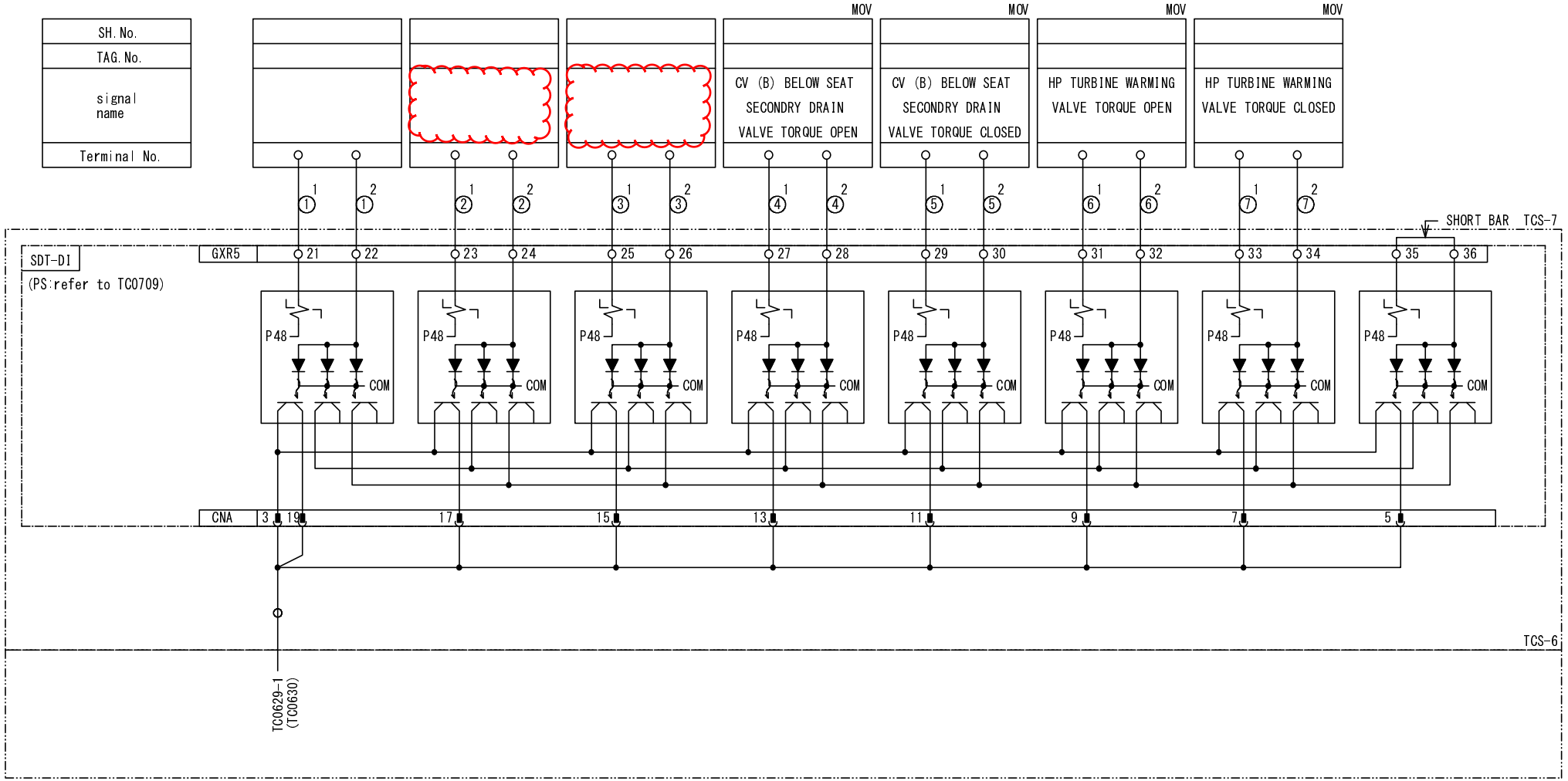
EXTERNAL INPUT COM DI(9) | SH. No. TC0729

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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2016/07/28 TC0729 XSMM-TC



◯								
⑦		2/2	MOV	TCS-7				
⑥		2/2	MOV	TCS-7				
⑤		2/2	MOV	TCS-7				
④		2/2		TCS-7				
③		2/2		TCS-7				
②		2/2		TCS-7				
①		2/2		TCS-7				
INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK	
			CABLE CONNECTION		RELATED SH. NO.			

EXTERNAL INPUT COM DI(10)

SH. No. TC0730

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

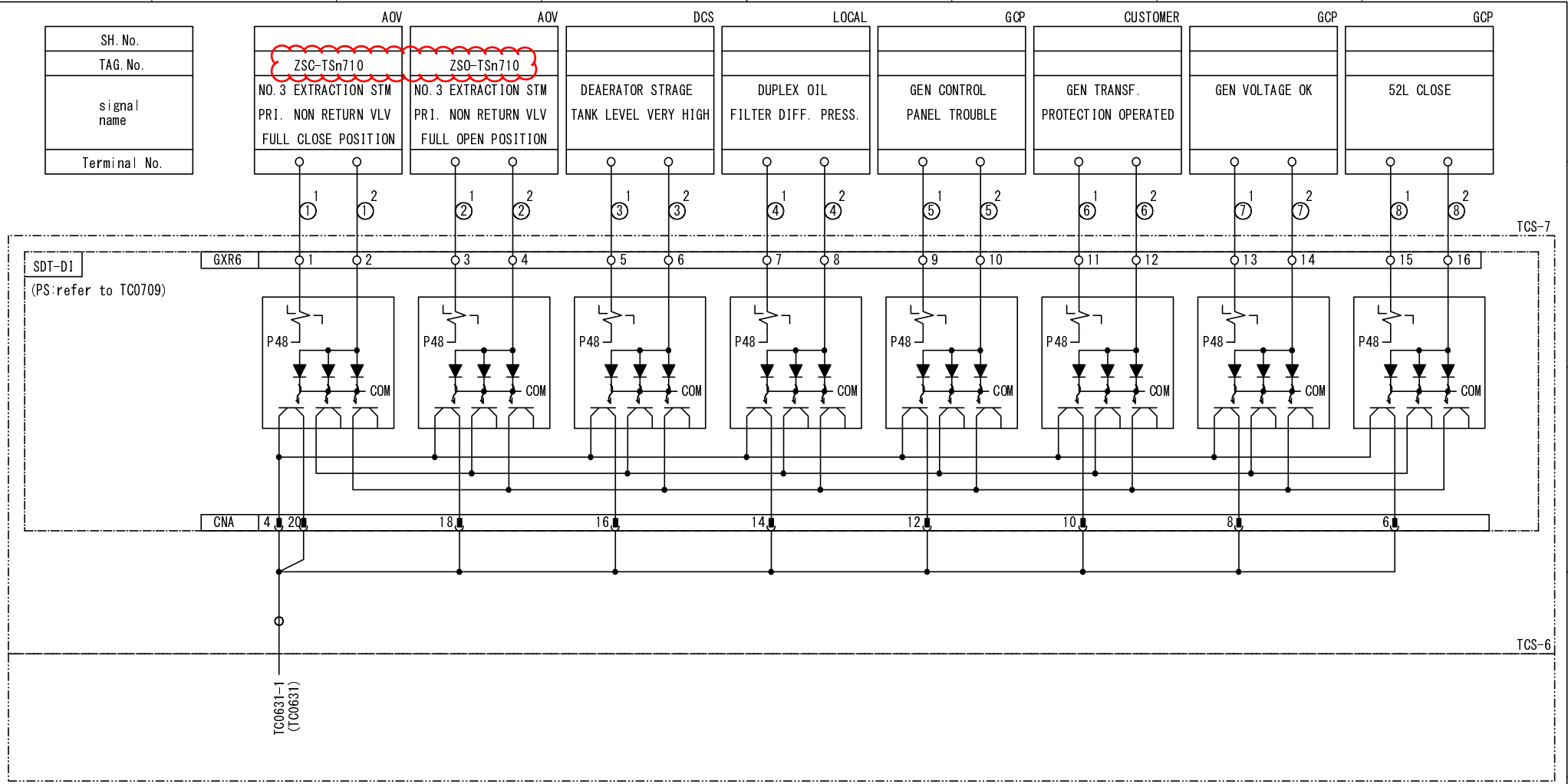
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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⑧		2/2	GCP	TCS-7			
⑦		2/2	GCP	TCS-7			
⑥		2/2	CUSTOMER	TCS-7			
⑤		2/2	GCP	TCS-7			
④		2/2	LOCAL	TCS-7			
③		2/2	DCS	TCS-7			
②		2/2	AOV	TCS-7			
①		2/2	AOV	TCS-7			
INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK

EXTERNAL INPUT COM DI(11)

SH. No. TC0731

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

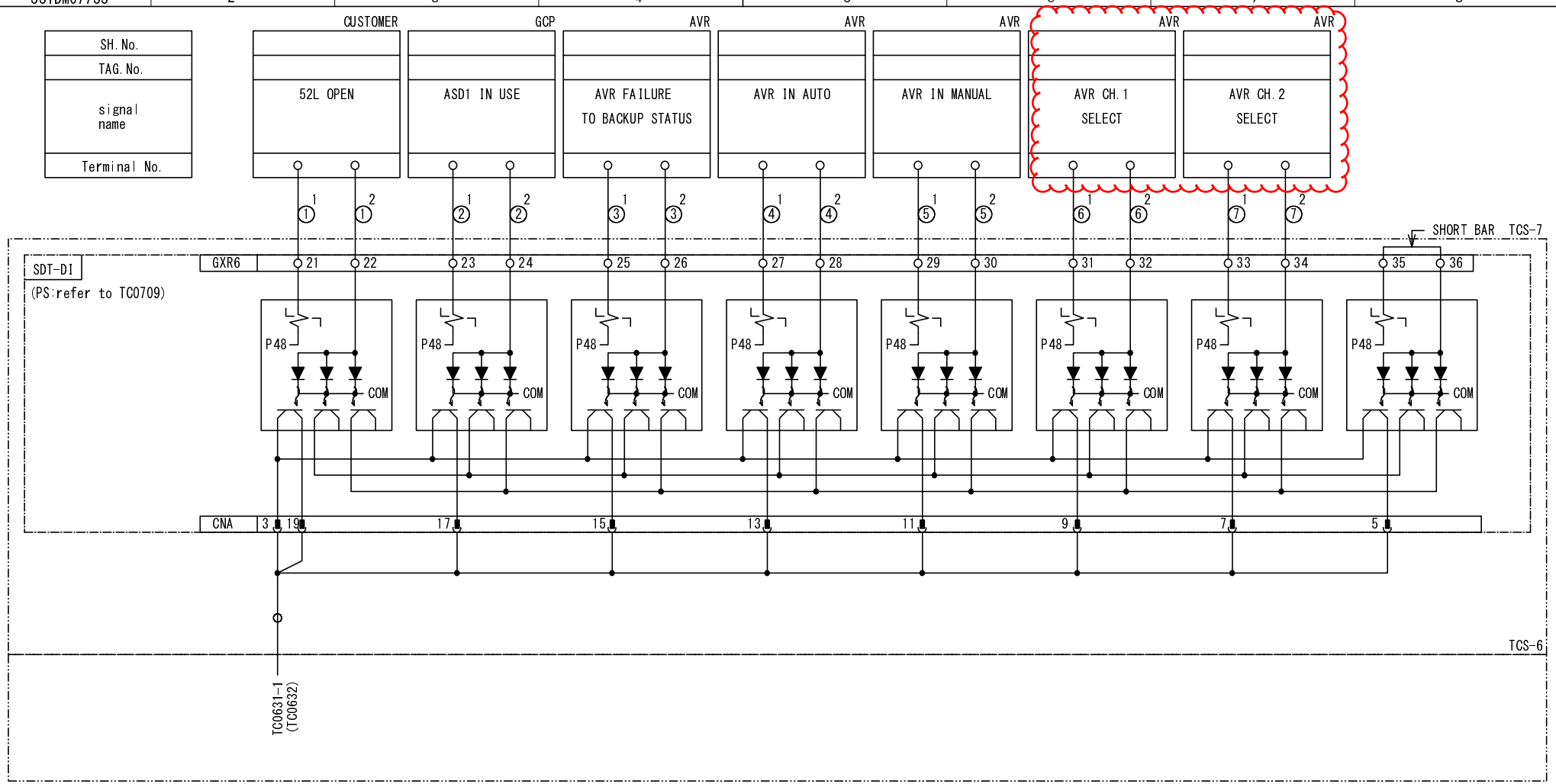
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0731 XSMM-TC



◯								
⑦		2/2	AVR	TCS-7				
⑥		2/2	AVR	TCS-7				
⑤		2/2	AVR	TCS-7				
④		2/2	AVR	TCS-7				
③		2/2	AVR	TCS-7				
②		2/2	GCP	TCS-7				
①		2/2	CUSTOMER	TCS-7				
INDEX	CABLE NO.	CORE NO.	CABLE CONNECTION FROM	CABLE CONNECTION TO	RELATED SH. NO. FROM	RELATED SH. NO. TO	REMARK	

EXTERNAL INPUT COM DI(12)

SH. No. TC0732

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

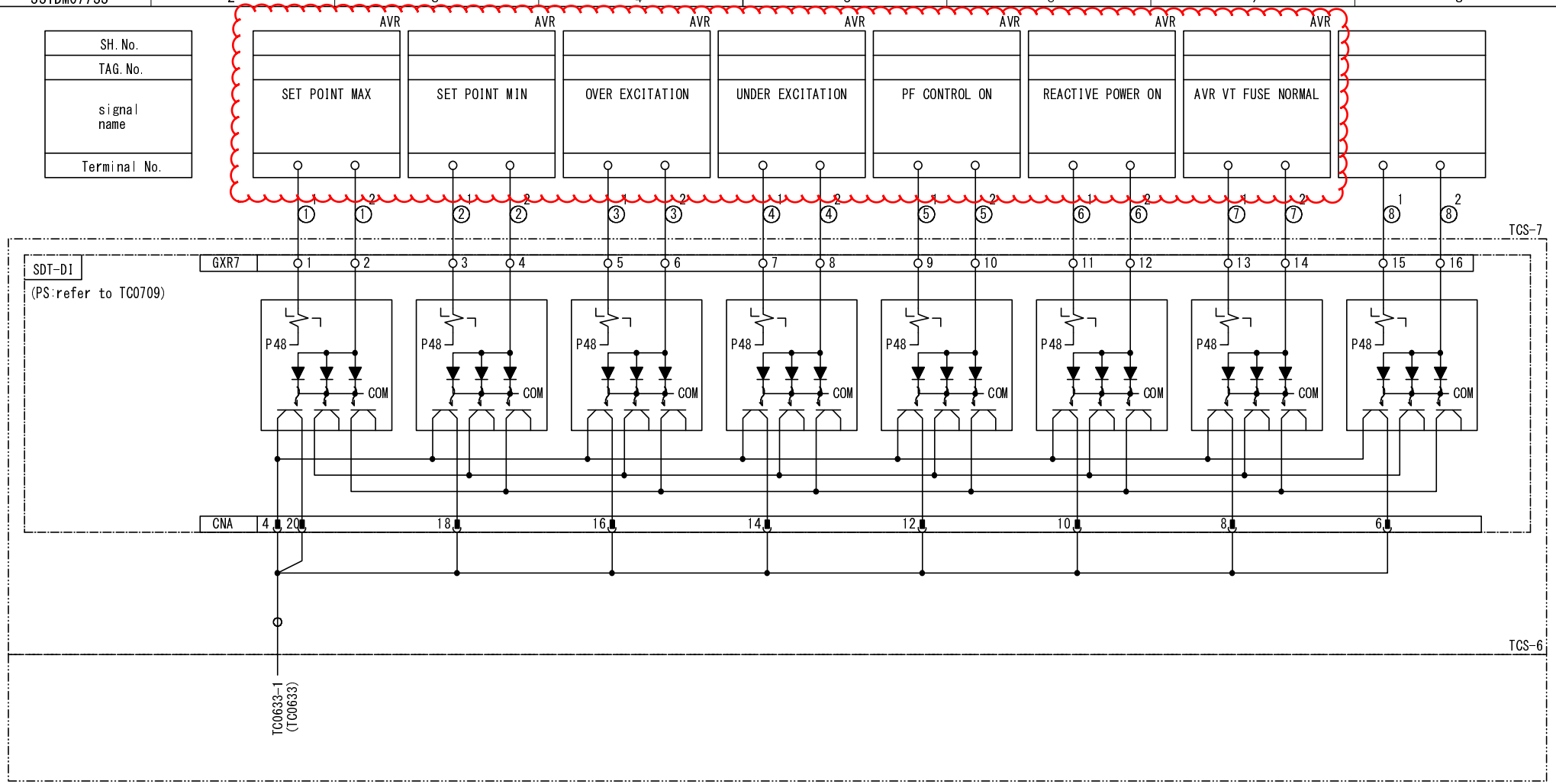
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0732 XSMM-TC



INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧			AVR	TCS-7			
⑦			AVR	TCS-7			
⑥			AVR	TCS-7			
⑤			AVR	TCS-7			
④			AVR	TCS-7			
③			AVR	TCS-7			
②			AVR	TCS-7			
①		2/2	AVR	TCS-7			

EXTERNAL INPUT COM DI(13)

SH. No. TC0733

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

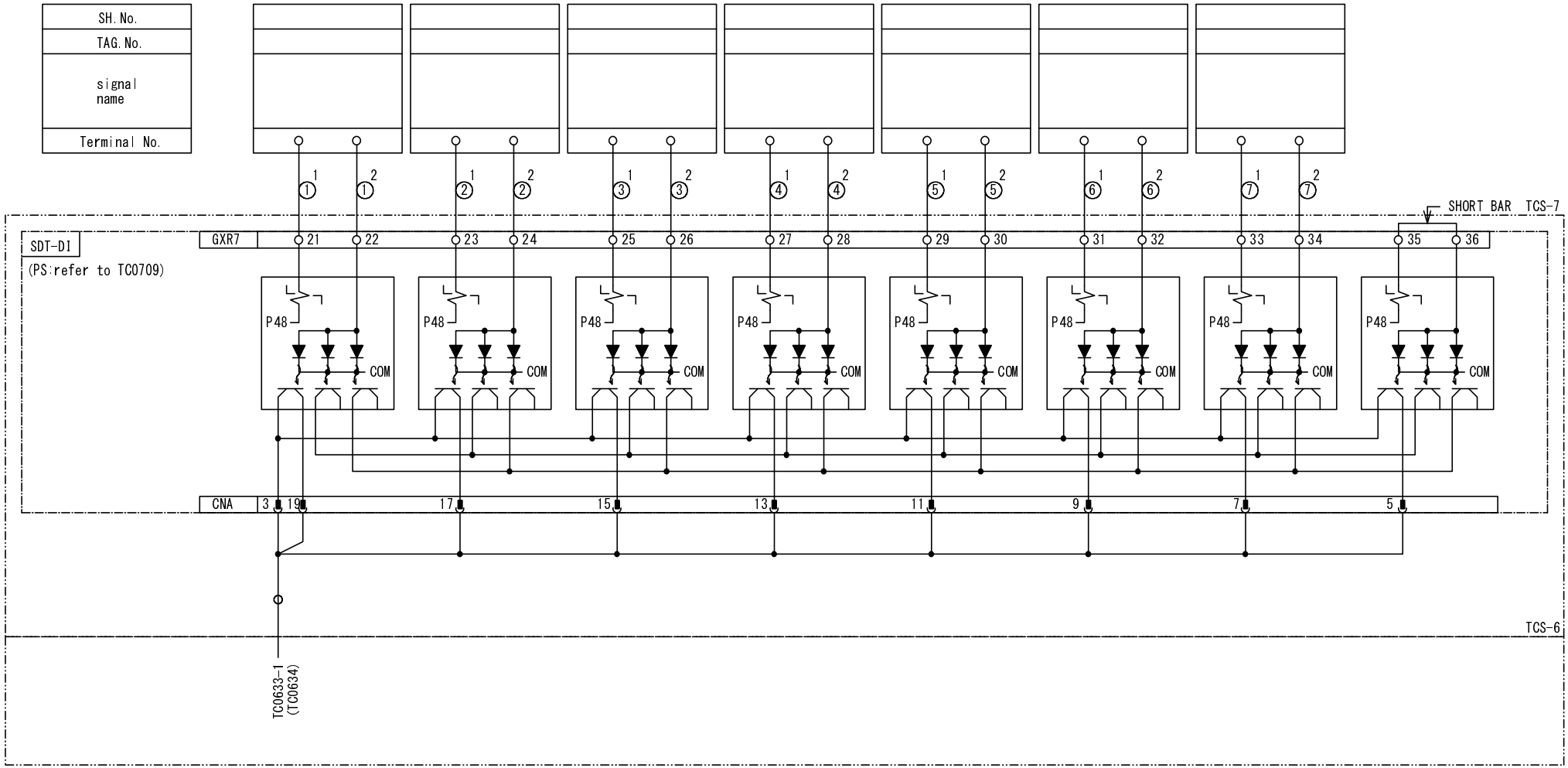
REGD. Hitachi, Ltd. Tokyo Japan

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2016/07/28 TC0733 XSMM-TC



◯							
⑦		2/2		TCS-7			
⑥		2/2		TCS-7			
⑤		2/2		TCS-7			
④		2/2		TCS-7			
③		2/2		TCS-7			
②		2/2		TCS-7			
①		2/2		TCS-7			
INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		

EXTERNAL INPUT COM DI(14)

SH. No. TC0734

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

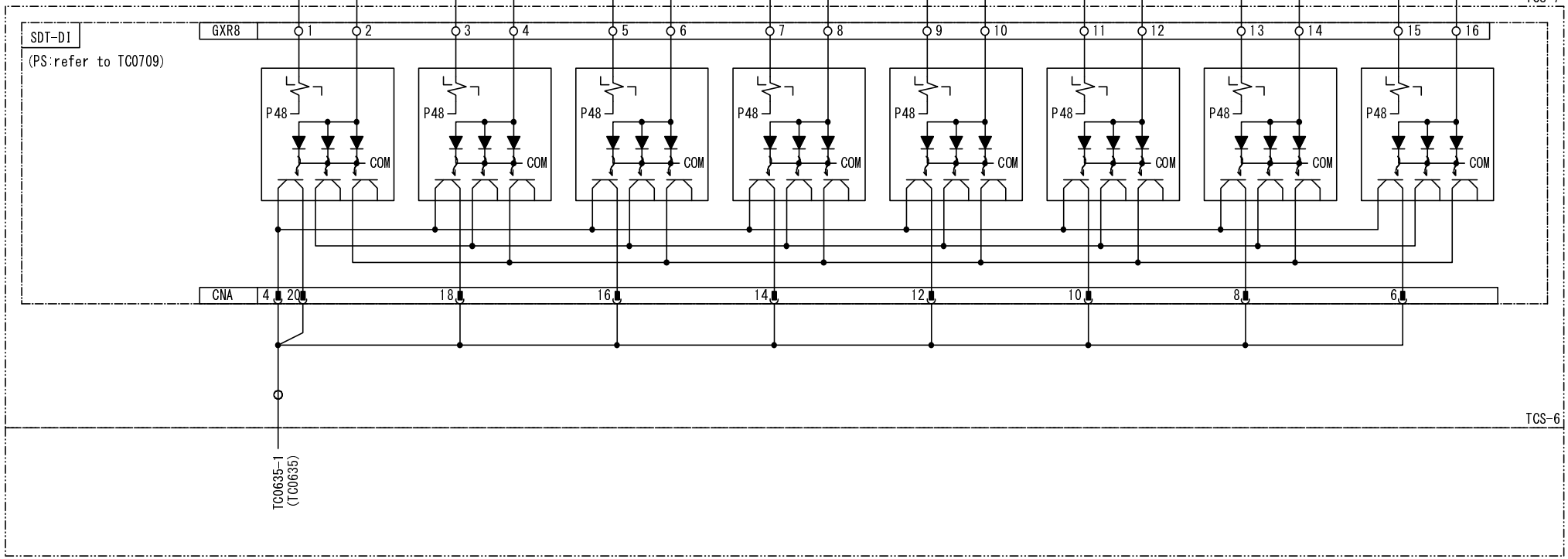
DWG. No. 331DM07733

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2016/07/28 TC0734 XSMM-TC

SH. No.	AOV	AOV	DCS	DCS	DCS	MCC	MCC	MCC
TAG. No.	ZSC-TSn740	ZSO-TSn740						
signal name	NO.3 EXTRACTION STM SEC. NON RETURN VLV FULL CLOSE POSITION	NO.3 EXTRACTION STM SEC. NON RETURN VLV FULL OPEN POSITION	HP TURBINE WARMING LINE DRAIN VALVE AUTO SELECTED	HP TURBINE WARMING LINE DRAIN VALVE OPEN FEEED BACK	HP TURBINE WARMING LINE DRAIN VALVE CLOSE FEED BACK	HP TURBINE WARMING VALVE ELECTRICAL TROUBLE	MAIN OIL TANK VAPOUR EXTRACTOR RUNNING	MAIN OIL TANK VAPOUR EXTRACTOR ELECTRICAL TROUBLE
Terminal No.	①-1 ①-2	②-1 ②-2	③-1 ③-2	④-1 ④-2	⑤-1 ⑤-2	⑥-1 ⑥-2	⑦-1 ⑦-2	⑧-1 ⑧-2



IN-DEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		2/2	MCC	TCS-7			
⑦		2/2	MCC	TCS-7			
⑥		2/2	MCC	TCS-7			
⑤		2/2	DCS	TCS-7			
④		2/2	DCS	TCS-7			
③		2/2	DCS	TCS-7			
②		2/2	AOV	TCS-7			
①		2/2	AOV	TCS-7			

EXTERNAL INPUT COM DI(15)

SH. No. TC0735

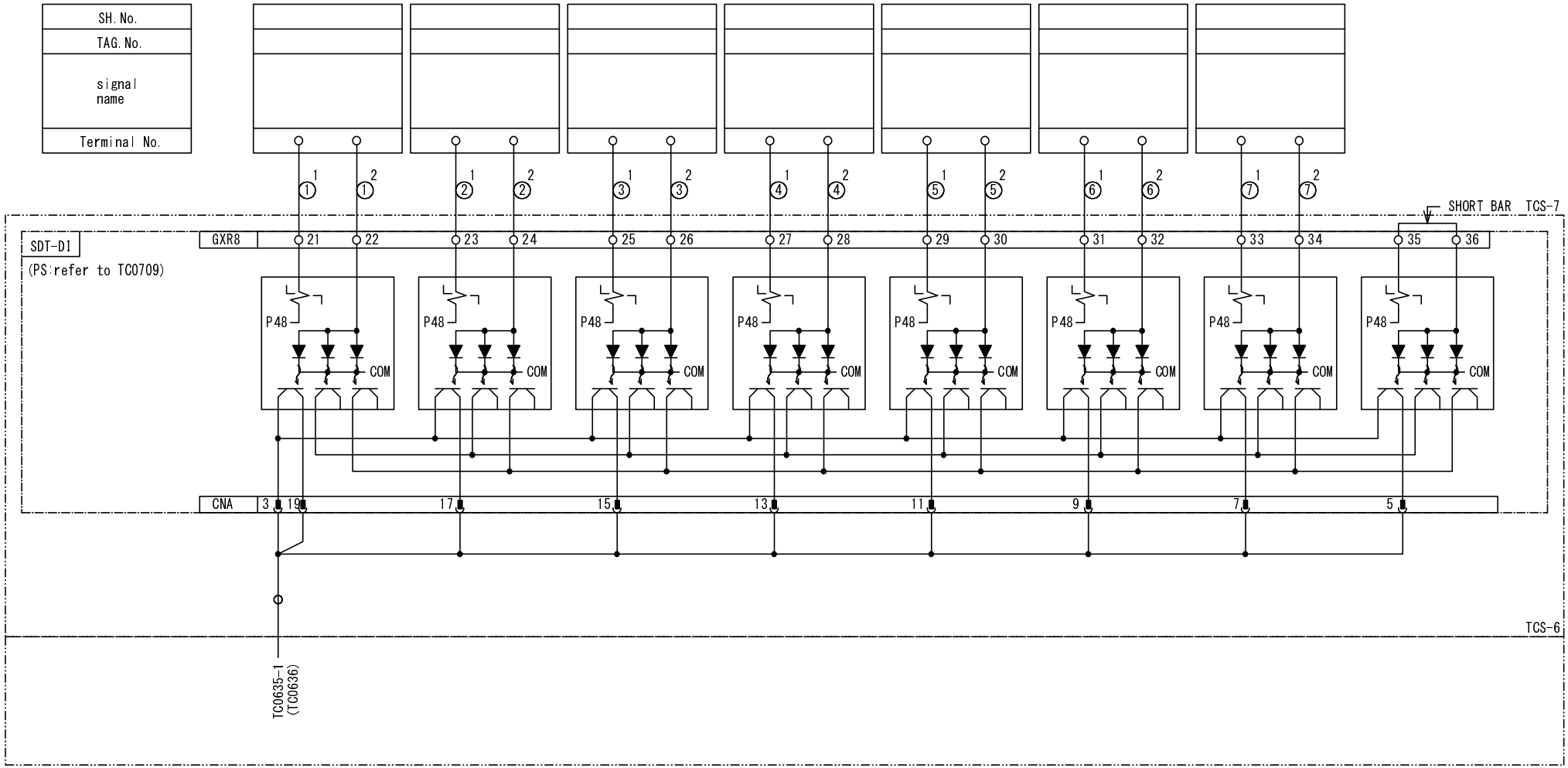
SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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2016/07/28 TC0735 XSMM-TC



○							
⑦		2/2		TCS-7			
⑥		2/2		TCS-7			
⑤		2/2		TCS-7			
④		2/2		TCS-7			
③		2/2		TCS-7			
②		2/2		TCS-7			
①		2/2		TCS-7			
IN-DEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		

EXTERNAL INPUT COM DI(16)

SH. No. TC0736

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

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2016/07/28 TC0736 XSMM-TC

SH. No.		TSI	TSI	AOV	AOV	FIELD	FIELD	FIELD	FIELD
TAG. No.				ZSC-CDn904	ZSO-CDn904	ZSC-CDn901	ZSO-CDn901	LSH-GEn001	DPIS-LOn942
signal name		NO.1 TURBINE SHAFT VIBRATION (X) HIGH	NO.3 GEN. SHAFT VIBRATION (Y) HIGH	LP EXHAUST HOOD WATER SPRAY VALVE FULL CLOSE POSITION	LP EXHAUST HOOD WATER SPRAY VALVE FULL OPEN POSITION	COLD REHEAT NON RETURN VALVE FULL CLOSE POSITION	COLD REHEAT NON RETURN VALVE FULL OPEN POSITION	GEN. WATER AND OIL LEAKAGE LEVEL HIGH	LUBE OIL CONDITIONER COALESCER FILTER DIFF. PRESS. HIGH
Terminal No.		1 2	1 2	1 2	1 2	1 2	1 2	1 2	1 2
		① ①	② ②	③ ③	④ ④	⑤ ⑤	⑥ ⑥	⑦ ⑦	⑧ ⑧

TCS-7

SDT-DI

(PS:refer to TC0710)

GXR9: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

P48 / COM

CNA: 4 20 18 16 14 12 10 8 6

TCS-6

TC0637-1 (TC0637)

IN-DEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		2/2	FIELD	TCS-7			
⑦		2/2	FIELD	TCS-7			
⑥		2/2	FIELD	TCS-7			
⑤		2/2	FIELD	TCS-7			
④		2/2	AOV	TCS-7			
③		2/2	AOV	TCS-7			
②		2/2	TSI	TCS-7			
①		2/2	TSI	TCS-7			

EXTERNAL INPUT COM DI(17)

SH. No. TC0737

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

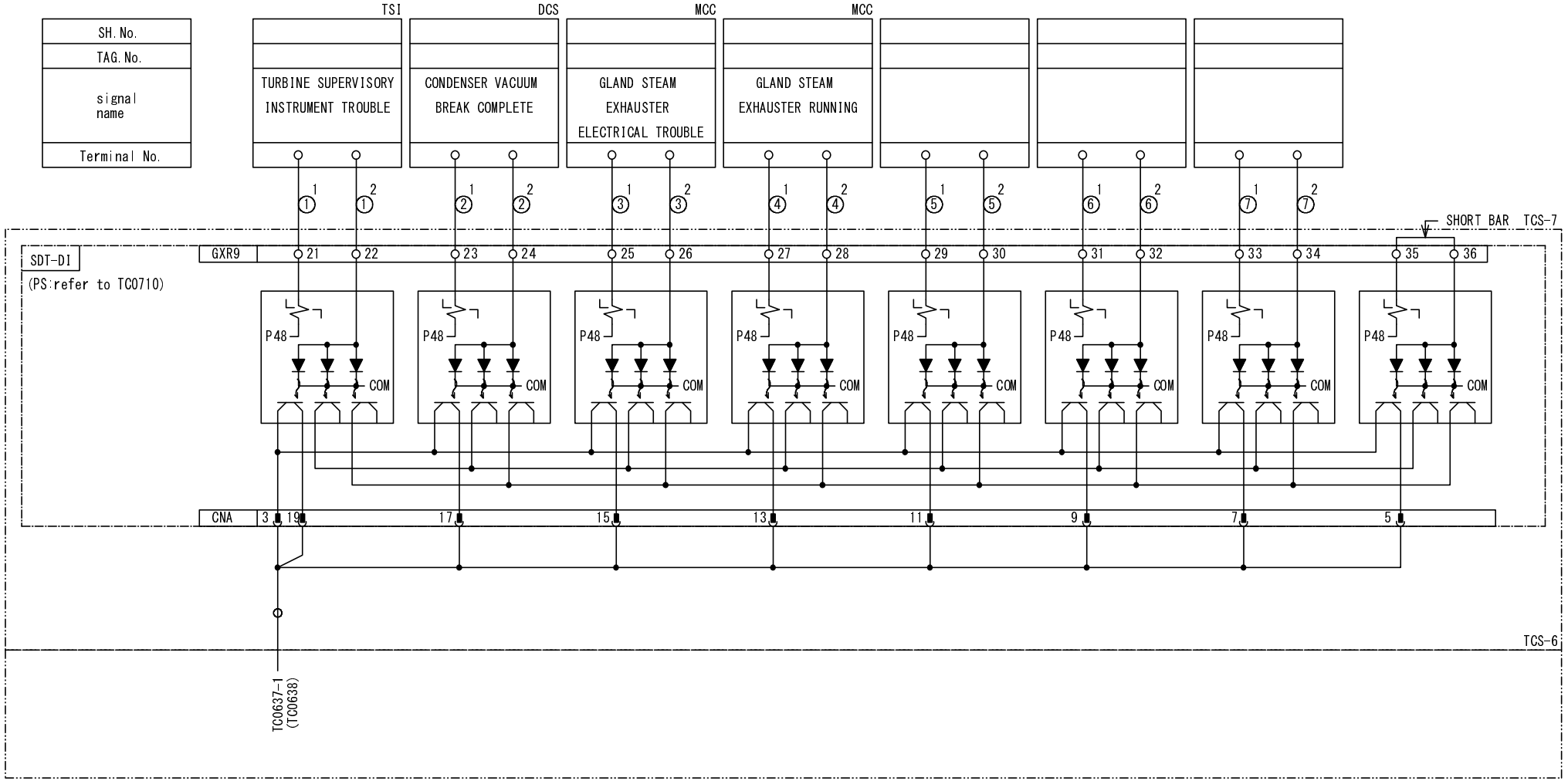
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
◯		2/2		TCS-7			
⑦		2/2		TCS-7			
⑥		2/2		TCS-7			
⑤		2/2		TCS-7			
④		2/2	MCC	TCS-7			
③		2/2	MCC	TCS-7			
②		2/2	DCS	TCS-7			
①		2/2	TSI	TCS-7			

EXTERNAL INPUT COM DI(18)

SH. No. TC0738

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

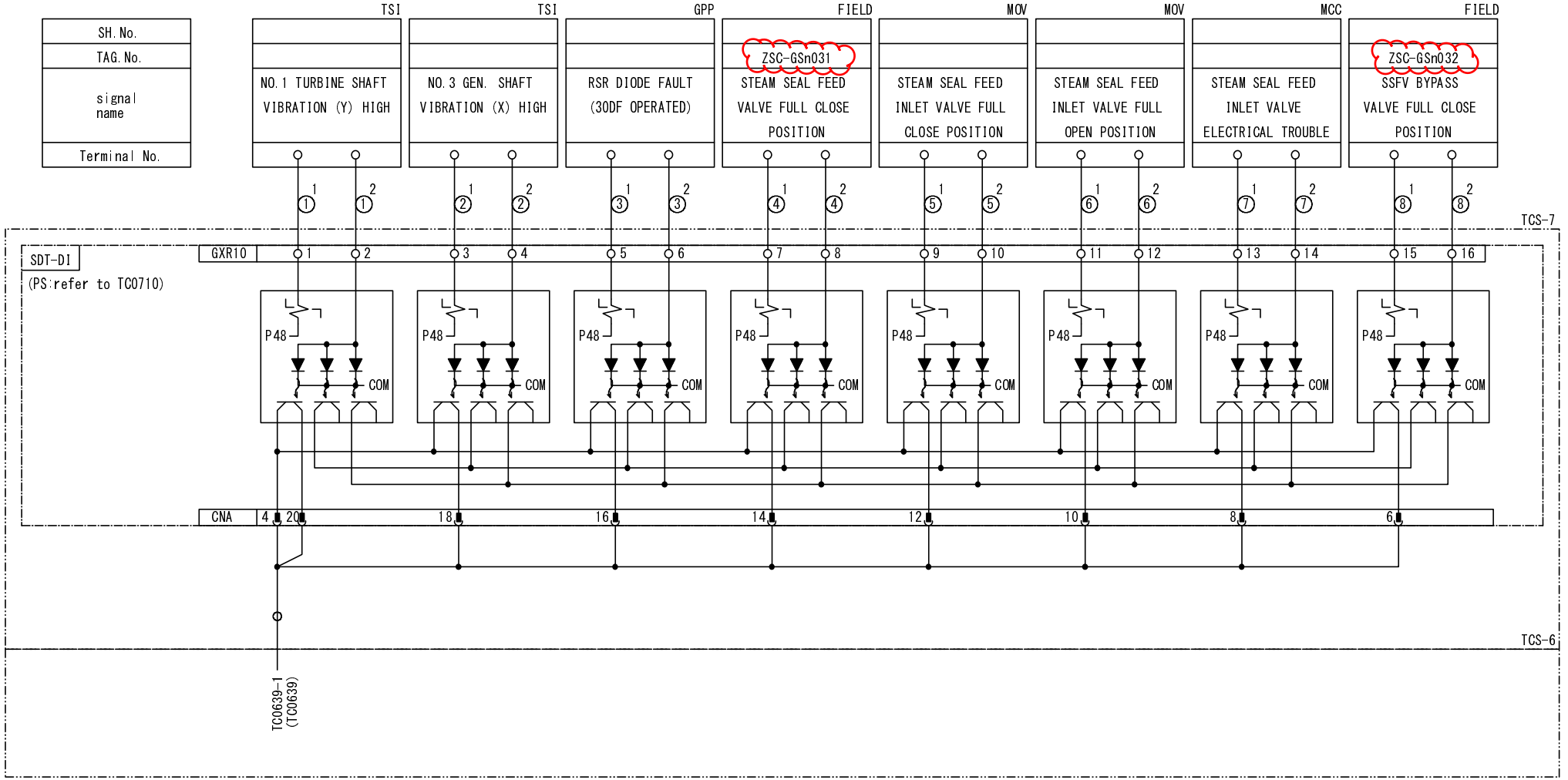
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	FIELD	TCS-7			
⑦		2/2	MCC	TCS-7			
⑥		2/2	MOV	TCS-7			
⑤		2/2	MOV	TCS-7			
④		2/2	FIELD	TCS-7			
③		2/2	GPP	TCS-7			
②		2/2	TSI	TCS-7			
①		2/2	TSI	TCS-7			

EXTERNAL INPUT COM DI(19)

SH. No. TC0739

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

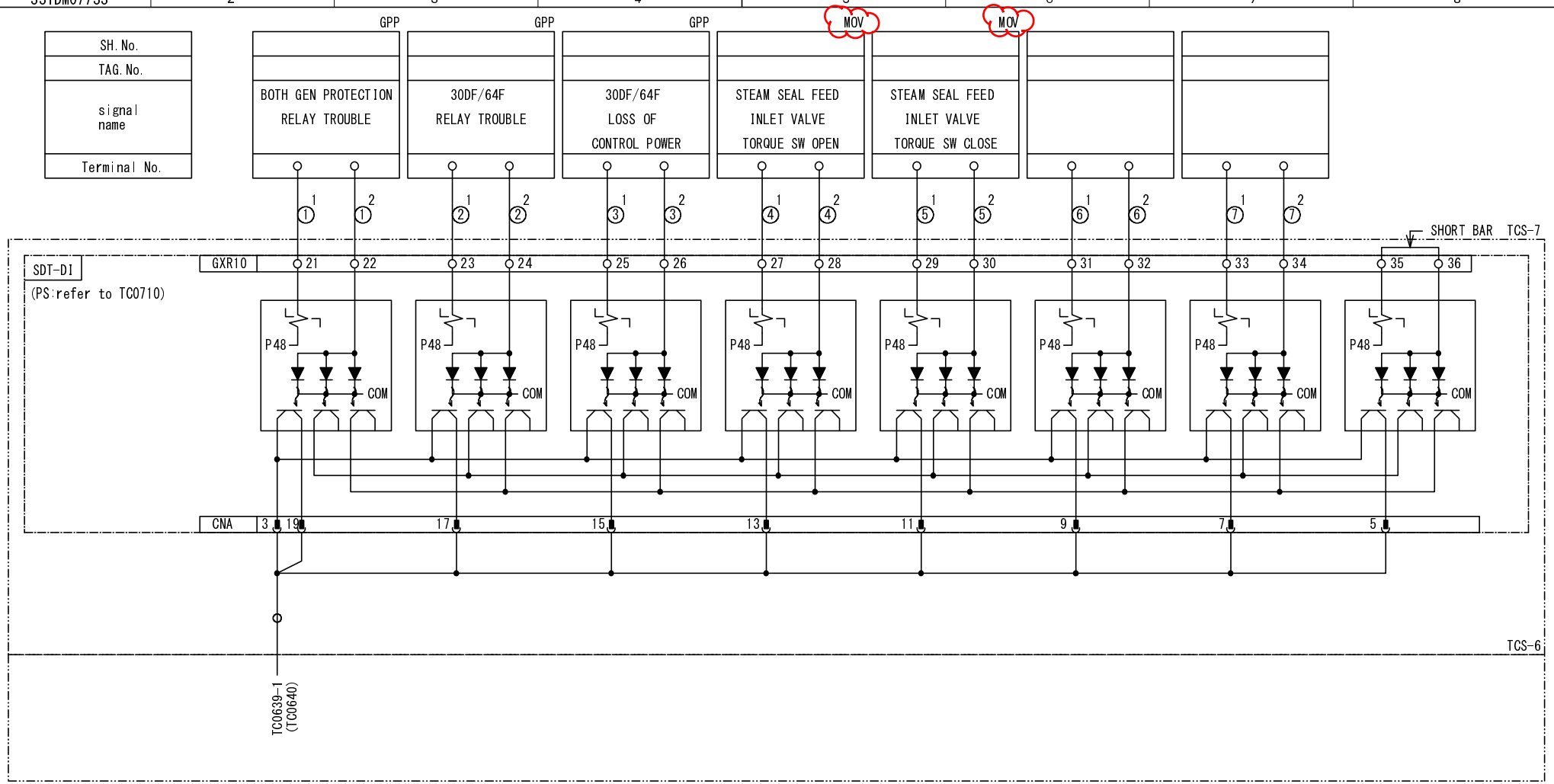
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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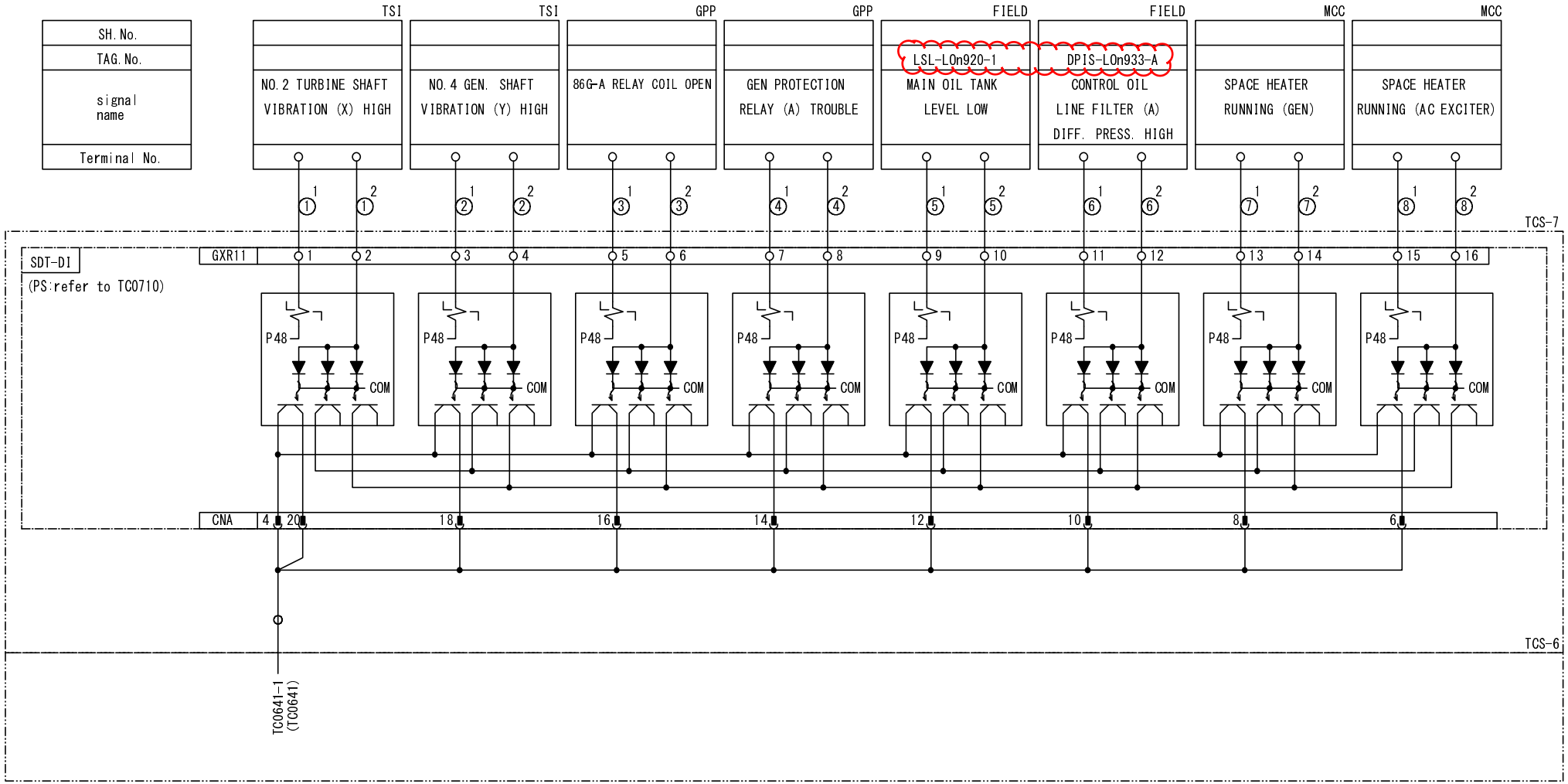
IN-DEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK
⑧		2/2		TCS-7			
⑦		2/2		TCS-7			
⑥		2/2		TCS-7			
⑤		2/2	MOV	TCS-7			
④		2/2	MOV	TCS-7			
③		2/2	GPP	TCS-7			
②		2/2	GPP	TCS-7			
①		2/2	GPP	TCS-7			

EXTERNAL INPUT COM DI(20) — SH. No. TC0740

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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IN-DEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	MCC	TCS-7			
⑦		2/2	MCC	TCS-7			
⑥		2/2	FIELD	TCS-7			
⑤		2/2	FIELD	TCS-7			
④		2/2	GPP	TCS-7			
③		2/2	GPP	TCS-7			
②		2/2	TSI	TCS-7			
①		2/2	TSI	TCS-7			

EXTERNAL INPUT COM DI(21)

SH. No. TC0741

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

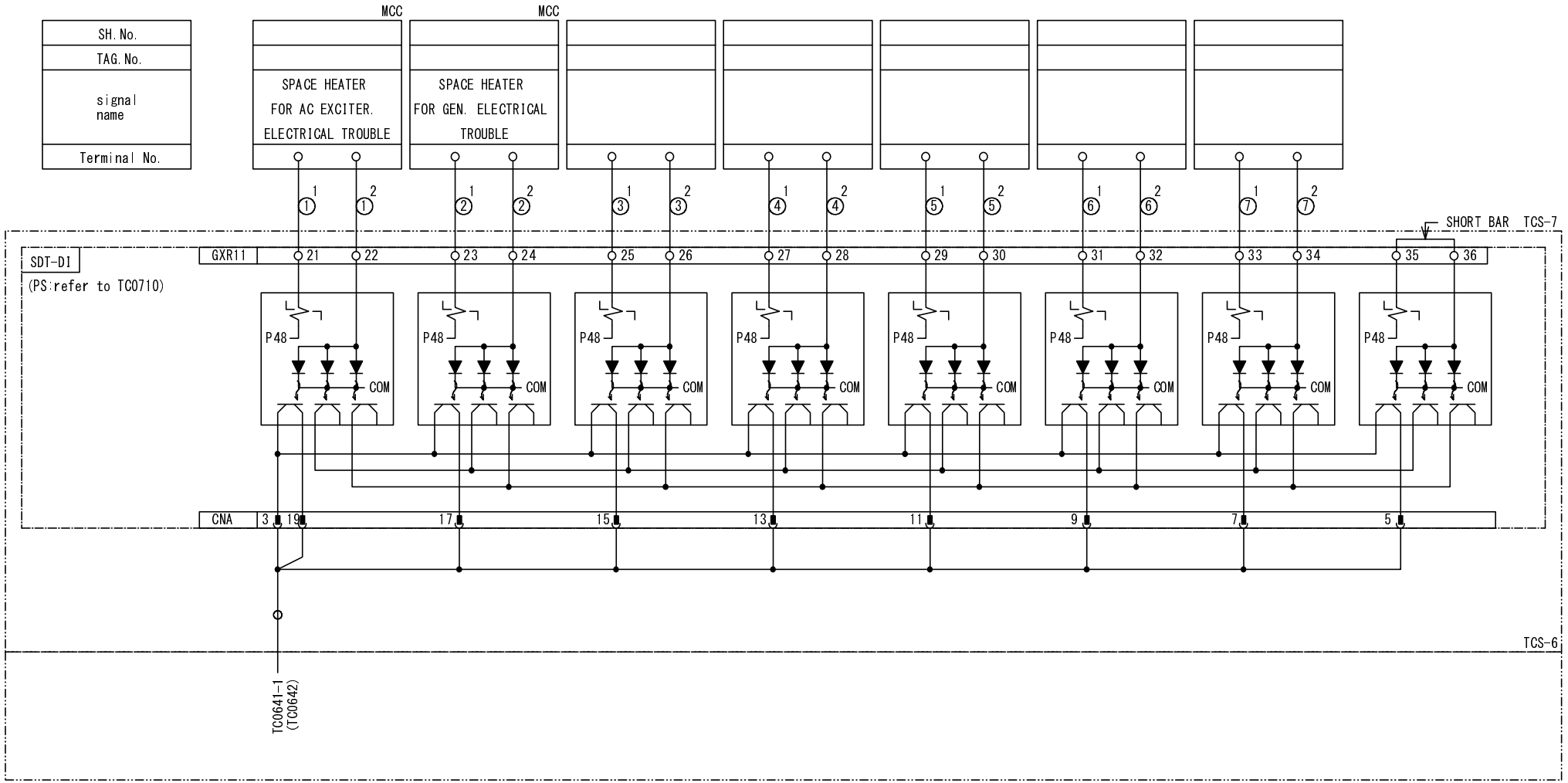
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0741 XSMM-TC



◯								
⑦		2/2		TCS-7				
⑥		2/2		TCS-7				
⑤		2/2		TCS-7				
④		2/2		TCS-7				
③		2/2		TCS-7				
②		2/2	MCC	TCS-7				
①		2/2	MCC	TCS-7				
INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK	
			CABLE CONNECTION		RELATED SH. NO.			

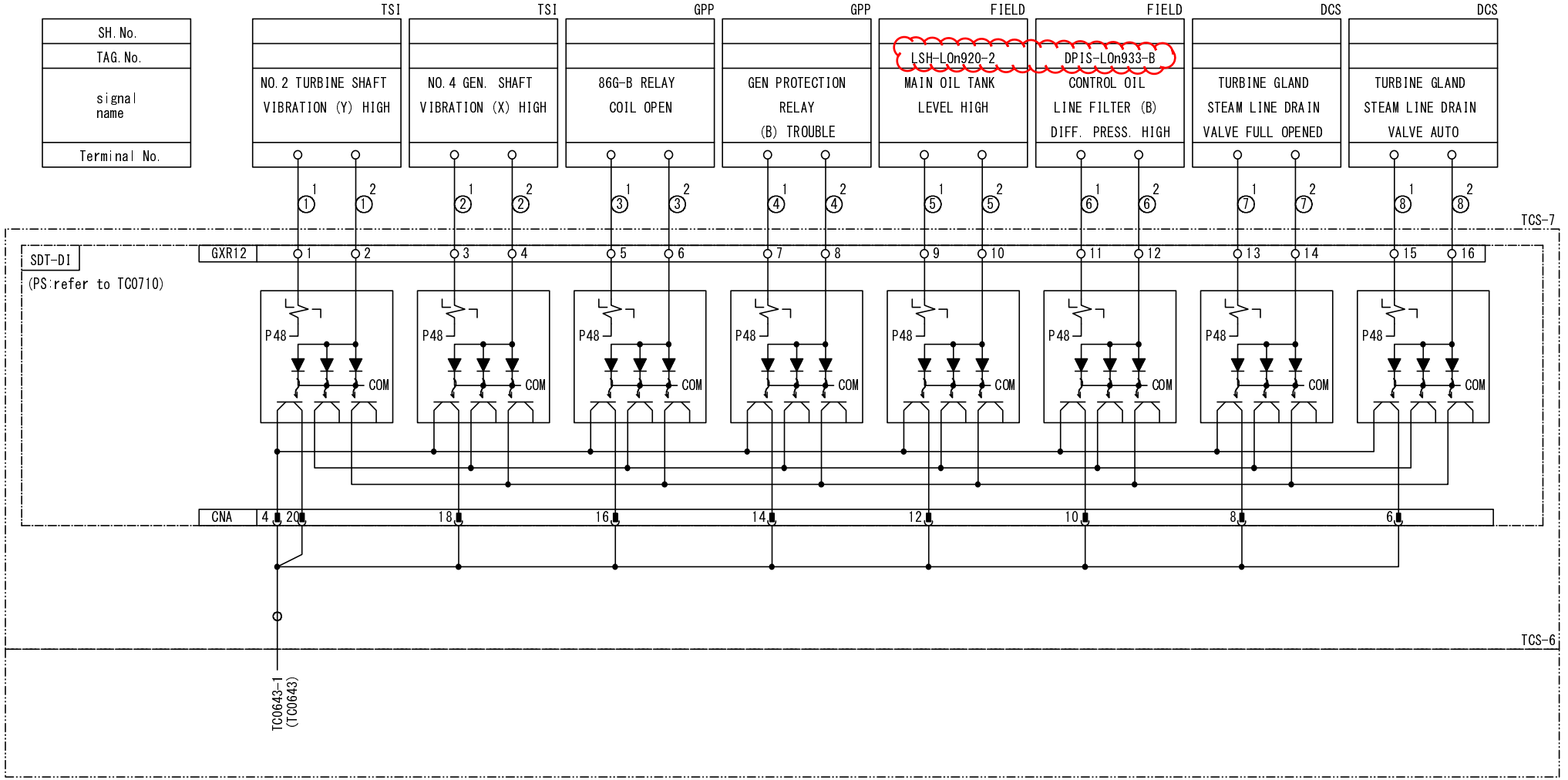
EXTERNAL INPUT COM DI(22) SH. No. TC0742

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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IN-DEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK
⑧		2/2	DCS	TCS-7			
⑦		2/2	DCS	TCS-7			
⑥		2/2	FIELD	TCS-7			
⑤		2/2	FIELD	TCS-7			
④		2/2	GPP	TCS-7			
③		2/2	GPP	TCS-7			
②		2/2	TSI	TCS-7			
①		2/2	TSI	TCS-7			

EXTERNAL INPUT COM DI(23)

SH. No. TC0743

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

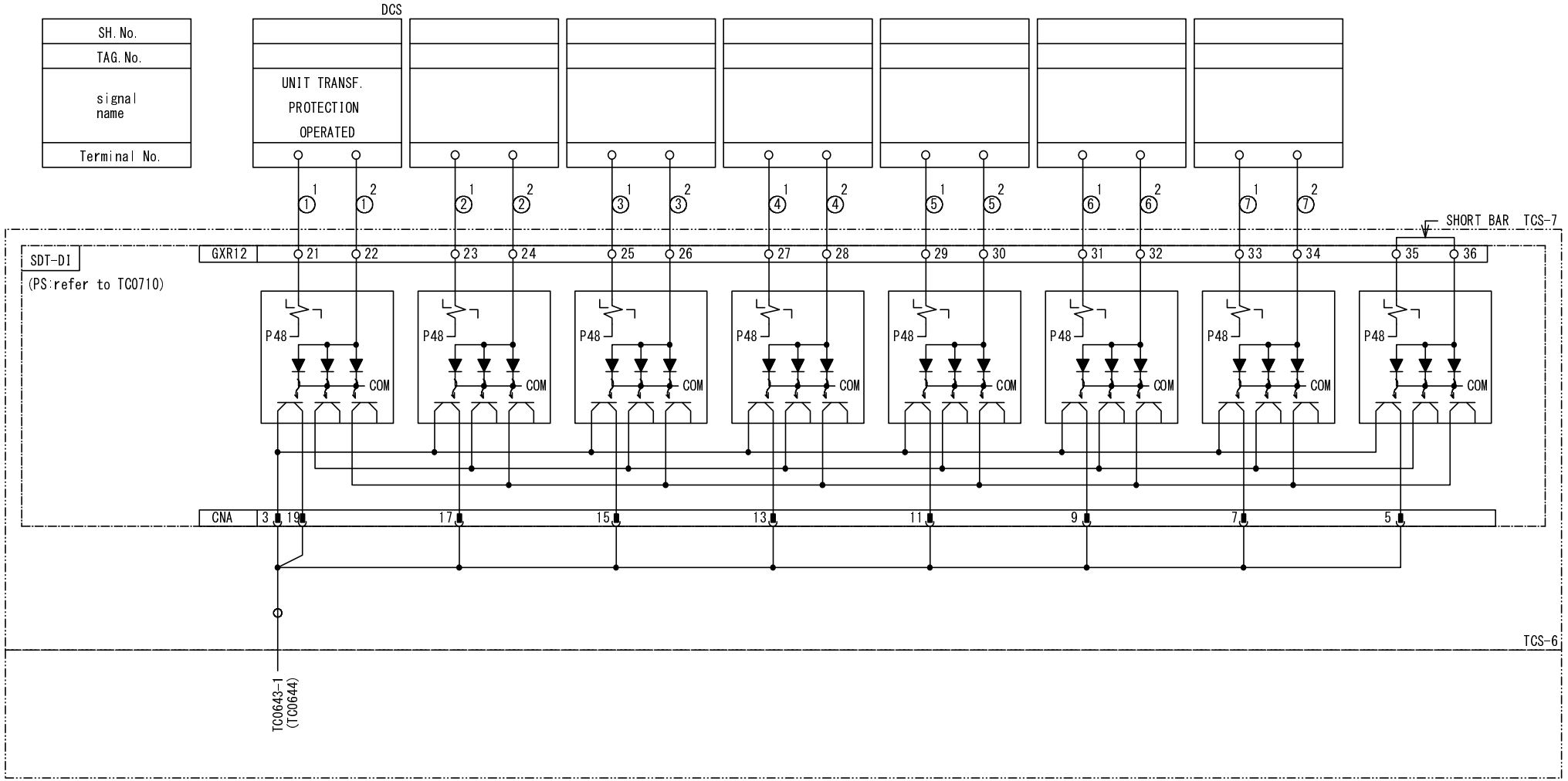
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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○							
⑦		2/2		TCS-7			
⑥		2/2		TCS-7			
⑤		2/2		TCS-7			
④		2/2		TCS-7			
③		2/2		TCS-7			
②		2/2		TCS-7			
①		2/2	DCS	TCS-7			
INDEX	CABLE NO.	CORE NO.	CABLE CONNECTION FROM	CABLE CONNECTION TO	RELATED SH. NO. FROM	RELATED SH. NO. TO	REMARK

EXTERNAL INPUT COM DI(24)

SH. No. TC0744

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

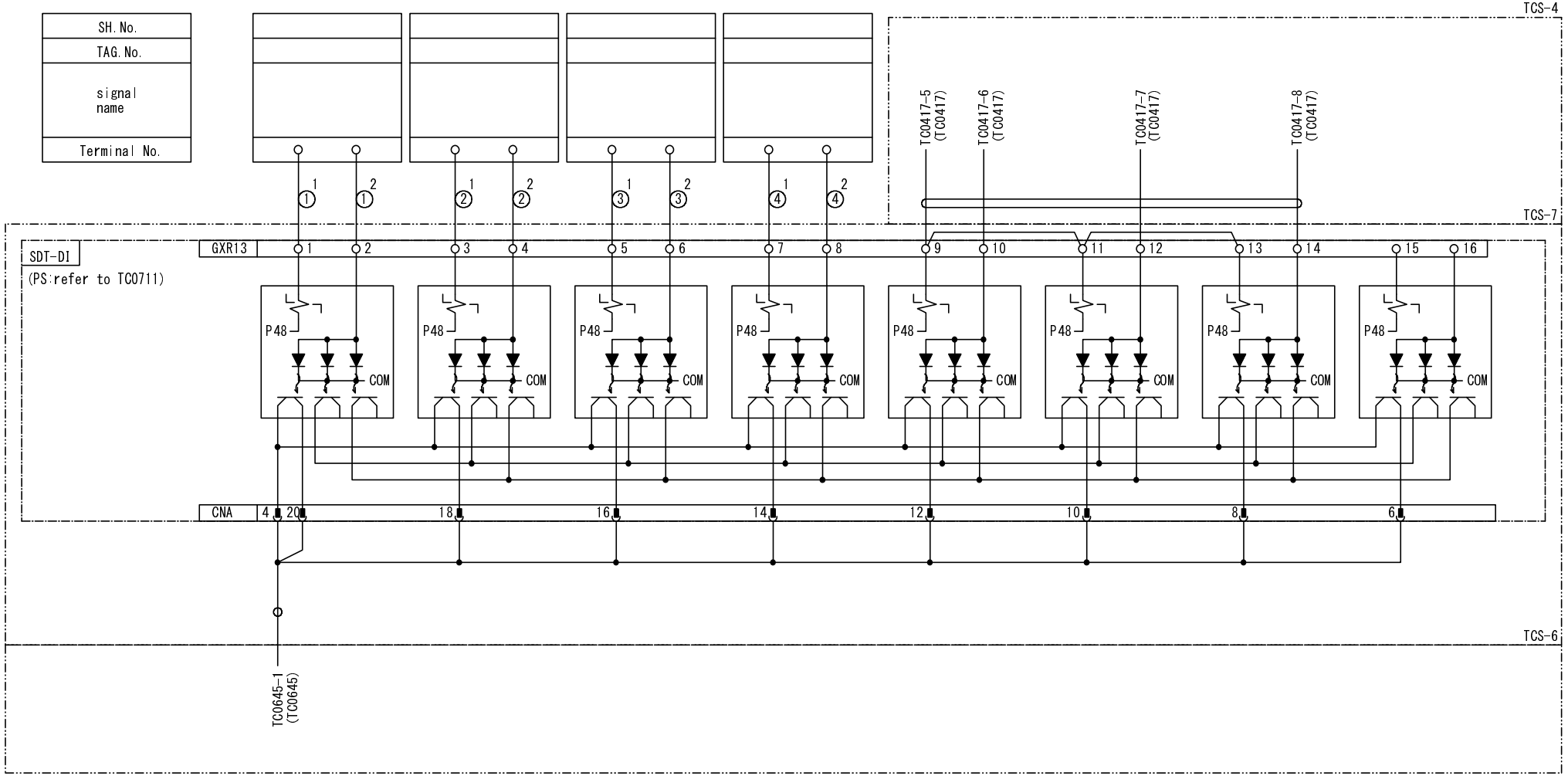
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0744 XSMM-TC



IN-DEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK
④		2/2		TCS-7			
③		2/2		TCS-7			
②		2/2		TCS-7			
①		2/2		TCS-7			

EXTERNAL INPUT COM DI(25)

SH. No. TC0745

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

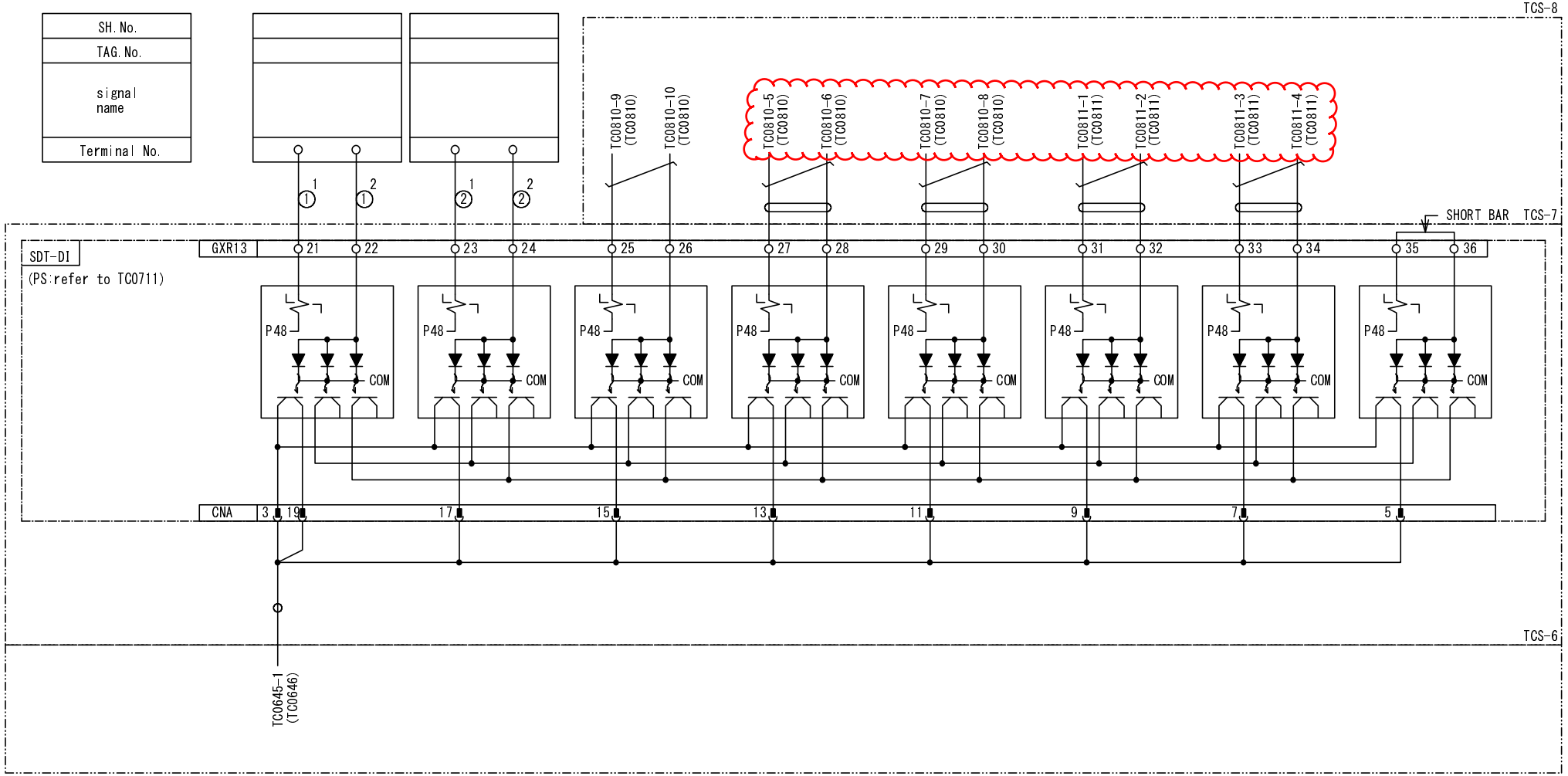
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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IN-DEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK
②		2/2		TCS-7			
①		2/2		TCS-7			

EXTERNAL INPUT COM DI(26)

SH. No. TC0746

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

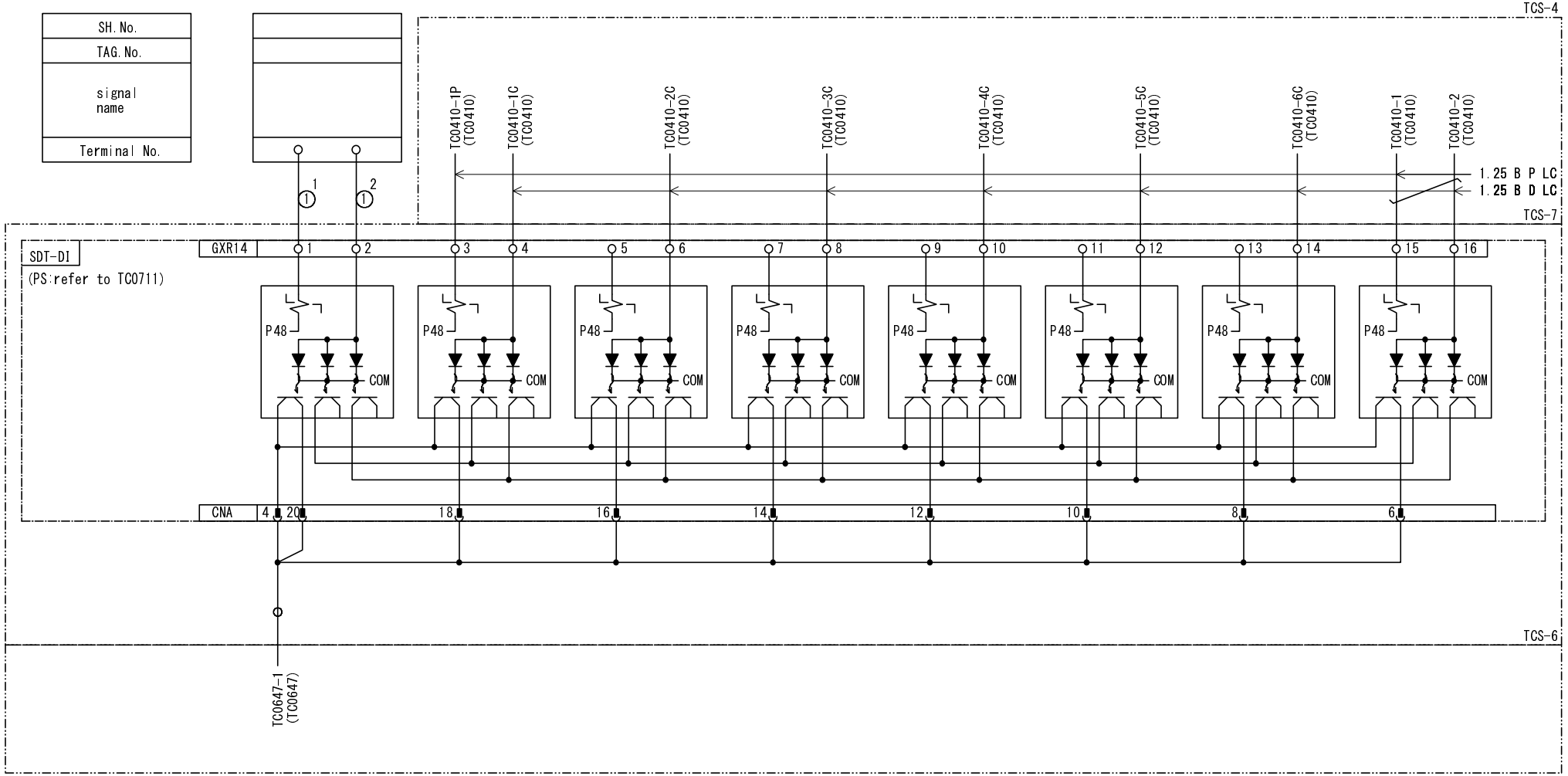
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/08/23 TC0746 XSMM-TC



INDEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK
①		2/2		TCS-7			

EXTERNAL INPUT COM DI(27)

SH. No. TC0747

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

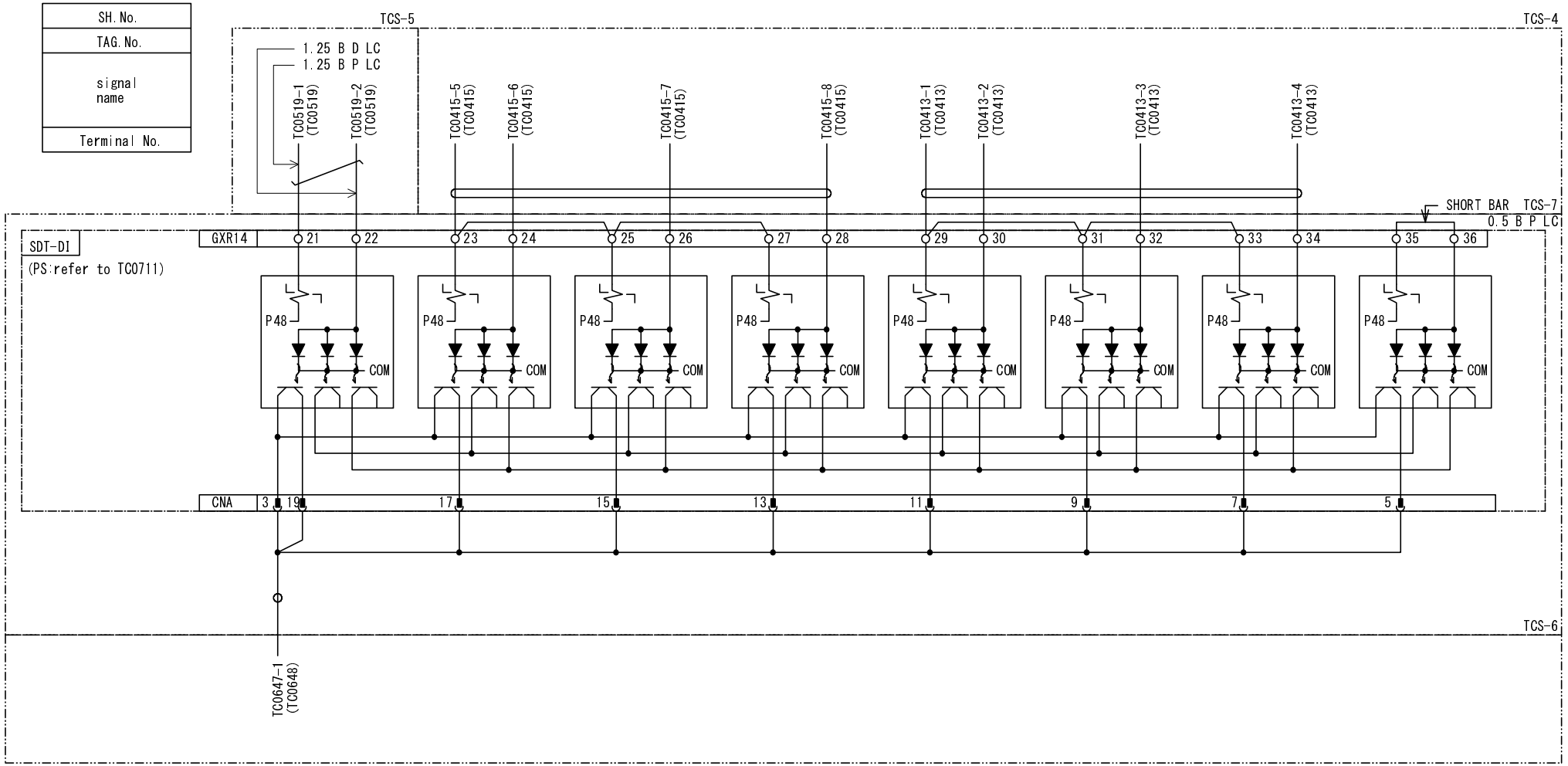
DWG. No. 331DM07733

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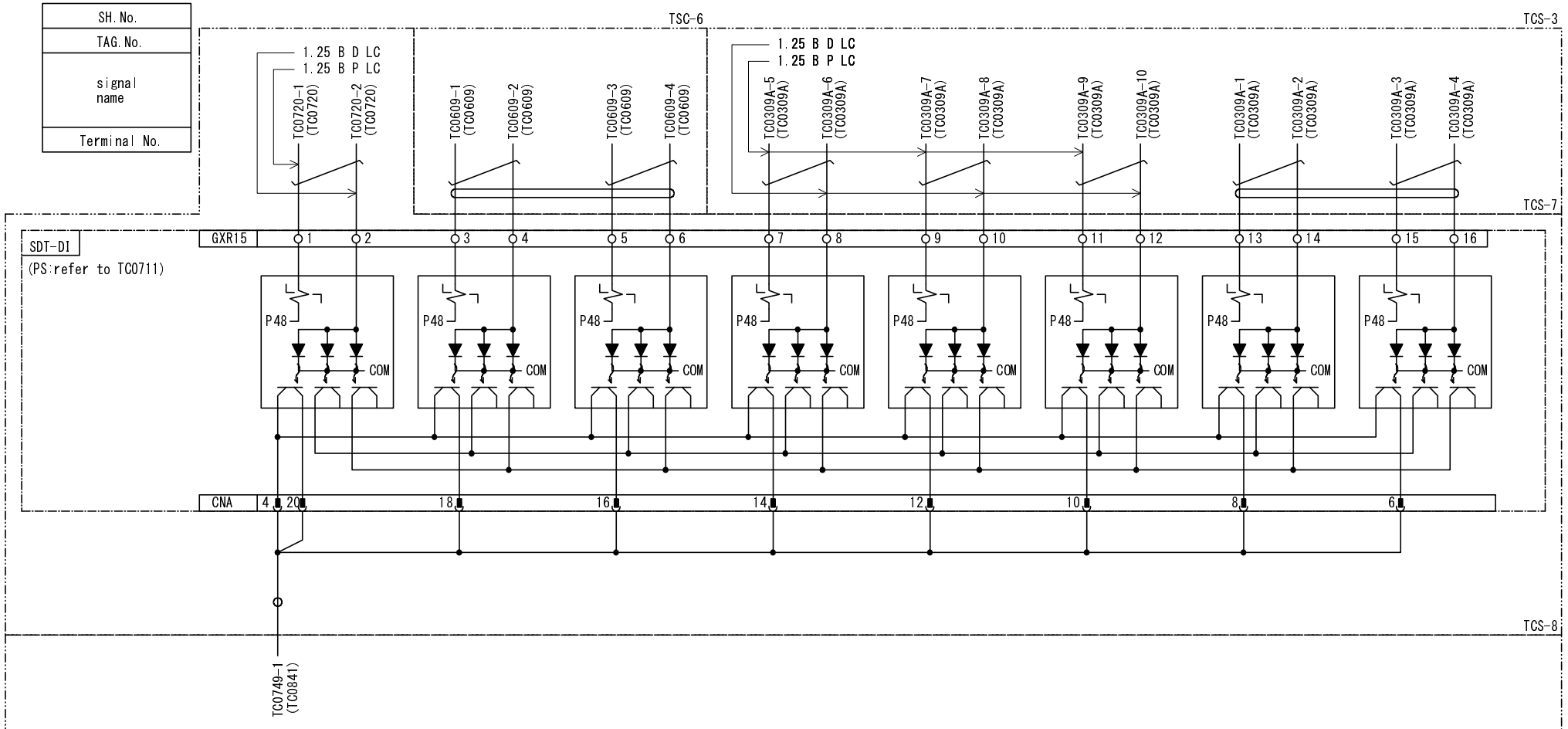
2016/07/28 TC0747 XSMM-TC

SH. No.
TAG. No.
signal name
Terminal No.



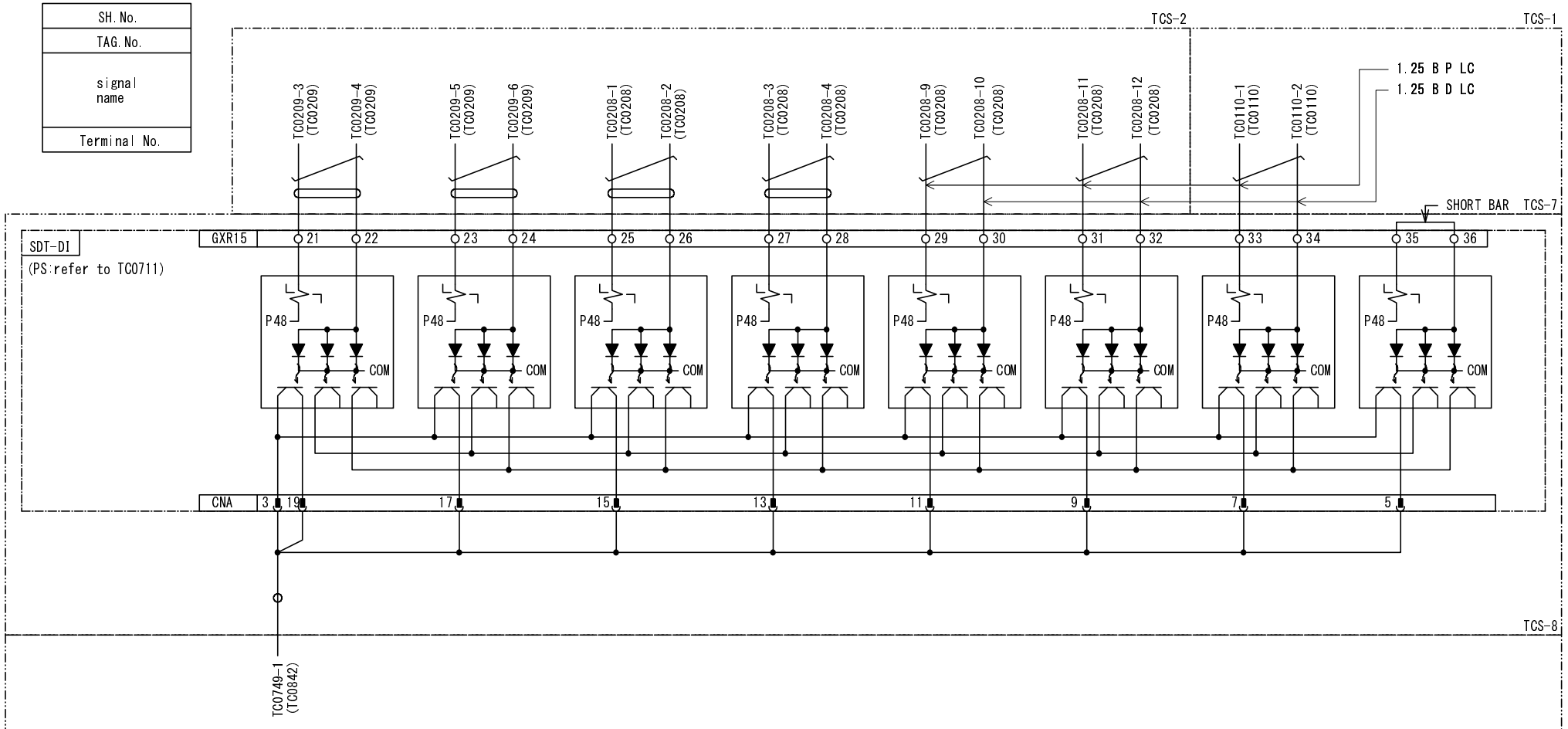
EXTERNAL INPUT COM DI(28)			SH. No.	TC0748
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 312	REV. 3

SH. No.
TAG. No.
signal name
Terminal No.

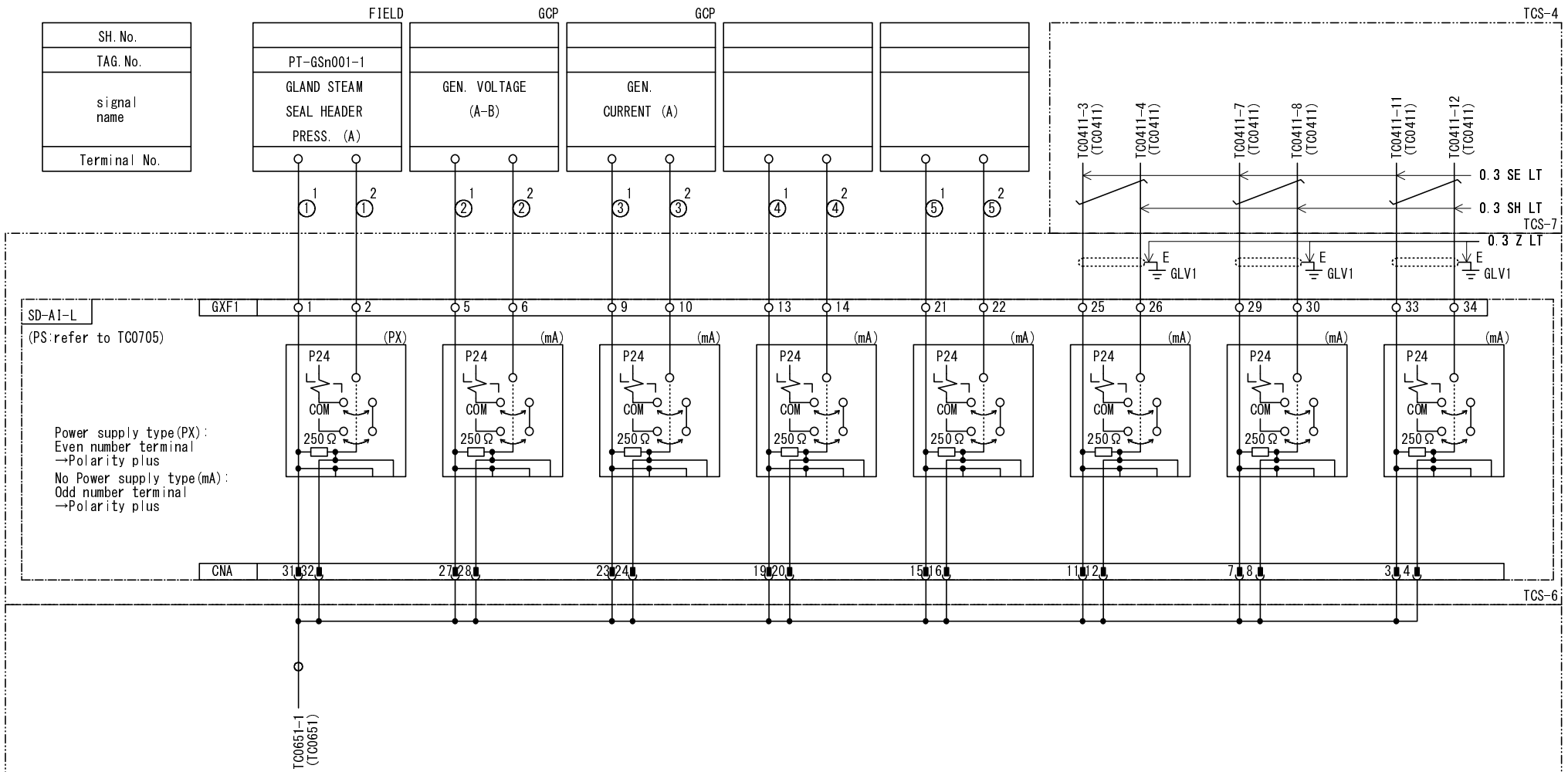


EXTERNAL INPUT COM DI(29)			SH. No.	TC0749
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 313	REV. 3

SH. No.
TAG. No.
signal name
Terminal No.



EXTERNAL INPUT COM DI(30)			SH. No.	TC0750
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 314	REV. 3



INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑤							
④							
③			GCP	TCS-4			
②			GCP	TCS-4			
①		2/2	FIELD	TCS-4			

EXTERNAL INPUT COM AI(1)

SH. No. TC0751

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

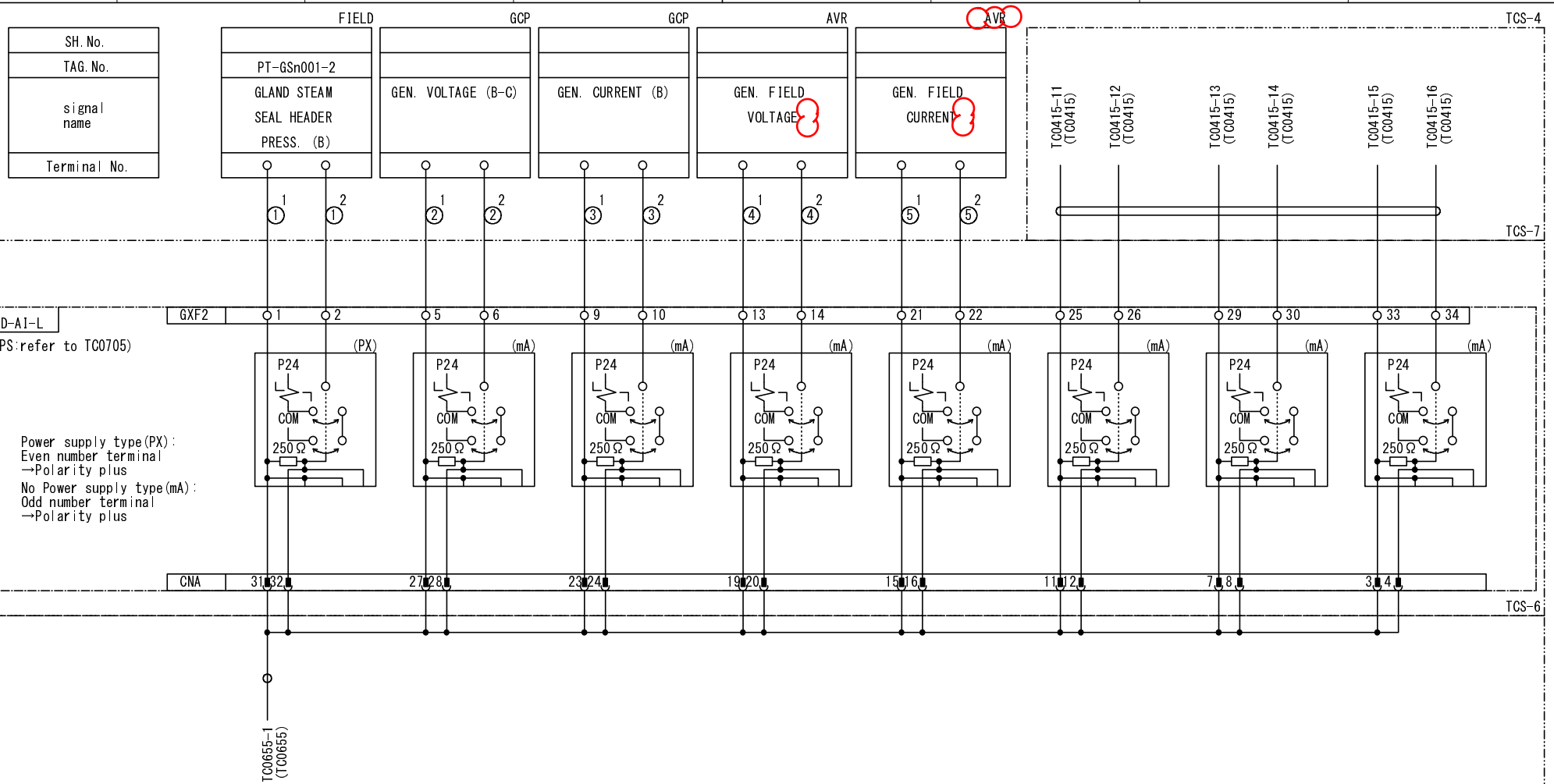
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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IN-DEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑤		2/2	AVR	TCS-4			
④		2/2	AVR	TCS-4			
③		2/2	GCP	TCS-4			
②		2/2	GCP	TCS-4			
①		2/2	FIELD	TCS-4			

EXTERNAL INPUT COM AI(2)

SH. No. TC0751A

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

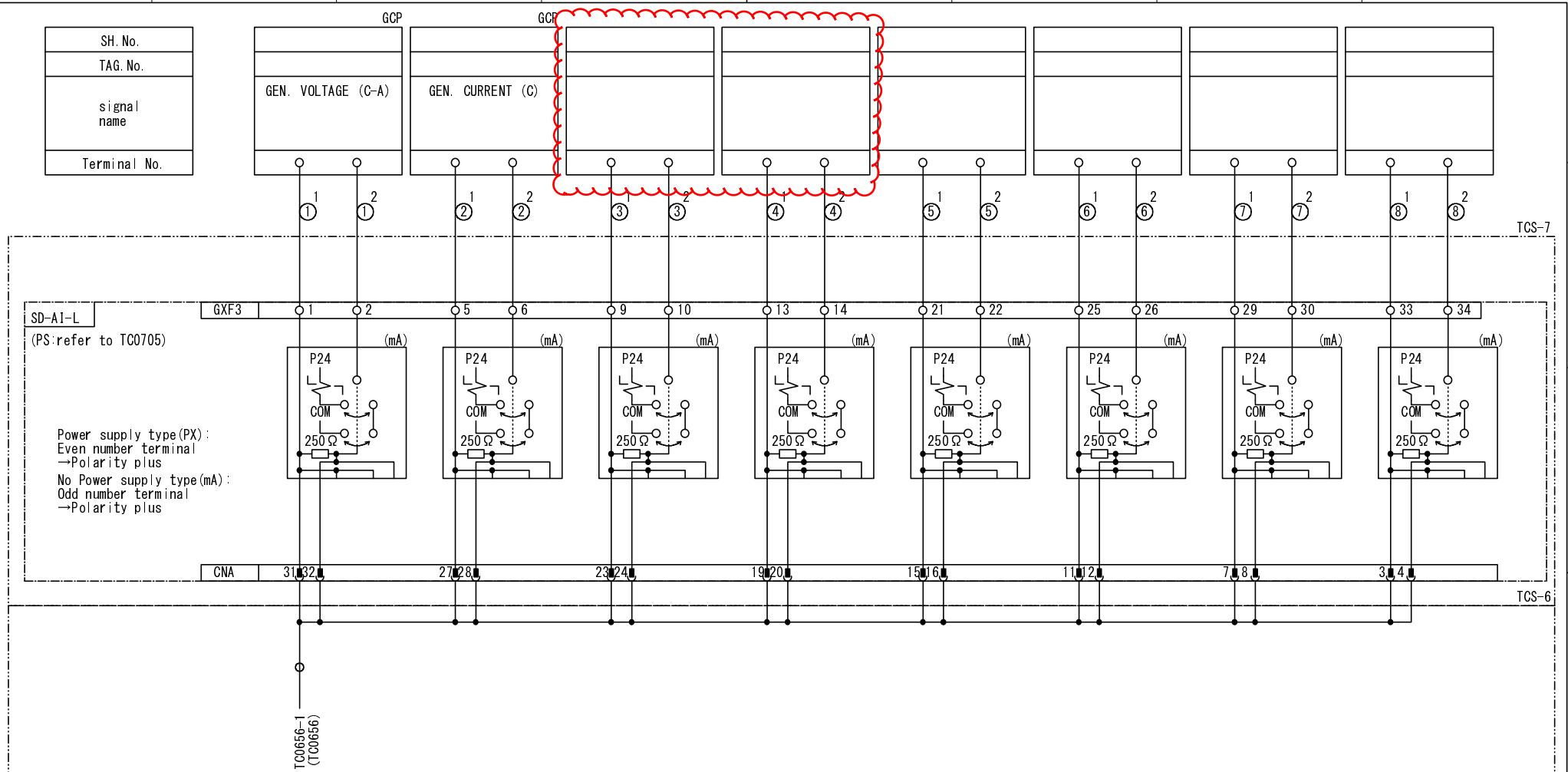
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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IN-DEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2		TCS-7			
⑦		2/2		TCS-7			
⑥		2/2		TCS-7			
⑤		2/2		TCS-7			
④		2/2		TCS-7			
③		2/2		TCS-7			
②		2/2	GCP	TCS-7			
①		2/2	GCP	TCS-7			

EXTERNAL INPUT COM AI(3)

SH. No. TC0751B

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

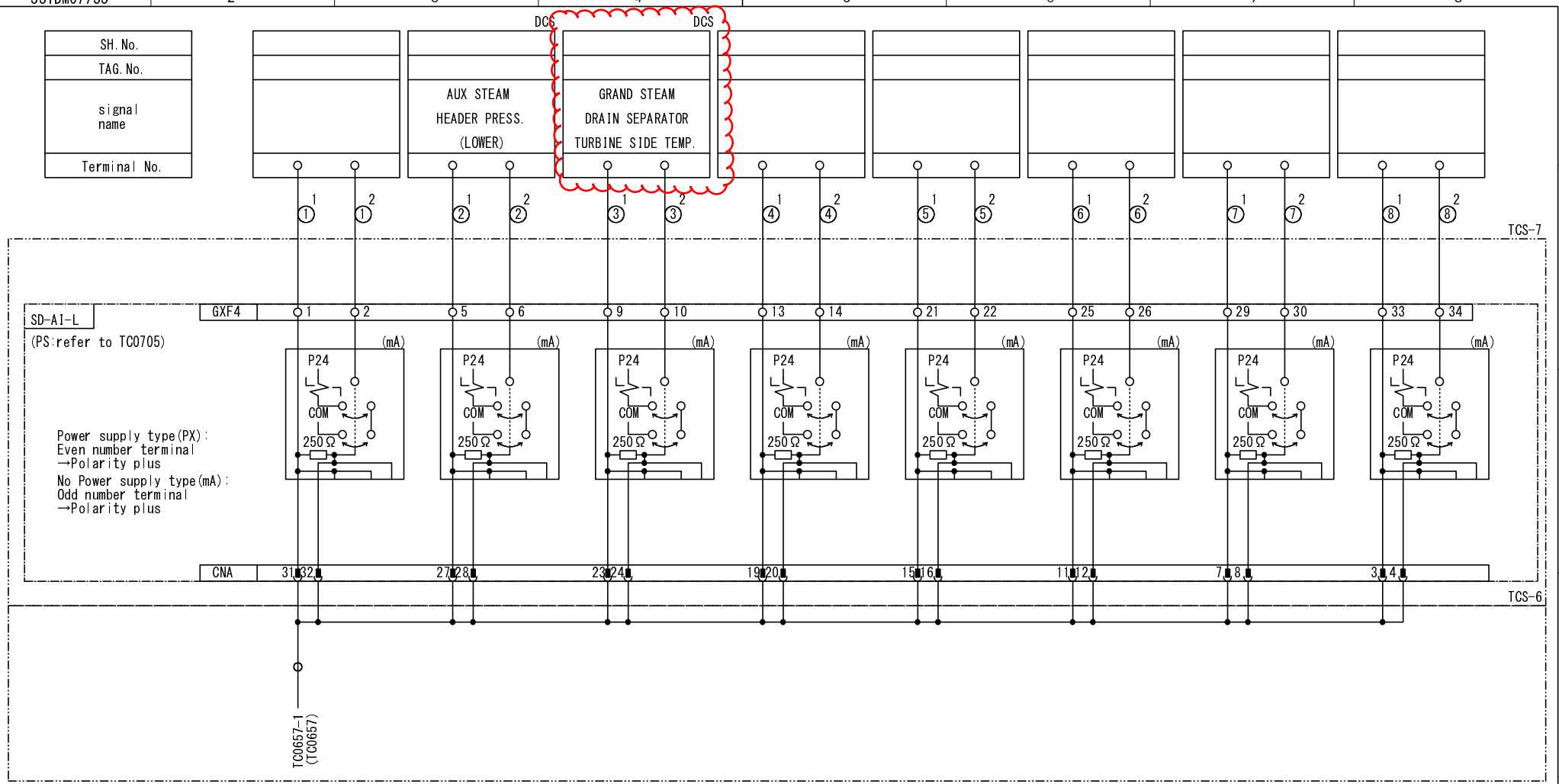
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		2/2		TCS-7			
⑦		2/2		TCS-7			
⑥		2/2		TCS-7			
⑤		2/2		TCS-7			
④		2/2		TCS-7			
③		2/2	DCS	TCS-7			
②		2/2	DCS	TCS-7			
①		2/2		TCS-7			

EXTERNAL INPUT COM AI(4)

SH. No. TC0751C

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

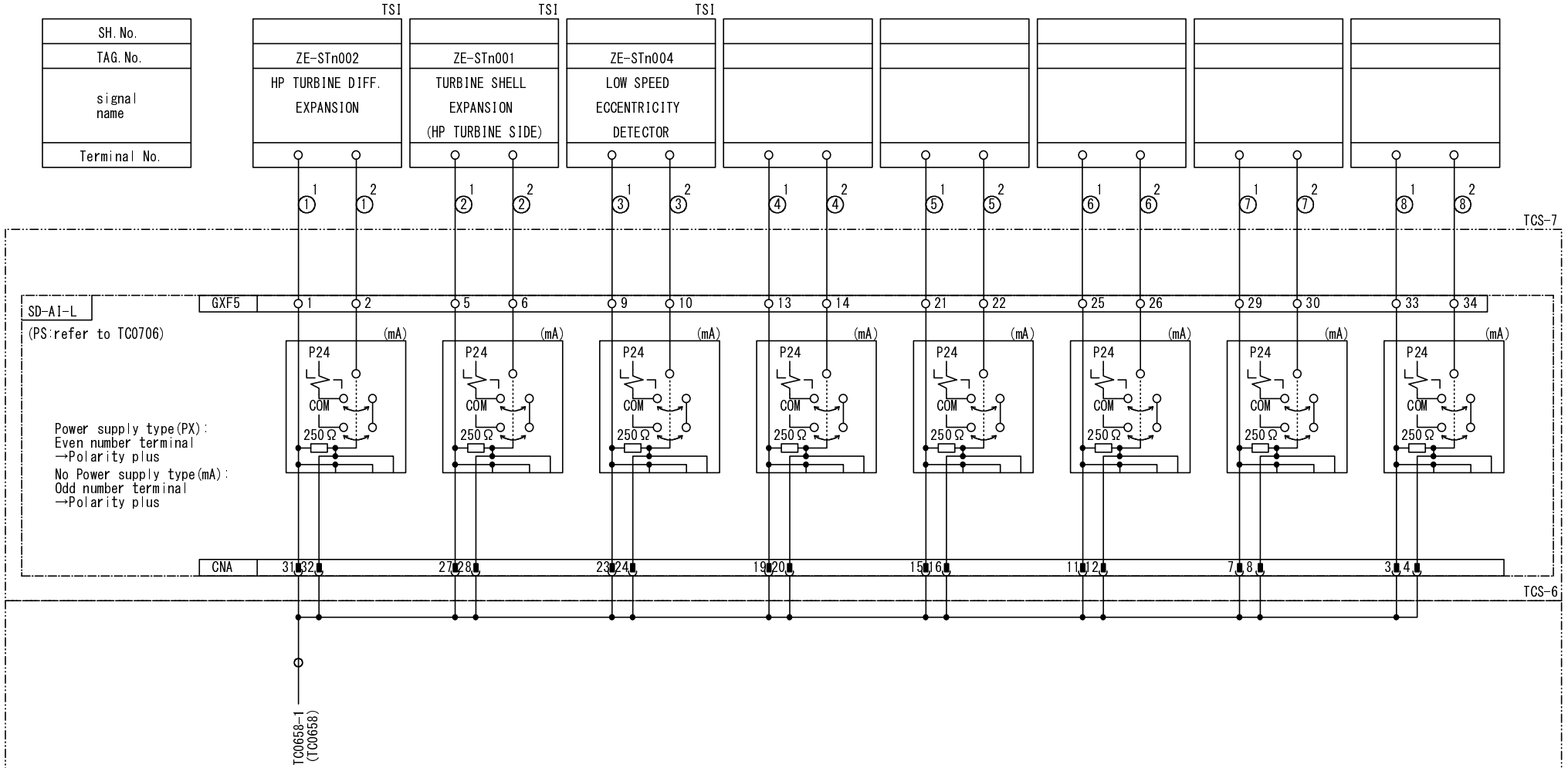
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/08/23 TC0751C XSMM-TC



IN-DEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2		TCS-7			
⑦		2/2		TCS-7			
⑥		2/2		TCS-7			
⑤		2/2		TCS-7			
④		2/2		TCS-7			
③		2/2	TSI	TCS-7			
②		2/2	TSI	TCS-7			
①		2/2	TSI	TCS-7			

EXTERNAL INPUT COM AI(5)

SH. No. TC0751D

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

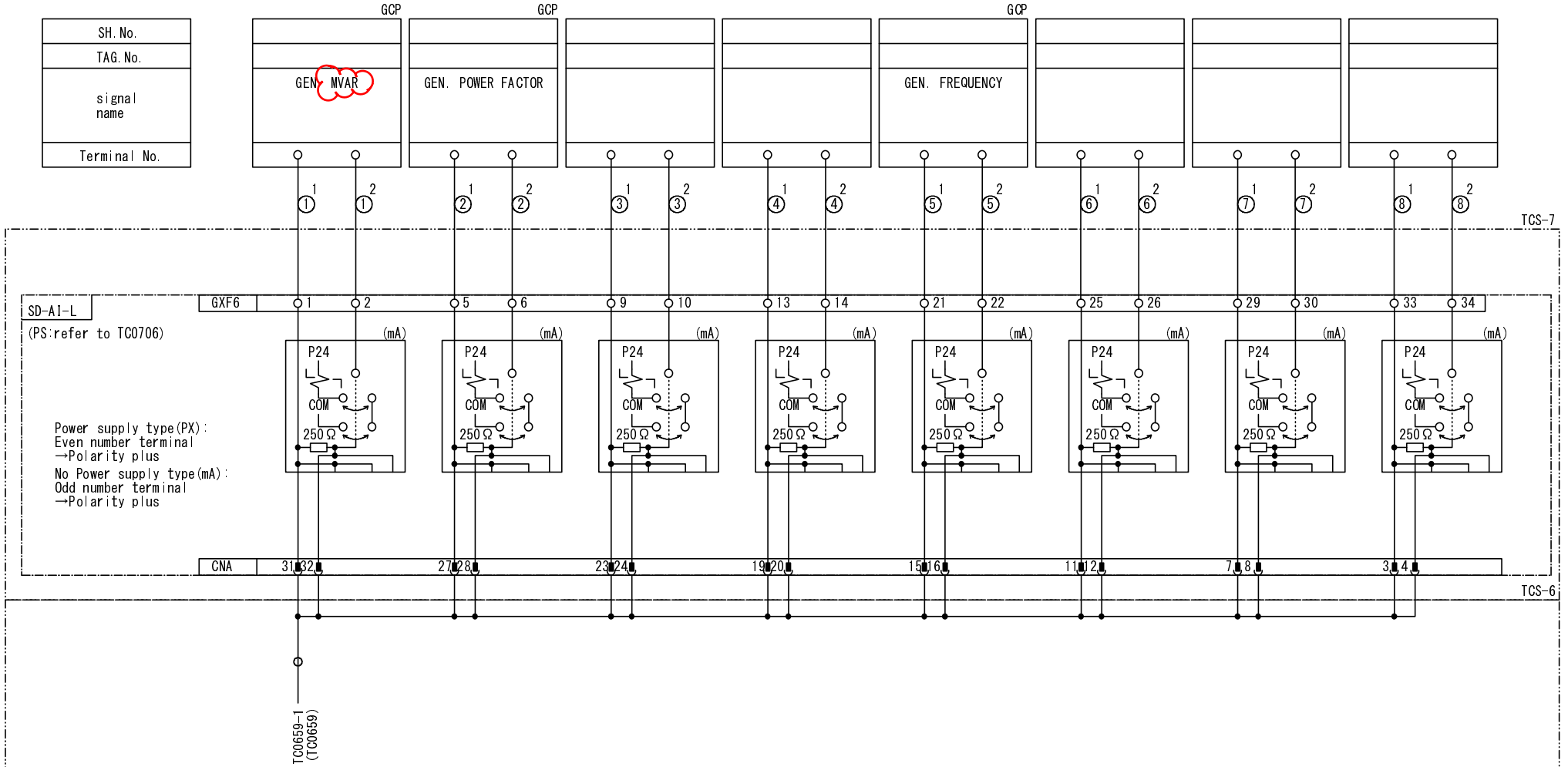
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0751D XSMM-TC



INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		2/2		TCS-7			
⑦		2/2		TCS-7			
⑥		2/2		TCS-7			
⑤		2/2	GCP	TCS-7			
④		2/2		TCS-7			
③		2/2		TCS-7			
②		2/2	GCP	TCS-7			
①		2/2	GCP	TCS-7			

EXTERNAL INPUT COM AI(6)

SH. No. TC0751E

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

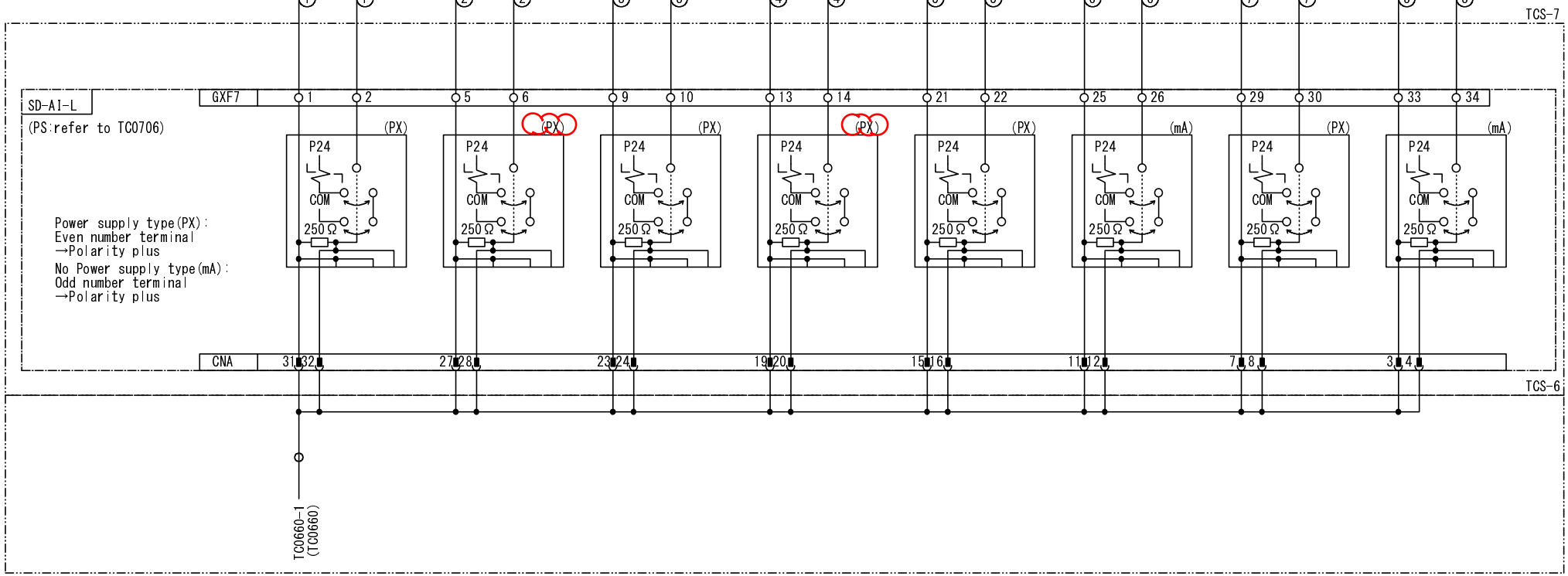
DWG. No. 331DM07733

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2016/07/28 TC0751E XSMM-TC

SH. No.							
TAG. No.	ZT-GSn031	ZT-GSn034	ZT-STn008	ZT-ASn011			
signal name	STEAM SEAL FEED VALVE POSITION	STEAM PACKING UNLOADING VALVE POSITION	CV (B) BELOW SEAT SECONDARY WARMING DRAIN VALVE POSITION	HP TURBINE WARMING VALVE POSITION			
Terminal No.							
	FIELD	FIELD	MOV	MOV			



INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		2/2		TCS-7			
⑦		2/2		TCS-7			
⑥		2/2		TCS-7			
⑤		2/2		TCS-7			
④		2/2	MOV	TCS-7			
③		2/2	MOV	TCS-7			
②		2/2	FIELD	TCS-7			
①		2/2	FIELD	TCS-7			

EXTERNAL INPUT COM AI(7)

SH. No. TC0751F

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

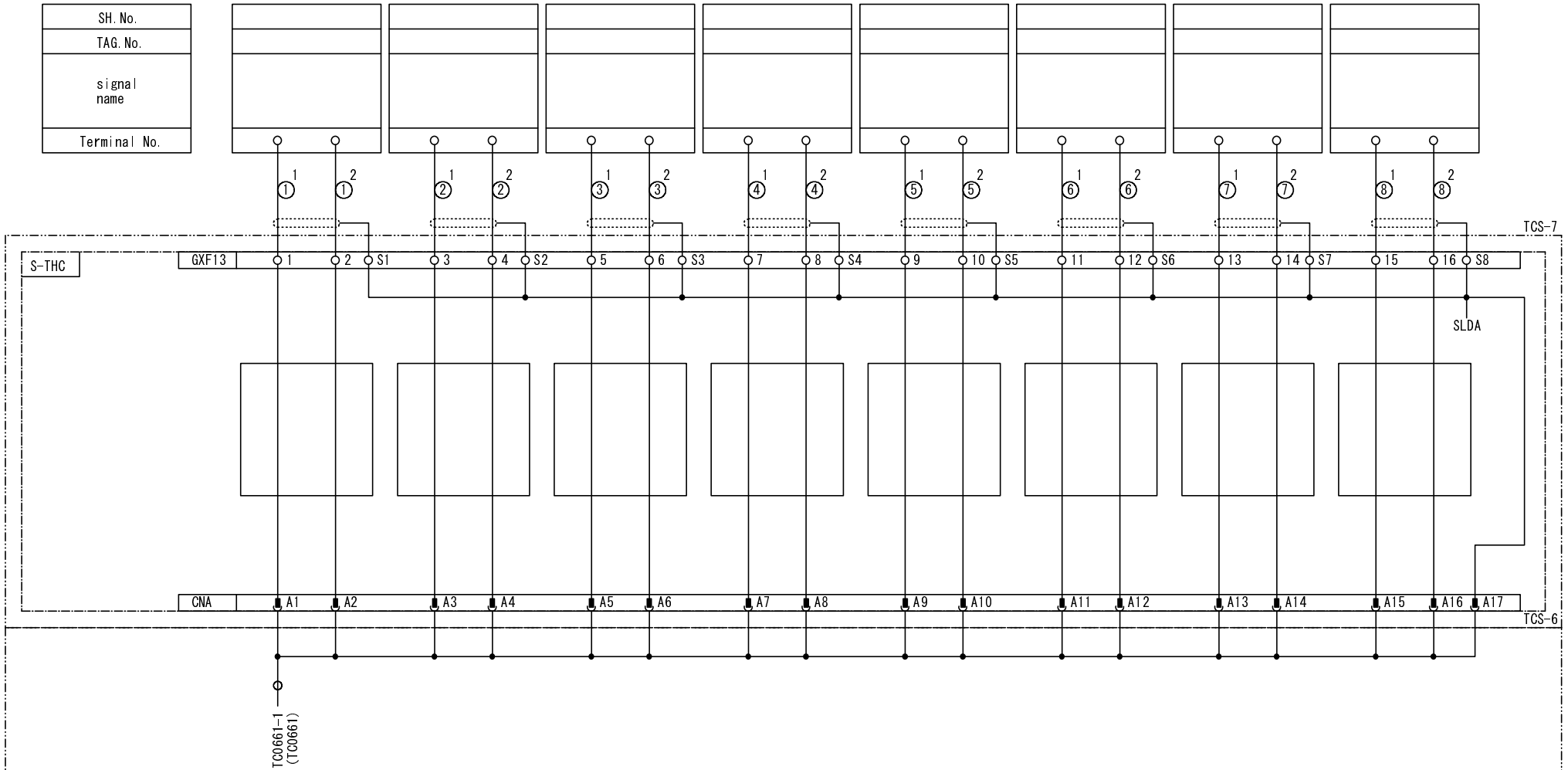
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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INDEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK
⑧		2/2		TCS-7			
⑦		2/2		TCS-7			
⑥		2/2		TCS-7			
⑤		2/2		TCS-7			
④		2/2		TCS-7			
③		2/2		TCS-7			
②		2/2		TCS-7			
①		2/2		TCS-7			

EXTERNAL INPUT COM THC/RTD AI(1)

SH. No. TC0755

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

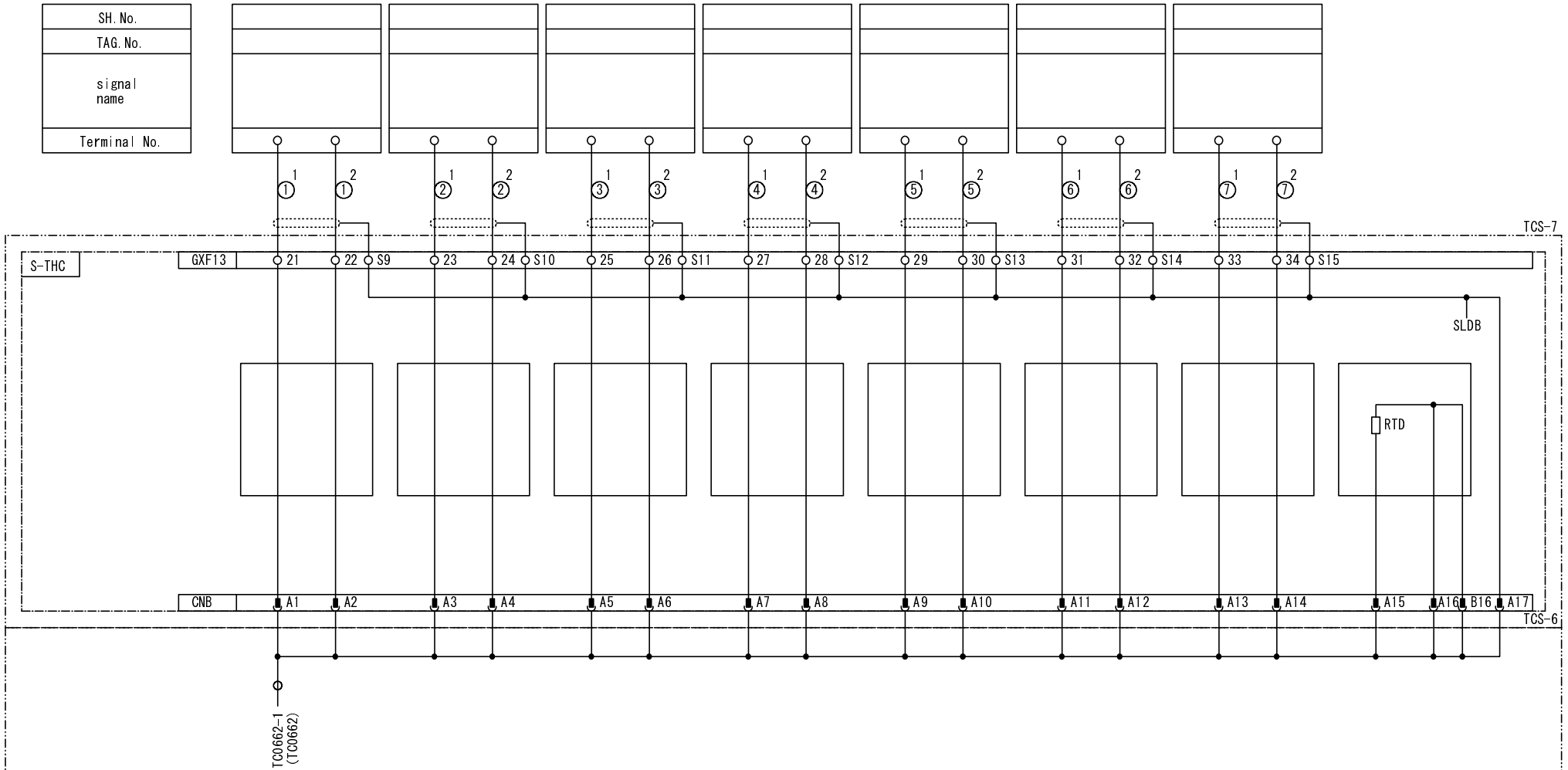
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0755 XSMM-TC



◯							
⑦		2/2		TCS-7			
⑥		2/2		TCS-7			
⑤		2/2		TCS-7			
④		2/2		TCS-7			
③		2/2		TCS-7			
②		2/2		TCS-7			
①		2/2		TCS-7			
IN-DEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		

EXTERNAL INPUT COM THC/RTD AI(2)

SH. No. TC0756

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

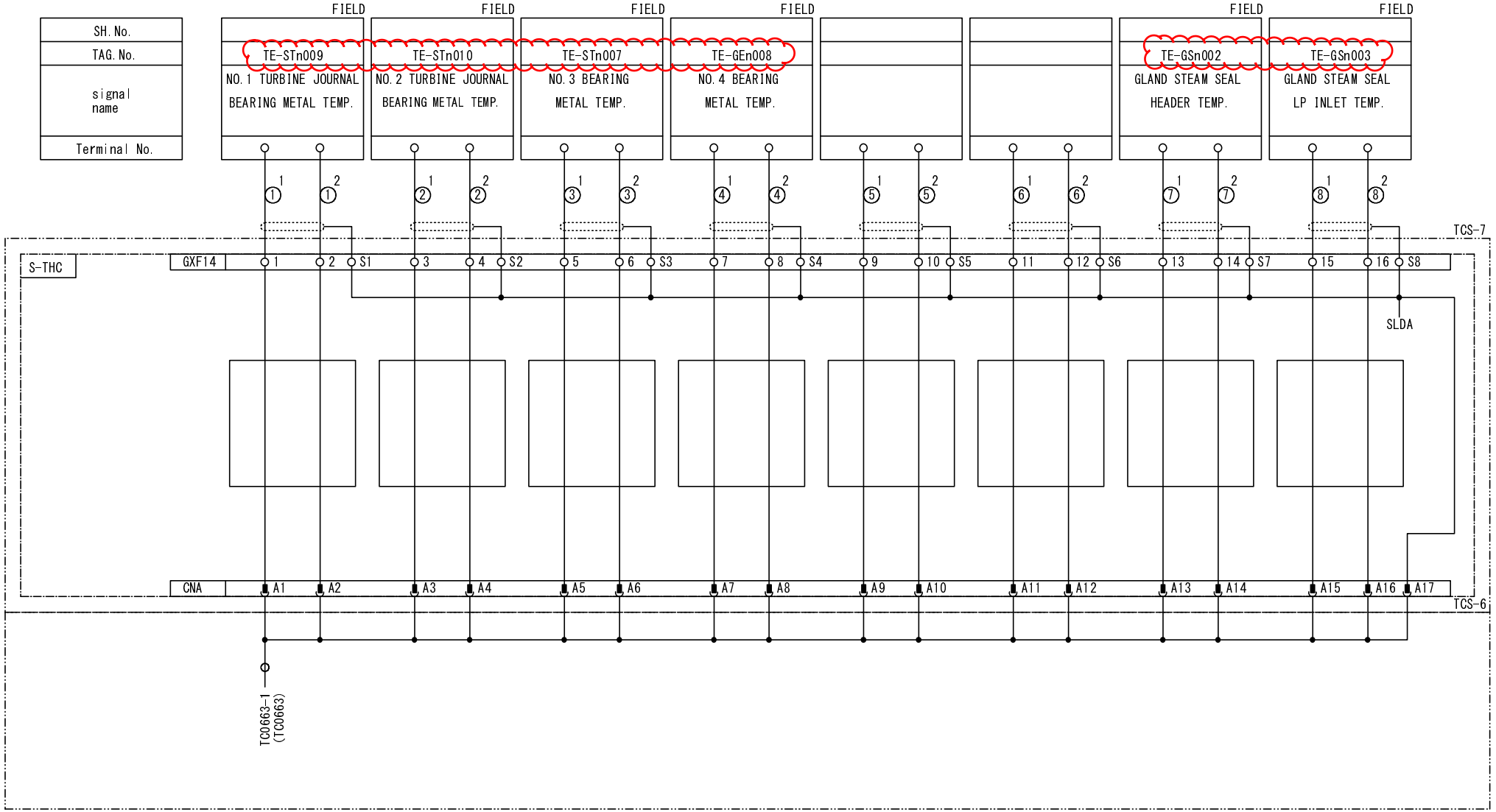
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0756 XSMM-TC



IN-DEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		2/2	FIELD	TCS-7			
⑦		2/2	FIELD	TCS-7			
⑥		2/2		TCS-7			
⑤		2/2		TCS-7			
④		2/2	FIELD	TCS-7			
③		2/2	FIELD	TCS-7			
②		2/2	FIELD	TCS-7			
①		2/2	FIELD	TCS-7			

EXTERNAL INPUT COM THC/RTD AI(3)

SH. No. TC0757

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

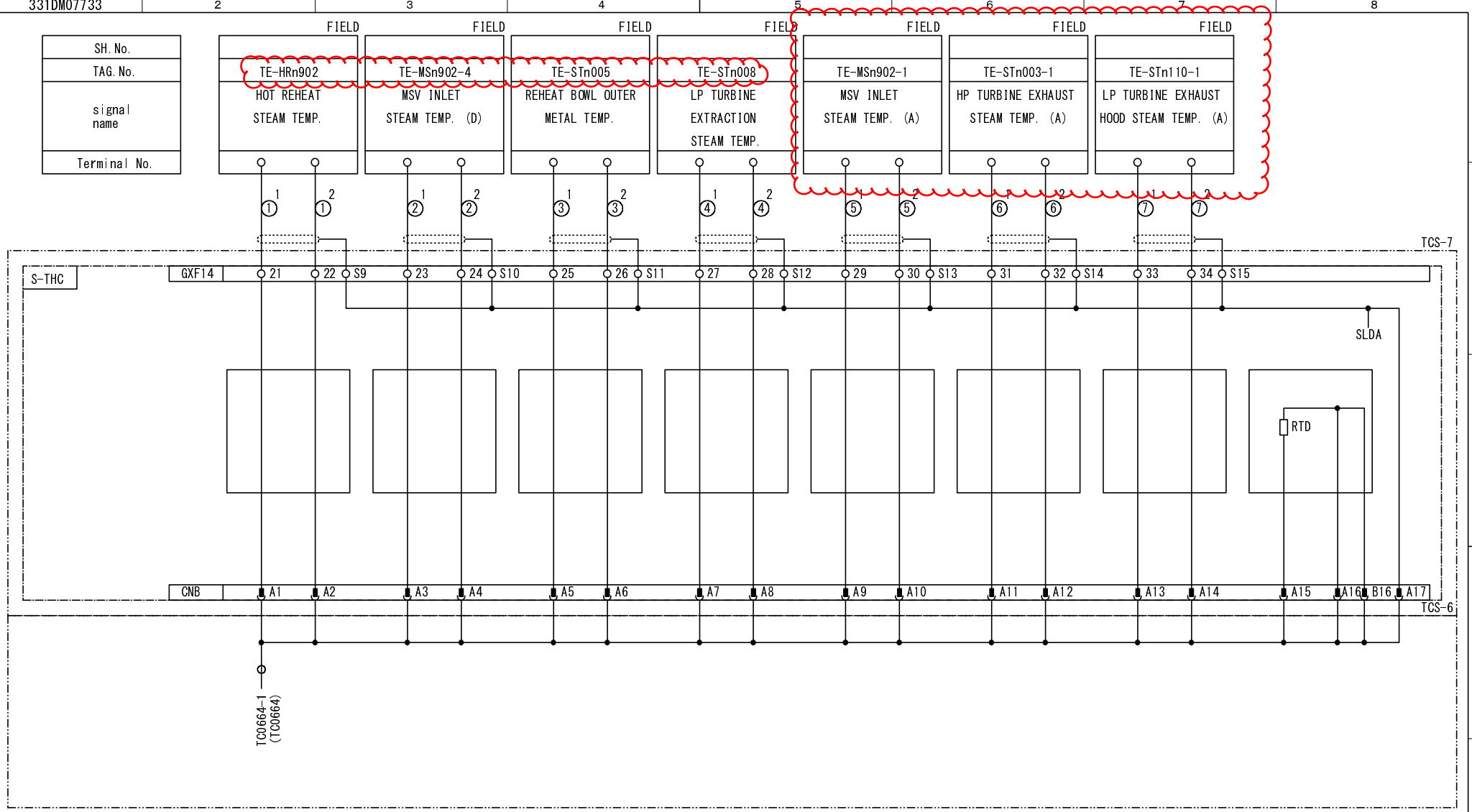
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0757 XSMM-TC



INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		2/2	FIELD	TCS-7			
⑦		2/2	FIELD	TCS-7			
⑥		2/2	FIELD	TCS-7			
⑤		2/2	FIELD	TCS-7			
④		2/2	FIELD	TCS-7			
③		2/2	FIELD	TCS-7			
②		2/2	FIELD	TCS-7			
①		2/2	FIELD	TCS-7			

EXTERNAL INPUT COM THC/RTD AI(4)

SH. No. TC0758

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

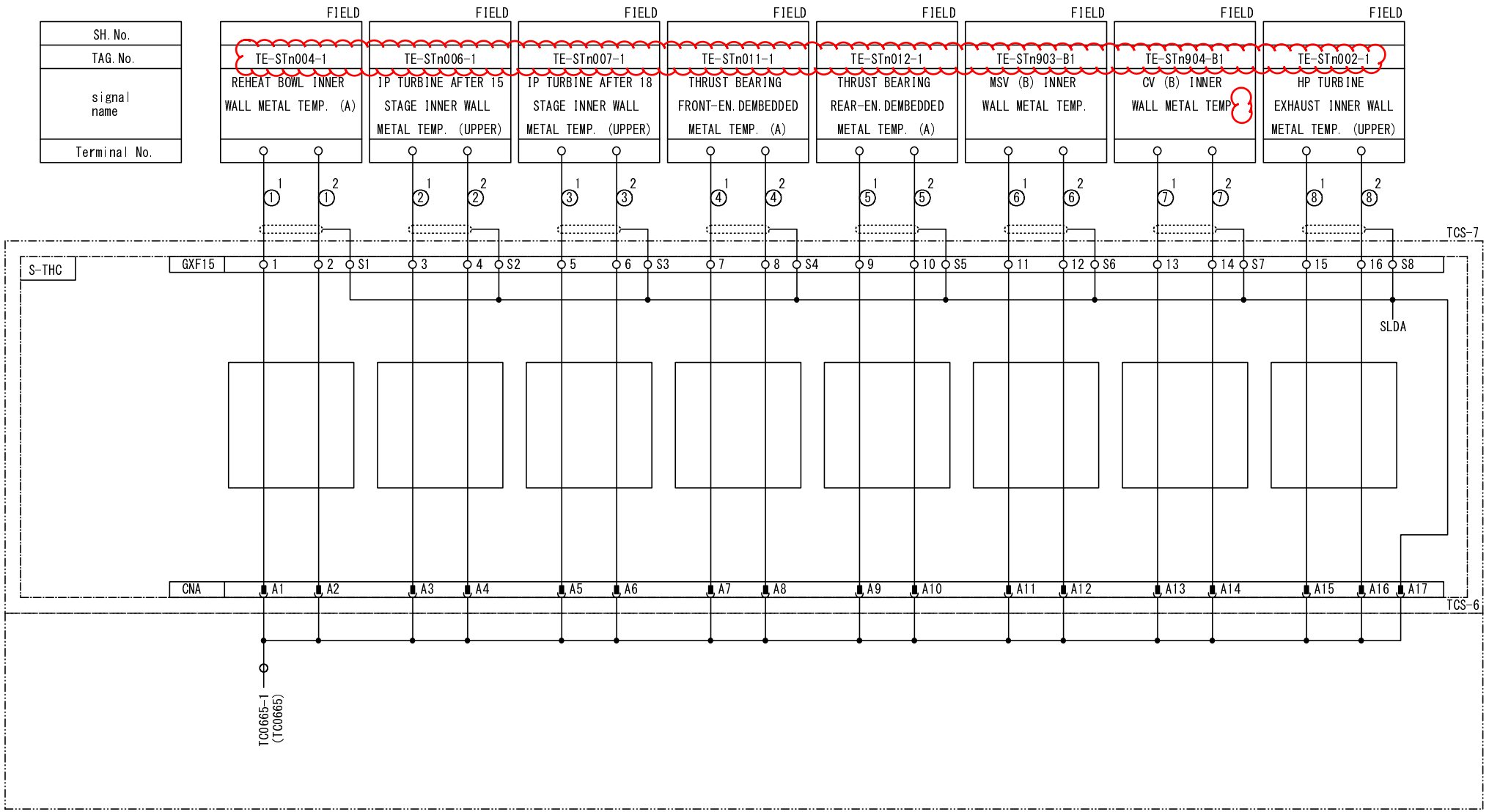
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0758 XSMM-TC



IN-DEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧			FIELD	TCS-7			
⑦			FIELD	TCS-7			
⑥			FIELD	TCS-7			
⑤			FIELD	TCS-7			
④			FIELD	TCS-7			
③			FIELD	TCS-7			
②			FIELD	TCS-7			
①		2/2	FIELD	TCS-7			

EXTERNAL INPUT COM THC/RTD AI(5)

SH. No. TC0759

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

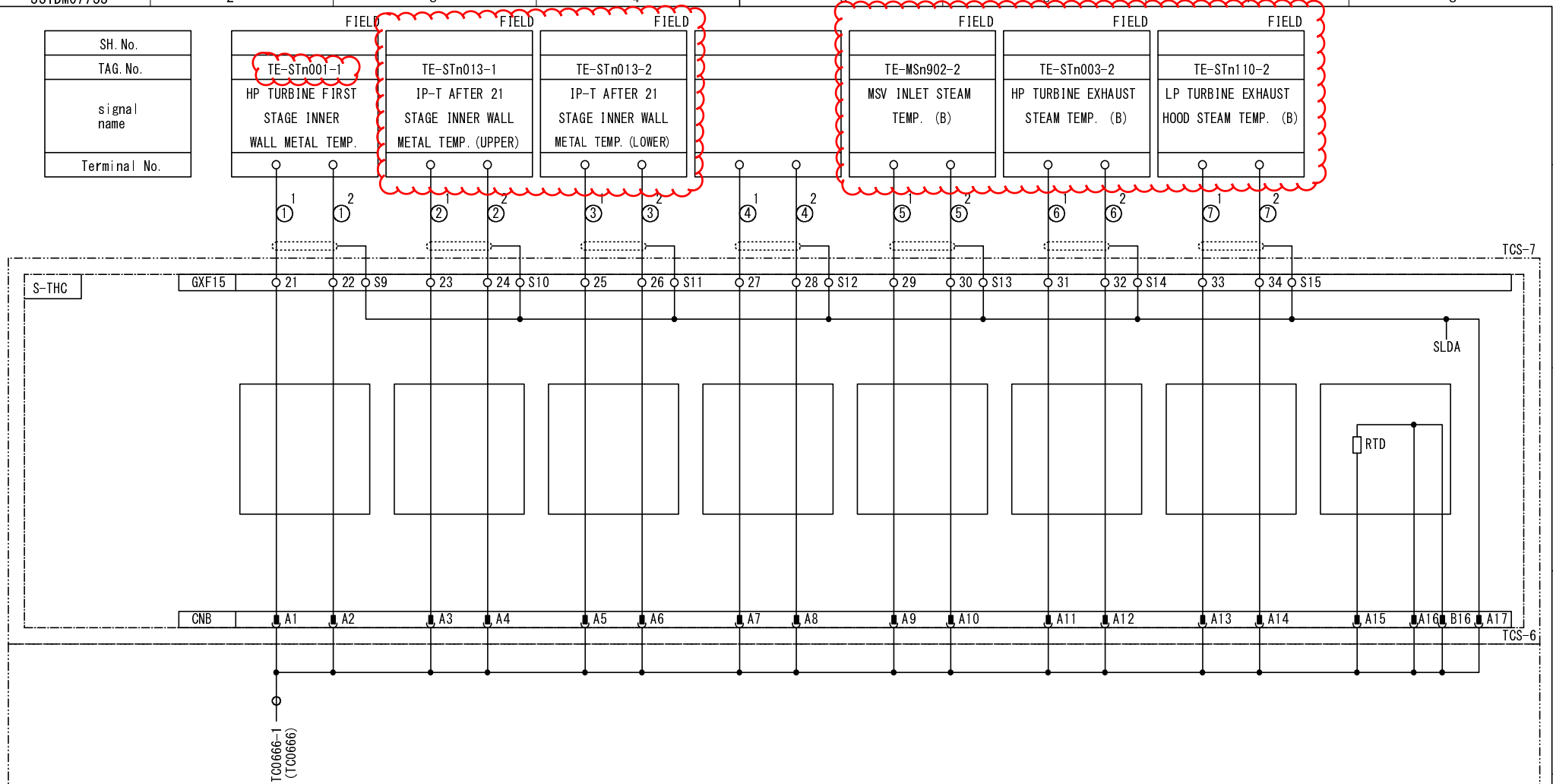
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0759 XSMM-TC



INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
◯				TCS-7			
⑦		2/2	FIELD	TCS-7			
⑥		2/2	FIELD	TCS-7			
⑤		2/2	FIELD	TCS-7			
④		2/2		TCS-7			
③		2/2	FIELD	TCS-7			
②		2/2	FIELD	TCS-7			
①		2/2	FIELD	TCS-7			

EXTERNAL INPUT COM THC/RTD AI(6)

SH. No. TC0760

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

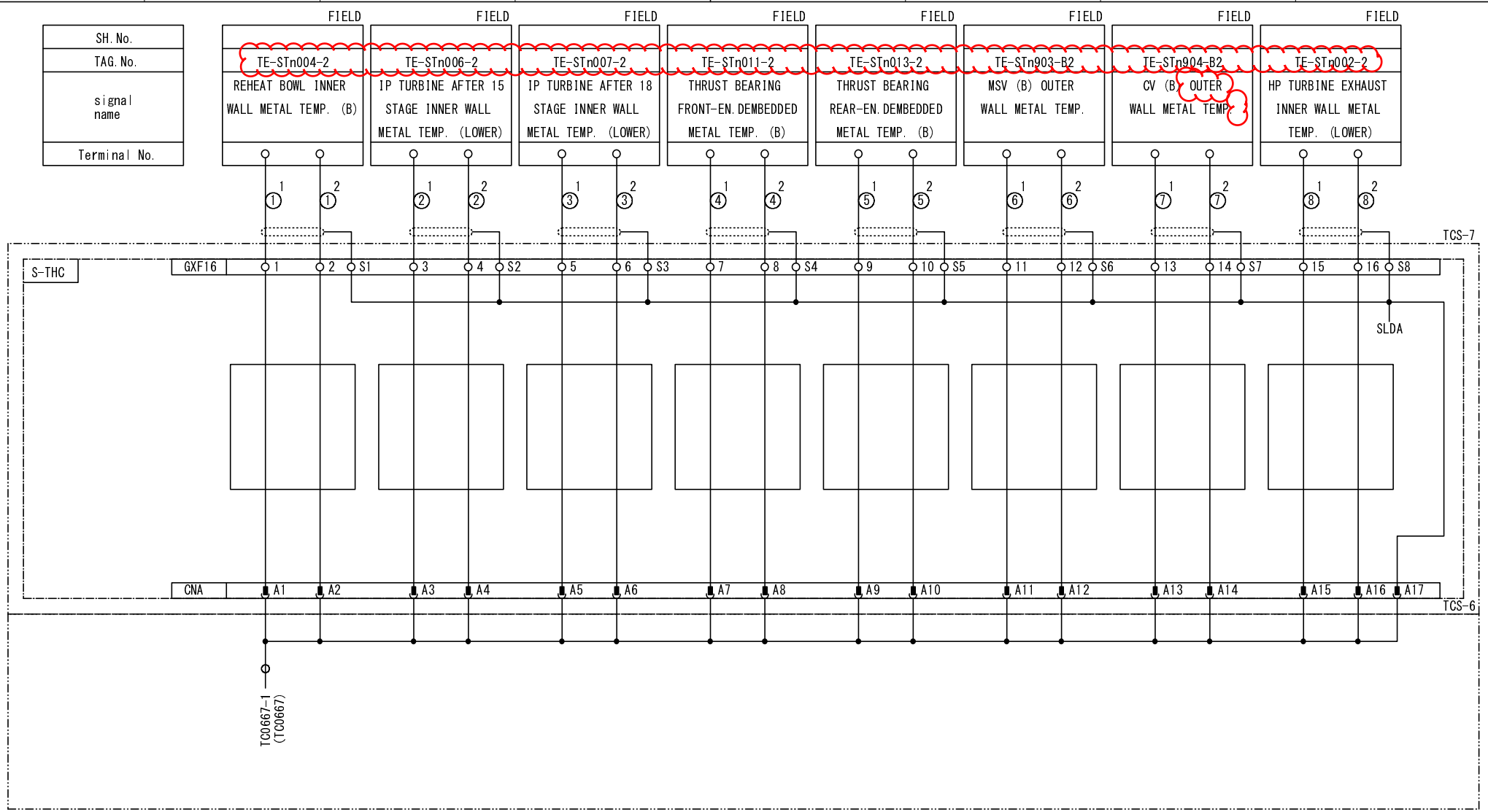
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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IN-DEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		2/2	FIELD	TCS-7			
⑦		2/2	FIELD	TCS-7			
⑥		2/2	FIELD	TCS-7			
⑤		2/2	FIELD	TCS-7			
④		2/2	FIELD	TCS-7			
③		2/2	FIELD	TCS-7			
②		2/2	FIELD	TCS-7			
①		2/2	FIELD	TCS-7			

EXTERNAL INPUT COM THC/RTD AI(7)

SH. No. TC0761

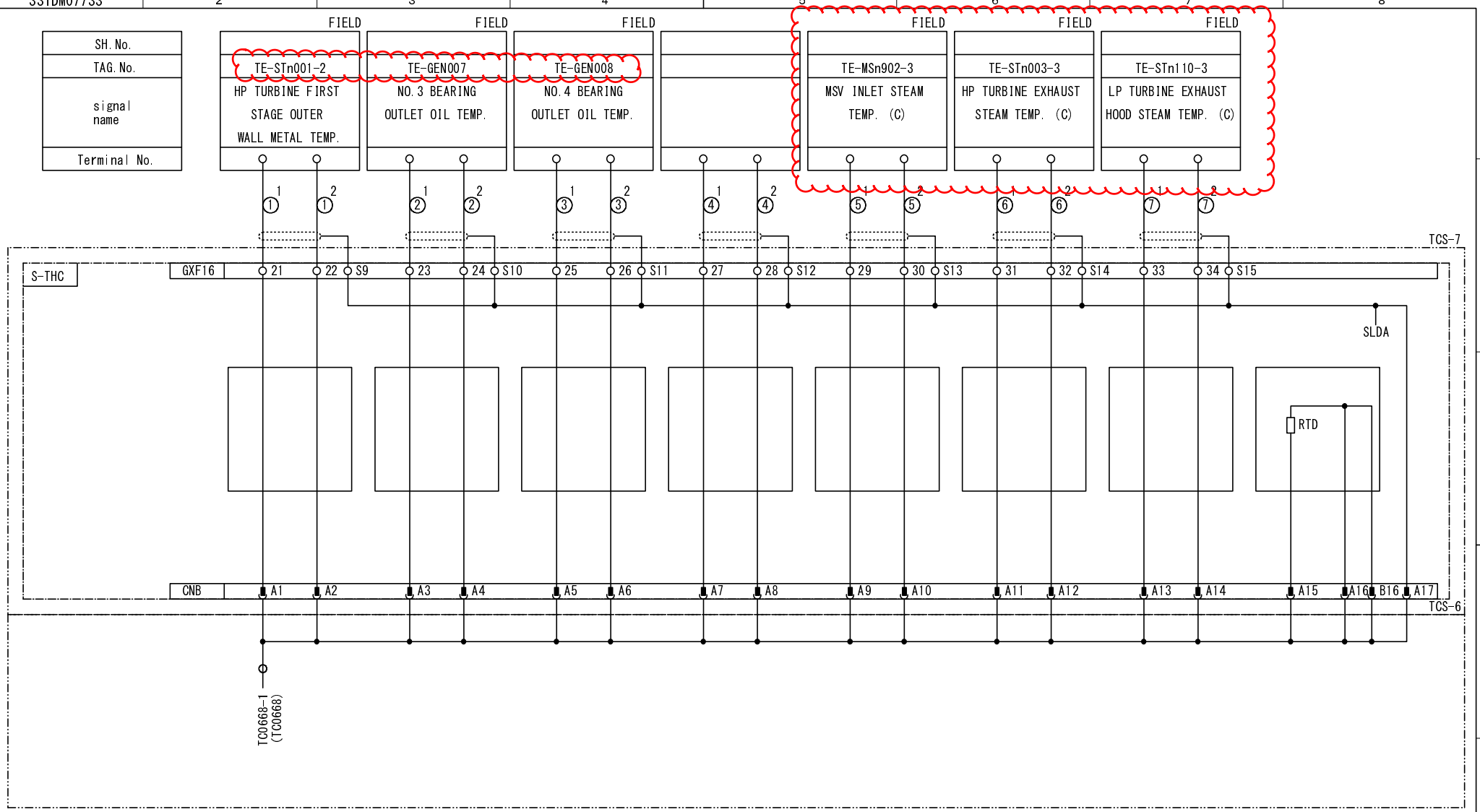
	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		2/2	FIELD	TCS-7			
⑦		2/2	FIELD	TCS-7			
⑥		2/2	FIELD	TCS-7			
⑤		2/2	FIELD	TCS-7			
④		2/2		TCS-7			
③		2/2	FIELD	TCS-7			
②		2/2	FIELD	TCS-7			
①		2/2	FIELD	TCS-7			

EXTERNAL INPUT COM THC/RTD AI(8)

SH. No. TC0762

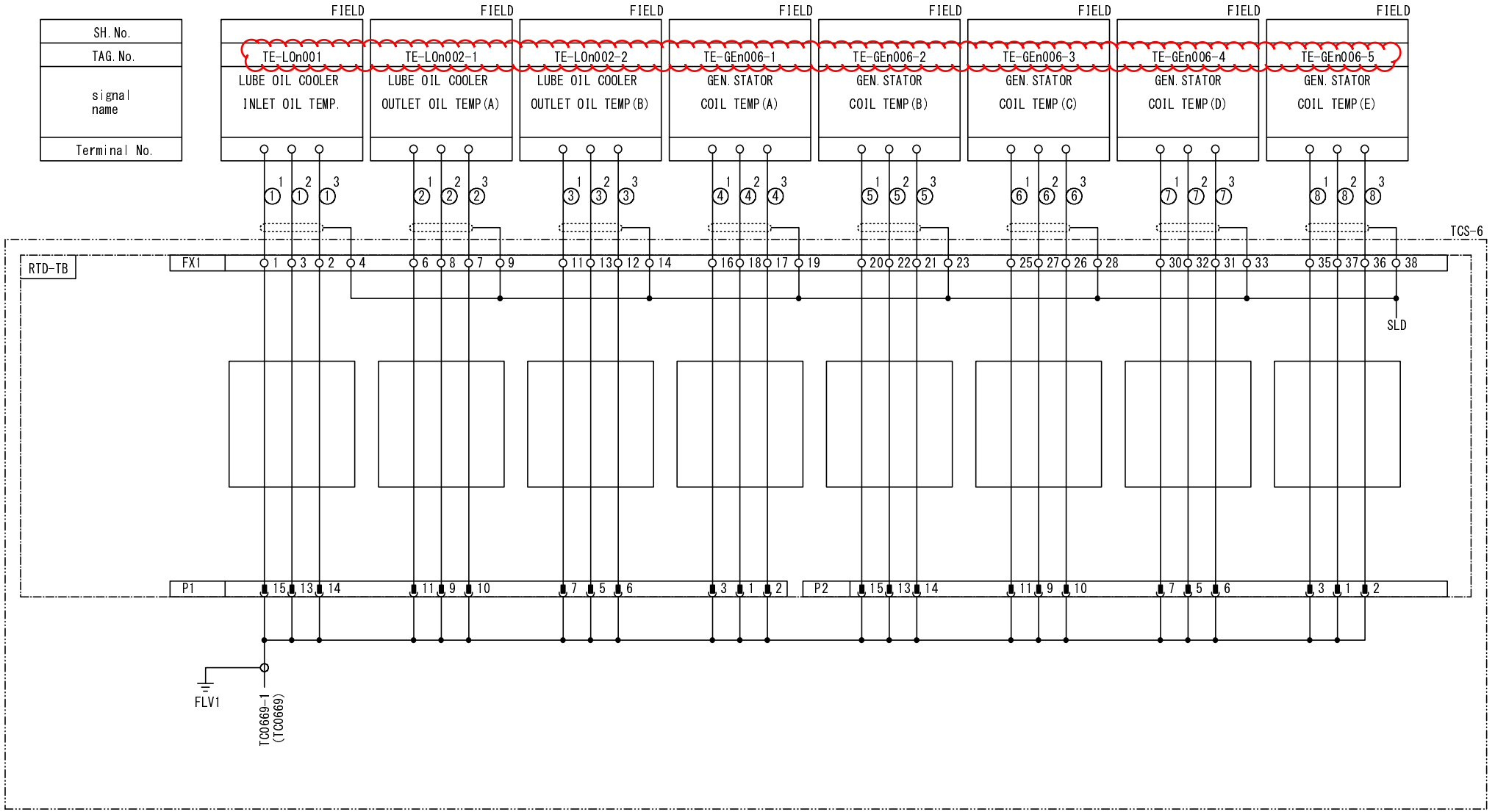
	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

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2016/07/28 TC0762 XSMM-TC



IN-DEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		3/3	FIELD	TCS-6			
⑦		3/3	FIELD	TCS-6			
⑥		3/3	FIELD	TCS-6			
⑤		3/3	FIELD	TCS-6			
④		3/3	FIELD	TCS-6			
③		3/3	FIELD	TCS-6			
②		3/3	FIELD	TCS-6			
①		3/3	FIELD	TCS-6			

EXTERNAL INPUT COM THC/RTD AI(9)

SH. No. TC0763

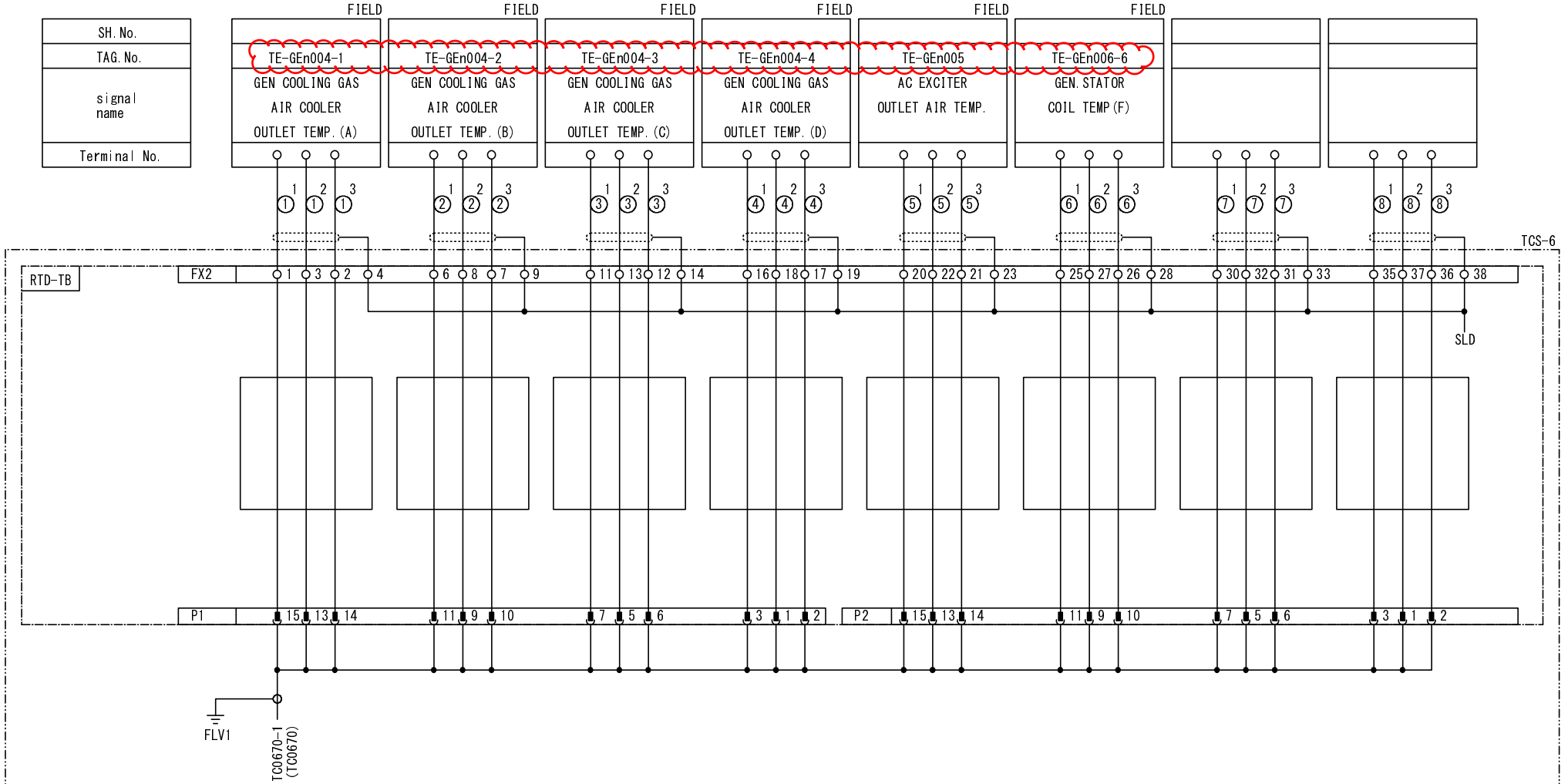
	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733 | PAGE 330 | REV. 3

2016/07/28 TC0763 XSMM-TC



INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	FIELD	TCS-6			
⑦		2/2	FIELD	TCS-6			
⑥		2/2	FIELD	TCS-6			
⑤		2/2	FIELD	TCS-6			
④		2/2	FIELD	TCS-6			
③		2/2	FIELD	TCS-6			
②		2/2	FIELD	TCS-6			
①		2/2	FIELD	TCS-6			

EXTERNAL INPUT COM THC/RTD AI(10)

SH. No. TC0764

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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DWG. No. 331DM07733

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TCS-6

TCS-7

TC0791-1
(TC0691)

SD-AO

CNA1 33 34 29 30 25 28 21 22 17 18 13 14 9 10 5 6

GXF11 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Terminal No.								
signal name	LUBE OIL COOLER OUTLET OIL TEMP. (A)	STEAM SEEL FEED VALVE POSITION DEMAND	HP TURBINE WARMING VALVE POSITION	STEAM SEAL FEED VALVE MANIPULATION VALUE	GEN. CURRENT (A)	GEN. VOLTAGE (A-B)		
TAG. No.	TE-LOn002-1	PV-GSn031	ZT-ASn011					
SH. No.								
	DCS	FIELD	DCS	DCS	DCS	DCS		

INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	TCS-7				
⑦		2/2	TCS-7				
⑥		2/2	TCS-7	DCS			
⑤		2/2	TCS-7	DCS			
④		2/2	TCS-7	DCS			
③		2/2	TCS-7	DCS			
②		2/2	TCS-7	FIELD			
①		2/2	TCS-7	DCS			

EXTERNAL OUTPUT COM AO(1)

SH. No. TC0791

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE
ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0791 XSMM-TC

TCS-6

TCS-7

TC0792-1
(TC0692)

SD-AO

CNA1 33 34 29 30 25 28 21 22 17 18 13 14 9 10 5 6

GXF12 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Terminal No.								
signal name	LUBE OIL COOLER OUTLET OIL TEMP. (B)	SPUV POSITION DEMAND	GEN. MVAR	GEN. CURRENT (B)	GEN. VOLTAGE (B-C)	GEN. POWER FACTOR	FREQUENCY	
TAG. No.	TE-LOn002-2	PV-GSn034						
SH. No.								
	DCS	FIELD	DCS	DCS	DCS	DCS	DCS	

INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	TCS-7				
⑦		2/2	TCS-7	DCS			
⑥		2/2	TCS-7	DCS			
⑤		2/2	TCS-7	DCS			
④		2/2	TCS-7	DCS			
③		2/2	TCS-7	DCS			
②		2/2	TCS-7	FIELD			
①		2/2	TCS-7	DCS			

EXTERNAL OUTPUT COM AO(2)

SH. No. TC0792

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

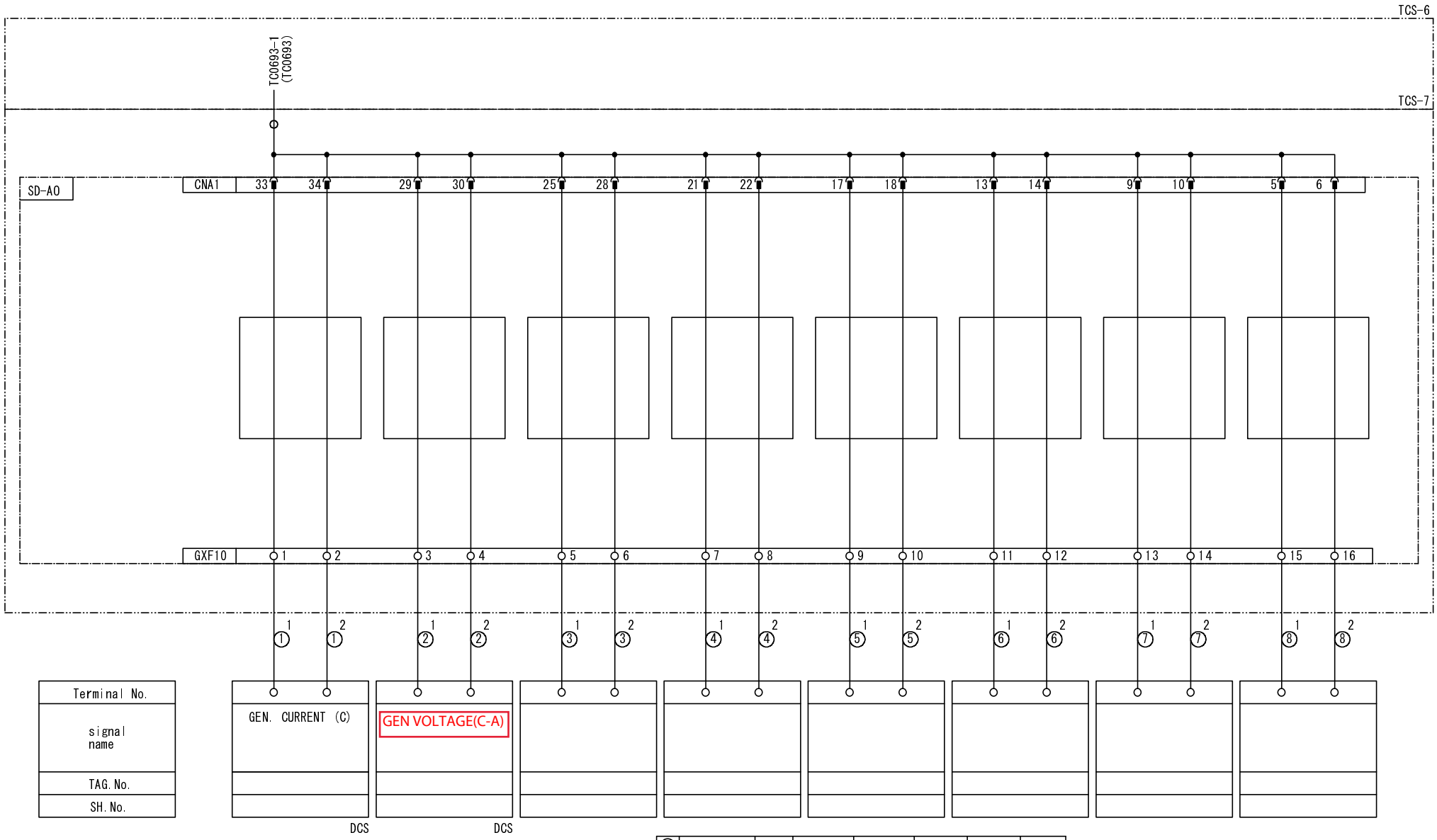
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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REV. 3

2016/07/28 TC0792 XSMM-TC



INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	TCS-7				
⑦		2/2	TCS-7				
⑥		2/2	TCS-7				
⑤		2/2	TCS-7				
④		2/2	TCS-7				
③		2/2	TCS-7				
②		2/2	TCS-7	DCS			
①		2/2	TCS-7	DCS			

EXTERNAL OUTPUT COM AO(3)

SH. No. TC0793

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

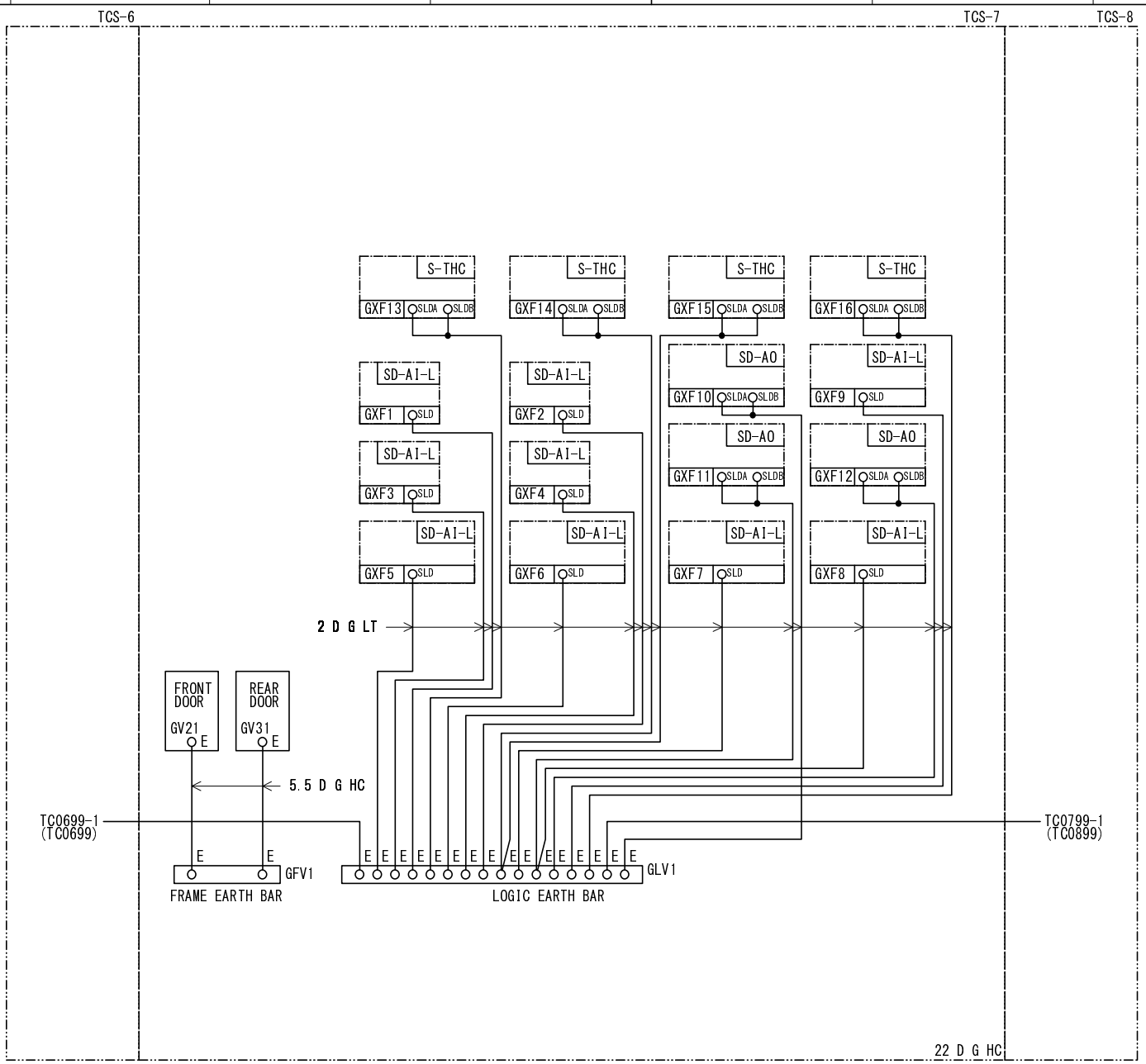
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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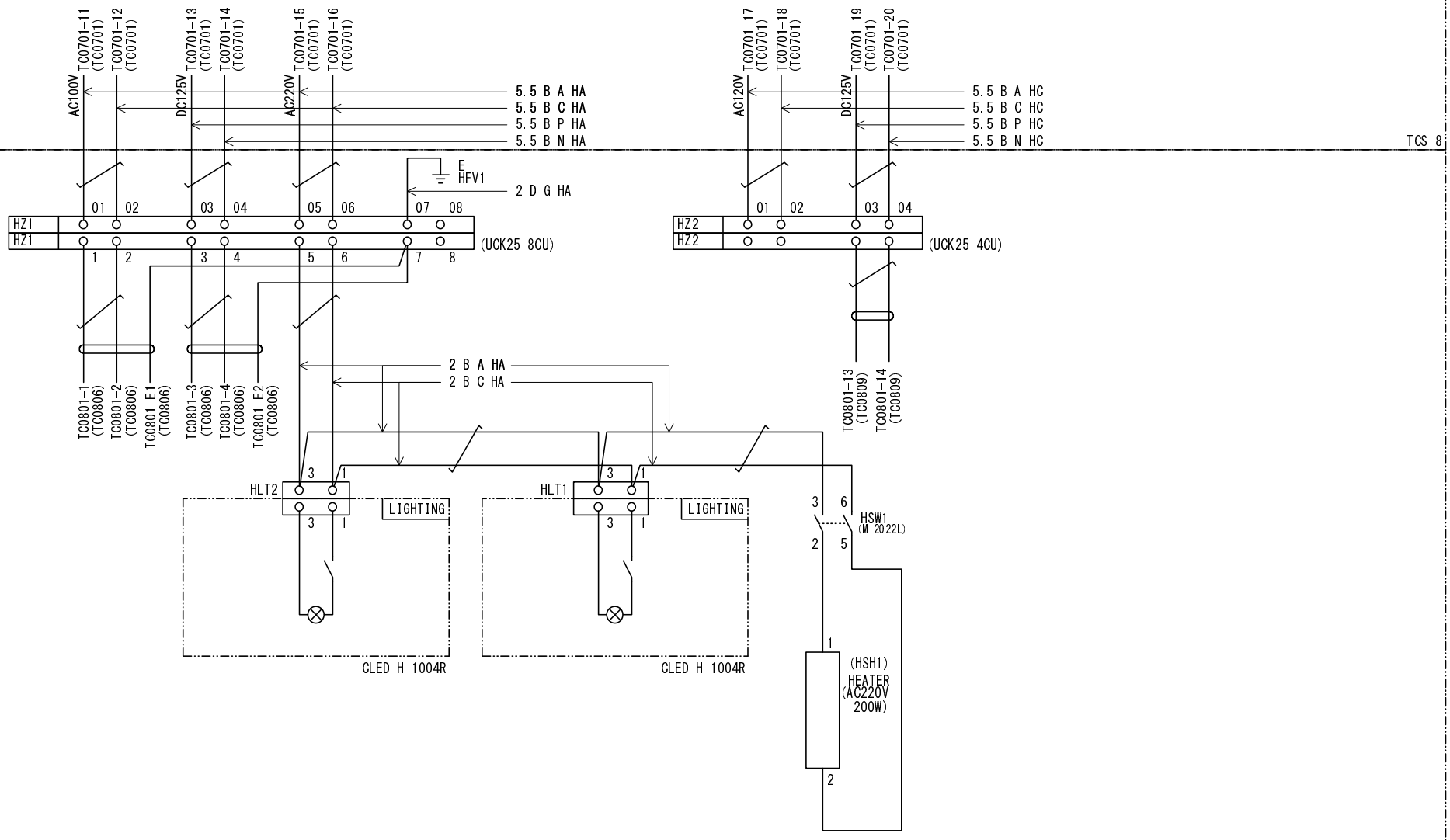
2016/07/28 TC0793 XSMM-TC



POWER SUPPLY SYSTEM EARTH CIRCUIT		SH. No.	TC0799
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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TCS-7



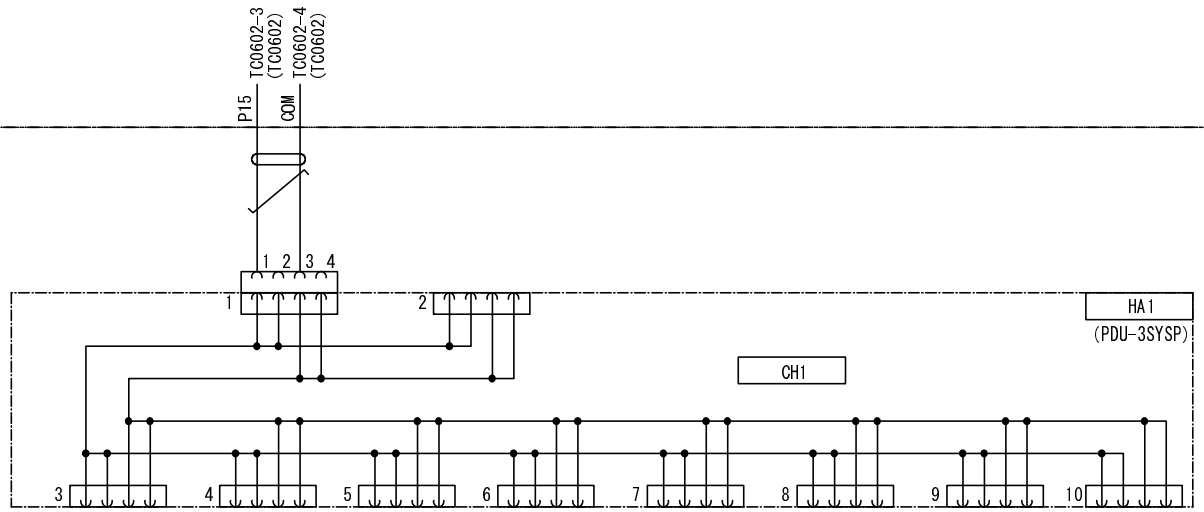
TCS-8

POWER SUPPLY SYSTEM(1)			SH. No.	TC0801
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 335	REV. 3

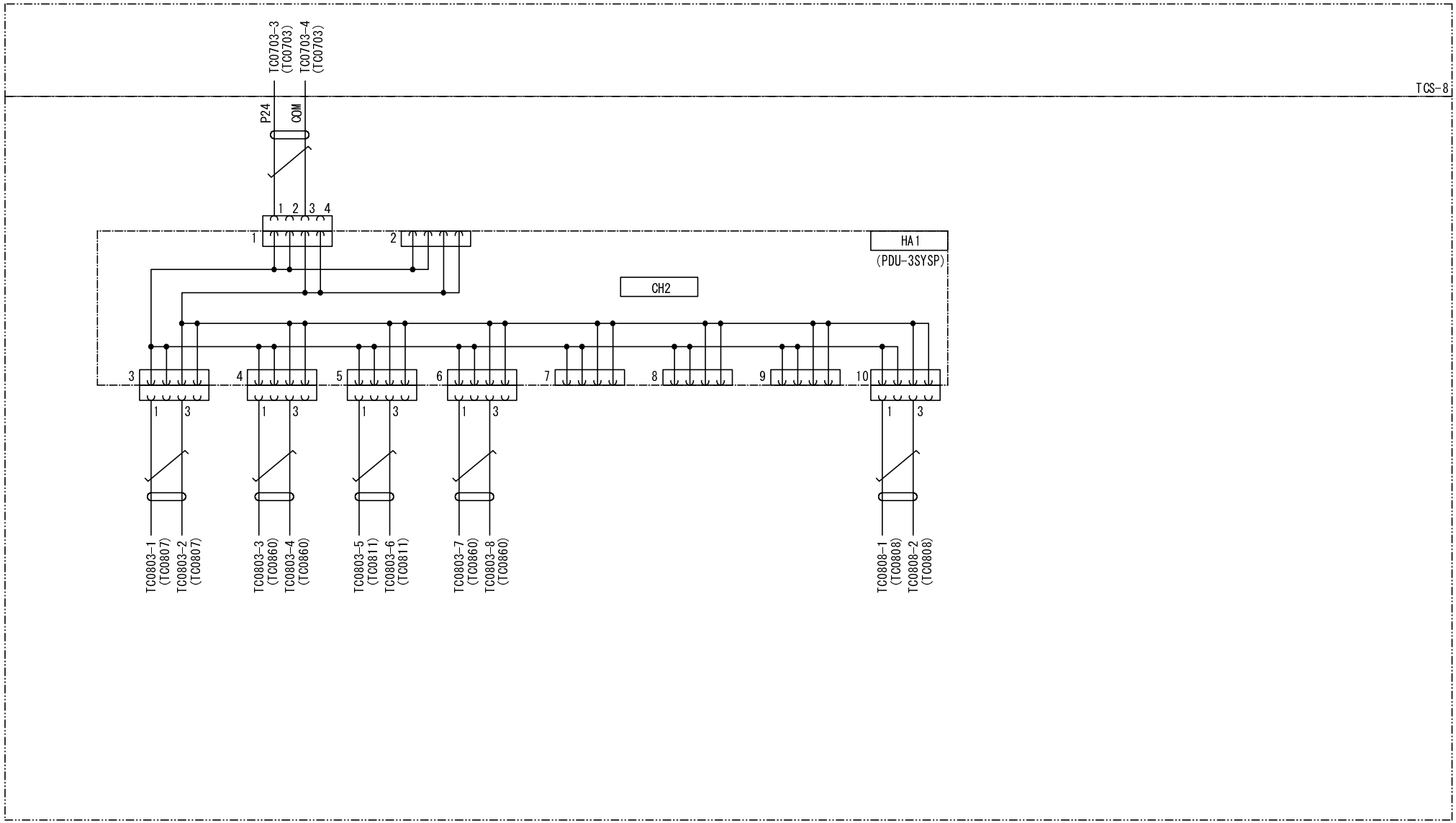
2016/07/28 TC0801 XSMM-TC

TCS-6

TCS-8



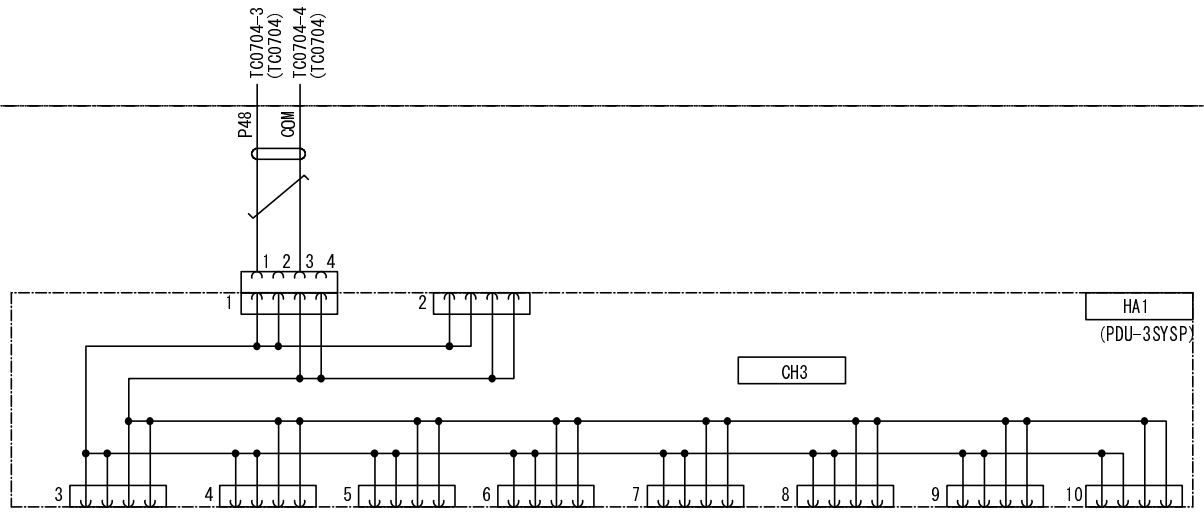
POWER SUPPLY SYSTEM(2)		SH. No.	TC0802
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 336	REV. 3



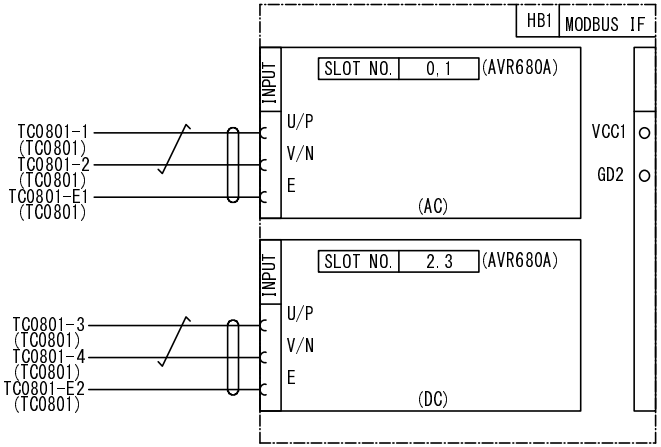
POWER SUPPLY SYSTEM(3)		SH. No.	TC0803
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 337	REV. 3

TCS-7

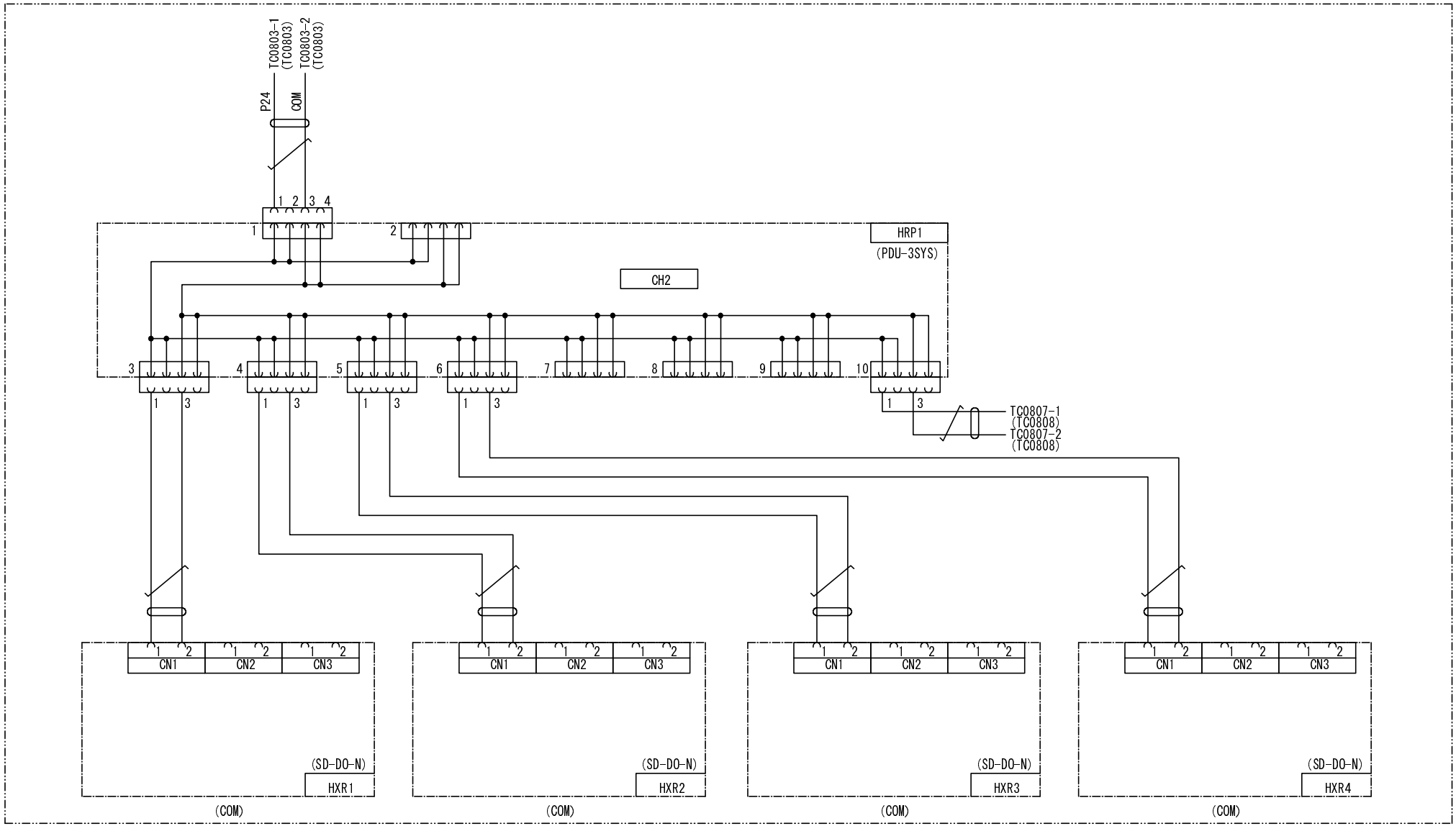
TCS-8



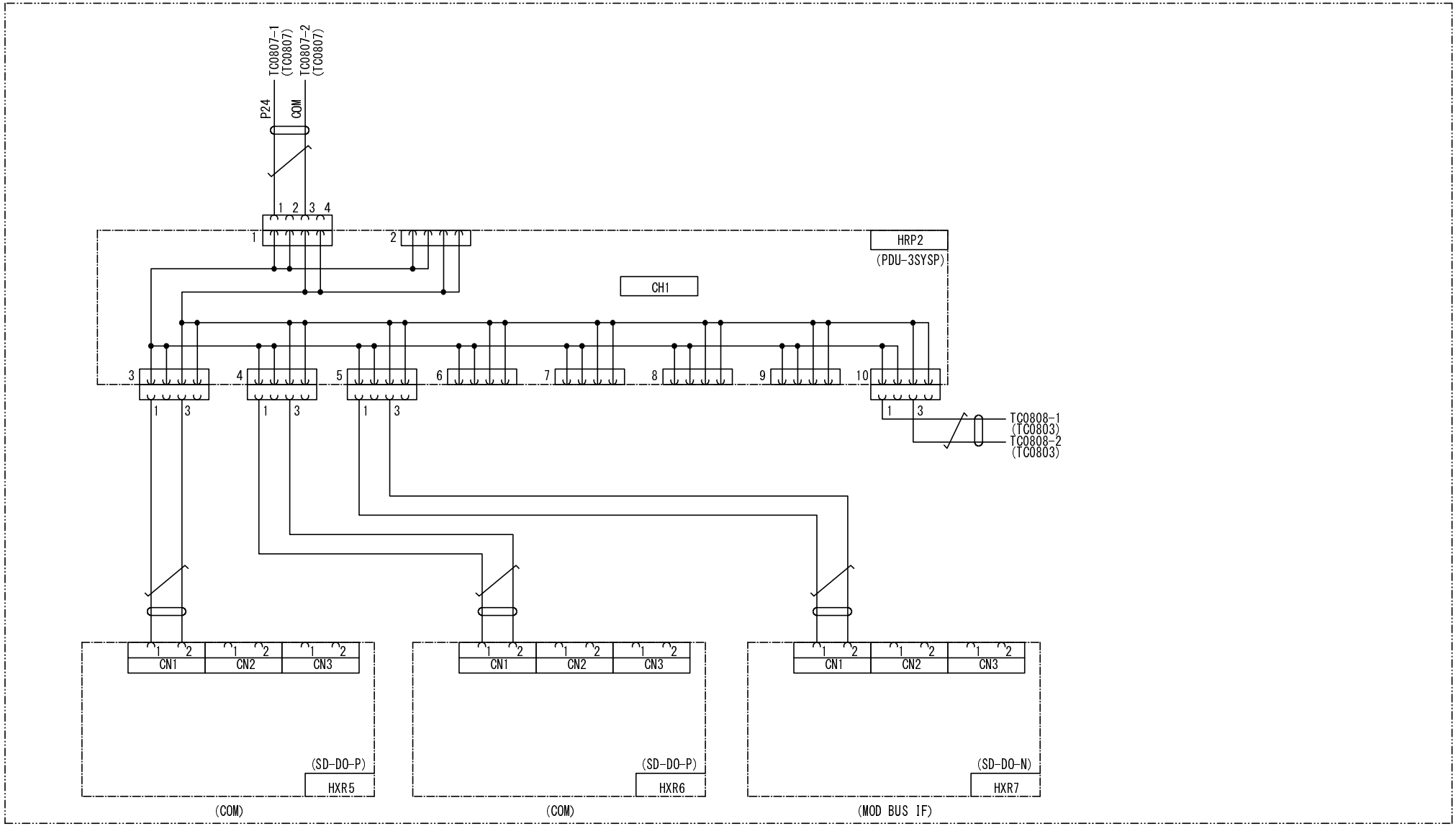
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SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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POWER SUPPLY SYSTEM(5)		SH. No.	TC0806
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
REGD. Hitachi,Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 339	REV. 3

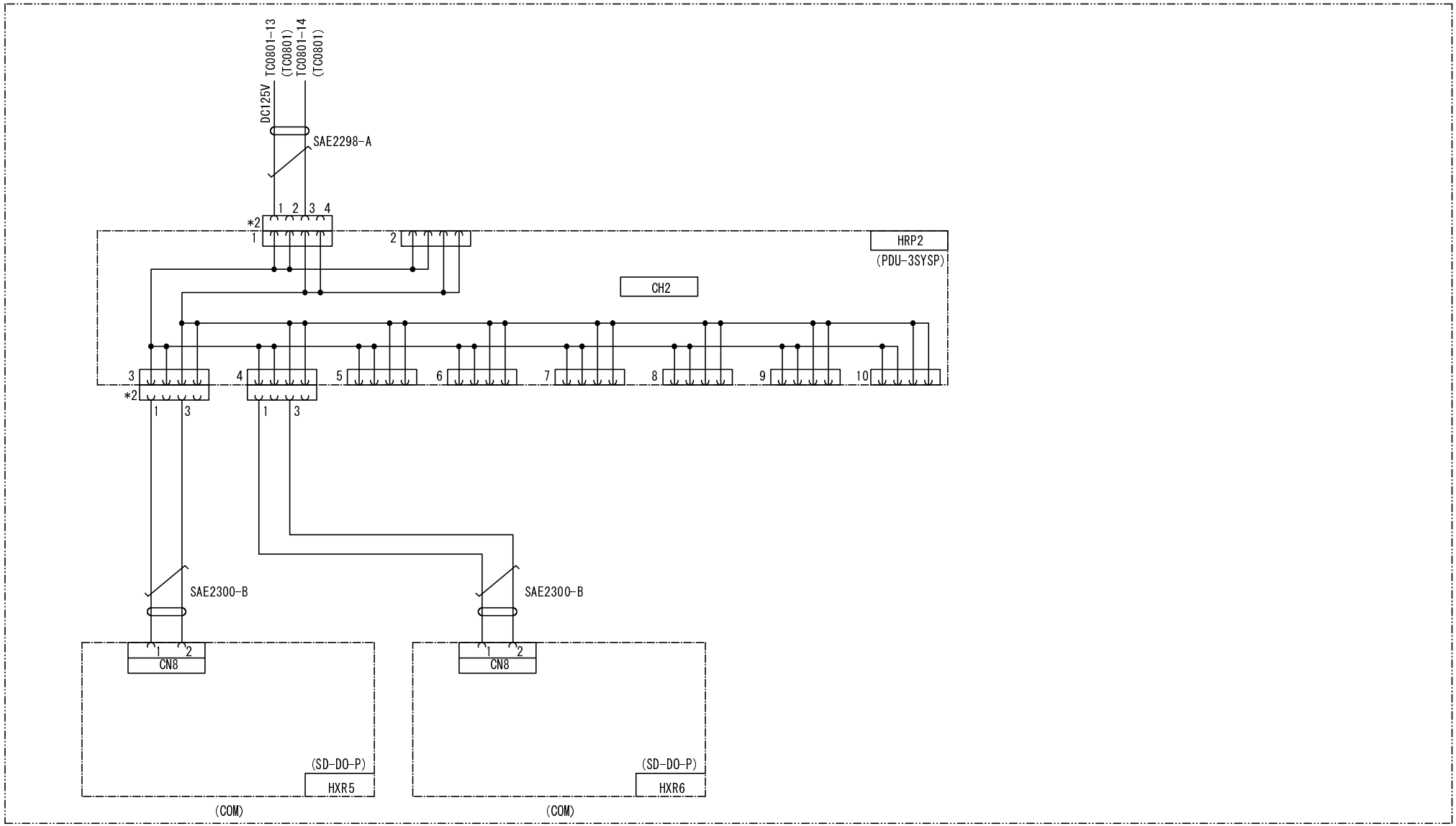


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SIGNATURE		DATE	TITLE
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM
CHKD.	T. Kawata	2014-04-18	
APPD.	Y. Fukushima	2014-04-18	
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 340 / REV. 3



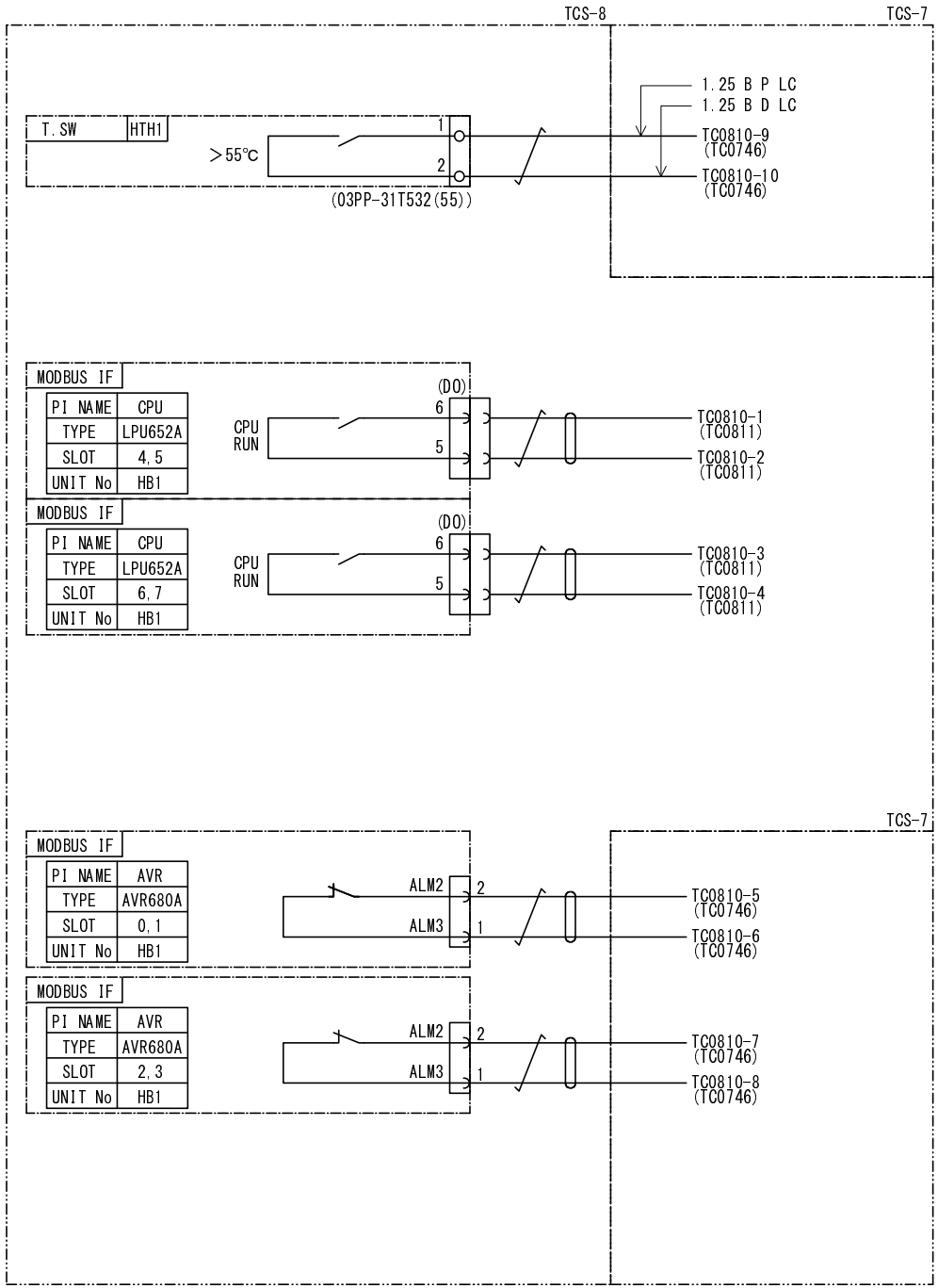
POWER SUPPLY SYSTEM(7)		SH. No.	TC0808
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
REGD. Hitachi,Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 341	REV. 3

TCS-8



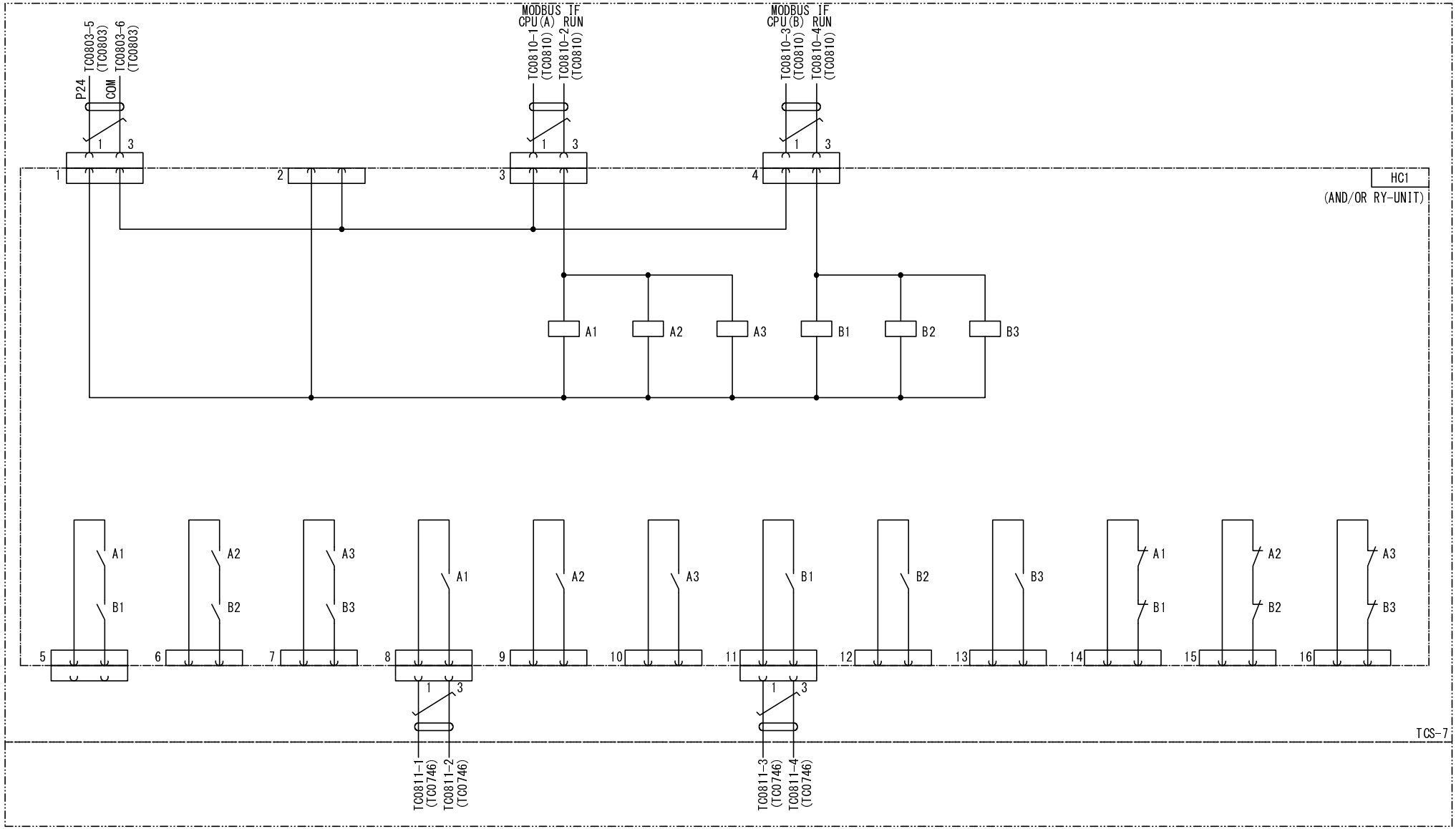
POWER SUPPLY SYSTEM(8)		SH. No.	TC0809
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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DIAGNOSIS CIRCUIT(1)		SH. No.	TC0810
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
REGD. Hitachi,Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 343	REV. 3

TCS-8



TCS-7

DEV.	a	b	DEV.	a	b

DIAGNOSIS CIRCUIT(2)		SH. No.	TC0811
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 344	REV. 3

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TCS-6

TCS-8

TC0671-1
(TC0671)

SD-DO-N
(PS:refer to TC0807)

CNA 21 3 20 18 16 14 12 10 8 6

P24
C24

P24 P24 P24 P24 P24 P24 P24 P24

HXR1 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Terminal No.								
signal name	CRV (A) BELOW SEAT SECONDARY DRAIN VALVE CLOSE COMMAND	CRV (A) BELOW SEAT SECONDRY DRAIN VALVE OPEN COMMAND	CV (A) ABOVE SEAT SECONDRY DRAIN VALVE OPEN COMMAND	CV (A) ABOVE SEAT SECONDRY DRAIN VALVE CLOSE COMMAND	CV (B) BELOW SEAT SEC. DRAIN VLV CLOSE CMD.	CV (B) BELOW SEAT SEC. DRAIN VLV OPEN CMD.	MSV (A) ABOVE SEAT SECONDRY DRAIN VALVE OPEN COMMAND	MSV (A) ABOVE SEAT SECONDRY DRAIN VALVE CLOSE COMMAND
TAG. No.								
SH. No.								
	MCC	MCC	MCC	MCC	MCC	MCC	MCC	MCC

INDEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK
⑧		2/2	TCS-8	MCC			
⑦		2/2	TCS-8	MCC			
⑥		2/2	TCS-8	MCC			
⑤		2/2	TCS-8	MCC			
④		2/2	TCS-8	MCC			
③		2/2	TCS-8	MCC			
②		2/2	TCS-8	MCC			
①		2/2	TCS-8	MCC			

EXTERNAL OUTPUT COM DO(1)

SH. No. TC0821

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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TCS-6

TCS-8

TC0671-1
(TC0672)

SD-DO-N
(PS:refer to TC0807)

CNA 22 4 19 17 15 13 11 9 7 5

P24
C24

P24 P24 P24 P24 P24 P24 P24 P24

HXR1 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36

Terminal No.								
signal name	STEAM SEAL FEED INLET VALVE CLOSE COMMAND	STEAM SEAL FEED INLET VALVE OPEN COMMAND	TURBINE GLAND STEAM LINE DRAIN VALVE OPEN COMMAND	TURBINE GLAND STEAM LINE DRAIN VALVE CLOSE COMMAND	STEAM SEAL FEED VALVE CLOSE		AMB COLD MODE TEMP.	MODE JUGEMENT INSTRUMENT NORMAL
TAG. No.								
SH. No.								
	MCC	MCC	DCS	DCS	DCS		DCS	DCS

INDEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK
⑧		2/2	TCS-8	DCS			
⑦		2/2	TCS-8	DCS			
⑥		2/2	TCS-8				
⑤		2/2	TCS-8	DCS			
④		2/2	TCS-8	DCS			
③		2/2	TCS-8	DCS			
②		2/2	TCS-8	MCC			
①		2/2	TCS-8	MCC			

EXTERNAL OUTPUT COM DO(2)

SH. No. TC0822

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE
ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0822 XSMM-TC

TCS-6

TCS-8

TC0673-1
(TC0673)

SD-DO-N
(PS:refer to TC0807)

CNA 21 3 20 18 16 14 12 10 8 6

P24
C24

P24 P24 P24 P24 P24 P24 P24 P24

HXR2 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Terminal No.	① 1	① 2	② 1	② 2	③ 1	③ 2	④ 1	④ 2	⑤ 1	⑤ 2	⑥ 1	⑥ 2	⑦ 1	⑦ 2	⑧ 1	⑧ 2
signal name	TURBINE START-UP AMBIENT COLD MODE		TURBINE START-UP COLD 1 MODE		TURBINE START-UP COLD 2 MODE		TURBINE START-UP HOT MODE		TURBINE START-UP WARM MODE		TURBINE LOAD HOLD				3-41 OPEN COMMAND	
TAG. No.																
SH. No.																
	DCS		DCS		DCS		DCS		DCS		DCS				AVR	

IN-DEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	TCS-8	AVR			
⑦		2/2	TCS-8				
⑥		2/2	TCS-8	DCS			
⑤		2/2	TCS-8	DCS			
④		2/2	TCS-8	DCS			
③		2/2	TCS-8	DCS			
②		2/2	TCS-8	DCS			
①		2/2	TCS-8	DCS			

EXTERNAL OUTPUT COM DO(3)

SH. No. TC0823

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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TCS-6

TCS-8

TC0673-1
(TC0674)

SD-DO-N
(PS:refer to TC0807)

CNA 22 41 19 17 15 13 11 9 7 5

P24
C24

P24 P24 P24 P24 P24 P24 P24 P24

HXR2 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36

1 2 1 2 1 2 1 1 1 1 1 1

Terminal No.		41 OPEN COMAND FROM ATS	PSS ON	PSS OFF	PF CONTROL ON	PF CONTROL OFF	TURBINE RATED SPEED	ST TRIP
signal name								
TAG. No.								
SH. No.								
		AVR	AVR	AVR	AVR	AVR	AVR	AVR

INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	TCS-8	AVR			
⑦		2/2	TCS-8	AVR			
⑥		2/2	TCS-8	AVR			
⑤		2/2	TCS-8	AVR			
④		2/2	TCS-8	AVR			
③		2/2	TCS-8	AVR			
②		2/2	TCS-8	AVR			
①		2/2	TCS-8				

EXTERNAL OUTPUT COM DO(4)

SH. No. TC0824

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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2016/07/28 TC0824 XSMM-TC

TCS-6

TCS-8

TC0675-1
(TC0675)

SD-DO-N
(PS:refer to TC0807)

CNA 21 3 20 18 16 14 12 10 8 6

P24
C24

P24 P24 P24 P24 P24 P24 P24 P24

HXR3 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Terminal No.								
signal name	CRV (B) BELOW SEAT SECONDARY DRAIN VALVE OPEN COMMAND	CRV (B) BELOW SEAT SECONDRY DRAIN VALVE CLOSE COMMAND				MSV (B) ABOVE SEAT SECONDRY DRAIN VALVE OPEN COMMAND	MSV (B) ABOVE SEAT SECONDRY DRAIN VALVE CLOSE COMMAND	HP TURBINE WARMING LINE DRAIN VALVE CLOSE COMMAND
TAG. No.								
SH. No.								
	MCC	MCC				MCC	MCC	DCS

INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	TCS-8	DCS			
⑦		2/2	TCS-8	MCC			
⑥		2/2	TCS-8	MCC			
⑤		2/2	TCS-8				
④		2/2	TCS-8				
③		2/2	TCS-8				
②		2/2	TCS-8	MCC			
①		2/2	TCS-8	MCC			

EXTERNAL OUTPUT COM DO(5)

SH. No. TC0825

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE
ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

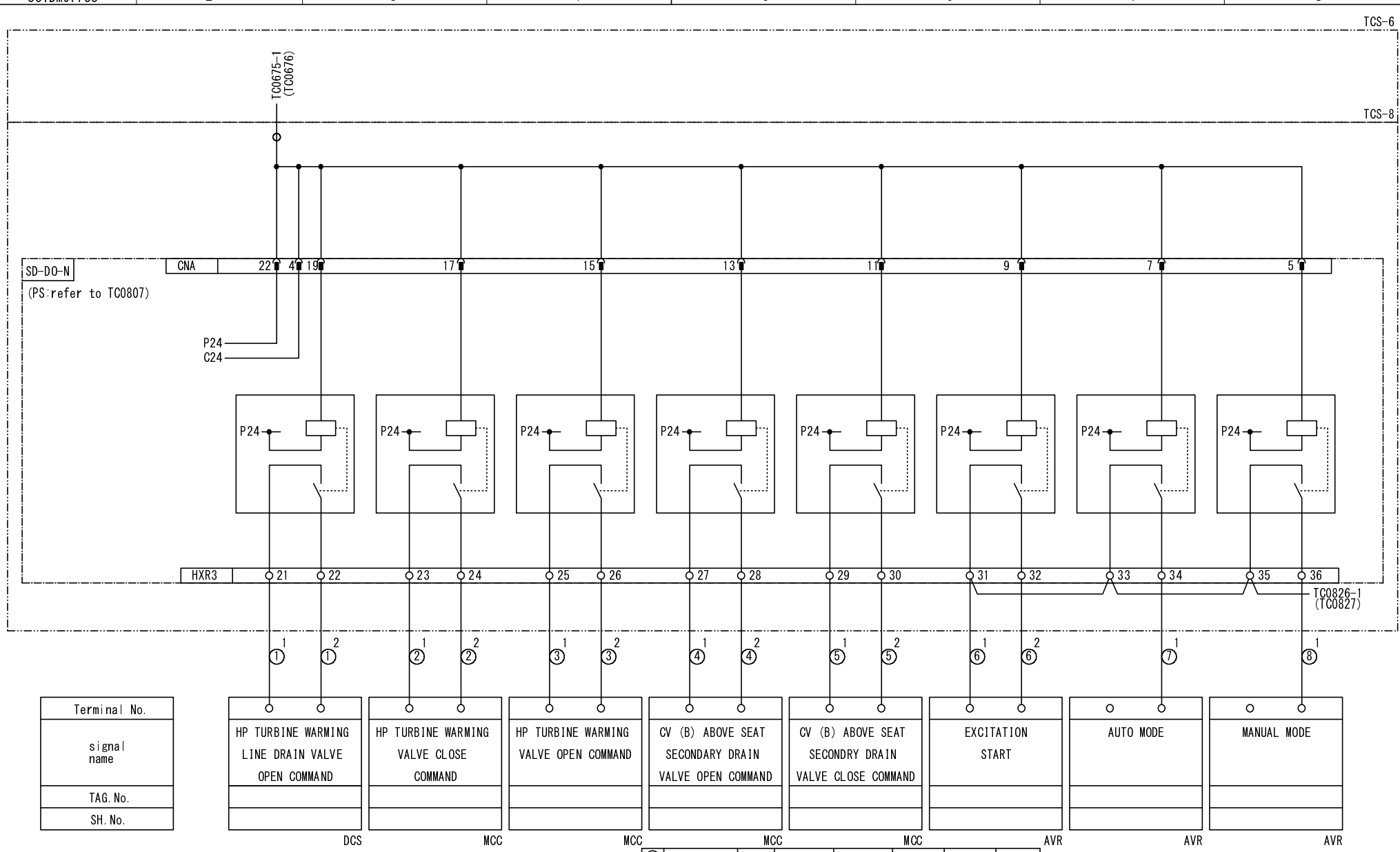
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		2/2	TCS-8	AVR			
⑦		2/2	TCS-8	AVR			
⑥		2/2	TCS-8	AVR			
⑤		2/2	TCS-8	MCC			
④		2/2	TCS-8	MCC			
③		2/2	TCS-8	MCC			
②		2/2	TCS-8	MCC			
①		2/2	TCS-8	DCS			

EXTERNAL OUTPUT COM DO(6)

SH. No. TC0826

	SIGNATURE	DATE
DWN.	(S.Kikuchi)	2014-04-18
CHKD.	T.Kawata	2014-04-18
APPD.	Y.Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

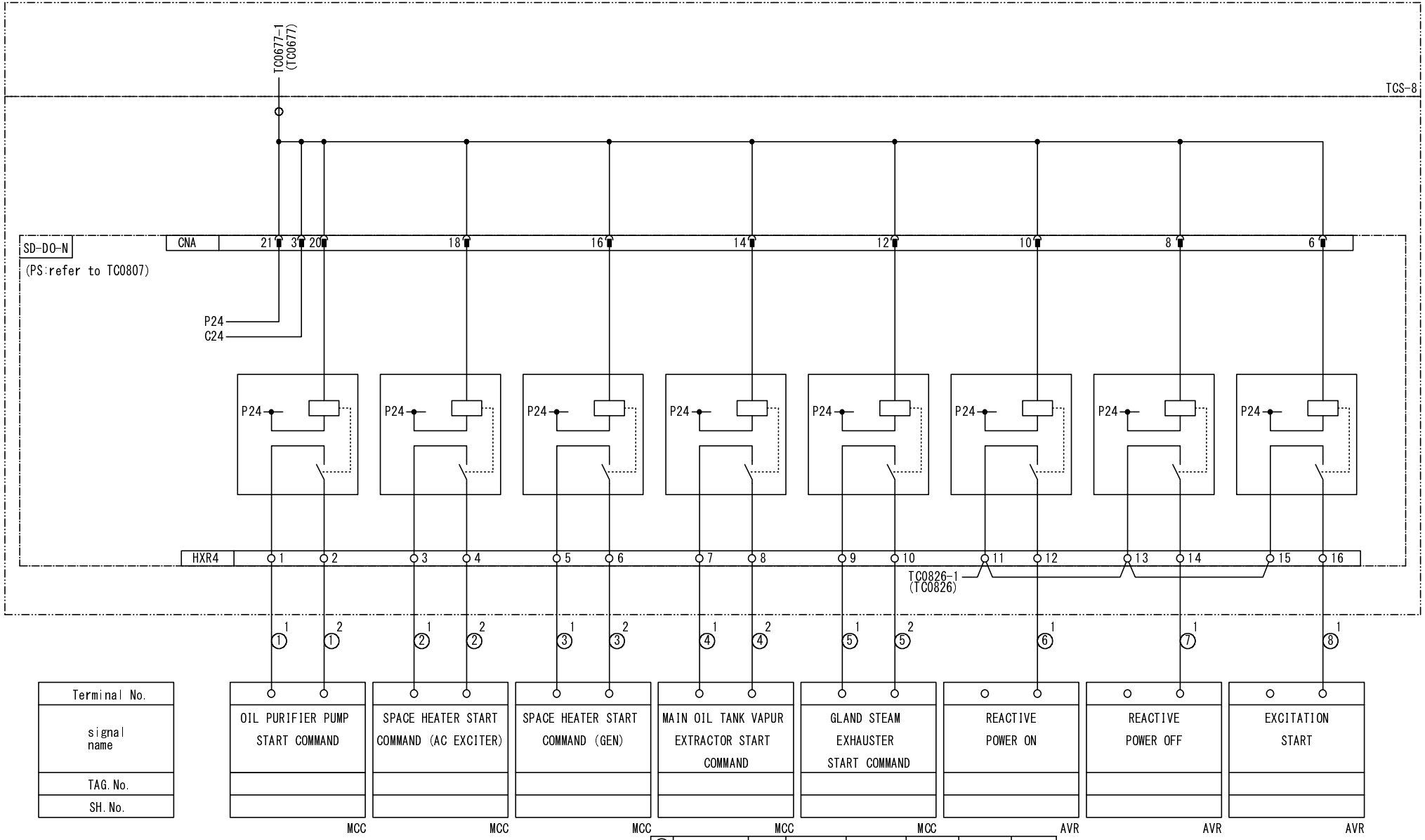
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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INDEX	CABLE NO.	CORE NO.	FROM (CABLE CONNECTION)	TO (CABLE CONNECTION)	FROM (RELATED SH. NO.)	TO (RELATED SH. NO.)	REMARK
⑧		2/2	TCS-8	AVR			
⑦		2/2	TCS-8	AVR			
⑥		2/2	TCS-8	AVR			
⑤		2/2	TCS-8	MCC			
④		2/2	TCS-8	MCC			
③		2/2	TCS-8	MCC			
②		2/2	TCS-8	MCC			
①		2/2	TCS-8	MCC			

EXTERNAL OUTPUT COM DO(7)

SH. No. TC0827

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

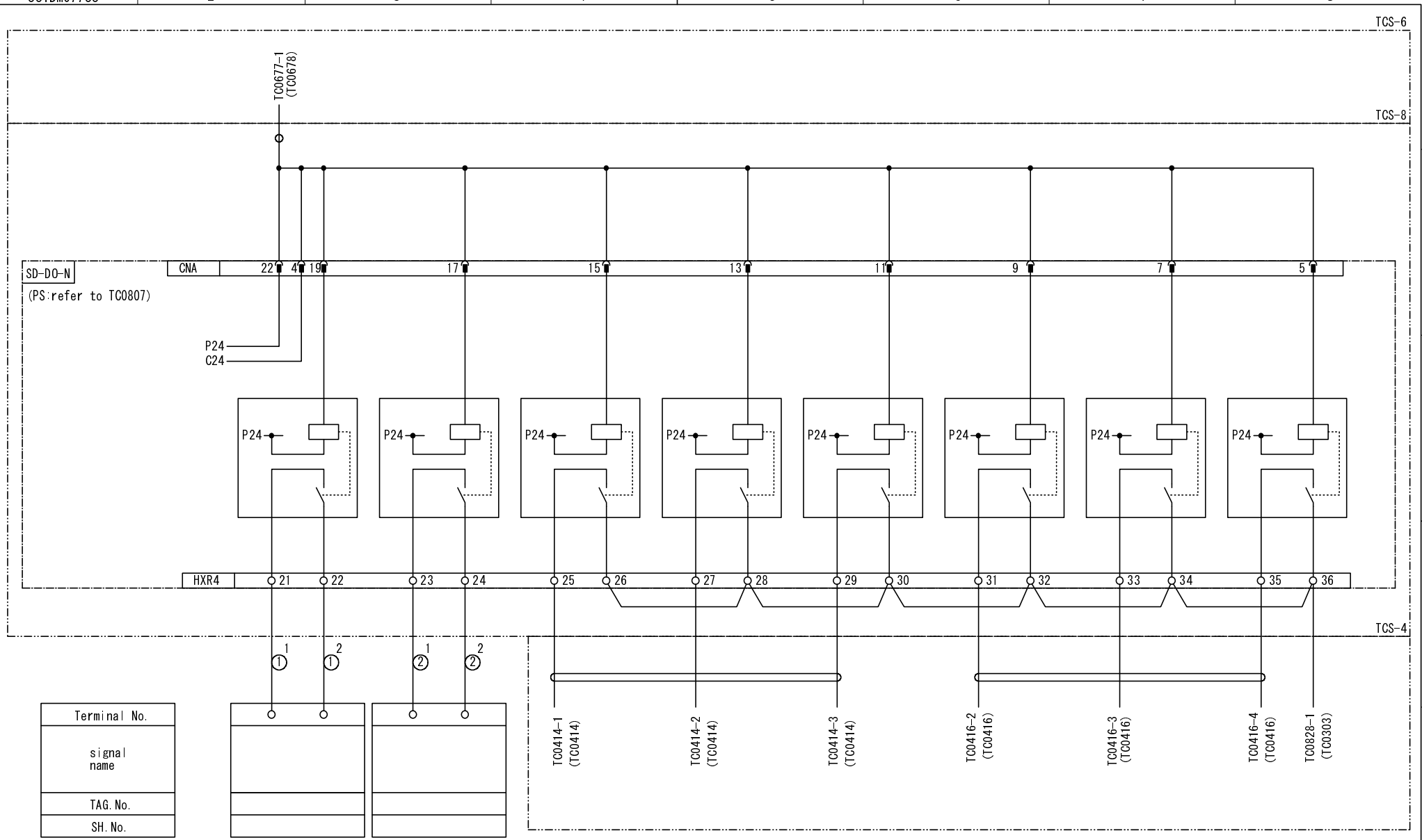
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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○								
○								
○								
②		2/2	TCS-8					
①		2/2	TCS-8					
IN-DEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK	
			CABLE CONNECTION		RELATED SH. NO.			

EXTERNAL OUTPUT COM DO(8)

SH. No. TC0828

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0828 XSMM-TC

TCS-6

TC0679-1 (TC0679)

TCS-8

SD-DO-P

(PS:refer to TC0809)

CNA 21 3 20 18 16 14 12 10 8 6

P24

C24

OUTPUT VOLTAGE DC125V

P24 P(A) N(C) ×8

HXR5 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Terminal No.	① 1 2	② 1 2	③ 1 2	④ 1 2	⑤ 1 2	⑥ 1 2	⑦ 1 2	⑧ 1 2
signal name	COLD REHEAT STEAM NON RETURN VALVE SOLENOID VALVE	NO.3 EXTRACTION STEAM NON RETURN VALVE ENERGIZE COMMAND	NO.2 EXTRACTION STEAM NON RETURN VALVE ENERGIZE COMMAND	STEAM SEAL DIVERTING VALVE SOLENOID VALVE ENERGIZE COMMAND				
TAG.No.	SV-CRn901	SV-TSn740	SV-STn700	SV-GSn036				
SH.No.								
	SV	FIELD	FIELD	FIELD				

INDEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK
⑧		2/2	TCS-8				
⑦		2/2	TCS-8				
⑥		2/2	TCS-8				
⑤		2/2	TCS-8				
④		2/2	TCS-8	FIELD			
③		2/2	TCS-8	FIELD			
②		2/2	TCS-8	FIELD			
①		2/2	TCS-8	SV			

EXTERNAL OUTPUT COM DO(9)

SH. No. TC0829

	SIGNATURE	DATE
DWN.	(S.Kikuchi)	2014-04-18
CHKD.	T.Kawata	2014-04-18
APPD.	Y.Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi,Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0829 XSMM-TC

TCS-6

TC0679-1
(TC0680)

TCS-8

SD-DO-P

(PS:refer to TC0809)

CNA 22 4 19 17 15 13 11 9 7 5

P24
C24

OUTPUT
VOLTAGE
DC125V

P24 P(A) N(C)

HXR5 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36

Terminal No.															
signal name															
TAG. No.															
SH. No.															

INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	TCS-8				
⑦		2/2	TCS-8				
⑥		2/2	TCS-8				
⑤		2/2	TCS-8				
④		2/2	TCS-8				
③		2/2	TCS-8				
②		2/2	TCS-8				
①		2/2	TCS-8				

EXTERNAL OUTPUT COM DO(10)

SH. No. TC0830

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE
ELEMENTARY DIAGRAM
OF
TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0830 XSMM-TC

TCS-6

TCS-8

TC0681-1 (TC0681)

SD-DO-P

(PS:refer to TC0809)

CNA 21 3 20 18 16 14 12 10 8 6

P24

C24

OUTPUT VOLTAGE DC125V

P(A) N(C)

HXR6 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Terminal No.								
signal name	LP EXHAUST HOOD WATER SPRAY VLV SOLENOID VLV ENERGIZE CMD.	NO.5 EXTRACTION STM NON RETURN VLV ENERGIZE COMMAND	NO.3 EXTRACTION STM PRI. NON RETURN VLV ENERGIZE CMD.	NO.4 EXTRACTION STEAM NON RETURN VALVE ENERGIZE COMMAND				
TAG.No.	SV-CDn904	SV-TSn730	SV-TSn710	SV-TSn720				
SH.No.								
	FIELD	FIELD	FIELD	FIELD				

INDEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK
⑧		2/2	TCS-8				
⑦		2/2	TCS-8				
⑥		2/2	TCS-8				
⑤		2/2	TCS-8				
④		2/2	TCS-8	FIELD			
③		2/2	TCS-8	FIELD			
②		2/2	TCS-8	FIELD			
①		2/2	TCS-8	FIELD			

EXTERNAL OUTPUT COM DO(11)

SH. No. TC0831

SIGNATURE		DATE
DWN.	(S.Kikuchi)	2014-04-18
CHKD.	T.Kawata	2014-04-18
APPD.	Y.Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi,Ltd. Tokyo Japan

DWG. No. 331DM07733

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2016/07/28 TC0831 XSMM-TC

TCS-6

TC0681-1
(TC0682)

TCS-8

SD-DO-P

(PS:refer to TC0809)

CNA 22 4 19 17 15 13 11 9 7 5

P24
C24

OUTPUT
VOLTAGE
DC125V

P24 P(A) N(C)

HXR6 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36

Terminal No.															
signal name															
TAG. No.															
SH. No.															

INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	TCS-8				
⑦		2/2	TCS-8				
⑥		2/2	TCS-8				
⑤		2/2	TCS-8				
④		2/2	TCS-8				
③		2/2	TCS-8				
②		2/2	TCS-8				
①		2/2	TCS-8				

EXTERNAL OUTPUT COM DO(12)

SH. No. TC0832

SIGNATURE		DATE
DWN.	(S.Kikuchi)	2014-04-18
CHKD.	T.Kawata	2014-04-18
APPD.	Y.Fukushima	2014-04-18

TITLE
ELEMENTARY DIAGRAM
OF
TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

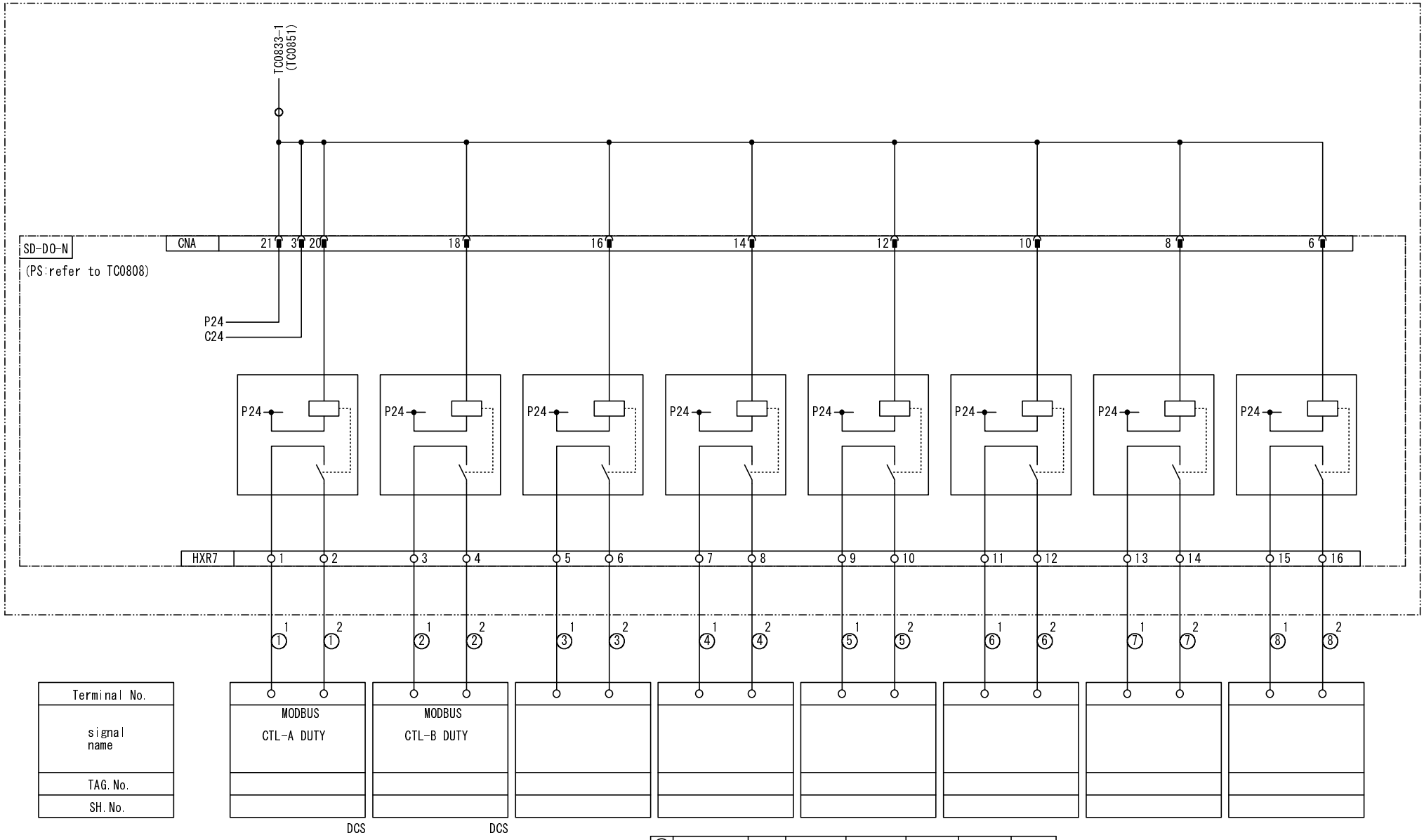
DWG. No. 331DM07733

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2016/07/28 TC0832 XSMM-TC

TCS-8



INDEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
⑧		2/2	TCS-8				
⑦		2/2	TCS-8				
⑥		2/2	TCS-8				
⑤		2/2	TCS-8				
④		2/2	TCS-8				
③		2/2	TCS-8				
②		2/2	TCS-8				
①		2/2	TCS-8				

EXTERNAL OUTPUT COM DO(13)

SH. No. TC0833

	SIGNATURE	DATE
DWN.	(S.Kikuchi)	2014-04-18
CHKD.	T.Kawata	2014-04-18
APPD.	Y.Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

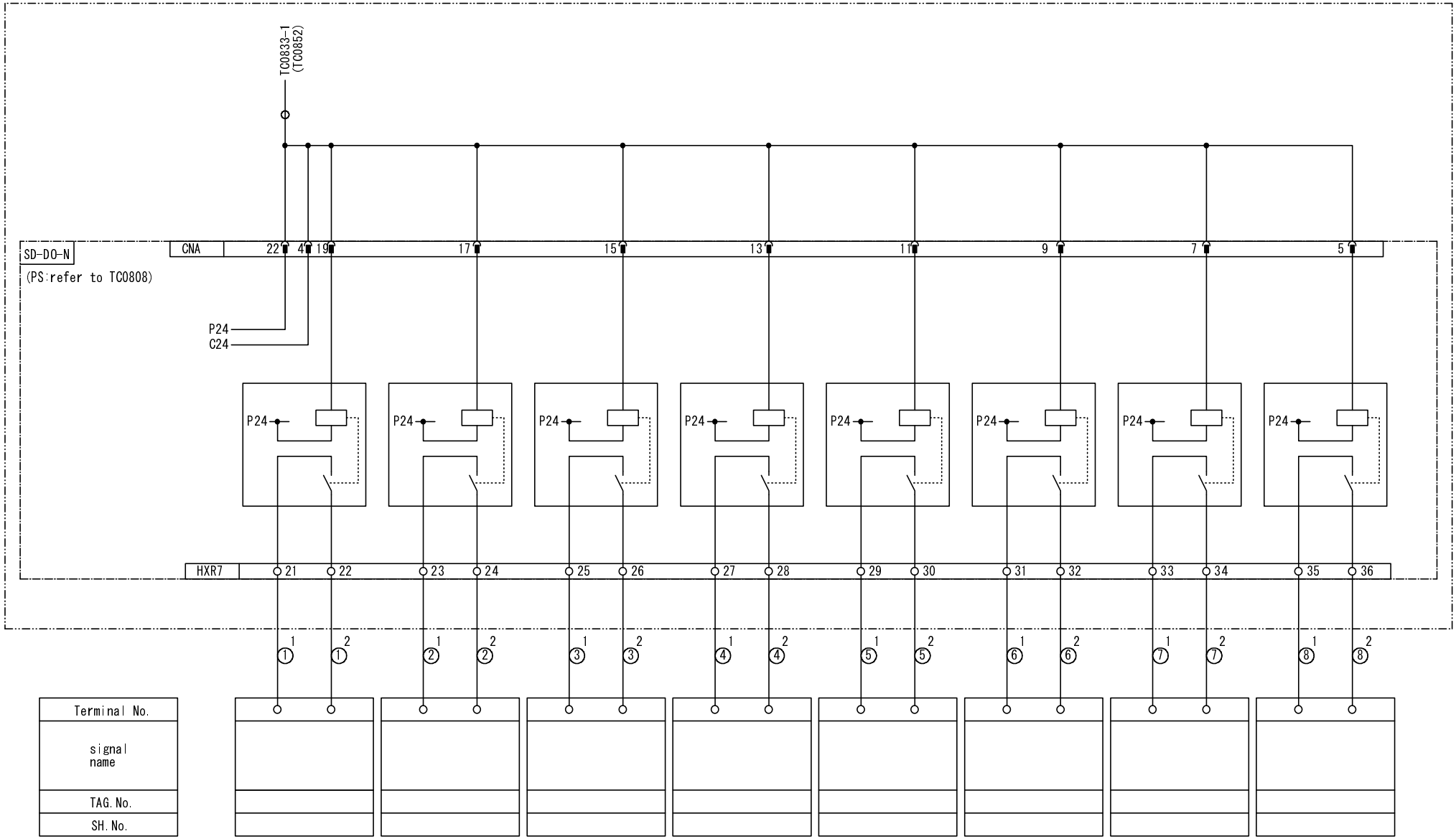
REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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TCS-8



IN-DEX	CABLE NO.	CORE NO.	FROM CABLE CONNECTION	TO CABLE CONNECTION	FROM RELATED SH. NO.	TO RELATED SH. NO.	REMARK
⑧		2/2	TCS-8				
⑦		2/2	TCS-8				
⑥		2/2	TCS-8				
⑤		2/2	TCS-8				
④		2/2	TCS-8				
③		2/2	TCS-8				
②		2/2	TCS-8				
①		2/2	TCS-8				

EXTERNAL OUTPUT COM DO(14)

SH. No. TC0834

	SIGNATURE	DATE
DWN.	(S.Kikuchi)	2014-04-18
CHKD.	T.Kawata	2014-04-18
APPD.	Y.Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

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TCS-7

TCS-8

TC0749-1
(TC0749)

(LYD050A)

unit NO.	slot NO.	4 20	18	16	14	12	10	8	6 (S)
HB1	8								
CTL	CH. NO.	DI 0	DI 1	DI 2	DI 3	DI 4	DI 5	DI 6	DI 7
	signal range/level								
logic SH. NO.									
DI	signal name	TCS-7 TEMP. HIGH	TCS-6 FAN TROUBLE	TCS-6 TEMP. HIGH	TPC(A) P.S. ABNORMAL	TPC(B) P.S. ABNORMAL	TPC(C) P.S. ABNORMAL	TCS-3 FAN TROUBLE	TCS-3 TEMP. HIGH

MODBUS IF DIGITAL INPUT(1)		SH. No.	TC0841
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
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TCS-7

TCS-8

TC0749-1
(TC0750)

(LYD050A)

unit NO.	slot NO.	3 19	17	15	13	11	9	7	5 (S)
HB1	8								
CTL	CH. NO.	DI 8	DI 9	DI A	DI B	DI C	DI D	DI E	DI F
	signal range/level								
logic SH. NO.									
DI	signal name	EHG CPU(A) NORMAL	EHG CPU(B) NORMAL	TCS-2 FAN TROUBLE	TCS-2 TEMP. HIGH	EHG P.S.(A) ABNORMAL	EHG P.S.(B) ABNORMAL	TCS-1 TEMP. HIGH	PS NORMAL

MODBUS IF DIGITAL INPUT(2)		SH. No.	TC0842
SIGNATURE	DATE	TITLE	
DWN. (S.Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T.Kawata	2014-04-18		
APPD. Y.Fukushima	2014-04-18		
REGD. Hitachi,Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 360	REV. 3

2016/07/28 TC0842 XSMM-TC

(LYD105A)

DO / logic SH. NO.	signal name								
	signal range/level								
CTL									
	CH. NO.	DO 0	DO 1	DO 2	DO 3	DO 4	DO 5	DO 6	DO 7
unit NO.	slot NO.								
HB1	9	21 3 20	18	16	14	12	10	8	6 (S)

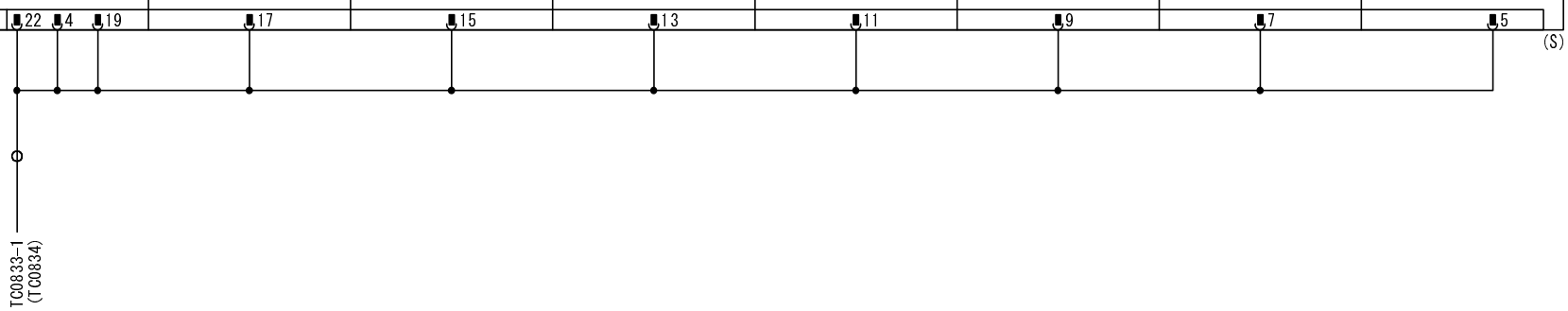
TC0833-1
(TC0833)

MODBUS IF DIGITAL OUTPUT(1)			SH. No.	TC0851
SIGNATURE		DATE	TITLE	
DWN.	(S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD.	T. Kawata	2014-04-18		
APPD.	Y. Fukushima	2014-04-18		
REGD.	Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 361	REV. 3

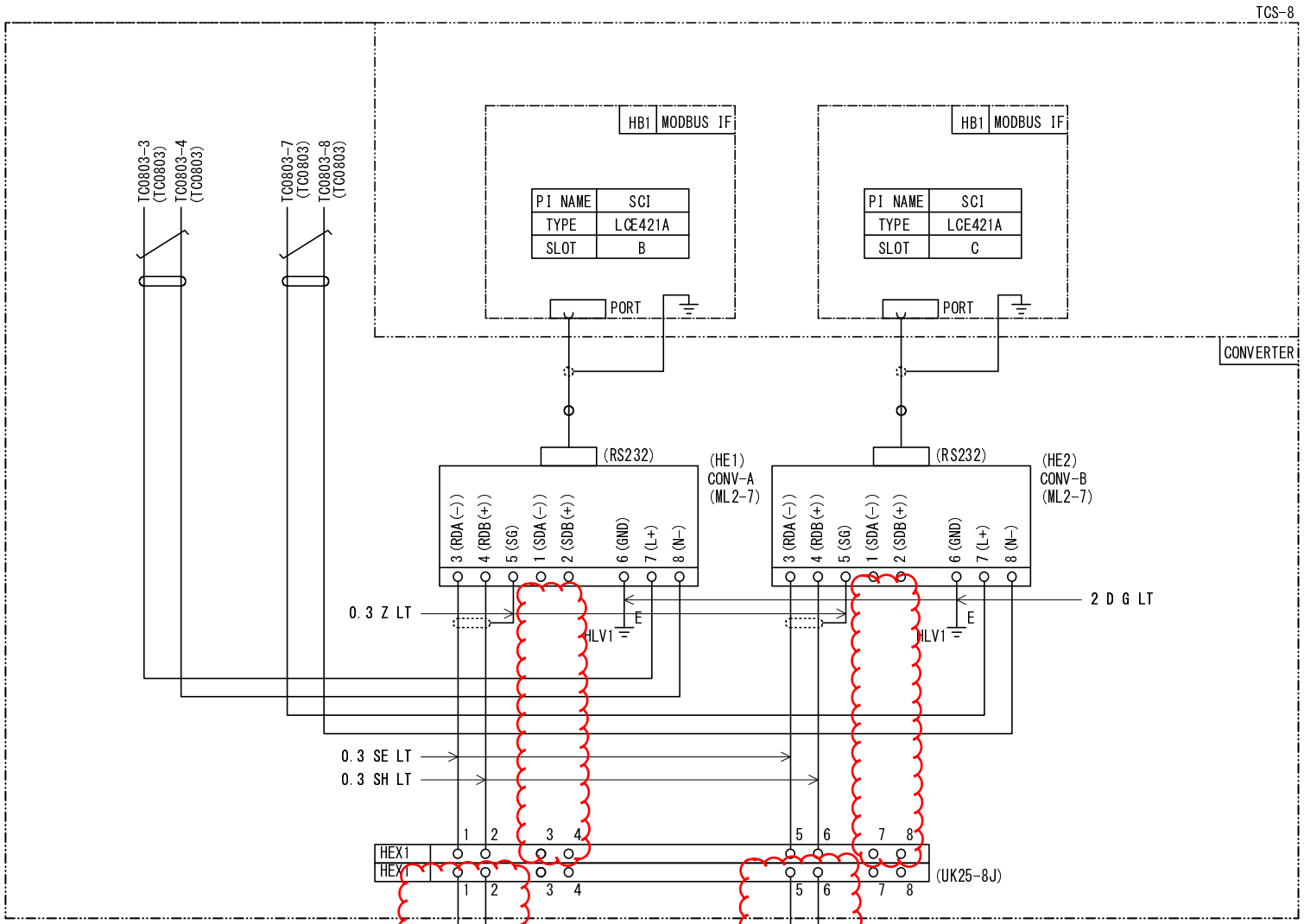
2016/07/28 TC0851 XSMM-TC

(LYD105A)

DO / logic SH. NO.	signal name									
	signal range/level									
CTL										
	CH. NO.	DO 8	DO 9	DO A	DO B	DO C	DO D	DO E	DO F	
unit NO.	slot NO.									
HB1	9	22 4 19	17	15	13	11	9	7	5	(S)



MODBUS IF DIGITAL OUTPUT(2)		SH. No.	TC0852
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 362	REV. 3



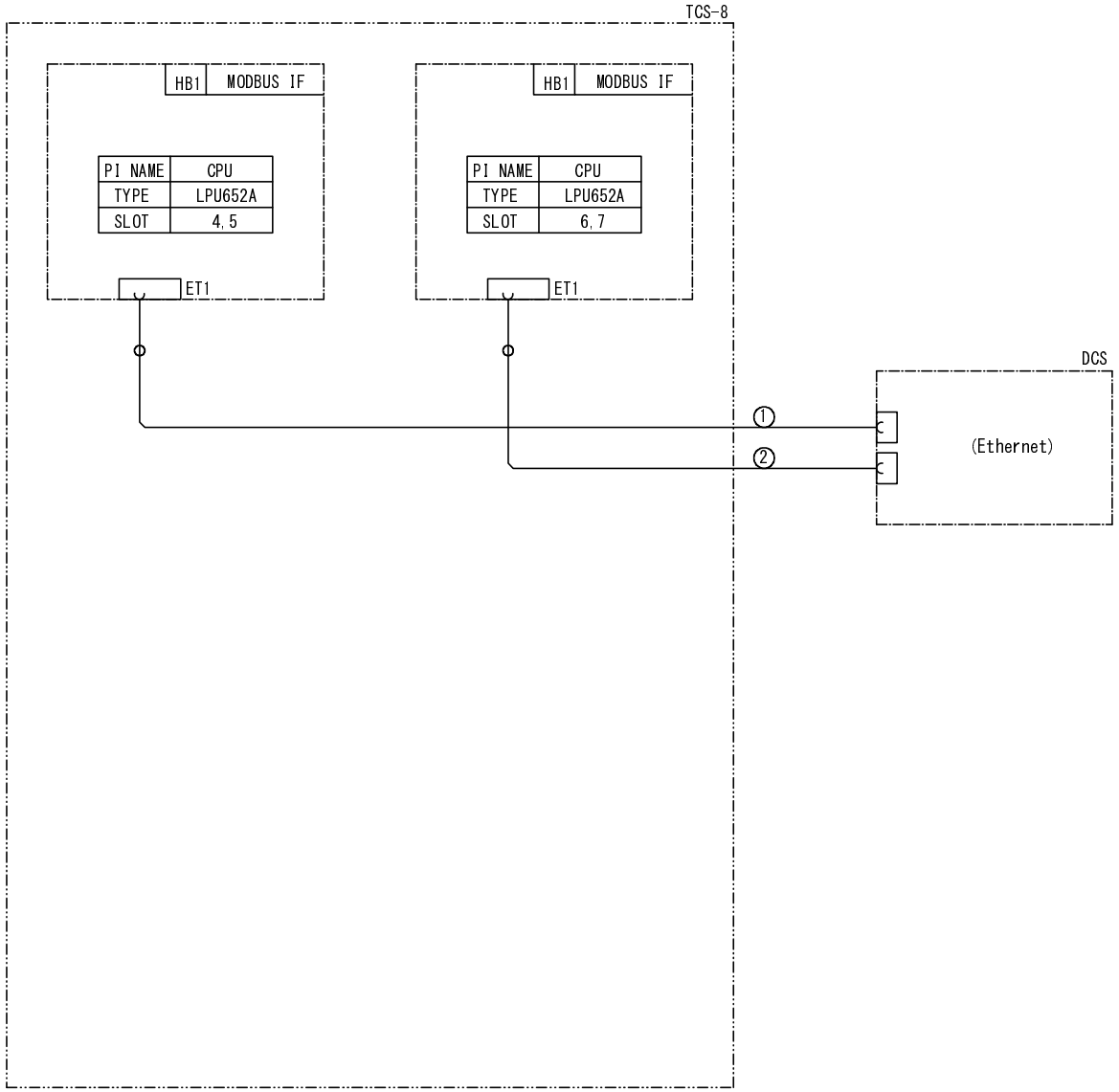
IN-DEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK
			CABLE CONNECTION		RELATED SH. NO.		
②		2/2	TCS-8	GPP			
①		2/2	TCS-8	GPP			

TRANSMISSION CABLE(1) SH. No. TC0860

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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○								
○								
○								
○								
○								
○								
②		1/1	TCS-8					
①		1/1	TCS-8					
IN-DEX	CABLE NO.	CORE NO.	FROM	TO	FROM	TO	REMARK	
			CABLE CONNECTION		RELATED SH. NO.			

TRANSMISSION CABLE(2)

SH. No. TC0861

	SIGNATURE	DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

REGD. Hitachi, Ltd. Tokyo Japan

DWG. No. 331DM07733

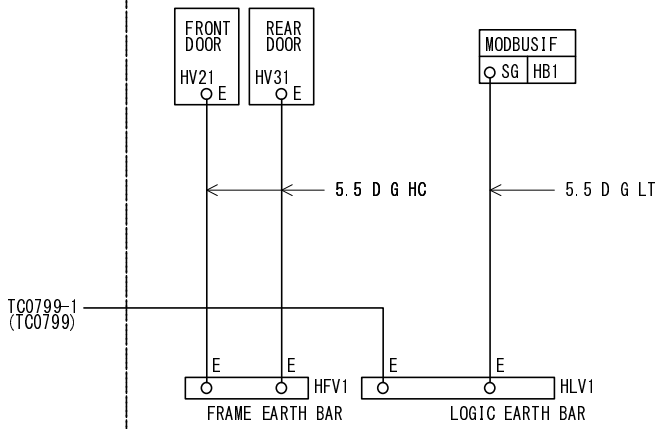
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TCS-7

TCS-8



22 D G HC

POWER SUPPLY SYSTEM EARTH CIRCUIT		SH. No.	TC0899
SIGNATURE	DATE	TITLE	
DWN. (S. Kikuchi)	2014-04-18	ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM	
CHKD. T. Kawata	2014-04-18		
APPD. Y. Fukushima	2014-04-18		
REGD. Hitachi, Ltd. Tokyo Japan	DWG. No. 331DM07733	PAGE 365	REV. 3

TCS-2

EHG CPU BE1

PI NAME	CPU
TYPE	LPU652A
SLOT	0, 1

PI NAME	CPU
TYPE	LPU652A
SLOT	2, 3

A B DCM A B DCM

TO EWS

TCS-6

COM CPU FE1

PI NAME	CPU
TYPE	LPU652A
SLOT	0, 1

PI NAME	CPU
TYPE	LPU652A
SLOT	2, 3

A B DCM A B DCM

TCS-8

MOD BUS IF HB1

PI NAME	CPU
TYPE	LPU652A
SLOT	4, 5

PI NAME	CPU
TYPE	LPU652A
SLOT	6, 7

A B DCM A B DCM

TO HMI

EHG CPU BE1

PI NAME	CE
TYPE	LCE250B
SLOT	D

PI NAME	CE
TYPE	LCE250B
SLOT	E

PI NAME	CE
TYPE	LCE250B
SLOT	F

PI NAME	CE
TYPE	LCE250B
SLOT	Z

R R R R

TCS-3

N N N N

PI NAME	UD
TYPE	LUD700A
SLOT	F

PI NAME	UD
TYPE	LUD070A
SLOT	UD

PI NAME	UD
TYPE	LUD070A
SLOT	UD

PI NAME	UD
TYPE	LUD070A
SLOT	UD

EHG PI/O BF1 TPC(A) CD1 TPC(B) CF1 TPC(C) CH1

COM CPU FE1

PI NAME	CE
TYPE	LCE250B
SLOT	Z

R

N

PI NAME	UD
TYPE	LUD700A
SLOT	F

COM PI/O FF1

OPTICAL CABLE UD CABLE CONECTION

SH. No. TC0900

SIGNATURE		DATE
DWN.	(S. Kikuchi)	2014-04-18
CHKD.	T. Kawata	2014-04-18
APPD.	Y. Fukushima	2014-04-18

TITLE: ELEMENTARY DIAGRAM OF TURBINE CONTROL SYSTEM

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2016/07/28 TC0900 XSMM-TC